

Workshop Manual

Amarok 2011 ➤, Amarok 2017 ➤,
Caddy 2016 ➤, Caddy 2021 ➤,
Crafter 2006 ➤, Crafter 2017 ➤,
Transporter 2016 ➤, Transporter 2020 ➤

Factory-installed accessories and conversions

Edition 09.2021



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Technical information should always be available to the foremen and mechanics, because their careful and constant adherence to the instructions is essential to ensure vehicle road-worthiness and safety. In addition, the normal basic safety precautions for working on motor vehicles must, as a matter of course, be observed.



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00 – Technical data

1 Tipper body (Henschel)

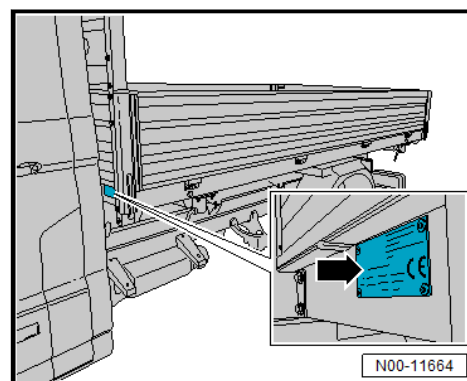
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1.1 Data

⇒ [p1.1.1 late", page 1](#)

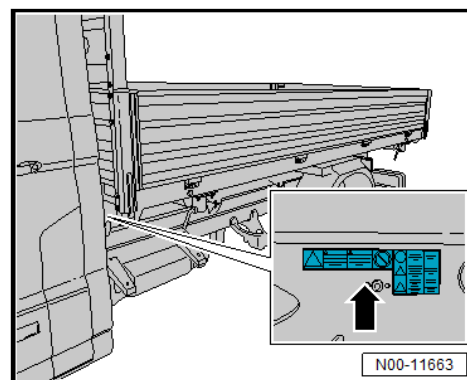
⇒ [a1.1.2 nd maintenance instructions", page 1](#)

1.1.1 Type plate



The type plate -arrow- is located at the front left on the bulk-head.

1.1.2 Operating and maintenance instructions



The operating and maintenance instructions -arrow- are located on the left on the frame of the tipper body.



2 Box body (Junge)

2.1 Data

⇒ [p2.1.1 late", page 2](#)

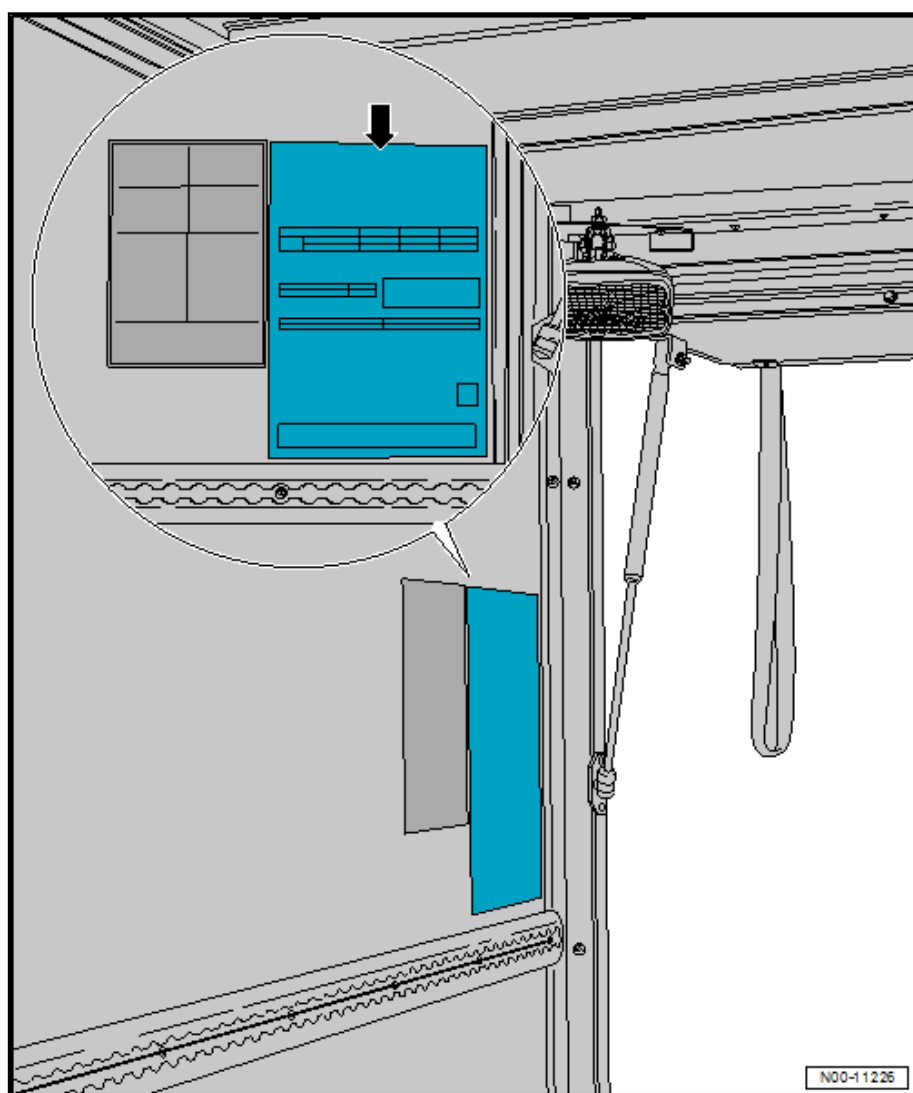
⇒ [a2.1.2 nd maintenance instructions", page 2](#)

2.1.1 Type plate



The type plate -arrow- is located at the front right of the box body behind the vehicle door.

2.1.2 Operating and maintenance instructions



Operating and maintenance instructions -arrow- are located at rear right on inside of box body.



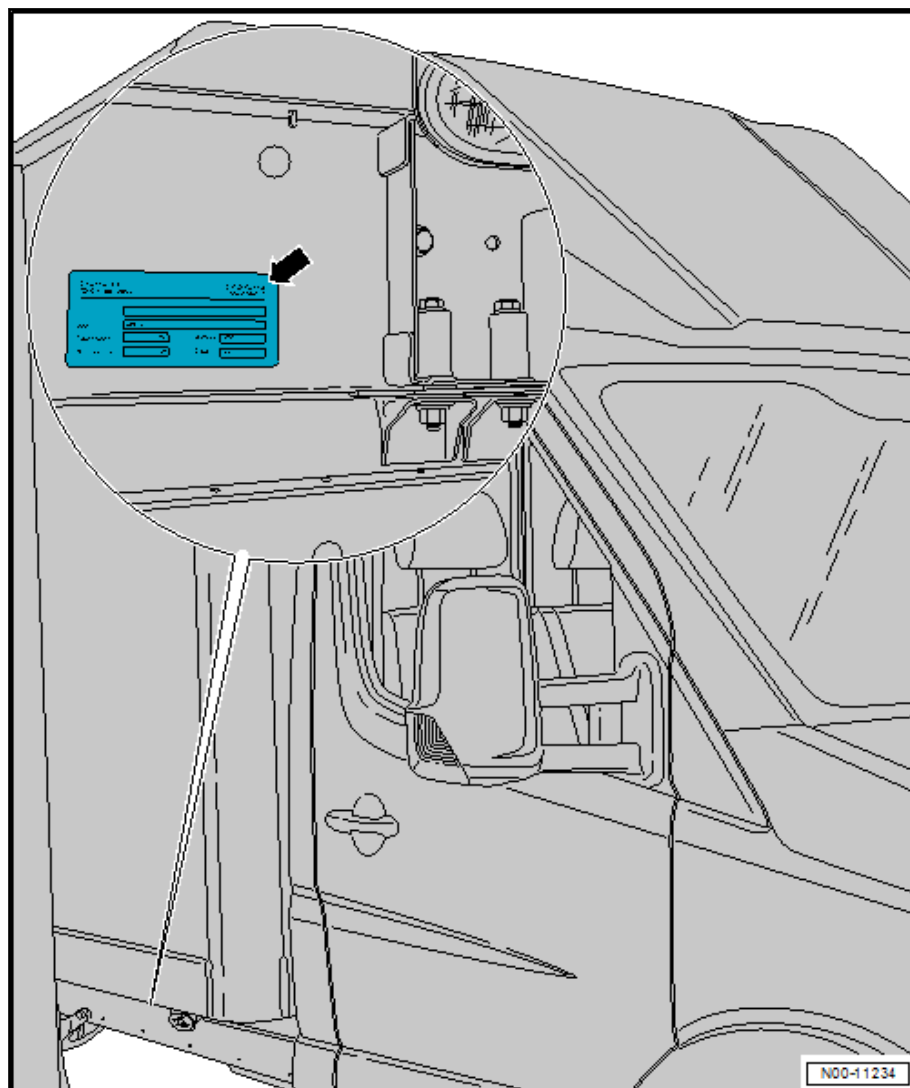
3 Box body (Spier)

3.1 Data

⇒ [p3.1.1 late", page 4](#)

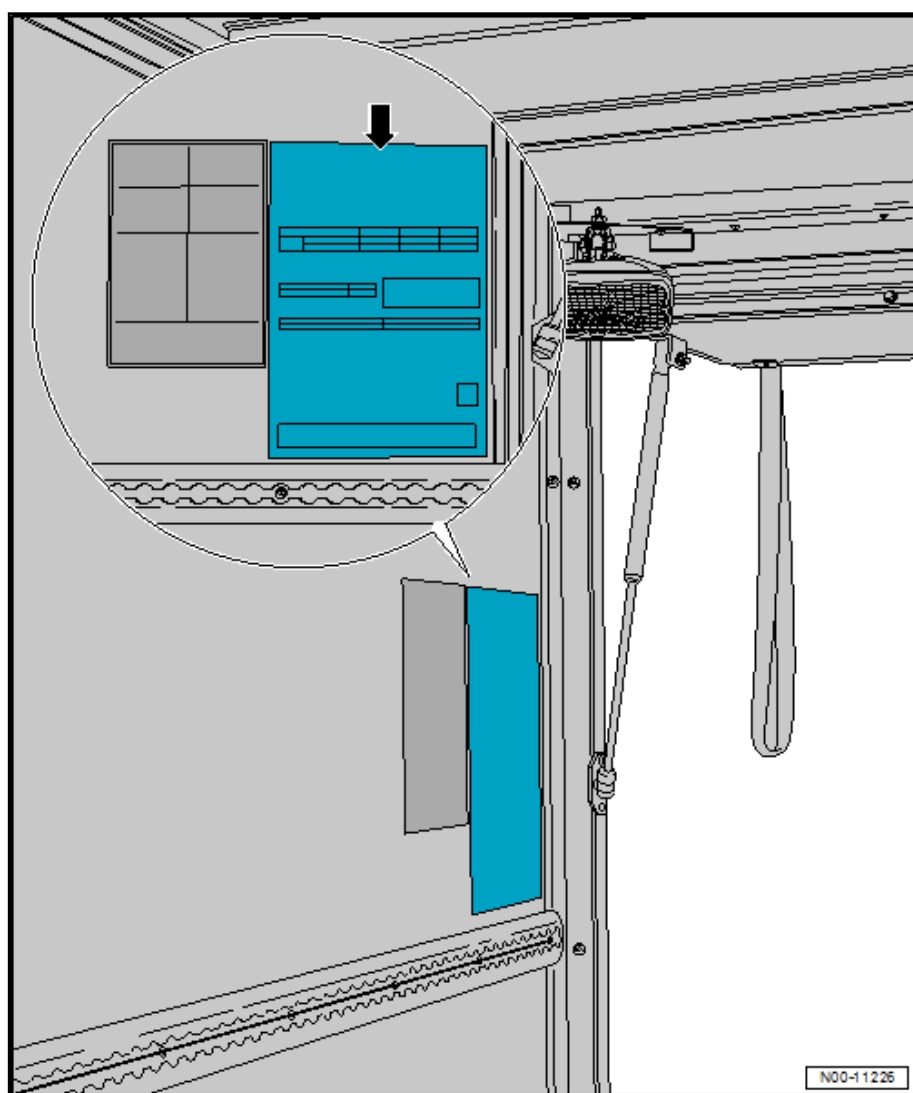
⇒ [a3.1.2 nd maintenance instructions", page 4](#)

3.1.1 Type plate



The type plate -arrow- is located at the front right on the vehicle frame.

3.1.2 Operating and maintenance instructions



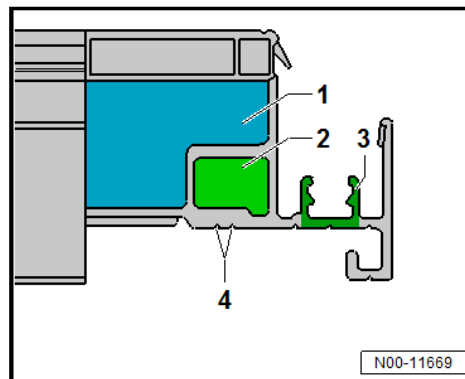
Operating and maintenance instructions -arrow- are located at rear right on inside of box body.



4 Box body (Schmitz Cargobull)

4.1 General information

The profiles of the floor plate feature several chambers that are separate from each other.



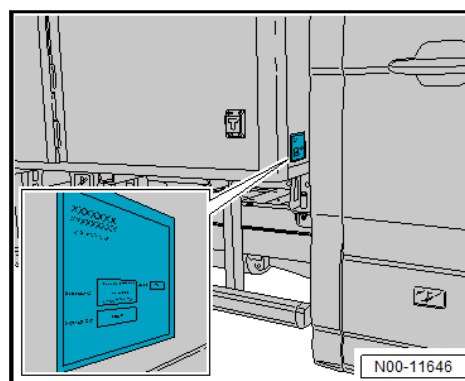
- ◆ Wiring harnesses are routed in chamber -1-.
- ◆ Chamber -2- is the securing channel, which protects the wiring harnesses.
- ◆ Chamber -2- is marked by 2 grooves -4- on the exterior.
- ◆ Only in the area of the grooves -4- is it permissible to drill holes for securing accessories.
- ◆ The clamp groove -3- serves as a means of securing the side panel.

4.2 Data

⇒ [p4.2.1 late", page 6](#)

⇒ [a4.2.2 nd maintenance instructions", page 7](#)

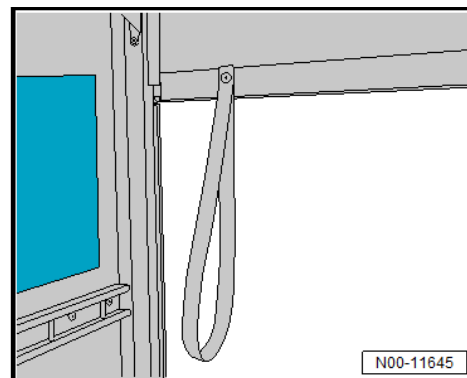
4.2.1 Type plate



The type plate is located at the front right on the box body.



4.2.2 Operating and maintenance instructions



The operating and maintenance instructions are located at the rear right on the inside of the box body.



5 Tail lift (Sörensen)

⇒ [i5.1 nformation", page 8](#)

⇒ [5.2 , page 9](#)

5.1 General information

- ◆ Before commencing with the repair measures, observe the manufacturer's general notes.
- ◆ Modifications to the tail lift may not be carried out unless written approval from the manufacturer's engineering department has been given.



WARNING

Injury hazard due to improper handling of the tail lift.

- The tail lift may only be operated, serviced and repaired by persons who have received relevant induction training on the technology and, in particular, have been instructed about the hazards associated with operation.
- The accident prevention regulations must be strictly adhered to, especially BGR 500 lifting platforms (VBG 14) in this case. The main regulations can be gleaned from the inspection booklet of the tail lift.

Safety precautions

Before commencing with repairs and maintenance measures, secure the tail lift to prevent it from moving unintentionally.

The tail lift must be observed at all times while functions are carried out using the controls.

The entire area of tail lift movement is to be kept clear of persons and objects.

Pay particular attention to crushing and shearing areas between the platform and body as well as between the platform and road surface.

Electric motor for hydraulic unit

To prevent damage to the electric motor and power relay, a sufficient supply of voltage must be assured for operation of the tail lift.

If it becomes apparent that the electric motor is struggling, the lifting procedure must be stopped immediately and the batteries must be charged.

Foot switch

Using the foot switch on the platform, which is installed protected in a pressure cast aluminium housing, the tail lift can be lowered down to the ground and raised up to the height of the vehicle bed.

Tilting of the platform to the horizontal position from floor level is initiated automatically.

The foot switch has priority over other commands. When the tail lift is operated via the foot switch, any other means of controlling it (operating panel, control box) are automatically blocked electrically.

Hydraulic system

The hydraulic system is bled automatically during operation of the tail lift.



Damaged hydraulic hoses are to be replaced without delay.

Lifting mechanism

The lifting mechanism (flange console as well as lifting arm, closing arm and skid plate) leaves the factory with the surface completely finished and painted black (RAL 9005).

All bearing points are maintenance-free and do not have to be lubricated. Lubrication measures are only necessary in the event of repairs when the pins are inserted in the bearing points.

Platform

The platform consists of interlinked hollow aluminium profile sections that are given their ultimate level of stability by means of pointed and final profiles.

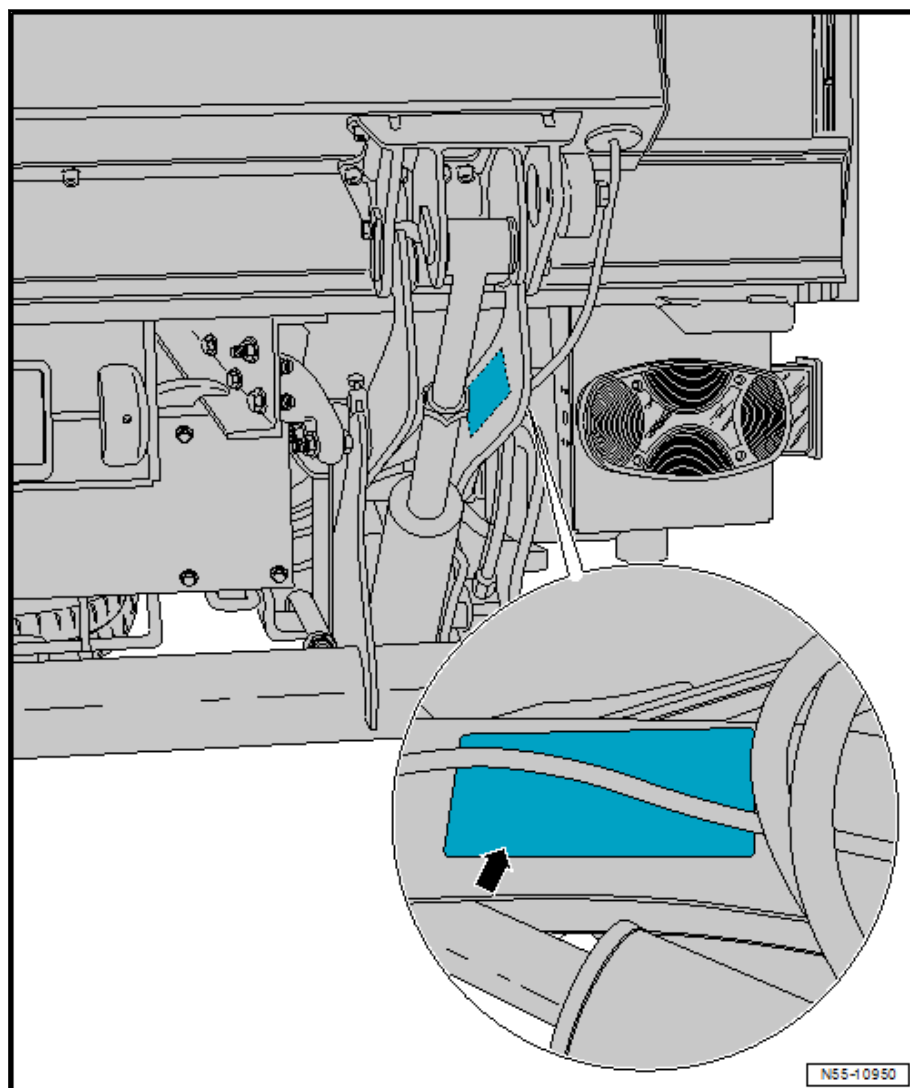
5.2 Data

⇒ [p5.2.1 late", page 9](#)

⇒ [i5.2.2 nformation", page 10](#)

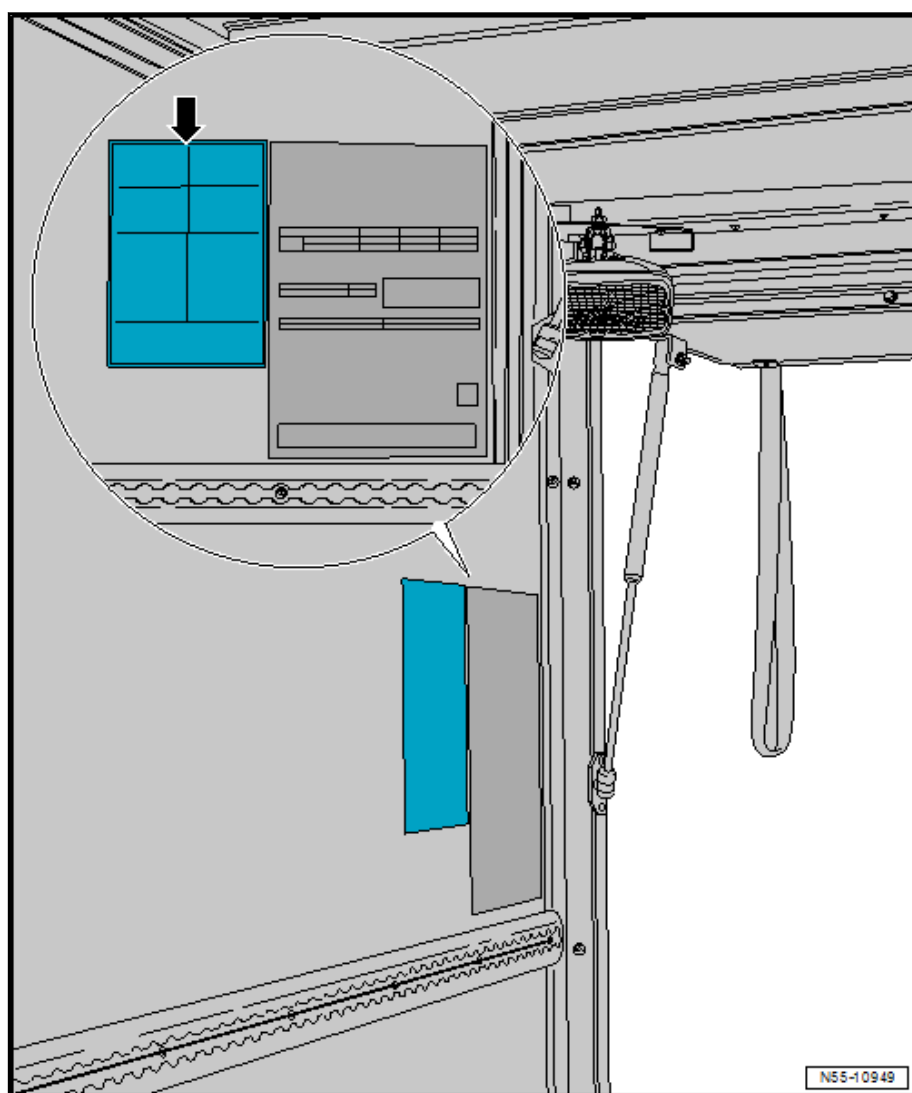
5.2.1 Type plate

- ◆ The factory number is bonded to the type plate on the closing arm, on the right as seen in the direction of travel. A second type plate is bonded to the inside of the assembly hood, on the right as seen in the direction of travel.
- ◆ In addition, the equipment number is stamped in the securing flange, on the left as seen in the direction of travel.



Type plate -arrow- with factory number is located on closing arm of tail lift on right (as seen in direction of travel).

5.2.2 Safety information



Safety instructions -arrow- are located at rear right on inside of box body.

- Observe specifications of tail lift manufacturer.
- Observe the maximum load of the tail lift platform for every operating state (raising and lowering).



6 Tail lift (BÄR - Cargolift)

⇒ [i6.1 nformation", page 12](#)

⇒ [6.2 , page 13](#)

6.1 General information

- ◆ Before commencing with the repair measures, observe the manufacturer's general notes.
- ◆ Modifications to the tail lift may not be carried out unless written approval from the manufacturer's engineering department has been given.



WARNING

Injury hazard due to improper handling of the tail lift.

- The tail lift may only be operated, serviced and repaired by persons who have received relevant induction training on the technology and, in particular, have been instructed about the hazards associated with operation.
- The accident prevention regulations must be strictly adhered to, especially BGR 500 lifting platforms (VBG 14) in this case. The main regulations can be gleaned from the inspection booklet of the tail lift.

Safety precautions

Before commencing with repairs and maintenance measures, secure the tail lift to prevent it from moving unintentionally.

The tail lift must be observed at all times while functions are carried out using the controls.

Solenoid valves may only be removed when the platform is open and lying on the floor.

The entire area of tail lift movement is to be kept clear of persons and objects.

Pay particular attention to crushing and shearing areas between the platform and body as well as between the platform and road surface.

Electric motor for hydraulic unit

To prevent damage to the electric motor and power relay, a sufficient supply of voltage must be assured for operation of the tail lift.

If it becomes apparent that the electric motor is struggling, the lifting procedure must be stopped immediately and the batteries must be charged.

Foot switch

Using the foot switch on the platform, which is installed protected in a pressure cast aluminium housing, the tail lift can be lowered down to the ground and raised up to the height of the vehicle bed.

Tilting of the platform to the horizontal position from floor level is initiated automatically.

The foot switch has priority over other commands. When the tail lift is operated via the foot switch, any other means of controlling it (operating panel, control box) are automatically blocked electrically.

Hydraulic system

The hydraulic system is bled automatically during operation of the tail lift.

Damaged hydraulic hoses are to be replaced without delay.

Lifting mechanism

The bearing points are not maintenance-free and must be lubricated.

Platform

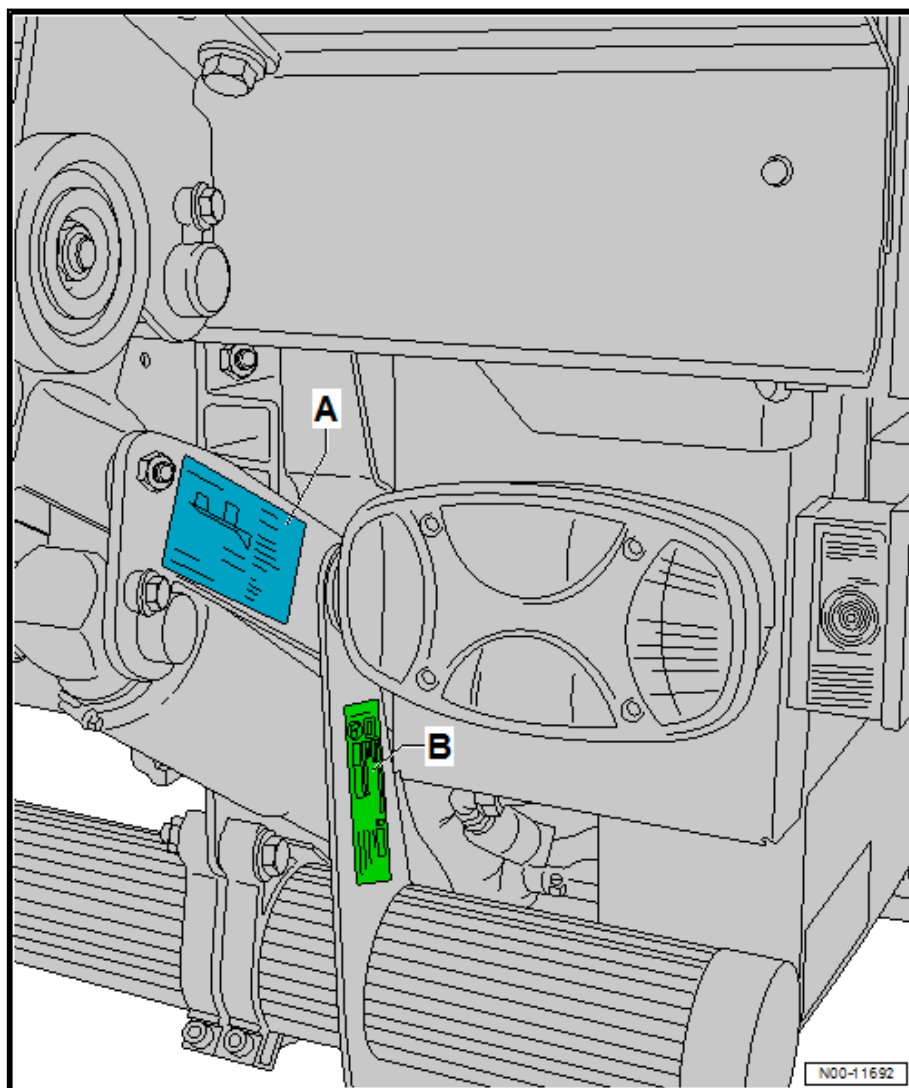
The platform consists of interlinked hollow aluminium profile sections that are given their ultimate level of stability by means of pointed and final profiles.

6.2 Data

⇒ [p6.2.1 late", page 13](#)

⇒ [i6.2.2 nformation", page 14](#)

6.2.1 Type plate

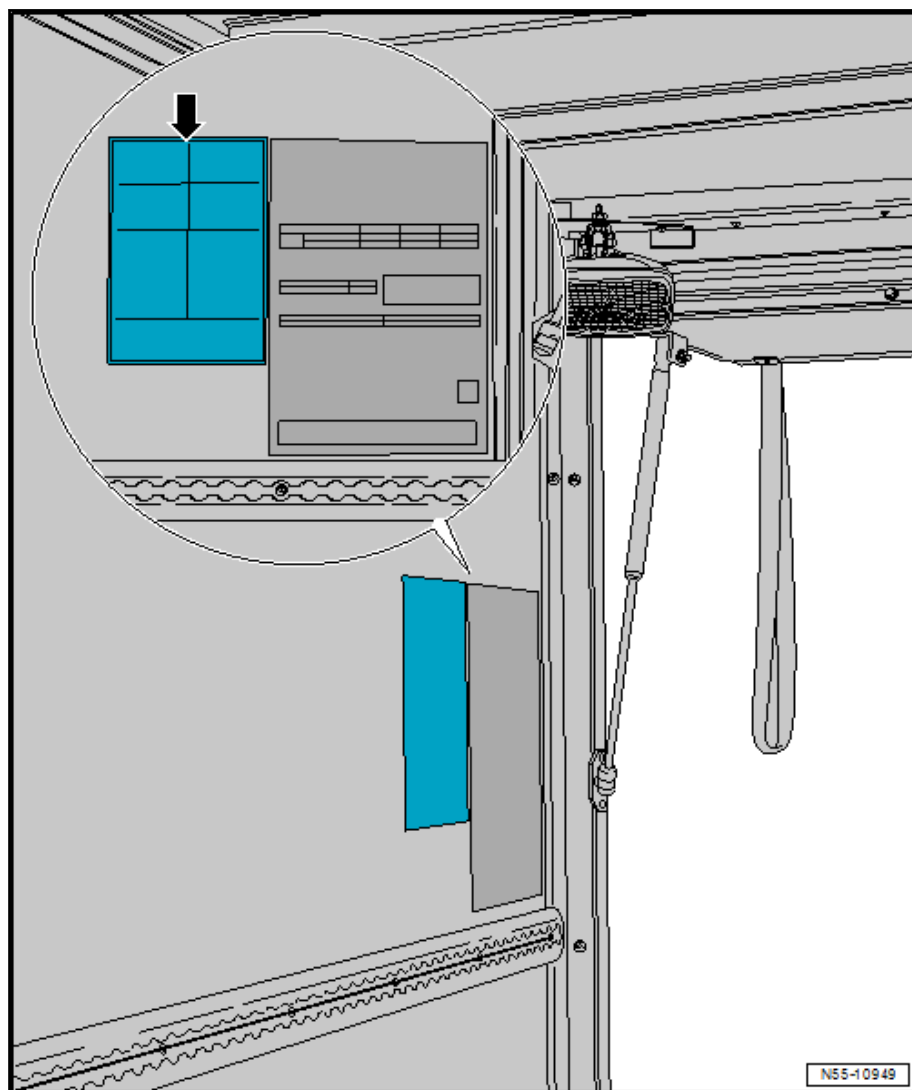


The serial number is located next to the type plate on the support tube -arrow A- and also in the following locations:



- ◆ Underbody guard -arrow B-
- ◆ Inspection book
- ◆ eBC controller
- ◆ In addition, the equipment number is stamped in the securing flange, on the left as seen in the direction of travel.

6.2.2 Safety information



Safety instructions -arrow- are located at rear right on inside of box body.

- Observe specifications of tail lift manufacturer.
- Observe maximum load of tail lift platform for every operating state (raising and lowering).



7 Tail lift (Palfinger)

⇒ [7.1 information", page 15](#)

⇒ [7.2, page 16](#)

7.1 General information

- ◆ Before commencing with the repair measures, observe the manufacturer's general notes.
- ◆ Modifications to the tail lift may not be carried out unless written approval from the manufacturer's engineering department has been given.



WARNING

Injury hazard due to improper handling of the tail lift.

- The tail lift may only be operated, serviced and repaired by persons who have received relevant induction training on the technology and, in particular, have been instructed about the hazards associated with operation.
- The accident prevention regulations must be strictly adhered to, especially BGR 500 lifting platforms (VBG 14) in this case. The main regulations can be gleaned from the inspection booklet of the tail lift.

Safety precautions

Before commencing with repairs and maintenance measures, secure the tail lift to prevent it from moving unintentionally.

The tail lift must be observed at all times while functions are carried out using the controls.

Solenoid valves may only be removed when the platform is open and lying on the floor.

The entire area of tail lift movement is to be kept clear of persons and objects.

Pay particular attention to crushing and shearing areas between the platform and body as well as between the platform and road surface.

Electric motor for hydraulic unit

To prevent damage to the electric motor and power relay, a sufficient supply of voltage must be assured for operation of the tail lift.

If it becomes apparent that the electric motor is struggling, the lifting procedure must be stopped immediately and the batteries must be charged.

Foot switch

Using the foot switch on the platform, which is installed protected in a pressure cast aluminium housing, the tail lift can be lowered down to the ground and raised up to the height of the vehicle bed.

Tilting of the platform to the horizontal position from floor level is initiated automatically.

Hydraulic system

The hydraulic system is bled automatically during operation of the tail lift.

Damaged hydraulic hoses are to be replaced without delay.



Lifting mechanism

All bearing points are maintenance-free and do not have to be lubricated.

Platform

The platform consists of interlinked hollow aluminium profile sections that are given their ultimate level of stability by means of pointed and final profiles.

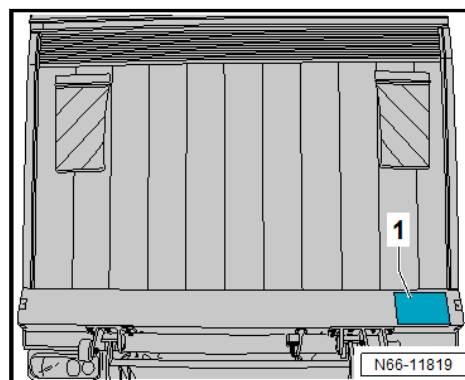
7.2 Data

⇒ p7.2.1 late", page 16

⇒ i7.2.2 nformation", page 16

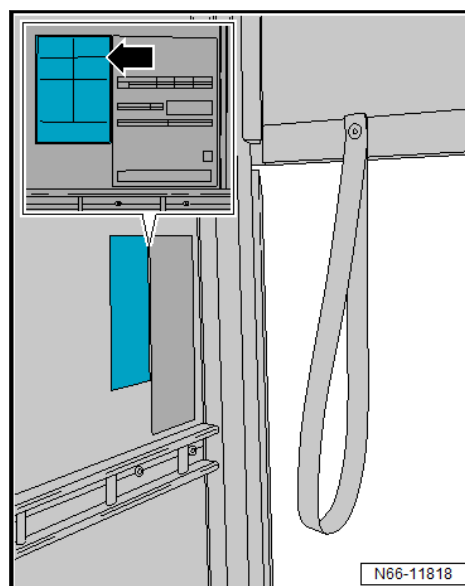
7.2.1 Type plate

The type plate -1- stating the factory number is located on the right of the tail lift (as seen in direction of travel).



7.2.2 Safety information

Safety instructions -arrow- are located at rear right on inside of box body.



- Observe maximum load of tail lift platform for every operating state (raising and lowering).
- Observe specifications of tail lift manufacturer.



8 Design double tube bar

⇒ [i8.1 nformation", page 17](#)

8.1 Safety information

The work and safety instructions in these installation instructions must be strictly observed.

Damage caused by non-observance is excluded from any liability.



9 Hazard classification of high-voltage system

DANGER

The vehicle's high-voltage system and the high-voltage battery are dangerous and can cause burns or other injuries and even lead to a fatal electric shock.

- Any work on the high-voltage system, or on systems which could be indirectly affected by it, may only be carried out by properly trained and qualified expert personnel.
- In the event of queries or uncertainties regarding the terms "high-voltage technician" or "high-voltage expert", or those concerning the high-voltage system, the responsible importer must be contacted prior to the commencement of any work.
- Any repair work must be performed in accordance with applicable laws and regulations, the recognised engineering practices, any relevant accident prevention regulations (in Germany, including but not limited to the Information of the German Social Accident Insurance (DGUV) 200-005 – Qualification training for work on vehicles with high-voltage systems), as well as this workshop manual.

The classification of the qualification levels, i.e. the respective explanations, as well as the tabular listing of the various tasks performed on the high-voltage system can be found here:

- ◆ Risk category of the high-voltage system ⇒ Electric motor (270, LS2); Rep. gr. 00; Risk category of high-voltage system.



27 – Starter, current supply, CCS

1 Preparation for second battery

⇒ [o1.1 overview – preparation for second battery”, page 19](#)

⇒ [o1.2 fitting locations – preparation for second battery”, page 20](#)

⇒ [a1.3 and installing second battery charging circuit relayJ713”, page 22](#)

⇒ [o1.4 function of second battery charging circuit relayJ713”, page 23](#)

1.1 System overview – preparation for second battery



1 - Dash panel insert -KX2-

- ❑ Removing and installing ⇒ Electrical system; Rep. gr. 90; Dash panel insert; Removing and installing dash panel insert KX2

2 - CAN bus wire

3 - Special vehicle control unit -J608-

- ❑ Removing and installing ⇒ Electrical system; Rep. gr. 97; Control units; Removing and installing special vehicle control unit -J608-

4 - E-LIN bus wire 2

5 - Battery monitor control unit 2 -J934-

6 - Second battery

- ❑ Different fitting locations, depending on particular special vehicle

7 - Electrical system connection for body builder

8 - Second battery charging circuit relay -J713-

- ❑ Capacity: 150 A
- ❑ The contact in the middle is intended for the body builder.
- ❑ Fuse for B+ is not inserted.
- ❑ Notes on function of second battery charging circuit relay -J713- ⇒ [page 23](#)
- ❑ Removing and installing ⇒ [page 22](#)

9 - Alternator

- ❑ Removing and installing ⇒ Electrical system; Rep. gr. 27; Alternator; Removing and installing alternator
- ❑ Checking ⇒ Electrical system; Rep. gr. 27; Alternator; Checking alternator

10 - Battery

- ❑ Removing and installing ⇒ Electrical system; Rep. gr. 27; Battery; Removing and installing battery

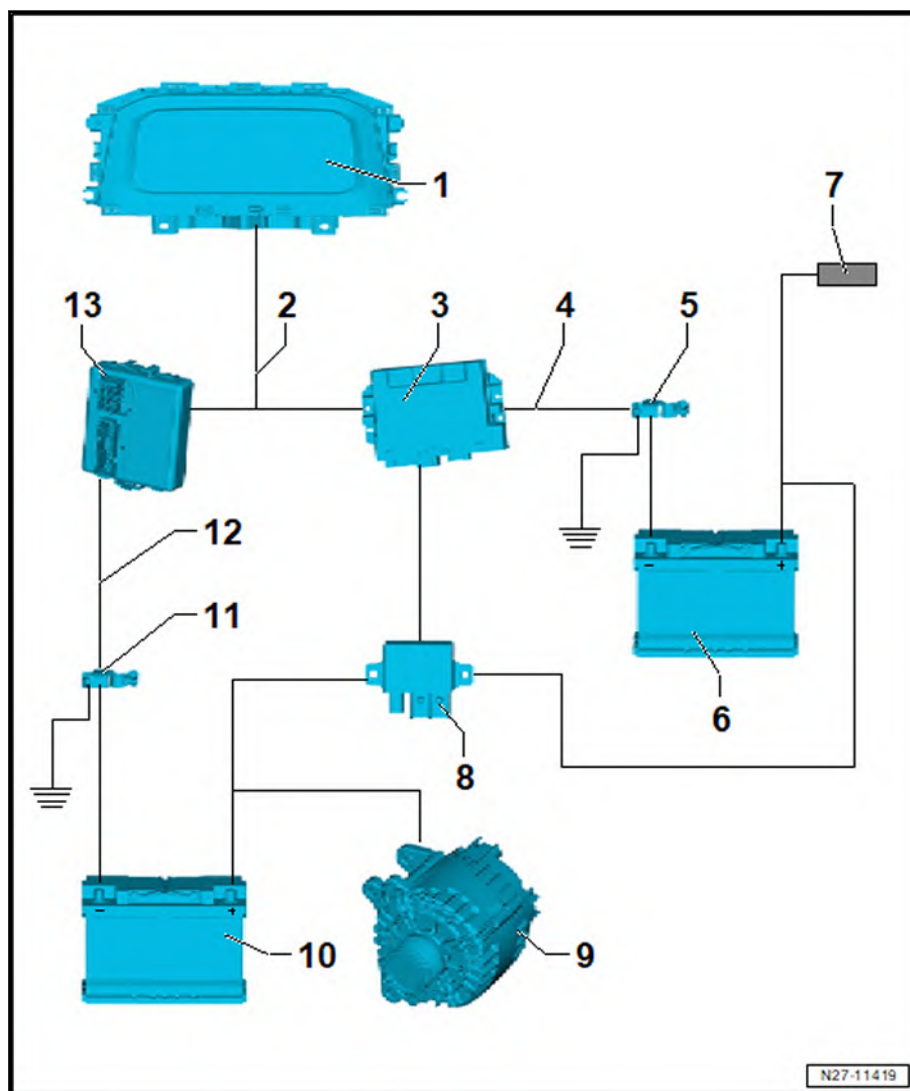
11 - Battery monitor control unit -J367-

- ❑ Removing and installing ⇒ Electrical system; Rep. gr. 27; Battery; Removing and installing earth wire with battery monitor control unit -J367-

12 - E-LIN bus wire 1

13 - Data bus diagnostic interface -J533-

- ❑ Removing and installing ⇒ Electrical system; Rep. gr. 97; Control units; Removing and installing data bus diagnostic interface -J533-



1.2 Overview of fitting locations – preparation for second battery

1 - Dash panel insert -KX2-

- ☐ Removing and installing ⇒ Electrical system; Rep. gr. 90; Dash panel insert; Removing and installing dash panel insert KX2

2 - Special vehicle control unit -J608-

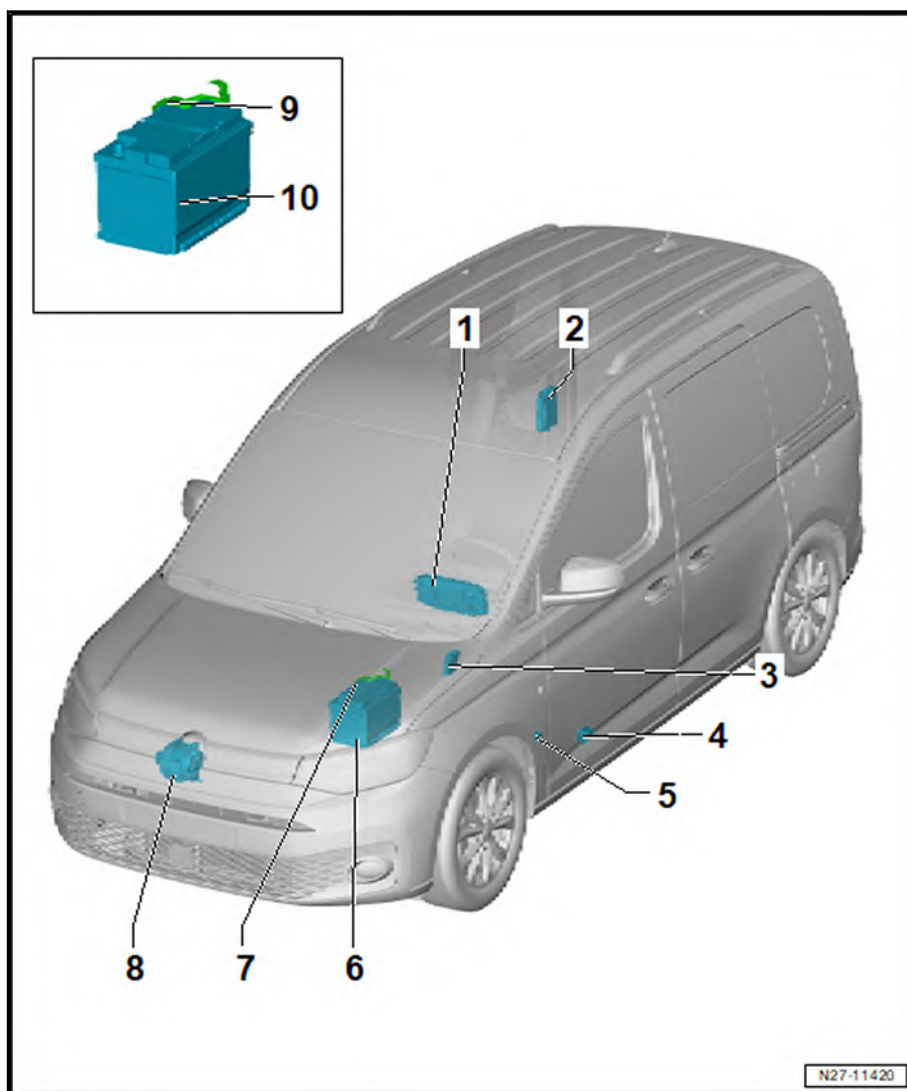
- ☐ Removing and installing ⇒ Electrical system; Rep. gr. 97; Control units; Removing and installing special vehicle control unit -J608-

3 - Data bus diagnostic interface -J533-

- ☐ Removing and installing ⇒ Electrical system; Rep. gr. 97; Control units; Removing and installing data bus diagnostic interface -J533-

4 - Second battery charging circuit relay -J713-

- ☐ On inside of outer seat cross member
- ☐ Fuse for B+ is not inserted.
- ☐ With LIN bus wire coming from special vehicle control unit -J608-
- ☐ Removing and installing ⇒ [page 22](#)



5 - Electrical connector

- ☐ B+ coupling point for body builder, behind lower A-pillar trim
- ☐ Removing and installing A-pillar trim ⇒ General body repairs, interior; Rep. gr. 70; Interior trims; Removing and installing A-pillar trim

6 - Battery

- ☐ Removing and installing ⇒ Electrical system; Rep. gr. 27; Battery; Removing and installing battery

7 - Battery monitor control unit -J367-

- ☐ Removing and installing ⇒ Electrical system; Rep. gr. 27; Battery; Removing and installing earth wire with battery monitor control unit -J367-

8 - Alternator

- ☐ Removing and installing ⇒ Electrical system; Rep. gr. 27; Alternator; Removing and installing alternator

9 - Battery monitor control unit 2 -J934-

- ☐ Different fitting locations, depending on particular special vehicle

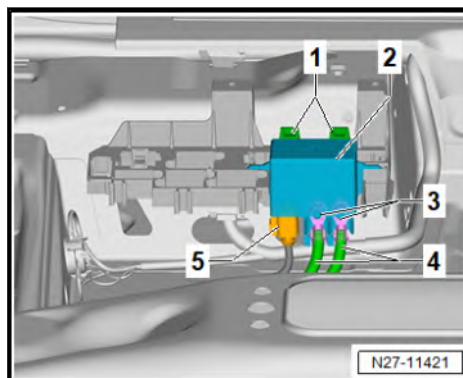
10 - Second battery

- ☐ Different fitting locations, depending on particular special vehicle



1.3 Removing and installing second battery charging circuit relay -J713-

Removing



- Remove left front seat ⇒ General body repairs, interior; Rep. gr. 72; Front seats; Removing and installing front seat.
- Remove sill panel moulding ⇒ General body repairs, exterior; Rep. gr. 66; Mouldings/trims/extensions; Removing and installing sill panel moulding.
- Disconnect battery in engine compartment ⇒ Electrical system; Rep. gr. 27; Battery; Disconnecting and connecting battery.
- Disconnect second battery ⇒ Electrical system; Rep. gr. 27; Battery; Disconnecting and connecting battery.
- Fold back carpet far enough to gain access to second battery charging circuit relay -J713- on inner side of outer seat cross member.
- Press catches -1-. While doing so, pull out second battery charging circuit relay -J713- -2-.
- Unscrew nuts -3-.
- Remove wiring harnesses -4-.
- Mark electrical wiring harnesses -4- according to their allocation to the both connections.
- Disconnect electrical connector -5-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ Battery; Assembly overview – battery ⇒ Electrical system; Rep. gr. 27; Battery; Assembly overview – battery
- ♦ Front seats; Assembly overview – front seat ⇒ General body repairs, interior; Rep. gr. 72; Front seats; Assembly overview – front seat

Component	Specified torque
Nuts for electrical wiring on second battery charging circuit relay -J713-	6 Nm



1.4 Notes on function of second battery charging circuit relay -J713-

Switching states of second battery charging circuit relay -J713-

- ◆ Closed when engine is running safely and there is no energy-critical condition.
- ◆ Closed when external charging is detected with engine not running.
- ◆ Open with engine running and energy recovery active, when charge level of second battery is more than 90% and only low current flows.
- ◆ Open when engine is not running or when engine start-stop system is in stop mode.
- ◆ Open for a period of one minute when external charging is detected with engine not running and ignition switched on.
- ◆ Open at engine start.



53 – Body - rear

1 Storage well liner and load compartment trim for rear lid

⇒ [o1.1 verview - storage well liner", page 24](#)

⇒ [a1.2 nd installing storage well liner", page 25](#)

⇒ [a1.3 nd installing load compartment for rear lid", page 26](#)

1.1 Assembly overview - storage well liner

1 - Storage well liner

- ☐ Removing and installing ⇒ [page 25](#)

2 - Washers

- ☐ Left and right
- ☐ Qty. 8

3 - Fastening rings

- ☐ Left and right
- ☐ Folding
- ☐ Qty. 2

4 - Bolts

- ☐ Left and right
- ☐ Qty. 8
- ☐ 20 Nm

5 - Fastening rings

- ☐ Left and right
- ☐ Rigid
- ☐ Qty. 2

6 - Load compartment trim

- ☐ Removing and installing ⇒ [page 26](#)

7 - Plugs

- ☐ Left and right
- ☐ Qty. 2

8 - Bolts

- ☐ Left and right
- ☐ Qty. 10
- ☐ 2 Nm

9 - Cover

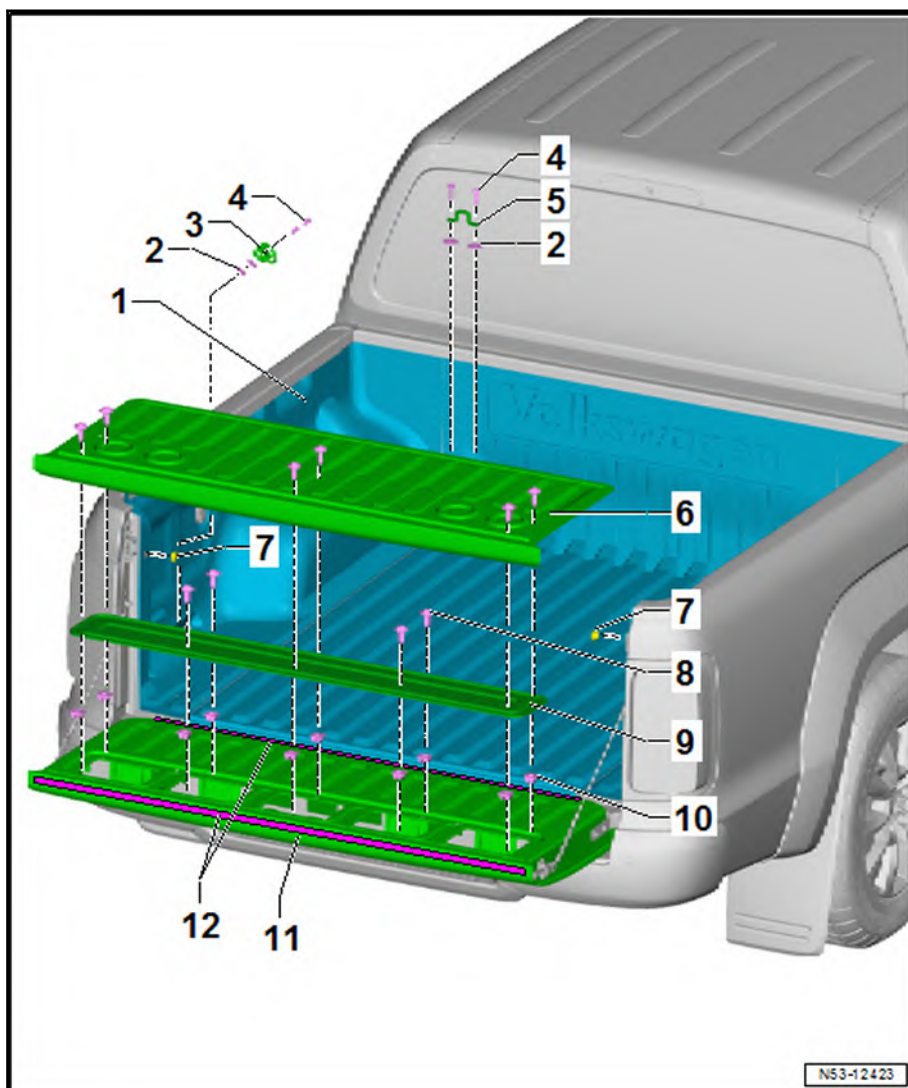
10 - Expanding nuts

- ☐ Qty. 10

11 - Tailboard

12 - Adhesive tape

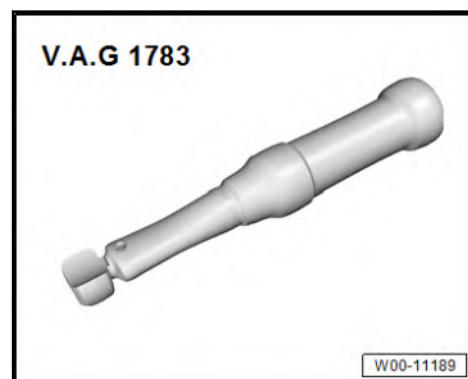
- ☐ Double-sided
- ☐ Qty. 2



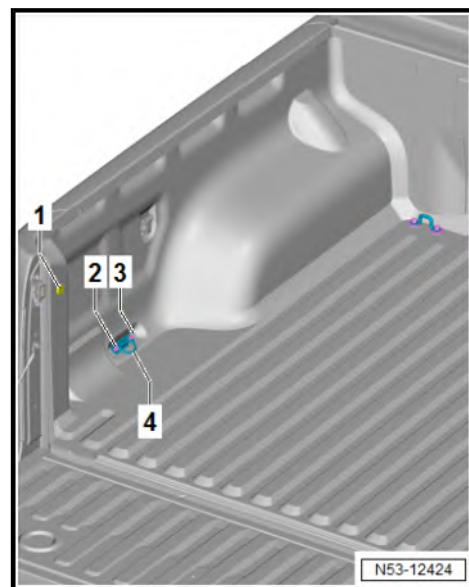
1.2 Removing and installing storage well liner

Special tools and workshop equipment required

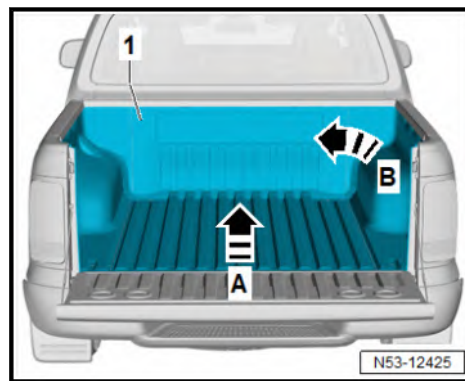
- ◆ Torque wrench -V.A.G 1783-



Removing



- Open tailboard.
- Remove plugs -1-.
- Unscrew bolts -2-.
- Remove fastening ring -4-.
- Remove washers -3-.
- Remove remaining fastening rings on storage well liner in same way.
- Push centre part of storage well liner -1- upwards in direction of -arrow A-.



- Remove storage well liner -1- in direction of -arrow B-.

Installing

Installation is carried out in reverse order.

1.3 Removing and installing load compartment for rear lid

Special tools and workshop equipment required

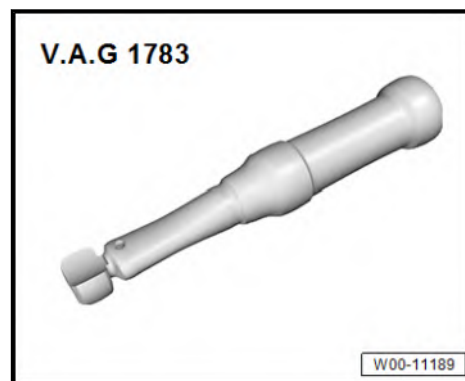
- ♦ Hot-air blower -V.A.G 1416-



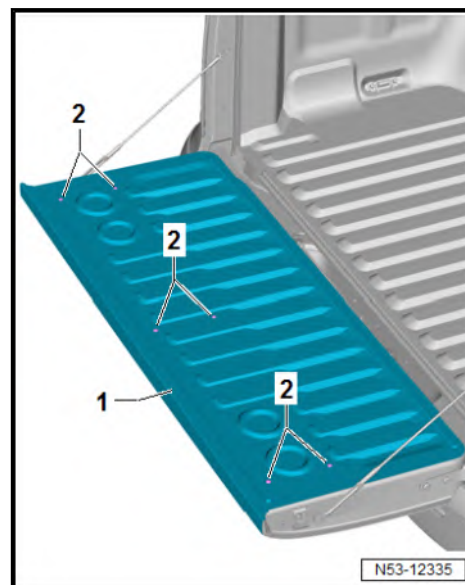
- ♦ Adhesive strip remover -VAS 6349-



◆ Torque wrench -V.A.G 1783-



Removing

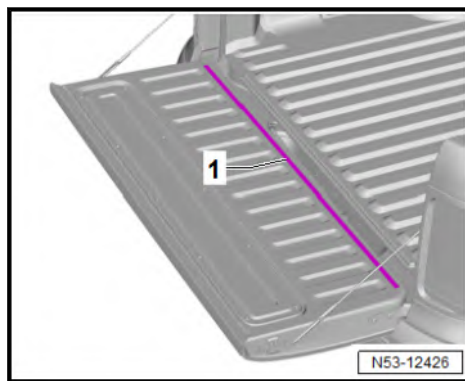


- Open rear lid.
- Unscrew bolts -2-.
- Remove load compartment trim for rear lid -1-.

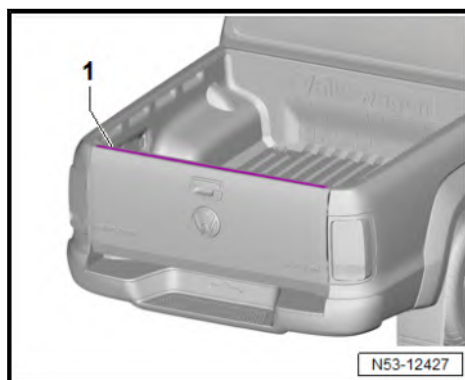
Installing

Install in reverse order of removal, observing the following:

- Heat-up double-sided adhesive tape using hot air blower -V.A.G 1416-.
- Remove double-sided adhesive tape.
- Remove any adhesive residue of double-sided adhesive tape with adhesive strip remover -VAS 6349-.
- Clean bonding surfaces -1- with cleaning solution. While doing this, observe flash-off time of cleaning solution.



- Apply double-sided adhesive tape -1- to inside of rear lid.
- Clean bonding surfaces -1- with cleaning solution. While doing this, observe flash-off time of cleaning solution.



- Apply double-sided adhesive tape -1- to outside of rear lid.



2 Aluminium load compartment trim

⇒ [n2.1 otes", page 29](#)

⇒ [o2.2 verview - aluminium load compartment trim", page 29](#)

⇒ [a2.3 nd installing load compartment trim, aluminium", page 31](#)

2.1 General notes

If the load bed, headboard, side panels or tailgate have been renewed during an accident repair, please use the installation instructions supplied with the vehicle to install the load compartment trim ⇒ Original Accessories; aluminium load compartment trim; Amarok 2010 ➤, 2H7 061 164 D.

The following description applies to non-renewed sheet metal parts in which the holes for fastening the load compartment trim are still present ⇒ [page 31](#) .

The bolts for fastening the load compartment trim are self-tapping. If they are unscrewed, it is not possible to screw in new bolts at these points. The bolts would no longer build up any tightening force. For this reason, removed load compartment trims must be secured with rivets when they are reinstalled. The bolts for fastening the fastening rings are not affected by this.

However, the bolts for fastening the load compartment trim can be retightened if necessary, specified torque ⇒ [page 29](#) .

2.2 Assembly overview - aluminium load compartment trim



1 - Bolts

- ☐ Qty. 48
- ☐ Replace by rivets after removal
- ☐ 4 Nm

2 - Load compartment trim

- ☐ Front
- ☐ Removing and installing ⇒ [page 31](#)

3 - Bolts

- ☐ Qty. 4
- ☐ Specified torque
⇒ General body repairs, exterior; Rep. gr. 55; Load bed; Assembly overview - fastening rings

4 - Fastening rings

- ☐ Qty. 4

5 - Spacer sleeves

- ☐ Qty. 4

6 - Side panel trim

- ☐ Right
- ☐ Removing and installing ⇒ [page 31](#)

7 - Bolts

- ☐ Qty. 4
- ☐ Specified torque
⇒ General body repairs, exterior; Rep. gr. 55; Load bed; Assembly overview - fastening rings

8 - Plugs

- ☐ Qty. 2

9 - Spacers

- ☐ Qty. 4

10 - Load compartment trim

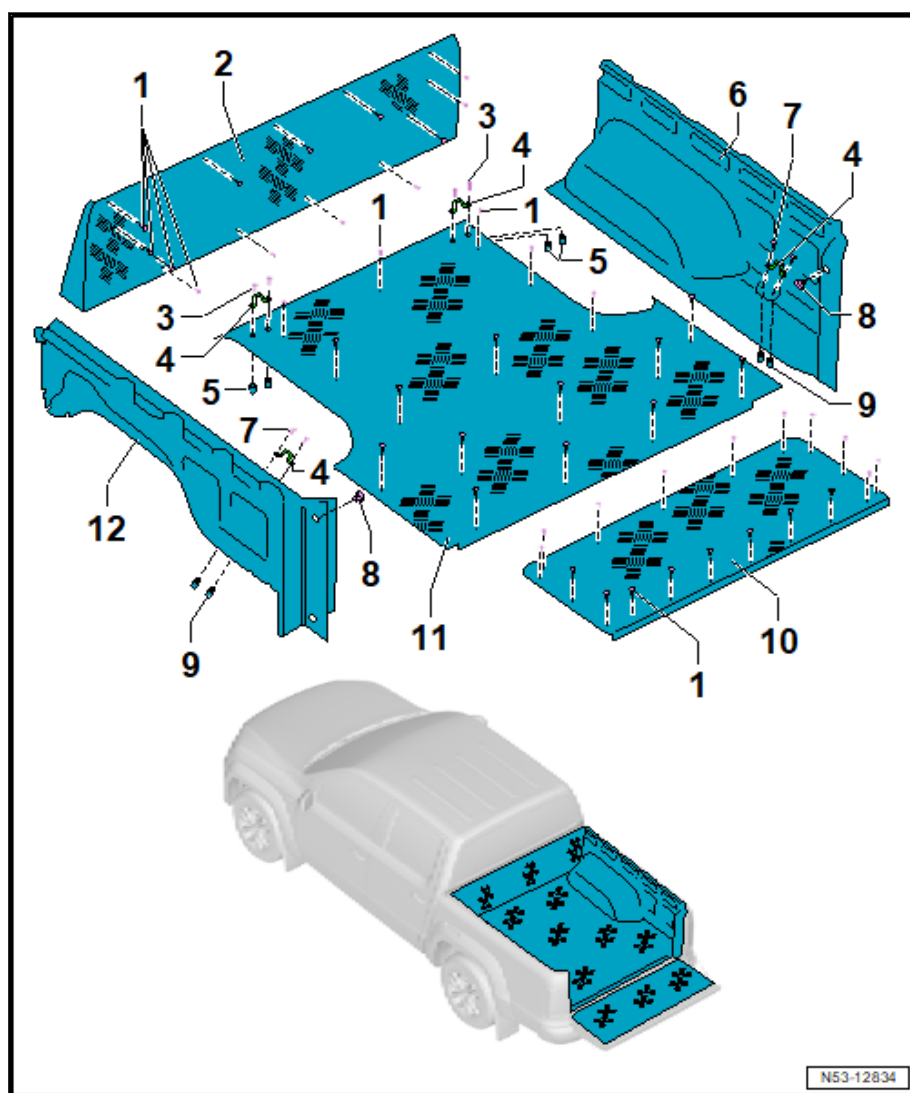
- ☐ On rear lid
- ☐ Removing and installing ⇒ [page 31](#)

11 - Load compartment trim

- ☐ On floor
- ☐ Removing and installing ⇒ [page 31](#)

12 - Side panel trim

- ☐ Left
- ☐ Removing and installing ⇒ [page 31](#)



2.3 Removing and installing load compartment trim, aluminium

⇒ [a2.3.1 nd installing load compartment trim, aluminium, front", page 31](#)

⇒ [a2.3.2 nd installing load compartment trim, aluminium, floor", page 32](#)

⇒ [a2.3.3 nd installing load compartment trim, aluminium, tail-gate", page 34](#)

⇒ [a2.3.4 nd installing load compartment trim, aluminium, side panel trims", page 35](#)

2.3.1 Removing and installing load compartment trim, aluminium, front

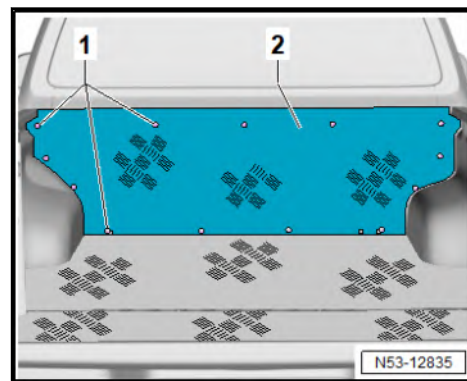
Special tools and workshop equipment required

- ◆ Pop rivet pliers -VAS 5072A-



Removing

- Unscrew bolts -1-.



- Wear safety gloves.
- The following work step must be performed with the aid of a second mechanic.
- Remove load compartment trim -2-.

Installing

Install in reverse order of removal, observing the following:

The bolts for fastening the load compartment trim must not be reused. They must be replaced by rivets.



⚠ CAUTION

**Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.**

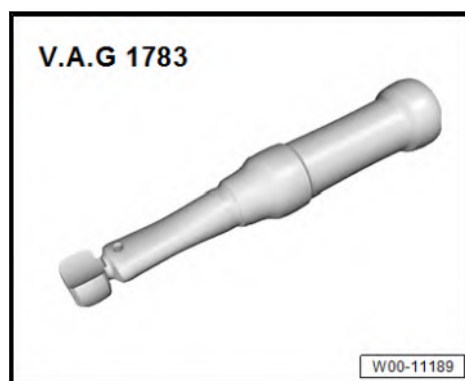
- **Wear protective goggles.**
- **Wear protective gloves!**

- Drill open existing holes in headboard to a diameter of 5.5 mm.
- Seal holes with corrosion protection and paint ⇒ Body; General paintwork information; Technical data; General information; Paintwork in Aftersales.
- Wear safety gloves.
- The following work step must be performed with the aid of a second mechanic.
- Fit load compartment trim to headboard.
- Draw in rivets of 5 mm in diameter using pop rivet pliers -VAS 5072A-.
- On the opposite side of the tailgate towards the cabin, apply corrosion protection measures using spray wax at all riveted points.

2.3.2 Removing and installing load compartment trim, aluminium, floor

Special tools and workshop equipment required

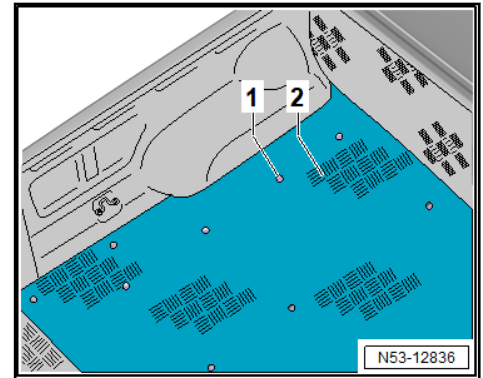
- ◆ Torque wrench -V.A.G 1783-



- ◆ Pop rivet pliers -VAS 5072A-



Removing



- Remove fastening rings in front of front load compartment trim ⇒ General body repairs, exterior; Rep. gr. 55; Load bed; Removing and installing fastening rings.
- Unscrew bolts -1-.
- Remove front load compartment trim ⇒ [page 31](#) .
- Wear safety gloves.
- The following work step must be performed with the aid of a second mechanic.
- Remove load compartment trim -2-.

Installing

Install in reverse order of removal, observing the following:

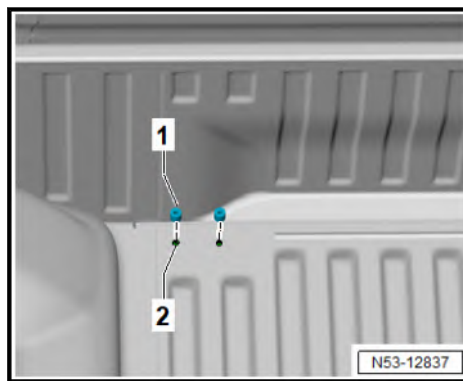
The bolts for fastening the load compartment trim must not be reused. They must be replaced by rivets.

CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- **Wear protective goggles.**
- **Wear protective gloves!**

- Drill open existing holes in floor panel to a diameter of 5.5 mm.
- Clean and dry bed thoroughly.
- Seal holes with corrosion protection and paint ⇒ Body; General paintwork information; Technical data; General information; Paintwork in Aftersales.
- Insert spacer sleeves -1- into threads -2- on left and right side of load bed floor panel, and fix them in position.



- Wear safety gloves.
- The following work step must be performed with the aid of a second mechanic.
- Place load compartment trim onto load bed.
- Draw in rivets of 5 mm in diameter using pop rivet pliers -VAS 5072A-.
- Carry out corrosion protection measures on the underbody at all riveted points using spray wax.

Specified torques

- ♦ ⇒ [o2.2 verview - aluminium load compartment trim", page 29](#)
- ♦ Load bed; Assembly overview - fastening rings ⇒ General body repairs, exterior; Rep. gr. 55; Load bed; Assembly overview - fastening rings

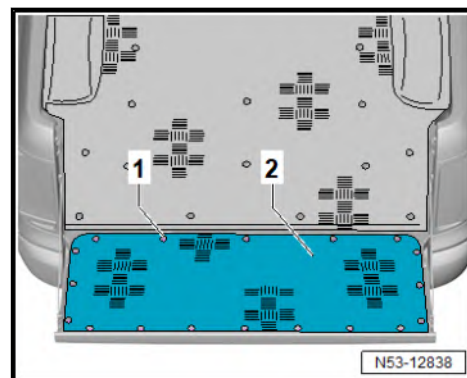
2.3.3 Removing and installing load compartment trim, aluminium, tailgate

Special tools and workshop equipment required

- ♦ Pop rivet pliers -VAS 5072A-



Removing



- Unscrew bolts -1-.
- Wear safety gloves.
- Remove load compartment trim -2-.

Installing

Install in reverse order of removal, observing the following:

The bolts for fastening the load compartment trim must not be reused. They must be replaced by rivets.

CAUTION

**Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.**

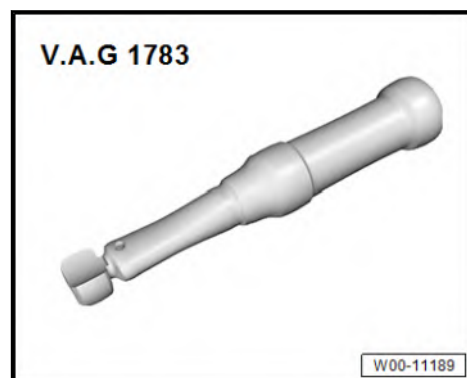
- Wear protective goggles.
- Wear protective gloves!

- Drill open existing holes in tailgate to a diameter of 5.5 mm.
- Seal holes with corrosion protection and paint ⇒ Body; General paintwork information; Technical data; General information; Paintwork in Aftersales.
- Wear safety gloves.
- Fit load compartment trim to tailgate.
- Draw in rivets of 5 mm in diameter using pop rivet pliers -VAS 5072A-.

2.3.4 Removing and installing load compartment trim, aluminium, side panel trims

Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1783-

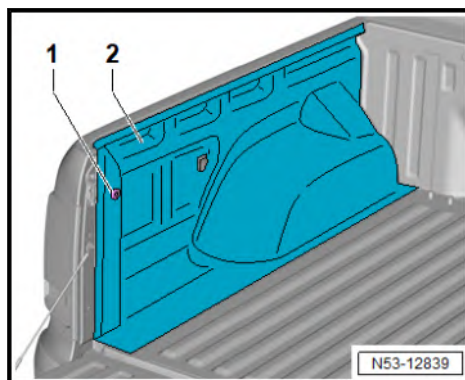




◆ Pop rivet pliers -VAS 5072A-

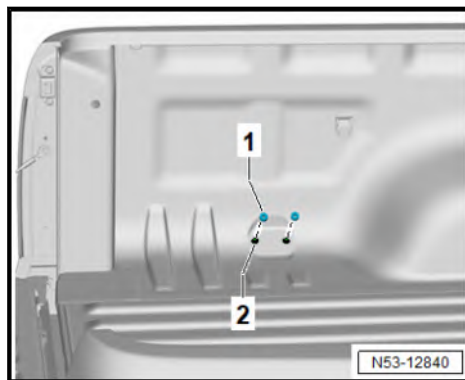


Removing



- Remove front load compartment trim ⇒ [page 31](#) .
- Remove load compartment trim on floor ⇒ [page 32](#) .
- Remove fastening rings at side ⇒ General body repairs, exterior; Rep. gr. 55; Load bed; Removing and installing fastening rings.
- Unclip plug -1-.
- Remove side panel trim -2-.

Installing



Install in reverse order of removal, observing the following:

- Fit spacers -1- into threads -2-, and fix them in position.
- Insert side panel trim.

Specified torque

- ◆ ⇒ [o2.2 verview - aluminium load compartment trim](#)”, [page 29](#)



- ◆ Load bed; Assembly overview - fastening rings ⇒ General body repairs, exterior; Rep. gr. 55; Load bed; Assembly overview - fastening rings



3 Bracket and wing for single and twin tyres, tipper body

⇒ [o3.1 verviews - wing bracket", page 38](#)

⇒ [a3.2 nd installing wing", page 41](#)

⇒ [a3.3 nd installing wing bracket", page 42](#)

3.1 Assembly overviews - wing bracket

⇒ [o3.1.1 verview - wing bracket, single tyres, ►03.2019", page 38](#)

⇒ [o3.1.2 verview - wing bracket, single tyres, 04.2019 ►", page 39](#)

⇒ [o3.1.3 verview - wing bracket, twin tyres, ►03.2019", page 39](#)

⇒ [o3.1.4 verview - wing bracket, twin tyres, 04.2019 ►", page 40](#)

3.1.1 Assembly overview - wing bracket, single tyres, ►03.2019

1 - Bracket

- For wing

2 - Nuts

- Qty. 6
- M8

3 - Bolts

- Qty. 3
- M8x35
- 30 Nm

4 - Bracket

- L-section

5 - Bolts

- Qty. 4
- M6x20
- 10 Nm

6 - Nuts

- Qty. 4
- M6

7 - Nuts

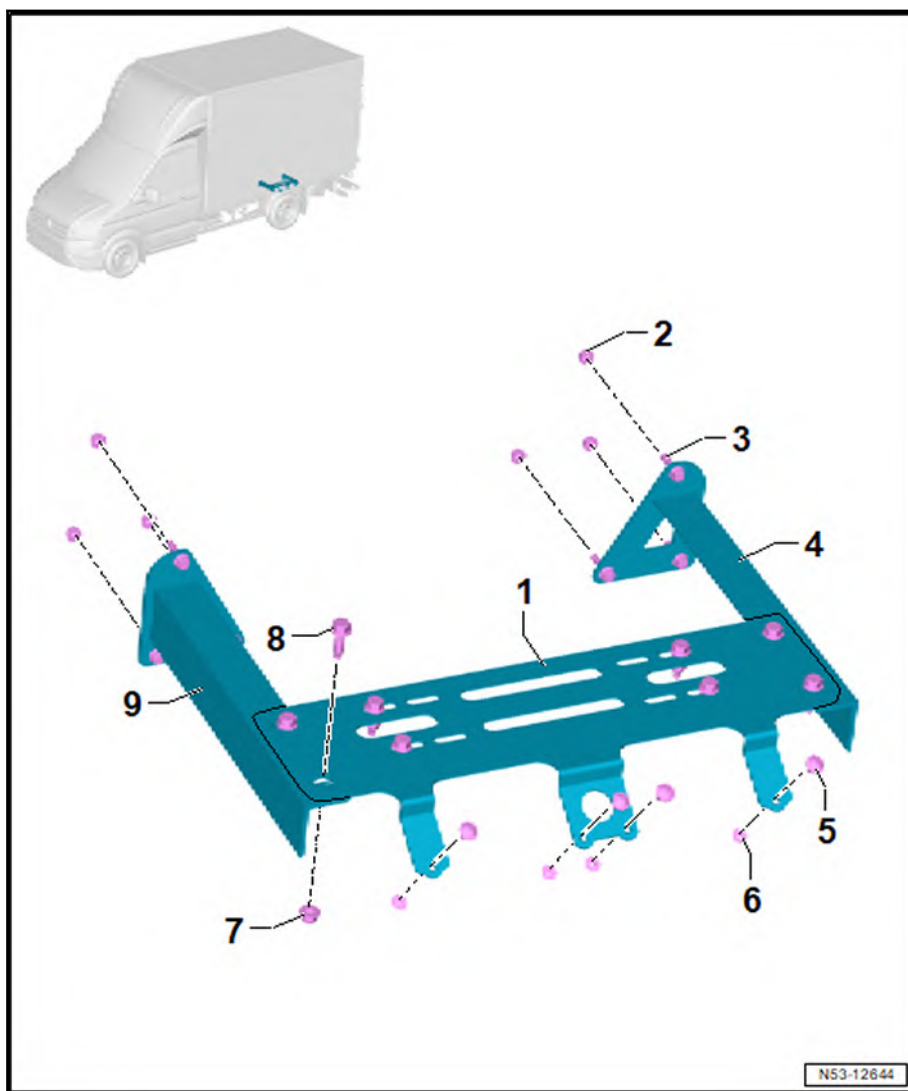
- Qty. 8
- M10

8 - Bolts

- Qty. 8
- M10x35
- 40 Nm

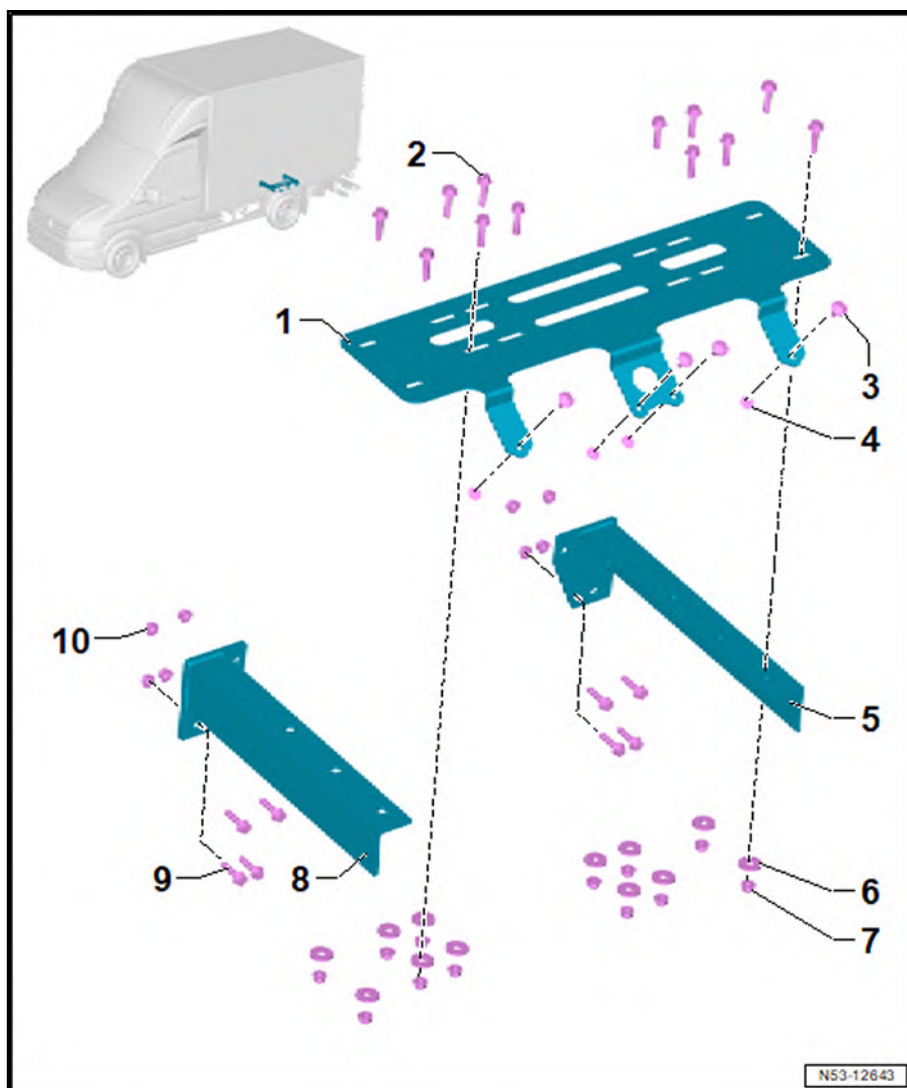
9 - Bracket

- L-section



3.1.2 Assembly overview - wing bracket, single tyres, 04.2019 ➤

- 1 - Bracket**
 - ☐ For wing
- 2 - Bolts**
 - ☐ Qty. 12
 - ☐ M8x35
 - ☐ 30 Nm
- 3 - Bolts**
 - ☐ Qty. 4
 - ☐ M6x20
 - ☐ 10 Nm
- 4 - Nuts**
 - ☐ Qty. 4
 - ☐ M6
- 5 - Bracket**
 - ☐ L-section
- 6 - Washers**
 - ☐ Qty. 8
- 7 - Nuts**
 - ☐ Qty. 12
 - ☐ M8
- 8 - Bracket**
 - ☐ L-section
- 9 - Bolts**
 - ☐ Qty. 6
 - ☐ M8x35
 - ☐ 30 Nm
- 10 - Nuts**
 - ☐ Qty. 6



3.1.3 Assembly overview - wing bracket, twin tyres, ➤03.2019



1 - Bracket

- ☐ For wing

2 - Bolts

- ☐ Qty. 12
- ☐ M10x35
- ☐ 50 Nm

3 - Bolts

- ☐ Qty. 4
- ☐ M6x20
- ☐ 10 Nm

4 - Nuts

- ☐ Qty. 4
- ☐ M6

5 - Bracket

- ☐ L-section

6 - Nuts

- ☐ Qty. 12
- ☐ M10

7 - Bracket

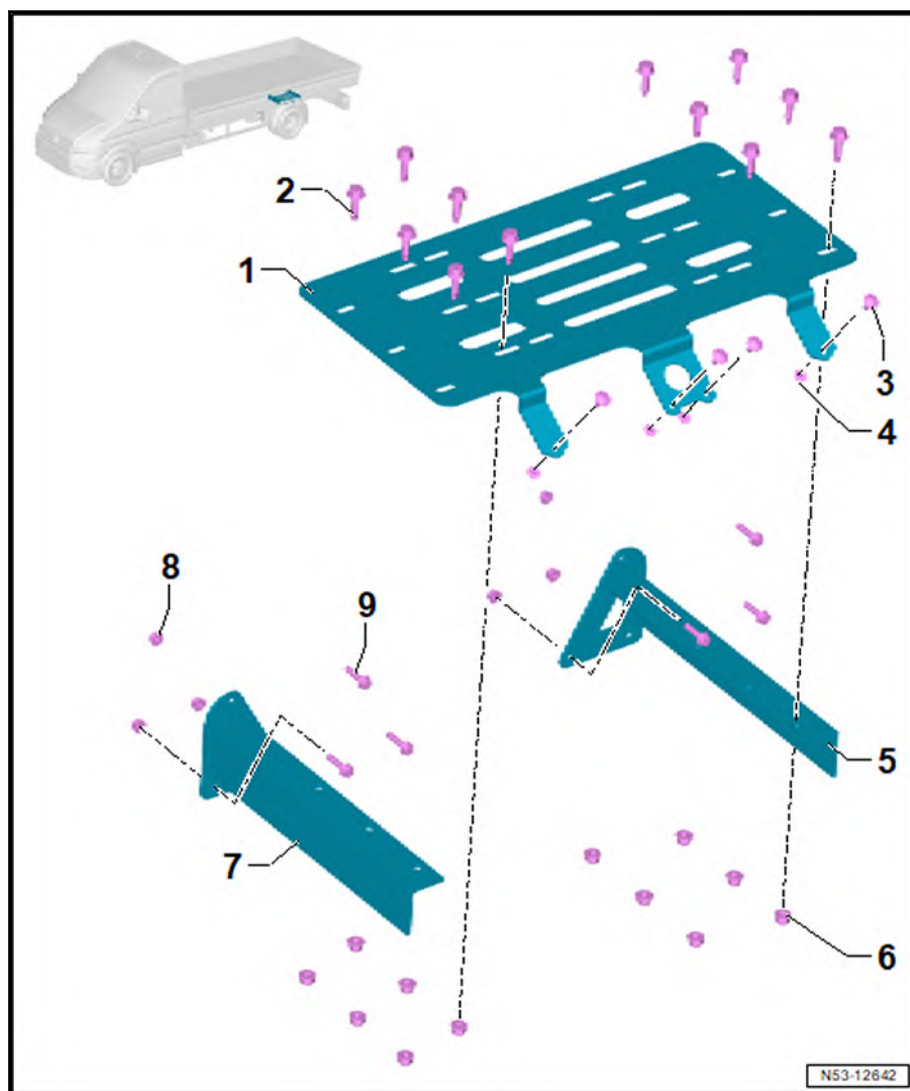
- ☐ L-section

8 - Nuts

- ☐ Qty. 6
- ☐ M8

9 - Bolts

- ☐ Qty. 6
- ☐ M8x35
- ☐ 30 Nm



3.1.4 Assembly overview - wing bracket, twin tyres, 04.2019 ➤



1 - Bracket

- ☐ For wing

2 - Bolts

- ☐ Qty. 18
- ☐ M8x35
- ☐ 30 Nm

3 - Bolts

- ☐ Qty. 4
- ☐ M6x20
- ☐ 10 Nm

4 - Nuts

- ☐ Qty. 4
- ☐ M6

5 - Bracket

- ☐ L-section

6 - Washers

- ☐ Qty. 12

7 - Nuts

- ☐ Qty. 18
- ☐ M8

8 - Bracket

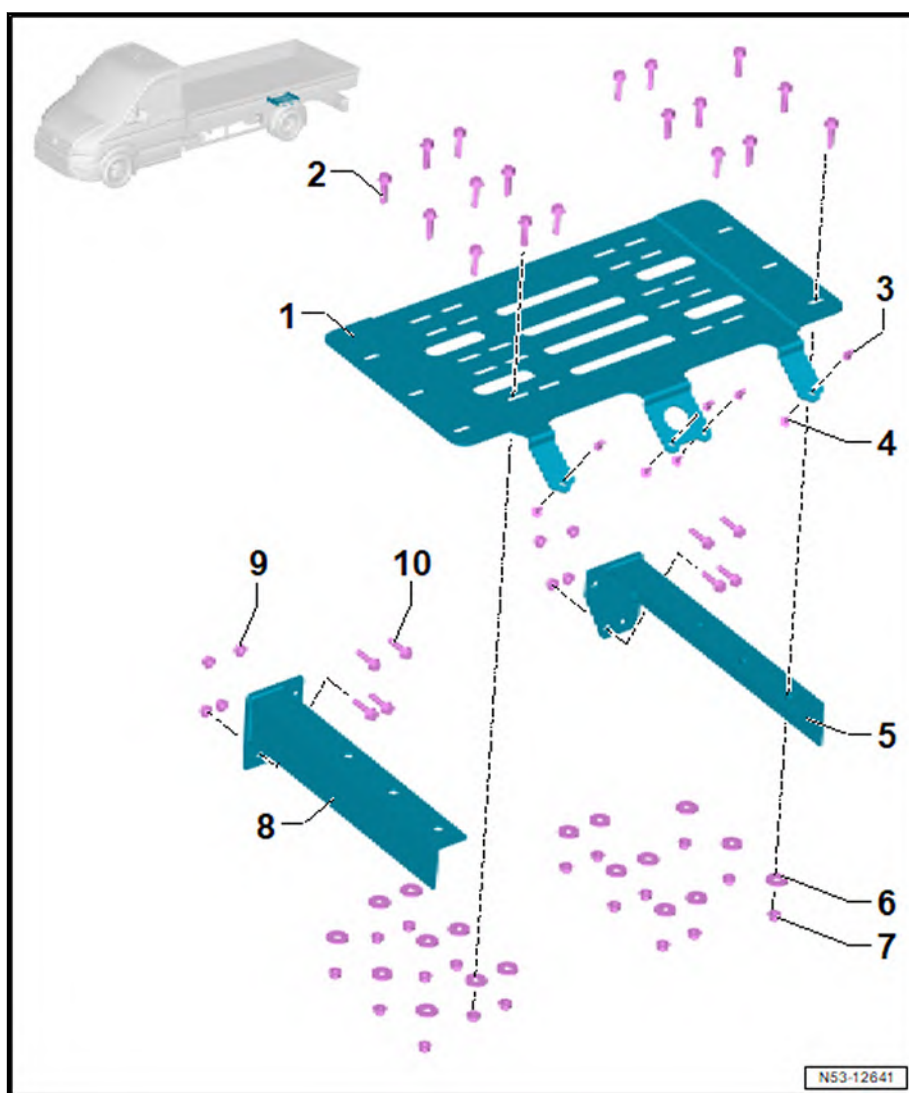
- ☐ L-section

9 - Nuts

- ☐ Qty. 6
- ☐ M8

10 - Bolts

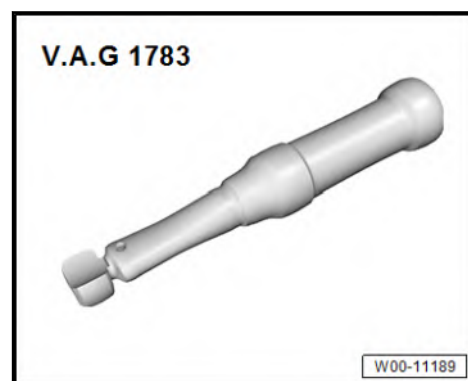
- ☐ Qty. 6
- ☐ M8x35
- ☐ 30 Nm



3.2 Removing and installing wing

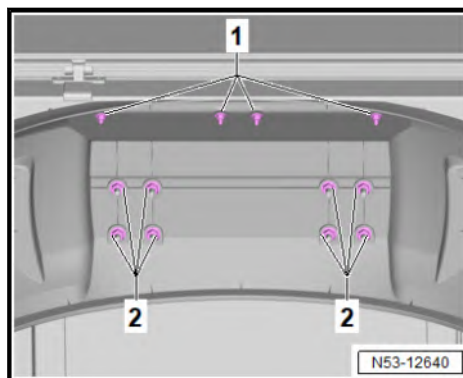
Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1783-





Removing



- Remove rear wheel ⇒ Running gear, axles, steering; Rep. gr. 44; Wheels, tyres; Wheel change.
- Unscrew bolts -1-.
- Unscrew nuts -2-. Remove wing.

Installing

Install in reverse order of removal, observing the following:

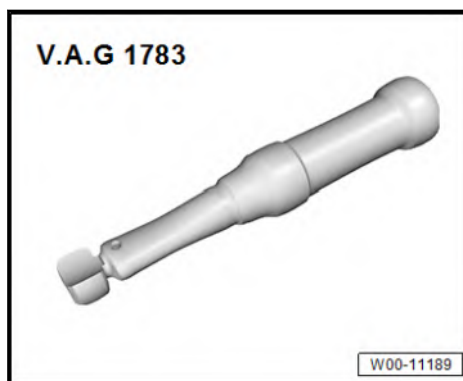
Specified torques

- ♦ ⇒ [o3.1 verviews - wing bracket", page 38](#)
- ♦ Wheels, tyres; Specified torque for wheel bolts ⇒ Running gear, axles, steering; Rep. gr. 44; Wheels, tyres; Specified torque for wheel bolts

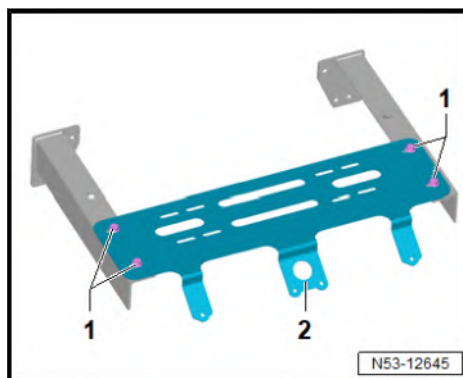
3.3 Removing and installing wing bracket

Special tools and workshop equipment required

- ♦ Torque wrench -V.A.G 1783-



Removing





- Remove wing ➤ [page 41](#) .
- Unscrew bolts -1-.
- Remove for wing bracket -2-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➤ [o3.1 verviews - wing bracket", page 38](#)



61 – Convertible roof, hardtop, canopy

1 Hardtop

⇒ [o1.1 verview - hardtop", page 44](#)

⇒ [a1.2 nd installing hardtop", page 44](#)

⇒ [a1.3 dhesive film", page 48](#)

⇒ [s1.4 eals", page 49](#)

⇒ [a1.5 nd installing gas strut", page 50](#)

⇒ [a1.6 nd installing vent window", page 50](#)

1.1 Assembly overview - hardtop

1 - Hardtop

- ☐ Removing and installing ⇒ [page 44](#)

2 - Vent window

- ☐ Qty. 2
- ☐ Removing and installing ⇒ [page 50](#)

3 - Rear lid

4 - Seals

- ☐ Qty. 2
- ☐ Removing and installing ⇒ [page 49](#)

5 - Seal

- ☐ Depending on equipment
- ☐ Removing and installing ⇒ [page 49](#)

6 - Cover

7 - Gas strut

- ☐ Qty. 6
- ☐ Removing and installing ⇒ [page 50](#)

8 - Bolts

- ☐ Qty. 7
- ☐ 7 Nm

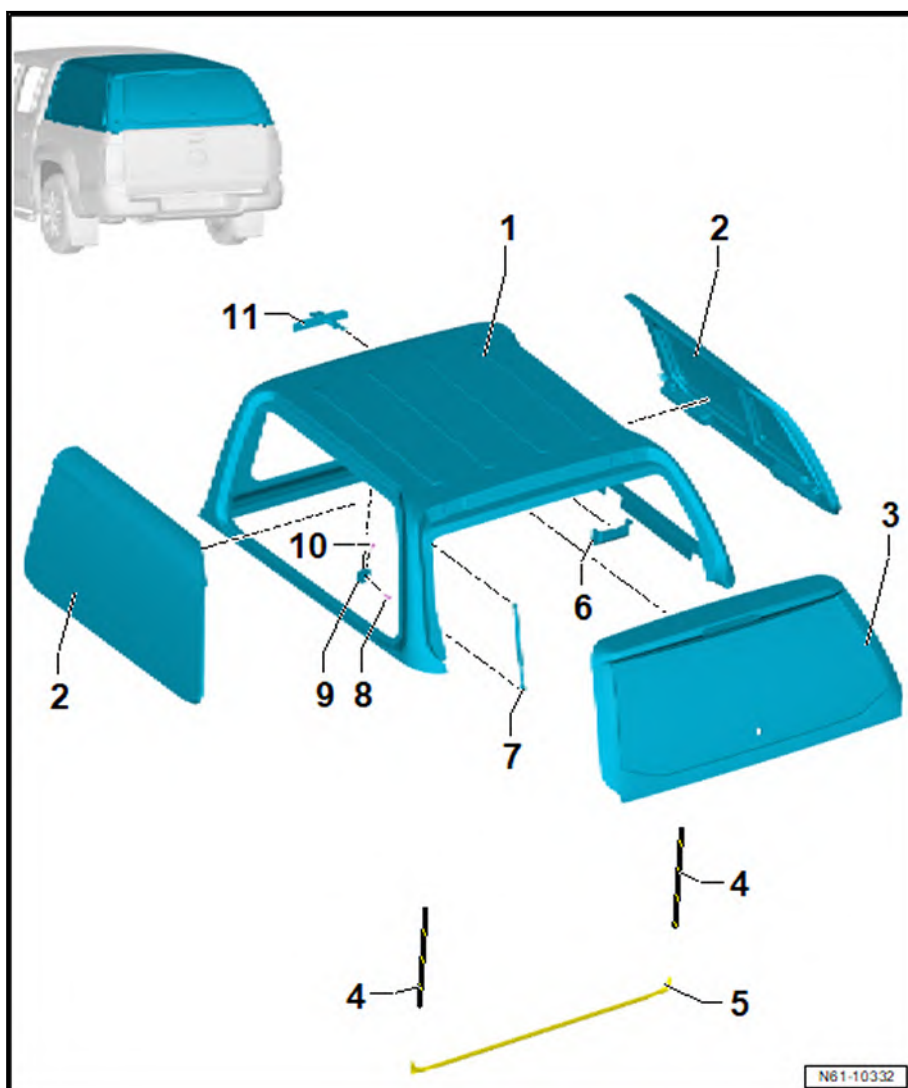
9 - Clamping element

- ☐ Qty. 7

10 - Bolts

- ☐ Qty. 7
- ☐ 7 Nm

11 - Adapter line



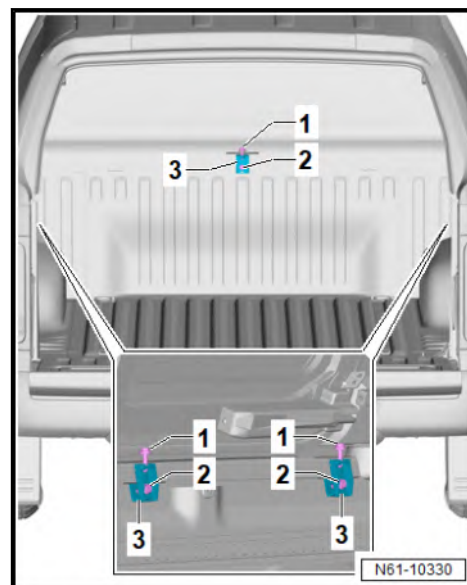
1.2 Removing and installing hardtop

Special tools and workshop equipment required

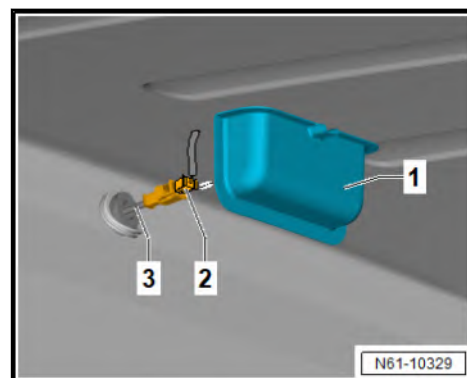
- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



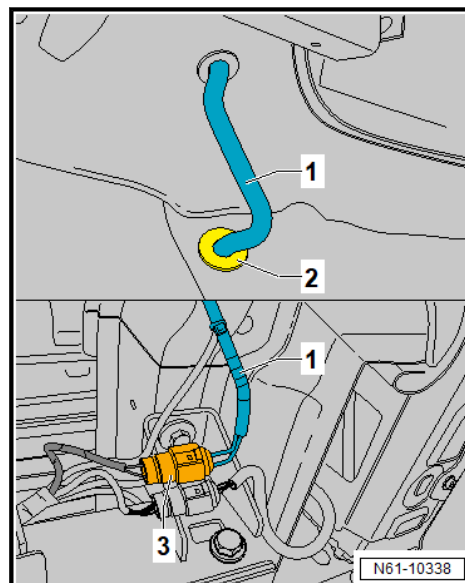
Removing



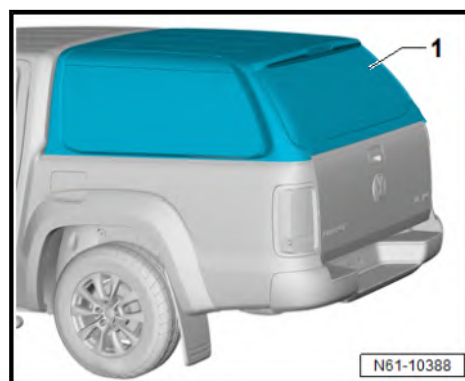
- If fitted, detach protective caps.
- Unscrew bolts -1-.
- Unscrew bolts -2-. Remove clamping element -3- when doing so.
- Detach cover -1-.



- Disconnect connector -2-.
- Free off wiring harness -3-.
- Disconnect connector -3- under vehicle.

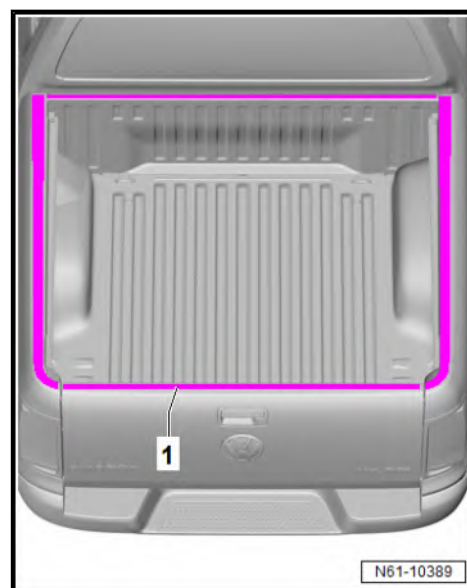


- Loosen rubber grommet -2-.
- Free off wiring harness -1-.
- The following work step must be performed by four mechanics.
- Check that all wiring harnesses have been freed off.



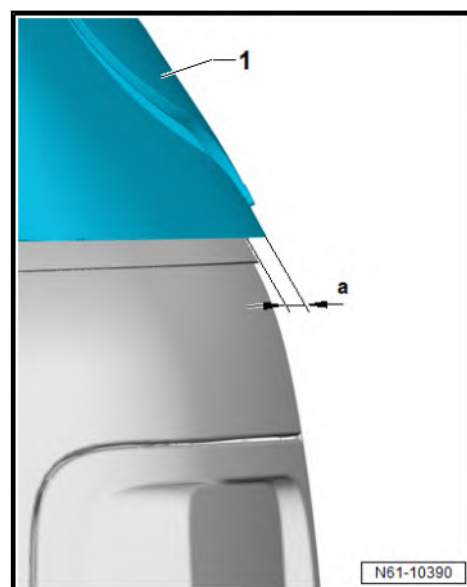
- Remove hardtop -1- with four mechanics.

Installing



Install in reverse order of removal, observing the following:

- Thoroughly clean area of adhesive protective film -1- on load bed.
- Check adhesive protective film -1- for damage.
- If necessary, bond on new adhesive protective film ➔ [page 48](#).
- Check seals for damage.
- If necessary, renew seals ➔ [page 49](#).
- Fit hardtop -1- with four mechanics. Take note of dimensions when doing this and centre hardtop.



- Dimension -a- = 11 mm $\pm$ 1 mm

Specified torques

- ♦ ➔ [o1.1 verview - hardtop", page 44](#)



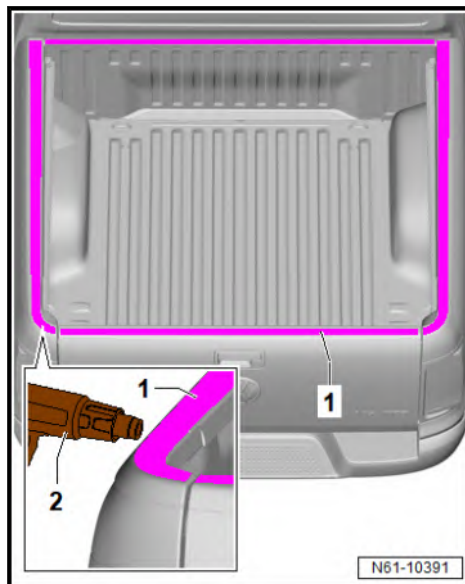
1.3 Renewing adhesive film

Special tools and workshop equipment required

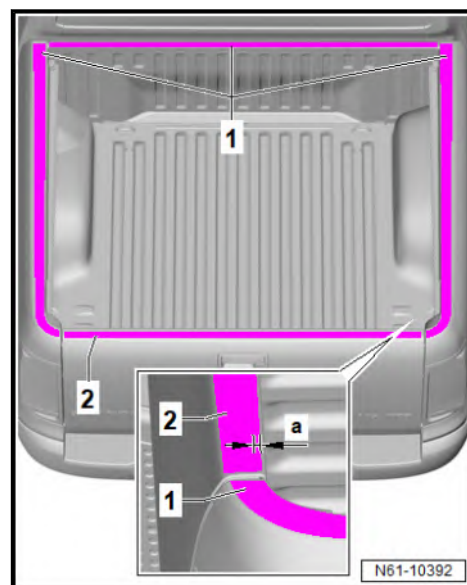
- ♦ Hot-air blower -V.A.G 1416-



Procedure



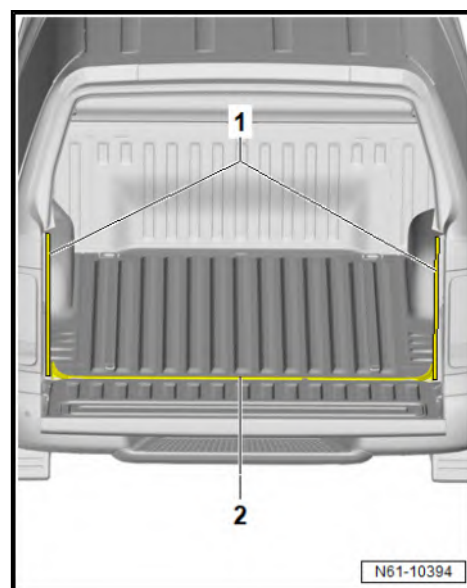
- Remove hardtop ➔ [page 44](#) .
- Set hot air blower -V.A.G 1416- -2- to 130°.
- Hold hot air blower -V.A.G 1416- -2- at a distance of approx. 20 cm.
- Heat up adhesive film -1- using hot air blower -V.A.G 1416- -2-. Pull off adhesive film when doing this.
- Clean bonding surfaces using cleaning solution and a micro-fibre cloth.



- Wear gloves.
- Observe flash-off time of cleaning solution.
- Attach adhesive film -1-.
- Close load bed.
- Spray adhesive film -2- and cleaned tailboard with a mixture of water and detergent.
- Attach adhesive film -2-.
- Attach adhesive film at a distance of approx. 1 cm from the corner of the handrail, dimension -a-.
- Use a squeegee to smooth down the adhesive film -2- and remove fluid and bubbles.
- Install hardtop ➤ [page 44](#) .

1.4 Renewing seals

Procedure



- Open load compartment flap.



- Pull off seal -1- and -2-.
- Clean bonding surfaces with cleaning solution.

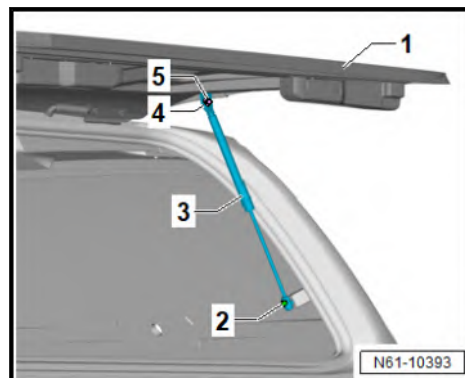
Observe flash-off time of cleaning solution.

Installing

Installation is carried out in reverse order.

1.5 Removing and installing gas strut

Removing



- Open vent window -1-.
- The following work steps require the assistance of a second mechanic.
- Release clip -2-. Have a second mechanic secure vent window -1-.
- Remove circlip -4-.
- Remove pin -5-. Remove gas strut -3-.

Installing

Installation is carried out in reverse order.

1.6 Removing and installing vent window

Special tools and workshop equipment required

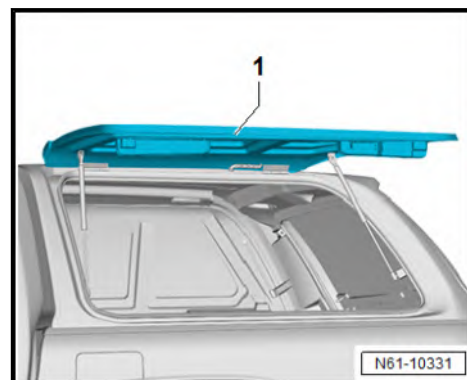
- ♦ Torque wrench 6-50 Nm -V.A.G 1331-



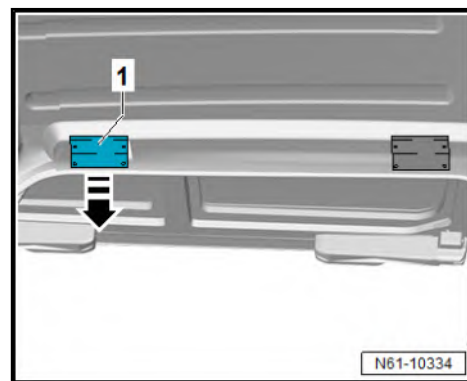
◆ Wiring harness repair set (220V) -VAS 1978B-



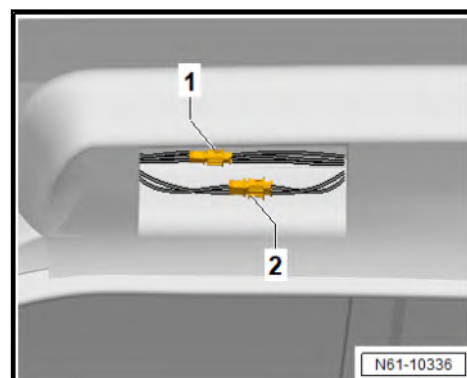
Removing



- Open vent window -1-.
- Disconnect battery ⇒ Electrical system; Rep. gr. 27; Battery; Disconnecting and reconnecting battery.
- Remove left cover -1- in direction of -arrow-.

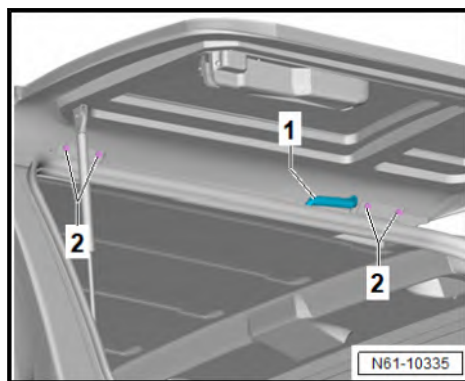


- Separate electrical connectors -1- and -2-.





- Make a note of the wiring allocation.
- Disconnect wiring harness with wiring harness repair set (220V) -VAS 1978B-.
- The following work steps require the assistance of a second mechanic.
- Remove gas strut ⇒ [page 50](#) .



- Detach bellows -1-.
- Pull out wiring harness.
- Unscrew bolts -2-. Remove vent window with aid of second mechanic when doing this.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ⇒ [o1.1 verview - hardtop", page 44](#)

66 – Exterior equipment

1 Tipper body (Henschel)

⇒ [b1.1 ed", page 53](#)

⇒ [s1.2 ystem", page 63](#)

1.1 Load bed

⇒ [a1.1.1 nd installing bulkhead attachment", page 53](#)

⇒ [a1.1.2 nd installing bulkhead", page 54](#)

⇒ [a1.1.3 nd installing panels \(side panels and tailboard\)", page 55](#)

⇒ [a1.1.4 nd installing hinge", page 56](#)

⇒ [a1.1.5 nd installing rear lid", page 57](#)

⇒ [a1.1.6 nd installing rear lock for tail lift", page 57](#)

⇒ [a1.1.7 nd installing wing", page 58](#)

⇒ [a1.1.8 nd installing tipper bearing", page 59](#)

⇒ [a1.1.9 nd installing safety support", page 60](#)

⇒ [a1.1.10 nd installing safety cable", page 61](#)

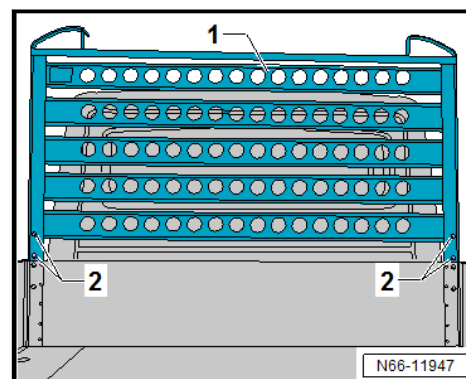
1.1.1 Removing and installing bulkhead attachment

Special tools and workshop equipment required

- ◆ Torque wrench, 40 to 200 Nm -V.A.G 1332A-



Removing



- Unscrew nuts -2-. While doing this, remove bulkhead attachment -1-.



Installing

Install in reverse order of removal, observing the following:

- Renew self-locking standard parts.

Specified torques

Component	Specified torque
Nuts	57 Nm

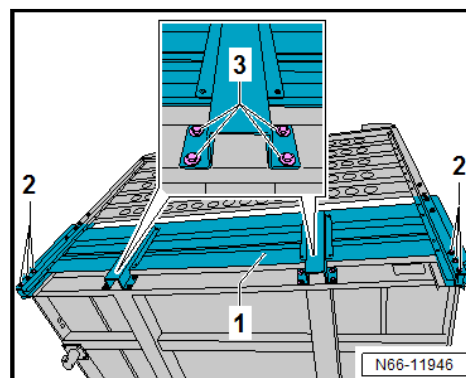
1.1.2 Removing and installing bulkhead

Special tools and workshop equipment required

- ♦ Torque wrench, 40 to 200 Nm -V.A.G 1332A-



Removing



- If fitted, remove bulkhead attachment ➔ [page 53](#) .
- Unscrew bolts -2-.
- Have a second mechanic secure bulkhead -1-.
- Unscrew bolts -3-.
- Remove bulkhead -1-.

Installing

Install in reverse order of removal, observing the following:

- Renew self-locking standard parts.

Specified torques

Component	Specified torque
Bolts -2-	57 Nm
Bolts -3-	20 Nm

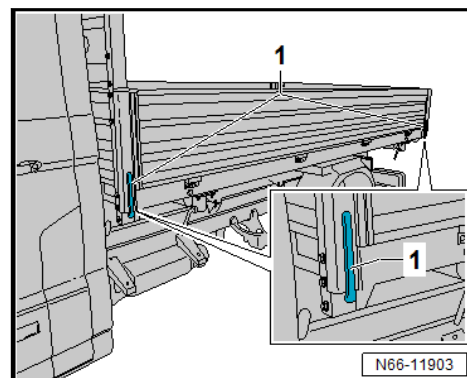
1.1.3 Removing and installing panels (side panels and tailboard)

Special tools and workshop equipment required

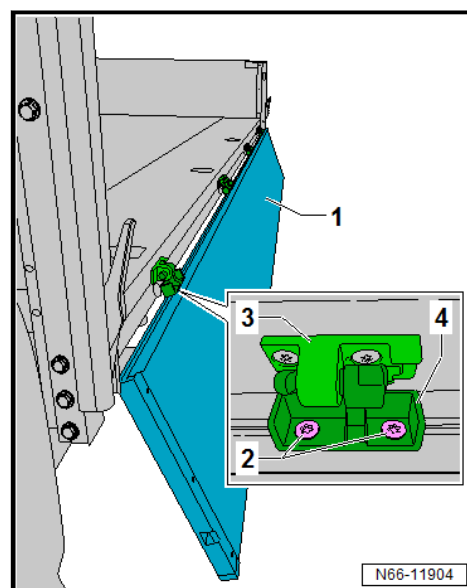
- ◆ Torque wrench -V.A.G 1410A-



Removing



- Open tailboard via quick-release fasteners -1-.
- Loosen bolts -2-.



- Have a second mechanic secure tailboard -1-.
- Push hinge -4- out of bracket -3-.
- Remove sideboard -1-.



Installing

Install in reverse order of removal, observing the following:

- Align panel centrally.

Specified torques

Component	Specified torque
Bolt	20 Nm

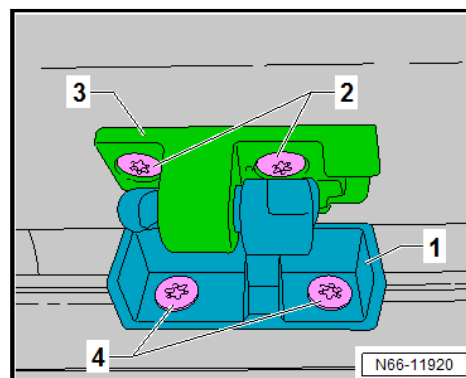
1.1.4 Removing and installing hinge

Special tools and workshop equipment required

- ♦ Torque wrench -V.A.G 1410A-



Removing



- Remove tailboard ⇒ [page 55](#) .
- Unscrew bolts -2-.
- Remove bracket -3-.
- Unscrew bolts -4-.
- Pull off hinge -1-.

Installing

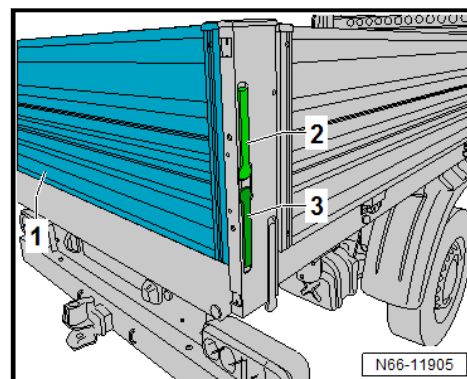
Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolt -2-	20 Nm
Bolt -4-	20 Nm

1.1.5 Removing and installing rear lid

Removing



- Have a second mechanic secure rear lid -1-.
- Open lock -2- on both sides.
- Open lock -3- on both sides.
- Remove rear lid -1-.

Installing

Installation is carried out in reverse order.

1.1.6 Removing and installing rear lock for tail lift

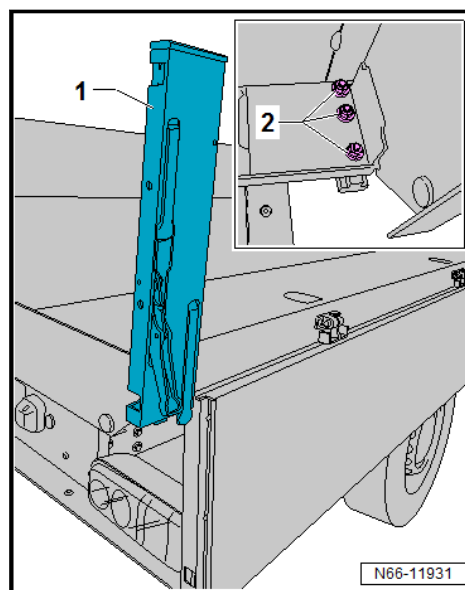
Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1410A-





Removing



- Remove rear lid ➤ [page 57](#) .
- Open tail lift.
- Unscrew nuts -2-.
- Take off lock -1-.

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Nut	20 Nm

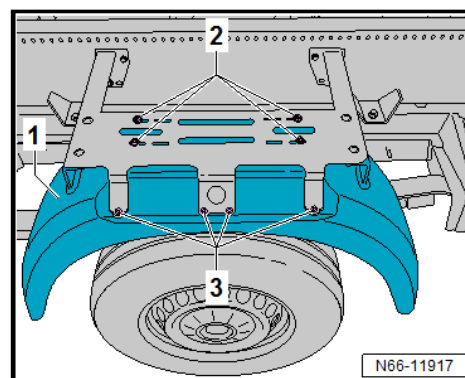
1.1.7 Removing and installing wing

Special tools and workshop equipment required

- ♦ Torque wrench -V.A.G 1410A-



Removing



- Unscrew bolts -3-.
- Have a second mechanic secure wing -1-.
- Unscrew bolts -2-.
- Remove wing -1-.

Installing

Install in reverse order of removal, observing the following:

- Renew self-locking standard parts.

Specified torques

Component	Specified torque
Bolts -2-	12 Nm
Bolts -3-	6 Nm

1.1.8 Removing and installing tipper bearing

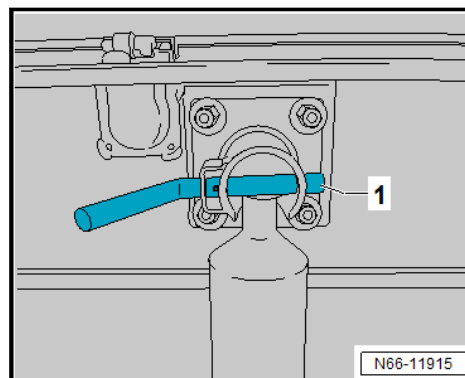
Special tools and workshop equipment required

- ◆ Torque wrench, 40 to 200 Nm -V.A.G 1332A-

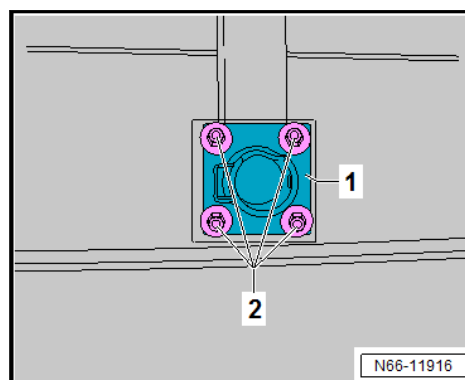




Removing



- Pull out pin -1-.
- Insert pin -1- in opposite tipper bearing.
- Lift load bed.
- Secure load bed against falling using provided safety support.
- Switch off tipper controls.
- Unscrew nuts -2-.



- Take off tipper bearing -1-.

Installing

Install in reverse order of removal, observing the following:

- Renew self-locking standard parts.

Specified torques

Component	Specified torque
Nuts	41 Nm

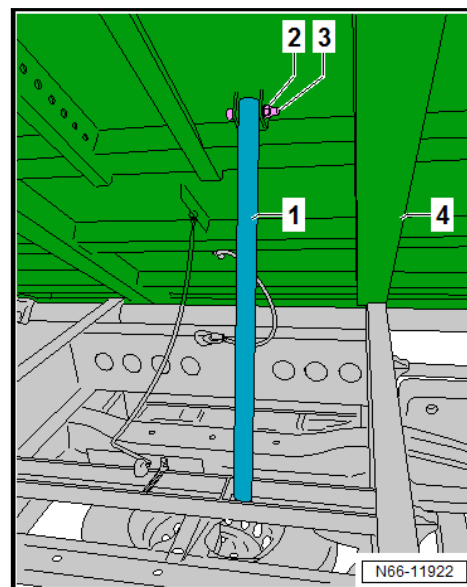
1.1.9 Removing and installing safety support

Special tools and workshop equipment required

◆ Torque wrench -V.A.G 1410A-



Removing



- Lift load bed -4-.
- Secure load bed -4- against falling using suitable workshop equipment.
- Switch off tipper controls.
- Unscrew nut -2-.
- Pull out bolt -3-. While doing this, remove safety support -1-.

Installing

Install in reverse order of removal, observing the following:

- Renew self-locking standard parts.

Specified torques

Component	Specified torque
Nut -2-	20 Nm

1.1.10 Removing and installing safety cable

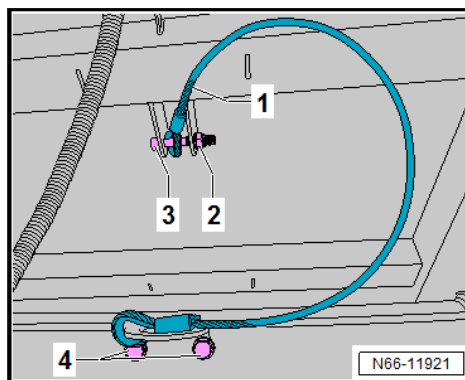
Special tools and workshop equipment required



- ◆ Torque wrench, 40 to 200 Nm -V.A.G 1332A-



Removing



- Lift load bed.
- Secure load bed against falling using provided safety support.
- Switch off tipper controls.
- Unscrew nut -2-.
- Pull out bolt -3-.
- Unscrew bolts -4-.
- Take off safety cable -1-.

Installing

Install in reverse order of removal, observing the following:

- Renew self-locking standard parts.

Specified torques

Component	Specified torque
Nut -2-	50 Nm
Bolt -4-	99 Nm

1.2 Hydraulic system

⇒ [a1.2.1 nd installing pop-up cylinder", page 63](#)

⇒ [a1.2.2 nd installing shaft ring", page 64](#)

⇒ [a1.2.3 nd installing hydraulic hose", page 65](#)

⇒ [a1.2.4 nd installing hydraulic pump", page 67](#)

⇒ [h1.2.5 ydraulic fluid", page 70](#)

⇒ [t1.2.6 he hydraulic system", page 72](#)

1.2.1 Removing and installing pop-up cylinder

Special tools and workshop equipment required

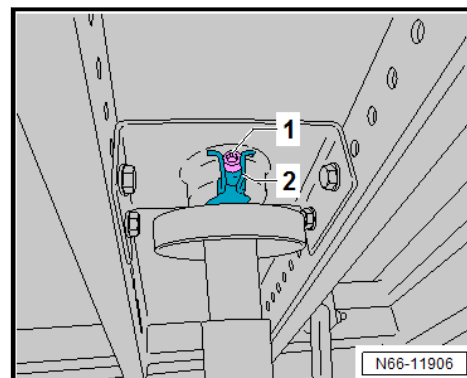
- ◆ Torque wrench, 40 to 200 Nm -V.A.G 1332A-



- ◆ Drip tray for workshop hoist -VAS 6208-



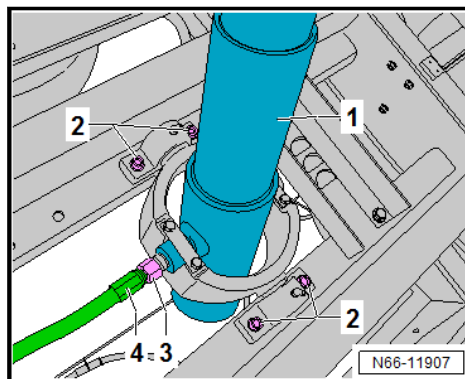
Removing



- Lift load bed.
- Secure load bed against falling using provided safety support.



- Switch off tipper controls.
- Have an additional mechanic secure hydraulic cylinder against falling down.
- Unscrew bolt -1-.
- Press together spring -2-. Remove spring -2-.
- Place drip tray for workshop hoist -VAS 6208- underneath.



CAUTION

Injury hazard from residual pressure in the hydraulic system.

- Wear safety goggles.
- Wear protective gloves!

- Unscrew union nut -3-.
- Detach hydraulic line -4-.
- Carefully compress hydraulic cylinder -1-. While so, collect escaping hydraulic fluid in drip tray for workshop hoist -VAS 6208-.
- Unscrew bolts -2-. While doing this, remove pop-up cylinder -1- with aid of second mechanic.

Installing

Install in reverse order of removal, observing the following:

- Fill hydraulic fluid.
- Move hydraulic cylinder to lift position.
- Check hydraulic fluid level, replenish as necessary.
- Remove support.
- Bleed hydraulic system ➔ [page 72](#) .

Specified torques

Component	Specified torque
Bolt -1-	57 Nm
Union nut, hydraulic line	60 Nm
Bolts -2-	99 Nm

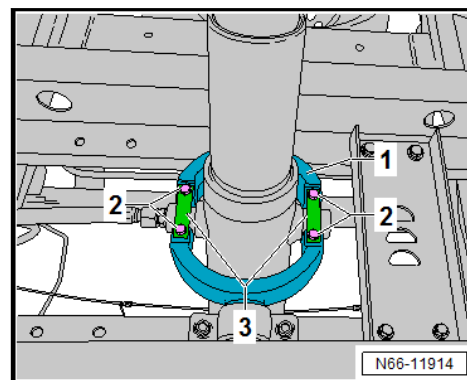
1.2.2 Removing and installing shaft ring

Special tools and workshop equipment required

◆ Torque wrench -V.A.G 1410A-



Removing



- Lift load bed.
- Secure load bed against falling using provided safety support.
- Switch off tipper controls.
- Remove pop-up cylinder ➔ [page 63](#) .
- Unscrew bolts -2-.
- Detach locking plates -3-.
- Remove shaft ring -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolts -2-	29 Nm

1.2.3 Removing and installing hydraulic hose

Special tools and workshop equipment required



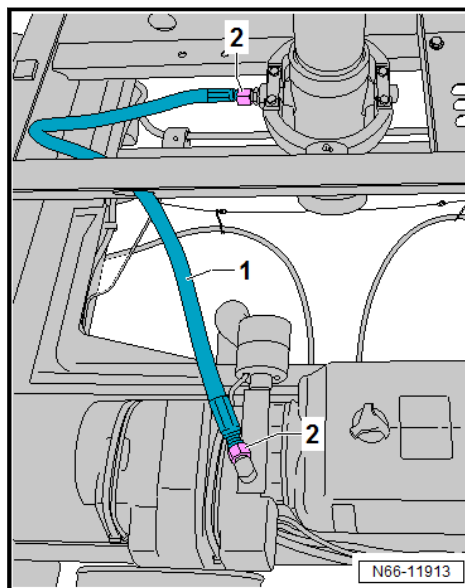
- ◆ Torque wrench, 40 to 200 Nm -V.A.G 1332A-



- ◆ Drip tray for workshop hoist -VAS 6208-



Removing



- Lift load bed.
- Secure load bed against falling using provided safety support.
- Place drip tray for workshop hoist -VAS 6208- underneath.

CAUTION

Injury hazard from residual pressure in the hydraulic system.

- Wear safety goggles.
- Wear protective gloves!



- Unscrew union nuts -2-.
- Carefully detach hydraulic hose -1-. While so, collect escaping hydraulic fluid in drip tray for workshop hoist -VAS 6208-.

Installing

Install in reverse order of removal, observing the following:

- Check hydraulic fluid level after installing hydraulic hose, replenish fluid as necessary.
- Bleed hydraulic system ⇒ [page 72](#) .

Specified torques

Component	Specified torque
Union nuts -2-	60 Nm

1.2.4 Removing and installing hydraulic pump

Special tools and workshop equipment required

- ◆ Torque wrench, 40 to 200 Nm -V.A.G 1332A-

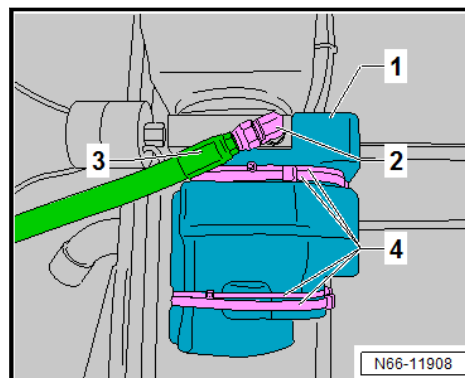


- ◆ Drip tray for workshop hoist -VAS 6208-





Removing

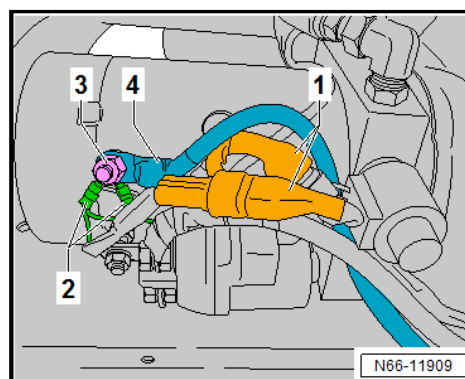


- Lift load bed.
- Secure load bed against falling using provided safety support.
- Disconnect battery ⇒ Electrical system; Rep. gr. 27; Battery; Disconnecting and reconnecting battery.
- Place drip tray for workshop hoist -VAS 6208- underneath.

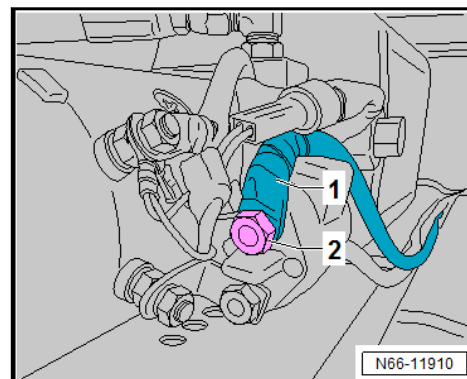
CAUTION

Injury hazard from residual pressure in the hydraulic system.

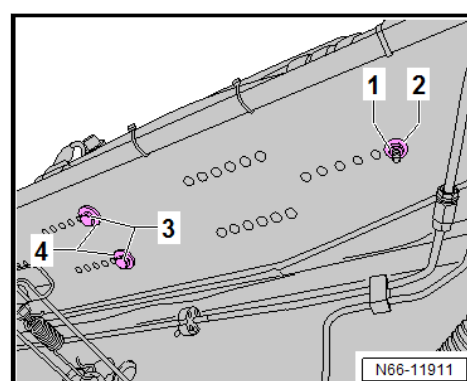
- Wear safety goggles.
 - Wear protective gloves!
-
- Unscrew union nut -2-.
 - Carefully detach hydraulic hose -3-. While so, collect escaping hydraulic fluid in drip tray for workshop hoist -VAS 6208-.
 - Cut through cable tie -4-.
 - Detach cover -1-.
 - Disconnect electrical connectors -1-.



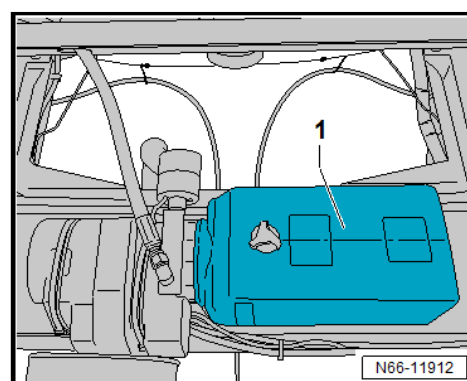
- Unscrew nut -3-.
- Detach wiring harness -2-.
- Detach wiring harness -4-.
- Unscrew nut -2-.



- Detach wiring harness -1-.
- Unscrew nut -1-. Remove washer -2- when doing this.



- Unscrew bolts -3-. Remove shims -4-.
- Detach hydraulic unit -1-.



Installing

Install in reverse order of removal, observing the following:

- Fill hydraulic fluid.
- Move hydraulic cylinder to lift position.
- Check hydraulic fluid level, replenish as necessary.
- Remove support.
- Bleed hydraulic system ➔ [page 72](#) .

Specified torques

Component	Specified torque
Bolt of hydraulic pump -3-	41 Nm
Nuts of wiring harness -2-	20 Nm



Component	Specified torque
Nuts of wiring harness -3-	20 Nm
Nuts of hydraulic pump -1-	20 Nm
Union nut of hydraulic hose -2-	60 Nm

1.2.5 Renewing hydraulic fluid

Special tools and workshop equipment required

- ◆ Torque wrench, 40 to 200 Nm -V.A.G 1332A-



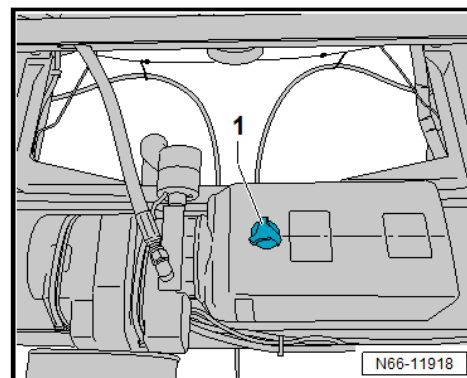
- ◆ Used oil collection and extraction unit -VAS 6622A-



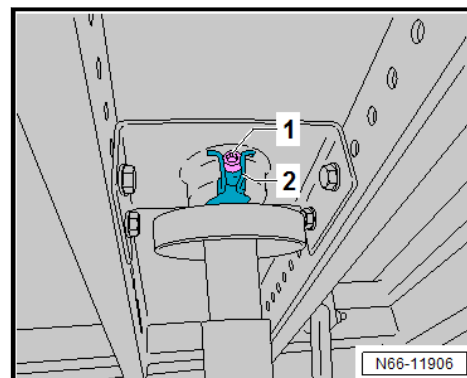
- ◆ Drip tray for workshop hoist -VAS 6208-



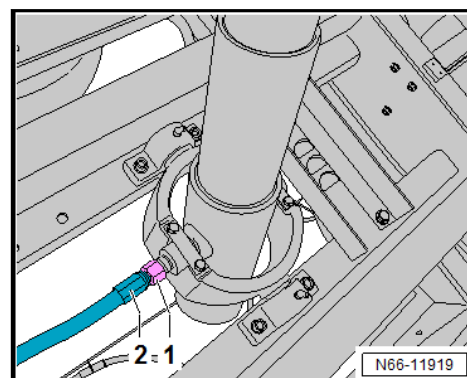
Sequence of operations



- Lift load bed.
- Secure load bed against falling using provided safety support.
- Switch off tipper controls.
- Unbolt cover -1-.
- Extract hydraulic fluid using used oil collection and extraction unit -VAS 6622A-.
- Have an additional mechanic secure hydraulic cylinder against falling down.



- Unscrew bolt -1-.
- Press together spring -2-. Remove spring -2-.
- Place drip tray for workshop hoist -VAS 6208- underneath.





⚠ CAUTION

Injury hazard from residual pressure in the hydraulic system.

- **Wear safety goggles.**
- **Wear protective gloves!**

- Unscrew union nut -1-.
- Remove hydraulic hose -2-.
- Carefully compress pop-up cylinder. While so, collect escaping hydraulic fluid in drip tray for workshop hoist -VAS 6208-.
- Fit hydraulic hose -2-.
- Tighten union nut -1-.
- Fill with new hydraulic fluid in accordance with manufacturer's instructions.
- Bleed hydraulic system ⇒ [page 72](#) .

Specified torques

Component	Specified torque
Union nut, hydraulic line	60 Nm

1.2.6 Bleeding the hydraulic system

Sequence of operations

- Raise and lower load bed several times.
- Check oil level, replenish as necessary.



2 Tipper body (Schoon)

⇒ [b2.1 ed", page 73](#)

⇒ [s2.2 ystem", page 76](#)

2.1 Load bed

⇒ [a2.1.1 nd installing bulkhead", page 73](#)

⇒ [a2.1.2 nd installing panels \(side panels and tailboard\)", page 74](#)

⇒ [a2.1.3 nd installing wing \(single tyres\)", page 75](#)

2.1.1 Removing and installing bulkhead

Special tools and workshop equipment required

◆ Pop rivet pliers -VAS 5072A-

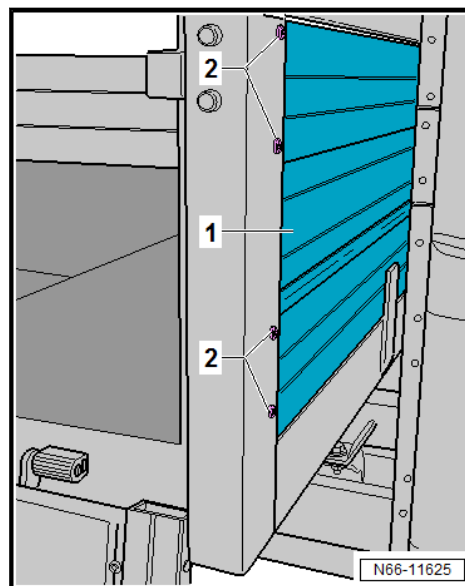


◆ Pin punch -VAS 5513-





Removing



CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out rivet head -2- on both sides of bulkhead with hand drill and drill bit for bulkhead panel -1-.
- Drive out rivet shank -2- with drift -VAS 5513- on both sides.
- Remove swarf.
- Remove bulkhead panels -1- inwards.

Installing

Install in reverse order of removal, observing the following:

- Use rivet pliers -VAS 5072A- to install new pop rivet.

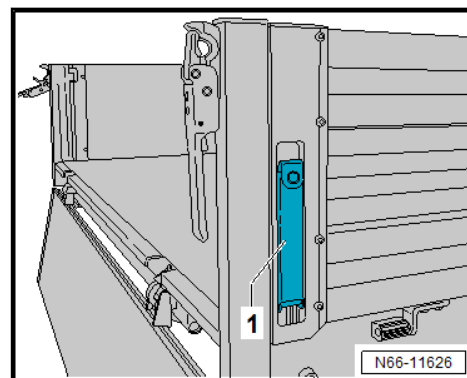
2.1.2 Removing and installing panels (side panels and tailboard)

Special tools and workshop equipment required

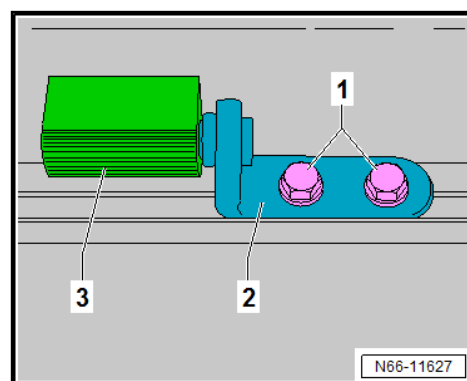
- ♦ Torque wrench 6-50 Nm -V.A.G 1331-



Removing



- Open quick-release fasteners -1- on both sides of respective panel.
- Loosen bolts -1-. Do not unscrew the bolts from the panel completely.



- Have a second mechanic secure the panel.
- Push hinge -2- out of bracket -3-.
- Remove panel from load bed.

Installing

Install in reverse order of removal, observing the following:

- Align panel centrally.

Specified torques

Component	Specified torque
Bolts -1-	27 Nm

2.1.3 Removing and installing wing (single tyres)

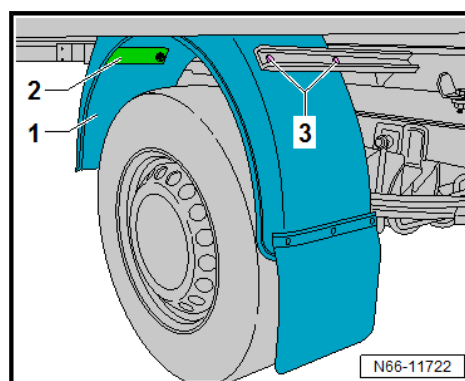
Special tools and workshop equipment required



- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



Removing



- Unscrew bolts -3- on both sides of wing -1-.
- Have a second mechanic secure the wing.
- Remove reinforcement -2-.
- Remove wing -1-.

Installing

Install in reverse order of removal, observing the following:

- Renew self-locking standard parts.

Specified torques

Component	Specified torque
Bolts -3- (renew self-locking nuts)	27 Nm

2.2 Hydraulic system

⇒ [p2.2.1 op-up cylinder \(stroke: 875 mm/1150 mm\)", page 76](#)

⇒ [a2.2.2 nd installing hydraulic hose", page 78](#)

⇒ [a2.2.3 nd installing hydraulic unit", page 80](#)

⇒ [h2.2.4 ydraulic fluid", page 82](#)

⇒ [t2.2.5 he hydraulic system", page 84](#)

2.2.1 5-stage pop-up cylinder (stroke: 875 mm/1150 mm)

Special tools and workshop equipment required



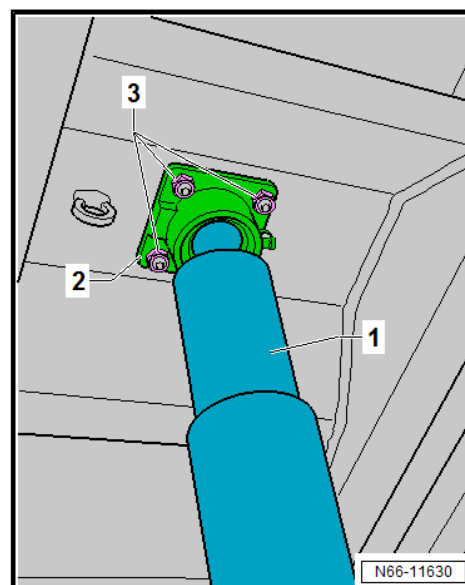
- ◆ Torque wrench, 40 to 200 Nm -V.A.G 1332A-



- ◆ Drip tray for workshop hoist -VAS 6208-



Removing



- Secure load bed against falling using provided safety support.
- Unscrew nuts -3- at upper mounting -2- of pop-up cylinder -1-.

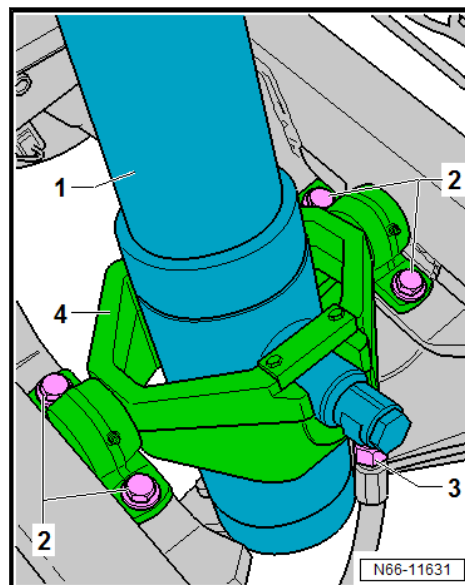
CAUTION

Injury hazard from residual pressure in the hydraulic system.

- Wear safety goggles.
- Wear protective gloves!



- Place drip tray for workshop hoist -VAS 6208- beneath pop-up cylinder.



- Unscrew union nut -3-. Allow hydraulic fluid to drain out into drip tray or similar.
- Unscrew bolts -2- from lower mounting -4-.
- Remove pop-up cylinder -1-.
- When removing, have a second mechanic securing the pop-up cylinder against falling.

Installing

Install in reverse order of removal, observing the following:

- Fill hydraulic fluid.
- Move hydraulic cylinder to lift position.
- Check hydraulic fluid level and replenish as necessary.
- Remove support.
- Bleed hydraulic system ➔ [page 84](#) .

Specified torques

Component	Specified torque
Upper bearing nuts -3-	93 Nm
Hydraulic line union nut -3-	50 Nm
Lower bearing bolts -2-	148 Nm

2.2.2 Removing and installing hydraulic hose

Special tools and workshop equipment required



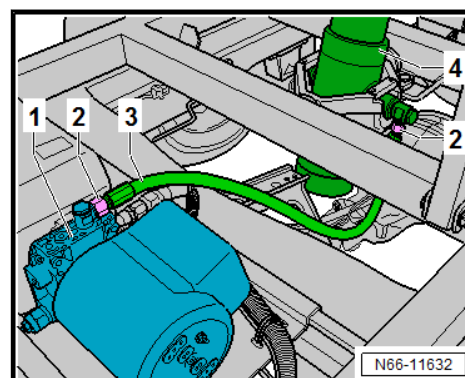
- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



- ◆ Drip tray for workshop hoist -VAS 6208-



Removing



- Secure load bed against falling using provided safety support.

CAUTION

Injury hazard from residual pressure in the hydraulic system.

- Wear safety goggles.
- Wear protective gloves!

- Place drip tray for workshop hoist -VAS 6208- underneath.
- Unscrew union nut -2- from pop-up cylinder -4-. Allow hydraulic fluid to drain out into drip tray or similar.
- Unscrew union nut union nut -2- from solenoid valve -1-.
- Remove hydraulic hose -3-.

Installing

Install in reverse order of removal, observing the following:



- Check hydraulic fluid level after installing the hydraulic hose, replenish fluid as necessary.
- Bleed hydraulic system ➤ [page 84](#) .

Specified torques

Component	Specified torque
Union nuts -2-	50 Nm

2.2.3 Removing and installing hydraulic unit

Special tools and workshop equipment required

- ◆ Torque wrench, 40 to 200 Nm -V.A.G 1332A-



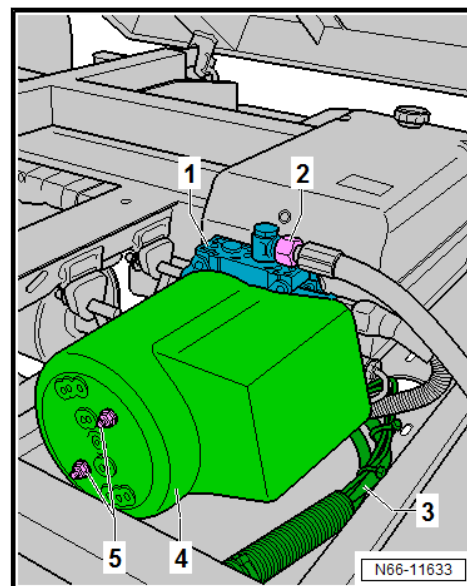
- ◆ Torque wrench -V.A.G 1410A-



- ◆ Drip tray for workshop hoist -VAS 6208-



Removing

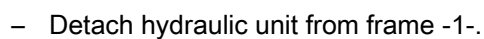
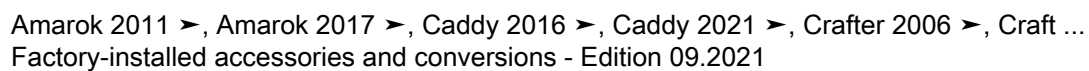


- Secure load bed against falling using provided safety support.
- Disconnect battery ⇒ Electrical system; Rep. gr. 27; Battery; Disconnecting and connecting battery.

CAUTION

Injury hazard from residual pressure in the hydraulic system.

- Wear safety goggles.
 - Wear protective gloves!
-
- Place drip tray for workshop hoist -VAS 6208- underneath.
 - Unscrew union nut -2- from solenoid valve -1-. Allow hydraulic fluid to drain out into drip tray or similar.
 - Unscrew nuts -5-.
 - Detach cover -4-.
 - Disconnect wiring harness -3-.
 - Unscrew bolts -2-. Remove shims -3-.



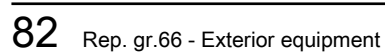
Install in reverse order of removal, observing the following:

- Check hydraulic fluid level after installing the hydraulic hose, replenish fluid as necessary.
- Bleed hydraulic system ⇒ [page 84](#) .

Component	Specified torque
Bolts -2-	54 Nm
Nuts -5-	2.7 Nm
Union nut -2-	50 Nm

Special tools and workshop equipment required

- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



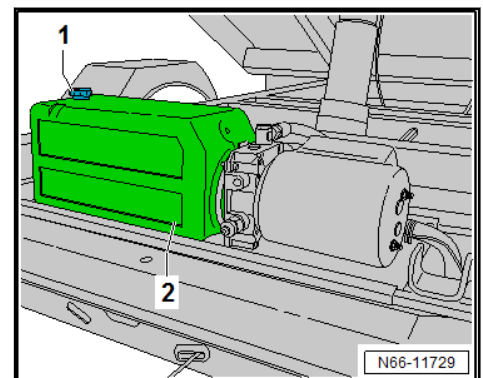
◆ Used oil collection and extraction unit -VAS 6622A-



◆ Drip tray for workshop hoist -VAS 6208-



Procedure



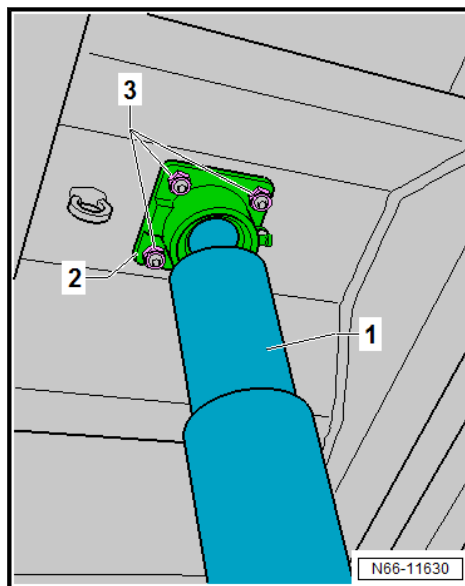
- Secure load bed against falling using provided safety support.

⚠ CAUTION

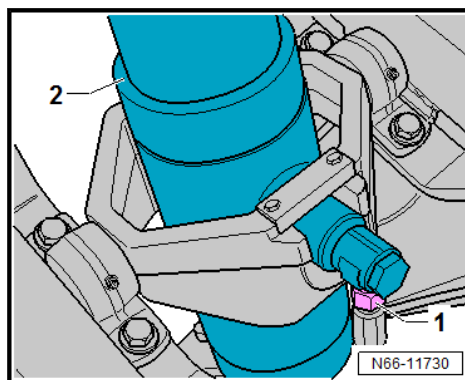
Injury hazard from residual pressure in the hydraulic system.

- Wear safety goggles.
- Wear protective gloves!

- Unscrew cap -1- from hydraulic reservoir -2-.
- Extract hydraulic fluid using used oil collection and extraction unit -VAS 6622A-.
- Have an additional mechanic secure hydraulic cylinder -1- against falling down.



- Unscrew nuts -3- at upper mounting -2-.
- Place drip tray for workshop hoist -VAS 6208- beneath hydraulic cylinder.



- Unscrew union nut -1-.
- Carefully compress hydraulic cylinder -2-. Allow remainder of hydraulic fluid to drain out into drip tray or similar.
- Tighten union nut -1-.
- Fill hydraulic fluid.
 - Observe manufacturer's specifications for the hydraulic fluid.
- Move hydraulic cylinder to lift position.
- Check hydraulic fluid level and replenish as necessary.
- Remove safety support.
- Bleed hydraulic system ➤ [page 84](#) .

Specified torques

Component	Specified torque
Union nut -1-	50 Nm
Nuts -3-	93 Nm

2.2.5 Bleeding the hydraulic system

- Raise and lower the load bed several times to bleed the hydraulic system.



- After bleeding, check fluid level, and replenish fluid if necessary.



3 Box body (Junge)

⇒ [b3.1 ody, exterior", page 86](#)

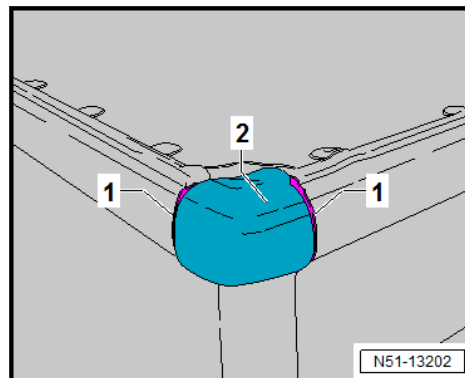
⇒ [b3.2 ody, rear", page 86](#)

3.1 Box body, exterior

⇒ [a3.1.1 nd installing roof corner", page 86](#)

3.1.1 Removing and installing roof corner

Removing



- Detach sealed and bonded joint -1- around circumference.
- Detach bonded joint between roof corner -2- and box body.
- Remove roof corner from box body.
- Remove sealing and adhesive compound residue.

Installing

Install in reverse order of removal, observing the following:

- Lightly sand down bonding surfaces and clean.
- Apply assembly adhesive.
- Fit roof corner and secure in position.
- Seal all around with sealing and adhesive compound.

3.2 Box body, rear

⇒ [a3.2.1 nd installing gas strut", page 86](#)

⇒ [a3.2.2 nd installing tail flap", page 87](#)

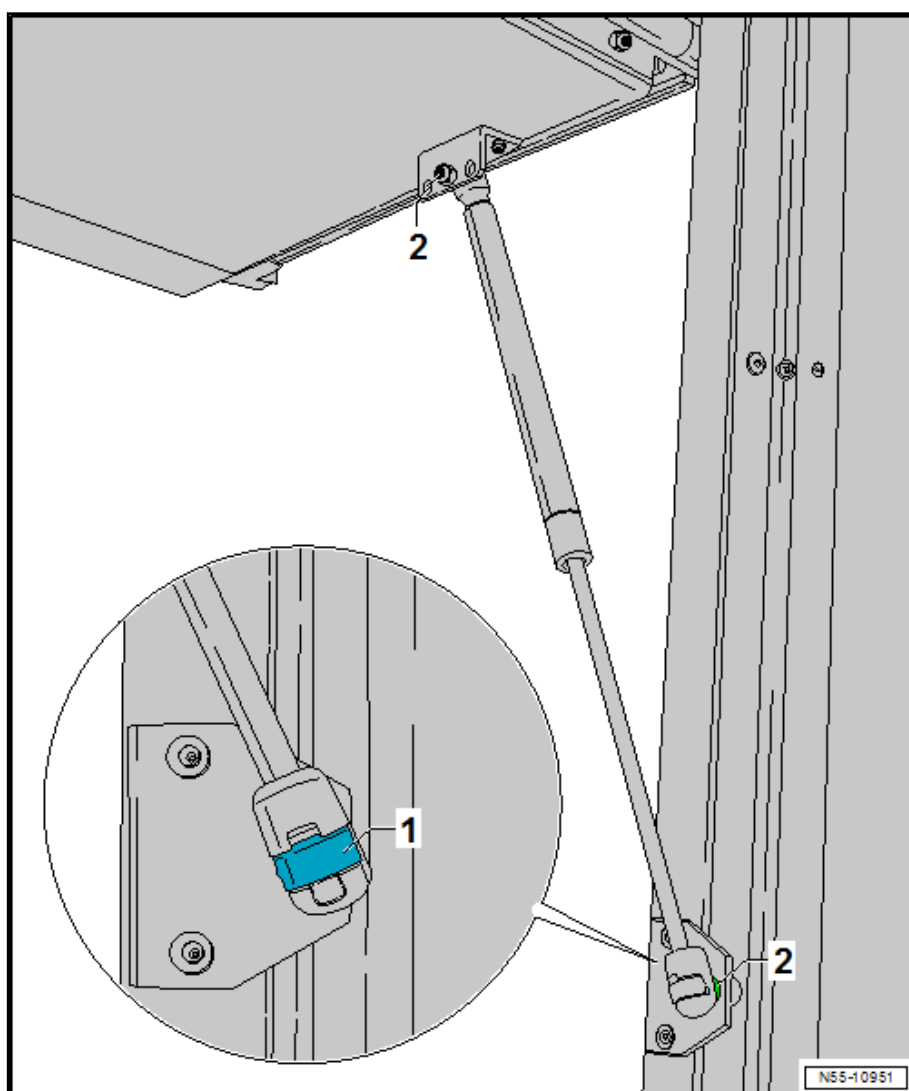
⇒ [a3.2.3 nd installing tail flap seal", page 88](#)

⇒ [a3.2.4 nd installing box body seals", page 89](#)

⇒ [a3.2.5 nd installing reversing camera", page 90](#)

3.2.1 Removing and installing gas strut

Removing



- Open tail lift.
- Open tail flap.
- Prop up tail flap and secure.
- Detach both metal clips -1- at top and bottom with a screw-driver.
- Pull off gas strut from ball heads -2-.

Installing

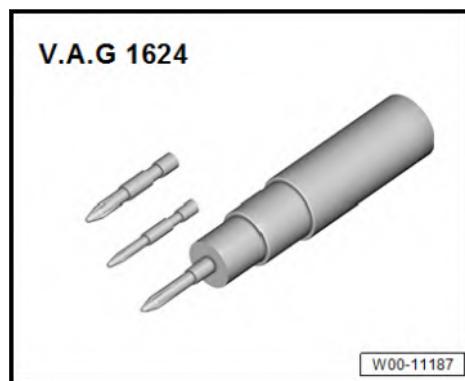
Installation is carried out in reverse order.

3.2.2 Removing and installing tail flap

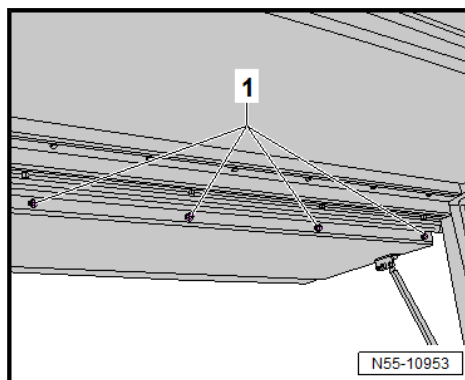
Special tools and workshop equipment required



◆ Torque screwdriver -V.A.G 1624-



Removing



- Open tail lift.
- Open tail flap.
- Prop up tail flap and secure.
- Remove gas struts of tail flap ➔ [page 86](#) .
- Unscrew bolts -1-.
- Remove tail flap from box body.

Installing

Installation is carried out in reverse order.

Specified torques

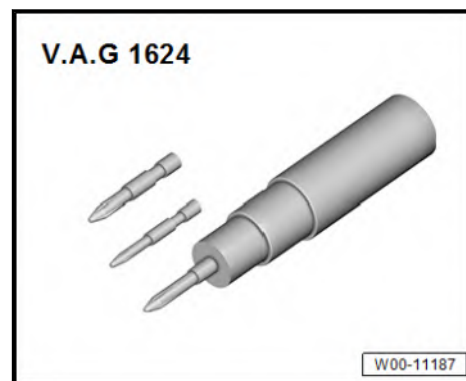
Components	Specified torque
Bolts -1-	5 Nm

3.2.3 Removing and installing tail flap seal

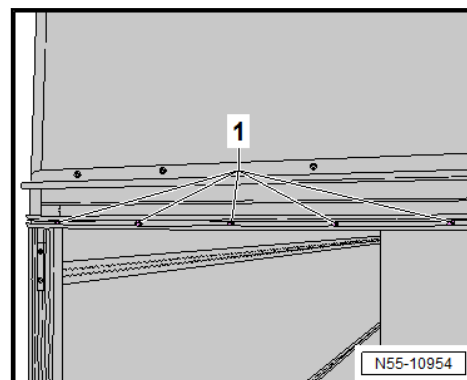
Special tools and workshop equipment required



- ◆ Torque screwdriver -V.A.G 1624-



Removing



- Remove bolts -1- of seal.
- Remove seal from tail flap.

Installing

Installation is carried out in reverse order.

Specified torques

Components	Specified torque
Bolts -1-	5 Nm

3.2.4 Removing and installing box body seals

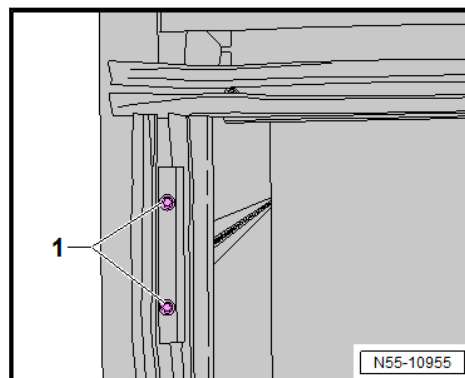
Special tools and workshop equipment required

- ◆ Torque wrench 6-50 Nm -V.A.G 1331-





Removing



- Unscrew bolts -1- of plastic block.
- Remove plastic block from box body.
- Unscrew all bolts in seal.
- Remove seal from box body.

Installing

Installation is carried out in reverse order.

Specified torques

Components	Specified torque
Bolts -1-	25 Nm

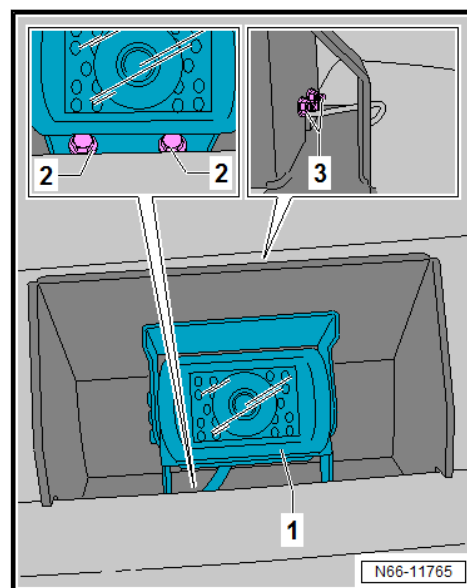
3.2.5 Removing and installing reversing camera

Special tools and workshop equipment required

- ◆ Torque screwdriver -VAS 6494-



Removing



- Disconnect wiring harness on reversing camera -1-.
- Unscrew nuts -3-.
- Remove bolts -2- towards front.
- Detach reversing camera -1-.

Installing

Install in reverse order of removal, observing the following:

- Repair wiring harness ⇒ Electrical system, General information; Rep. gr. 97; Wiring; Repairing wiring harnesses and connectors.
- Renew self-locking nuts.

Specified torques

Component	Specified torque
Nuts -3-	2 Nm



4 Box body (Spier)

⇒ [b4.1 ody, exterior", page 92](#)

⇒ [b4.2 ody, rear", page 95](#)

4.1 Box body, exterior

⇒ [a4.1.1 nd installing front roof corner", page 92](#)

⇒ [a4.1.2 nd installing rear roof corner", page 93](#)

4.1.1 Removing and installing front roof corner

Special tools and workshop equipment required

◆ Hot-air blower -V.A.G 1416-



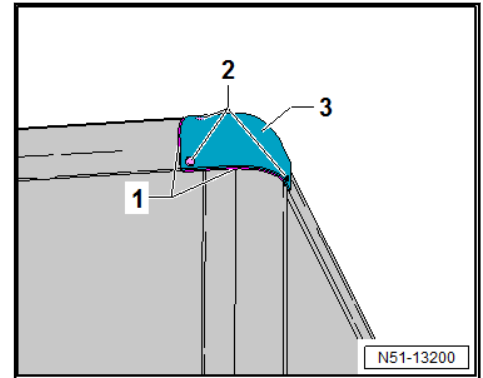
◆ Pop rivet pliers -VAS 5072A-



◆ Pin punch -VAS 5513-



Removing



CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out rivet head -2- using drill and drill bit.
- Drive out rivet shank -2- with drift -VAS 5513-.
- Remove swarf.
- Detach sealed and bonded joint -1- around circumference.
- Heat up bonded joint between roof corner -3- and box body using hot air blower -V.A.G 1416- and carefully detach.
- Remove roof corner from box body.
- Remove sealing and adhesive compound residue.

Installing

Install in reverse order of removal, observing the following:

- Lightly sand down bonding surfaces and clean.
- Apply SIKA activator 100 to the bonding surface of the new part and the sidewall.
- Observe flash-off period of at least 20 minutes.
- Apply SIKA adhesive 552 to new part.
- Fit roof corner.
- Use rivet pliers -VAS 5072A- to install new pop rivet.
- Seal all around with SIKA adhesive 552.
- The minimum curing period of 24 hours at room temperature (at least 18°C) must be adhered to. During this time, the vehicle must be standing on a level surface.

4.1.2 Removing and installing rear roof corner

Special tools and workshop equipment required



◆ Hot-air blower -V.A.G 1416-



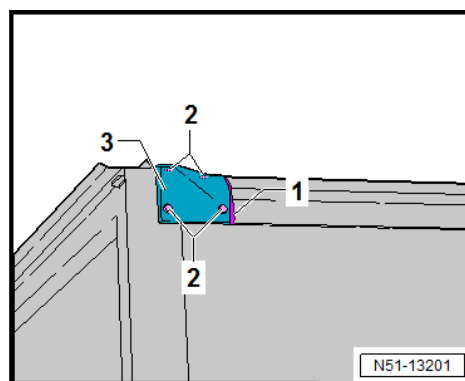
◆ Pop rivet pliers -VAS 5072A-



◆ Pin punch -VAS 5513-



Removing





⚠ CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out rivet head -2- using drill and drill bit.
- Drive out rivet shank -2- with drift -VAS 5513-.
- Remove swarf.
- Detach sealed and bonded joint -1- around circumference.
- Heat up bonded joint between roof corner -3- and box body using hot air blower -V.A.G 1416- and carefully detach.
- Remove roof corner from box body.
- Remove sealing and adhesive compound residue.

Installing

Install in reverse order of removal, observing the following:

- Lightly sand down bonding surfaces and clean.
- Apply SIKA activator 100 to the bonding surface of the new part and the sidewall.
- Observe flash-off period of at least 20 minutes.
- Apply SIKA adhesive 552 to new part.
- Fit roof corner.
- Use rivet pliers -VAS 5072A- to install new pop rivet.
- Seal all around with SIKA adhesive 552.
- The minimum curing period of 24 hours at room temperature (at least 18°C) must be adhered to. During this time, the vehicle must be standing on a level surface.

4.2 Box body, rear

⇒ [a4.2.1 nd installing gas strut", page 95](#)

⇒ [a4.2.2 nd installing tail flap", page 97](#)

⇒ [a4.2.3 nd installing tail flap seals", page 98](#)

⇒ [a4.2.4 nd installing rear box body seals", page 98](#)

⇒ [a4.2.5 nd installing reversing camera", page 98](#)

4.2.1 Removing and installing gas strut

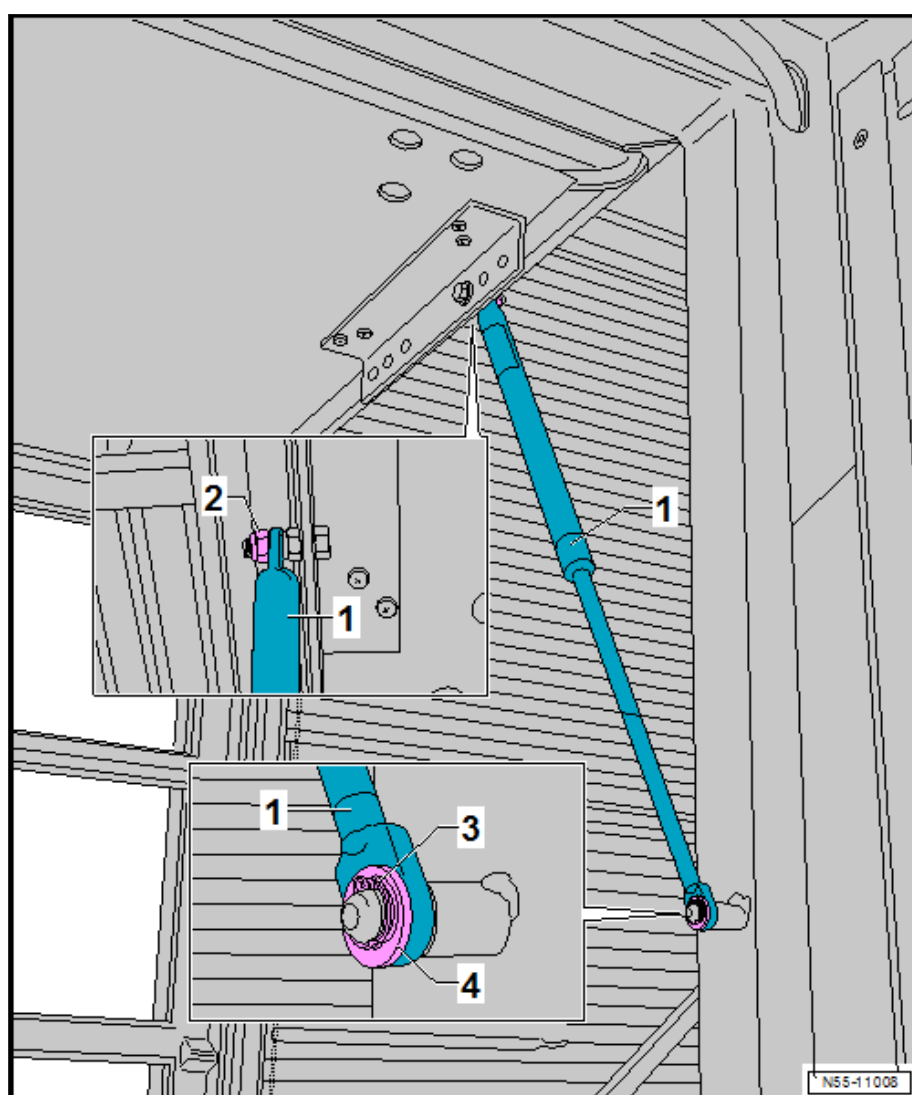
Special tools and workshop equipment required



◆ Torque wrench -VAS 6253A-



Removing



- Open tail lift.
- Open tail flap.
- Prop up tail flap and secure.
- Remove split pin -3- with washer -4-.
- Unscrew nut -2-.



- Remove gas strut -1- from box body.

Installing

- Renew split pin.
- Secure nut with split pin.

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Nut -2-	1 Nm

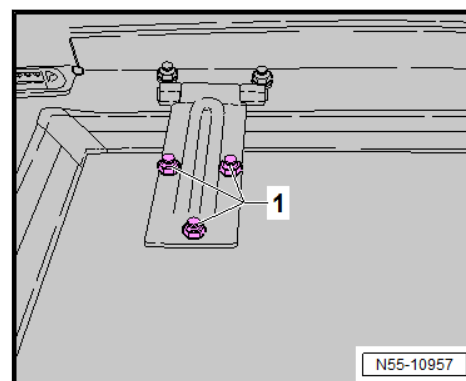
4.2.2 Removing and installing tail flap

Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1410A-



Removing



- Open tail lift.
- Prop up tail flap and secure.

CAUTION

Injury hazard from residual pressure in the hydraulic system.

- Wear safety goggles.
- Wear protective gloves!

- Remove gas struts of tail flap ➔ [page 95](#) .
- Unscrew nuts -1- from all hinges.
- Remove tail flap from box body.

Installing

Install in reverse order of removal, observing the following:



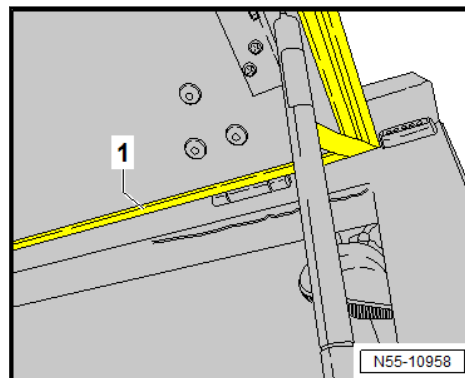
Specified torques

Component	Specified torque
Nuts -1-	15 Nm

4.2.3 Removing and installing tail flap seals

Removing

- Detach seal -1- from all around tail flap.



- Remove seal from tail flap.

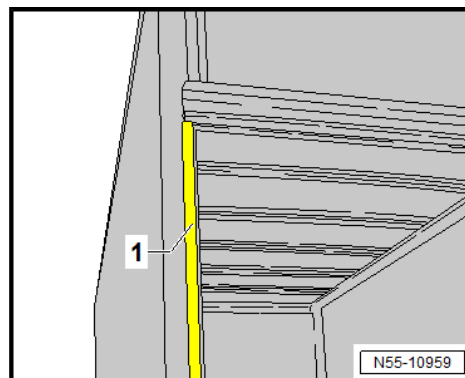
Installing

Installation is carried out in reverse order.

4.2.4 Removing and installing rear box body seals

Removing

- Pull out seal from metal rail.



Installing

Install in reverse order of removal, observing the following:

- Insert seal in metal rail using commercially available lubricant.

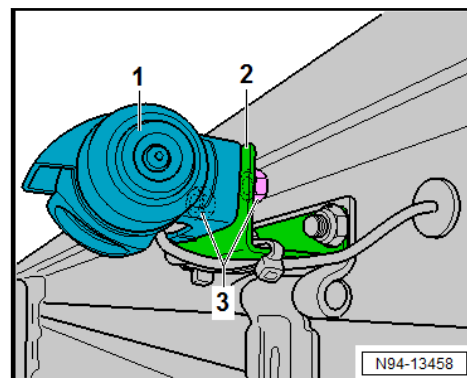
4.2.5 Removing and installing reversing camera

Special tools and workshop equipment required

◆ Torque wrench -V.A.G 1410A-



Removing



- Separate electrical connector on camera.
- Unscrew nuts -3-.
- Remove reversing camera -1- from bracket -2-.

Installing

Installation is carried out in reverse order.

Specified torques

Component	Specified torque
Nuts -2-	10 Nm



5 Box body (Schmitz Cargobull)

⇒ [b5.1 ody, exterior", page 100](#)

⇒ [b5.2 ody, rear", page 105](#)

5.1 Box body, exterior

⇒ [a5.1.1 nd installing roof corner, front", page 100](#)

⇒ [a5.1.2 nd installing roof corner, rear", page 102](#)

⇒ [a5.1.3 nd installing stirrup", page 103](#)

5.1.1 Removing and installing roof corner, front

Special tools and workshop equipment required

◆ Electric cutter -V.A.G 1561A-



◆ Cutting cord -VAS 861 001/1A-



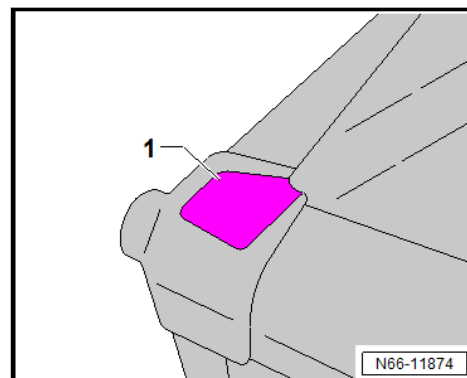
◆ Pull handle -V.A.G 1351/1-

◆ Suction lifter -VAS 6801-

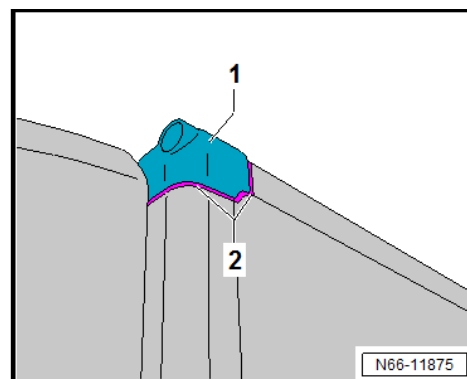
Materials

◆ 1-pack MS assembly adhesive, see ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE

Removing



- Remove front upper marker light ➤ [page 480](#) .
- Separate bonded joint -1- with electric cutter -V.A.G 1561A-.
- Separate seal and bonded joint -2- with cutting cord -VAS 861 001/1A- and pull handle -V.A.G 1351/1- all around.

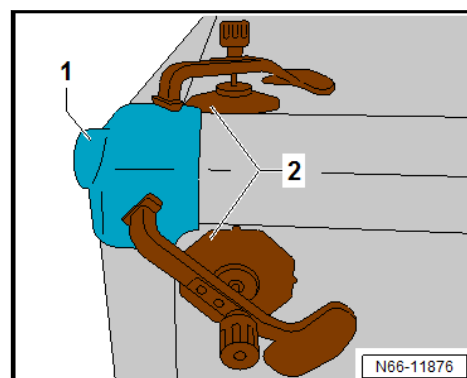


- Remove roof corner -1-.
- Remove sealing and adhesive compound residue.

Installing

Install in reverse order of removal, observing the following:

- Clean bonding surfaces.
- Apply assembly adhesive 1-pack MS assembly adhesive all around.
- Observe processing instructions, see data sheet 1-pack MS assembly adhesive.
- Fit roof corner.
- Secure roof corner -1- with suction lifter -VAS 6801- -2-.





5.1.2 Removing and installing roof corner, rear

Special tools and workshop equipment required

- ◆ Electric cutter -V.A.G 1561A-

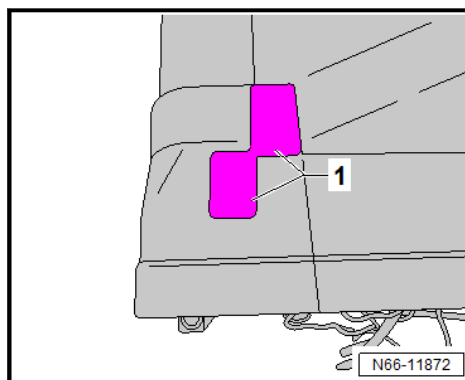


- ◆ Cutting cord -VAS 861 001/1A-

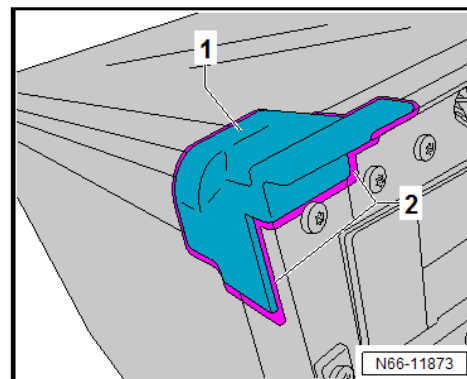


- ◆ Pull handle -V.A.G 1351/1-
- ◆ Suction lifter -VAS 6801-
- ◆ 1-pack MS assembly adhesive, see ➔ Electronic parts catalogue (ETKA) or ➔ webMANTIS for MAN TGE

Removing



- Separate bonded joint -1- with electric cutter -V.A.G 1561A-.
- Separate seal and bonded joint -2- with cutting cord -VAS 861 001/1A- and pull handle -V.A.G 1351/1- all around.



- Remove roof corner -1-.
- Remove sealing and adhesive compound residue.

Installing

Install in reverse order of removal, observing the following:

- Clean bonding surfaces.
- Apply assembly adhesive 1-pack MS assembly adhesive all around.
- Observe processing instructions, see data sheet 1-pack MS assembly adhesive.
- Fit roof corner.
- Secure roof corner with suction lifter -VAS 6801-.

5.1.3 Removing and installing stirrup

Special tools and workshop equipment required

- ◆ Pop rivet pliers -VAS 5072A-

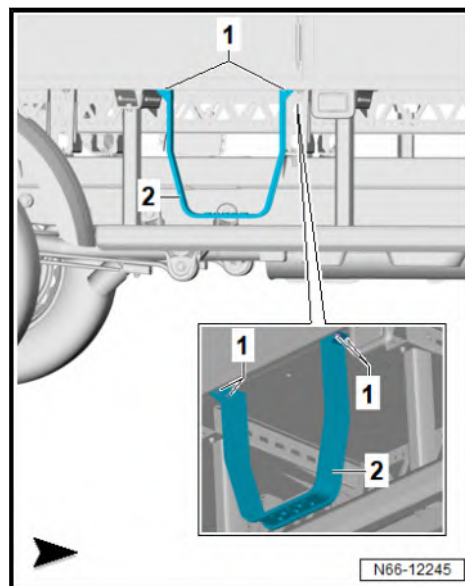


- ◆ Pin punch -VAS 5513-





Removing



CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out rivet heads -1-.
- Drive out rivet shanks -1- with drift -VAS 5513-.
- Remove swarf.
- Detach stirrup -2-.

Installing

Install in reverse order of removal, observing the following:

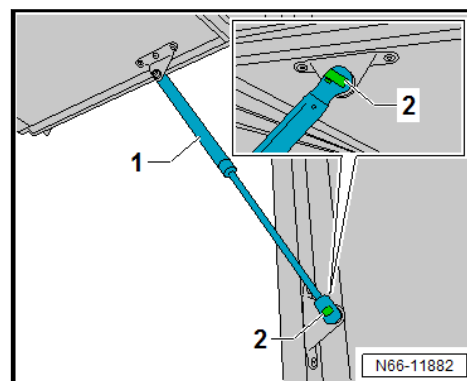
- Use rivet pliers -VAS 5072A- to install new pop rivets.

5.2 Box body, rear

- ⇒ [a5.2.1 nd installing gas strut", page 105](#)
- ⇒ [a5.2.2 nd installing tail flap", page 105](#)
- ⇒ [a5.2.3 nd installing side door", page 107](#)
- ⇒ [a5.2.4 nd installing wing door", page 107](#)
- ⇒ [a5.2.5 nd installing wing door/tail flap seals", page 109](#)
- ⇒ [a5.2.6 nd installing rear box body seals", page 110](#)
- ⇒ [a5.2.7 nd installing reversing camera", page 111](#)
- ⇒ [a5.2.8 nd installing grab handle", page 112](#)
- ⇒ [a5.2.9 nd installing lock case", page 113](#)

5.2.1 Removing and installing gas strut

Removing



- Open tail lift.
- Prop up tail flap and secure.
- Remove clips -2- at top and bottom.
- Remove gas strut -1-.

Installing

Installation is carried out in reverse order.

5.2.2 Removing and installing tail flap

Special tools and workshop equipment required

- ◆ Torque screwdriver -VAS 6494-

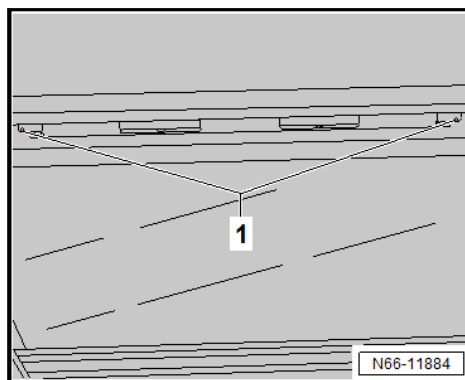




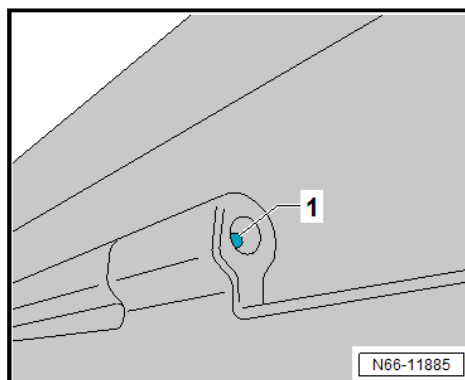
◆ Pin punch -VAS 5513-



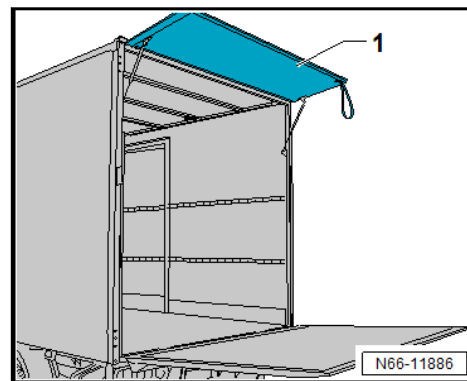
Removing



- Open tail lift.
- Prop up tail flap and secure.
- Remove gas struts of tail flap ➔ [page 105](#) .
- Unscrew bolts -1-.
- Drive out hinge pin -1- with drift -VAS 5513-.



- Remove tail flap -1-.



Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolts -1-	2 Nm

5.2.3 Removing and installing side door

Removal and installation are identical to the tail flap ➔ [page 105](#).

5.2.4 Removing and installing wing door

Special tools and workshop equipment required

- ◆ Adhesive strip remover -VAS 6349-

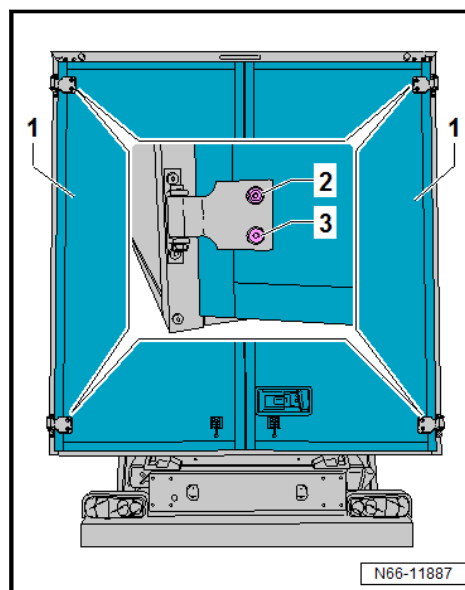


- ◆ Torque wrench 6-50 Nm -V.A.G 1331-

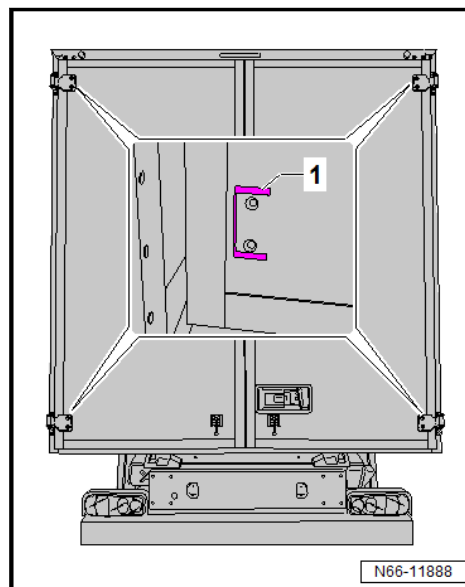




Removing



- Open door.
- Prevent door from falling down.
- Unscrew bolts -2-.
- Unscrew bolts -3-.
- Remove rear wing door -1-.
- Remove adhesive strip residue -1- with adhesive strip remover -VAS 6349-.



Installing

Install in reverse order of removal, observing the following:

- Renew self-locking screws.
- Renew foam rubber seal.

Specified torques

Component	Specified torque
Bolts -2-	40 Nm



Component	Specified torque
Bolts -3-	40 Nm

5.2.5 Removing and installing wing door/tail flap seals

Special tools and workshop equipment required

- ◆ Adhesive strip remover -VAS 6349-

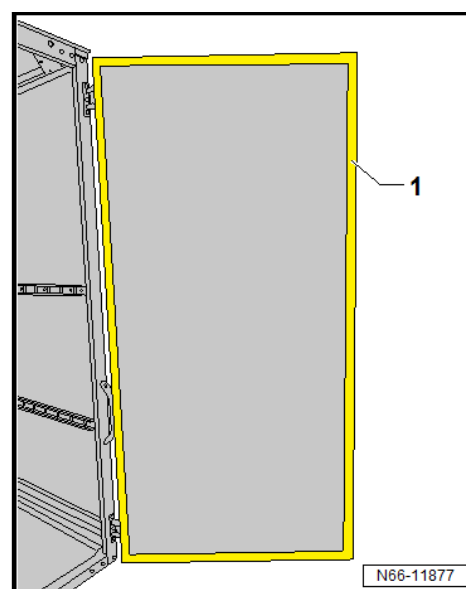


- ◆ Scraper set -VAS 6845-



- ◆ 1-pack MS assembly adhesive, see ➔ Electronic parts catalogue (ETKA) or ➔ webMANTIS for MAN TGE

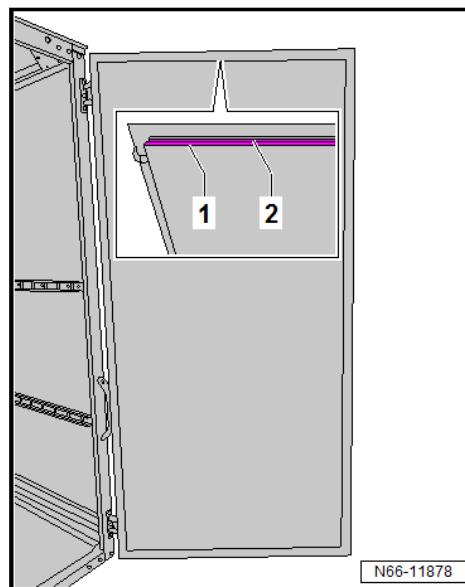
Removing



- Detach seal -1- all round.

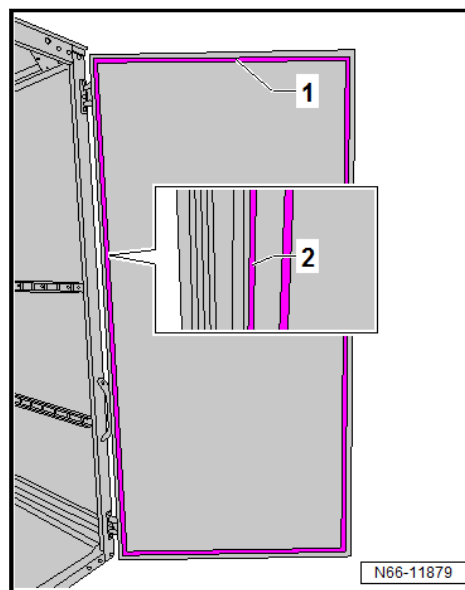


- Remove seal -1-.
- Remove adhesive strip residue -1- with adhesive strip remover -VAS 6349-.



- Remove adhesive residue -2- using scraper set -VAS 6845-.

Installing



Install in reverse order of removal, observing the following:

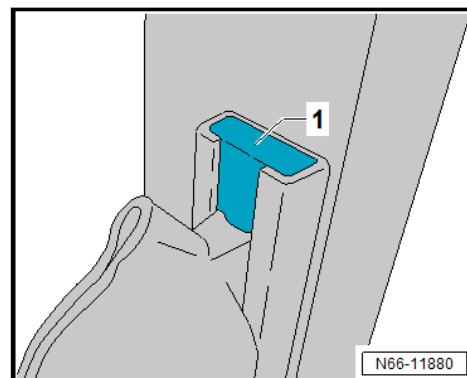
- Apply double-sided adhesive tape -1- all around.
- Apply 1-component MS assembly adhesive -2- all around in a bead of 2 to 3 mm max. height.
- Observe processing instructions, see data sheet 1-pack MS assembly adhesive.

5.2.6 Removing and installing rear box body seals

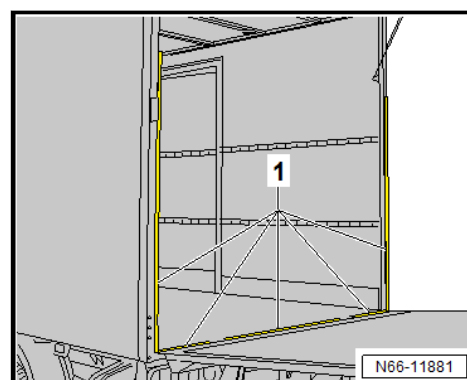
Materials

- ♦ 1-pack MS assembly adhesive, see ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE

Removing



- Remove end piece -1-.



- Pull out seal -1-.

Installing

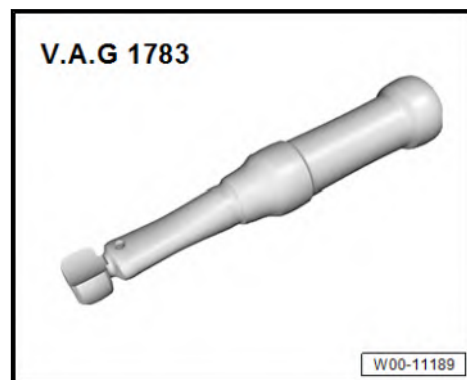
Install in reverse order of removal, observing the following:

- Insert seal in metal rail using commercially available lubricant.
- Clean end piece.
- Bond in end piece with 1-pack MS assembly adhesive.
- Observe processing instructions, see data sheet 1-pack MS assembly adhesive.

5.2.7 Removing and installing reversing camera

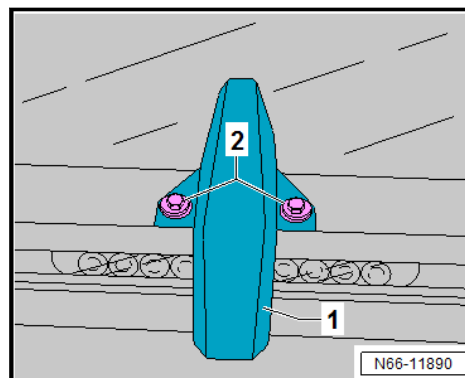
Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1783-

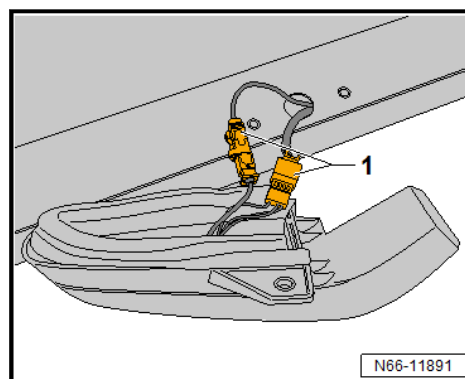




Removing



- Unscrew bolts -2-.
- Detach reversing camera -1-.
- Disconnect electrical connectors -1-.



Installing

Install in reverse order of removal, observing the following:

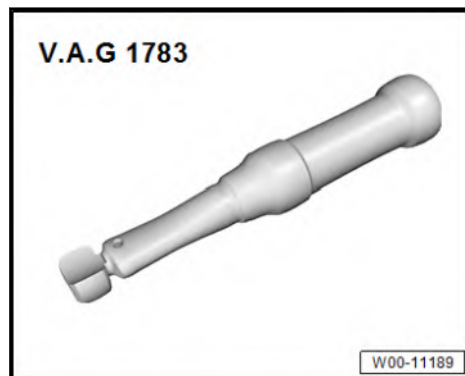
Specified torques

Component	Specified torque
Bolts -2-	8 Nm

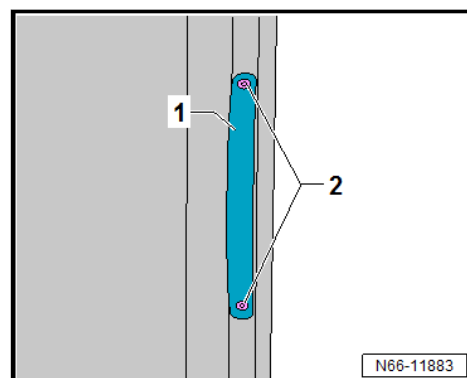
5.2.8 Removing and installing grab handle

Special tools and workshop equipment required

- ♦ Torque wrench -V.A.G 1783-



Removing



- Unscrew bolts -2-.
- Remove handle -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolts -2-	10 Nm

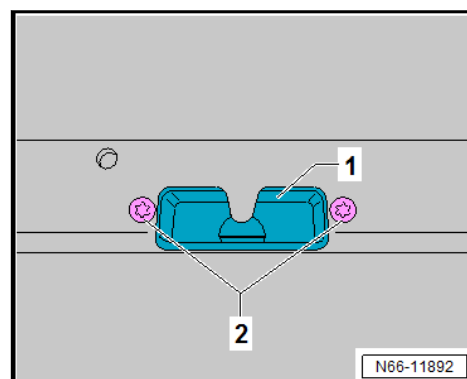
5.2.9 Removing and installing lock case

Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1410A-



Removing



- Open door.
- Unscrew bolts -2-.
- Remove lock case -1-.



Installing

- Renew self-locking screws.

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolts -2-	11 Nm

6 Tail lift (Sörensen)

⇒ [6.1 , page 115](#)

⇒ [s6.2 ystem", page 120](#)

6.1 Platform

⇒ [a6.1.1 nd installing platform", page 115](#)

⇒ [p6.1.2 latform to floor of vehicle", page 116](#)

⇒ [p6.1.3 latform to body", page 117](#)

⇒ [p6.1.4 latform angle", page 119](#)

⇒ [a6.1.5 nd installing warning flag", page 119](#)

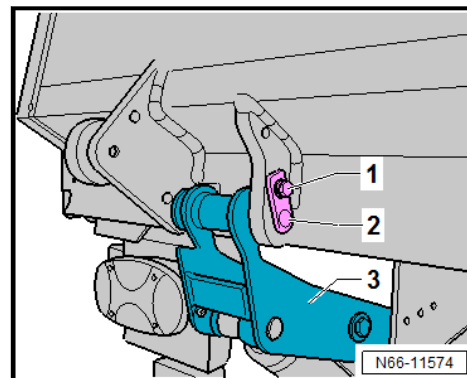
6.1.1 Removing and installing platform

Special tools and workshop equipment required

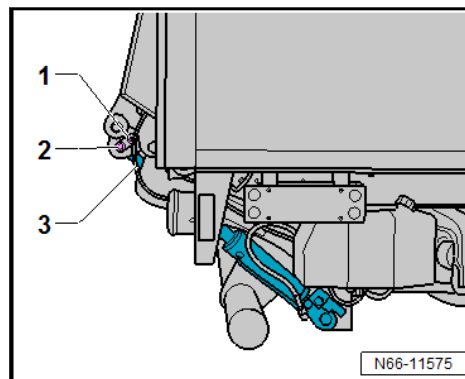
- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



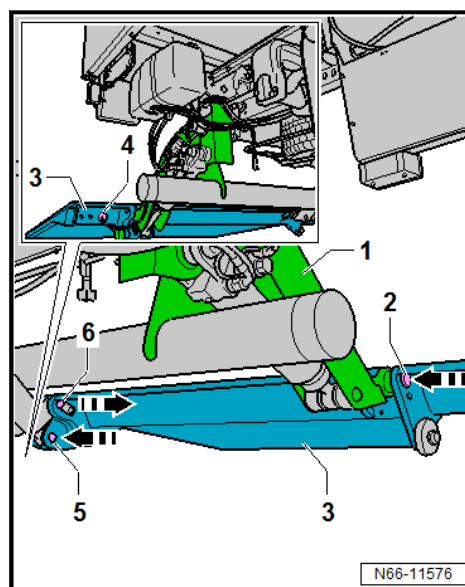
Removing



- Before starting any work, secure the platform against accidental movement.
- Before removal, open the platform via hydraulic system far enough to allow for the pins to be driven out.
- Unscrew the retaining bolt -1- for the pin -2- of the lifting lever -3- on both sides of the platform.
- Unscrew retaining bolt -1- for pin -2- of tilt cylinder -3-.



- Lower the tail lift down to ground level and secure it.
- Select off position on switch for tail lift operation ⇒ [page 522](#).
- Detach cap -4- on platform -3-, pull out and separate electrical connector.



- Drive out pins -2-, -5- and -6- in -direction of arrow-.
- Remove platform -3- from lifting mechanism -1-.

Installing

Install in reverse order of removal, observing the following:

- Before reassembly, apply special assembly grease onto the lifting lever and closing cylinder bearings.

Specified torques

Component	Specified torque
Bolts	46 Nm

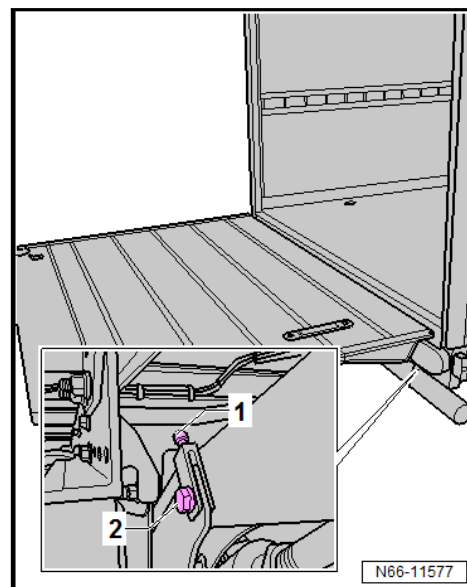
6.1.2 Adjusting platform to floor of vehicle

Special tools and workshop equipment required

- ◆ Torque wrench, 40 to 200 Nm -V.A.G 1332A-



Procedure



- Move the lifting mechanism hydraulically behind the rear frame with the platform open. In the process, make sure that the closing arms do not come into contact with the rear frame.
- Create a gap of approx. 10 mm.
- Loosen bolt of right closing arm -2-.
- Turn the setting bolt -1- until the rear frame and platform are parallel to one another.
 - Turning the setting bolt -1- clockwise moves the closing arm higher up to the rear frame.
 - Turning the setting bolt -1- anti-clockwise lowers the closing arm further away from the rear frame.
- Tighten bolt -2-.

Specified torques

Component	Specified torque
Bolt -2-	195 Nm

6.1.3 Adjusting platform to body

Special tools and workshop equipment required



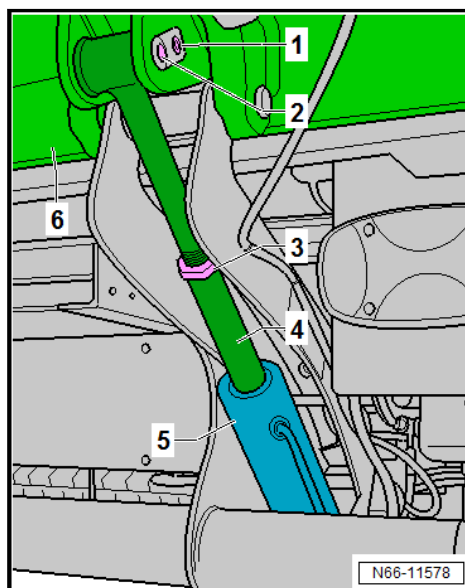
- ◆ Torque wrench, 40 to 200 Nm -V.A.G 1332A-



- ◆ Slugging ring spanner AF46 -VAS 273 003-

Procedure

- ◆ During the hydraulic closing procedure, the cylinder must move against the stop (restriction in cylinder) when the platform is moved lightly against the rear portal or 90° to the floor of the vehicle.
- ◆ If the platform moves with full pressure against the body, the piston rod of the tilt cylinder must be adjusted.
- To relieve tension from the tilt cylinder -5-, open the platform by 10°-15°.



- Secure platform -6- against falling using suitable workshop equipment.
- Loosen lock nut -3- on piston rod -4- using slugging ring spanner AF46 -VAS 273 003-.
- Remove retaining bolt -1- from pin -2-.
- Drive out pin -2-. Secure tilt cylinder -5- when doing this to prevent it from falling down.

Adjusting

- Grease the pin -2- before any trial run in the platform -6-, drive it in and secure it with the retaining bolt -1- to prevent it from turning.
- Screw piston rod head in or out accordingly until a light preload is built up during hydraulic closing of platform -6-.

- Secure the pin -2- with retaining bolt -1-.
- Tighten lock nut -3- on piston rod -4- using slugging ring spanner AF46 -VAS 273 003-.

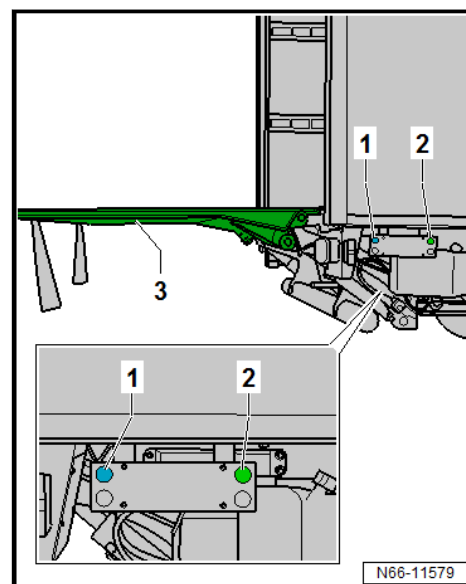
Specified torques

Component	Specified torque
Bolt -1-	46 Nm
Nut -3-	80 Nm

6.1.4 Adjusting platform angle

- ◆ Adjustment of the platform angle is carried out by means of a key combination on the operating panel.
- ◆ The set position is moved to each time until re-programming is carried out. This is indicated by a brief interruption of the warning lights.

Adjustment of the platform



- Move platform -3- to horizontal position using manual switch.
- Press button -1- three times.
- Afterwards, press button -2- three times.
- Readjust the tilting angle of the platform following every repair or assembly of the tail lift.

6.1.5 Removing and installing warning flag

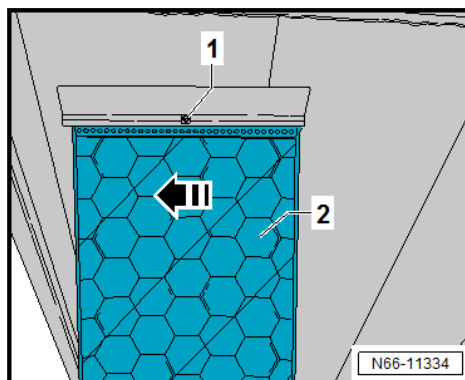
Special tools and workshop equipment required



◆ Torque screwdriver -VAS 6494-



Removing



- Unscrew bolt -1-.
- Pull out warning flag -2- from side -arrow-.

Installing

Install in reverse order of removal, observing the following:

- Before installing the warning flag, note correct warning direction.

Specified torques

Component	Specified torque
Bolt	2 Nm

6.2 Hydraulic system

⇒ [a6.2.1 nd installing pop-up cylinder", page 120](#)

⇒ [a6.2.2 nd installing lock cylinder", page 122](#)

⇒ [a6.2.3 nd installing hydraulic pump", page 124](#)

⇒ [t6.2.5 he hydraulic system", page 125](#)

⇒ [a6.2.6 nd installing tail lift pressure switchF559", page 125](#)

⇒ [a6.2.7 nd installing solenoid valveN729/N730", page 127](#)

6.2.1 Removing and installing pop-up cylinder

Special tools and workshop equipment required



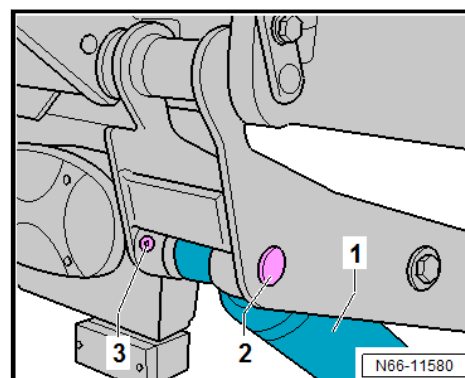
- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



- ◆ Drip tray for workshop hoist -VAS 6208-



Removing

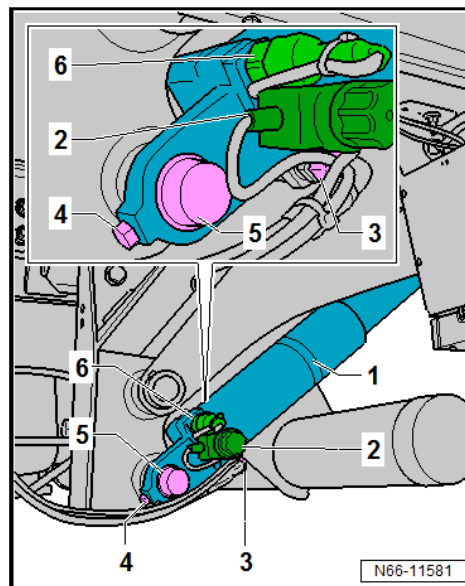


- Before removing the pop-up cylinders, the platform must be supported and secured against falling down with suitable workshop means.

CAUTION

Injury hazard from residual pressure in the hydraulic system.

- Wear safety goggles.
 - Wear protective gloves!
-
- Drive out split pin -3- from upper pin -2- of the pop-up cylinder -1-.
 - Carefully loosen threaded connection of hydraulic line -3-. While so, collect escaping hydraulic fluid in drip tray for workshop hoist -VAS 6208-.



- Remove tail lift pressure switch -F559- -6- ➤ [page 125](#) .
- Remove pop-up cylinder solenoid valve -N729--2- ➤ [page 127](#) .
- Unscrew bolt -4- from lower pin.
- Remove protective cap -5- from pin.
- Drive out upper and lower pin. When doing this, prevent the pop-up cylinder -1- from falling down.
- Remove pop-up cylinder -1- from lifting mechanism.

Installing

Install in reverse order of removal, observing the following:

- Bleed hydraulic system by raising and lowering several times.
- Check hydraulic fluid level and replenish as necessary.

Specified torques

Component	Specified torque
Bolt	20 Nm

6.2.2 Removing and installing lock cylinder

Special tools and workshop equipment required

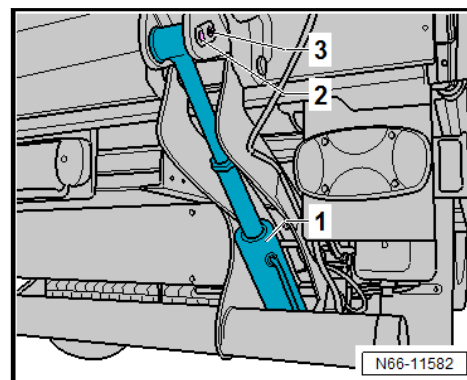
- ♦ Torque wrench 6-50 Nm -V.A.G 1331-



- ◆ Drip tray for workshop hoist -VAS 6208-



Removing

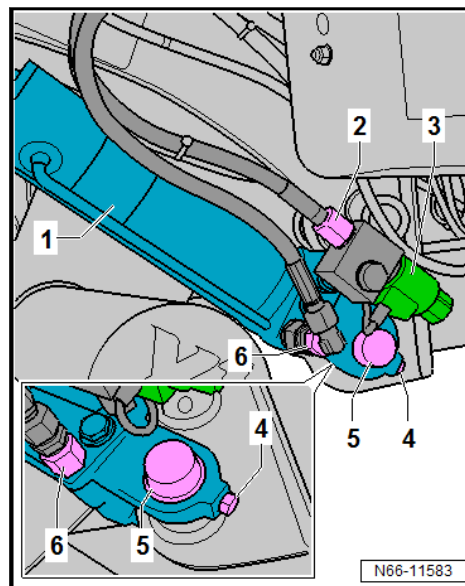


- Before removing the closing cylinder, the platform must be supported and secured against falling down with suitable workshop means.

CAUTION

Injury hazard from residual pressure in the hydraulic system.

- Wear safety goggles.
 - Wear protective gloves!
-
- Unscrew the bolt -3- for the upper pin -2- of the closing cylinder -1-.
 - Carefully loosen threaded connection of hydraulic line -2-. While so, collect escaping hydraulic fluid in drip tray for workshop hoist -VAS 6208-.



- Carefully loosen threaded connection of hydraulic line with angled piece -6-. While so, collect escaping hydraulic fluid in drip tray for workshop hoist -VAS 6208-.
- Remove closing cylinder solenoid valve -N730--3- ➔ [page 127](#).
- Unscrew bolt -4- from lower pin -5-.
- Drive out upper and lower pin. When doing this, prevent lock cylinder -1- from falling down.
- Remove closing cylinder -1- from lifting mechanism.

Installing

Install in reverse order of removal, observing the following:

- Bleed hydraulic system by opening and closing several times.
- Check hydraulic fluid level and replenish as necessary.

Specified torques

Component	Specified torque
Union nut -6-	20 Nm
Union nut -2-	Tighten onto stop. Then, turn 90° further.

6.2.3 Removing and installing hydraulic pump

For technical reasons, instructions for this repair group could not be completed in time for publication in this workshop manual. The repair procedures will be published in a later revision.

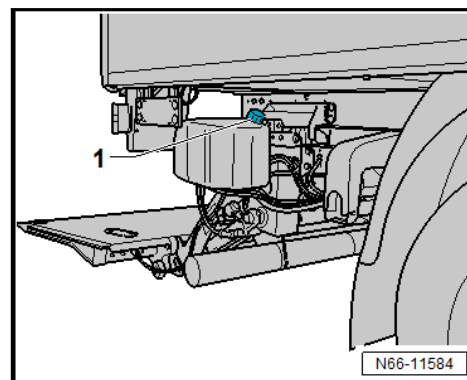
6.2.4 Changing hydraulic fluid

Special tools and workshop equipment required

◆ Used oil collection and extraction unit -VAS 6622A-



- Open tail lift, and lower it to the ground.



- Unscrew cap with dipstick -1- from hydraulic fluid reservoir, and pull it out.
- Extract hydraulic fluid from hydraulic fluid reservoir using used oil collection and extraction unit -VAS 6622A-.
- Fill new hydraulic fluid.
- Observe manufacturer's notes and filling quantity specifications.

The maximum oil level is reached when the dipstick is submerged approx. one centimetre in the oil.

To clean the filter, the hydraulic fluid reservoir needs to be removed.

- Bleed hydraulic system ⇒ [page 125](#) .

6.2.5 Bleeding the hydraulic system

- Raise, lower, open and close tail lift several times to bleed hydraulic system.
- Check oil level at cap with dipstick, and replenish oil if required.

6.2.6 Removing and installing tail lift pressure switch -F559-

Special tools and workshop equipment required



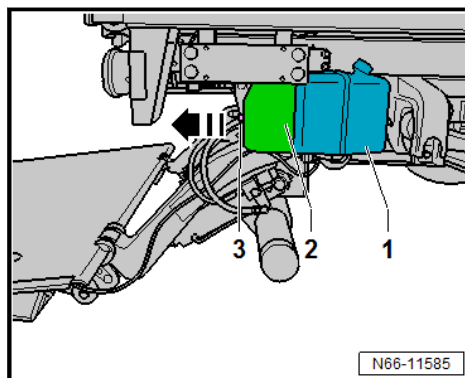
- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



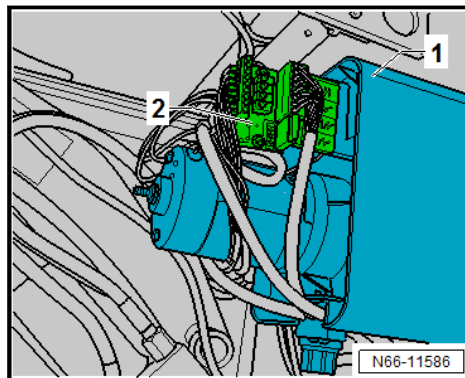
- ◆ Drip tray for workshop hoist -VAS 6208-



Removing



- Lower the platform to the ground.
- Unscrew nut -3- from hydraulic unit -1- and remove cap -2- in -direction of arrow-.
- Separate electrical connector on service switch -2- in hydraulic unit -1- (2 x pin S4).



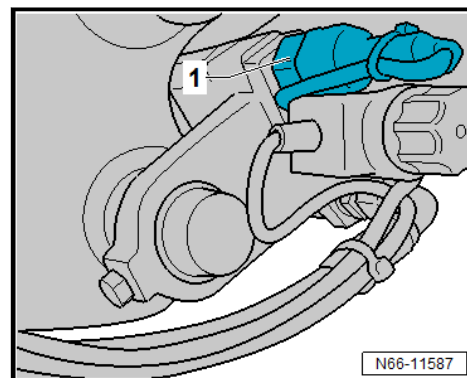


- Detach wiring harness as far as tail lift pressure switch -F559-.

⚠ CAUTION

Injury hazard from residual pressure in the hydraulic system.

- **Wear safety goggles.**
 - **Wear protective gloves!**
- Carefully unscrew tail lift pressure switch -F559- -1-. While so, collect escaping hydraulic fluid in drip tray for workshop hoist -VAS 6208-.



Installing

Install in reverse order of removal, observing the following:

- Use new seal.
- Check hydraulic fluid level after installing the tail lift pressure switch -F559-, replenish as necessary.
- Bleed hydraulic system ➔ [page 125](#).

Specified torques

Component	Specified torque
Tail lift pressure switch -F559- -1-	20 Nm
Nut -3-	9.5 Nm

6.2.7 Removing and installing solenoid valve -N729/N730-

Special tools and workshop equipment required

- ◆ Torque wrench 6-50 Nm -V.A.G 1331-

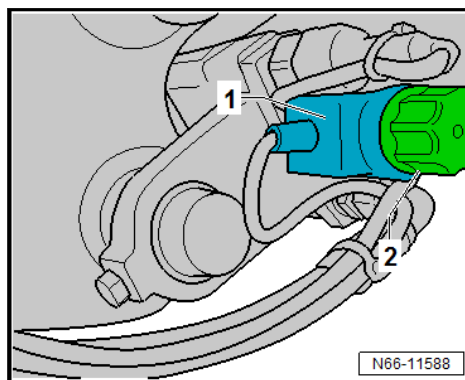




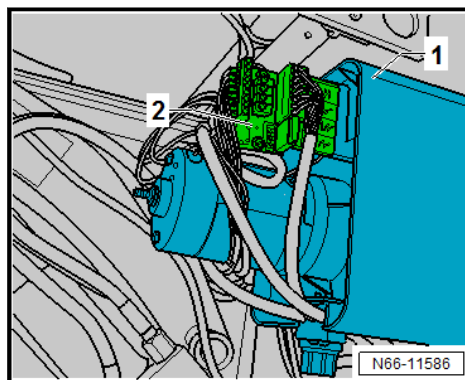
◆ Drip tray for workshop hoist -VAS 6208-



Removing



- Lower the platform to the ground.
- Unscrew cap nut -2- from solenoid valve -N729/N730- -1-.
- Pull off solenoid from valve insert and detach cable up to service switch in hydraulic unit.
- Separate electrical connector on service switch -2- in hydraulic unit -1-.



Notes on valve types:

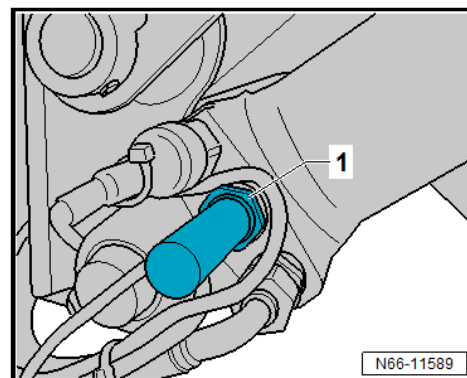
- ◆ Right lock cylinder = PIN Y1 + engine earth
- ◆ Right pop-up cylinder = PIN Y3 + engine earth
- ◆ Directional valve on hydraulic unit = YA + engine ground



⚠ CAUTION

Injury hazard from residual pressure in the hydraulic system.

- **Wear safety goggles.**
 - **Wear protective gloves!**
-
- Carefully unscrew valve insert -1-. While so, collect escaping hydraulic fluid in drip tray for workshop hoist -VAS 6208-.



Installing

Install in reverse order of removal, observing the following:

- Use new seal.
- Check hydraulic fluid level after installing the solenoid valve -N729/N730-, replenish as necessary.
- Bleed hydraulic system ➔ [page 125](#) .

Specified torques

Component	Specified torque
Cap nut -2-	Hand-tight
Valve insert -1-	25 Nm



7 Tail lift (BÄR - Cargolift)

⇒ [7.1, page 130](#)

⇒ [s7.2 ystem", page 134](#)

7.1 Platform

⇒ [a7.1.1 nd installing platform", page 130](#)

⇒ [p7.1.2 latform to body", page 131](#)

⇒ [a7.1.3 nd installing warning flag", page 133](#)

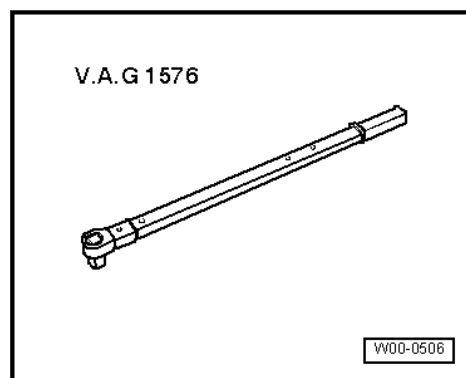
7.1.1 Removing and installing platform

Special tools and workshop equipment required

◆ Torque wrench -V.A.G 1410A-

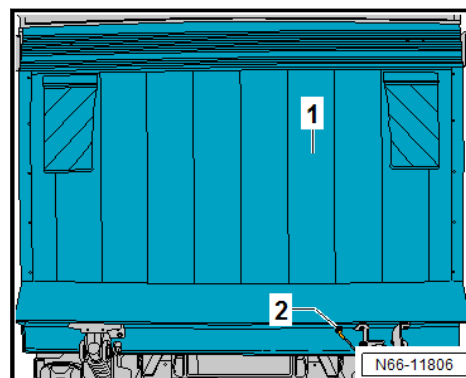


◆ Torque wrench 80-400 Nm -V.A.G 1576-

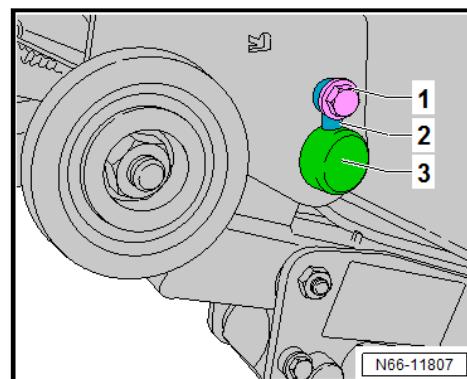


◆ Grease, see ⇒ Electronic parts catalogue (ETKA) or ⇒ web-MANTIS for MAN TGE

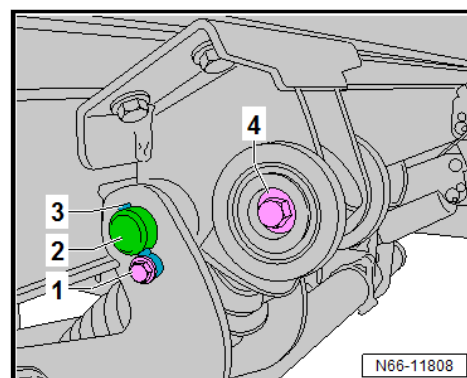
Removing



- Secure platform -1- against falling using suitable workshop equipment.
- Switch off tail lift operation switch.
- Disconnect connector -2-.
- Unscrew bolts -1- on both sides.



- Remove locking pin -2- on both sides.
- Drive out pin -3- using a commercially available soft-head hammer.
- Unscrew bolts -1- on both sides.



- Remove locking pin -3- on both sides.
- Unscrew bolt -4-.
- Drive out pin -2-.

Installing

Install in reverse order of removal, observing the following:

- Clean and grease insides of bearings.
- Renew self-locking nuts and bolts.
- After installing the platform, the bearing points must be lubricated via the grease nipples until grease escapes on both sides and a band of grease can be seen.

Specified torques

Component	Specified torque
Bolts -1-	10 Nm
Bolt -4-	⇒ page 138

7.1.2 Adjusting platform to body

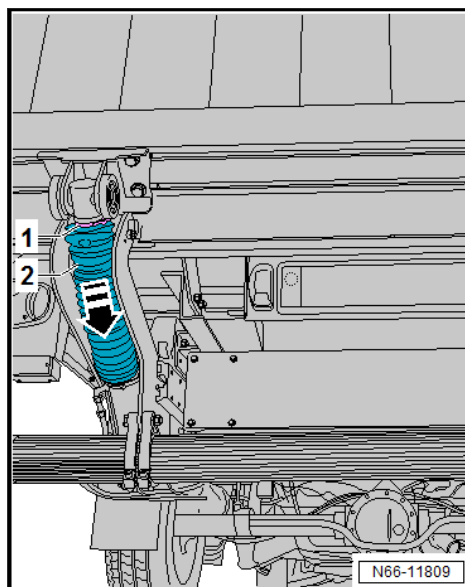
Special tools and workshop equipment required



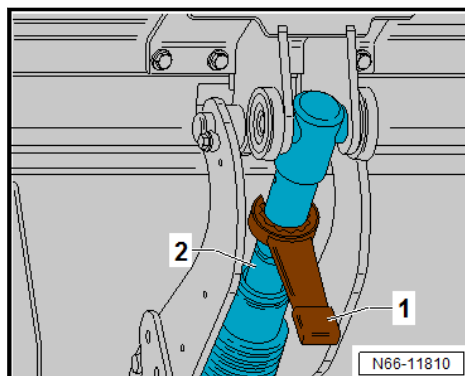
- ◆ Slugging ring spanner AF60 -VAS 273 001- for vehicles ➤ 2016
- ◆ Slugging ring spanner AF41 -VAS 273 005- for vehicles 2017 ➤
- ◆ Torque screwdriver -VAS 6494-



Procedure



- Shut the platform.
- Switch off tail lift operation switch.
- Release hose clip -1-.
- Push back bellows -2- in direction of arrow.
- Loosen lock nut with slugging ring spanner AF60 -VAS 273 001- or slugging ring spanner AF41 -VAS 273 005- -1-.





- Set tilt cylinder -2- to slight preload.

Specified torques

Component	Specified torque
Hose clip -1-	3 Nm

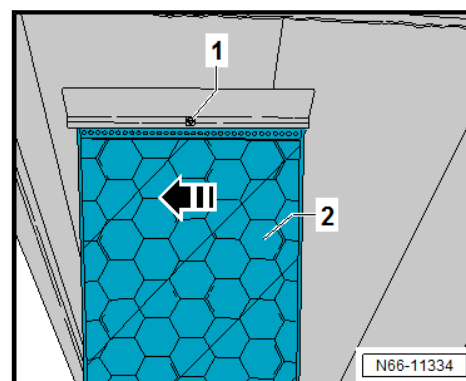
7.1.3 Removing and installing warning flag

Special tools and workshop equipment required

- ◆ Torque wrench -VAS 6253A-



Removing



- Unscrew bolt -1-.
- Remove warning flag -2- in direction of arrow.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolt -1-	1 Nm



7.2 Hydraulic system

⇒ [a7.2.1 nd installing pop-up cylinder", page 134](#)

⇒ [a7.2.2 nd installing lock cylinder", page 136](#)

⇒ [a7.2.3 nd installing tail lift hydraulics motorV642", page 138](#)

⇒ [a7.2.4 nd installing solenoid valveN729/N730", page 140](#)

⇒ [a7.2.5 nd installing tail lift pressure switchF559", page 142](#)

⇒ [h7.2.6 ydraulic fluid", page 143](#)

⇒ [t7.2.7 he hydraulic system", page 144](#)

7.2.1 Removing and installing pop-up cylinder

Special tools and workshop equipment required

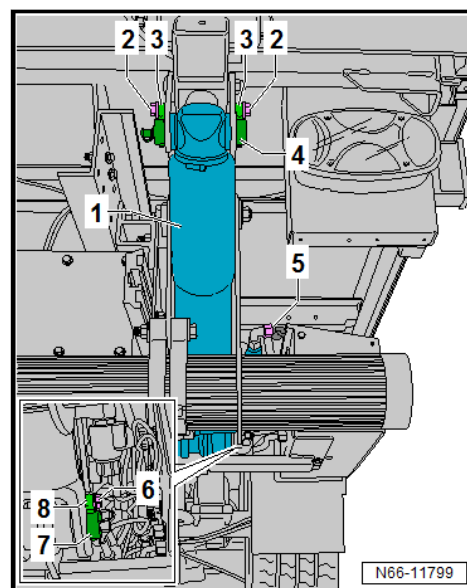
- ◆ Torque wrench -V.A.G 1410A-



- ◆ Drip tray for workshop hoist -VAS 6208-



Removing



- Secure platform against falling using suitable workshop equipment.
- Switch off tail lift operation switch.
- Unscrew bolts -2- on both sides.
- Pull out locking pins -3- on both sides.
- Drive out pin -4- using a commercially available soft-head hammer.

CAUTION

Injury hazard from residual pressure in the hydraulic system.

- Wear safety goggles.
- Wear protective gloves!

- Unscrew union nut -5-. While so, collect escaping hydraulic fluid in drip tray for workshop hoist -VAS 6208-.
- If necessary, remove tail lift pressure switch -F559- ➔ [page 142](#).
- If necessary, remove pop-up cylinder solenoid valve -N729- ➔ [page 140](#).
- Unscrew bolts -6- on both sides.
- Pull out locking pins -7- on both sides.
- Drive out pin -8-.
- Remove pop-up cylinder -1-.

Installing

Install in reverse order of removal, observing the following:

- Fill hydraulic fluid.
- Observe manufacturer's specifications.
- Bleed hydraulic system.



Specified torques

Component	Specified torque
Locking pin bolts -2 and 6-	10 Nm
Hydraulic line union nut -5-	20 Nm

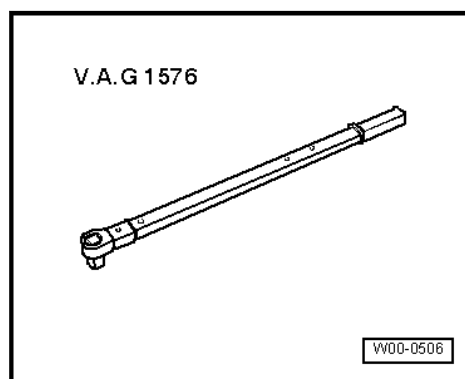
7.2.2 Removing and installing lock cylinder

Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1410A-



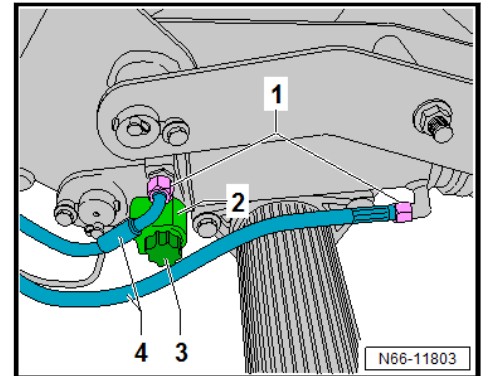
- ◆ Torque wrench 80-400 Nm -V.A.G 1576-



- ◆ Drip tray for workshop hoist -VAS 6208-



Removing

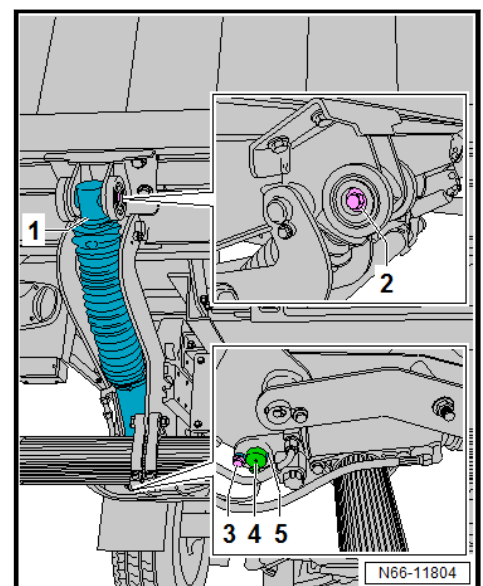


- Secure platform against falling using suitable workshop equipment.
- Switch off tail lift operation switch.

CAUTION

Injury hazard from residual pressure in the hydraulic system.

- Wear safety goggles.
 - Wear protective gloves!
- Carefully unscrew union nuts -1-. While so, collect escaping hydraulic fluid in drip tray for workshop hoist -VAS 6208-.
 - Detach hydraulic lines -4-.
 - Unscrew cap nut -3-.
 - Remove solenoid -2-.
 - If necessary, remove closing cylinder solenoid valve -N730- ➔ [page 140](#).
 - Unscrew bolt -2-.



- Unscrew bolts -3- on both sides.
- Remove locking pin -5- on both sides.
- Drive out pin -4- using a commercially available soft-head hammer.



- Remove closing cylinder -1-.

Installing

Install in reverse order of removal, observing the following:

- Fill hydraulic fluid in accordance with manufacturer's instructions.
- Refer to operating instructions and service handbook.
- Bleed hydraulic system.

Specified torques

Component	Specified torque
Locking pin bolts -3-	10 Nm
Hydraulic line union nuts -1-	20 Nm
Lock cylinder bolt -2-	310 Nm

7.2.3 Removing and installing tail lift hydraulics motor -V642-

Special tools and workshop equipment required

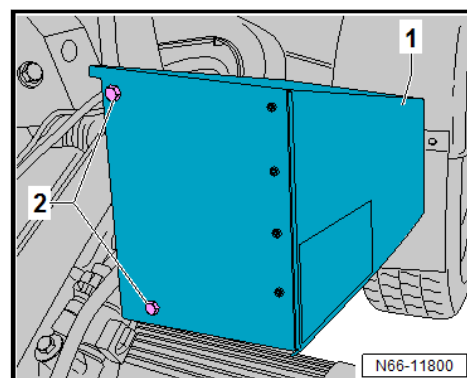
- ♦ Torque wrench -V.A.G 1410A-



- ♦ Drip tray for workshop hoist -VAS 6208-



Removing



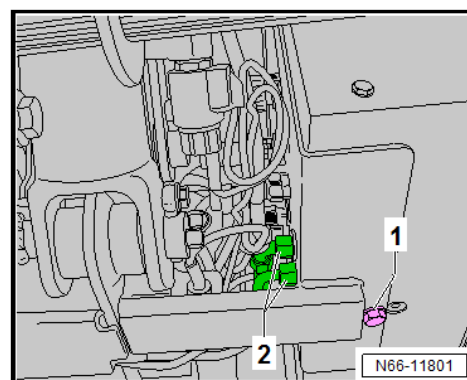
- Switch off tail lift operation switch.
- Disconnect battery ⇒ Electrical system; Rep. gr. 27; Battery; Disconnecting and reconnecting battery.
- Unscrew bolts -2- on both sides.
- Detach cover -1-.



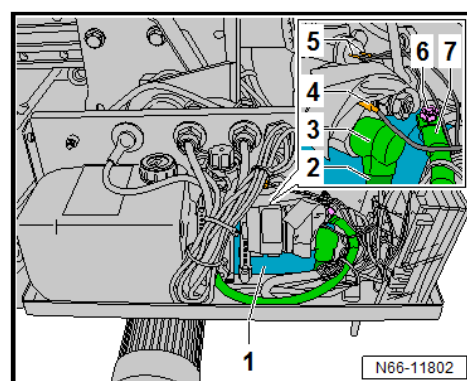
CAUTION

Injury hazard from residual pressure in the hydraulic system.

- Wear safety goggles.
 - Wear protective gloves!
- Unscrew bolt -1-. While so, collect escaping hydraulic fluid in drip tray for workshop hoist -VAS 6208-.



- Loosen union nut -2-. While so, collect escaping hydraulic fluid in drip tray for workshop hoist -VAS 6208-.
- Detach cap -3- together with nut located underneath.





- Detach electrical wire -2-.
- Disconnect connector -4-.
- Disconnect connector -5-.
- Unscrew nut -6-.
- Detach electrical wire -7-.
- Remove tail lift hydraulics motor -V642- -1-.

Installing

Install in reverse order of removal, observing the following:

- Fill hydraulic fluid in accordance with manufacturer's instructions.
- Refer to operating instructions and service handbook.
- Bleed hydraulic system ⇒ [page 143](#) .

Specified torques

Component	Specified torque
Cover bolt -1-	10 Nm
Hydraulic line union nut -2-	20 Nm
Electrical wire nut -6-	7 Nm

7.2.4 Removing and installing solenoid valve -N729/N730-

Special tools and workshop equipment required

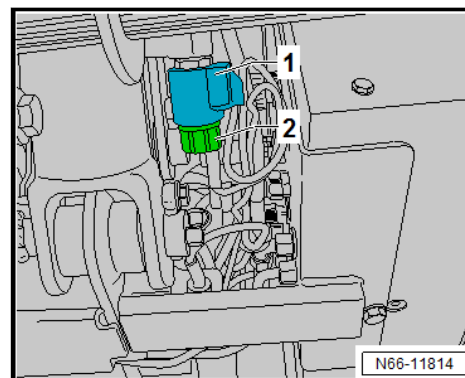
- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



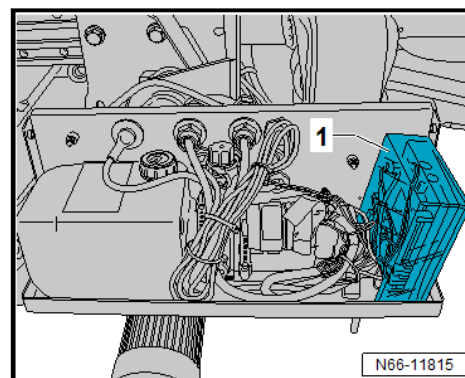
- ◆ Drip tray for workshop hoist -VAS 6208-



Removing



- Lower platform onto floor.
- Switch off tail lift operation switch.
- Unscrew cap nut -2-.
- Pull off solenoid -1-.
- Detach wiring harness up to hydraulic unit.
- Disconnect electrical connector on central electrics -1-.

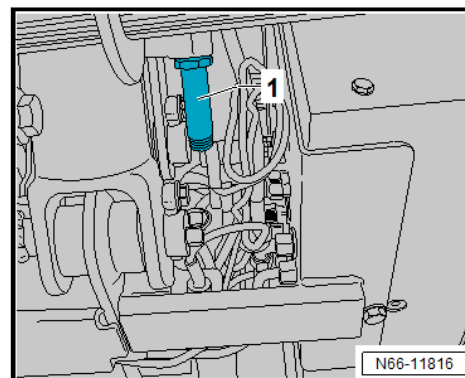


CAUTION

Injury hazard from residual pressure in the hydraulic system.

- Wear safety goggles.
- Wear protective gloves!

- Unscrew valve insert -1-. While so, collect escaping hydraulic fluid in drip tray for workshop hoist -VAS 6208-.





Installing

Install in reverse order of removal, observing the following:

- Check hydraulic fluid level after installing the solenoid valve -N729/N730-, replenish as necessary.
- Refer to operating instructions and service handbook.
- Bleed hydraulic system.

Specified torques

Component	Specified torque
Cap nut -2-	6 Nm
Valve insert -1-	25 Nm

7.2.5 Removing and installing tail lift pressure switch -F559-

Special tools and workshop equipment required

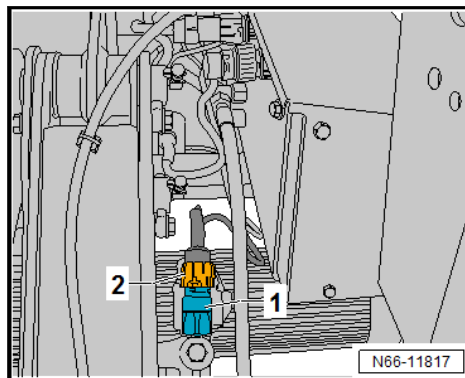
- ◆ Torque wrench -V.A.G 1410A-



- ◆ Drip tray for workshop hoist -VAS 6208-



Removing



- Lower platform onto floor.
- Switch off tail lift operation switch.
- Disconnect connector -2-.

⚠ CAUTION

Injury hazard from residual pressure in the hydraulic system.

- Wear safety goggles.
- Wear protective gloves!

- Unscrew tail lift pressure switch -F559- -1-. While so, collect escaping hydraulic fluid in drip tray for workshop hoist -VAS 6208-.

Installing

Install in reverse order of removal, observing the following:

- Check hydraulic fluid level after installing the solenoid valve, replenish as necessary.
- Refer to operating instructions and service handbook.
- Bleed hydraulic system.

Specified torques

Component	Specified torque
Tail lift pressure switch -F559- -1-	15 Nm

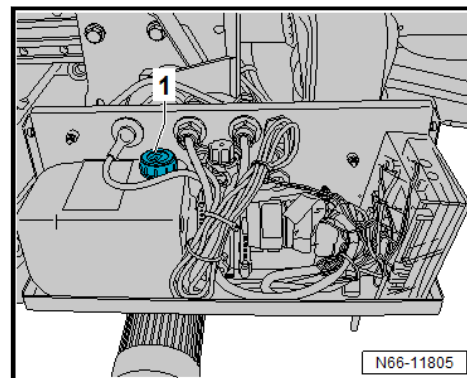
7.2.6 Renewing hydraulic fluid

Special tools and workshop equipment required

- ◆ Used oil collection and extraction unit -VAS 6622A-



Removing



- Open cover -1-.



- Extract hydraulic fluid from hydraulic fluid reservoir using used oil collection and extraction unit -VAS 6622A-.

Installing

Install in reverse order of removal, observing the following:

- Replenish hydraulic fluid after installation of hydraulic unit.
- Refer to operating instructions and service handbook.
- Bleed hydraulic system.
- After bleeding, check fluid level, and replenish fluid if necessary.

7.2.7 Bleeding the hydraulic system

Sequence of operations

- Open and close platform several times.
- After bleeding, check fluid level, and replenish fluid if necessary.
- Refer to operating instructions and service handbook.

8 Tail lift (Palfinger)

⇒ [8.1 , page 145](#)

⇒ [s8.2 ystem", page 150](#)

8.1 Platform

⇒ [a8.1.1 nd installing platform", page 145](#)

⇒ [t8.1.2 ilt switch", page 147](#)

⇒ [p8.1.3 latform to body", page 147](#)

⇒ [a8.1.4 nd installing warning flag", page 149](#)

8.1.1 Removing and installing platform

Special tools and workshop equipment required

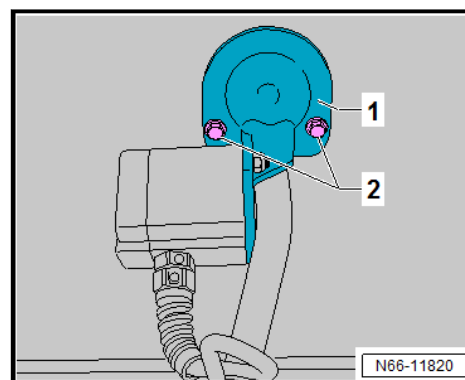
◆ Torque wrench -V.A.G 1410A-



◆ Torque wrench, 40 to 200 Nm -V.A.G 1332A-



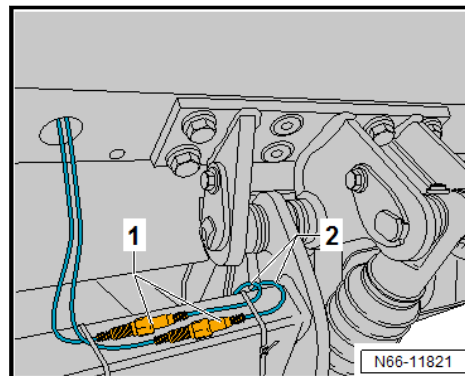
Removing



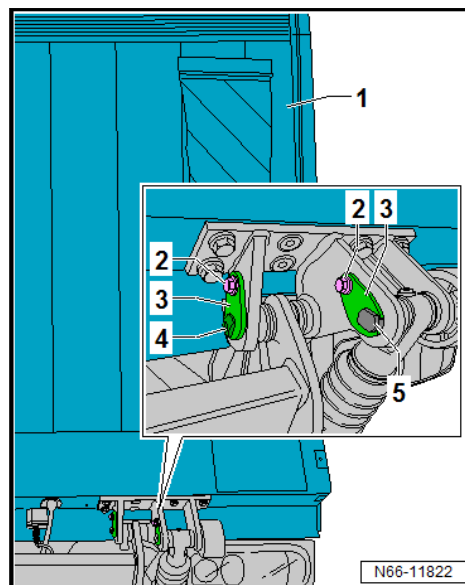
- Secure platform against falling using suitable workshop equipment.



- Remove inclination sensor ➔ [page 497](#) .
- Unscrew bolts -2-.
- Detach strain relief -1-.
- Pull out wiring harness -2-.



- Disconnect connector -1-.
- Unscrew bolts -2- on both sides.



- Detach securing plates -3- on both sides.
- Drive out pin -4- on both sides using a commercially available soft-head hammer.
- Drive out pin -5- using a commercially available soft-head hammer.
- Detach platform -1-.

Installing

Install in reverse order of removal, observing the following:

- Renew self-locking nuts and bolts.

Specified torques

Component	Specified torque
Pull relief bolts -2-	6 Nm
Tab washer bolts -2-	75 Nm

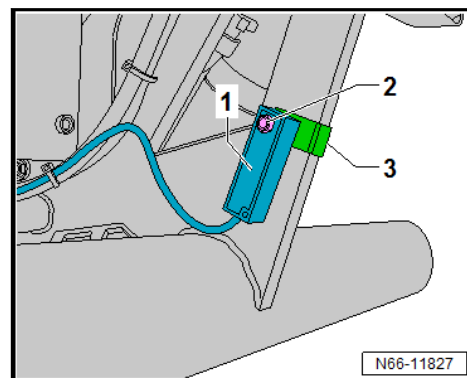
8.1.2 Adjusting tilt switch

Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1410A-



Procedure



- Open platform approx. 250 mm above floor.
- Loosen nut -2-.
- Release tilt switch -1-.
- Move tilt switch to horizontal position.
- Tighten nut -2-.
- Bend securing plate -3- around edge.

Specified torques

Component	Specified torque
Self-locking nut -2-	9 Nm

8.1.3 Adjusting platform to body

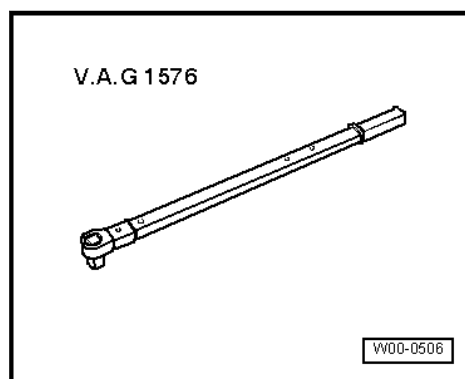
Special tools and workshop equipment required



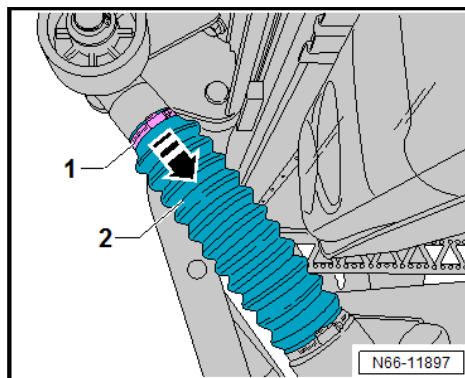
◆ Torque screwdriver -VAS 6494-



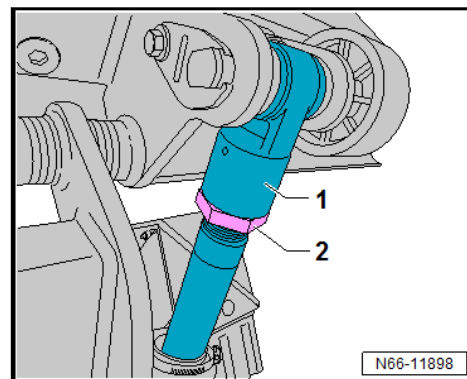
◆ Torque wrench 80-400 Nm -V.A.G 1576-



Procedure



- Close platform as far as possible.
- Relieve tilt cylinder with button.
- Loosen hose clip -1-.
- Press bellows -2- downwards in direction of arrow.
- Loosen lock nut -2- with impact ring spanner.



- Adjust tilt cylinder -1- to desired end position by turning piston rod.
- Relief tilt cylinder by opening platform slightly.
- Tighten lock nut -2-.

Specified torques

Component	Specified torque
Hose clip -1-	3 Nm
Tilt cylinder lock nut -2-	300 Nm

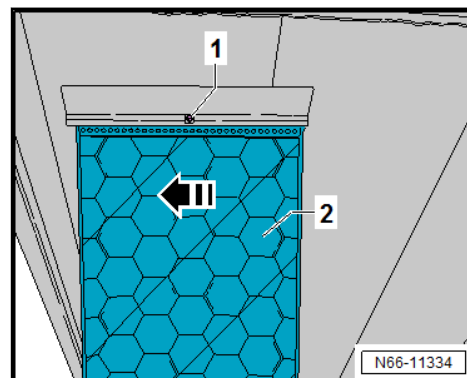
8.1.4 Removing and installing warning flag

Special tools and workshop equipment required

- ◆ Torque wrench -VAS 6253A-



Removing



- Unscrew bolt -1-.
- Remove warning flag -2- in direction of arrow.



Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolt -1-	1 Nm

8.2 Hydraulic system

⇒ [a8.2.1 nd installing pop-up cylinder", page 150](#)

⇒ [a8.2.2 nd installing tilt cylinder", page 152](#)

⇒ [a8.2.3 nd installing hydraulic pump", page 154](#)

⇒ [a8.2.4 nd installing solenoid valve", page 157](#)

⇒ [h8.2.5 ydraulic fluid", page 159](#)

⇒ [t8.2.6 he hydraulic system", page 160](#)

8.2.1 Removing and installing pop-up cylinder

Special tools and workshop equipment required

- ◆ Drip tray for workshop hoist -VAS 6208-



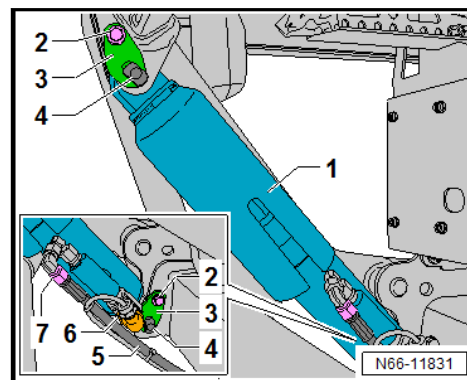
- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



◆ Torque wrench -VAS 6253A-



Removing



- Secure platform against falling using suitable workshop equipment.
- Switch off tail lift switch.

CAUTION

Injury hazard from residual pressure in the hydraulic system.

- Wear safety goggles.
- Wear protective gloves!

- Carefully unscrew union nuts -7-. While so, collect escaping hydraulic fluid in drip tray for workshop hoist -VAS 6208-.
- Detach hydraulic line -5-.
- Disconnect connector -6-.
- If necessary, remove solenoid valve ➔ [page 157](#) .
- Unscrew bolts -2-.
- Detach locking plates -3-.
- Drive out pin -4- using a commercially available soft-head hammer.
- Remove pop-up cylinder -1-.

Installing

Install in reverse order of removal, observing the following:

- Fill hydraulic fluid in accordance with manufacturer's instructions.
- Refer to operating instructions and service handbook
- Bleed hydraulic system ➔ [page 160](#) .



Specified torques

Component	Specified torque
Tab washer bolts -2-	75 Nm
Hydraulic line union nut -7-	35 Nm

8.2.2 Removing and installing tilt cylinder

Special tools and workshop equipment required

- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



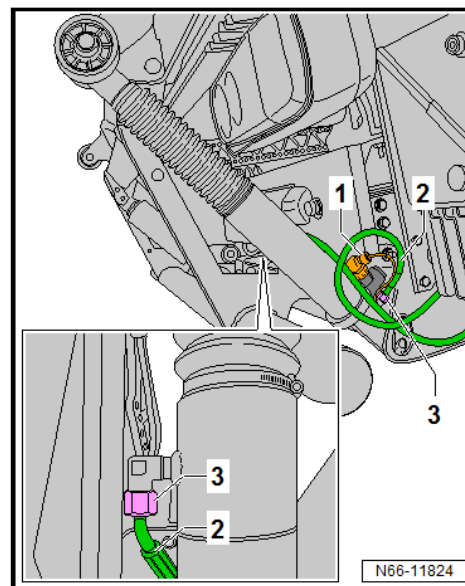
- ◆ Torque wrench -VAS 6253A-



- ◆ Drip tray for workshop hoist -VAS 6208-



Removing



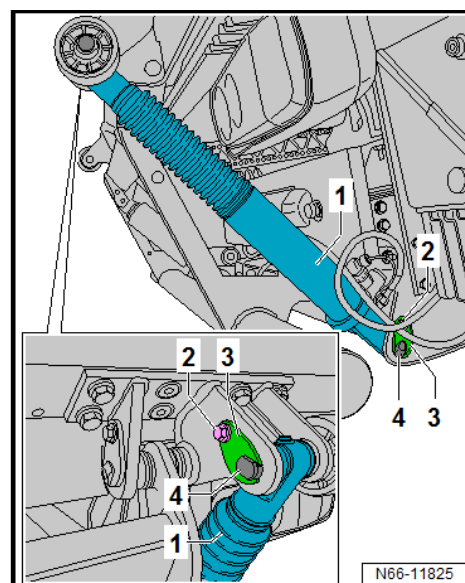
- Secure platform against falling using suitable workshop equipment.
- Switch off tail lift switch.

CAUTION

Injury hazard from residual pressure in the hydraulic system.

- Wear safety goggles.
- Wear protective gloves!

- Unscrew union nuts -3-.
- Carefully detach hydraulic lines -2-. While so, collect escaping hydraulic fluid in drip tray for workshop hoist -VAS 6208-.
- Disconnect connector -1-.
- If necessary, remove solenoid valve ➔ [page 157](#) .
- Unscrew bolts -2-.





- Detach locking plates -3-.
- Drive out pin -4- using a commercially available soft-head hammer.
- Detach tilt cylinder -1-.

Installing

Install in reverse order of removal, observing the following:

- Fill hydraulic fluid in accordance with manufacturer's instructions.
- Refer to operating instructions and service handbook
- Bleed hydraulic system ➔ [page 160](#) .

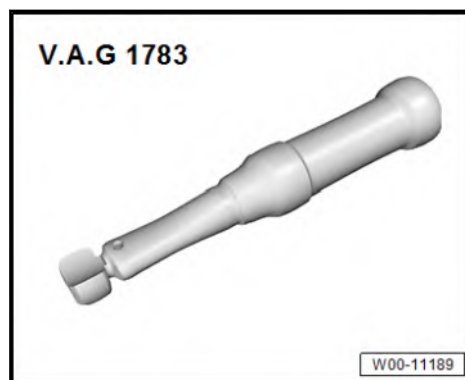
Specified torques

Component	Specified torque
Tab washer bolts -2-	75 Nm
Hydraulic line union nut -3-	35 Nm

8.2.3 Removing and installing hydraulic pump

Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1783-



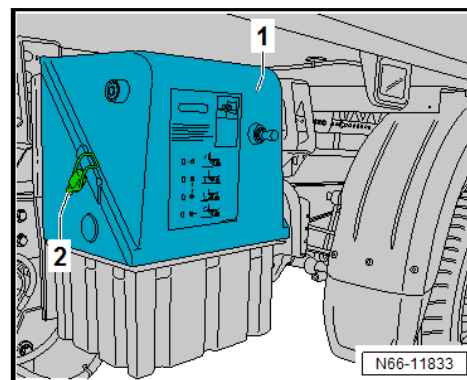
- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



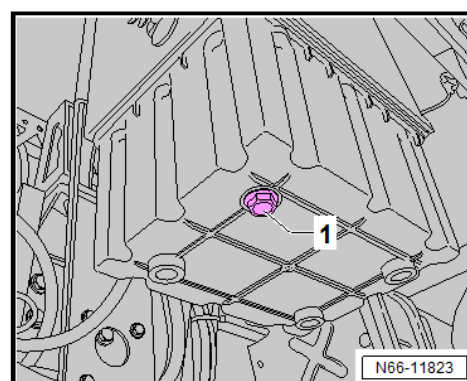
◆ Drip tray for workshop hoist -VAS 6208-



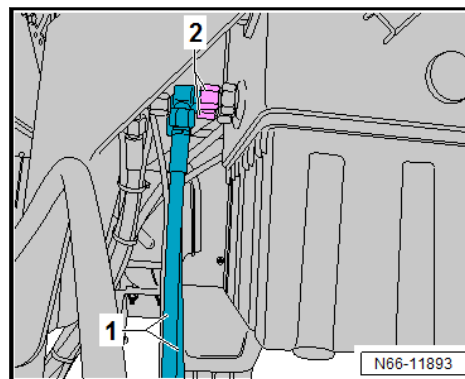
Removing



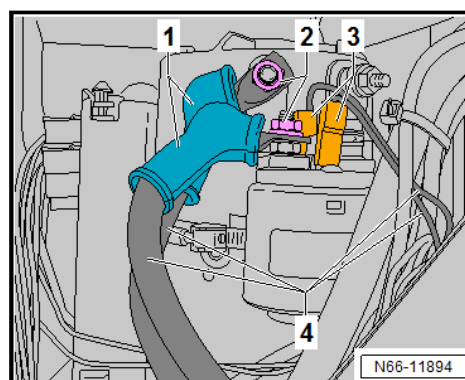
- Switch off tail lift switch.
- Disconnect battery ⇒ Electrical system; Rep. gr. 27; Battery; Disconnecting and reconnecting battery.
- Open clip -2- on both sides.
- Detach cover -1-.
- Place drip tray for workshop hoist -VAS 6208- underneath.
- Unscrew oil drain plug -1-.



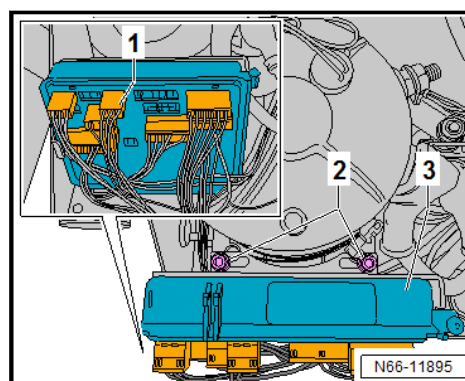
- Unscrew union nuts -2-.



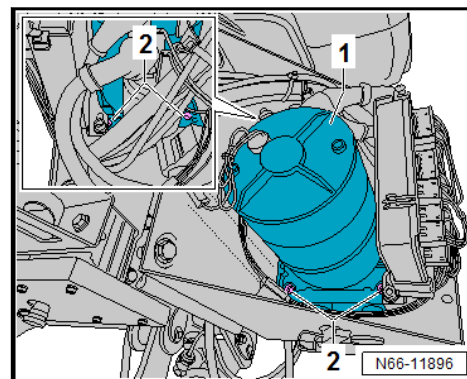
- Detach hydraulic lines -1-.
- Pull back protective caps -1-.



- Unscrew nuts -2-.
- Disconnect electrical connectors -3-.
- Remove wiring harnesses -4-.
- Disconnect electrical connectors -1-.



- Unscrew bolts -2-.
- Remove control unit -3-.
- Unscrew bolts -2-.



- Remove hydraulic pump -1-.

Installing

Install in reverse order of removal, observing the following:

- Fill hydraulic fluid in accordance with manufacturer's instructions.
- Refer to operating instructions and service handbook
- Bleed hydraulic system ⇒ [page 160](#) .

Specified torques

Component	Specified torque
Control unit bolts -2-	30 Nm
Nuts -2-	9 Nm
Union nut -2-	35 Nm
Hydraulic pump bolts -2-	10 Nm
Oil drain plug -1-	2.5 Nm

8.2.4 Removing and installing solenoid valve

Special tools and workshop equipment required

- ◆ Torque wrench 6-50 Nm -V.A.G 1331-

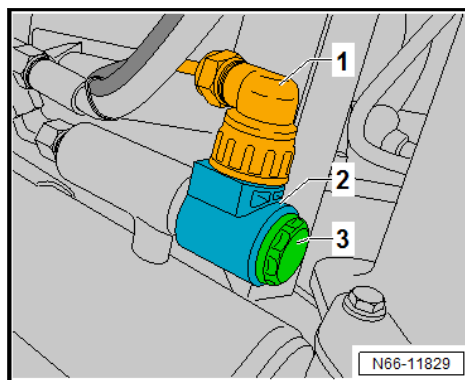




◆ Drip tray for workshop hoist -VAS 6208-



Removing

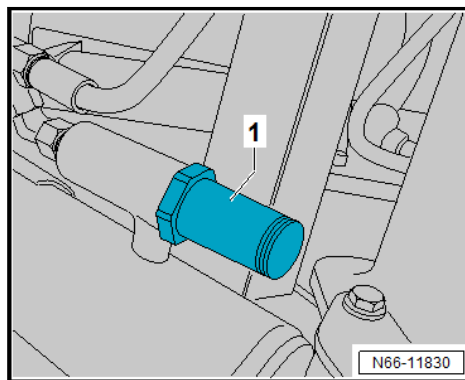


- Lower platform onto floor.
- Switch off tail lift switch.
- Disconnect connector -1-.
- Unscrew cap nut -3-.
- Pull off solenoid -2-.

CAUTION

Injury hazard from residual pressure in the hydraulic system.

- Wear safety goggles.
 - Wear protective gloves!
-
- Unscrew valve insert -1-. While so, collect escaping hydraulic fluid in drip tray for workshop hoist -VAS 6208-.



Installing

Install in reverse order of removal, observing the following:



- Check hydraulic fluid level after installing the solenoid valve, replenish as necessary.
- Refer to operating instructions and service handbook
- Bleed hydraulic system ⇒ [page 160](#) .

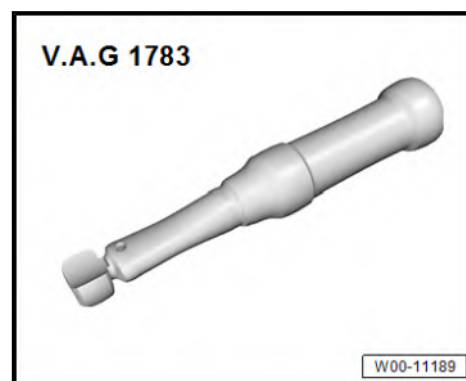
Specified torques

Component	Specified torque
Cap nut -3-	6 Nm
Valve insert -1-	30 Nm

8.2.5 Renewing hydraulic fluid

Special tools and workshop equipment required

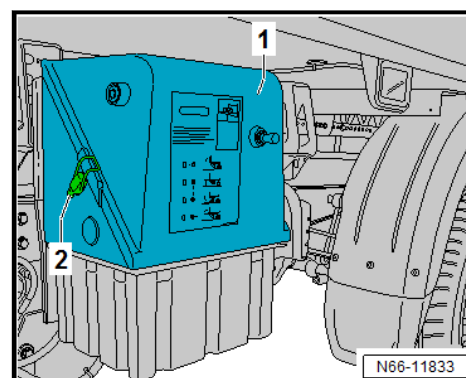
- ◆ Torque wrench -V.A.G 1783-



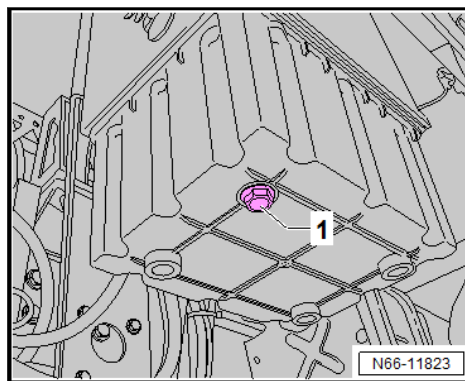
- ◆ Drip tray for workshop hoist -VAS 6208-



Sequence of operations



- Open clip -2- on both sides.
- Detach cover -1-.
- Place drip tray for workshop hoist -VAS 6208- underneath.



- Unscrew oil drain plug -1-.
- Screw in oil drain plug -1-. Tighten oil drain plug -1- using torque wrench -V.A.G 1783-.
- Fill hydraulic fluid in accordance with manufacturer's instructions.
- Refer to operating instructions and service handbook
- Bleed hydraulic system ➤ [page 160](#) .

Specified torques

Component	Specified torque
Bolt -1-	2.5 Nm

8.2.6 Bleeding the hydraulic system

Sequence of operations

- Open and close platform several times.
- After bleeding, check fluid level, and replenish fluid if necessary.
- Refer to operating instructions and service handbook



9 Repairing box body

⇒ [p9.1 plywood panel", page 161](#)

⇒ [s9.2 sandwich panel", page 165](#)

⇒ [t9.3 transparent roof", page 181](#)

9.1 Repairing plywood panel

Special tools and workshop equipment required

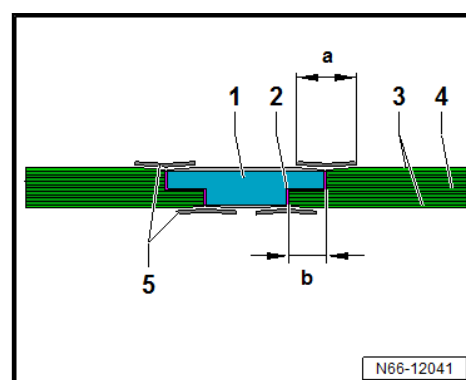
- ◆ Sanding system -VAS 6677-
- ◆ Hand-cartridge gun -V.A.G 1628-



- ◆ Extraction system -VAS 6572/2-



Overview cross-sectional diagram of side panel repair



Item	Designation
-1-	Repair section
-2-	2-pack assembly adhesive
-3-	GRP cover layer
-4-	Plywood



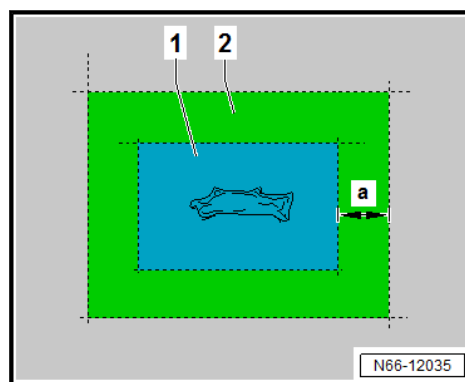
Item	Designation
-5-	GRP laminate
-a-	60 mm
-b-	40 mm

Replacement parts, materials

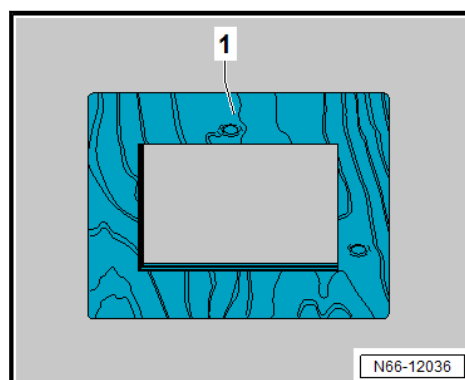
For part numbers of materials, refer to ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

- ◆ 2-pack assembly adhesive
- ◆ 2-pack polyester resin
- ◆ Fibreglass cloth
- ◆ Fibreglass filler
- ◆ Fine filler
- ◆ Cleaner for plastics -D.195.850.A1-

Processing damaged area

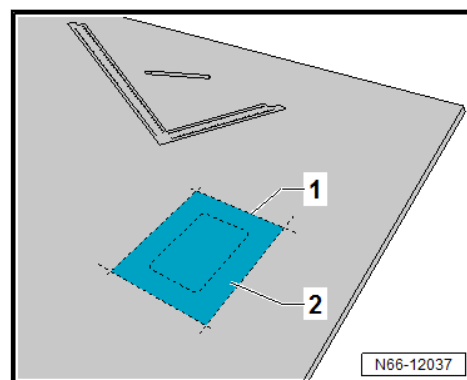


- Mark inner repair area -1- all around.
- Mark outer repair area -2- all around to dimension -a- = 40 mm.
- Cut out inner repair area -1- all around using commercially available jigsaw.
- Set to half material thickness on depth stop of commercially available router.



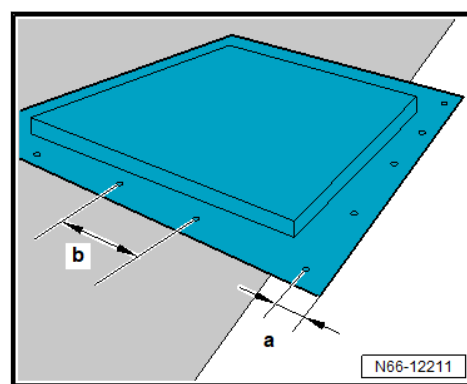
- Route outer repair area -1- all around to half material thickness.
- Clean repair area with extraction unit -VAS 6572/2-.

Preparing repair section



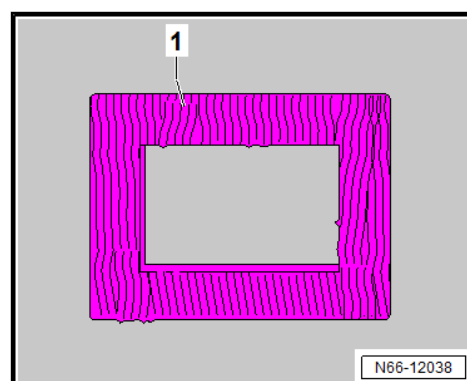
It must be ensured that an adhesive gap of approx. 2 mm is guaranteed between the edges and bonding surfaces when manufacturing the repair piece.

- Mark repair section -1-.
- Cut out repair section -1- using commercially available jig-saw.
- Set to half material thickness on depth stop of commercially available router plus approx. 1 mm.
- Route plywood -2- all around using commercially available router.
- Drill holes in repair section -1- according to dimensions -a- = approx. 10 mm and -b- = approx. 100 mm (diameter: 4.2 mm).



- Clean repair section -1- of dust.
- Adapt repair section -1- to fit.

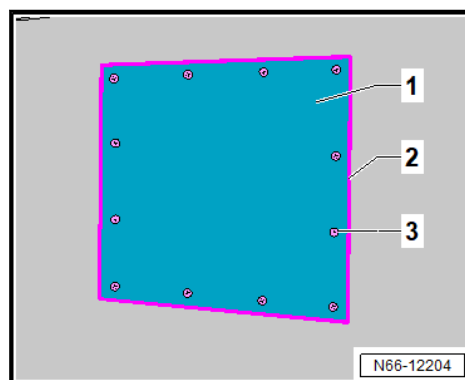
Bonding in repair section





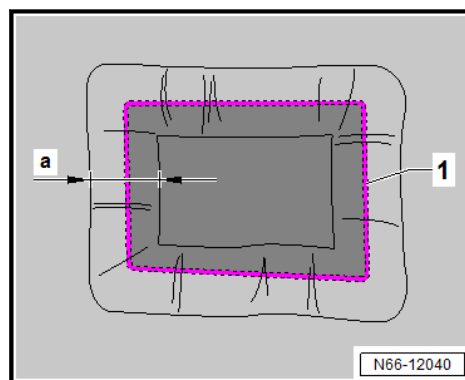
Observe processing instructions on packaging and in technical bulletins of 2-pack assembly adhesive.

- Apply 2-pack assembly adhesive -1- using hand-cartridge gun -V.A.G 1628-.
- Spread 2-pack assembly adhesive -1- evenly using commercially available toothed scraper.
- Bond in repair section -1-.

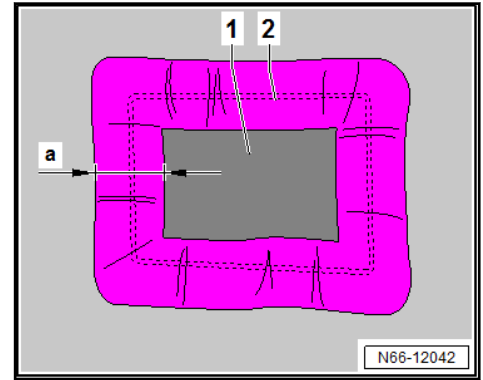


- Make sure to observe the processing time of the 2-pack assembly adhesive which is 45 minutes.
- Fix repair section in position with commercially available wood screws -3-, ensuring that length of screw is suitable for material thickness.
- Fill in adhesive gap -2- all around with 2-pack assembly adhesive.
- Remove excess 2-pack assembly adhesive.

Treating repair area

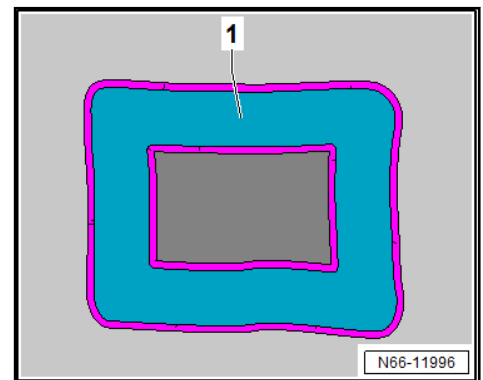


- After drying time of 2-pack assembly adhesive is over, unscrew bolts.
- Sand down adhesive gap -1- all around to dimension -a- = 60 mm outwards tapering out to plywood respectively.
- Clean repair area -1- using extraction unit -VAS 6572/2-.



- Clean repair area -1- with cleaner for plastics - D.195.850.A1-.
- Make sure to observe the flash-off time of the cleaner for plastics -D.195.850.A1- which is at least 10 minutes.
- Observe processing instructions on packaging and in technical bulletins of 2-pack polyester resin.
- Fill screw holes with 2-pack polyester resin.
- Apply 2-pack polyester resin-2- with commercially available paint roller in area measuring -a- = 60 mm.

Observe processing instructions on packaging and in technical bulletins of 2-pack polyester resin and fibreglass filler.



- Position fibreglass cloth -1- with coarse structure centrally.
- Apply 2-pack polyester resin to fibreglass cloth -1-.
- Sand down repair area with grinding/sanding system -VAS 6677-.
- Fill repair area with fine filler compound.
- Sand down repair area with grinding/sanding system -VAS 6677-.
- Prime repair area.
- Paint repair area.

9.2 Repairing sandwich panel

⇒ [G9.2.1 RP cover layer", page 165](#)

⇒ [o9.2.2 ne-sided damage", page 167](#)

⇒ [t9.2.3 wo-sided damage", page 174](#)

9.2.1 Repairing GRP cover layer

Special tools and workshop equipment required



- ◆ Sanding system -VAS 6677-
- ◆ Extraction system -VAS 6572/2-

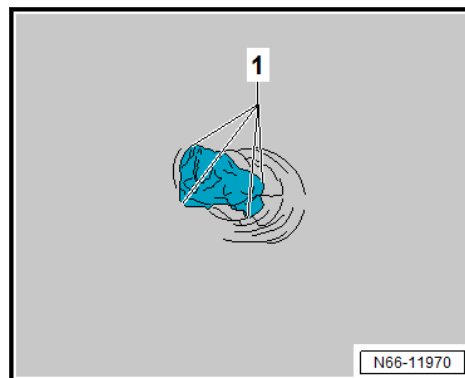


Replacement parts, materials

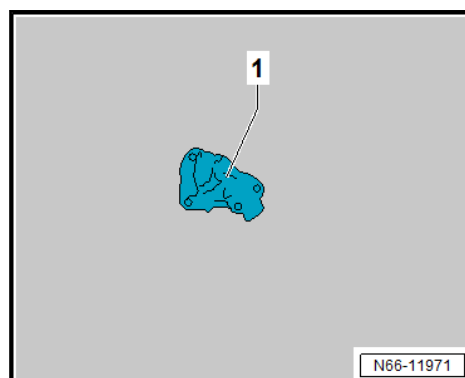
For part numbers of materials, refer to ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

- ◆ Fibreglass filler

Processing damaged area

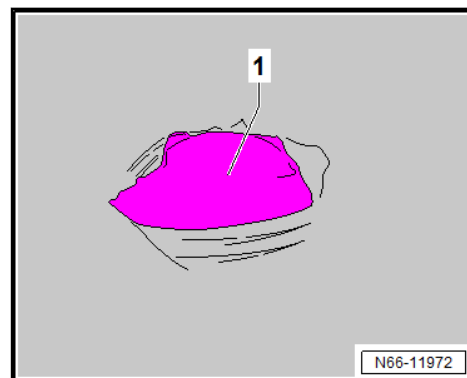


- Set stop of drill to material thickness of GRP cover layer.
- Drill cracks -1- with drill.
- Sand down damaged area -1- with grinding/sanding system -VAS 6677- to hard foam core.



- Remove loose items from GRP cover layer.

Filling repair area



- Observe processing instructions on packaging and in technical bulletins of fibreglass filler.
- Apply fibreglass filler -1-.
- Smooth over fibreglass filler with squeegee.
- Sand down fibreglass filler once hardened using grinding/sanding system -VAS 6677-.
- Apply fine filler compound.
- Sand down fine filler compound once hardened using grinding/sanding system -VAS 6677-.
- Prime repair area.
- Paint repair area.

9.2.2 Repairing one-sided damage

Special tools and workshop equipment required

- ◆ Flat scraper -VAS 5517-
- ◆ Sanding system -VAS 6677-
- ◆ Hand-cartridge gun -V.A.G 1628-





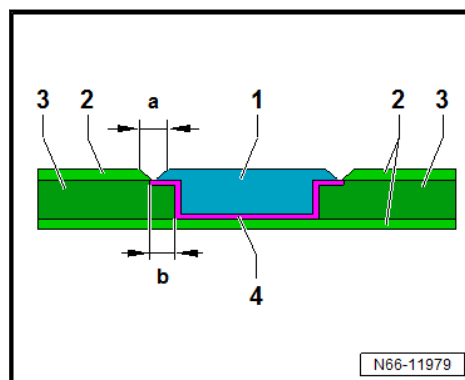
◆ Electric cutter -V.A.G 1561A-



◆ Extraction system -VAS 6572/2-



Overview cross-sectional diagram of side panel repair



Item	Designation
-1-	Repair section
-2-	GRP cover layer
-3-	Hard foam core
-4-	2-pack assembly adhesive
-a-	50 mm
-b-	40 mm

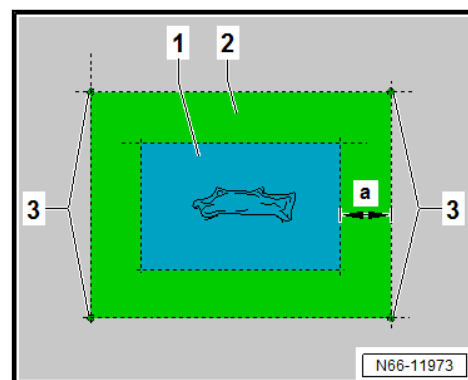
Replacement parts, materials

For part numbers of materials, refer to ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

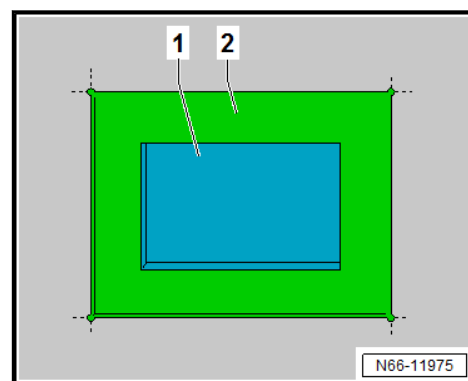
- ◆ Repair panel
- ◆ Self-tapping screw (4x16) -N 907 178 02-
- ◆ 2-pack assembly adhesive

- ◆ 2-pack polyester resin
- ◆ Fibreglass cloth
- ◆ Fibreglass filler
- ◆ Fine filler
- ◆ Paint roller
- ◆ Cleaner for plastics -D 195 850 A1-

Cutting out damaged area



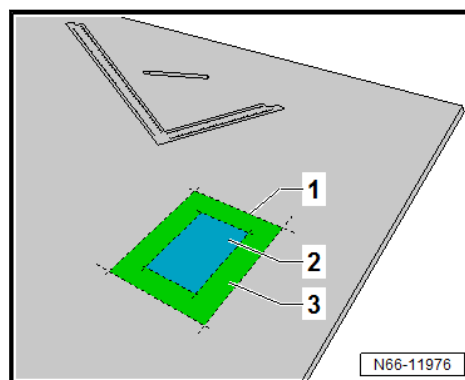
- Mark repair area -1- all around.
- Mark repair area -2- all around to dimension -a- = 40 mm.
- Set stop of drill to material thickness of GRP cover layer.
- Drill corners -3- with drill (8 mm bit).
- Set stop of electric cutter -V.A.G 1561A- to material thickness of hard foam core.



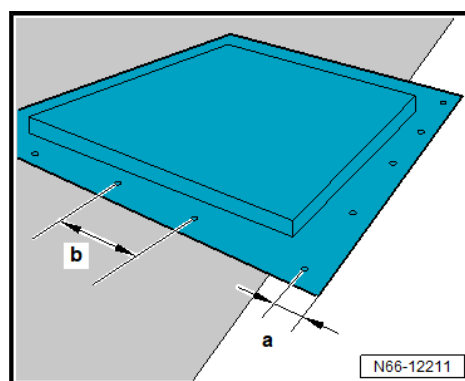
- Cut out repair area -1- all around with electric cutter -V.A.G 1561A-.
- Remove hard foam core with flat scraper -VAS 5517-.
- Set electric cutter -V.A.G 1561A- stop to material thickness of GRP cover layer.
- Cut out repair area -2- all around with electric cutter -V.A.G 1561A-.
- Remove GRP cover layer -2-.



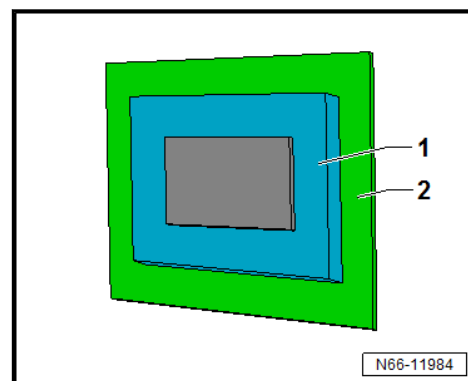
Preparing repair section



- Transfer dimensions for repair section -1- from box body to repair panel.
- Cut out repair section -1- with electric cutter -V.A.G 1561A-.
- Remove GRP cover layer -2- on one side.
- Set stop of electric cutter -V.A.G 1561A- to material thickness of hard foam core -3-.
- Cut out hard foam core -3- with electric cutter -V.A.G 1561A-.
- Remove hard foam core -3- with flat scraper -VAS 5517-.
- Sand down hard foam core -3- on new part and repair area by approx. 0.5 mm using P 60-80 sandpaper.
- Check that repair section fits in box body, adjust if necessary.
- Mark installation position on repair section and on box body.
- Drill holes in repair section -1- according to dimensions -a- = approx. 10 mm and -b- = approx. 100 mm (diameter: 4.2 mm).

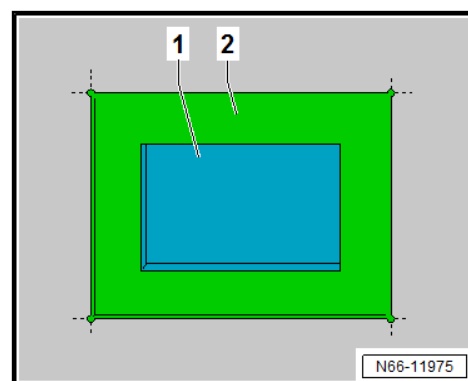


- Clean bonding surface -1- of dust.

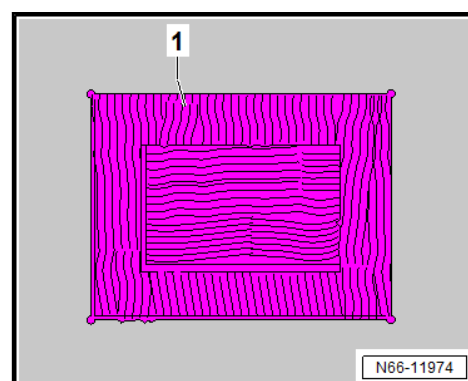


- Clean bonding surface -2- using cleaning sponge -000.012.300- and cleaner for plastics -D 195 850 A1-.
- Make sure to observe the flash-off time of the cleaner for plastics -D 195 850 A1- which is at least 10 minutes.

Bonding in repair section



- Clean bonding surfaces -1- using cleaning sponge -000.012.300- and cleaner for plastics -D 195 850 A1-.
- Clean bonding surfaces -2- of dust.
- Make sure to observe the flash-off time of the cleaner for plastics -D 195 850 A1- which is at least 10 minutes.



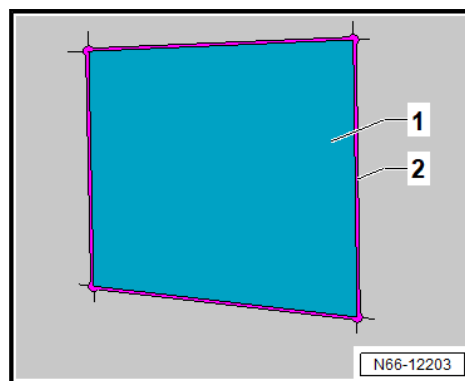
- Observe processing instructions on packaging and in technical bulletins of 2-pack assembly adhesive.

Make sure that 2-pack assembly adhesive is applied without air bubbles in a layer with a thickness of at least 1 mm.

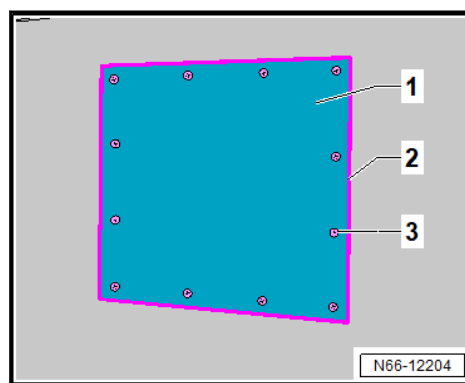
- Apply 2-pack assembly adhesive -1- using hand-cartridge gun -V.A.G 1628-.



- Spread 2-pack assembly adhesive -1- evenly using commercially available scraper.
- Insert repair section -1- applying even pressure until 2-pack assembly adhesive starts emerging from bonding gap -2-. Note installation position while doing so.



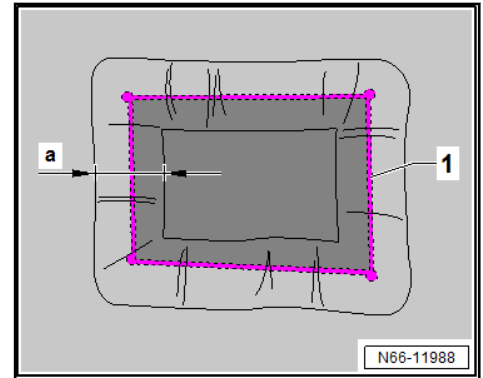
Fixing repair section in position



Make sure that no cordless screwdriver is being used for screwing in the bolts -3-.

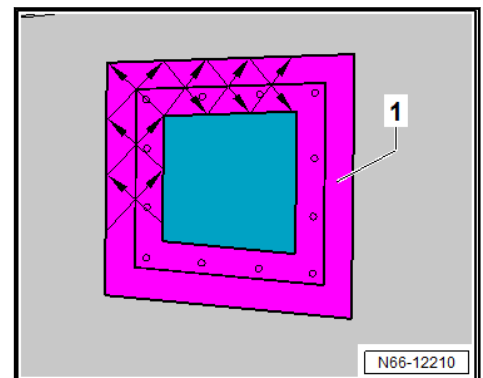
- Fix repair section -1- evenly in position using bolts -3-. While doing so, make sure that the outer GRP cover layer of the repair section -1- is flush with the GRP cover layer of the repair area -4-.
- Remove emerged 2-pack assembly adhesive from bonding gap -2- using scraper.
- If necessary, fill bonding gap -2- with 2-pack assembly adhesive, and go after the adhesive with a scraper.
- Make sure to observe the processing time of the 2-pack assembly adhesive which is 45 minutes.
- Then, remove adhesive residue using cleaner for plastics -D 195 850 A1- and a cloth.
- Make sure to observe the flash-off time of the cleaner for plastics -D 195 850 A1- which is at least 10 minutes.
- After drying time of 2-pack assembly adhesive is over, unscrew bolts.

Processing repair area



- Sand down bonding gap -1- all around to dimension -a- = 50 mm outwards, tapering out respectively using sanding/grinding system -VAS 6677- and sandpaper (P60).
- Clean repair area with extraction unit -VAS 6572/2-.
- Clean repair area with cleaning sponge -000.012.300- and cleaner for plastics -D 195 850 A1-.
- Make sure to observe the flash-off time of the cleaner for plastics -D 195 850 A1- which is at least 10 minutes.

Laminate repair area



- Observe processing instructions on packaging and in technical bulletins of 2-pack polyester resin and fibreglass filler.
- Cut fibreglass cloth into 50 mm wide strips with a length as required according to circumference of repair area.
- Mix 2-pack polyester resin.
- Apply 2-pack polyester resin -2- onto repair area all around using paint roll.
- Position structured fibreglass cloth -1- centrally.
- Apply 2-pack polyester resin -2- to fibreglass cloth -1- in a cross pattern as shown in illustration. Make sure that there are no air pockets.
- If necessary, apply another layer of fibreglass cloth and 2-pack polyester resin.
- Make sure to observe the processing time of the 2-pack assembly adhesive.
- Sand down repair area with grinding/sanding system -VAS 6677-.
- Fill repair area with fine filler compound.



- Sand down repair area with grinding/sanding system -VAS 6677-.
- Repeat repair procedure on inside.
- Prime repair area.
- Paint repair area.

9.2.3 Repairing two-sided damage

Special tools and workshop equipment required

- ◆ Flat scraper -VAS 5517-
- ◆ Sanding system -VAS 6677-
- ◆ Hand-cartridge gun -V.A.G 1628-



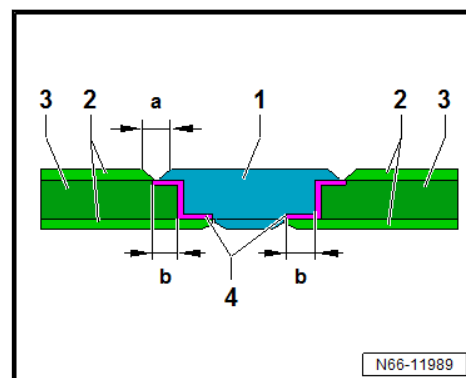
- ◆ Electric cutter -V.A.G 1561A-



- ◆ Extraction system -VAS 6572/2-



Overview cross-sectional diagram of side panel repair



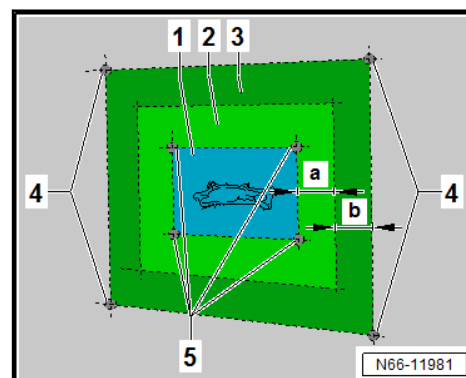
Item	Designation
-1-	Repair section
-2-	GRP cover layer
-3-	Hard foam core
-4-	2-pack assembly adhesive
-a-	50 mm
-b-	40 mm

Replacement parts, materials

For part numbers of materials, refer to ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

- ◆ Repair panel
- ◆ Self-tapping screw (4x16) -N 907 178 02-
- ◆ 2-pack assembly adhesive
- ◆ 2-pack polyester resin
- ◆ Fibreglass cloth
- ◆ Fibreglass filler
- ◆ Fine filler
- ◆ Paint roller
- ◆ Cleaner for plastics -D 195 850 A1-

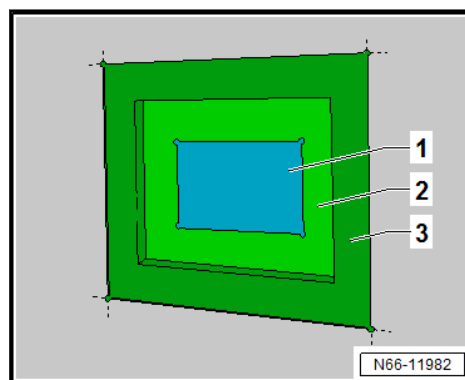
Cutting out damaged area



- If necessary, remove fastening rail ⇒ [page 325](#) .
- Mark repair area -1- all around.

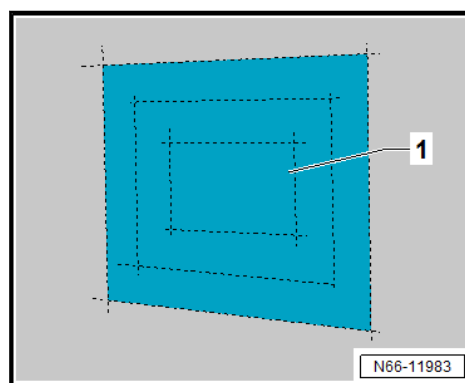


- Mark repair area -2- all around to dimension -a- 40 mm.
- Mark repair area -3- all around to dimension -b- 40 mm.
- Set stop of drill to material thickness of GRP cover layer.
- Drill corners -4- with drill from outside (8 mm bit).
- Set stop of drill to material thickness of GRP cover layer.
- Drill corners -5- with drill from inside (8 mm bit).
- Cut out repair area -1- all around with electric cutter -V.A.G 1561A-.

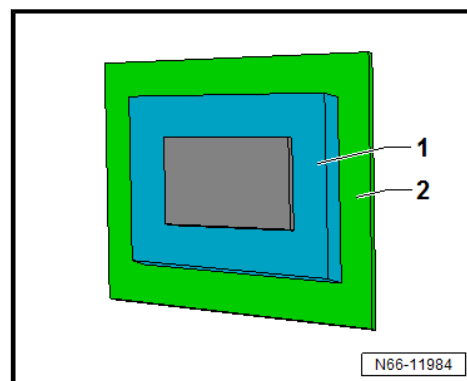


- Set electric cutter -V.A.G 1561A- stop to material thickness of GRP cover layer and hard foam core.
- Cut out repair area -2- all around with electric cutter -V.A.G 1561A-.
- Remove hard foam core with flat scraper -VAS 5517-.
- Set electric cutter -V.A.G 1561A- stop to material thickness of GRP cover layer.
- Cut out repair area -3- all around with electric cutter -V.A.G 1561A-.
- Remove GRP cover layer.

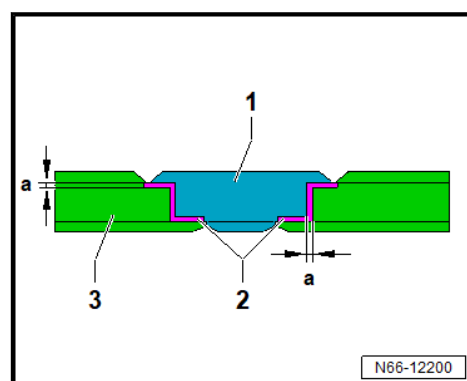
Preparing repair section



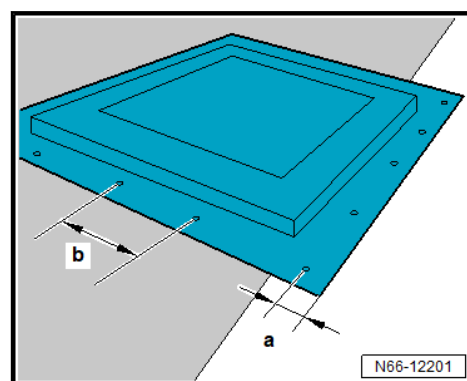
- Transfer dimensions for repair section -1- from box body to repair panel.
- Cut out repair section -1- with electric cutter -V.A.G 1561A-.
- Set electric cutter -V.A.G 1561A- stop to material thickness of GRP cover layer.



- Cut out GRP cover layer -1- on rigid foam core.
- Remove GRP cover layer -1- from rigid foam core using flat scraper -VAS 5517-.
- Set stop of electric cutter -V.A.G 1561A- to material thickness of rigid foam core -2- on GRP cover layer.
- Cut out rigid foam core using electric cutter -V.A.G 1561A-.
- Remove rigid foam core from GRP cover layer -2- using flat scraper -VAS 5517-.
- For bonding gap, sand new part -1- (or old part) as per dimension -a- = approx. 1 mm using sandpaper (P 60 - P80).

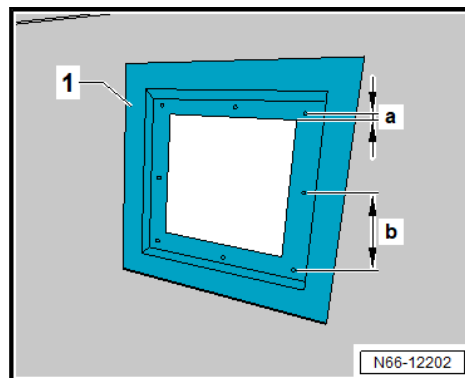


- Check repair section for fit with repair area from inside and outside, and adjust it if necessary.
- Mark installation position on repair section and at repair area.
- Drill holes in repair section -1- according to dimensions -a- = approx. 10 mm and -b- = approx. 100 mm (diameter: 4.2 mm).



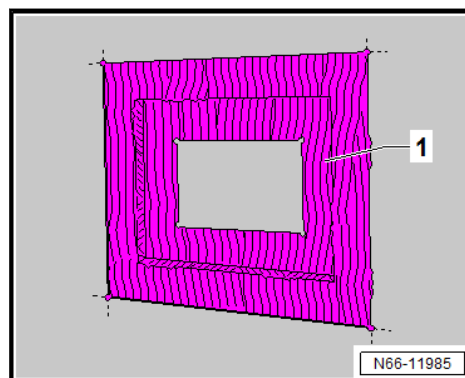


- Drill holes in repair area -1- according to dimensions -a- = approx. 10 mm and -b- = approx. 100 mm (diameter: 4.2 mm).



- Completely remove emery residue from repair area and repair section.
- Clean bonding surfaces in repair area and on repair section using cleaning sponge -000.012.300- and cleaner for plastics -D.195.850.A1-.
- Observe flash-off period of at least 10 minutes.

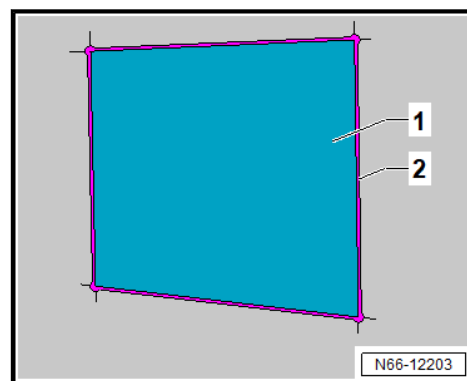
Bonding in repair section



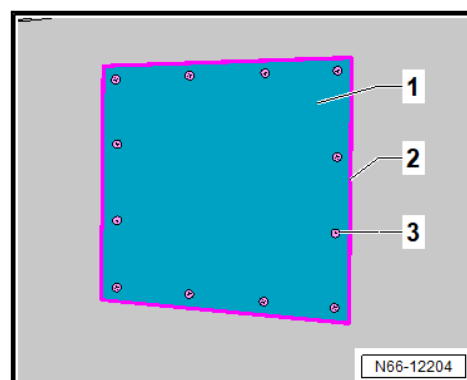
- Observe processing instructions on packaging and in technical bulletins of 2-pack assembly adhesive.

Make sure that 2-pack assembly adhesive is applied without air bubbles in a layer with a thickness of at least 1 mm.

- Apply 2-pack assembly adhesive -1- using hand-cartridge gun -V.A.G 1628-.
- Spread 2-pack assembly adhesive -1- evenly using commercially available scraper.
- Insert repair section -1- applying even pressure until 2-pack assembly adhesive starts emerging from bonding gap -2-. Note installation position while doing so.



Fixing repair section in position

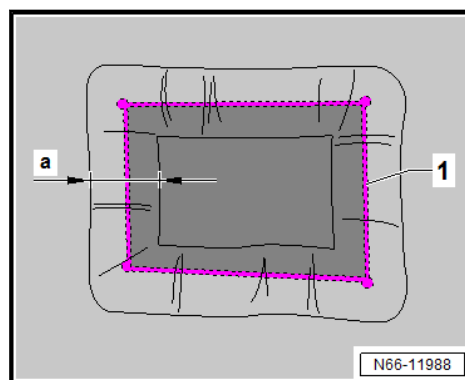


Make sure that no cordless screwdriver is being used for screwing in the bolts.

- Fix position of repair section -1- with bolts -3- evenly on inside and on outside. While doing so, make sure that the outer and inner GRP cover layers of the repair section are flush with the GRP cover layer of the repair area.
- Remove emerged 2-pack assembly adhesive from bonding gap -2- using scraper.
- If necessary, fill bonding gap -2- with 2-pack assembly adhesive, and go after the adhesive with a scraper.
- Make sure to observe the processing time of the 2-pack assembly adhesive which is 45 minutes.
- Then, remove adhesive residue using cleaner for plastics -D 195 850 A1- and a cloth.
- Make sure to observe the flash-off time of the cleaner for plastics -D 195 850 A1- which is at least 10 minutes.
- After drying time of 2-pack assembly adhesive is over, unscrew bolts.



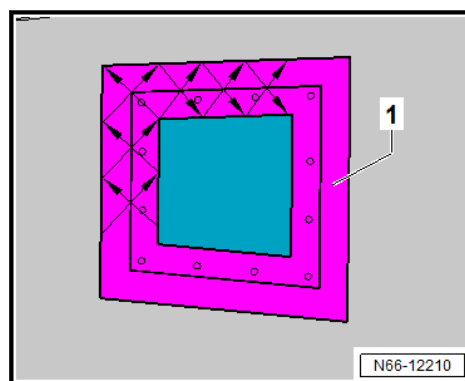
Processing repair area from inside and outside



The sanded down repair section creates space for application of the fiberglass cloth.

- Sand down bonding gap -1- all around to dimension -a- = 50 mm outwards, tapering out respectively using sanding/grinding system -VAS 6677- and sandpaper (P60).
- Clean repair area with extraction unit -VAS 6572/2-.
- Clean repair area with cleaning sponge -000.012.300- and cleaner for plastics -D 195 850 A1-.
- Make sure to observe the flash-off time of the cleaner for plastics -D 195 850 A1- which is at least 10 minutes.

Laminate repair area



- Observe processing instructions on packaging and in technical bulletins of 2-pack polyester resin and fiberglass filler.
- Cut fiberglass cloth into 50 mm wide strips with a length as required according to circumference of repair area.
- Mix 2-pack polyester resin.
- Apply 2-pack polyester resin to sanded area -1- all around using paint roll.
- Position structured fiberglass cloth on applied 2-pack polyester resin centrally all around.
- Apply 2-pack polyester resin to fiberglass cloth all around in a cross pattern. Make sure that there are no air pockets.
- If necessary, apply another layer of fiberglass cloth and 2-pack polyester resin.
- Make sure to observe the processing time of the 2-pack assembly adhesive.
- Sand down repair area with grinding/sanding system -VAS 6677-.



- Fill repair area with fine filler compound.
- Sand down repair area with grinding/sanding system -VAS 6677-.
- Repeat repair procedure on inside.
- Prime repair area on inside and outside.
- Paint repair area on inside and outside.

9.3 Repairing transparent roof

Special tools and workshop equipment required

- ◆ File set -V.A.G 1523/5-
- ◆ Hand-cartridge gun -V.A.G 1628-



- ◆ Electric cutter -V.A.G 1561A-



- ◆ Extraction system -VAS 6572/2-



Replacement parts, materials

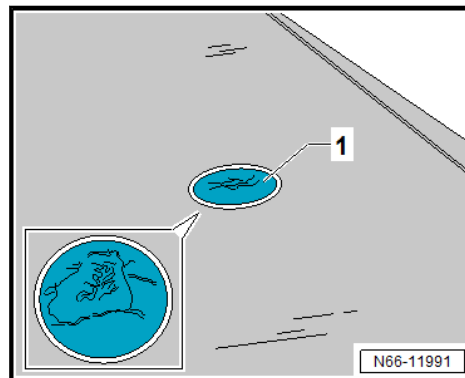
For part numbers of materials, refer to ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

- ◆ Transparent roof section
- ◆ 1-component MS assembly adhesive -D 172 120 M2-



- ◆ Cartridge nozzle for door rebates -000 012 279 A-
- ◆ Cleaning solution -D.009.404.014-
- ◆ Cleaning sponge -000.012.300-

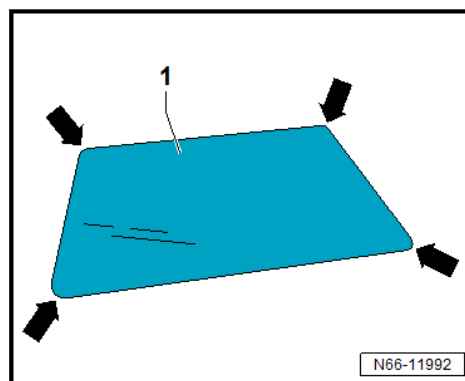
Cutting out damaged area



Cracks must be cut out during the repair.

- Mark a circle around repair area -1-.
- Cut out repair area -1- with electric cutter -V.A.G 1561A- using commercially available saw blade.
- Deburr cut edge with file set -V.A.G 1523/5-.
- Clean repair area -1- with cleaning sponge -000.012.300- and cleaning solution -D.009.404.014-.
- Make sure to observe the flash-off time of the cleaning solution -D.009.404.014- which is at least 10 minutes.

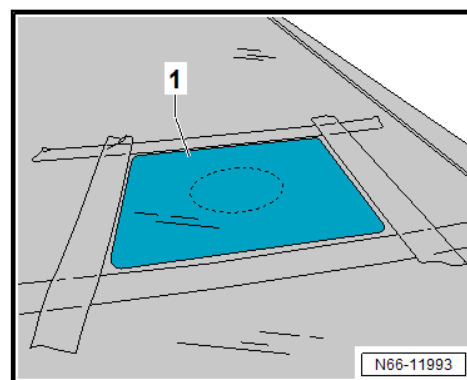
Preparing new part



The repair section -1- is made from part of the transparent roof. To do this, mark an area all around at least 50 mm larger from the outer edge of the repair area.

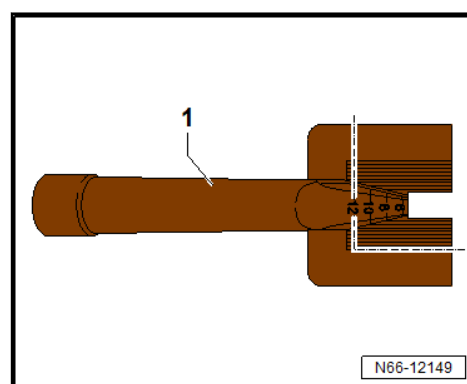
- Prepare repair section -1-.
- Round off corners of repair section -1- with file set -V.A.G 1523/5-.
- Deburr cut edges with file set -V.A.G 1523/5-.
- Clean repair section -1- with cleaning sponge -000.012.300- and cleaning solution -D.009.404.014-.

Masking off repair area



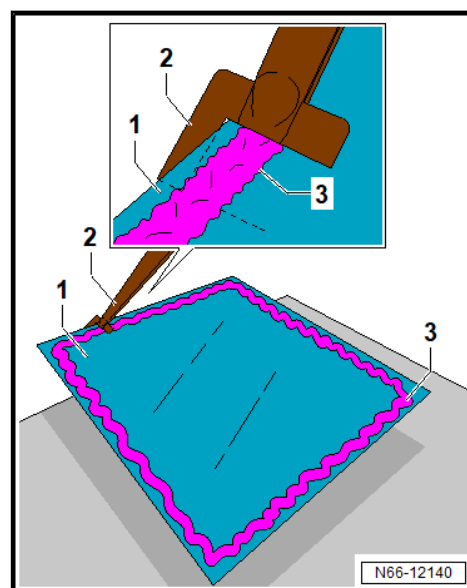
- Mask off repair area -1- as shown in illustration.

Preparing tools



- Cut out cartridge nozzle for door rebates -000 012 279 A- -1- with commercially available scissors at nozzle position 12 and at outer left groove.

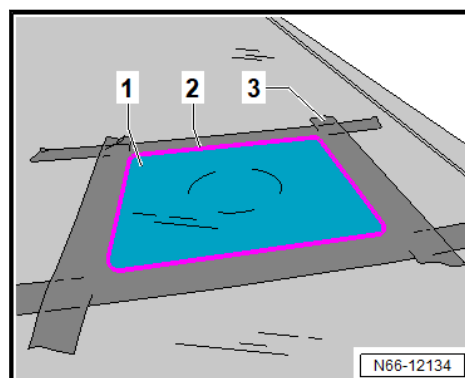
Bonding in repair section



- Apply 1-component MS assembly adhesive -D 172 120 M2- onto repair section -1- all around using cartridge nozzle for door rebates -000 012 279 A- and hand-cartridge gun -V.A.G 1628-.



- Observe processing instructions of 1-component MS assembly adhesive -D 172 120 M2-.
- Fit repair section -1- and press down.



Ensure when pressing down that no air is trapped in the adhesive.

- Smooth down emerging 1-component MS assembly adhesive -D 172 120 M2- with cleaning solution -D.009.404.014-.
- Ensure uniform gap -2- all around.
- If necessary, further rework/smooth with 1-component MS assembly adhesive -D 172 120 M2- and cleaning solution -D.009.404.014-.
- Remove masking tape -3-.
- Observe curing period of 24 hours. The vehicle must not be driven outside the workshop area during this period of time.
- Do not visit car wash earlier than 5 to 7 days after repair.

10 Trim film and protective film

⇒ [o10.1 verview - protective films", page 185](#)

⇒ [p10.2 roective films", page 187](#)

10.1 Assembly overview - protective films

⇒ [o10.1.1 verview - warning film, bonnet", page 185](#)

⇒ [o10.1.2 verview - warning film, rear wing doors", page 185](#)

⇒ [o10.1.3 verview - warning film, dropside", page 186](#)

10.1.1 Assembly overview - warning film, bonnet

1 - Warning film

- ☐ Left
- ☐ Replacing ⇒ [page 188](#)
- ☐ Dimensions ⇒ [page 190](#)

2 - Warning film

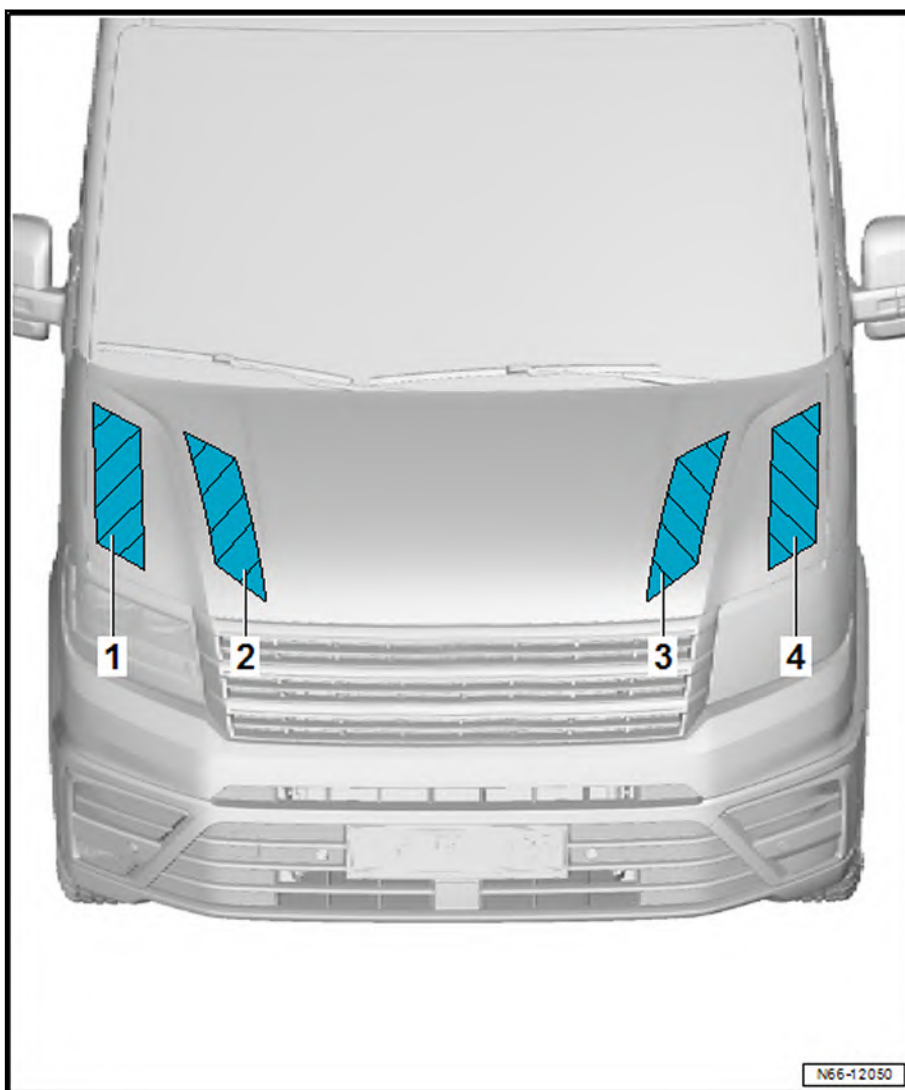
- ☐ Centre left
- ☐ Replacing ⇒ [page 188](#)
- ☐ Dimensions ⇒ [page 190](#)

3 - Warning film

- ☐ Centre right
- ☐ Replacing ⇒ [page 188](#)
- ☐ Dimensions ⇒ [page 190](#)

4 - Warning film

- ☐ Right
- ☐ Replacing ⇒ [page 188](#)
- ☐ Dimensions ⇒ [page 190](#)



10.1.2 Assembly overview - warning film, rear wing doors

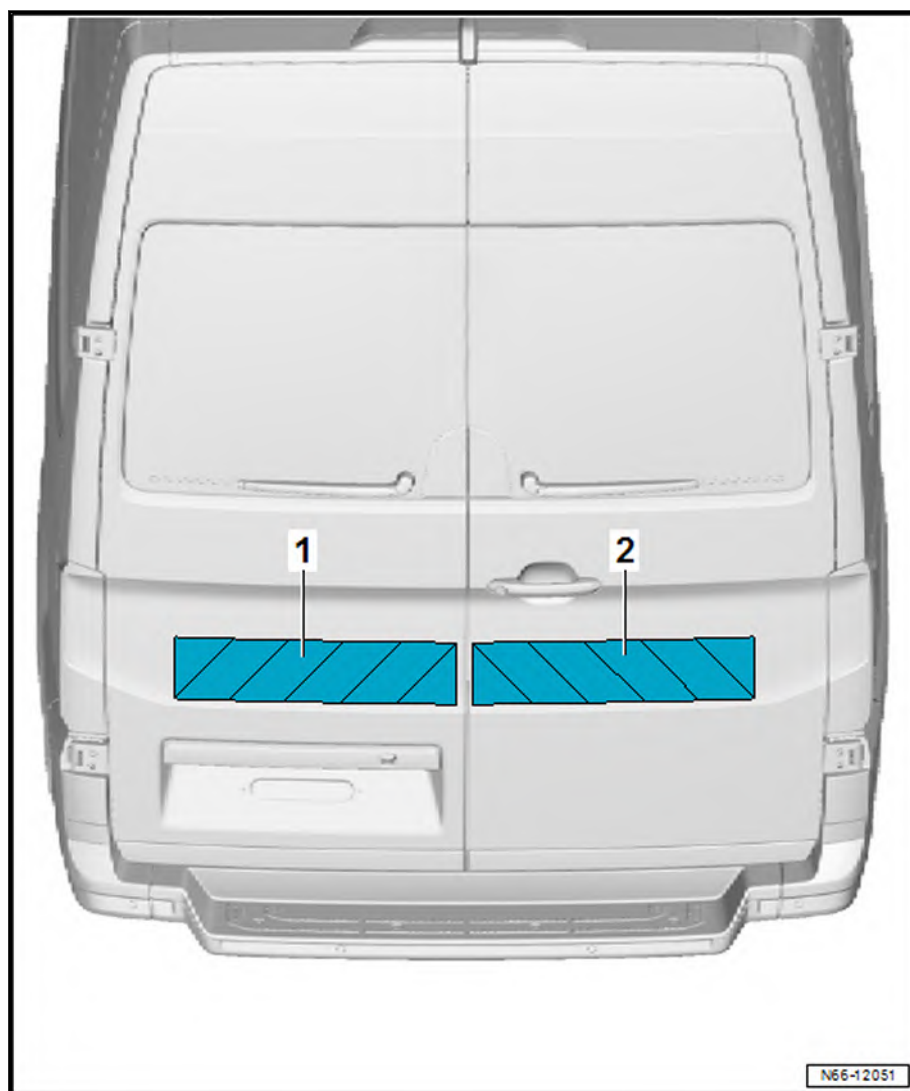


1 - Warning film

- ☐ Left
- ☐ Replacing ➤ [page 188](#)
- ☐ Dimensions ➤ [page 191](#)

2 - Warning film

- ☐ Right
- ☐ Replacing ➤ [page 188](#)
- ☐ Dimensions ➤ [page 191](#)



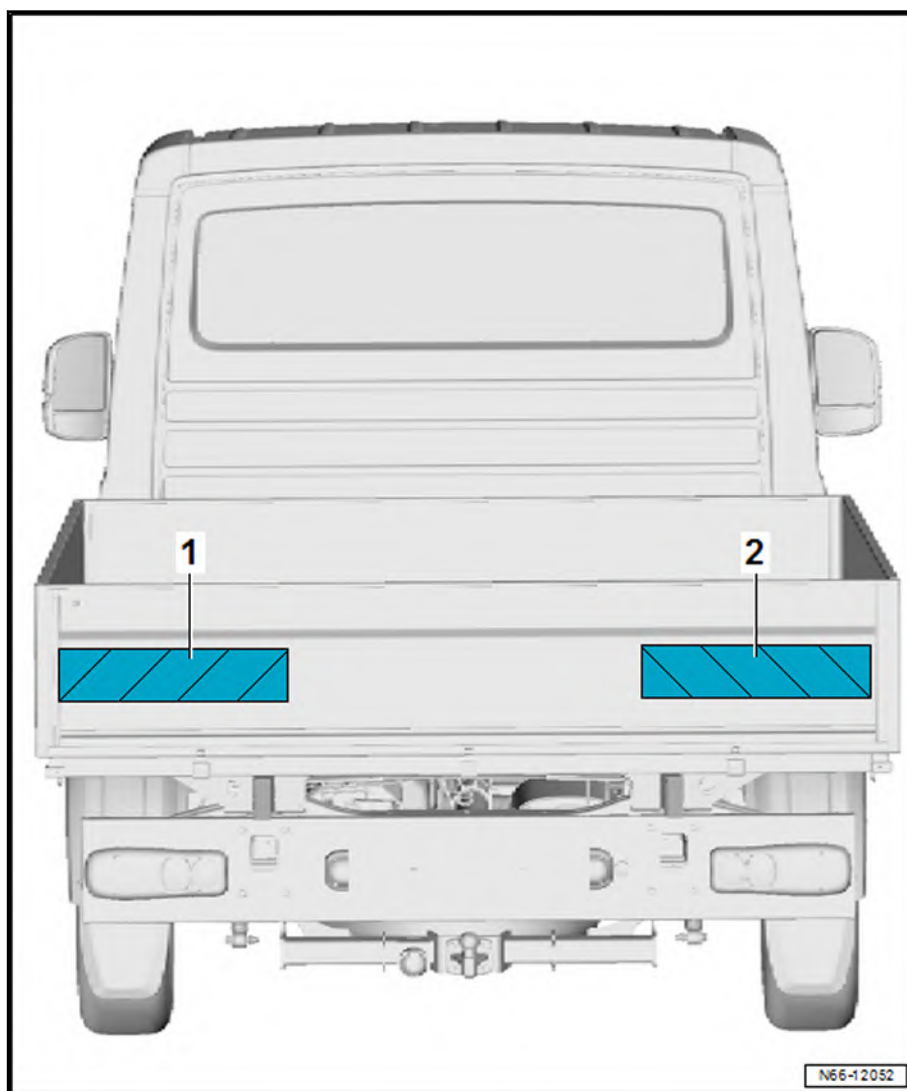
10.1.3 Assembly overview - warning film, dropside

1 - Warning film

- ☐ Left
- ☐ Replacing ➔ [page 188](#)
- ☐ Dimensions ➔ [page 191](#)

2 - Warning film

- ☐ Right
- ☐ Replacing ➔ [page 188](#)
- ☐ Dimensions ➔ [page 191](#)



10.2 Renewing protective films

➔ [p10.2.1 protective films", page 187](#)

➔ [w10.2.2 warning film", page 188](#)

➔ [-10.2.3 warning film, bonnet", page 190](#)

➔ [-10.2.4 warning film, rear wing doors", page 191](#)

➔ [-10.2.5 warning film, dropside", page 191](#)

10.2.1 Renewing protective films

Special tools and workshop equipment required



◆ Hot-air blower -V.A.G 1416-



- ◆ Adhesive remover as per ⇒ Electronic parts catalogue (ET-KA) or ⇒ webMANTIS for MAN TGE

Procedure

Protective films cannot be removed without being damaged.

- Before removing the protective film, heat it with a hot air blower -V.A.G 1416-.
- If protective film is removed and installed, use only adhesive remover to remove adhesive residue.
- Glue on protective film immediately after cleaning.

Fitting instructions

- If protective film is attached for the first time on a new vehicle or a newly painted surface, observe the procedure described in the ⇒ Paintwork manual ⇒ Paintwork repairs ⇒ Paint finish ⇒ Mouldings & film.
- Spray a thin film of soapy water onto exterior panel using a spray bottle.

Mixing ratio for soapy solution: 1 ml washing-up liquid to 1 l water. The mixture should be mixed well.

- Plastic squeegee wrapped with paper towel.



Note

Paper towel is required to prevent hard edge of plastic blade from scratching film.

- Do not remove protective film of bonding surface until immediately before installation.
- Apply and align protective film.
- Force out water from under protective film using plastic squeegee.
- Working temperature is approx. 21°C.
- Keep bonding surfaces clean and free of grease.

10.2.2 Renewing warning film

Special tools and workshop equipment required

◆ Hot-air blower -V.A.G 1416-



- ◆ Adhesive remover as per ⇒ Electronic parts catalogue (ET-KA) or ⇒ webMANTIS for MAN TGE

Procedure

Warning film cannot be removed without damage.

- Before removing warning film, heat it with hot air blower -V.A.G 1416-.
- When removing and reinstalling warning film, adhesive remover is the only aid that is allowed to be used to remove the adhesive residue.
- Affix warning film immediately after cleaning.

Fitting instructions

- If warning film is attached for the first time on a new vehicle or a newly painted surface, observe the procedure described in the ⇒ Paintwork manual ⇒ Paintwork repairs ⇒ Paint finish ⇒ Mouldings & film.
- Spray a thin film of soapy water onto exterior panel using a spray bottle.

Mixing ratio for soapy solution: 1 ml washing-up liquid to 1 l water. The mixture should be mixed well.

- Plastic squeegee wrapped with paper towel.



Note

Paper towel is required to prevent hard edge of plastic blade from scratching film.

- Do not remove protective backing from bonding surface until immediately before installation.
- Apply and align warning film.
- Force out water from under film using plastic squeegee.
- Working temperature is approx. 21°C.
- Keep bonding surfaces clean and free of grease.

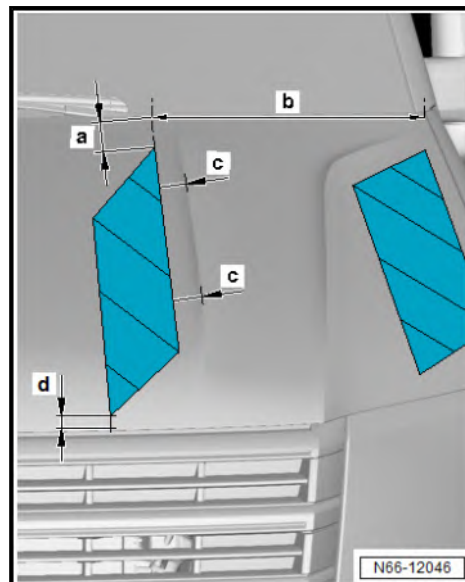


10.2.3 Dimensions - warning film, bonnet

Dimensions centre left ➔ [page 190](#)

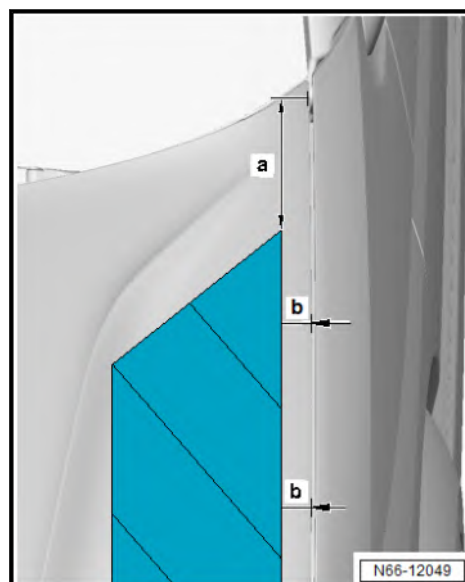
Dimensions left ➔ [page 190](#)

Dimensions centre left



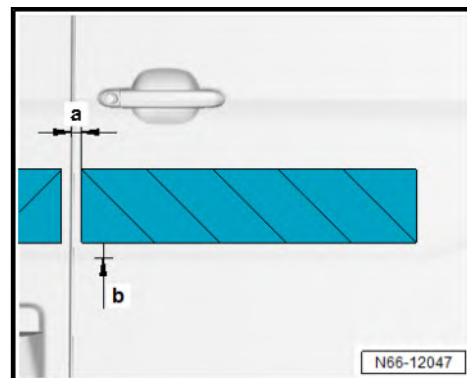
- ◆ Dimension a = 200 mm
- ◆ Dimension b = 415 mm
- ◆ Dimension c = 35 mm
- ◆ Dimension d = 25 mm

Dimensions left



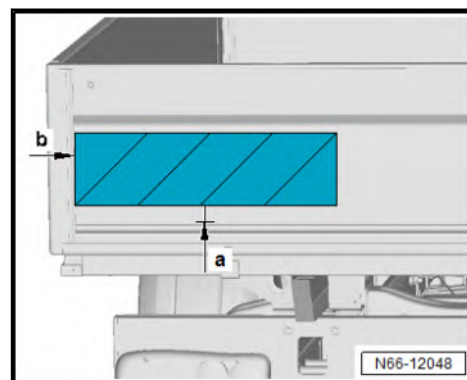
- ◆ Dimension a = 140 mm
- ◆ Dimension b = 25 mm

10.2.4 Dimensions - warning film, rear wing doors



- ◆ Dimension a = 10 mm
- ◆ Dimension b = 30 mm

10.2.5 Dimensions - warning film, dropside



- ◆ Dimension a = 80 mm
- ◆ Dimension b = edge of stop profile



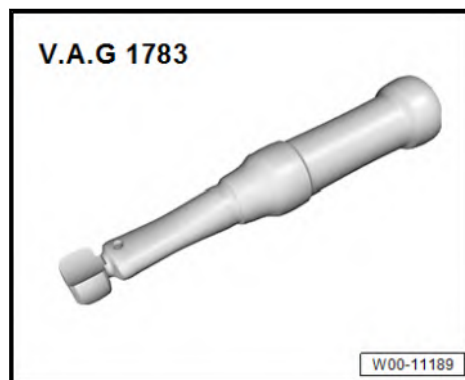
11 Taxi roof sign preparation

⇒ [a11.1 nd installing taxi roof sign preparation", page 192](#)

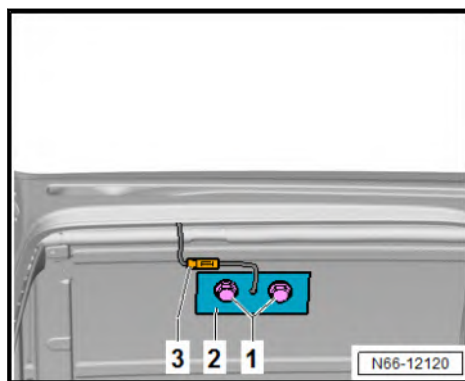
11.1 Removing and installing taxi roof sign preparation

Special tools and workshop equipment required

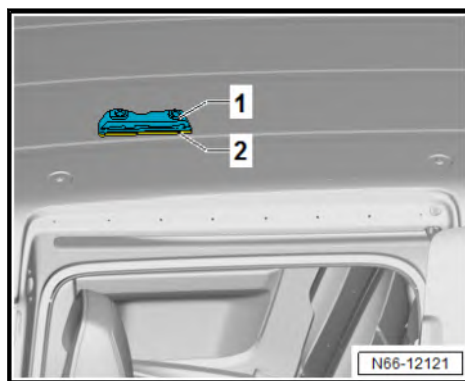
- ◆ Torque wrench -V.A.G 1783-



Removing



- Remove moulded headliner ⇒ General body repairs, interior; Rep. gr. 70; Roof trims; Removing and installing moulded headliner.
- Unscrew nuts -1-. Remove washer -2-.
- Disconnect connector -3-.
- Remove taxi roof sign preparation -1- along with seal -2-.



Installing

Install in reverse order of removal, observing the following:



- Check seal -2- for damage, renew if necessary.

Specified torques

Component	Specified torque
Nut -1-	7 Nm



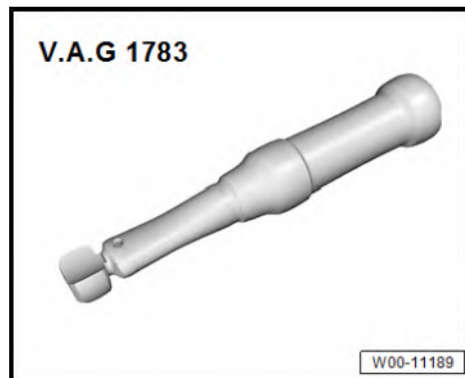
12 Taxi roof sign preparation

⇒ a12.1 nd installing taxi roof sign preparation", page 194

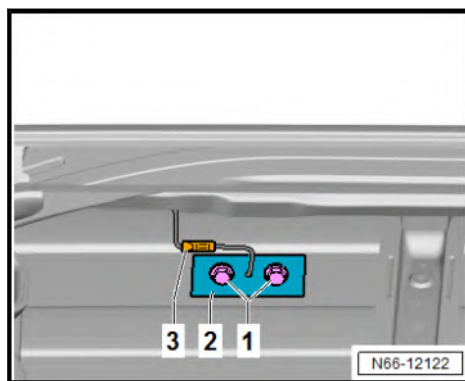
12.1 Removing and installing taxi roof sign preparation

Special tools and workshop equipment required

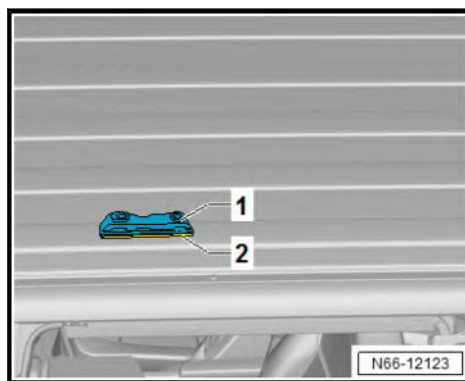
- ◆ Torque wrench -V.A.G 1783-



Removing



- Remove moulded headliner ⇒ General body repairs, interior; Rep. gr. 70; Roof trims; Removing and installing moulded headliner.
- Unscrew nuts -1-. Remove washer -2-.
- Disconnect connector -3-.
- Remove taxi roof sign preparation -1- along with seal -2-.



Installing

Install in reverse order of removal, observing the following:



- Check seal -2- for damage, renew if necessary.

Specified torques

Component	Specified torque
Nut -1-	7 Nm



13 Roof spoiler

⇒ a13.1 nd installing roof spoiler", page 196

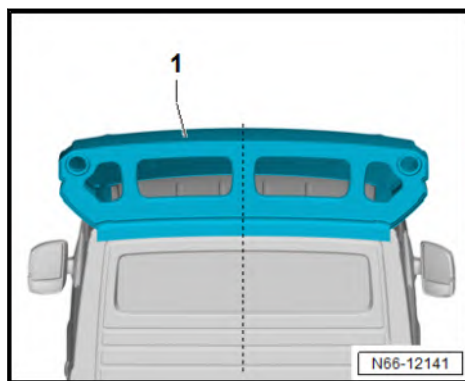
13.1 Removing and installing roof spoiler

Special tools and workshop equipment required

- ♦ Electric cutter -V.A.G 1561A-



Removing



- Secure roof spoiler -1- to prevent it from falling.
- Cut through bonded joints with electric cutter -V.A.G 1561A-.
- Remove roof spoiler -1- from body.

Installing

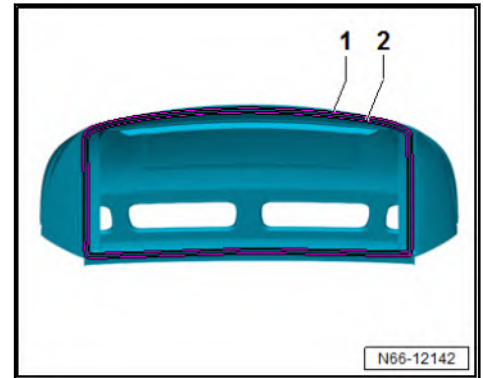
Install in reverse order of removal, observing the following:

Replacement parts, materials

For part numbers of materials, refer to ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

- ♦ Adhesive tape 3M 4225F
- ♦ Sika Flex 252
- ♦ Sika Primer 206 G+P
- ♦ Sika Activator 100 UN1866
- ♦ Cleaning solution
- ♦ Commercially available adhesive tape
- Place roof spoiler -1- on body and secure in place.
- Align roof spoiler centrally and mark position on vehicle.
- Mask off vehicle all around using commercially available adhesive tape.

- Remove roof spoiler from body.
- Clean roof spoiler with cleaning solution.
- Observe flash-off time of cleaning solution.
- Apply Sika Activator 100 UN1866 -1-, ensuring that processing instructions are followed when doing so.



- Apply Sika Primer 206 G+P -1-, ensuring that processing instructions are followed when doing so.
- Apply Sika Flex 252 -1-, ensuring that processing instructions are followed when doing so.
- Apply adhesive tape 3M 4225F -2-, ensuring that processing instructions are followed when doing so.
- Clean bonding surface on body using cleaning solution.
- Observe flash-off time of cleaning solution.
- Place roof spoiler on body as per marking and secure in place.
- Smooth down any excess adhesive all around.
- Check adhesive seam all around and smooth down with Sika Flex 252 if necessary.
- Carefully pull off adhesive tape all around.



14 Mounting bracket for towing bracket

⇒ o14.1 **view - mounting bracket for towing bracket**", page 198

⇒ a14.2 **nd installing mounting bracket for towing bracket**", page 198

14.1 Assembly overview - mounting bracket for towing bracket

1 - Mounting bracket

- ☐ Removing and installing ⇒ [page 198](#)

2 - Nuts

- ☐ Renew after removal
- ☐ Qty. 4
- ☐ 50 Nm +90°

3 - Ball head

4 - Bolts

- ☐ Renew after removal
- ☐ Qty. 4

5 - Nuts

- ☐ Qty. 2
- ☐ 8 Nm

6 - Bolts

- ☐ Qty. 3
- ☐ 3 Nm

7 - Socket

8 - Bracket for electric socket

9 - Bolts

- ☐ Qty. 2

10 - Fastener

11 - Electrical connector

12 - Seal

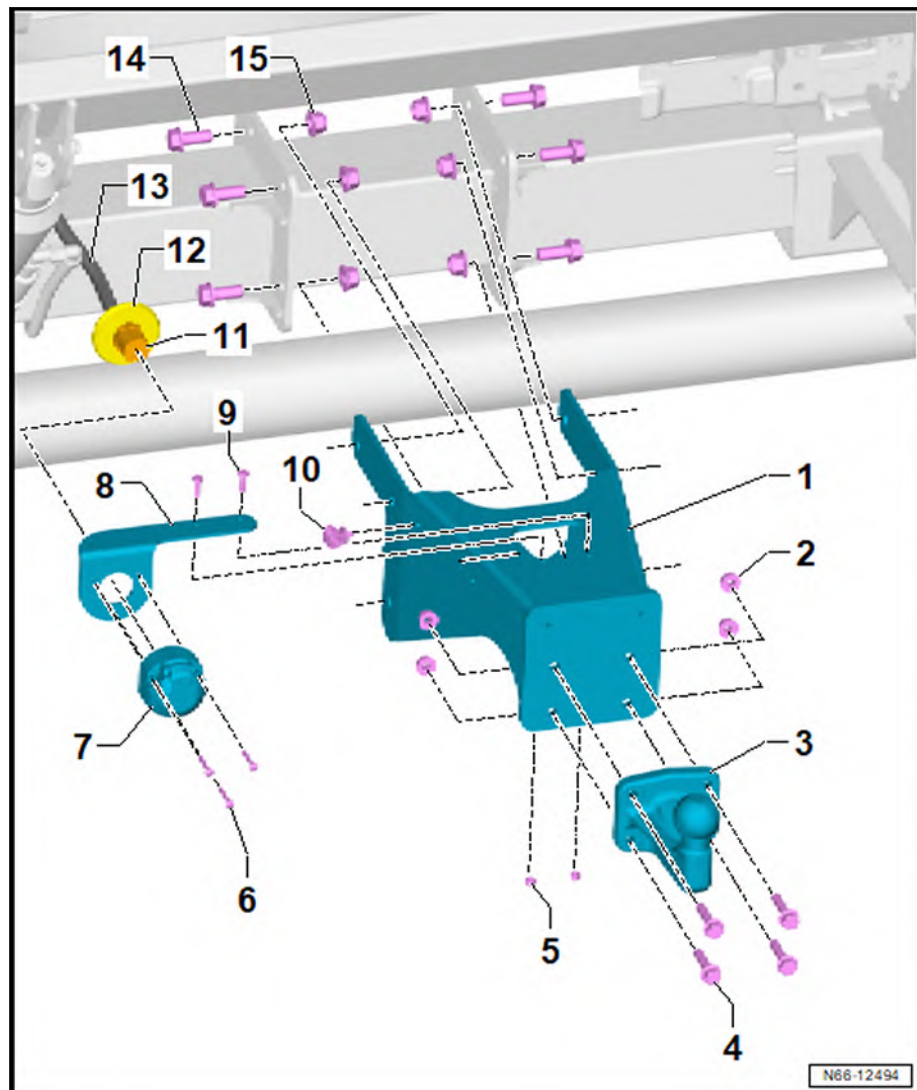
13 - Wiring harness

14 - Bolts

- ☐ Qty. 6

15 - Nuts

- ☐ Qty. 6
- ☐ 125 Nm



14.2 Removing and installing mounting bracket for towing bracket

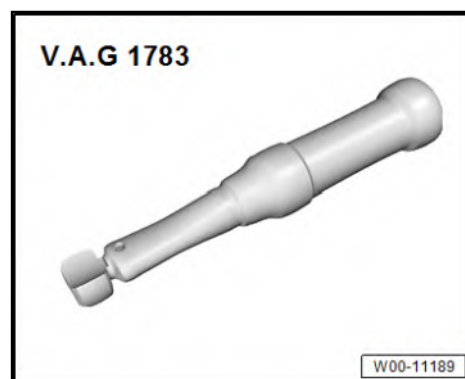
Special tools and workshop equipment required



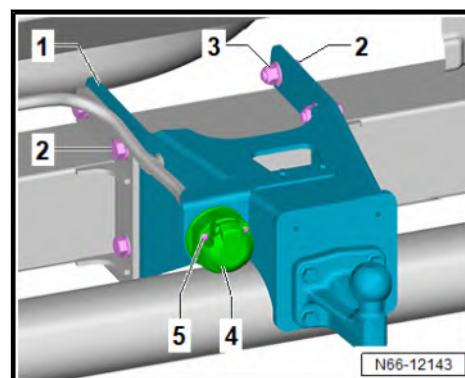
- ◆ Torque wrench, 40 to 200 Nm -V.A.G 1332A-



- ◆ Torque wrench -V.A.G 1783-



Removing



- If necessary, remove ball head ⇒ General body repairs, exterior; Rep. gr. 66; Towing bracket; Removing and installing towing bracket; Removing and installing ball head.
- Unscrew bolts -5-.
- Detach socket -4-. While doing this, disconnect electrical connector.
- Unclip wiring harness from mounting bracket -1-.
- Unscrew bolts -2-. Counterhold nuts -3- while doing this.
- Remove mounting bracket -1- with the aid of a second mechanic.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [o14.1 verview - mounting bracket for towing bracket", page 198](#)



15 Roof basket

⇒ [o15.1 verview - roof basket", page 200](#)

⇒ [a15.2 nd installing roof basket", page 202](#)

15.1 Assembly overview - roof basket



1 - Roof basket

- ☐ Removing and installing ⇒ [page 202](#)

2 - Cross member pins

- ☐ Left and right
- ☐ Qty. 16 ... 20 depending on equipment
- ☐ 17 Nm

3 - Washers

- ☐ Left and right
- ☐ Qty. 16 ... 20 depending on equipment

4 - Loading roller

- ☐ Removing and installing ⇒ [page 203](#)

5 - Cross member

- ☐ Qty. 8 ... 10 depending on equipment
- ☐ Removing and installing ⇒ [page 203](#)

6 - Side panels

- ☐ Left and right
- ☐ Qty. 2

7 - Bolts

- ☐ Left and right
- ☐ Qty. 32 ... 40 depending on equipment
- ☐ 20 Nm

8 - Washers

- ☐ Left and right
- ☐ Qty. 32 ... 40 depending on equipment

9 - Feet

- ☐ Left and right
- ☐ Qty. 16 ... 20 depending on equipment
- ☐ Removing and installing ⇒ [page 202](#)

10 - Washers

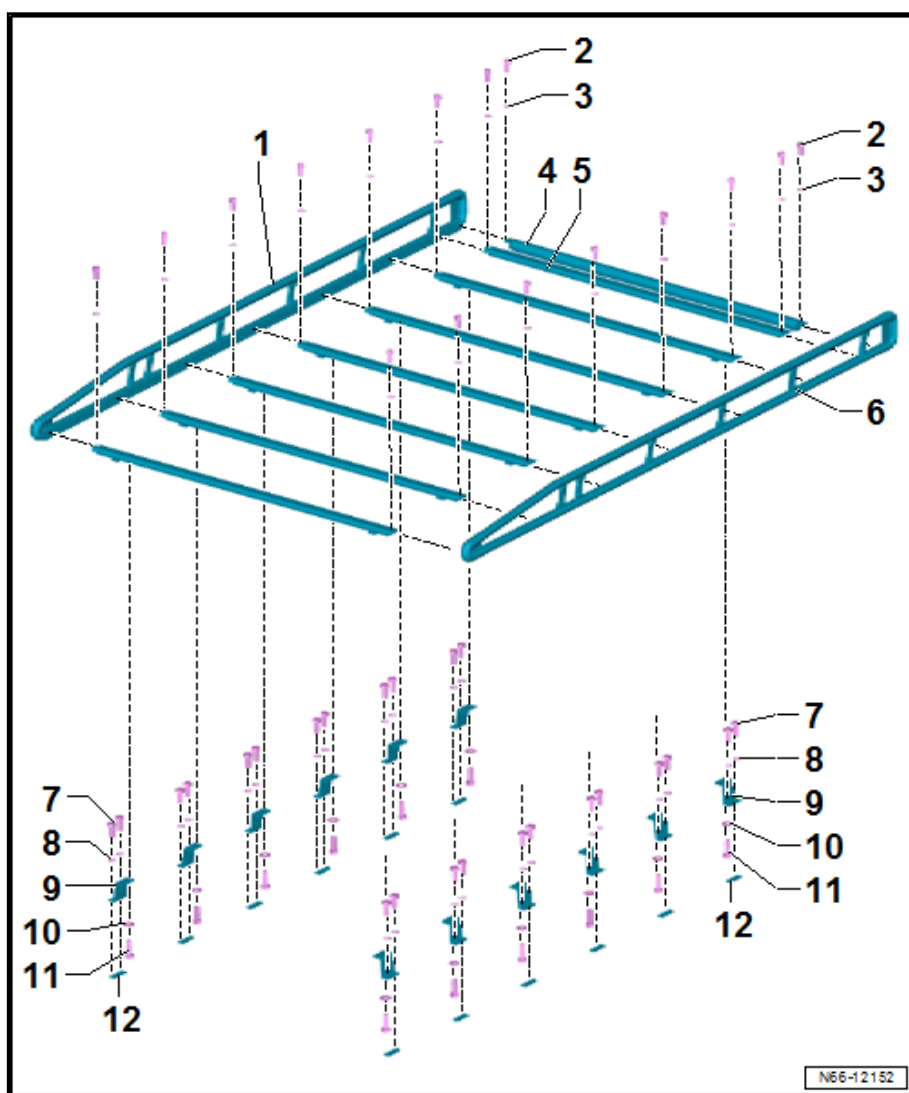
- ☐ Left and right
- ☐ Qty. 16 ... 20 depending on equipment

11 - Bolts

- ☐ Left and right
- ☐ Qty. 16 ... 20 depending on equipment
- ☐ 17 Nm

12 - T-slot nuts

- ☐ Left and right
- ☐ Qty. 16 ... 20 depending on equipment



N66-12152



15.2 Removing and installing roof basket

⇒ [a15.2.1 nd installing roof basket", page 202](#)

⇒ [a15.2.2 nd installing foot for roof basket", page 202](#)

⇒ [a15.2.3 nd installing loading roller and cross member for roof basket", page 203](#)

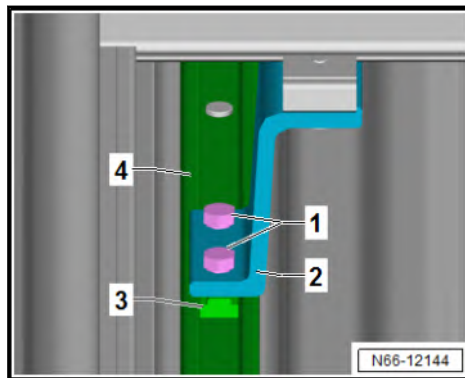
15.2.1 Removing and installing roof basket

Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1410A-



Removing



- Unscrew bolts -1-.
- Lift down roof basket with feet -2-.
- Remove T-slot nuts -3- from C-rail -4-.

Installing

Install in reverse order of removal, observing the following:

- Do not use electric tools for installation of the bolts.
- Apply assembly paste to all bolts.

Specified torques

- ◆ ⇒ [o15.1 verview - roof basket", page 200](#)

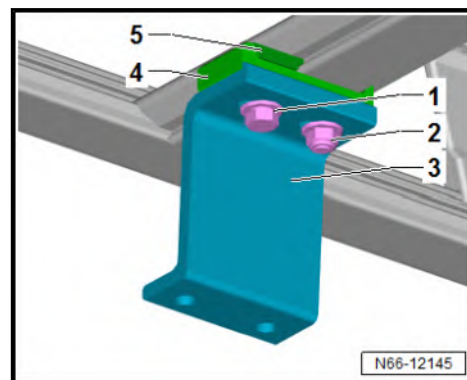
15.2.2 Removing and installing foot for roof basket

Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1410A-



Removing



- Remove roof basket ➔ [page 202](#) .
- Unscrew bolt -1-.
- Remove plates -5- from connection pieces -4-.
- Unscrew nut -2-.
- Remove feet -3-.

Installing

Install in reverse order of removal, observing the following:

- Do not use electric tools for installation of the bolts.
- Apply assembly paste to all bolts.

Specified torques

- ◆ ➔ [o15.1 verview - roof basket", page 200](#)

15.2.3 Removing and installing loading roller and cross member for roof basket

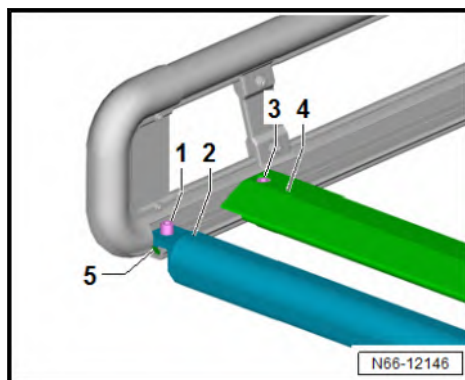
Special tools and workshop equipment required



- ◆ Torque wrench -V.A.G 1410A-



Removing

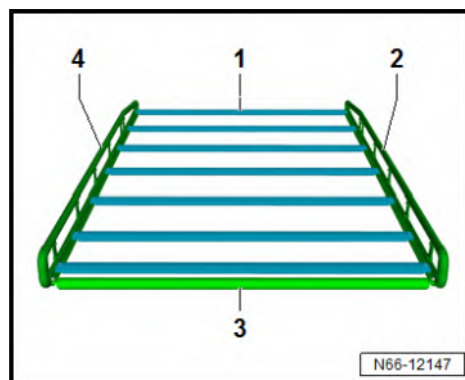


- Remove roof basket ⇒ [page 202](#) .
- Unscrew bolts -1-.
- Remove loading roller -2-.
- Remove T-slot nuts -5-.
- Unscrew cross member pin -3-
- Remove cross member -4-.

Installing

Install in reverse order of removal, observing the following:

- Do not use electric tools for installation of the bolts.
- Apply assembly paste to all bolts.
- Place cross member -1- on first side panel -2-.



- Screw in cross member pin.
- Place cross member -1- on second side panel -4-.



- Screw in cross member pin.
- Tighten cross member pin.

Specified torques

- ◆ [⇒ o15.1 verview - roof basket", page 200](#)



16 Storage box on drop-side

⇒ [o16.1 verview - storage box on drop-side", page 206](#)

⇒ [a16.2 nd installing storage box on drop-side", page 206](#)

16.1 Assembly overview - storage box on drop-side

1 - Storage box

- ☐ Removing and installing ⇒ [page 206](#)

2 - Bolts

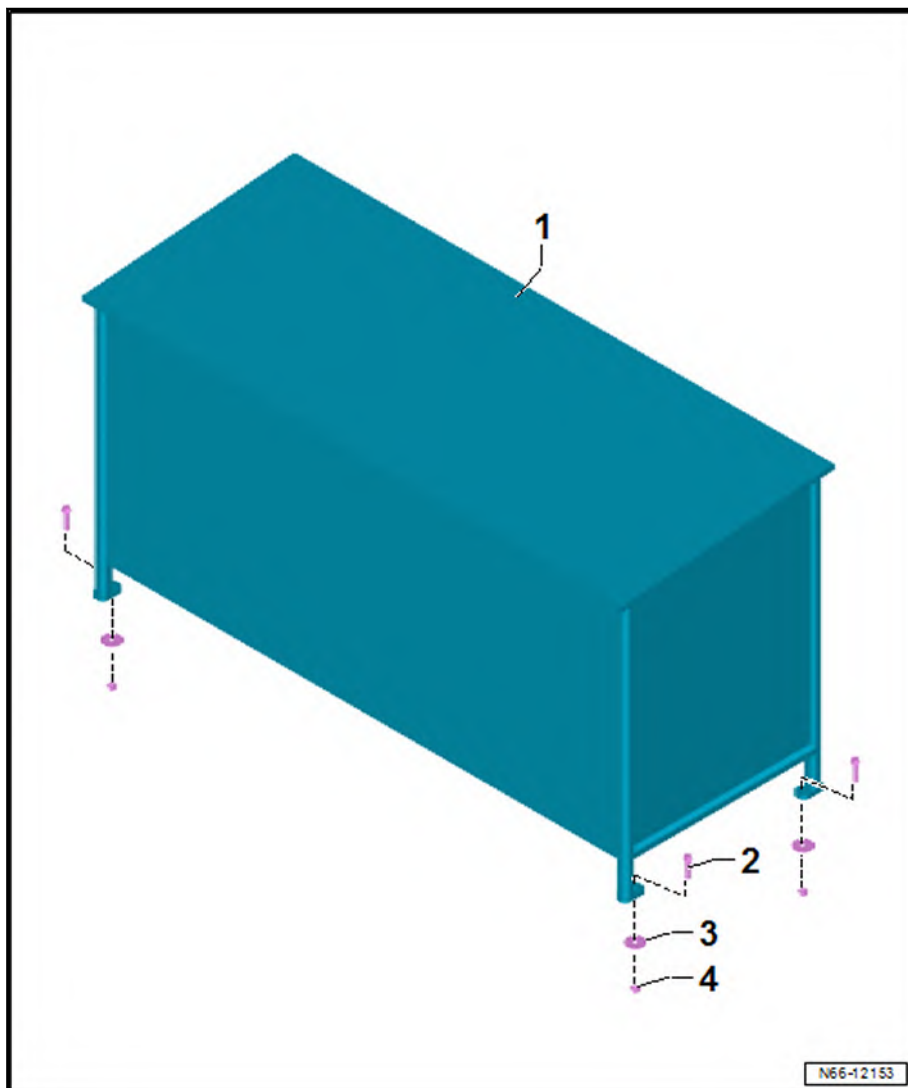
- ☐ Qty. 4
- ☐ 40 Nm

3 - Washers

- ☐ Qty. 4

4 - Nuts

- ☐ Qty. 4



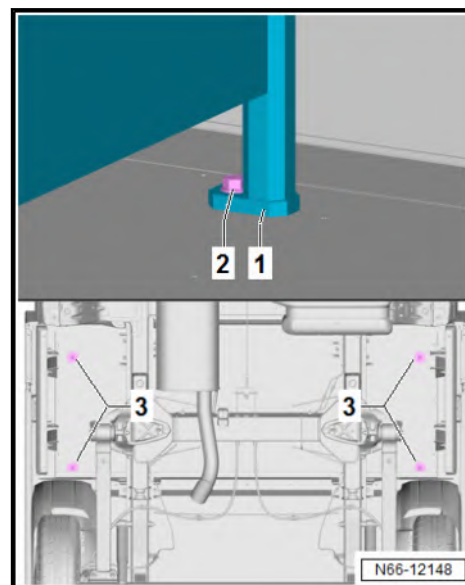
16.2 Removing and installing storage box on drop-side

Special tools and workshop equipment required

- ◆ Torque wrench 6-50 Nm -V.A.G 1331-

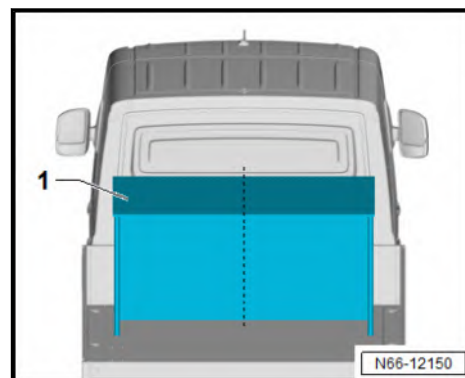


Removing



- Unscrew nuts -3-.
- Remove washers.
- Unscrew bolts -2- from storage box foot -1-.
- Remove storage box.

Installing



Install in reverse order of removal, observing the following:

- Align storage box -1-.

Specified torques

- ◆ ➔ [o16.1 verview - storage box on drop-side", page 206](#)



17 Cross strut

⇒ [o17.1 verview - cross strut", page 208](#)

⇒ [a17.2 nd installing cross strut", page 210](#)

17.1 Assembly overview - cross strut

1 - Cross strut

- ☐ If individual components of the cross strut are renewed, the respective components must be adjusted to the old dimension.
- ☐ Removing and installing ➔ [page 210](#)

2 - Bolts

- ☐ Left and right
- ☐ Front and rear
- ☐ Qty. 12
- ☐ 40 Nm

3 - Plugs

- ☐ Left and right
- ☐ Front and rear
- ☐ Qty. 12

4 - Feet

- ☐ Rear
- ☐ Qty. 2

5 - Nuts

- ☐ Left and right
- ☐ Qty. 6

6 - Bolts

- ☐ Left and right
- ☐ Qty. 12
- ☐ 40 Nm

7 - Bolts

- ☐ Left and right
- ☐ Qty. 4
- ☐ 40 Nm

8 - Feet

- ☐ Front
- ☐ Qty. 2

9 - Nuts

- ☐ Left and right
- ☐ Qty. 6

10 - Counterpieces

- ☐ Qty. 2
- ☐ With nuts welded on

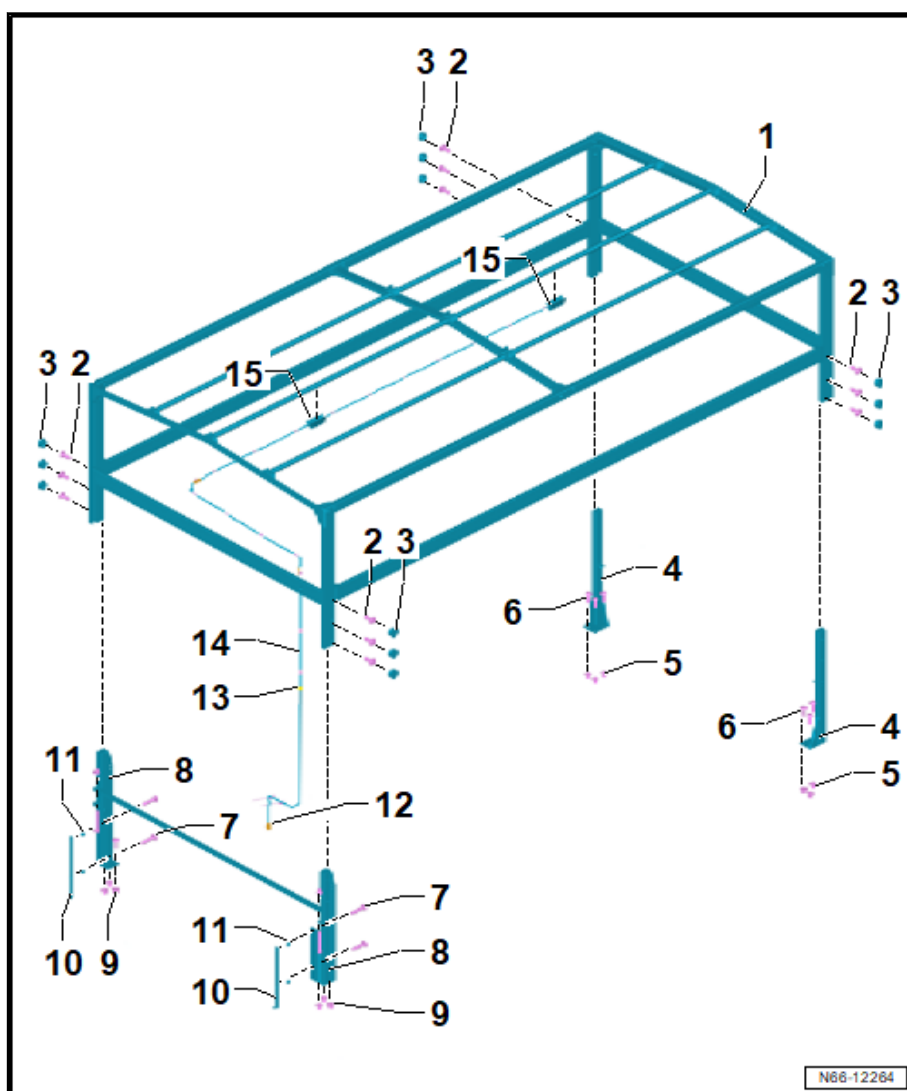
11 - Sleeves

- ☐ Left and right
- ☐ Qty. 4

12 - Electrical connector

13 - Plugs

14 - Wiring harness





15 - Cross strut light

17.2 Removing and installing cross strut

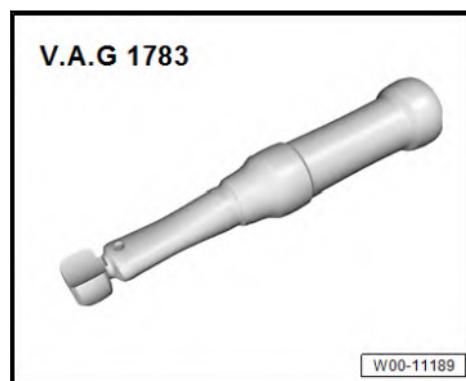
⇒ [a17.2.1 nd installing cross strut on drop-side", page 210](#)

⇒ [a17.2.2 nd installing cross strut on base frame for ladder transport", page 212](#)

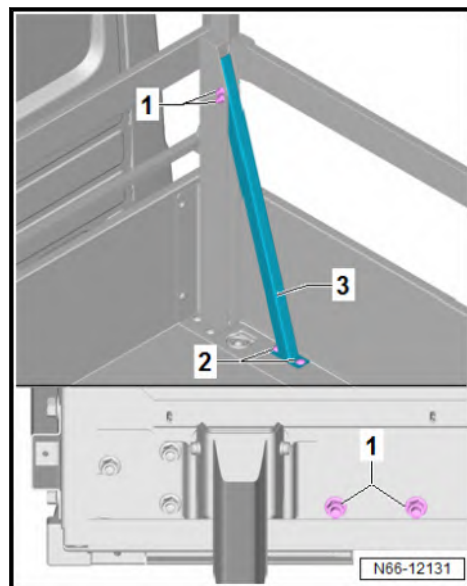
17.2.1 Removing and installing cross strut on drop-side

Special tools and workshop equipment required

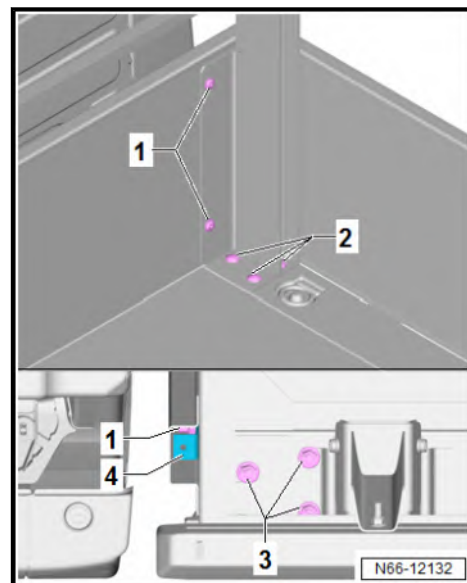
- ◆ Torque wrench -V.A.G 1783-



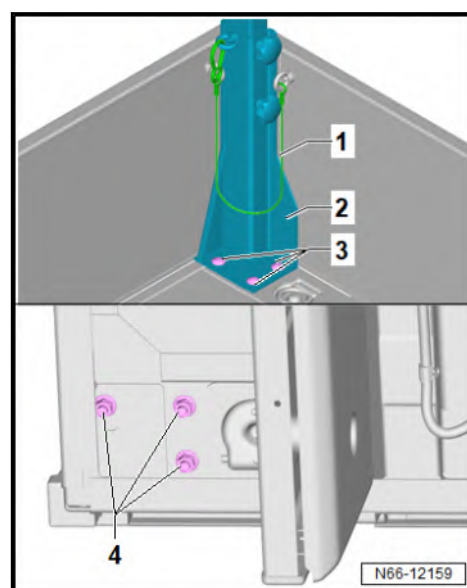
Removing



- Unscrew nuts -1-.
- Unscrew bolts -2-.
- Remove support -3-.
- Unscrew nuts -3-.



- Unscrew bolts -2-.
- Remove screws -1-. Secure counterpiece -4-.
- Remove counterpiece -4- and sleeves.
- Unhook securing cable -1- on foot -2-.



- Unscrew nuts -4-.
- Unscrew bolts -3-.

Installing

Install in reverse order of removal, observing the following:

If individual components of the cross strut are renewed, the respective components must be adjusted to the old dimension.

Specified torques

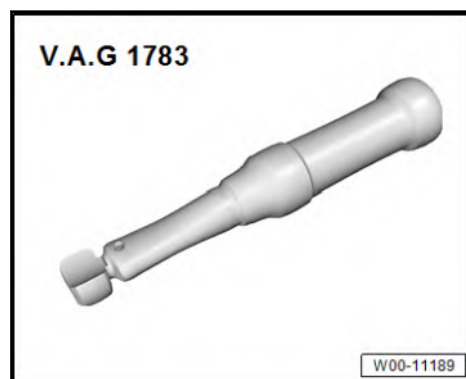
- ◆ ➤ [o17.1 verview - cross strut", page 208](#)



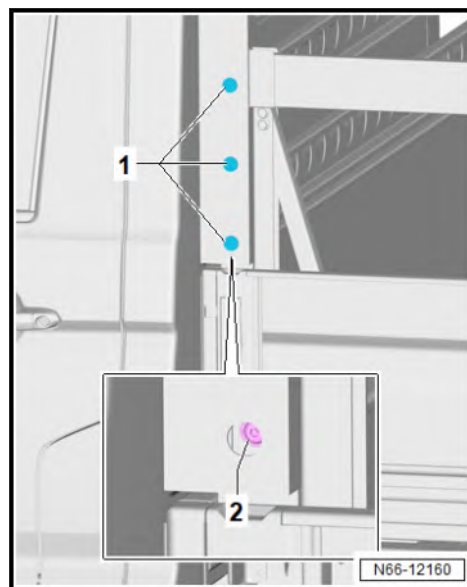
17.2.2 Removing and installing cross strut on base frame for ladder transport

Special tools and workshop equipment required

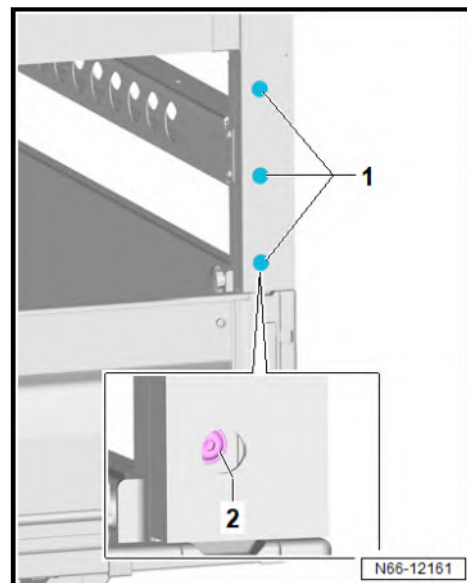
- ◆ Torque wrench -V.A.G 1783-



Removing



- Remove plugs -1-.
- Unscrew bolts -2-.
- Remove plugs -1-.



- Unscrew bolts -2-.

Installing

Installation is carried out in reverse order.

Specified torques

- ◆ ⇒ [o17.1 verview - cross strut", page 208](#)



18 Storage box under drop-side

⇒ [o18.1 verview - storage box bracket", page 214](#)

⇒ [a18.2 nd installing bracket for storage box", page 218](#)

18.1 Assembly overview - storage box bracket

⇒ [o18.1.1 verview - storage box bracket, version 1", page 214](#)

⇒ [o18.1.2 verview - storage box bracket, version 2", page 216](#)

18.1.1 Assembly overview - storage box bracket, version 1

1 - Bracket

- ☐ Left and right
- ☐ Qty. 2

2 - Nuts

- ☐ Left and right
- ☐ Qty. 4

3 - Bolts

- ☐ Left and right
- ☐ Qty. 6
- ☐ 20 Nm

4 - Nuts

- ☐ Left and right
- ☐ Qty. 6

5 - Bracket

- ☐ Top
- ☐ Removing and installing ➔ [page 218](#)
- ☐ Front and rear
- ☐ Qty. 2

6 - Nuts

- ☐ Front and rear
- ☐ Left and right
- ☐ Qty. 8

7 - Bracket

- ☐ Left and right
- ☐ Qty. 2

8 - Bolts

- ☐ Left and right
- ☐ Qty. 4
- ☐ 20 Nm

9 - Strip

- ☐ Top
- ☐ Left and right
- ☐ Qty. 2

10 - Bolts

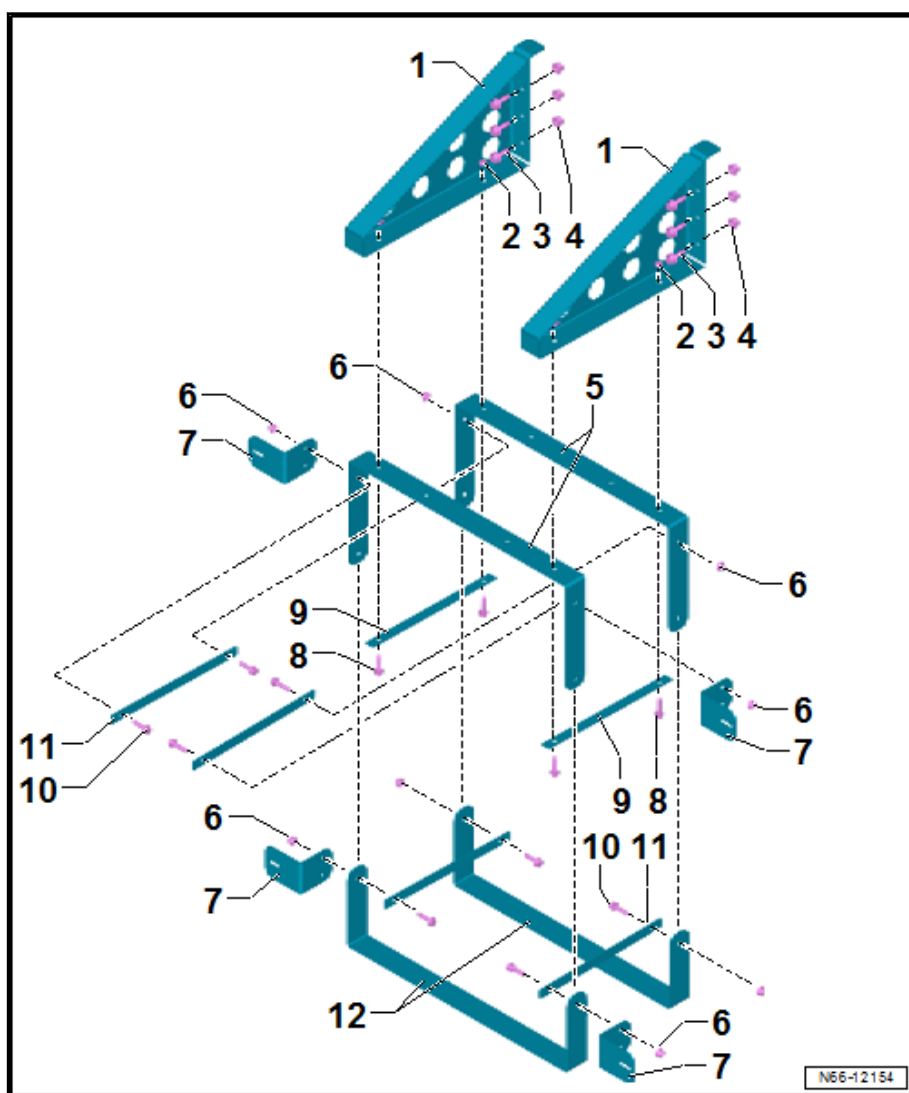
- ☐ Front and rear
- ☐ Left and right
- ☐ Qty. 8
- ☐ 20 Nm

11 - Strip

- ☐ On side
- ☐ Left and right
- ☐ Qty. 2

12 - Bracket

- ☐ Bottom
- ☐ Front and rear





□ Qty. 2

18.1.2 Assembly overview - storage box bracket, version 2

1 - Bolts

- ☐ Left and right
- ☐ Qty. 2
- ☐ 20 Nm

2 - Bracket

- ☐ Top
- ☐ Removing and installing ⇒ [page 218](#)
- ☐ Left and right
- ☐ Qty. 2

3 - Nuts

- ☐ Top
- ☐ Left and right
- ☐ Qty. 2

4 - Nuts

- ☐ Top
- ☐ Left and right
- ☐ Front and rear
- ☐ Qty. 4

5 - Bolts

- ☐ Rear
- ☐ Left and right
- ☐ Qty. 2
- ☐ 20 Nm

6 - Nuts

- ☐ Rear
- ☐ Left and right
- ☐ Qty. 2

7 - Nuts

- ☐ Top
- ☐ Left and right
- ☐ Front and rear
- ☐ Qty. 4

8 - Bolts

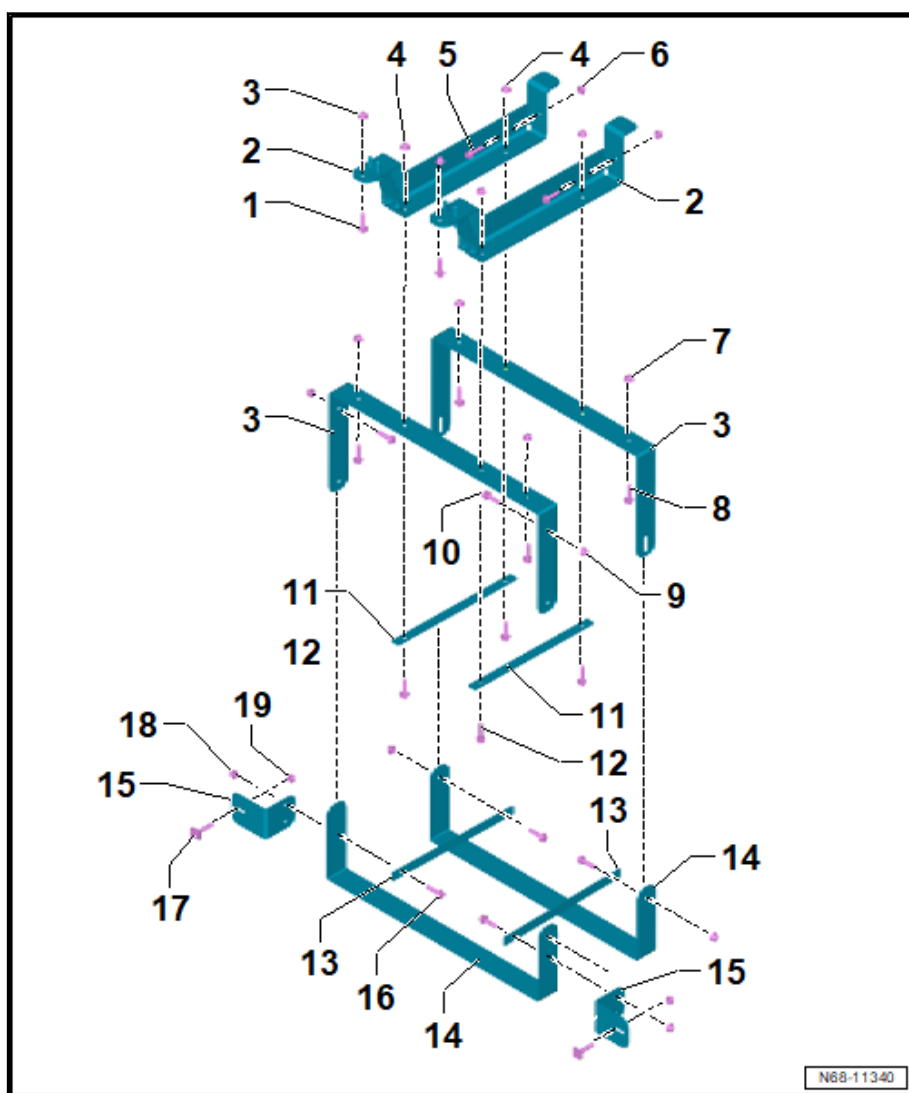
- ☐ Top
- ☐ Left and right
- ☐ Front and rear
- ☐ Qty. 4
- ☐ 20 Nm

9 - Nuts

- ☐ On side
- ☐ Left and right
- ☐ Front and rear
- ☐ Qty. 4

10 - Bolts

- ☐ On side
- ☐ Left and right





- ☐ Front and rear
- ☐ Qty. 4
- ☐ 20 Nm

11 - Strip

- ☐ Top
- ☐ Left and right
- ☐ Qty. 2

12 - Bolts

- ☐ Top
- ☐ Left and right
- ☐ Front and rear
- ☐ Qty. 4
- ☐ 20 Nm

13 - Strip

- ☐ On side
- ☐ Left and right
- ☐ Qty. 2

14 - Bracket

- ☐ Bottom
- ☐ Front and rear
- ☐ Qty. 2

15 - Bracket

- ☐ Left and right
- ☐ Qty. 2

16 - Bolts

- ☐ On side
- ☐ Left and right
- ☐ Front and rear
- ☐ Qty. 4
- ☐ 20 Nm

17 - T-slot nuts with fastener

- ☐ Left and right
- ☐ Qty. 2

18 - Nuts

- ☐ On side
- ☐ Left and right
- ☐ Front and rear
- ☐ Qty. 4

19 - Nuts

- ☐ Left and right
- ☐ Qty. 2
- ☐ 20 Nm

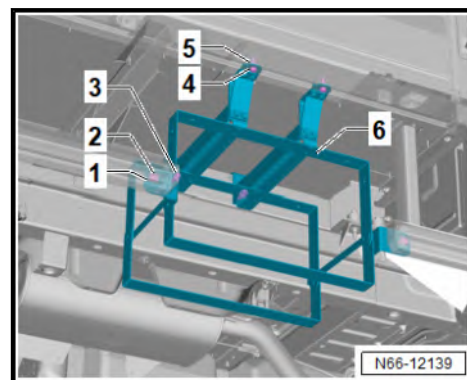
18.2 Removing and installing bracket for storage box

Special tools and workshop equipment required

- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



Removing



- Unscrew bolts -3-.
- Unscrew nuts -1- from T-slot nuts with fastener -2-.
- Unscrew bolts -4-. Counterhold nuts -5- when doing this.
- Remove bracket -6-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ [⇒ o18.1.2 verview - storage box bracket, version 2", page 216](#)



19 Side underbody guard

⇒ [o19.1 verview - side underbody guard", page 220](#)

⇒ [a19.2 nd installing side underbody guard", page 220](#)

19.1 Assembly overview - side underbody guard

1 - Skid plate

- ☐ On side
- ☐ Removing and installing ⇒ [page 220](#)
- ☐ Depending on equipment

2 - End caps

- ☐ At front and rear.
- ☐ Depending on equipment

3 - Nuts

- ☐ Depending on equipment
- ☐ 20 Nm

4 - T-slot nuts with fastener

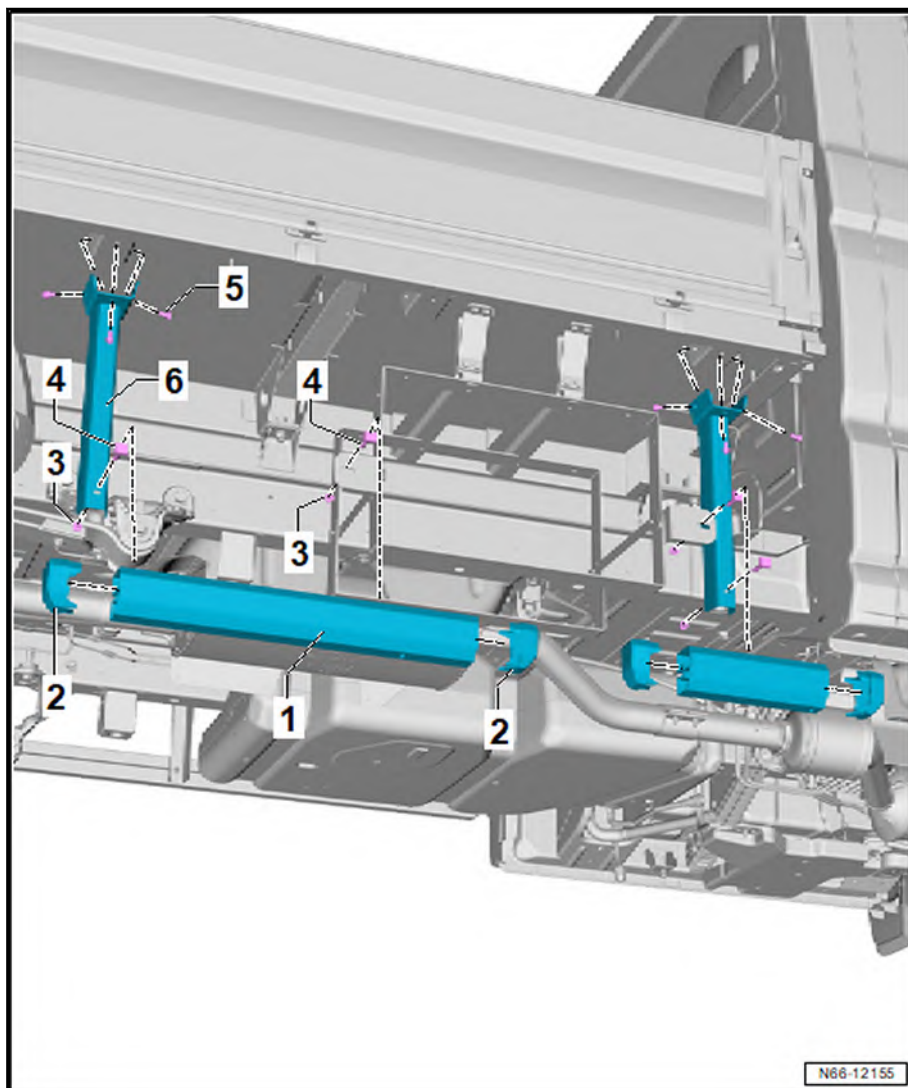
- ☐ Depending on equipment

5 - Bolts

- ☐ Top
- ☐ Depending on equipment
- ☐ 20 Nm

6 - Bracket

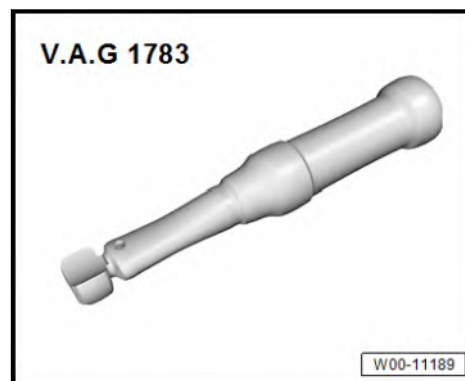
- ☐ Depending on equipment



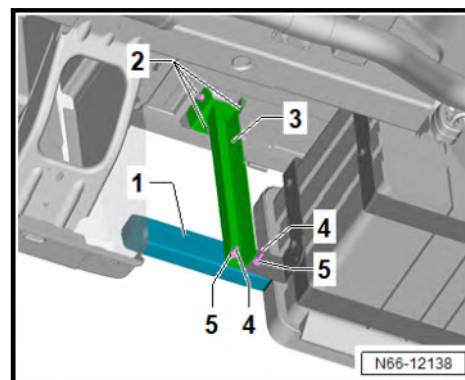
19.2 Removing and installing side underbody guard

Special tools and workshop equipment required

◆ Torque wrench -V.A.G 1783-



Removing



- Unscrew nuts -5- from T-slot nuts with fastener -4-.
- Remove skid plate -1-.
- Unscrew bolts -2-.
- Remove bracket -3-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ [⇒ o19.1 verview - side underbody guard", page 220](#)



20 Storage box

⇒ [o20.1 verview - storage compartment", page 222](#)

⇒ [a20.2 nd installing storage compartment", page 222](#)

20.1 Assembly overview - storage compartment

1 - Storage box

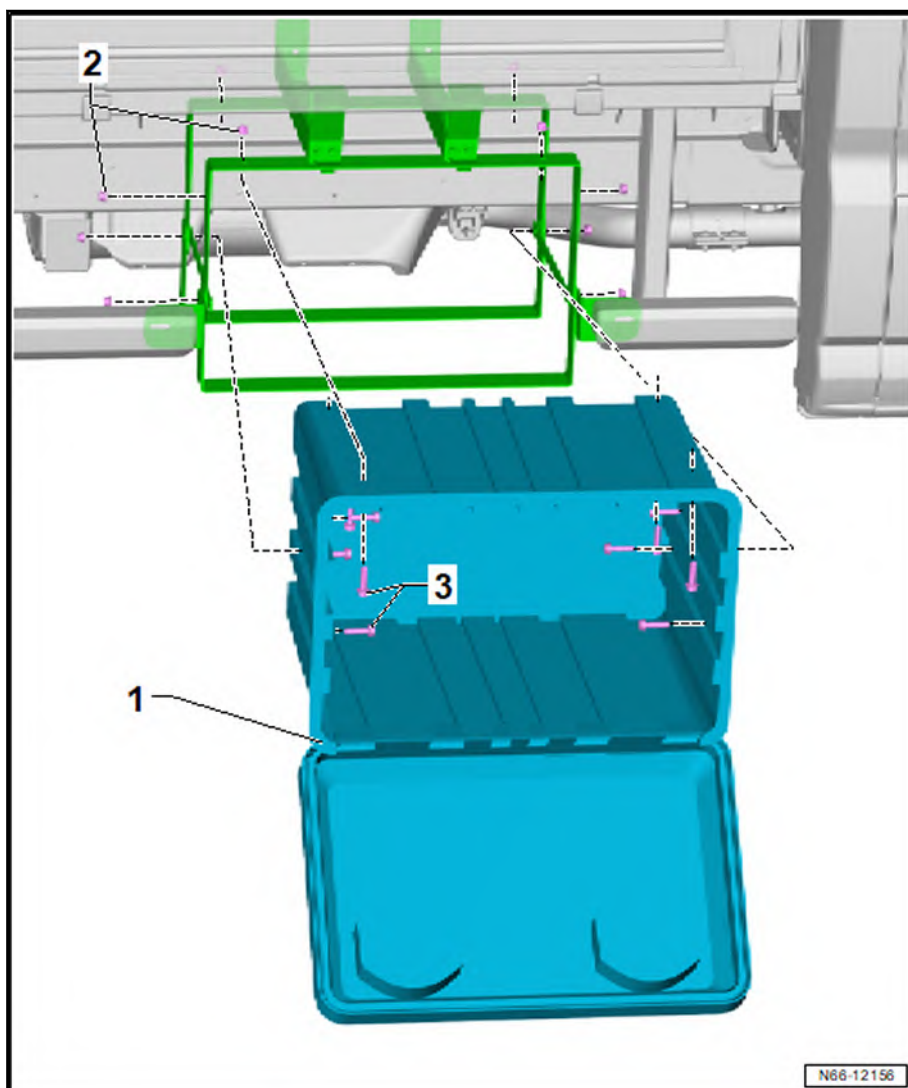
- ☐ Removing and installing ⇒ [page 222](#)
- ☐ Depending on equipment

2 - Nuts

- ☐ Top, left and right
- ☐ Qty. 10 per storage compartment

3 - Bolts

- ☐ Top, left and right
- ☐ Qty. 10 per storage compartment
- ☐ 20 Nm



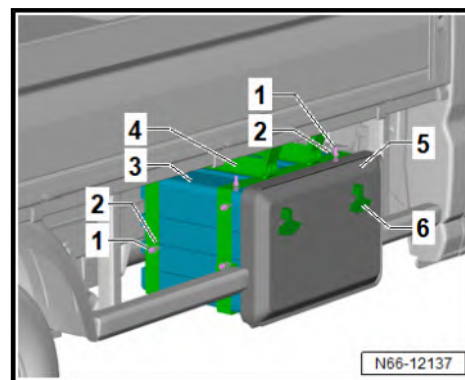
20.2 Removing and installing storage compartment

Special tools and workshop equipment required

- ◆ Torque wrench, 6–50 Nm -V.A.G 1331-



Removing



- Fold down handles -6- and turn by 90 degrees.
- Open cover -5- of storage compartment -3-.
- Unscrew nuts -1-.
- Unscrew bolts -2-.
- Pull storage compartment -3- out of storage compartment bracket -4-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ [⇒ o20.1 verview - storage compartment", page 222](#)



21 Ladder carrier

⇒ [o21.1 verview - ladder carrier", page 224](#)

⇒ [a21.2 nd installing ladder carrier", page 226](#)

21.1 Assembly overview - ladder carrier

⇒ [o21.1.1 verview - front ladder carrier", page 224](#)

⇒ [o21.1.2 verview - rear ladder carrier", page 225](#)

21.1.1 Assembly overview - front ladder carrier



1 - Ladder carrier

- ☐ Removing and installing ⇒ [page 228](#)

2 - Feet

3 - Nuts

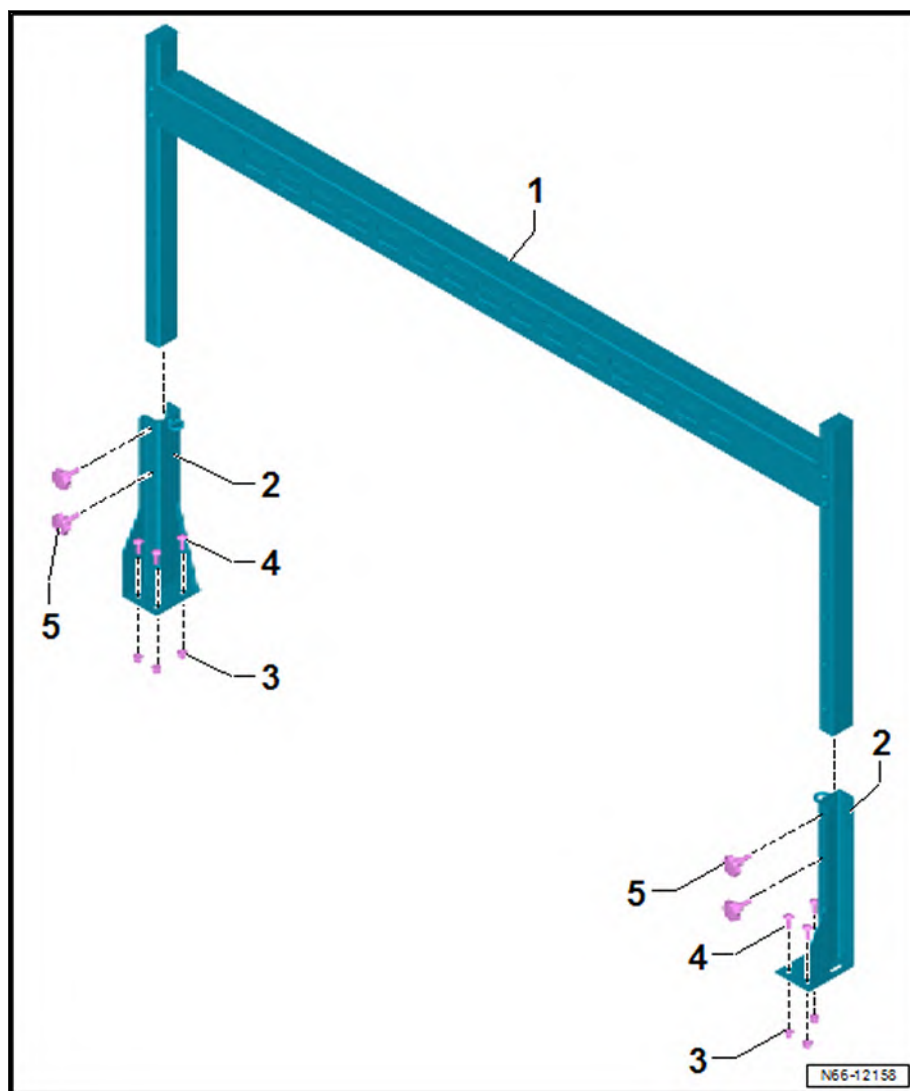
- ☐ Qty. 6

4 - Bolts

- ☐ Qty. 6
- ☐ 40 Nm

5 - Knurled screws

- ☐ Qty. 4



21.2 Removing and installing ladder carrier

⇒ [a21.2.1 nd installing front ladder carrier", page 226](#)

⇒ [a21.2.2 nd installing rear ladder carrier", page 228](#)

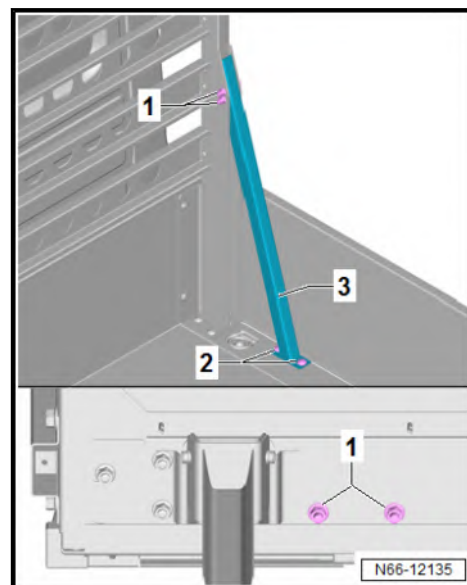
21.2.1 Removing and installing front ladder carrier

Special tools and workshop equipment required

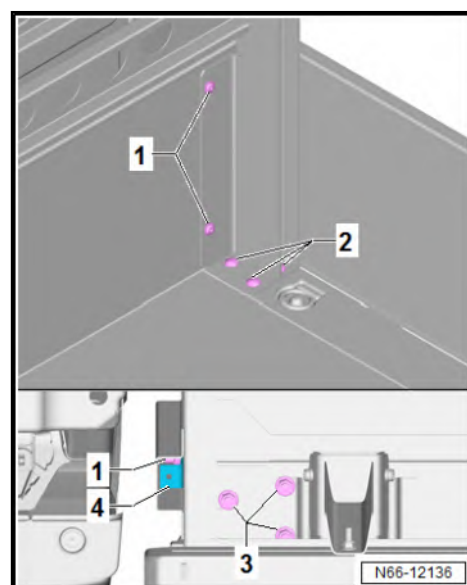
- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



Removing



- Unscrew nuts -1-.
- Unscrew bolts -2-.
- Remove support -3-.
- Unscrew nuts -3-.





- Unscrew bolts -2-.
- Unscrew bolts -1- and secure counterpiece -4-.
- Remove counterpiece -4- and sleeves.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ⇒ [o21.1.1 verview - front ladder carrier", page 224](#)

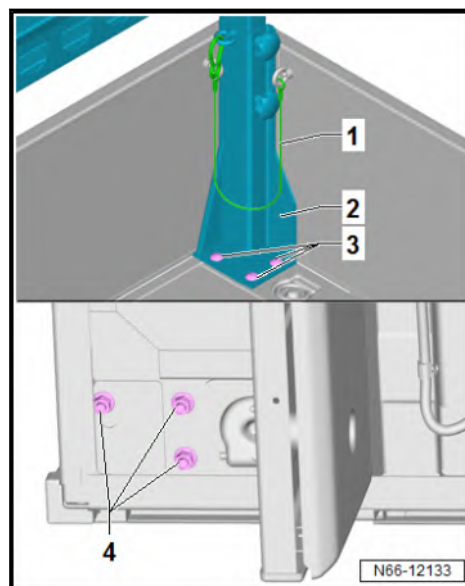
21.2.2 Removing and installing rear ladder carrier

Special tools and workshop equipment required

- ♦ Torque wrench 6-50 Nm -V.A.G 1331-



Removing



- Unhook securing cable -1- on foot -2-.
- Unscrew nuts -4-.
- Unscrew bolts -3-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ⇒ [o21.1.2 verview - rear ladder carrier", page 225](#)

22 Roof carrier system with securing device

⇒ [o22.1 verview - roof carrier system with securing device", page 229](#)

⇒ [a22.2 nd installing roof carrier system with securing device", page 229](#)

22.1 Assembly overview - roof carrier system with securing device

1 - Roof carrier

- ☐ With fastening device
- ☐ Removing and installing ⇒ [page 229](#)

2 - Bolts

- ☐ Left and right
- ☐ Qty. 8
- ☐ 20 Nm

3 - Connection pieces

- ☐ Left and right
- ☐ Qty. 4

4 - Bolts

- ☐ Left and right
- ☐ Qty. 4
- ☐ 17 Nm

5 - Washers

- ☐ Left and right
- ☐ Qty. 8

6 - Feet

- ☐ Left and right
- ☐ Qty. 4

7 - T-slot nuts

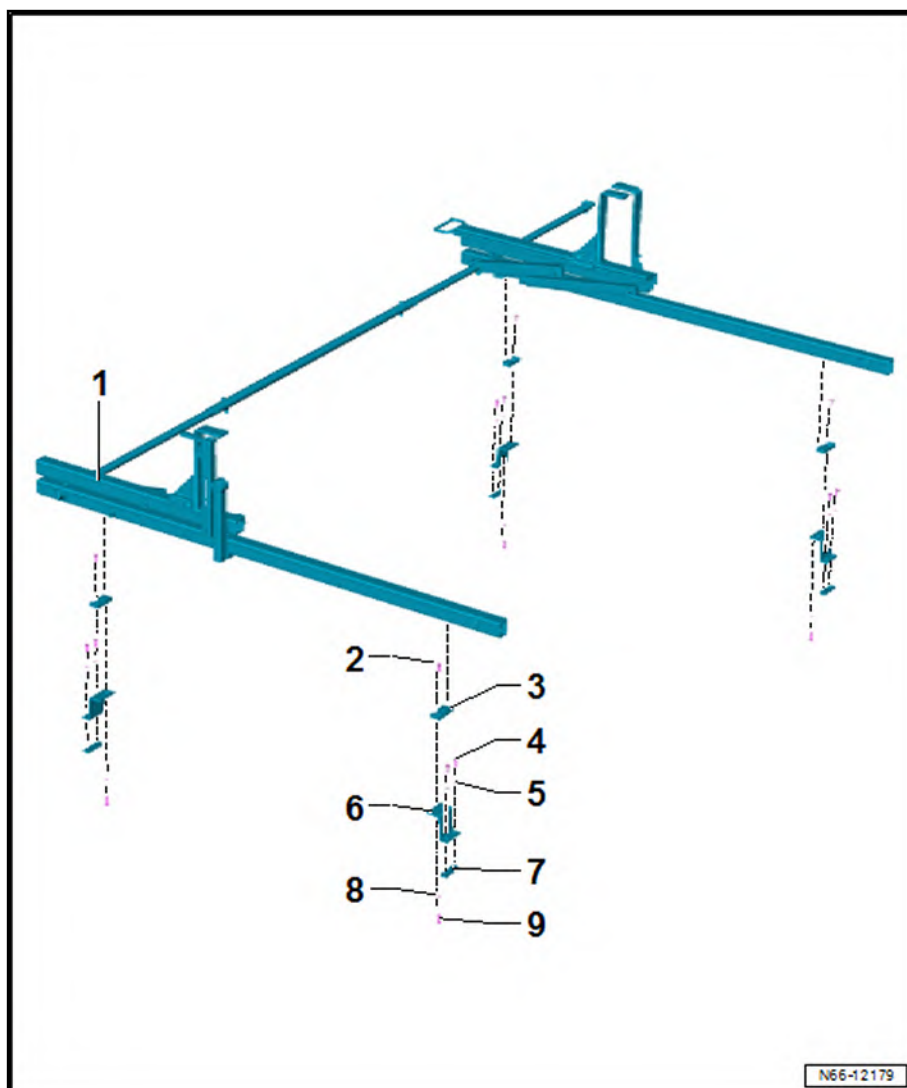
- ☐ Left and right
- ☐ Qty. 4

8 - Washers

- ☐ Left and right
- ☐ Qty. 4

9 - Bolts

- ☐ Left and right
- ☐ Qty. 4
- ☐ 17 Nm



22.2 Removing and installing roof carrier system with securing device

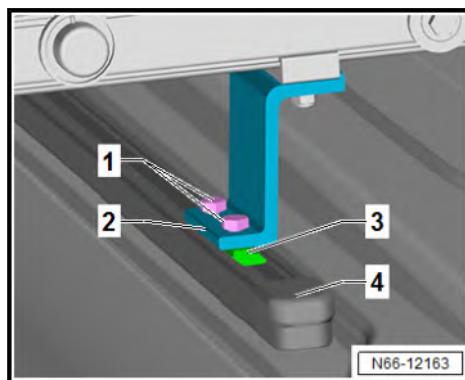
Special tools and workshop equipment required



- ◆ Torque wrench -V.A.G 1410A-



Removing



- Unscrew bolts -1-.
- Remove roof carrier system with securing device and feet -2-.
- Remove T-slot nuts -3- from C-rail -4-.

Installing

Install in reverse order of removal, observing the following:

- Do not use electric tools for installation of the bolts.
- Apply assembly paste to all bolts.

Specified torques

- ◆ [⇒ o22.1 verview - roof carrier system with securing device", page 229](#)

23 Easy Entry, right rear wing door

⇒ [o23.1 verview – Easy Entry, right rear wing door”, page 231](#)

⇒ [a23.2 nd installing Easy Entry, right rear wing door”, page 231](#)

23.1 Assembly overview – Easy Entry, right rear wing door

1 - Bracket

- ☐ Rear

2 - Bracket

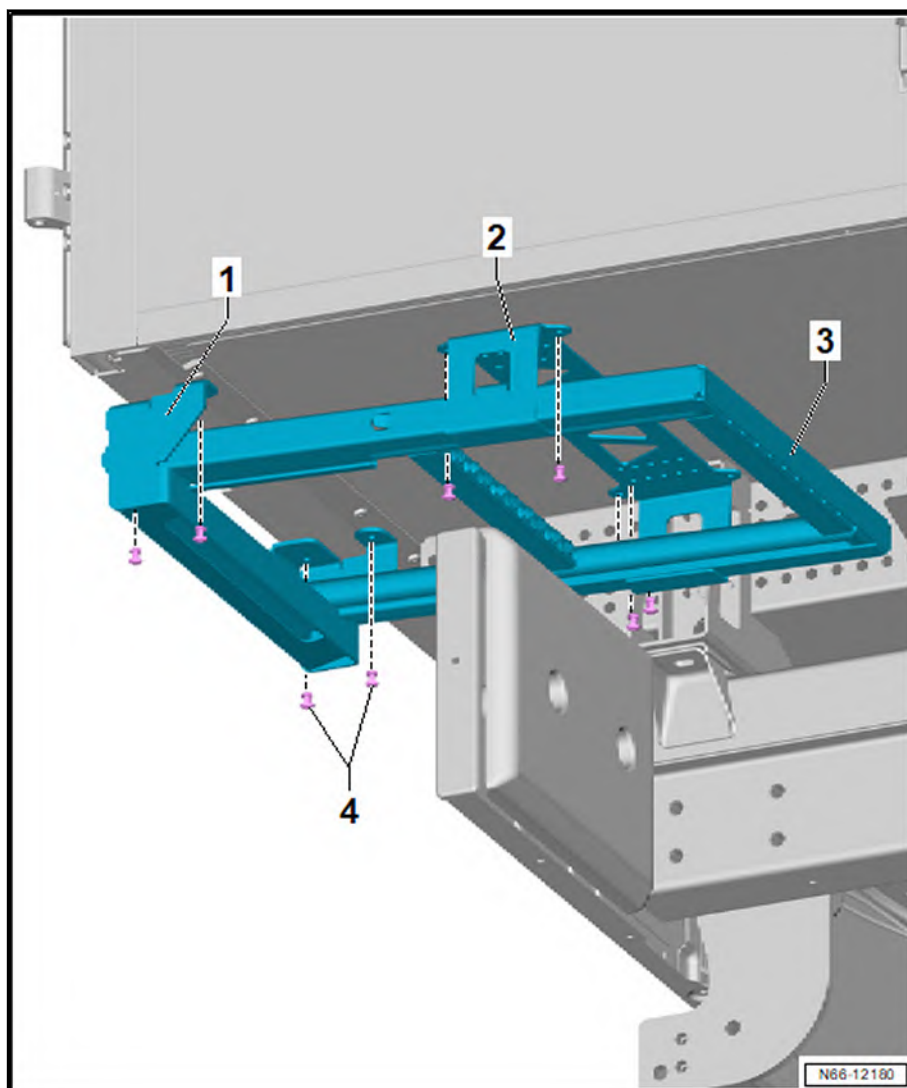
- ☐ Front

3 - Easy Entry

- ☐ Removing and installing ⇒ [page 231](#)

4 - Pop rivets

- ☐ Qty. 8

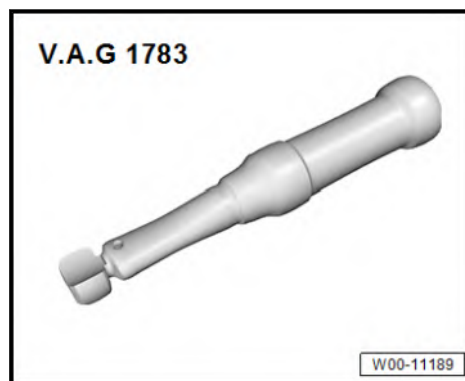


23.2 Removing and installing Easy Entry, right rear wing door

Special tools and workshop equipment required



- ◆ Torque wrench -V.A.G 1783-



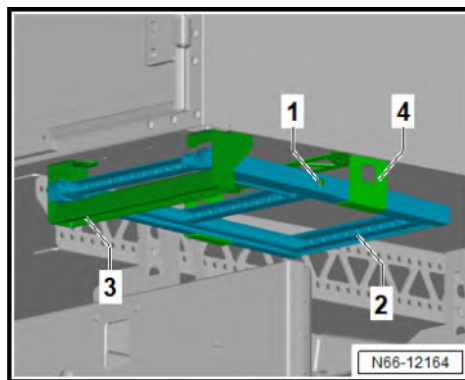
- ◆ Pop rivet pliers -VAS 5072A-



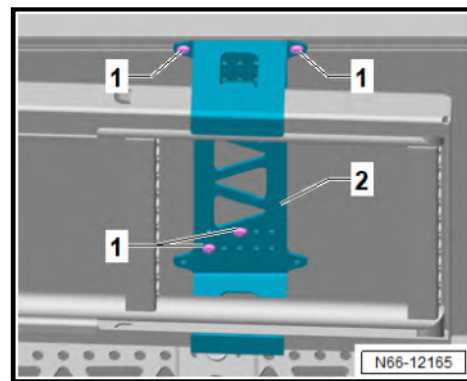
- ◆ Pin punch -VAS 5513-



Removing



- Press in securing tab -1-.
- Detach ladder -2- from bracket -3- and bracket -4-.

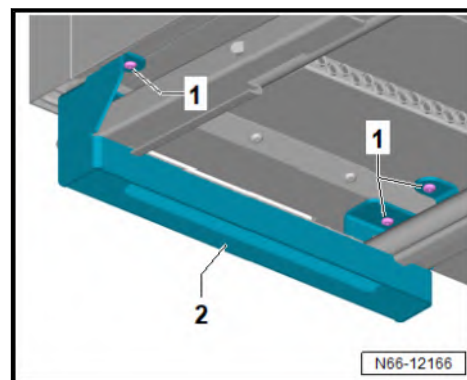


! CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out rivet heads -1-.
- Drive out rivet shanks -1- with drift -VAS 5513-.
- Remove swarf.
- Remove bracket -2-.



! CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out rivet heads -1- using drill and drill bit.
- Drive out rivet shanks -1- with drift -VAS 5513-.
- Remove swarf.
- Remove bracket -2-.

Installing

Install in reverse order of removal, observing the following:

- Use rivet pliers -VAS 5072A- to install new pop rivets.



Specified torques

- ◆ [⇒ o23.1 verview – Easy Entry, right rear wing door”, page 231](#)



24 Underbody guard (protection package)

⇒ [o24.1 verview - underbody guard \(protection package\)", page 235](#)

⇒ [a24.2 nd installing underbody guard \(protection package\)", page 237](#)

24.1 Assembly overview - underbody guard (protection package)



1 - Skid plate

- ☐ Centre
- ☐ Removing and installing ⇒ [page 239](#)

2 - Bolts

- ☐ Qty. 2
- ☐ 20 Nm

3 - Nut

- ☐ 20 Nm

4 - Bolts

- ☐ Renew after removal
- ☐ Qty. 3
- ☐ 20 Nm

5 - Bolts

- ☐ Qty. 2
- ☐ 40 Nm

6 - Bracket

- ☐ For skid plate
- ☐ Removing and installing ⇒ [page 238](#)

7 - Nuts

- ☐ Qty. 2
- ☐ 20 Nm

8 - Clamps

- ☐ Qty. 2
- ☐ 18 Nm

9 - Skid plate

- ☐ Rear
- ☐ Removing and installing ⇒ [page 240](#)

10 - Bolts

- ☐ Qty. 4
- ☐ 3 Nm

11 - Bolts

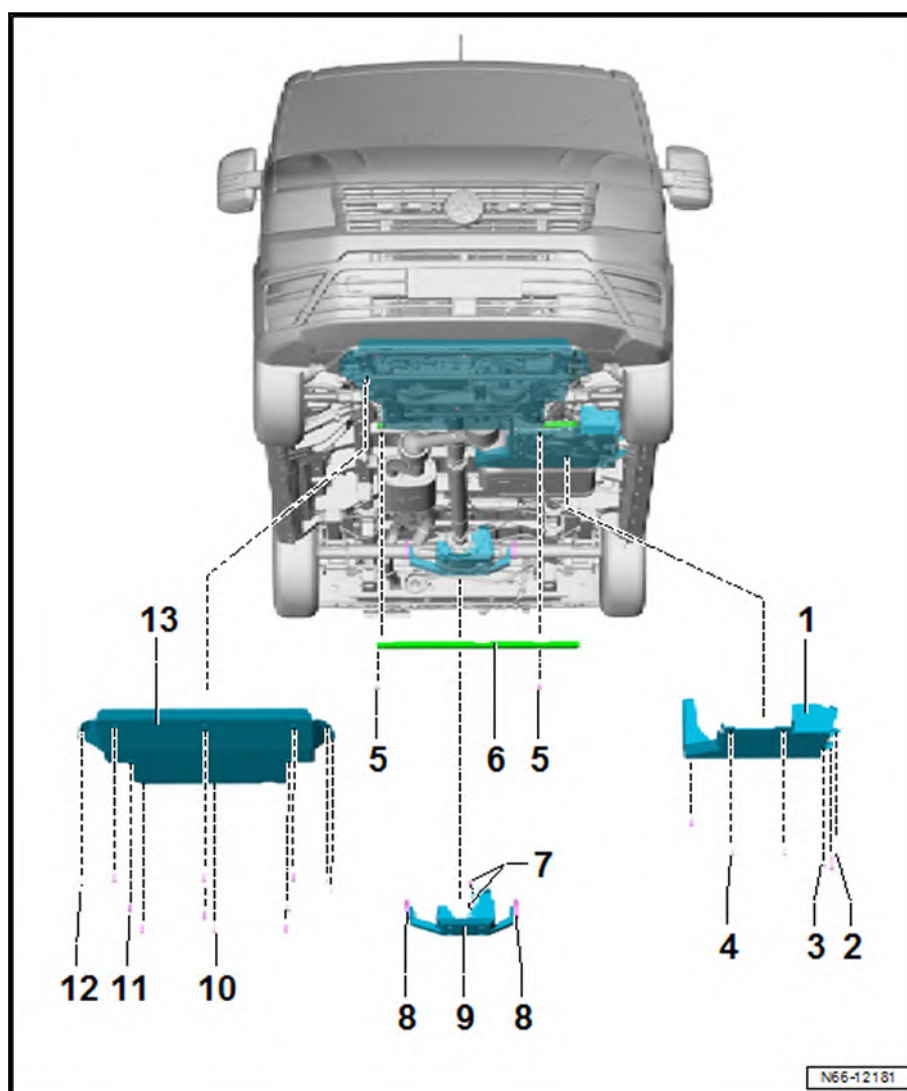
- ☐ Qty. 6
- ☐ 20 Nm

12 - Bolts

- ☐ Qty. 2
- ☐ 70 Nm

13 - Skid plate

- ☐ Front
- ☐ Removing and installing ⇒ [page 237](#)



24.2 Removing and installing underbody guard (protection package)

⇒ [a24.2.1 nd installing skid plate, front", page 237](#)

⇒ [a24.2.2 nd installing bracket for skid plate, front", page 238](#)

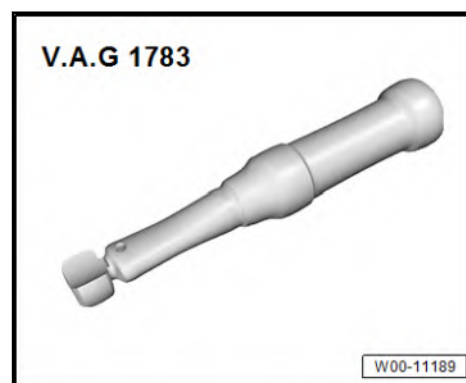
⇒ [a24.2.3 nd installing skid plate, centre", page 239](#)

⇒ [a24.2.4 nd installing skid plate, rear", page 240](#)

24.2.1 Removing and installing skid plate, front

Special tools and workshop equipment required

◆ Torque wrench -V.A.G 1783-



◆ Torque wrench 6-50 Nm -V.A.G 1331-

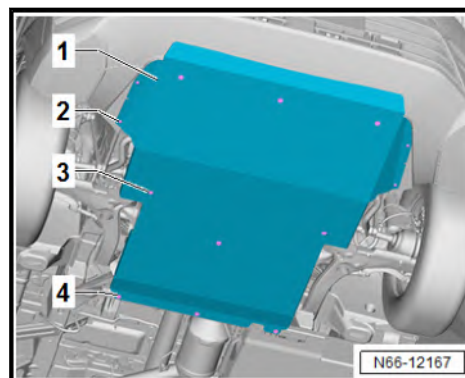


◆ Torque wrench, 40 to 200 Nm -V.A.G 1332A-





Removing



- Unscrew bolts -2-.
- Unscrew bolts -3-.
- Unscrew bolts -4-.
- Remove skid plate -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ [⇒ o24.1 overview - underbody guard \(protection package\)", page 235](#)

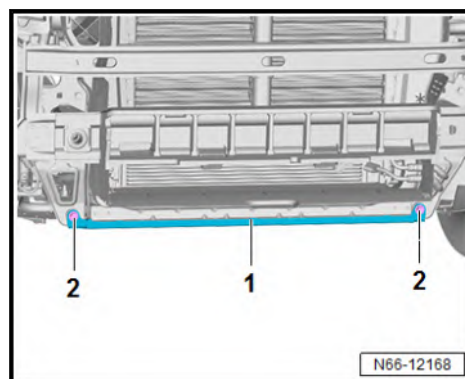
24.2.2 Removing and installing bracket for skid plate, front

Special tools and workshop equipment required

- ♦ Torque wrench 6-50 Nm -V.A.G 1331-

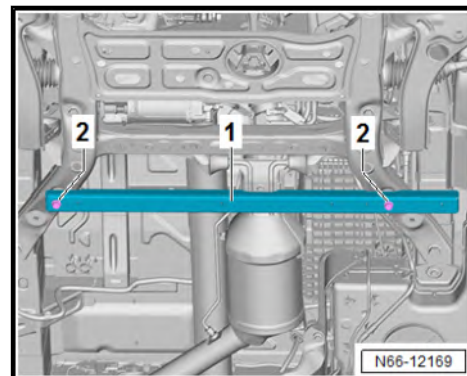


Removing





- Remove front skid plate ➔ [page 237](#) .
- Unscrew bolts -2-.
- Remove bracket -1-.
- Unscrew bolts -2-.



- Remove bracket -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➔ [o24.1 overview - underbody guard \(protection package\)", page 235](#)

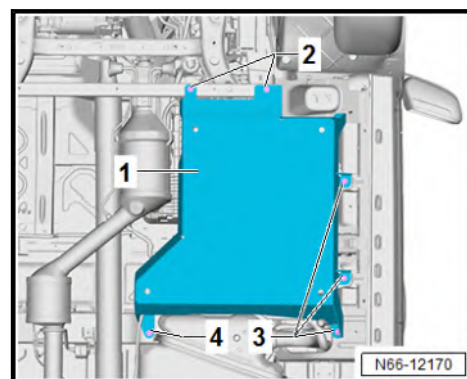
24.2.3 Removing and installing skid plate, centre

Special tools and workshop equipment required

- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



Removing





- Unscrew bolts -2-.
- Unscrew bolts/nut -3-.
- Unscrew bolt -4-.
- Remove skid plate -1- with the aid of a second mechanic.

Installing

Install in reverse order of removal, observing the following:

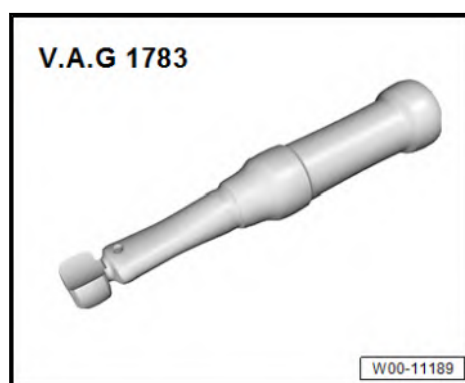
Specified torques

- ♦ [⇒ o24.1 verview - underbody guard \(protection package\)", page 235](#)

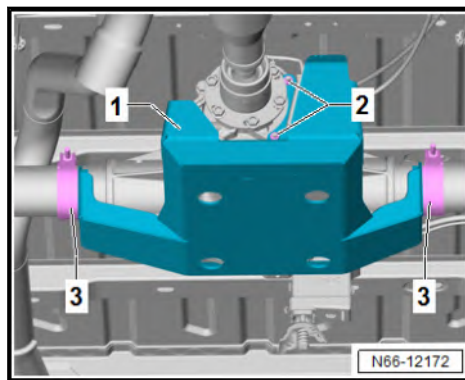
24.2.4 Removing and installing skid plate, rear

Special tools and workshop equipment required

- ♦ Torque wrench -V.A.G 1783-



Removing



- Release clips -3-.
- Unscrew bolts -2-.
- Remove skid plate -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ [⇒ o24.1 verview - underbody guard \(protection package\)", page 235](#)

25 Roof ventilator

⇒ [a25.1 verview - roof ventilator", page 241](#)

⇒ [a25.2 nd installing roof ventilator", page 241](#)

25.1 Assembly overview - roof ventilator

1 - Bolts

- ☐ Qty. 4
- ☐ 2 Nm

2 - Hood

3 - Bolts

- ☐ Qty. 9
- ☐ 4 Nm

4 - Roof ventilator

- ☐ Removing and installing with catch ⇒ [page 241](#)
- ☐ Removing and installing without catch ⇒ [page 243](#)

5 - Moulded foam insert

6 - Frame

7 - Nuts

- ☐ Qty. 9

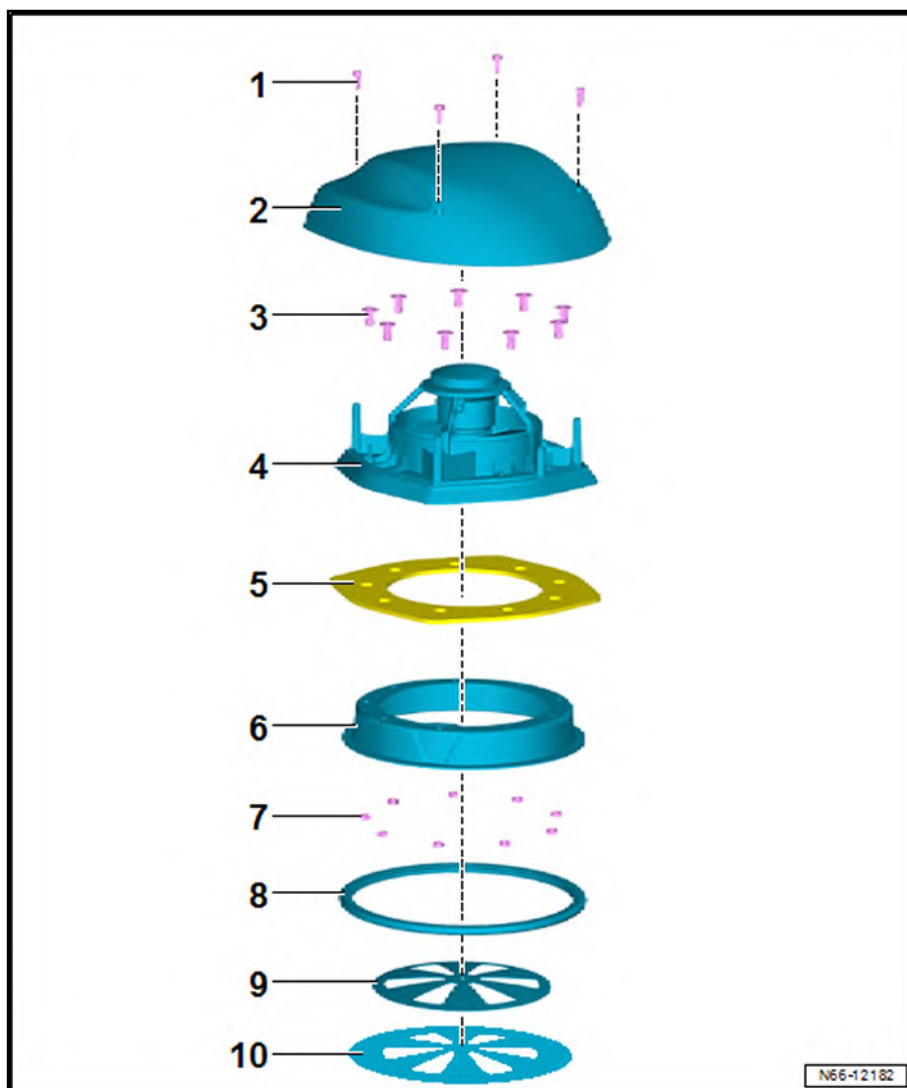
8 - Edge protector

9 - Air guide

- ☐ Upper part

10 - Air guide

- ☐ Lower part



25.2 Removing and installing roof ventilator

⇒ [a25.2.1 nd installing electric/mechanical roof ventilator with catch", page 241](#)

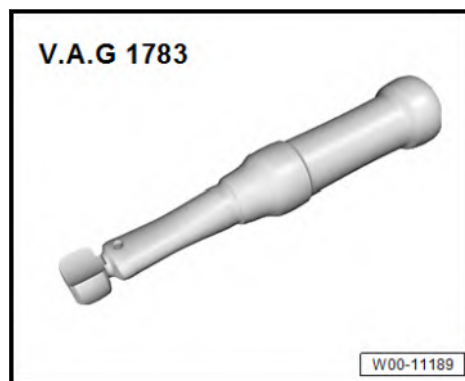
⇒ [a25.2.2 nd installing electric/mechanical roof ventilator without catch", page 243](#)

25.2.1 Removing and installing electric/mechanical roof ventilator with catch

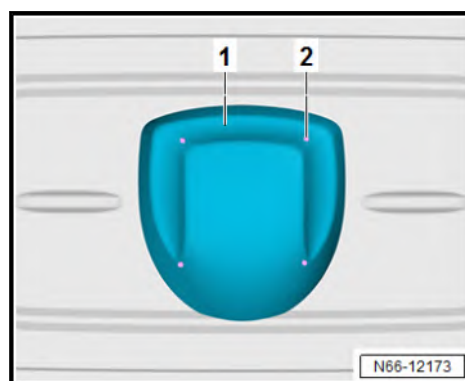
Special tools and workshop equipment required



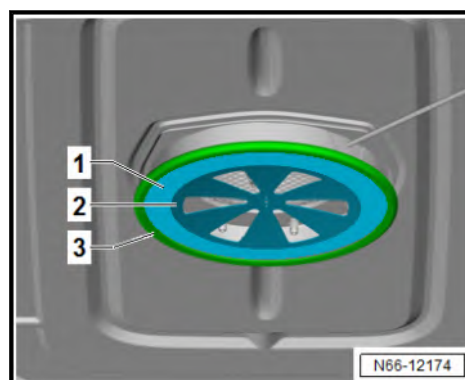
- ◆ Torque wrench -V.A.G 1783-



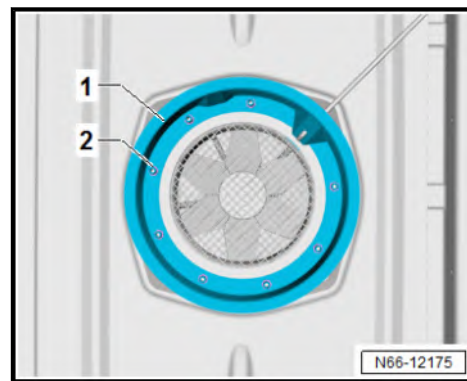
Removing



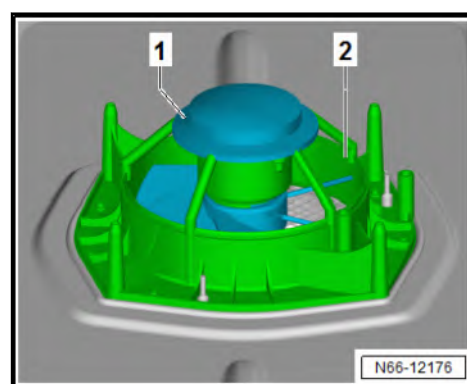
- Unscrew bolts -2-.
- Remove hood -1-.
- Remove edge protection -3-.



- Remove air guide upper part -1- and air guide lower part -2-.
- Unscrew nuts -2-.



- Remove frame -1-.
- If fitted, disconnect connector.
- Remove roof ventilator -1- with frame -2-.



Installing

Install in reverse order of removal, observing the following:

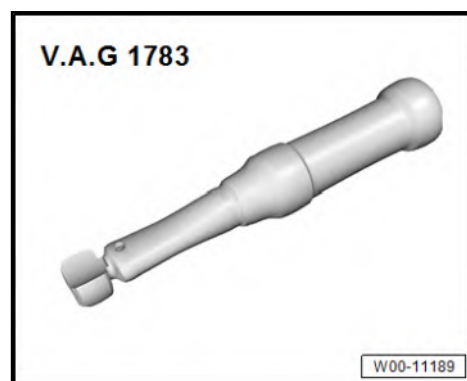
Specified torques

- ◆ [⇒ o25.1 verview - roof ventilator", page 241](#)

25.2.2 Removing and installing electric/mechanical roof ventilator without catch

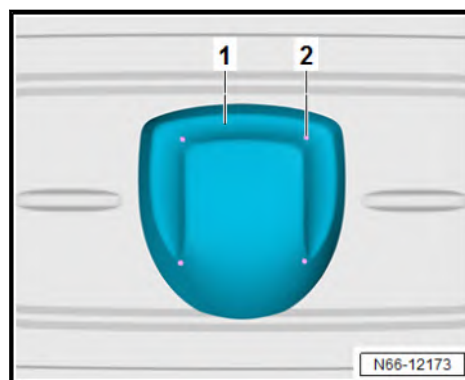
Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1783-

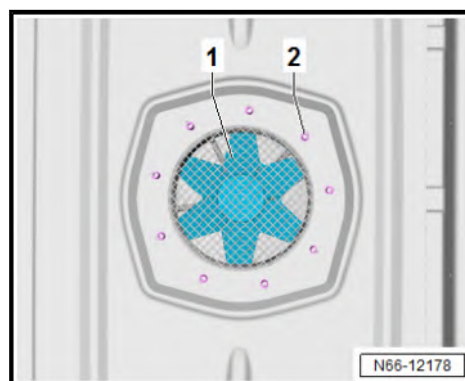




Removing



- Unscrew bolts -2-.
- Remove hood -1-.
- Unscrew nuts -2-.



- If fitted, disconnect connector.
- Remove roof ventilator -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ➔ [o25.1 verview - roof ventilator", page 241](#)

26 Ladder support

⇒ [o26.1 verview - ladder support", page 245](#)

⇒ [a26.2 nd installing ladder support", page 246](#)

26.1 Assembly overview - ladder support

1 - Ladder support

- ☐ Front
- ☐ Removing and installing ⇒ [page 246](#)

2 - Ladder support

- ☐ Rear
- ☐ Removing and installing ⇒ [page 247](#)

3 - Base mounting

- ☐ Qty. 2
- ☐ Removing and installing ⇒ [page 248](#)

4 - Bolts

- ☐ Left and right
- ☐ Qty. 16
- ☐ 20 Nm

5 - Bolts

- ☐ Renew after removal
- ☐ Left and right
- ☐ Qty. 4
- ☐ 10 Nm +45°

6 - Bolts

- ☐ Left and right
- ☐ Qty. 20
- ☐ 20 Nm

7 - Nuts

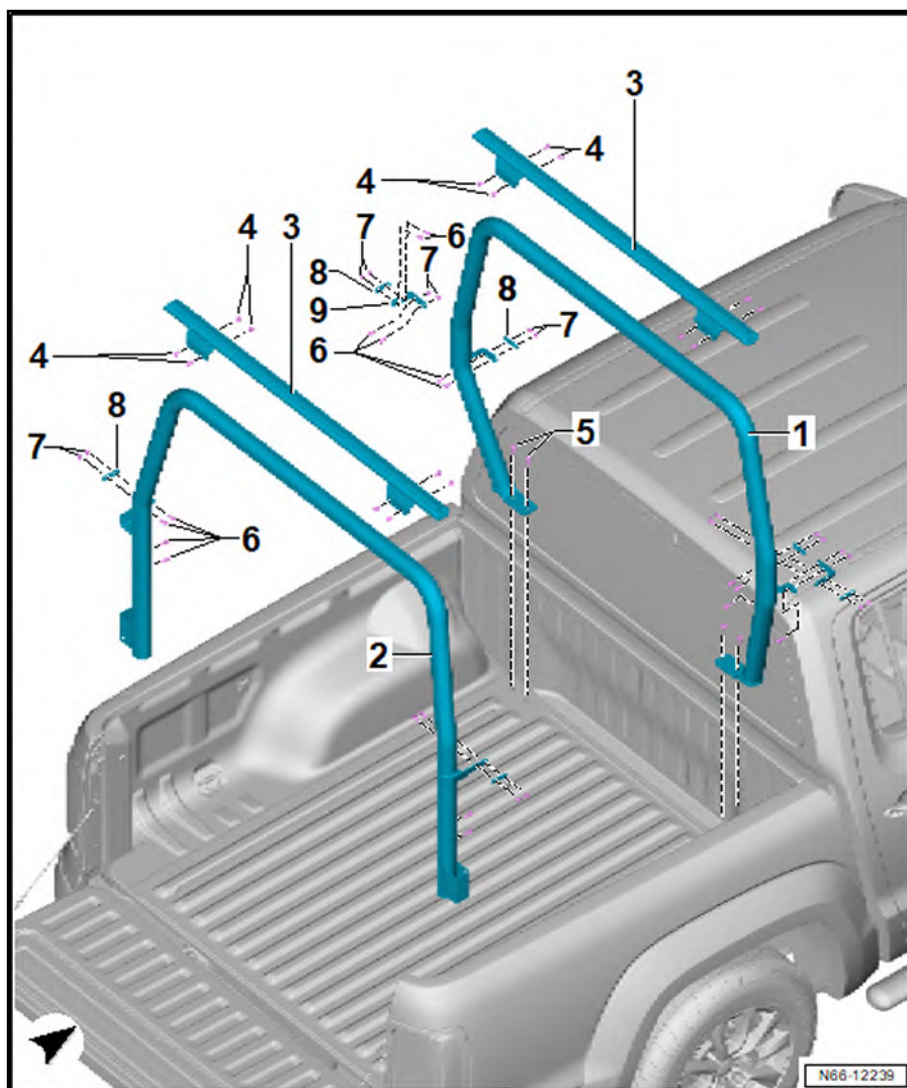
- ☐ Renew after removal
- ☐ Left and right
- ☐ Qty. 16

8 - Securing plate

- ☐ Left and right
- ☐ Qty. 4

9 - Reinforcement bracket

- ☐ Left and right
- ☐ Qty. 2





26.2 Removing and installing ladder support

⇒ [a26.2.1 nd installing ladder support, front", page 246](#)

⇒ [a26.2.2 nd installing ladder support, rear", page 247](#)

⇒ [a26.2.3 nd installing cross bars", page 248](#)

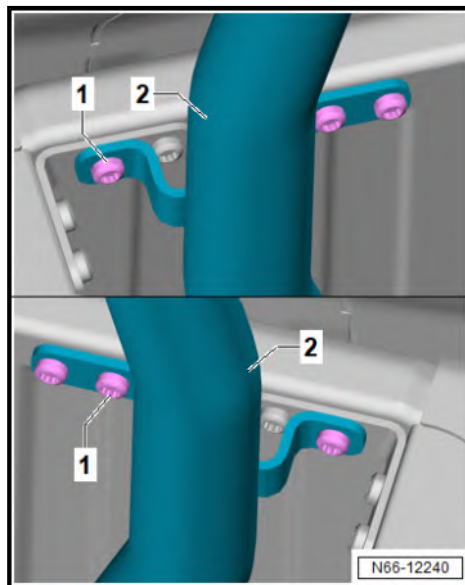
26.2.1 Removing and installing ladder support, front

Special tools and workshop equipment required

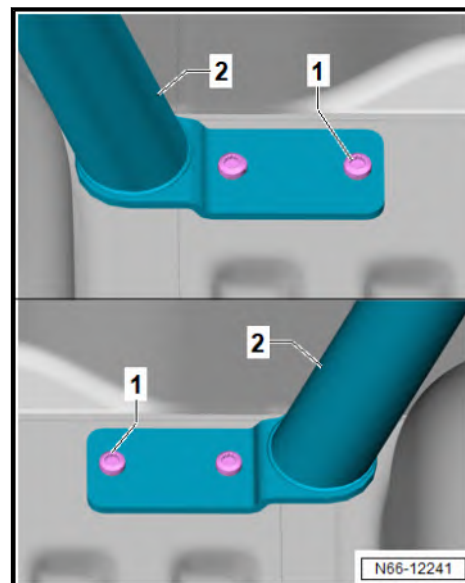
- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



Removing



- Unscrew bolts -1- from ladder carrier at front -2-. Counter-hold nuts ⇒ [Item 7 \(page 245\)](#) while doing this.
- Detach securing plate ⇒ [Item 8 \(page 245\)](#) .
- Unscrew bolts -1-.



- Remove front ladder carrier -2-.

Installing

Install in reverse order of removal, observing the following:

- Apply assembly paste to all bolts.

Specified torques

- ♦ [⇒ o26.1 verview - ladder support", page 245](#)

26.2.2 Removing and installing ladder support, rear

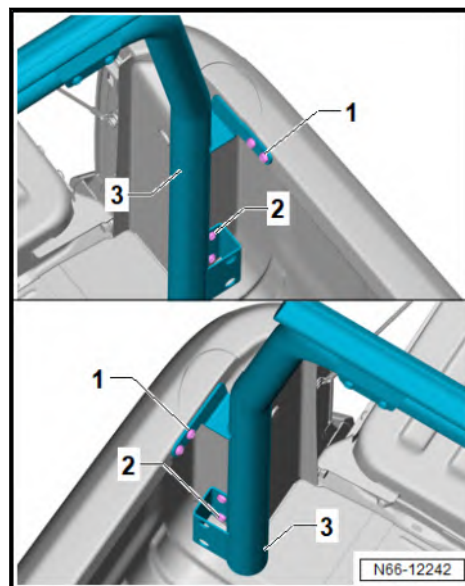
Special tools and workshop equipment required

- ♦ Torque wrench 6-50 Nm -V.A.G 1331-





Removing



- Unscrew bolts -1-. Counterhold nuts ⇒ [Item 7 \(page 245\)](#) while doing this.
- Unscrew bolts -2-.
- Remove front ladder carrier -3-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ⇒ [o26.1 verview - ladder support", page 245](#)

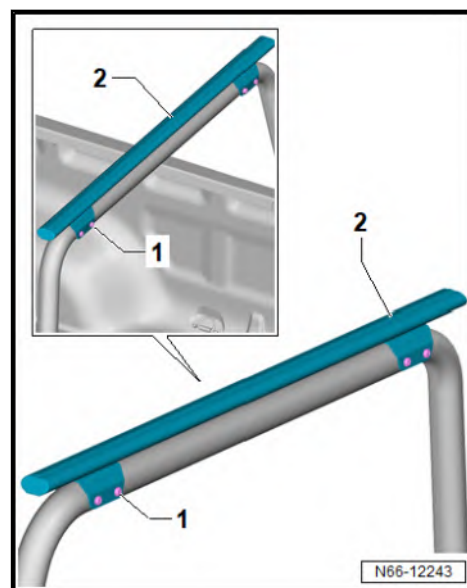
26.2.3 Removing and installing cross bars

Special tools and workshop equipment required

- ♦ Torque wrench 6-50 Nm -V.A.G 1331-



Removing



- Unscrew bolts -1-.
- Detach cross bars -2-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [o26.1 verview - ladder support", page 245](#)



27 Design double tube bar

⇒ [o27.1 verview - design double tube bar", page 250](#)

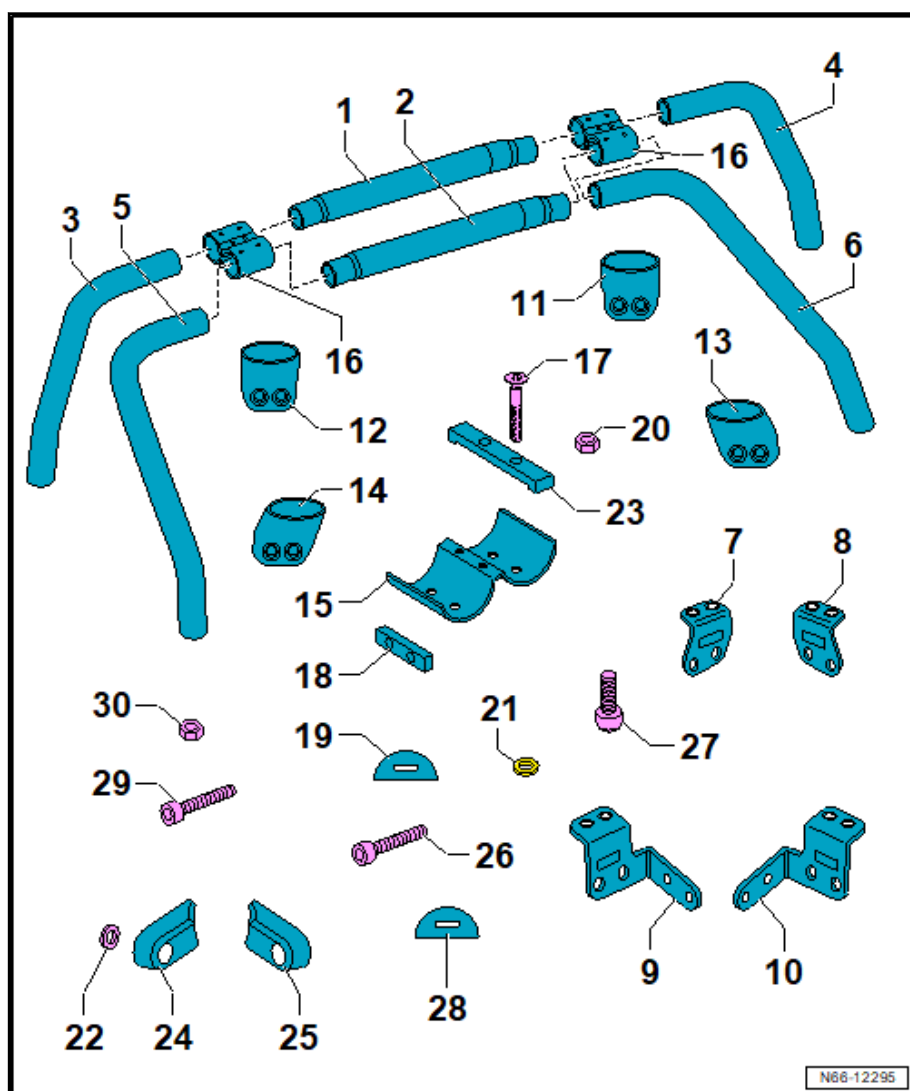
⇒ [a27.2 nd installing design double tube bar", page 252](#)

⇒ [i27.3 nstructions - design double tube bar", page 254](#)

27.1 Assembly overview - design double tube bar



- 1 - Tube**
 - ☐ Centre, front
- 2 - Tube**
 - ☐ Centre, rear
- 3 - Tube**
 - ☐ Left, front
- 4 - Tube**
 - ☐ Right, front
- 5 - Tube**
 - ☐ Left, rear
- 6 - Tube**
 - ☐ Right, rear
- 7 - Bracket**
 - ☐ Left, rear (R LH)
- 8 - Bracket**
 - ☐ Right, rear (R RH)
- 9 - Bracket**
 - ☐ Left, front (FRT LH)
- 10 - Bracket**
 - ☐ Right, front (FRT RH)
- 11 - Protective cap**
 - ☐ Right, front
- 12 - Protective cap**
 - ☐ Left, front
- 13 - Protective cap**
 - ☐ Right, rear
- 14 - Protective cap**
 - ☐ Left, rear
- 15 - Tube coupling**
 - ☐ Qty. 2
- 16 - Sleeves**
 - ☐ Qty. 2
- 17 - Bolts**
 - ☐ Qty. 4
 - ☐ Countersunk head bolt, M8x60
 - ☐ 2 Nm
- 18 - Counterplates**
 - ☐ Qty. 6
- 19 - Protective films**
 - ☐ Front left and right
 - ☐ Qty. 2
- 20 - Nuts**
 - ☐ Renew after removal
 - ☐ Qty. 12
 - ☐ M8, self-locking, with collar



N66-12295



21 - Spacer rings

- ☐ Qty. 10

22 - Washers

- ☐ Qty. 40
- ☐ M8

23 - Spacers

- ☐ For item 16
- ☐ Qty. 2

24 - Cover

- ☐ Right

25 - Cover

- ☐ Left

26 - Bolts

- ☐ Qty. 12
- ☐ Hexagon socket head bolt, M8x30
- ☐ 26 Nm

27 - Bolts

- ☐ Renew after removal
- ☐ Qty. 12
- ☐ Hexagon socket head bolt, M8x20, self-locking
- ☐ 20 Nm

28 - Protective films

- ☐ Left and right at rear
- ☐ Qty. 2

29 - Bolts

- ☐ Renew after removal
- ☐ Qty. 8
- ☐ Hexagon socket head bolt, M8x30, self-locking
- ☐ 26 Nm

30 - Nuts

- ☐ Renew after removal
- ☐ Qty. 4
- ☐ M8, self-locking

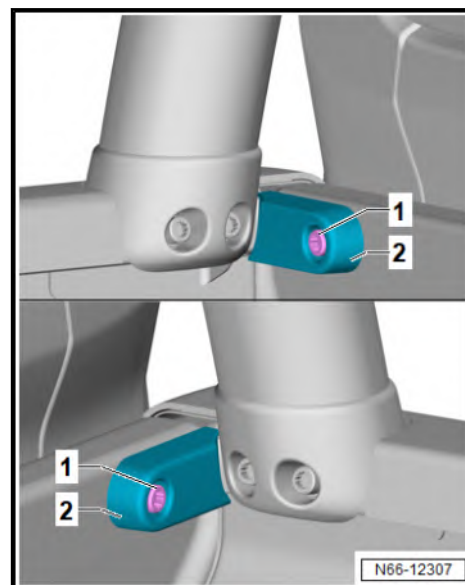
27.2 Removing and installing design double tube bar

Special tools and workshop equipment required

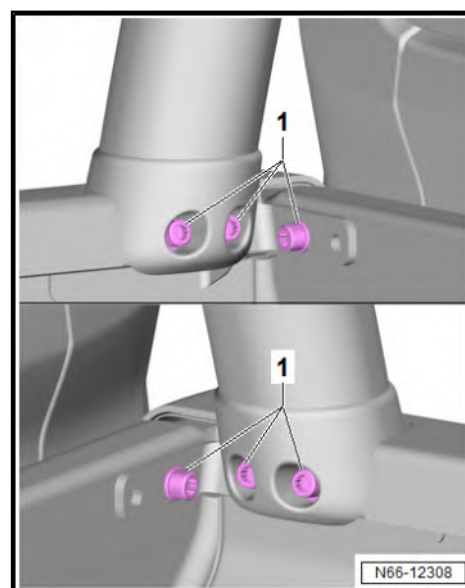
- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



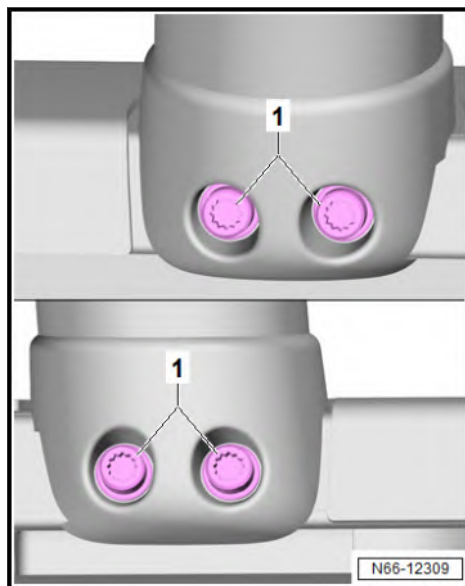
Removing



- Unscrew bolts -1-.
- Remove caps -2-.
- Unscrew bolts -1-.



- Unscrew bolts -1-.



The following work steps require the assistance of a second mechanic.

- Remove design double tube bar.
- Place design double tube bar on a clean surface.

Installing

Install in reverse order of removal, observing the following:

- Clean headboard and side panels in the area of the mounting points, e.g. using professional cloth VAS 6006.
- Note instructions on installing design double tube bar ➤ [page 260](#).

Specified torques

- ◆ Assembly overview - design double tube bar ➤ [page 250](#)

27.3 Installation instructions - design double tube bar

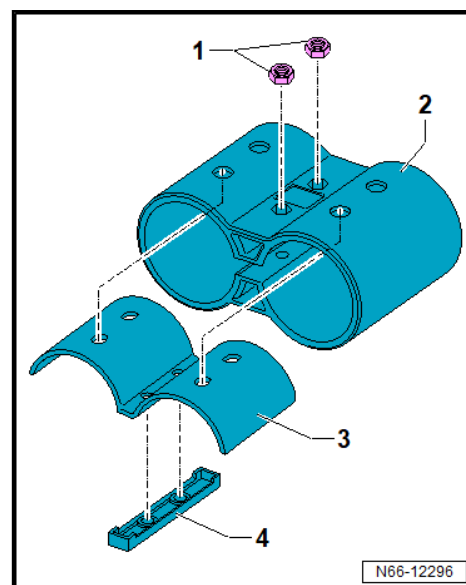
Special tools and workshop equipment required

- ◆ Torque wrench 6-50 Nm -V.A.G 1331-

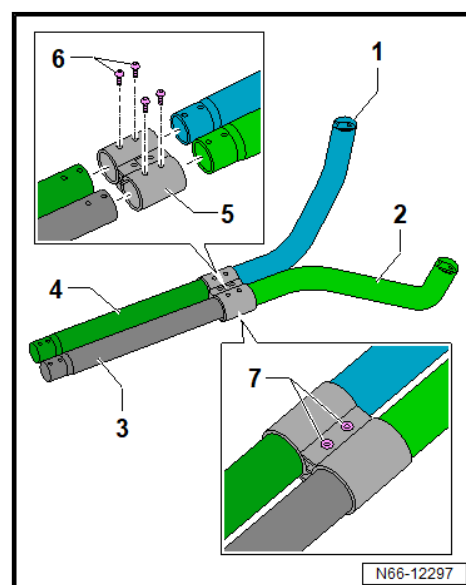


Work instructions

Pre-assembly of design double tube bar



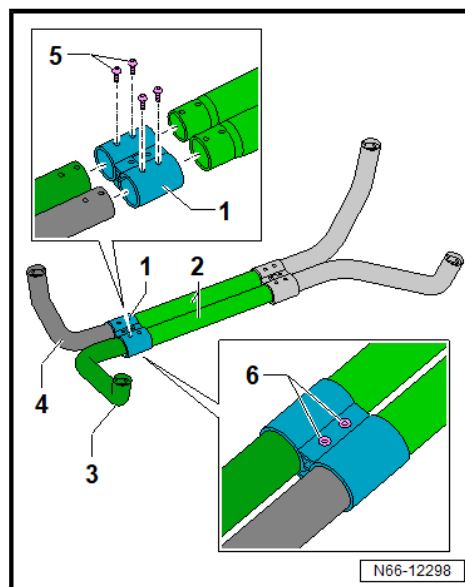
- Insert nuts -1- (➤ [Item 30 \(page 252\)](#)) into sleeve -2- (➤ [Item 16 \(page 251\)](#)).
- Note correct insertion direction (arrow in sleeve).
- Insert tube coupling -3- (➤ [Item 15 \(page 251\)](#)) with spacer -4- (➤ [Item 23 \(page 252\)](#)) into sleeve -2- (➤ [Item 16 \(page 251\)](#)).
- Repeat work steps at second sleeve.
- Insert tube (left, front) -1- (➤ [Item 3 \(page 251\)](#)) into sleeve -5- (➤ [Item 16 \(page 251\)](#)).



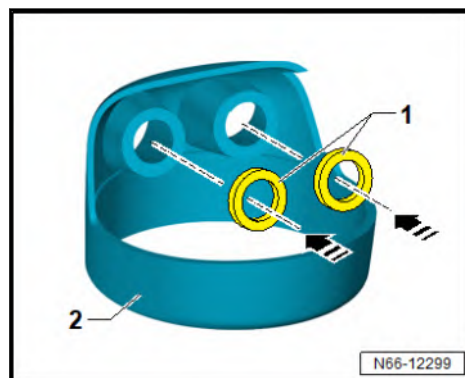
- Insert tube (centre, front) -4- (➤ [Item 1 \(page 251\)](#)) into sleeve -5- (➤ [Item 16 \(page 251\)](#)).
- Screw in bolts -6- (➤ [Item 17 \(page 251\)](#)) by two turns.
- Threaded holes are one above the other
- Insert tube (left, rear) -2- (➤ [Item 5 \(page 251\)](#)) into sleeve -5- (➤ [Item 16 \(page 251\)](#)).



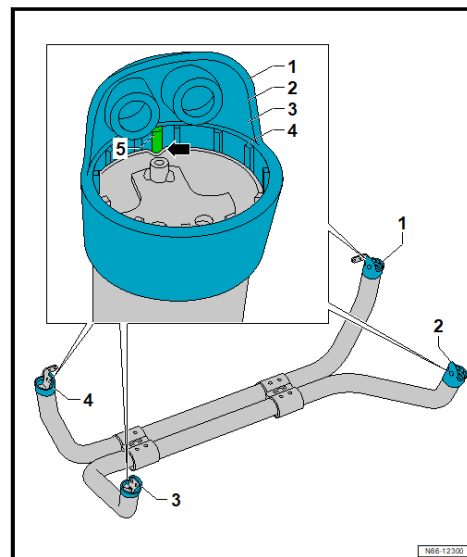
- Insert tube (centre, front) -3- (⇒ [Item 2 \(page 251\)](#)) into sleeve -5- (⇒ [Item 16 \(page 251\)](#)).
- Screw in bolts -6- (⇒ [Item 17 \(page 251\)](#)) by two turns.
- Threaded holes are one above the other
- Tighten bolts -7- (⇒ [Item 17 \(page 251\)](#)) by hand.
- Slide sleeve -1- (⇒ [Item 16 \(page 251\)](#)) onto tubes -2-.



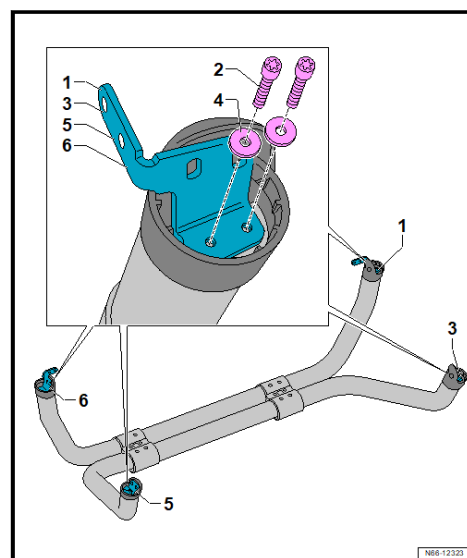
- Insert tube -3- (⇒ [Item 6 \(page 251\)](#)) into sleeve -1- (⇒ [Item 16 \(page 251\)](#)).
- Screw in bolts -6- (⇒ [Item 17 \(page 251\)](#)) by two turns.
- Threaded holes are one above the other
- Insert tube -4- (⇒ [Item 4 \(page 251\)](#)) into sleeve -1-.
- Screw in bolts -5- (⇒ [Item 27 \(page 252\)](#)) by two turns.
- Threaded holes are one above the other
- Tighten bolts -6- (⇒ [Item 17 \(page 251\)](#)) by hand.
- Insert spacer rings -1- (⇒ [Item 21 \(page 251\)](#)) flush into protective cap -2- (⇒ [Item 11 \(page 251\)](#)).



- Perform work step for protective caps ⇒ [Item 12 \(page 251\)](#) ,
⇒ [Item 13 \(page 251\)](#) and ⇒ [Item 14 \(page 251\)](#) .
- Fit protective caps -1, 2, 3 and 4-.



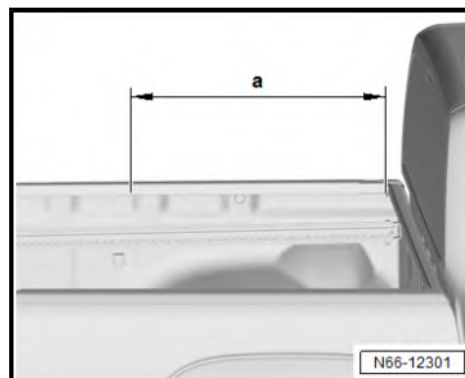
- ◆ -1- ➤ [Item 12 \(page 251\)](#)
- ◆ -2- ➤ [Item 14 \(page 251\)](#)
- ◆ -3- ➤ [Item 13 \(page 251\)](#)
- ◆ -4- ➤ [Item 11 \(page 251\)](#)
- Note installation position. Mounting -5- must be aligned with -arrow- in protective cap.
- Bolt bracket -1- to end of tube using bolts -2- and washers -4-.



- ◆ -1- ➤ [Item 9 \(page 251\)](#)
- ◆ -3- ➤ [Item 7 \(page 251\)](#)
- ◆ -5- ➤ [Item 8 \(page 251\)](#)
- ◆ -6- ➤ [Item 10 \(page 251\)](#)
- ◆ -2- ➤ [Item 29 \(page 252\)](#)
- ◆ -4- ➤ [Item 22 \(page 252\)](#)
- Tighten bolts -2- to specified torque ➤ [Item 29 \(page 252\)](#) .
- Place design double tube bar on a clean surface.

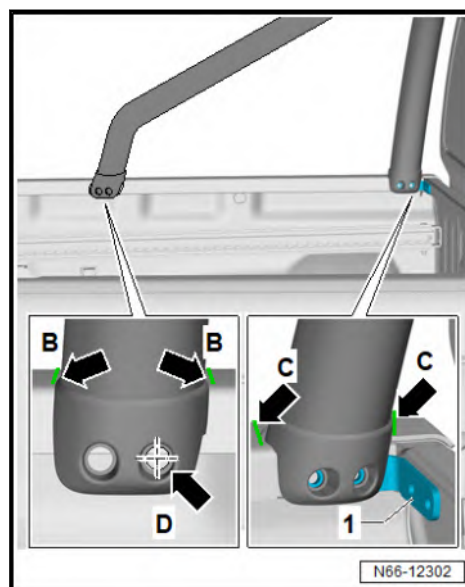


Preparing body



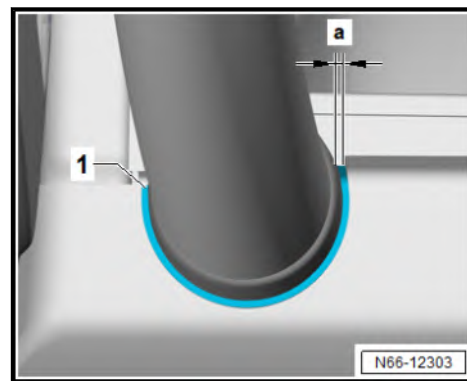
- Clean headboard and side panels in the area of the markings, e.g. using professional cloth VAS 6006.
- Mark vertical markings.
- -a- = 790 mm

The following work steps require the assistance of a second mechanic.

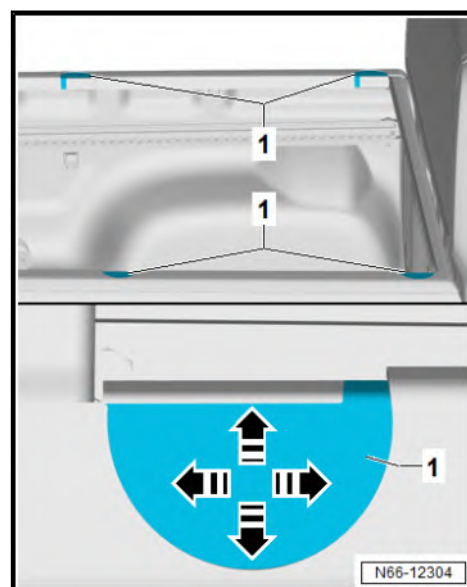


- Fit design double tube bar.
- Push design double tube bar to stop towards front on both sides.
- Align rear tubes with rear marking -arrow D-, as shown in illustration.
- Design double tube bar must rest evenly at all points.
- Mark six holes for each side using 8.5 mm drill bit.
- Mark installation position of protective caps all round with a distance of 2 mm -arrows B and C-.

Marking all round -1-



- -a- = 2 mm
- Remove design double tube bar, and place it on a clean surface.
- Attach protective films -1- (➤ [Item 19 \(page 251\)](#) and ➤ [Item 28 \(page 252\)](#)) centrally between the markings on both sides.

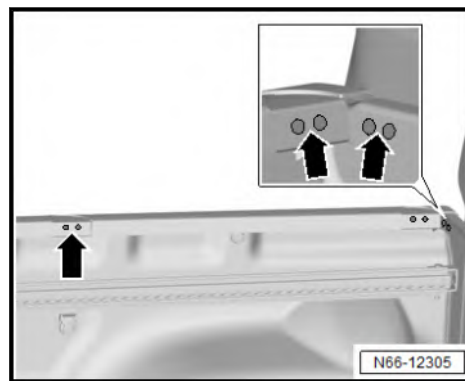


⚠ CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

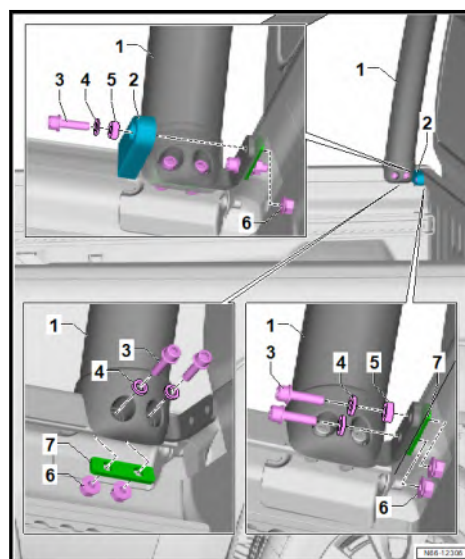
- Wear protective goggles.
- Wear protective gloves!

- Mark the 6 holes per side -arrows- with a punch mark.



- Pre-drill holes using 5 mm drill bit.
- Pre-drill holes using 8.5 mm drill bit.
- Drill holes using 10 mm drill bit.
- Deburr holes both on the inside and outside.
- Pick up/extract metal chips.
- Seal holes with corrosion protection and paint ⇒ Body; General paintwork information; Technical data; General information; Paintwork in Aftersales.
- Observe curing period of corrosion protection.
- Observe curing period of paint seal.

Installing design double tube bar



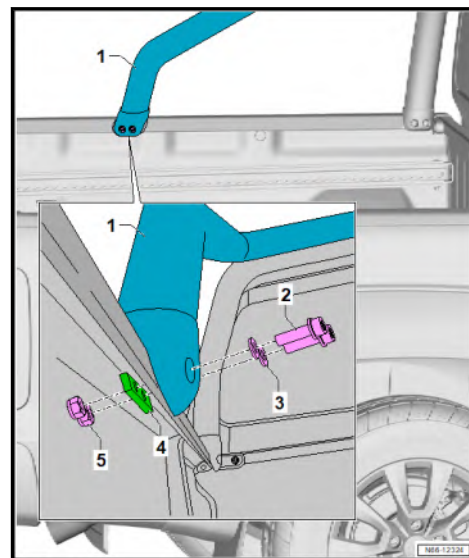
Fasteners at front

The following work step must be performed with the aid of a second mechanic.

- Fit design double tube bar -1-.
- Install design double tube bar -1-, as shown in illustration. Tighten all threaded connections only by hand.
- The covers -3- are installed in a later work step.
- ◆ -2- ⇒ [Item 25 \(page 252\)](#) and ⇒ [Item 24 \(page 252\)](#)
- ◆ -3- ⇒ [Item 26 \(page 252\)](#)
- ◆ -4- ⇒ [Item 22 \(page 252\)](#)

- ◆ -5- ➔ [Item 21 \(page 251\)](#)
- ◆ -6- ➔ [Item 20 \(page 251\)](#)
- ◆ -7- ➔ [Item 18 \(page 251\)](#)
- Tighten bolts (qty. 4) in side panel and headboard to specified torque ➔ [Item 26 \(page 252\)](#) .
- Unscrew the inner bolts again (qty. 2).
- Install covers -3- (➔ [Item 25 \(page 252\)](#)) and ➔ [Item 24 \(page 252\)](#)), as shown in illustration.

Rear fasteners



- Install design double tube bar -1-, as shown in illustration. Tighten all threaded connections only by hand.
- ◆ -2- ➔ [Item 26 \(page 252\)](#)
- ◆ -3- ➔ [Item 22 \(page 252\)](#)
- ◆ -4- ➔ [Item 18 \(page 251\)](#)
- ◆ -5- ➔ [Item 20 \(page 251\)](#)

All fasteners

- Tighten threaded connections to specified torque ➔ [page 250](#) .



28 Load compartment partition

⇒ [o28.1 verview - load compartment partition", page 262](#)

⇒ [a28.2 nd installing locking device", page 262](#)

⇒ [a28.3 nd installing grille", page 263](#)

⇒ [a28.4 nd installing frame", page 264](#)

28.1 Assembly overview - load compartment partition

1 - Frame

- ☐ Removing and installing ⇒ [page 264](#)

2 - Grille

- ☐ Qty. 2
- ☐ Removing and installing ⇒ [page 263](#)

3 - Nuts

- ☐ Qty. 2
- ☐ 20 Nm

4 - Nuts

- ☐ Qty. 4
- ☐ 20 Nm

5 - Bolts

- ☐ Qty. 4

6 - Locking device

- ☐ Qty. 2
- ☐ Removing and installing ⇒ [page 262](#)

7 - Blind rivet nut

- ☐ Qty. 14

8 - Bolts

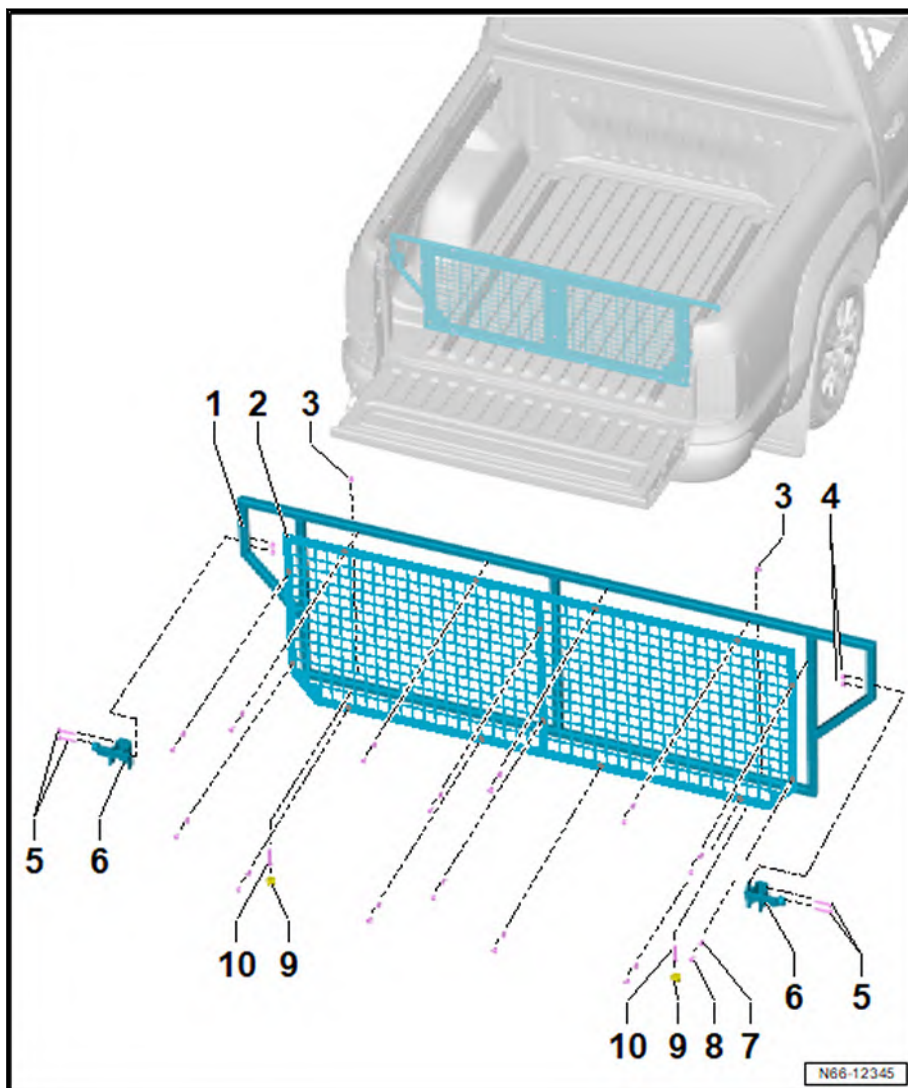
- ☐ Qty. 14
- ☐ 4 Nm

9 - Rollers

- ☐ Qty. 2
- ☐ Removing and installing ⇒ [page 263](#)

10 - Bolts

- ☐ Qty. 2

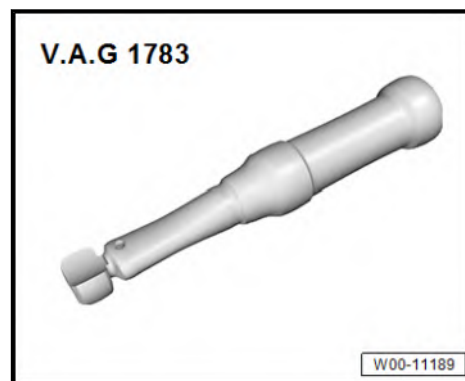


28.2 Removing and installing locking device

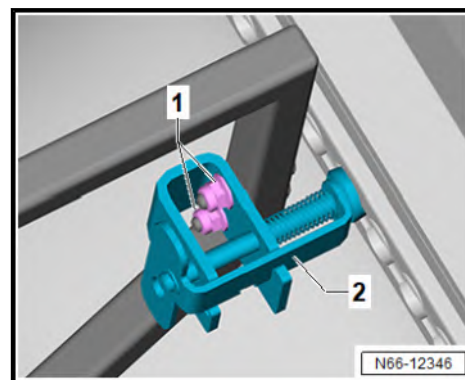
Special tools and workshop equipment required



- ◆ Torque wrench -V.A.G 1783-



Removing



- Unscrew nuts -1-.
- Remove locking device -2-.

Installing

Install in reverse order of removal, observing the following:

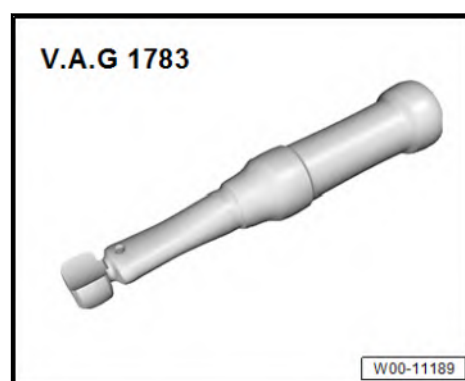
Specified torques

- ◆ Assembly overview - load compartment partition ⇒ [page 262](#)

28.3 Removing and installing grille

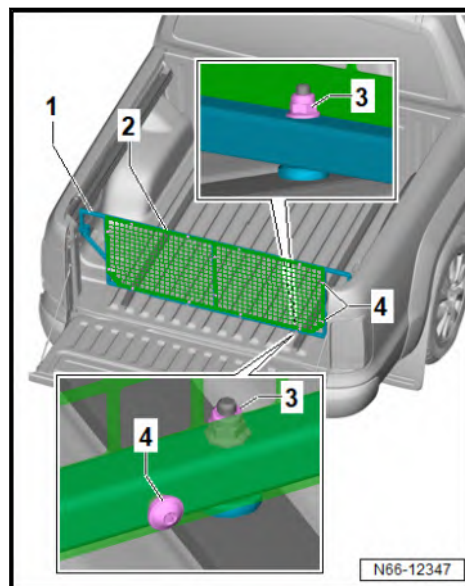
Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1783-





Removing



- Move frame -1- to rearmost position.
- Unscrew bolts -4-.
- Remove grille -2-.

Further work

Removing lower guide.

- Remove frame -1-.
- Unscrew nut -3-. Remove guides.

Installing

Install in reverse order of removal, observing the following:

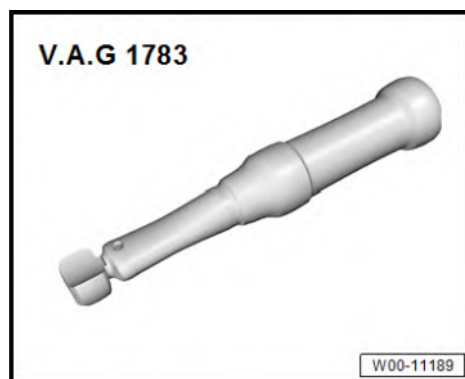
Specified torques

- ◆ Assembly overview - load compartment partition ➔ [page 262](#)

28.4 Removing and installing frame

Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1783-



Removing

- Remove frame.
- Remove locking device ➔ [page 262](#)
- Remove grille ➔ [page 263](#).



- Unscrew bolts ➔ [Item 10 \(page 262\)](#) .

Specified torques

- ◆ Assembly overview - load compartment partition ➔ [page 262](#)



29 Loading edge protection

⇒ [o29.1 verview - loading edge protection", page 266](#)

⇒ [a29.2 nd installing loading edge protection", page 267](#)

29.1 Assembly overview - loading edge protection

1 - Loading edge protection

- ☐ Left
- ☐ Removing and installing ⇒ [page 267](#)

2 - Loading edge protection

- ☐ Rear
- ☐ Removing and installing ⇒ [page 270](#)

3 - Loading edge protection

- ☐ Right
- ☐ Removing and installing ⇒ [page 269](#)

4 - Loading edge protection

- ☐ Left
- ☐ Vehicles with short design double tube bar
- ☐ Removing and installing ⇒ [page 267](#)

5 - Loading edge protection

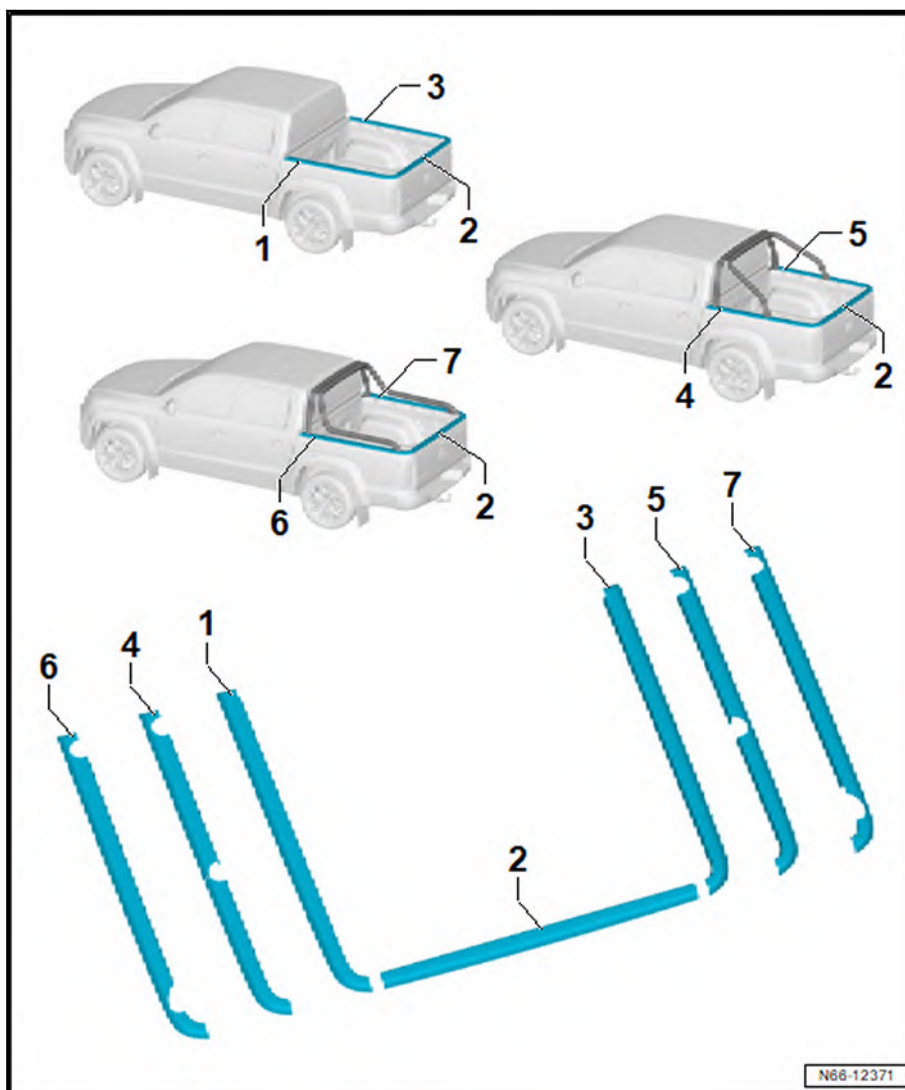
- ☐ Right
- ☐ Vehicles with short design double tube bar
- ☐ Removing and installing ⇒ [page 269](#)

6 - Loading edge protection

- ☐ Left
- ☐ Vehicles with long design double tube bar
- ☐ Removing and installing ⇒ [page 267](#)

7 - Loading edge protection

- ☐ Right
- ☐ Vehicles with long design double tube bar
- ☐ Removing and installing ⇒ [page 269](#)



29.2 Removing and installing loading edge protection

⇒ [a29.2.1 nd installing loading edge protection, left", page 267](#)

⇒ [a29.2.2 nd installing loading edge protection, right", page 269](#)

⇒ [a29.2.3 nd installing loading edge protection, rear", page 270](#)

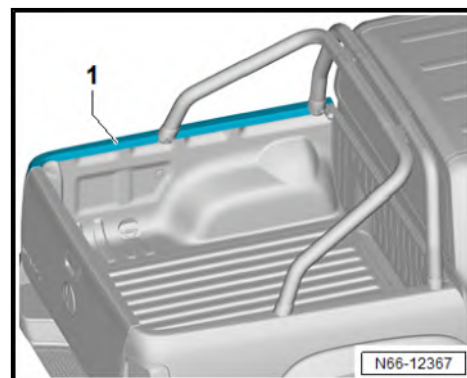
29.2.1 Removing and installing loading edge protection, left

Special tools and workshop equipment required

- ◆ Hot-air blower -V.A.G 1416-



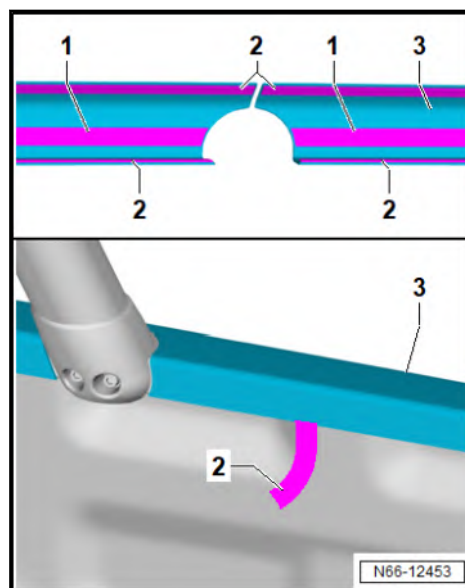
Removing



- Heat up loading edge protection -1- using hot air blower -V.A.G 1416-.
- Carefully pull off loading edge protection -1-.



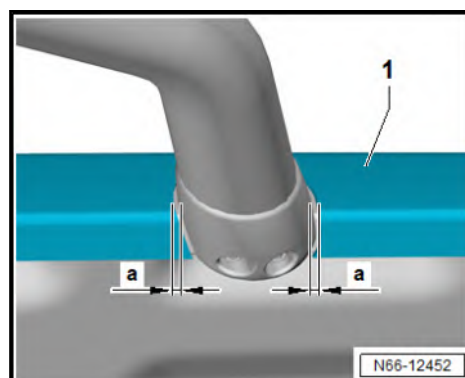
Installing



Install in reverse order of removal, observing the following:

- Heat up bonding surfaces using hot air blower -V.A.G 1416-.
- Remove adhesive residue.
- Clean bonding surfaces on body using cleaning solution and a microfibre cloth.
- Observe flash-off time of cleaning solution.
- Remove centre protective film -1-.
- Detach the other protective films -2- only at the edge.
- Do not touch adhesive surfaces once backing film has been removed.
- Fit loading edge protection -3-. Detach the remaining adhesive strips using pull-off aid -2-.
- Press on loading edge protection -3- evenly.

Vehicles with styling bar



- Align loading edge protection -1-.
- Dimension -a-: approx. 5 mm

Continued for all vehicles

- Press on loading edge protection -1- evenly.

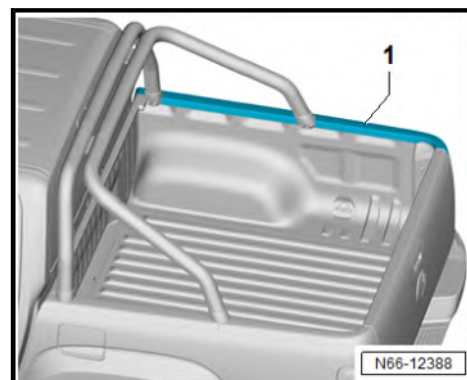
29.2.2 Removing and installing loading edge protection, right

Special tools and workshop equipment required

- ◆ Hot-air blower -V.A.G 1416-

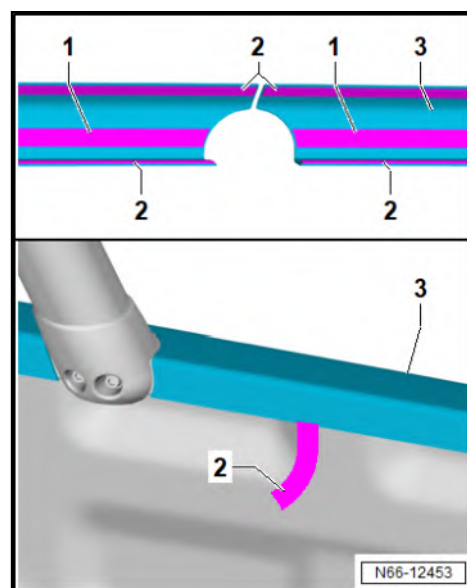


Removing



- Heat up loading edge protection -1- using hot air blower -V.A.G 1416-.
- Carefully pull off loading edge protection -1-.

Installing



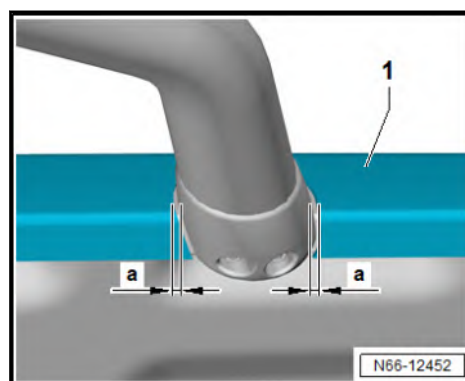
Install in reverse order of removal, observing the following:

- Heat up bonding surfaces using hot air blower -V.A.G 1416-.



- Remove adhesive residue.
- Clean bonding surfaces on body using cleaning solution and a microfibre cloth.
- Observe flash-off time of cleaning solution.
- Remove centre protective film -1-.
- Detach the other protective films -2- only at the edge.
- Do not touch adhesive surfaces once backing film has been removed.
- Fit loading edge protection -3-. Detach the remaining adhesive strips using pull-off aid -2-.
- Press on loading edge protection -3- evenly.

Vehicles with styling bar



- Align loading edge protection -1-.
- Dimension -a-: approx. 5 mm

Continued for all vehicles

- Press on loading edge protection -1- evenly.

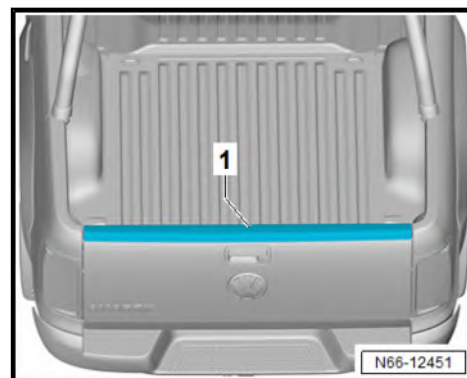
29.2.3 Removing and installing loading edge protection, rear

Special tools and workshop equipment required

- ◆ Hot-air blower -V.A.G 1416-

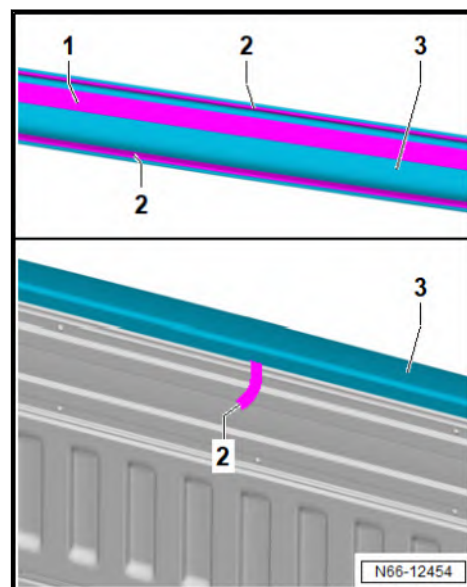


Removing



- Heat up loading edge protection -1- using hot air blower -V.A.G 1416-.
- Carefully pull off loading edge protection -1-.

Installing



Install in reverse order of removal, observing the following:

- Heat up bonding surfaces using hot air blower -V.A.G 1416-.
- Remove adhesive residue.
- Clean bonding surfaces on body using cleaning solution and a microfibre cloth.
- Observe flash-off time of cleaning solution.
- Remove centre protective film -1-.
- Detach the other protective films -2- only at the edge.
- Do not touch adhesive surfaces once backing film has been removed.
- Fit loading edge protection -3-. Detach the remaining adhesive strips using pull-off aid -2-.
- Press on loading edge protection -3- evenly.



30 Loading edge protection

⇒ [a30.1 nd installing loading edge protection", page 272](#)

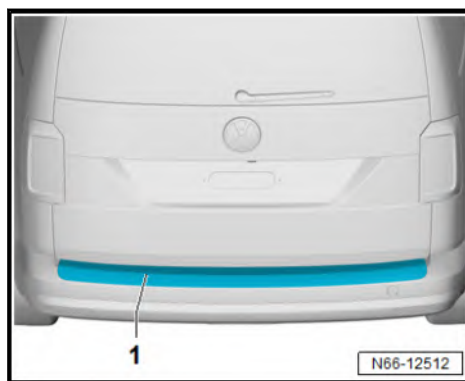
30.1 Removing and installing loading edge protection

Special tools and workshop equipment required

- ◆ Hot-air blower -V.A.G 1416-



Removing



- Heat up loading edge protection -1- using hot air blower -V.A.G 1416-.
- Pull off loading edge protection -1-.

Installing

Install in reverse order of removal, observing the following:

- Heat up bonding surfaces using hot air blower -V.A.G 1416-.
- Remove adhesive residue.
- Clean bonding surfaces on body using cleaning solution and a microfibre cloth.
- Observe flash-off time of cleaning solution.
- Do not touch adhesive surfaces once backing film has been removed.
- Centre loading edge protection -1-. To do this, mark ends of loading edge protection on bumper.
- Press on loading edge protection -1- evenly.

31 Load compartment cover, aluminium

⇒ [a31.1 verview - load compartment cover", page 273](#)

⇒ [a31.2 nd installing load compartment cover", page 273](#)

⇒ [a31.3 nd installing gas strut", page 275](#)

⇒ [a31.4 nd installing hinge for load compartment cover", page 276](#)

31.1 Assembly overview - load compartment cover

1 - Load compartment cover

- ☐ Removing and installing ⇒ [page 273](#)

2 - Gas strut

- ☐ Qty. 2
- ☐ Removing and installing ⇒ [page 275](#)

3 - Retaining clips

- ☐ Qty. 2

4 - Bolts

- ☐ Qty. 8
- ☐ 20 Nm

5 - Washers

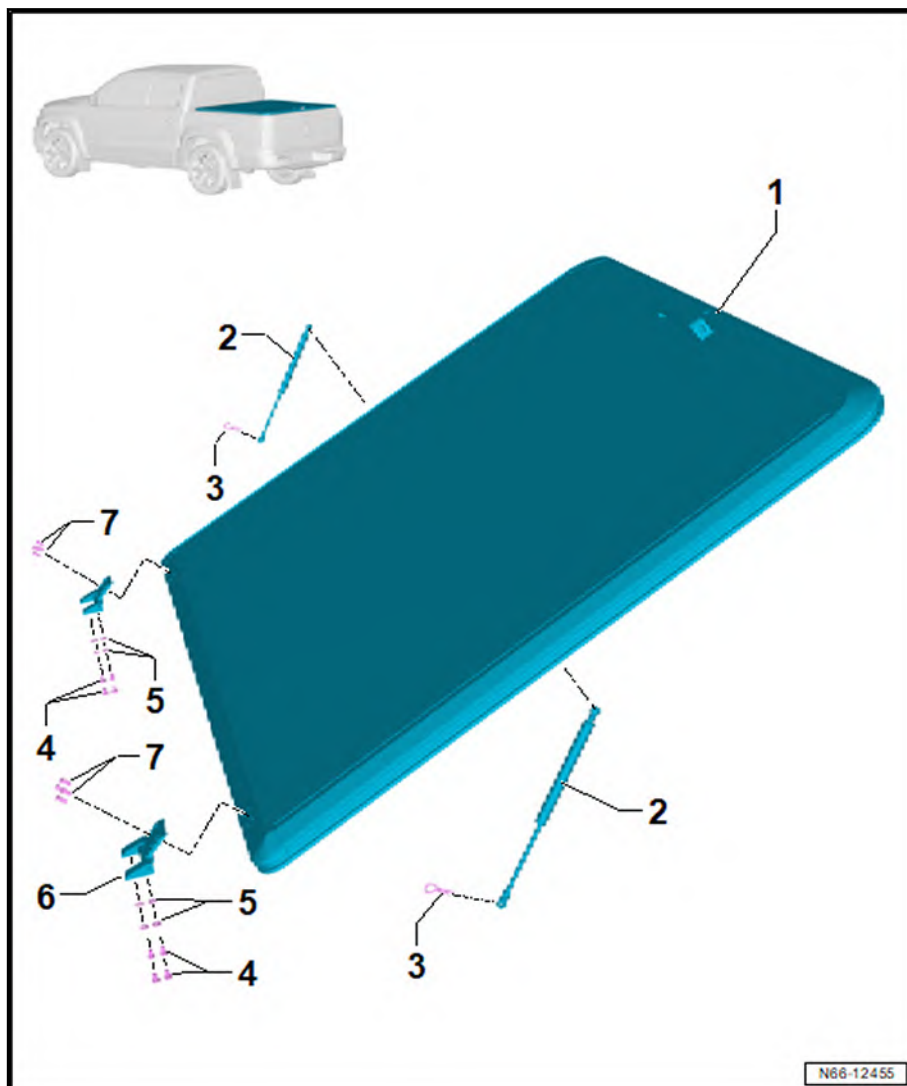
- ☐ Qty. 8

6 - Hinges

- ☐ Qty. 2
- ☐ Removing and installing ⇒ [page 276](#)

7 - Bolts

- ☐ Qty. 8
- ☐ 20 Nm



31.2 Removing and installing load compartment cover

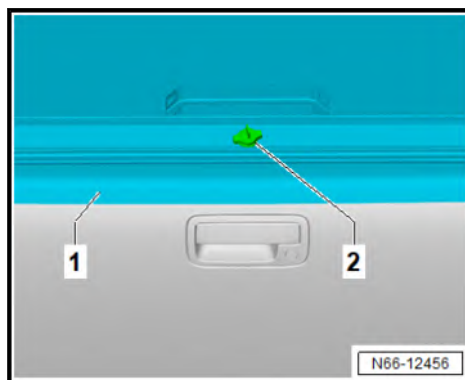
Special tools and workshop equipment required



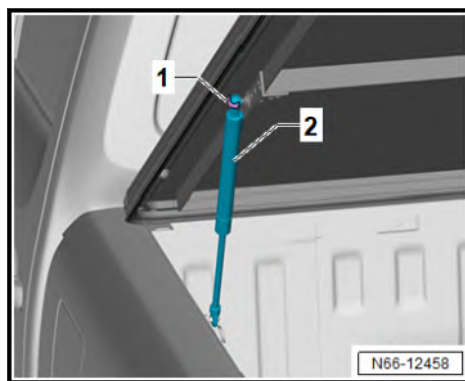
- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



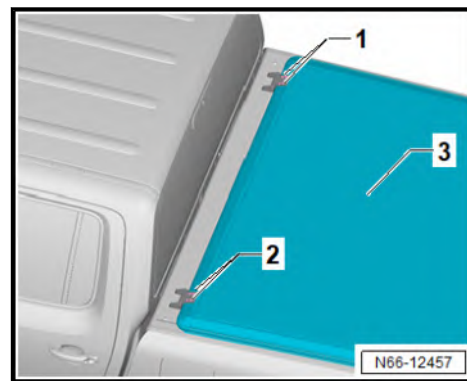
Removing



- Unlock lock cylinder -2-.
- Open load compartment cover -1-.
- Release clips -1- on left and right.



- The following work steps require the assistance of a second mechanic.
- Release gas strut -2- on left and right. When doing this, have a second mechanic hold the load compartment cover.
- Unscrew bolts -1-.



- Unscrew bolts -2-.
- Remove load compartment cover -3- together with a second mechanic.

Installing

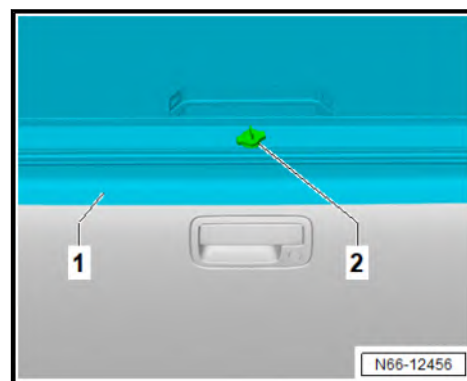
Install in reverse order of removal, observing the following:

Specified torques

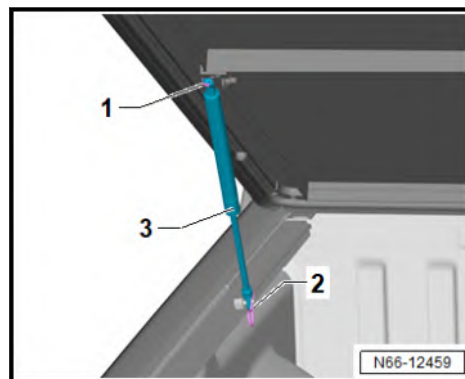
♦ ⇒ [o31.1 verview - load compartment cover](#)", page 273

31.3 Removing and installing gas strut

Removing



- Unlock lock cylinder -2-.
- Open load compartment cover -1-.
- The following work steps require the assistance of a second mechanic.
- Have a second mechanic secure load compartment cover -1-.
- Release clip -1-.



- Remove retaining clip -2-.
- Remove gas strut -3-. When doing this, have a second mechanic secure the load compartment cover.

Installing

Installation is carried out in reverse order.

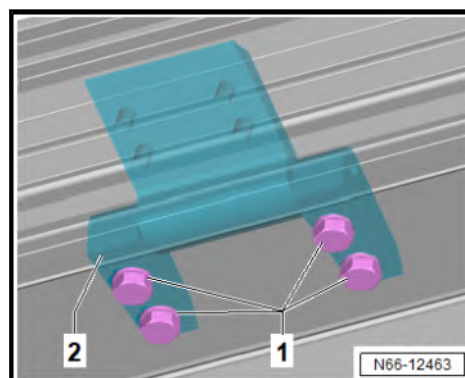
31.4 Removing and installing hinge for load compartment cover

Special tools and workshop equipment required

- ♦ Torque wrench 6-50 Nm -V.A.G 1331-



Removing



- Remove load compartment cover ➤ [page 273](#) .
- Unscrew bolts -1-.
- Pull off hinge -2-.

Installing

Install in reverse order of removal, observing the following:



Specified torques

- ◆ [⇒ o31.1 verview - load compartment cover", page 273](#)



32 Protective grille for rear window

⇒ [o32.1 verview - protective grille for rear window", page 278](#)

⇒ [a32.2 nd installing protective grille for rear window", page 278](#)

⇒ [a32.3 nd installing retaining bracket", page 279](#)

32.1 Assembly overview - protective grille for rear window

1 - Protective grille

- ☐ Removing and installing ⇒ [page 278](#)

2 - Bolts

- ☐ Qty. 6
- ☐ 20 Nm

3 - Washers

- ☐ Qty. 18

4 - Bolts

- ☐ Qty. 4
- ☐ 20 Nm

5 - Bolts

- ☐ Qty. 6
- ☐ 20 Nm

6 - Retaining bracket

- ☐ Right
- ☐ Removing and installing ⇒ [page 279](#)

7 - Nuts

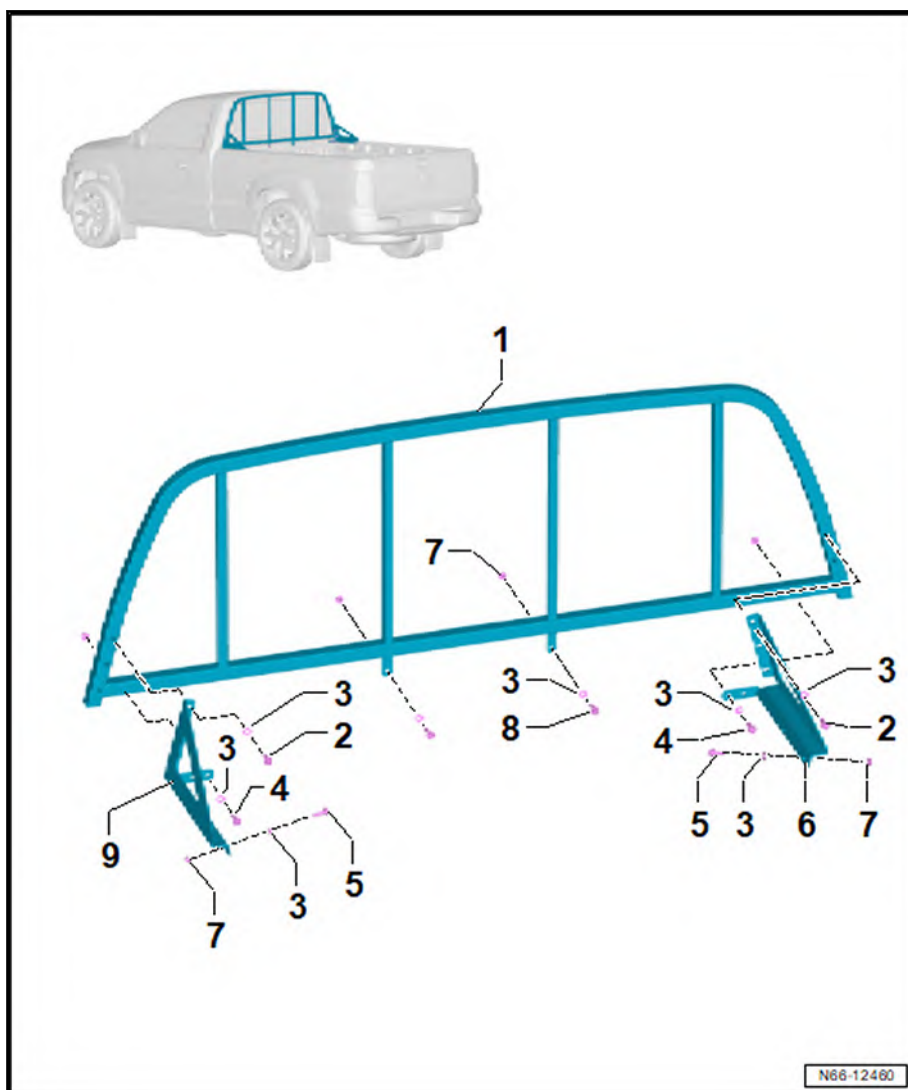
- ☐ Qty. 18

8 - Bolts

- ☐ Qty. 2
- ☐ 20 Nm

9 - Retaining bracket

- ☐ Left
- ☐ Removing and installing ⇒ [page 279](#)



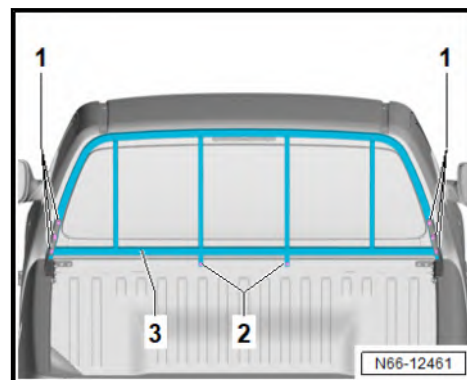
32.2 Removing and installing protective grille for rear window

Special tools and workshop equipment required

- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



Removing



- Unscrew bolts -1-. Counterhold nuts while doing this.
- The following work steps require the assistance of a second mechanic.
- Have a second mechanic secure protective grille -3-.
- Unscrew bolts -2-. Counterhold nuts while doing this.
- Remove protective grille -3- together with a second mechanic.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [o32.1 verview - protective grille for rear window", page 278](#)

32.3 Removing and installing retaining bracket

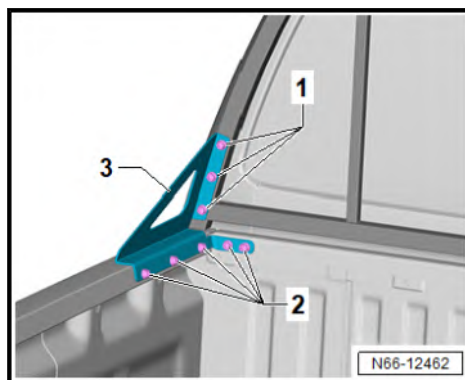
Special tools and workshop equipment required



- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



Removing



- Unscrew bolts -1-. Counterhold nuts while doing this.
- Unscrew bolts -2-. Counterhold nuts while doing this.
- Remove retaining bracket -3-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ [⇒ o32.1 verview - protective grille for rear window", page 278](#)



33 Rail system for load bed

⇒ [o33.1 verview - rail system", page 281](#)

⇒ [a33.2 nd installing cross bars", page 283](#)

⇒ [a33.3 nd installing retaining brackets for cross bars", page 283](#)

⇒ [a33.4 nd installing guide rails", page 284](#)

33.1 Assembly overview - rail system



1 - Cross bars

- ☐ Qty. 2
- ☐ Removing and installing ⇒ [page 283](#)

2 - Retaining bracket

- ☐ Qty. 4
- ☐ Removing and installing ⇒ [page 283](#)

3 - Washers

- ☐ Qty. 8

4 - Nuts

- ☐ Qty. 8

5 - Nuts

- ☐ Qty. 8
- ☐ 8 Nm

6 - Washers

- ☐ Qty. 8

7 - Guide rails

- ☐ Qty. 2
- ☐ On side
- ☐ Removing and installing ⇒ [page 286](#)

8 - Bolts

- ☐ Number depends on equipment level
- ☐ 40 Nm

9 - Covers

- ☐ Qty. 2
- ☐ Removing and installing ⇒ [page 284](#)

10 - Bolts

- ☐ Qty. 8

11 - Nuts

- ☐ Number depends on equipment level

12 - Washers

- ☐ Number depends on equipment level

13 - Retaining bracket

- ☐ Number depends on equipment level

14 - Guide rails

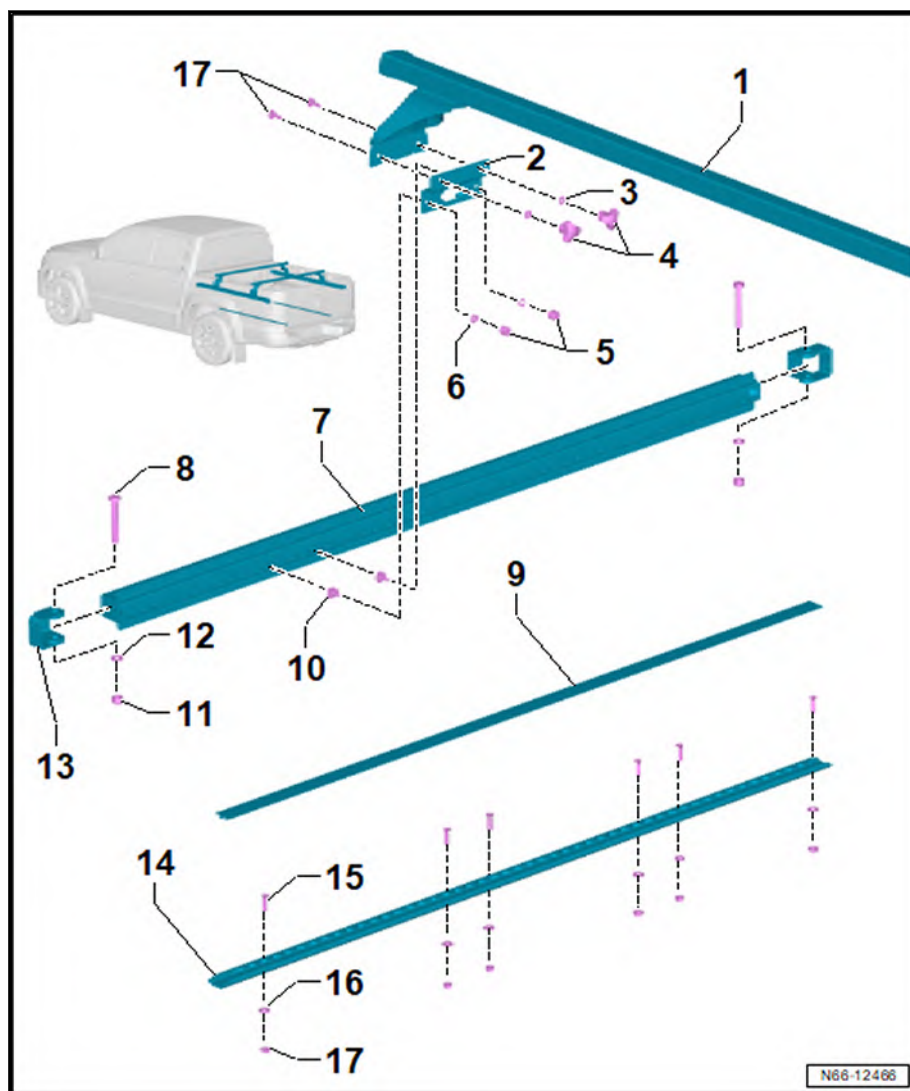
- ☐ Qty. 2
- ☐ Load bed
- ☐ Removing and installing ⇒ [page 284](#)

15 - Bolts

- ☐ Qty. 12
- ☐ 7 Nm

16 - Washers

- ☐ Qty. 6



17 - Nuts

□ Qty. 6

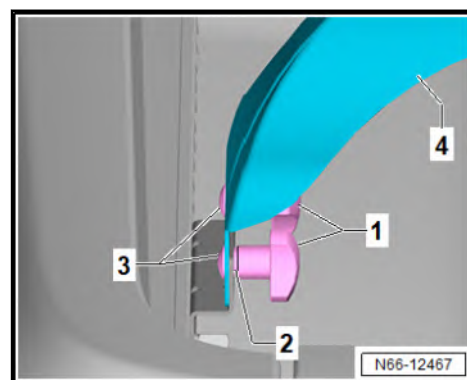
33.2 Removing and installing cross bars

Special tools and workshop equipment required

◆ Torque wrench 6-50 Nm -V.A.G 1331-



Removing



- Unscrew nuts -1- on left and right.
- Remove bolts -3- and washers -2- on left and right.
- Detach cross bars -4-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

◆ ⇒ [o33.1 verview - rail system", page 281](#)

33.3 Removing and installing retaining brackets for cross bars

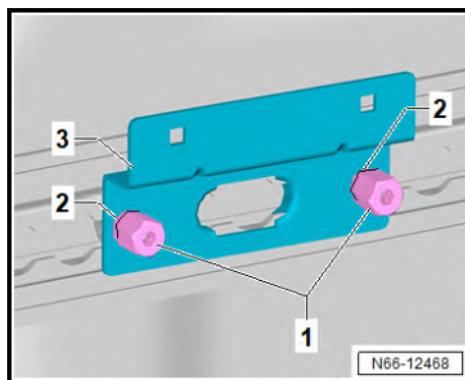
Special tools and workshop equipment required



- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



Removing



- Removing cross bars ⇒ [page 283](#) .
- Unscrew nuts -1-.
- Remove retaining bracket -3- and washers -2-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [o33.1 verview - rail system", page 281](#)

33.4 Removing and installing guide rails

⇒ [a33.4.1 nd installing guide rails, load bed", page 284](#)

⇒ [a33.4.2 nd installing guide rails, side", page 286](#)

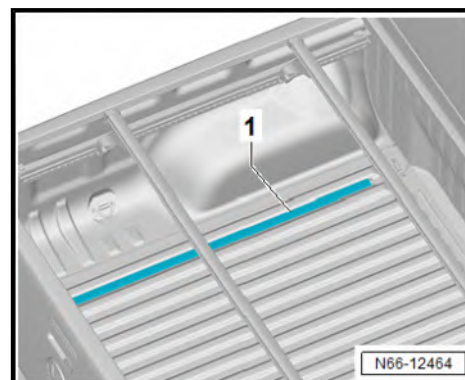
33.4.1 Removing and installing guide rails, load bed

Special tools and workshop equipment required

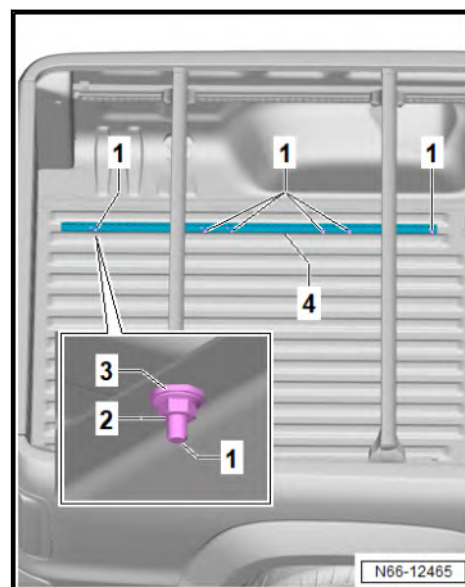
- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



Removing



- Unclip cover -1-.
- Unscrew bolts -1-. Have a second mechanic counterhold tool nuts -2- when doing this.



- Remove nuts -2- and washers -3-.
- Remove guide rail -4-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➔ [o33.1 verview - rail system", page 281](#)



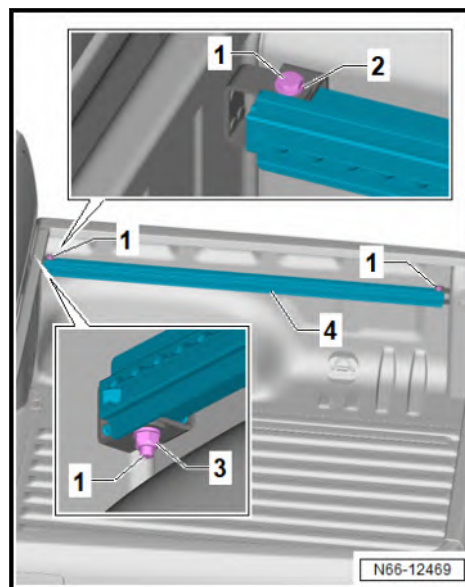
33.4.2 Removing and installing guide rails, side

Special tools and workshop equipment required

- ◆ Torque wrench, 40 to 200 Nm -V.A.G 1332A-



Removing



- Unscrew bolts -1-. Counterhold nuts -3- when doing this.
- Remove nuts -3- and washers -2-.
- Remove guide rail -4-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [o33.1 verview - rail system", page 281](#)



34 Side bar

⇒ [o34.1 overview - side bars", page 287](#)

⇒ [a34.2 nd installing side bar", page 289](#)

⇒ [a34.3 nd installing bracket for side bar", page 291](#)

⇒ [a34.4 nd installing step panel", page 292](#)

34.1 Assembly overview - side bars



1 - Sill panel bars

- ☐ Qty. 2
- ☐ With step panel
- ☐ With lighting, depending on equipment level
- ☐ Removing and installing ⇒ [page 290](#)

2 - Bolts

- ☐ Number depends on equipment level
- ☐ 24 Nm

3 - Protective caps

- ☐ Depending on equipment
- ☐ Qty. 3

4 - Nuts

- ☐ Qty. 6
- ☐ 35 Nm

5 - Washers

- ☐ Qty. 6

6 - Bracket

- ☐ Front
- ☐ Removing and installing ⇒ [page 291](#)

7 - Bracket

- ☐ Centre
- ☐ Removing and installing ⇒ [page 291](#)

8 - Bracket

- ☐ Rear
- ☐ Removing and installing ⇒ [page 291](#)

9 - Step panel

- ☐ Removing and installing ⇒ [page 292](#)

10 - Step panel

- ☐ Removing and installing ⇒ [page 292](#)

11 - Washers

- ☐ Qty. 6

12 - Nuts

- ☐ Qty. 6

13 - Sill panel bars

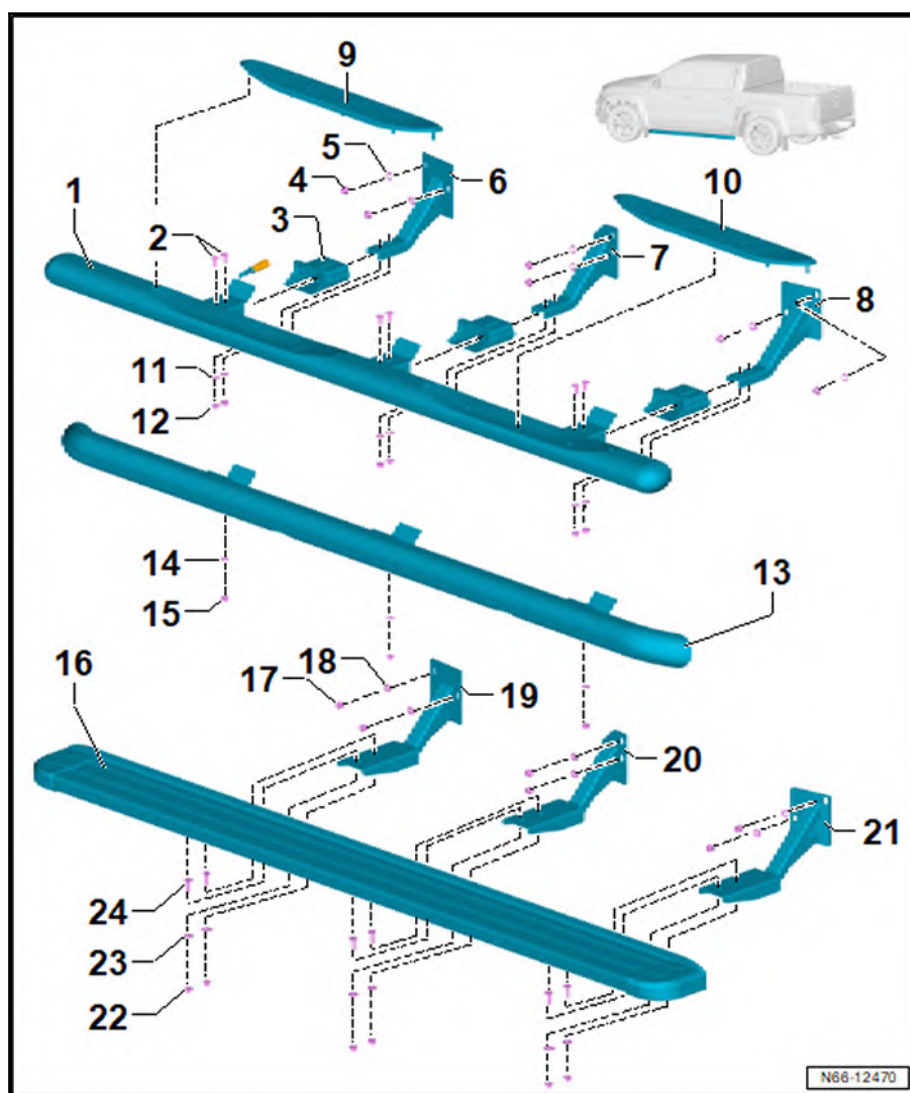
- ☐ Qty. 2
- ☐ Without step panel
- ☐ Removing and installing ⇒ [page 289](#)

14 - Washers

- ☐ Qty. 3

15 - Nuts

- ☐ Qty. 3
- ☐ 24 Nm





16 - Running boards

- ☐ Qty. 2
- ☐ Removing and installing ⇒ [page 289](#)

17 - Nuts

- ☐ Qty. 6
- ☐ 35 Nm

18 - Washers

- ☐ Qty. 6

19 - Bracket

- ☐ Front
- ☐ Removing and installing ⇒ [page 291](#)

20 - Bracket

- ☐ Centre
- ☐ Removing and installing ⇒ [page 291](#)

21 - Bracket

- ☐ Rear
- ☐ Removing and installing ⇒ [page 291](#)

22 - Nuts

- ☐ Qty. 6

23 - Washers

- ☐ Qty. 6

24 - Bolts

- ☐ Qty. 6
- ☐ 24 Nm

34.2 Removing and installing side bar

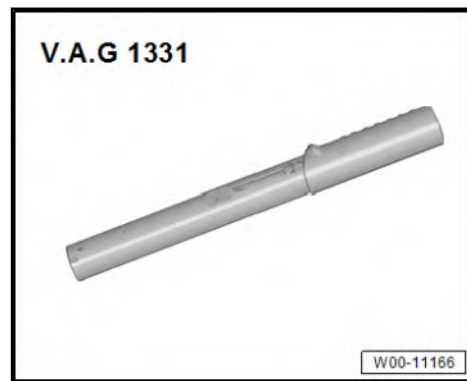
⇒ [a34.2.1 nd installing side bar", page 289](#)

⇒ [a34.2.2 nd installing bumper cover, with step panel", page 290](#)

34.2.1 Removing and installing side bar

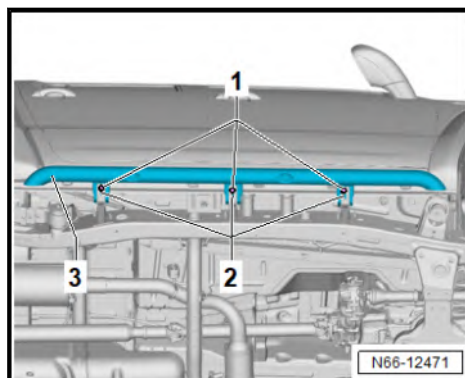
Special tools and workshop equipment required

- ◆ Torque wrench 6-50 Nm -V.A.G 1331-





Removing



- Unscrew nuts -1-. Remove washers.
- Remove side bar -2-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ [⇒ o34.1 verview - side bars", page 287](#)

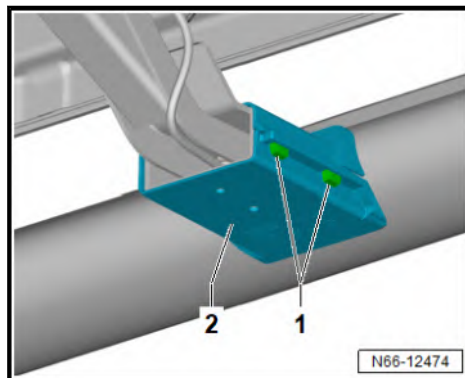
34.2.2 Removing and installing bumper cover, with step panel

Special tools and workshop equipment required

- ♦ Torque wrench 6-50 Nm -V.A.G 1331-



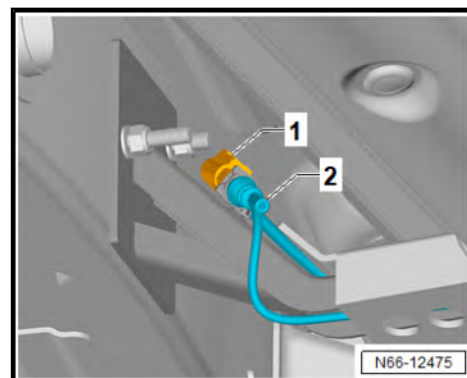
Removing



Vehicles with entry lighting

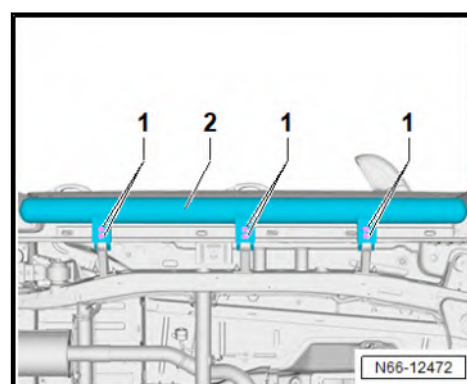
- Release locking lugs -1-.
- Detach cover -2-.

- Disconnect connector -1-.



- Move aside wiring harness -2-.

Continued for all vehicles



- Unscrew nuts -1-. Remove washers.
- Remove side bar with step panel -2-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ [⇒ o34.1 verview - side bars", page 287](#)

34.3 Removing and installing bracket for side bar

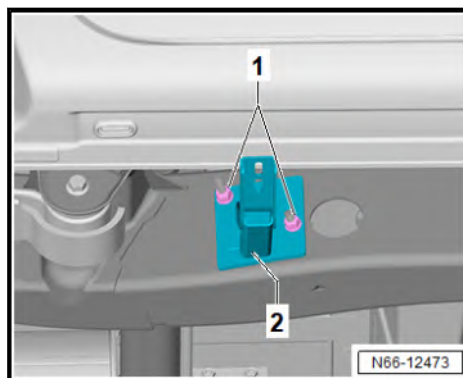
Special tools and workshop equipment required

- ♦ Torque wrench 6-50 Nm -V.A.G 1331-





Removing



- Remove side bar ➔ [page 289](#) .
- Unscrew nuts -1-.
- Remove bracket for side bar -2-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ➔ [o34.1 verview - side bars", page 287](#)

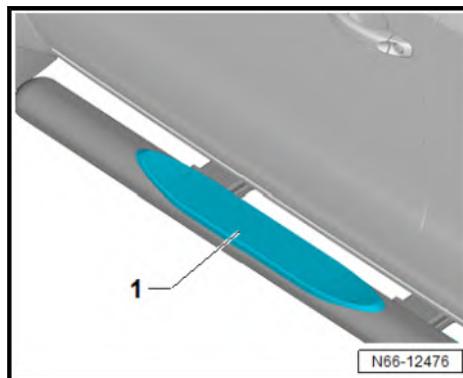
34.4 Removing and installing step panel

Special tools and workshop equipment required

- ♦ Scraper set -VAS 6845-



Removing



- Unclip step panel -1- using an appropriate wedge from scraper set -VAS 6845-, and remove it.



Installing

Installation is carried out in reverse order.



35 Fuel tank skid plate

⇒ [o35.1 verview - fuel tank skid plate", page 294](#)

⇒ [a35.2 nd installing underbody guard for fuel tank", page 294](#)

35.1 Assembly overview - fuel tank skid plate

1 - Mounting plate

2 - Bracket

- ☐ Front
- ☐ Left

3 - Washers

- ☐ Qty. 6

4 - Bolts

- ☐ Qty. 4
- ☐ 24 Nm

5 - Nuts

- ☐ Renew after removal
- ☐ Qty. 4
- ☐ 24 Nm

6 - Skid plate

- ☐ Removing and installing ⇒ [page 294](#)

7 - Bracket

- ☐ Rear

8 - Nut

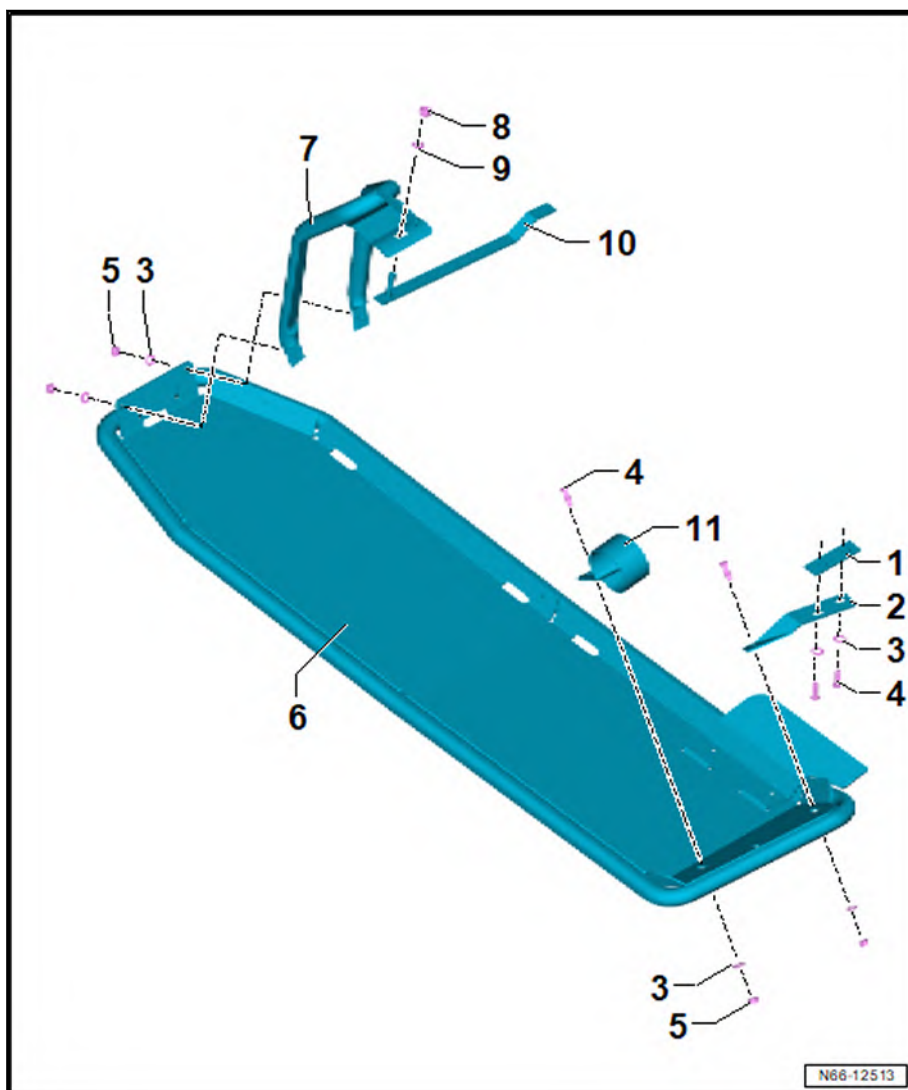
- ☐ Renew after removal
- ☐ 40 Nm

9 - Washer

10 - Bracket

11 - Bracket

- ☐ Front
- ☐ Right



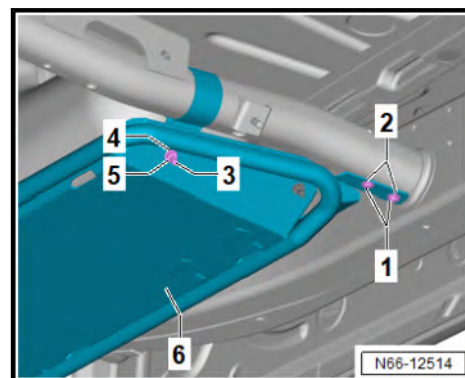
35.2 Removing and installing underbody guard for fuel tank

Special tools and workshop equipment required

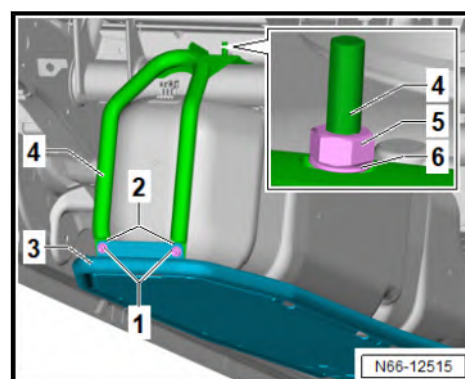
- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



Removing



- Unscrew bolts -1-. Remove washers -2-.
- The following work steps require the assistance of a second mechanic.
- Unscrew nut -5-. Remove washer -4- when doing this.
- Unscrew bolt -3-. Have a second mechanic secure skid plate -6-.
- Unscrew nuts -1-. Remove washer -2- when doing this.



- Remove skid plate -3- with the aid of a second mechanic.
- Unscrew nut -5-. Remove washer -6- when doing this.
- Remove retainer -4-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➔ [o35.1 verview - fuel tank skid plate", page 294](#)

36 Bumper trim

⇒ [o36.1 verview - bumper trim", page 296](#)

⇒ [a36.2 nd installing bumper trim", page 296](#)

36.1 Assembly overview - bumper trim

1 - Mounting plate

- ☐ Removing and installing ⇒ [page 296](#)

2 - Spreader rivets

- ☐ Qty. 15

3 - Bolts

- ☐ Qty. 11
- ☐ 2 Nm

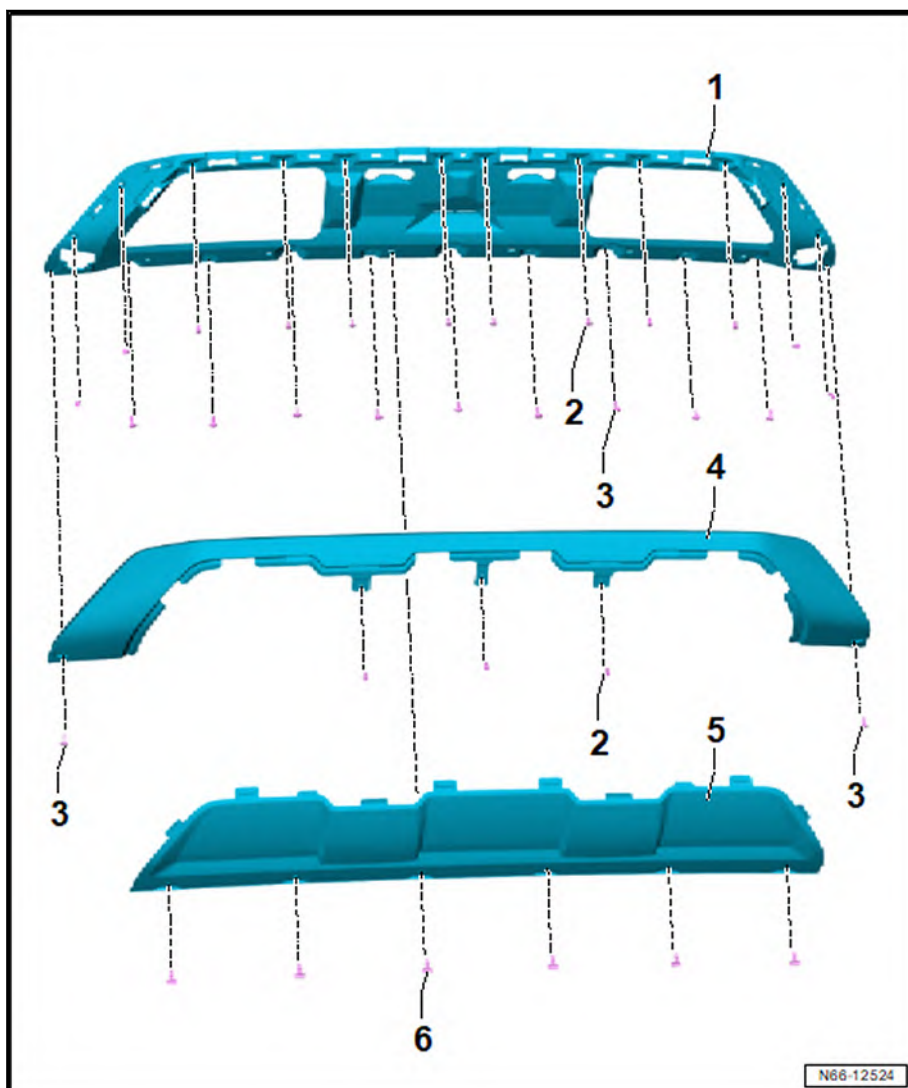
4 - Frame

- ☐ Removing and installing ⇒ [page 296](#)

5 - Flap

- ☐ Removing and installing ⇒ [page 296](#)

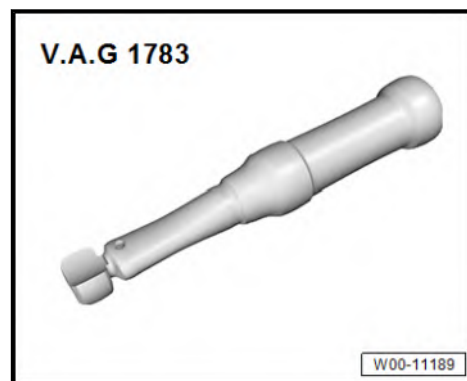
6 - Quick-release fasteners



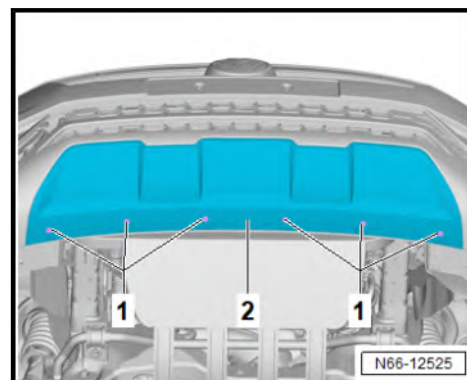
36.2 Removing and installing bumper trim

Special tools and workshop equipment required

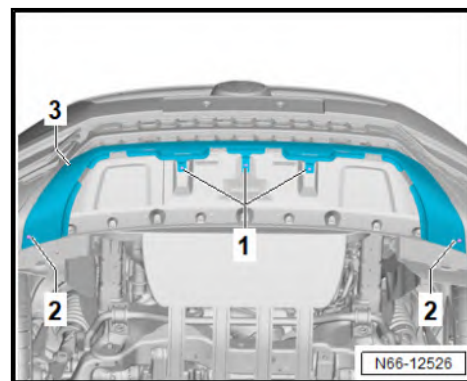
- ◆ Torque wrench -V.A.G 1783-



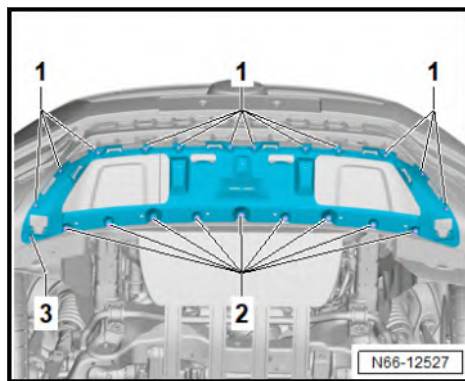
Removing



- Turn quick-release fasteners -1- 90° in anti-clockwise direction.
- Unclip flap -2-. Remove flap when doing this.
- Release spreader rivets -1-.



- Unscrew bolts -2-.
- Unclip frame -3-. Remove frame when doing this.
- Release spreader rivets -1-.



- Unscrew bolts -2-.
- Unclip carrier section -3-. Remove carrier section when doing this.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ⇒ [o36.1 verview - bumper trim", page 296](#)

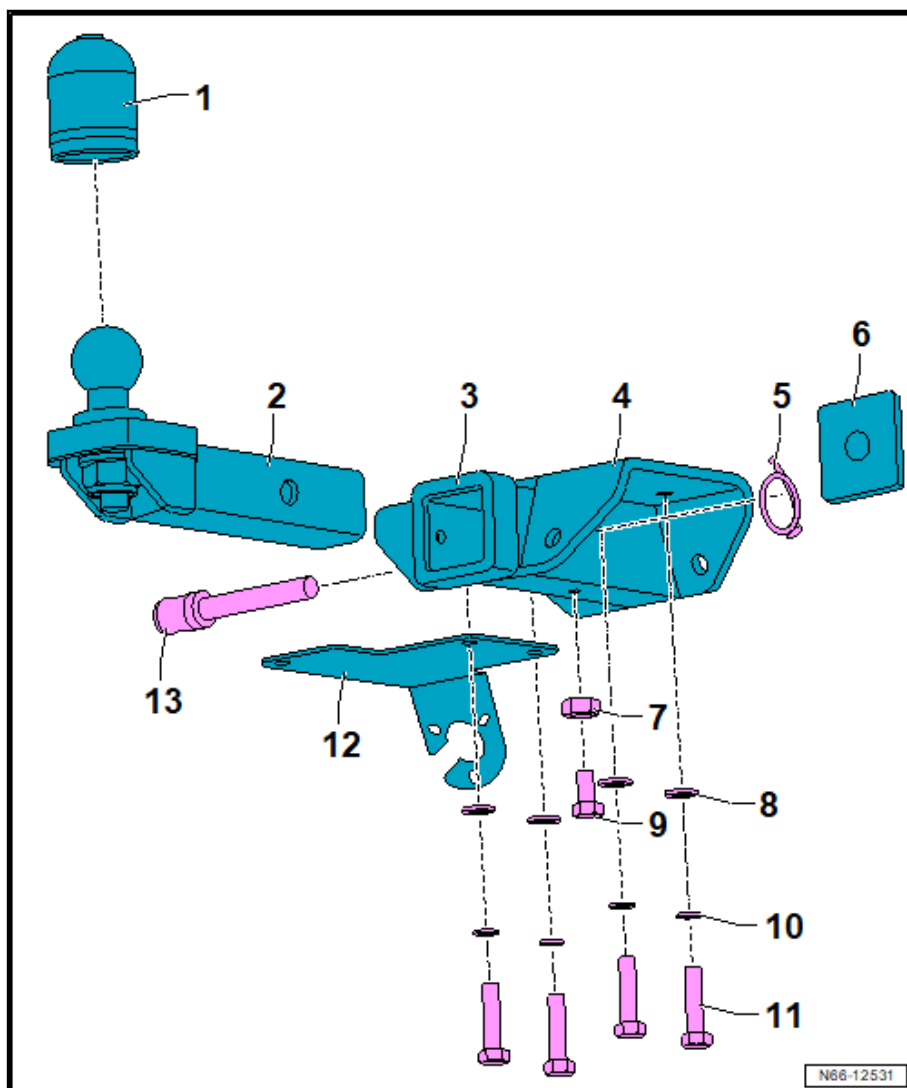
37 Towing bracket (for Australia only)

⇒ [o37.1 verview – towing bracket”, page 299](#)

⇒ [a37.2 nd installing towing bracket”, page 299](#)

37.1 Assembly overview – towing bracket

- 1 - Cap
- 2 - Towing bracket
 - ☐ Removing and installing ⇒ [page 299](#)
- 3 - Cover
- 4 - Bracket
 - ☐ Removing and installing ⇒ [page 299](#)
- 5 - Locking ring
- 6 - Cover
- 7 - Nut
- 8 - Washers
 - ☐ Qty. 4
- 9 - Bolt
 - ☐ For reducing vibrations on towing bracket.
- 10 - Washers
 - ☐ Qty. 4
- 11 - Bolts
 - ☐ Renew after removal
 - ☐ Qty. 4
 - ☐ 100 Nm +180°
- 12 - Bracket
 - ☐ For electric socket
- 13 - Pin



37.2 Removing and installing towing bracket

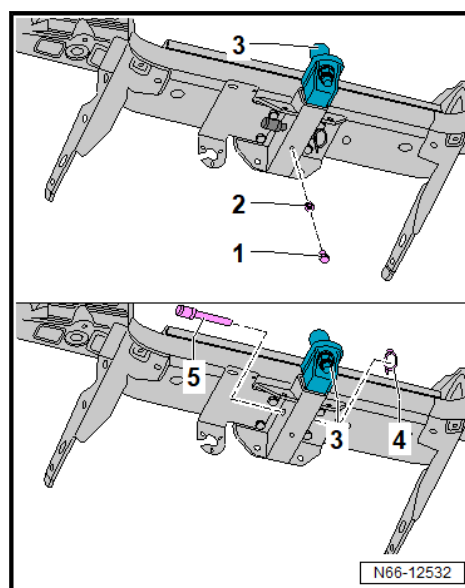
Special tools and workshop equipment required



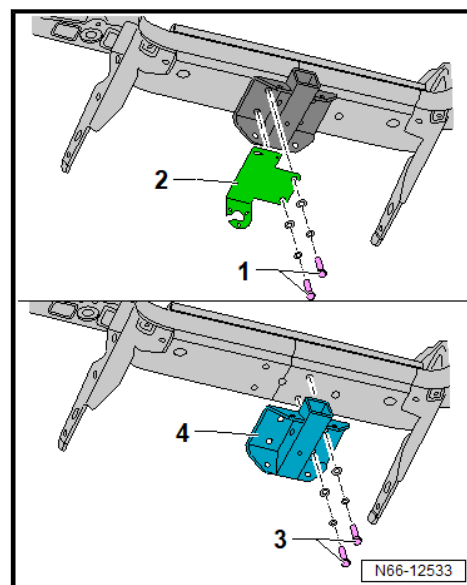
- ◆ Torque wrench, 40 to 200 Nm -V.A.G 1332A-



Removing



- Loosen bolt -1- and nut -2-.
- Remove circlip -4-.
- Pull out pin -5-.
- Pull out towing bracket -3-.
- Unscrew bolts -1-.



- Lay bracket -2- with wiring harness to one side.
- Unscrew bolts -3-. When doing this, remove bracket -4-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➤ [o37.1 verview – towing bracket”, page 299](#)



38 Roof load carrier

⇒ [o38.1 verview - roof rails/cross bars", page 302](#)

⇒ [o38.2 verview - ladder holder", page 303](#)

⇒ [o38.3 verview - loading roller with holder", page 304](#)

⇒ [a38.4 nd installing roof rails", page 305](#)

⇒ [a38.5 nd installing cross bars", page 306](#)

⇒ [a38.6 nd installing ladder holder", page 307](#)

⇒ [a38.7 nd installing loading roller with holder", page 308](#)

38.1 Assembly overview - roof rails/cross bars



1 - Cross bars

- ☐ Qty. 3
- ☐ Removing and installing ➤ [page 306](#)

2 - Bracket

- ☐ Qty. 6
- ☐ Removing and installing ➤ [page 306](#)

3 - Nuts

- ☐ Qty. 12
- ☐ 10 Nm

4 - Washers

- ☐ Qty. 12

5 - Covers

- ☐ Qty. 6

6 - Bolts

- ☐ Qty. 12
- ☐ 6 Nm

7 - Sliding blocks

- ☐ Qty. 12

8 - Bolts

- ☐ Number depends on equipment level
- ☐ 10 Nm

9 - Nuts

- ☐ Number depends on equipment level
- ☐ 10 Nm

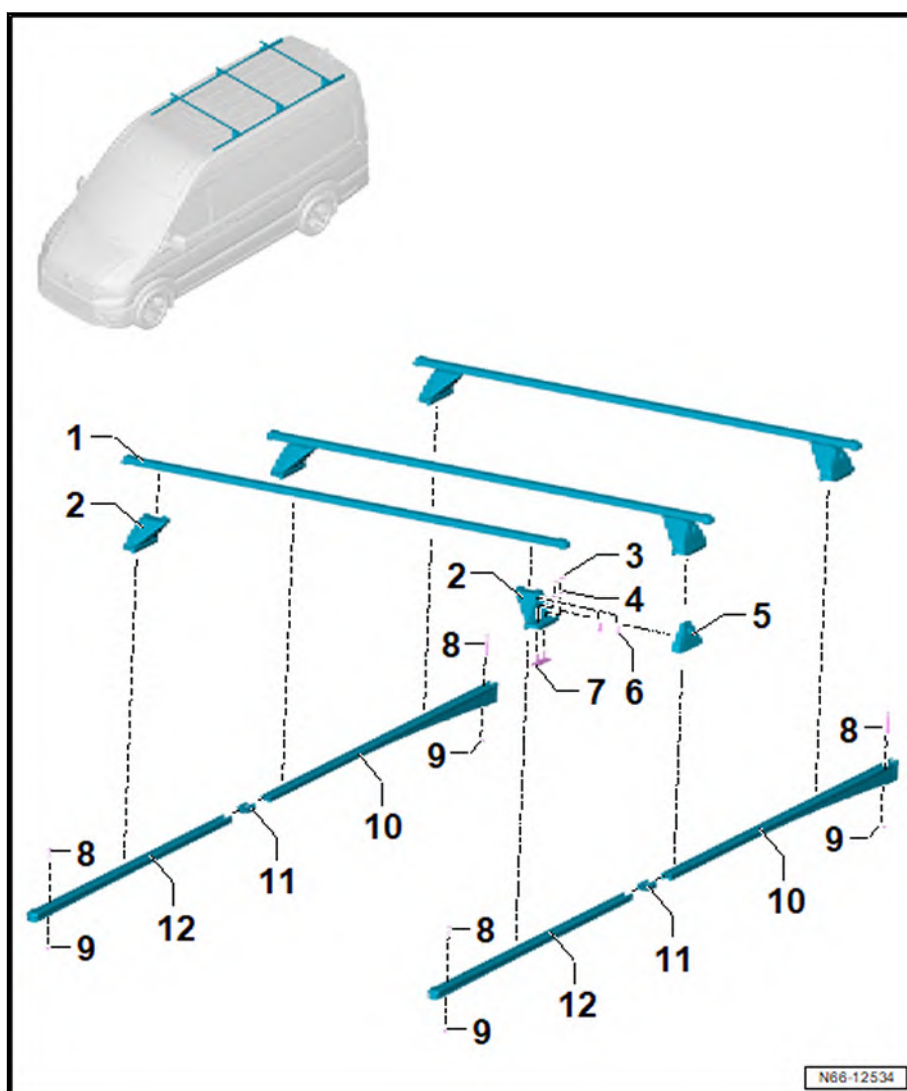
10 - Roof rails

- ☐ Rear
- ☐ Qty. 2
- ☐ Removing and installing ➤ [page 305](#)

11 - Adapter

12 - Roof rails

- ☐ Front
- ☐ Qty. 2
- ☐ Removing and installing ➤ [page 305](#)



38.2 Assembly overview - ladder holder



1 - Retaining bracket

- ☐ Qty. 4
- ☐ Top hole: cross piece, 20 mm high
- ☐ Bottom hole: cross piece, 30 mm high
- ☐ Removing and installing ⇒ [page 307](#)

2 - Tensioning rubbers

- ☐ Qty. 2

3 - Bar

- ☐ Removing and installing ⇒ [page 307](#)

4 - Bracket

- ☐ Qty. 2

5 - Washers

- ☐ Qty. 6

6 - Lock washers

- ☐ Qty. 2

7 - Nut

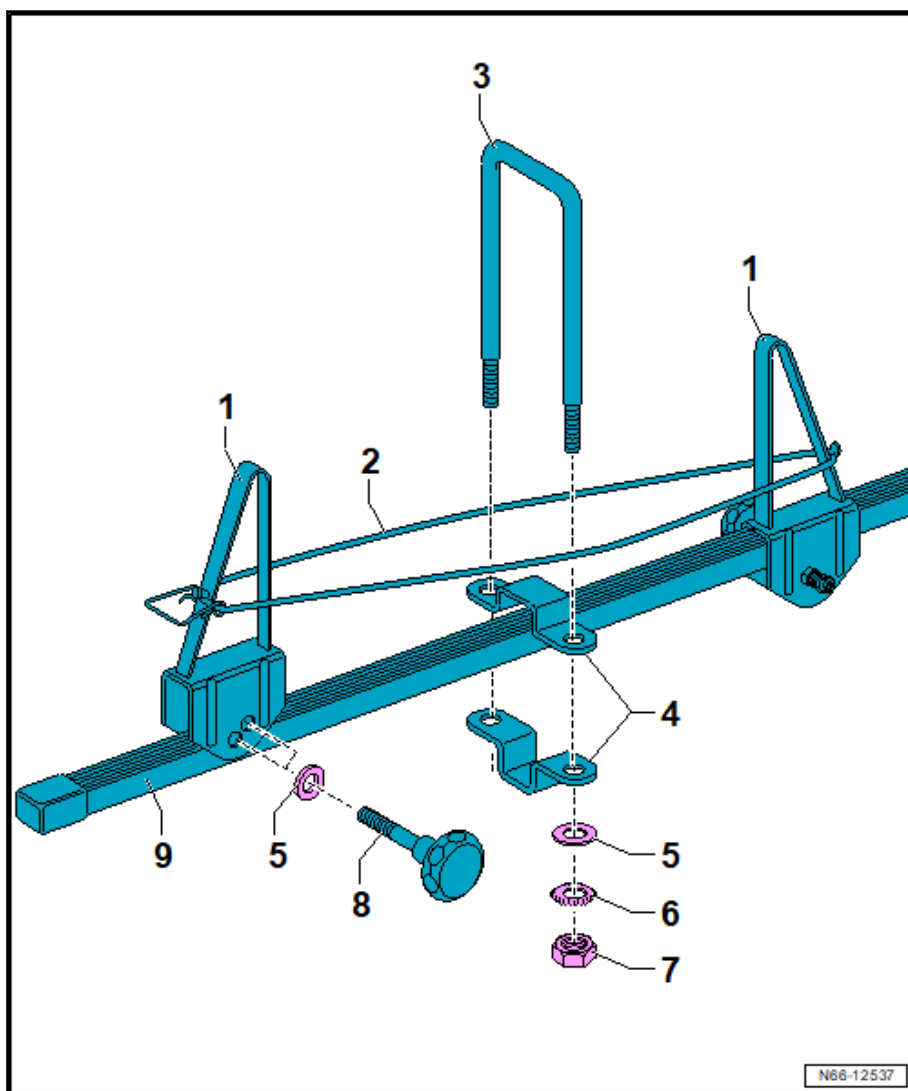
- ☐ 20 Nm

8 - Knurled screws

- ☐ Qty. 4

9 - Cross bars

- ☐ Assembly overview ⇒ [page 302](#)
- ☐ Removing and installing ⇒ [page 306](#)



38.3 Assembly overview - loading roller with holder



1 - Holder

☐ Qty. 2

2 - Washers

☐ Qty. 12

3 - Bolts

☐ Qty. 4

☐ 10 Nm

4 - Nuts

☐ Qty. 2

☐ 10 Nm

5 - Spring lock washers

☐ Qty. 2

6 - Washers

☐ Qty. 2

7 - Nuts

☐ Qty. 8

☐ 10 Nm

8 - Bracket

☐ Qty. 2

9 - Bolts

☐ Qty. 4

☐ 10 Nm

10 - Bracket

☐ Removing and installing ➤ [page 308](#)

11 - Nuts

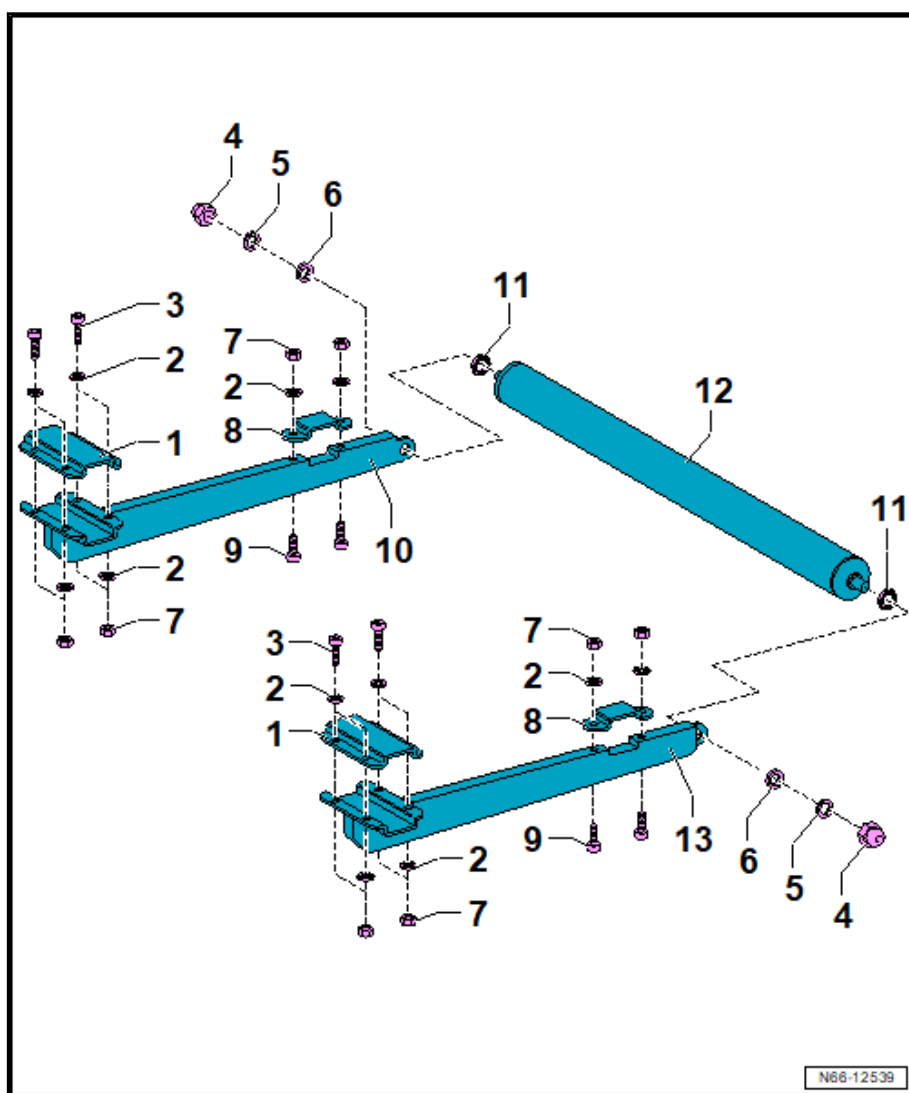
☐ Qty. 2

12 - Loading roller

☐ Removing and installing ➤ [page 308](#)

13 - Bracket

☐ Removing and installing ➤ [page 308](#)



38.4 Removing and installing roof rails

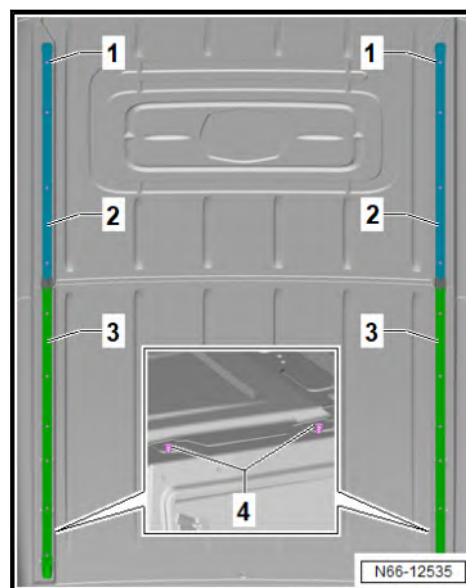
Special tools and workshop equipment required

- ◆ Torque wrench 6-50 Nm -V.A.G 1331-





Removing



- If fitted, remove cross bars ➔ [page 306](#) .
- The following work step must be performed with the aid of a second mechanic.
- Unscrew bolts -1-. Have a second mechanic counterhold nuts -4- when doing this.
- Pull apart roof rails -2- and -3-.
- Remove roof rails -2- and -3-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ➔ [o38.1 verview - roof rails/cross bars", page 302](#)

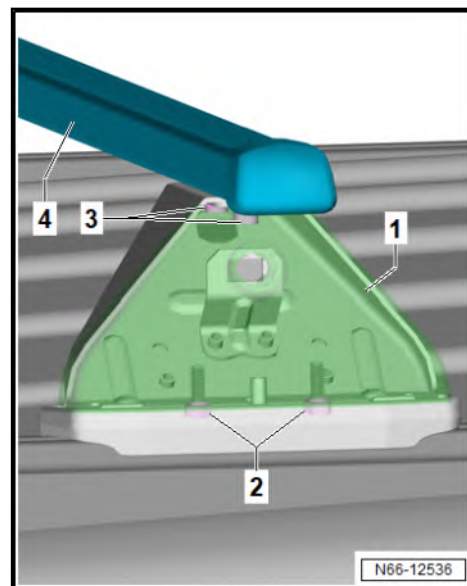
38.5 Removing and installing cross bars

Special tools and workshop equipment required

- ♦ Torque wrench 6-50 Nm -V.A.G 1331-



Removing



- Open cover -1-.
- Loosen bolts -3-.
- Unscrew nuts -2-. Remove washers.
- The following work step must be performed with the aid of a second mechanic.
- Remove cross bars -4- with aid of second mechanic.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ [⇒ o38.1 verview - roof rails/cross bars", page 302](#)

38.6 Removing and installing ladder holder

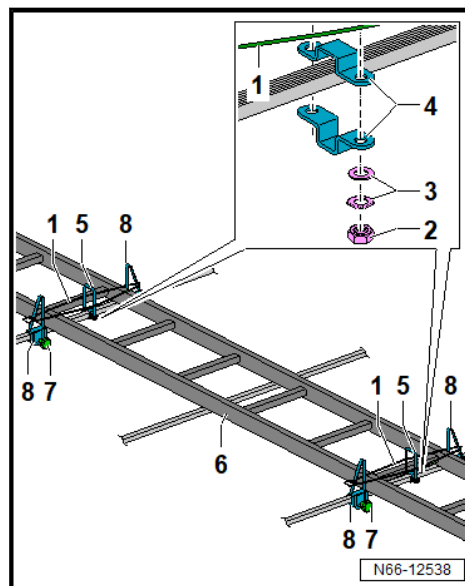
Special tools and workshop equipment required

- ◆ Torque wrench 6-50 Nm -V.A.G 1331-





Removing



- Release tensioning rubbers -1-.
- Unscrew nuts -2-. Remove washers -3- and holder -4- when doing this.
- Remove bar -5-.
- The following work step must be performed with the aid of a second mechanic.
- Remove ladder -6- with aid of second mechanic.
- Loosen knurled screws -7-.
- Remove retaining bracket -8-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ➔ [o38.2 verview - ladder holder", page 303](#)

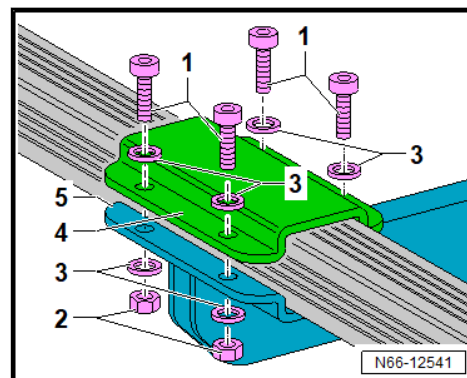
38.7 Removing and installing loading roller with holder

Special tools and workshop equipment required

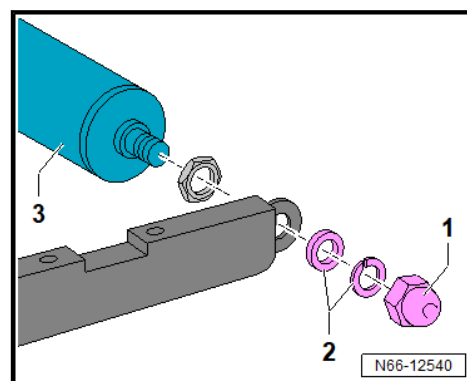
- ♦ Torque wrench 6-50 Nm -V.A.G 1331-



Removing



- Unscrew bolts -1-. Counterhold nuts -2- when doing this.
- Remove washers -3- and holder -4-.
- Remove holder with loading roller -5-.
- Unscrew nut -1-.



- Remove retaining ring and washer -2-.
- Remove loading roller -3-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ [⇒ o38.3 verview - loading roller with holder", page 304](#)



39 Skid plate

⇒ [o39.1 verview – skid plate”, page 310](#)

⇒ [a39.2 nd installing skid plate”, page 310](#)

39.1 Assembly overview – skid plate

1 - Skid plate

- ☐ Removing and installing ⇒ [page 310](#)

2 - Bolts

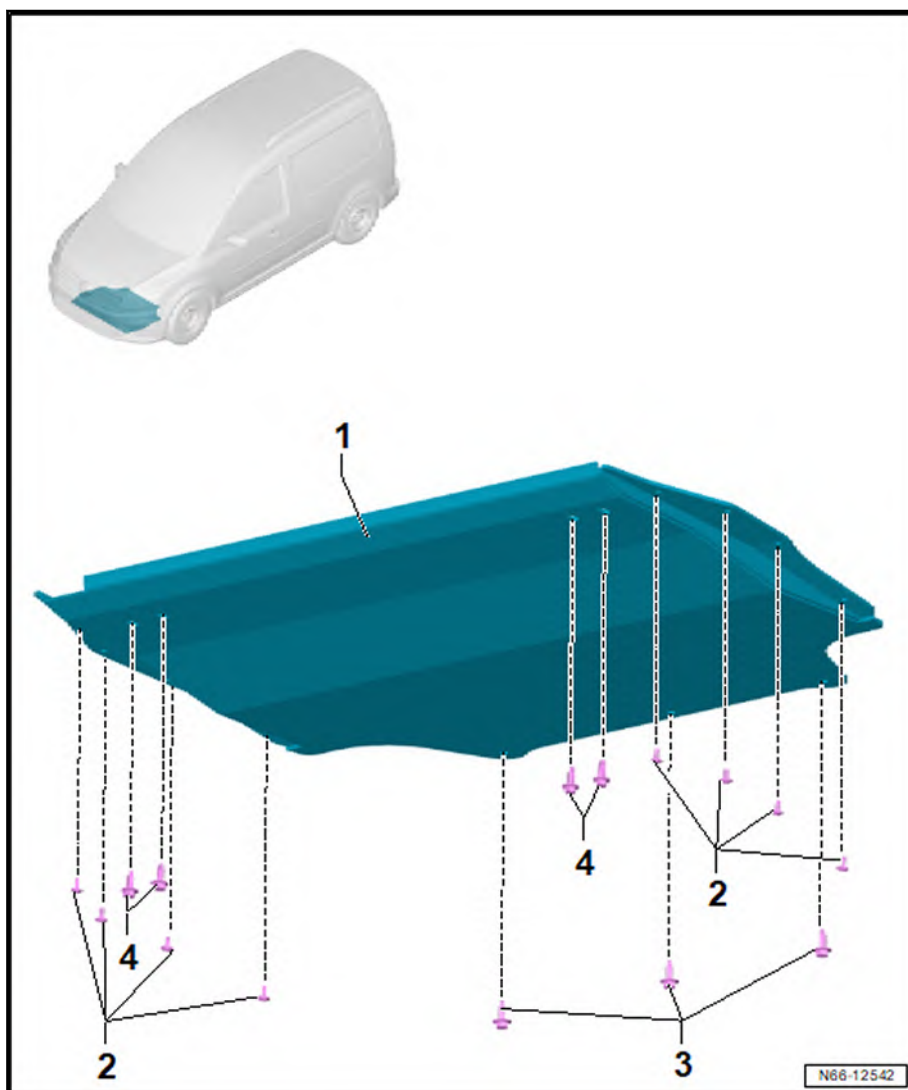
- ☐ Qty. 8
- ☐ 2 Nm

3 - Bolts

- ☐ Qty. 3
- ☐ M6: 6 Nm
- ☐ M8: 20 Nm

4 - Bolts

- ☐ Qty. 4
- ☐ 20 Nm



39.2 Removing and installing skid plate

Special tools and workshop equipment required



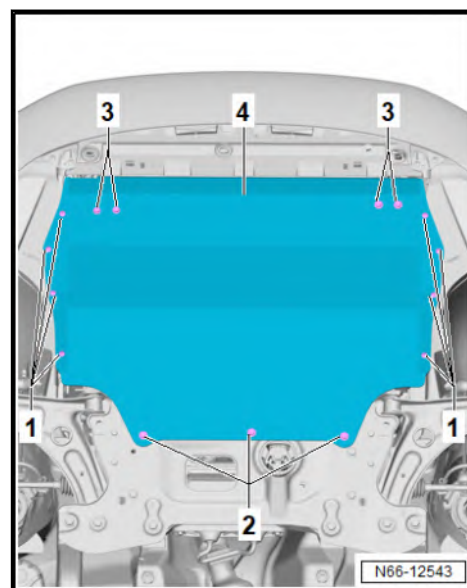
- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



- ◆ Torque screwdriver -V.A.G 1624-



Removing



- Unscrew bolts -1-.
- Unscrew bolts -2-.
- The following work steps require the assistance of a second mechanic.
- Unscrew bolts -3-.
- Remove skid plate -4- with the aid of a second mechanic.

Installing

Install in reverse order of removal, observing the following:



Specified torques

- ◆ ⇒ [o39.1 verview – skid plate”, page 310](#)

40 Cross bars

⇒ [o40.1 verview – cross bars”, page 313](#)

⇒ [a40.2 nd installing cross bars”, page 313](#)

40.1 Assembly overview – cross bars

1 - Cap

2 - Lock

3 - Pin

4 - Bolt

- ☐ Left and right
- ☐ 9 Nm

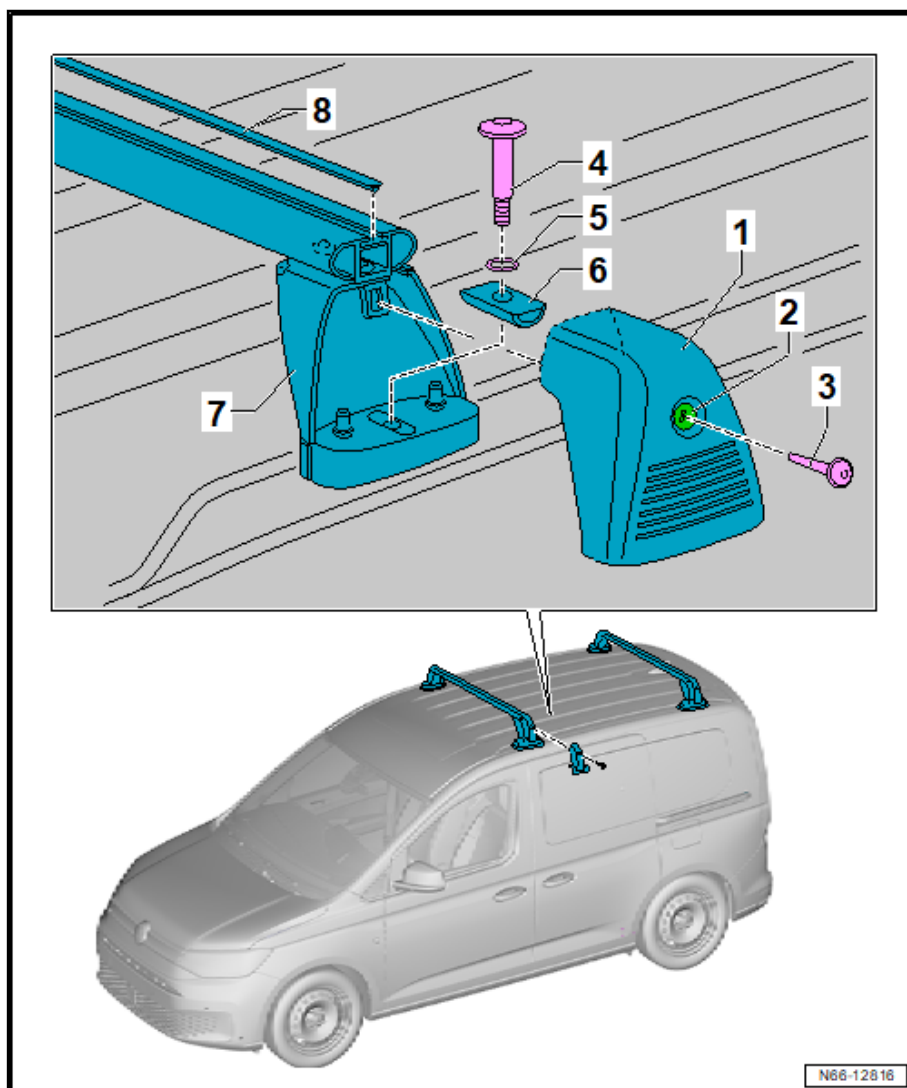
5 - Washer

6 - Positioning bush

7 - Cross bars

- ☐ Removing and installing ⇒ [page 313](#)

8 - Trim section



40.2 Removing and installing cross bars

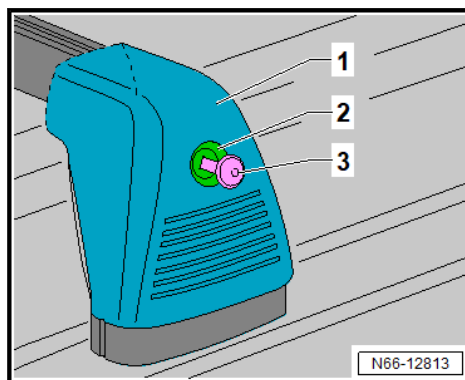
Special tools and workshop equipment required



- ◆ Torque wrench 6-50 Nm -V.A.G 1331-

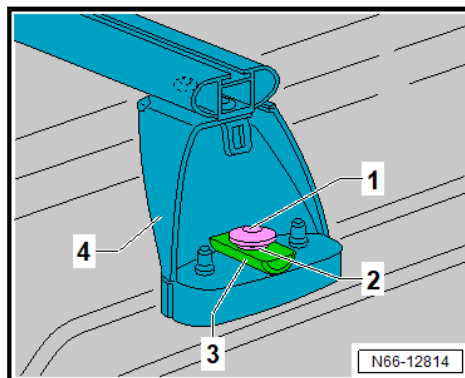


Removing



Perform the described work steps for the both cross bars.

- Turn pin -3- to open lock -2-.
- Remove cap -1-.
- Unscrew bolt -1-.



- Detach washer -2- together with positioning bush -3-.
- Detach cross bars -4-.

Installing

Install in reverse order of removal, observing the following:

- Contact surfaces must be free of dirt and grease.

Specified torques

- ◆ ⇒ [o40.1 verview – cross bars](#)”, page 313



68 – Interior equipment

1 Box body (Junge)

⇒ [a1.1 nd installing fastening rail", page 315](#)

⇒ [a1.2 nd installing securing strap recesses", page 316](#)

⇒ [a1.3 nd installing pull strap", page 317](#)

1.1 Removing and installing fastening rail

Special tools and workshop equipment required

◆ Pop rivet pliers -VAS 5072A-



◆ Pin punch -VAS 5513-

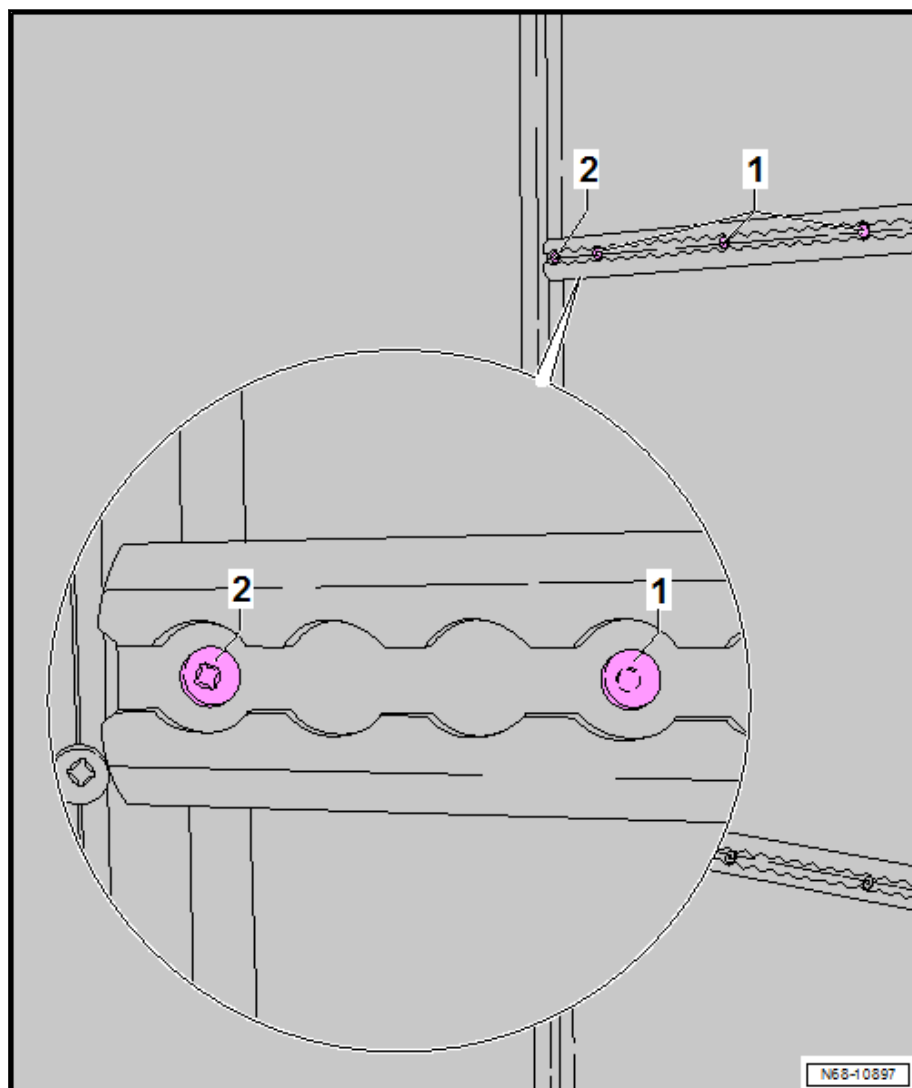


Removing

CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!



- Drill out rivet head -1- in fastening rail using drill and drill bit.
- Drive out rivet shank -1- with drift -VAS 5513-.
- Remove swarf.
- Unscrew bolts -2- at start and end of fastening rail.
- Remove fastening rail from box body.

Installing

Install in reverse order of removal, observing the following:

- Use rivet pliers -VAS 5072A- to install new pop rivets.

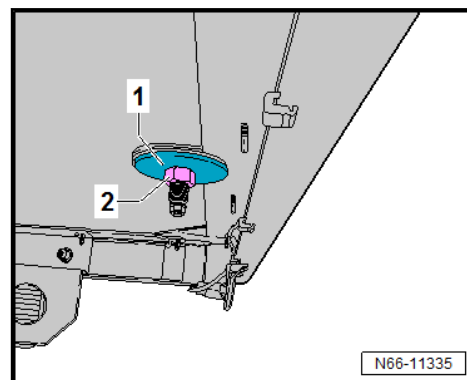
1.2 Removing and installing securing strap recesses

Special tools and workshop equipment required

◆ Torque wrench -V.A.G 1410A-



Removing



- Unscrew nut -2-.
- Detach washer -1- and remove.
- Release bonded joint.
- Remove securing strap recess from inside.
- Remove sealing and adhesive compound residue.

Installing

Install in reverse order of removal, observing the following:

- Lightly sand down bonding surfaces and clean.
- Apply sealing and adhesive compound.
- Fit securing strap recess from inside.
- Fit washer -1-.
- Tighten nut -2-.

Specified torques

Component	Specified torque
Nut	8 Nm

1.3 Removing and installing pull strap

Special tools and workshop equipment required



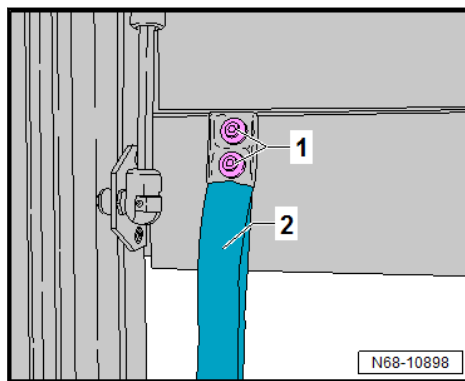
◆ Pop rivet pliers -VAS 5072A-



◆ Pin punch -VAS 5513-



Removing



CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out rivet head -1- using drill and drill bit.
- Drive out rivet shank -1- with drift -VAS 5513-.
- Remove swarf.
- Remove pull strap -2- from tail flap.

Installing

Install in reverse order of removal, observing the following:

- Use rivet pliers -VAS 5072A- to install new pop rivets.



2 Box body (Spier)

⇒ [a2.1 nd installing fastening rail", page 319](#)

⇒ [a2.2 nd installing securing strap recess", page 321](#)

⇒ [s2.3 trap", page 322](#)

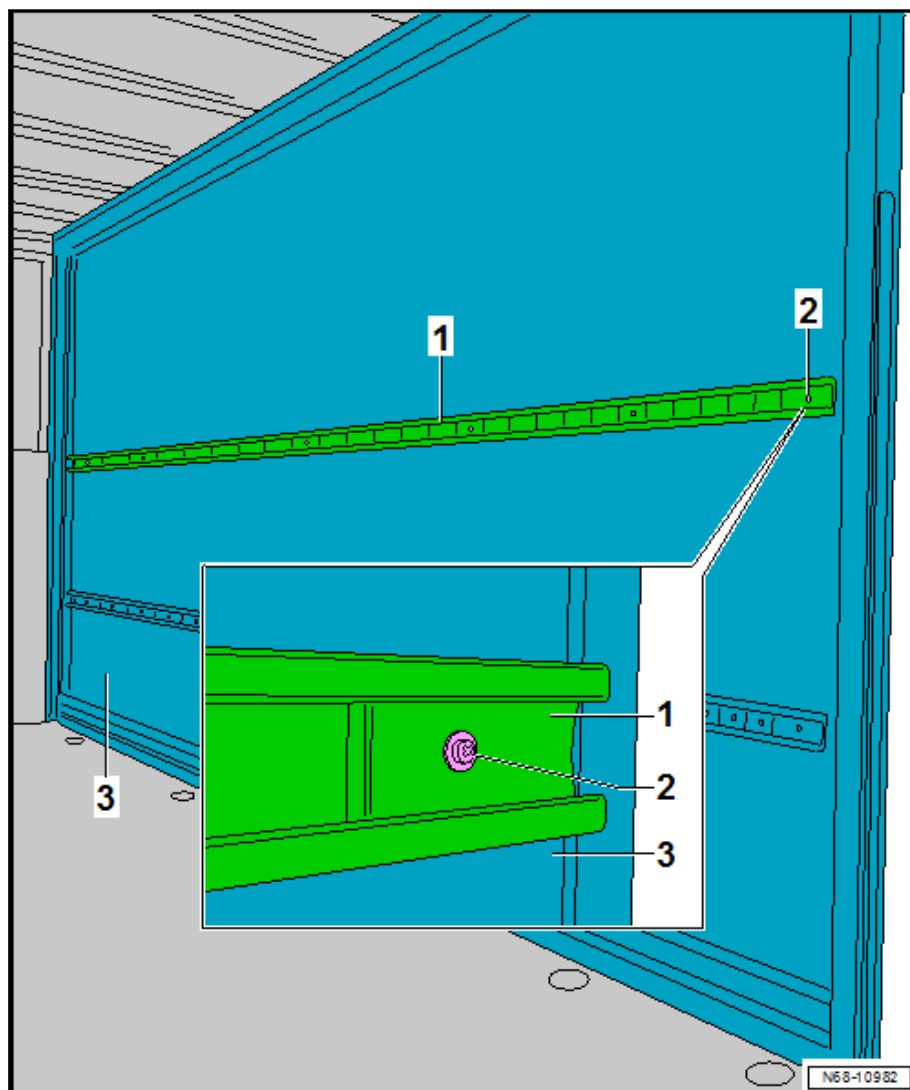
2.1 Removing and installing fastening rail

Special tools and workshop equipment required

- ◆ Torque screwdriver -VAS 6494-



Removing



- Unscrew bolts -2-.
- Release bonded joint between fastening rail -1- and sidewall -3-.
- Remove fastening rail from sidewall.
- Remove adhesive residue.
- Lightly sand down bonding surface and clean.

Installing

Install in reverse order of removal, observing the following:

The bolts serve purely to attach the fastening rail.

- Clean bonding surface.
- Apply SIKA activator 100 to the bonding surface of the new part and the sidewall.
- Observe flash-off period of at least 20 minutes.
- Apply SIKA adhesive 552 to new part.
- Fit fastening rail and secure with bolts -2-.
- Remove any excess adhesive after approx. 2 hours using a commercially available scraper.

- The minimum curing period of 48 hours at room temperature (at least 18°C) must be adhered to. During this time, the vehicle must be standing on a level surface.

Specified torques

Component	Specified torque
Bolts	1.5 Nm

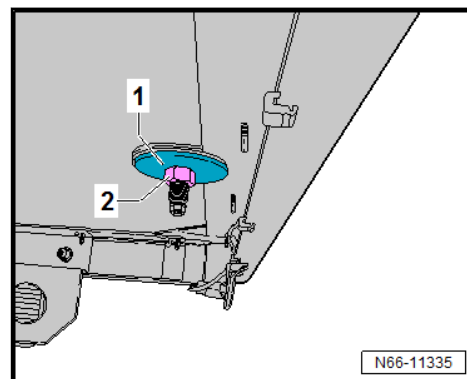
2.2 Removing and installing securing strap recess

Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1410A-



Removing



- Unscrew nut -2-.
- Detach washer -1- and remove.
- Release bonded joint.
- Remove securing strap recess from inside.

Installing

Install in reverse order of removal, observing the following:

- Remove loose adhesive residue.
- Clean bonding surfaces with cleaning solution.
- Observe flash-off time of cleaning solution.
- Apply SIKA-FLEX (brown) 221 assembly adhesive to new part.
- Adapt moulding from inside and spread adhesive evenly by turning.
- Remove excess adhesive.



- Fit washer -1-.
- Tighten nut -2-.

Specified torques

Component	Specified torque
Nut	10 Nm

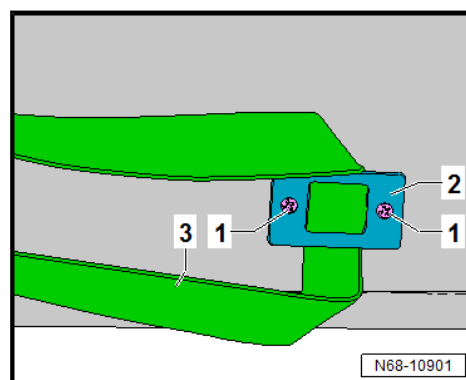
2.3 Pull strap

Special tools and workshop equipment required

- ◆ Torque screwdriver -VAS 6494-



Removing



- Unscrew bolts -1-.
- Remove holder -2- with pull strap -3- from tail flap.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolts	1.5 Nm



3 Box body (Schmitz Cargobull)

⇒ [a3.1 nd installing roof cross strut", page 323](#)

⇒ [a3.2 nd installing fastening rail", page 325](#)

⇒ [a3.3 nd installing pull strap", page 327](#)

3.1 Removing and installing roof cross strut

Special tools and workshop equipment required

◆ Adhesive strip remover -VAS 6349-



◆ Pop rivet pliers -VAS 5072A-

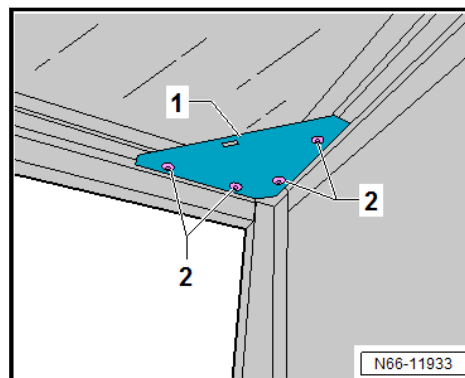


◆ Pin punch -VAS 5513-





Removing

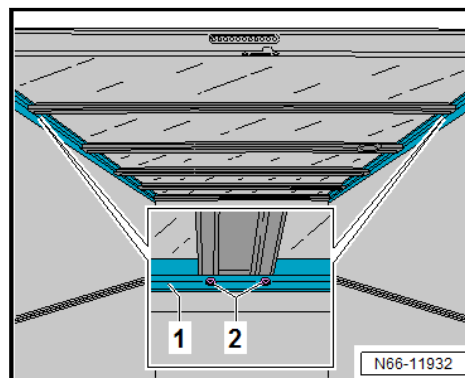


- If necessary, remove interior light ➔ [page 520](#) .

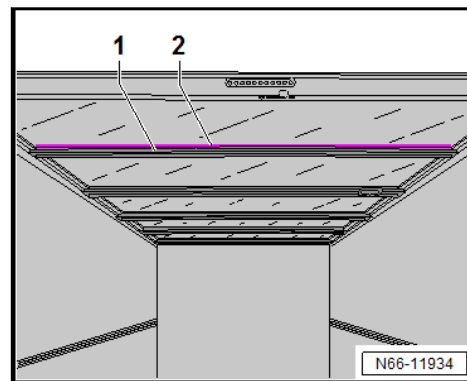
CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

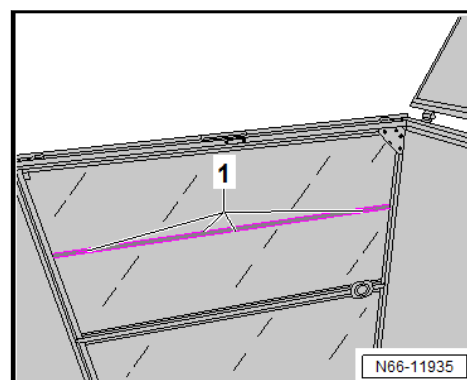
- Wear protective goggles.
 - Wear protective gloves!
-
- Drill out rivet head -2- using drill and drill bit.
 - Drive out rivet shank -2- with drift -VAS 5513-.
 - Remove swarf.
 - Remove reinforcement -1-.
 - Using hand drill and drill bit, drill out rivet head -2- on profile strip -1- on both sides.



- Drive out rivet shank -2- with drift -VAS 5513-.
- Remove swarf.
- Detach profile strip -1- on both sides.
- Detach bonded joint -2-.



- Remove roof cross strut -1-.
- Remove adhesive strip residue -1- with adhesive strip remover -VAS 6349-.



Installing

Install in reverse order of removal, observing the following:

- Apply double-sided adhesive tape to roof cross strut.
- Use rivet pliers -VAS 5072A- to install new pop rivets.

3.2 Removing and installing fastening rail

Special tools and workshop equipment required

- ◆ Pop rivet pliers -VAS 5072A-

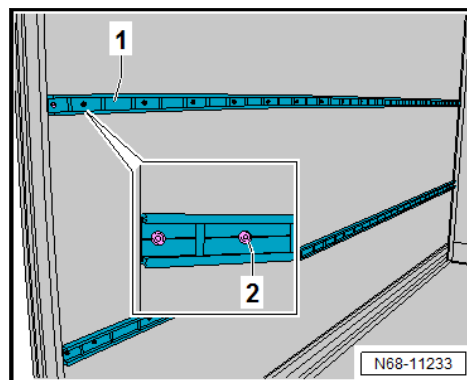




◆ Pin punch -VAS 5513-



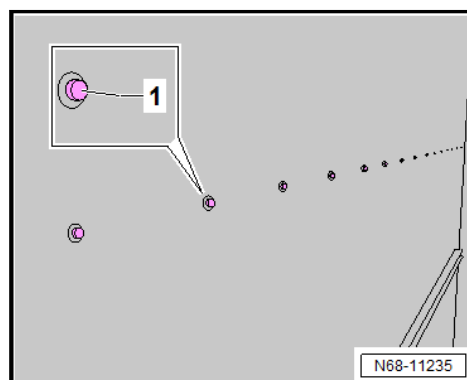
Removing



⚠ CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
 - Wear protective gloves!
-
- Drill out rivet head -2- in fastening rail using drill and drill bit.
 - Drive out rivet shank -2- with drift -VAS 5513-.
 - Remove swarf.
 - Remove fastening rail -1-.
 - Drill down remains of rivets -1- so that they are flush with sidewall.



Installing

Install in reverse order of removal, observing the following:



- ◆ Remains of rivets remain in the sidewall.
- Drill new holes for fastening rail.
- Use rivet pliers -VAS 5072A- to install new pop rivets.

3.3 Removing and installing pull strap

Special tools and workshop equipment required

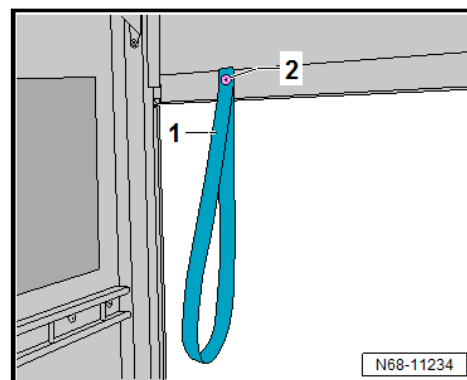
- ◆ Pop rivet pliers -VAS 5072A-



- ◆ Pin punch -VAS 5513-



Removing



CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out rivet -2-. While doing this, remove pull strap -1-.



Installing

Install in reverse order of removal, observing the following:

- Use rivet pliers -VAS 5072A- to install new pop rivets.

4 Equipment

⇒ [f4.1 loor", page 329](#)

⇒ [f4.2 loor panel", page 333](#)

⇒ [c4.3 arrier in load compartment", page 335](#)

⇒ [a4.4 nd installing grab handle, A-pillar", page 336](#)

⇒ [p4.5 anel \(Holland double cab\)", page 338](#)

4.1 Universal floor

⇒ [o4.1.1 verview - universal floor, longitudinal rails", page 329](#)

⇒ [o4.1.2 verview - universal floor, transverse rails", page 329](#)

⇒ [a4.1.3 nd installing universal floor", page 331](#)

4.1.1 Assembly overview - universal floor, longitudinal rails

1 - Universal floor

- ☐ Front
- ☐ Removing and installing ⇒ [page 331](#)

2 - Longitudinal rails

3 - Bolts

- ☐ Number depends on equipment level
- ☐ 40 Nm

4 - Bolts

- ☐ Qty. 7
- ☐ 1.3 Nm

5 - Strip

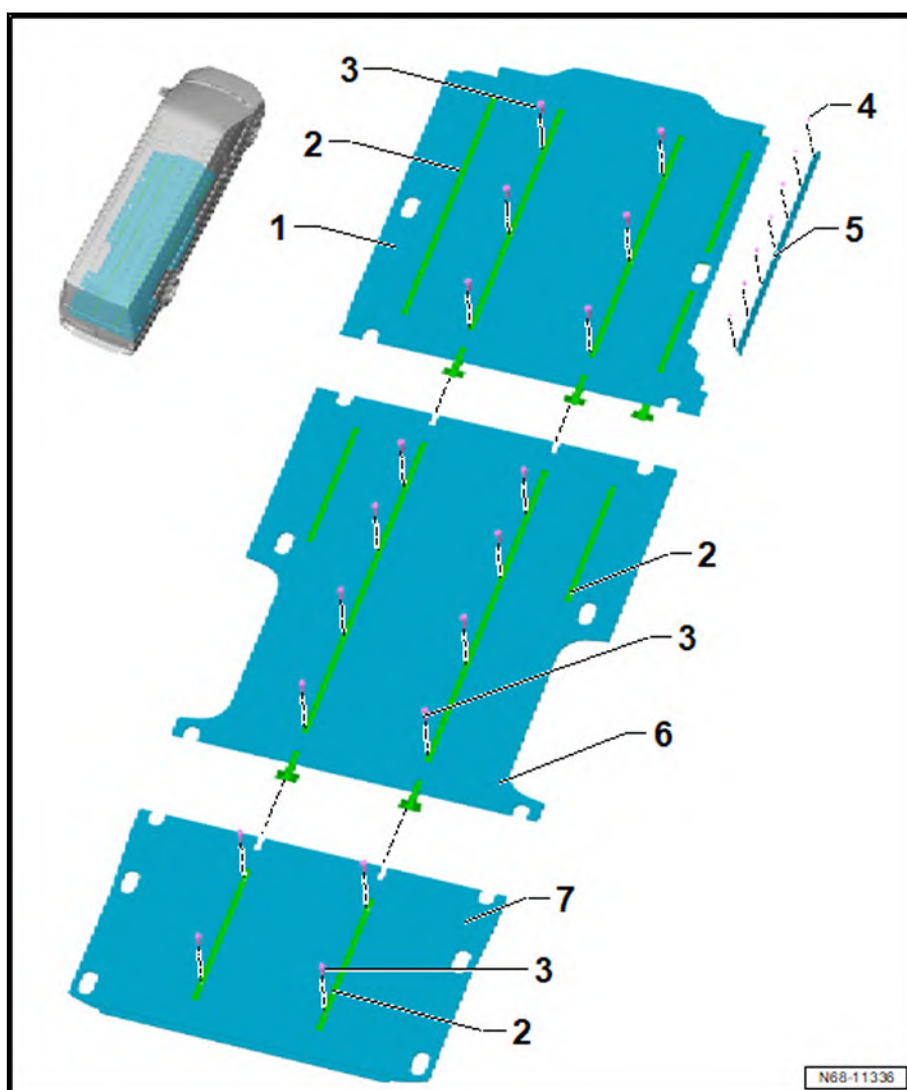
- ☐ Sliding door sill

6 - Universal floor

- ☐ Centre
- ☐ Removing and installing ⇒ [page 331](#)

7 - Universal floor

- ☐ Rear
- ☐ Removing and installing ⇒ [page 331](#)



4.1.2 Assembly overview - universal floor, transverse rails



1 - Strips

- ☐ Number depends on equipment level

2 - Bolts

- ☐ Number depends on equipment level
- ☐ 1.3 Nm

3 - Bolts

- ☐ Qty. 4
- ☐ 20 Nm

4 - Fastening rings

- ☐ Qty. 2

5 - Bracket

- ☐ Qty. 2

6 - Universal floor

- ☐ Front
- ☐ Removing and installing ⇒ [page 331](#)

7 - Transverse rails

- ☐ Number depends on equipment level

8 - Universal floor

- ☐ Centre
- ☐ Removing and installing ⇒ [page 331](#)

9 - Universal floor

- ☐ Rear
- ☐ Depending on equipment
- ☐ Removing and installing ⇒ [page 331](#)

10 - Strip

- ☐ Rear

11 - Bolts

- ☐ Qty. 6
- ☐ 1.3 Nm

12 - Caps

- ☐ Number depends on equipment level

13 - Fastening rings

- ☐ Number depends on equipment level

14 - Bolts

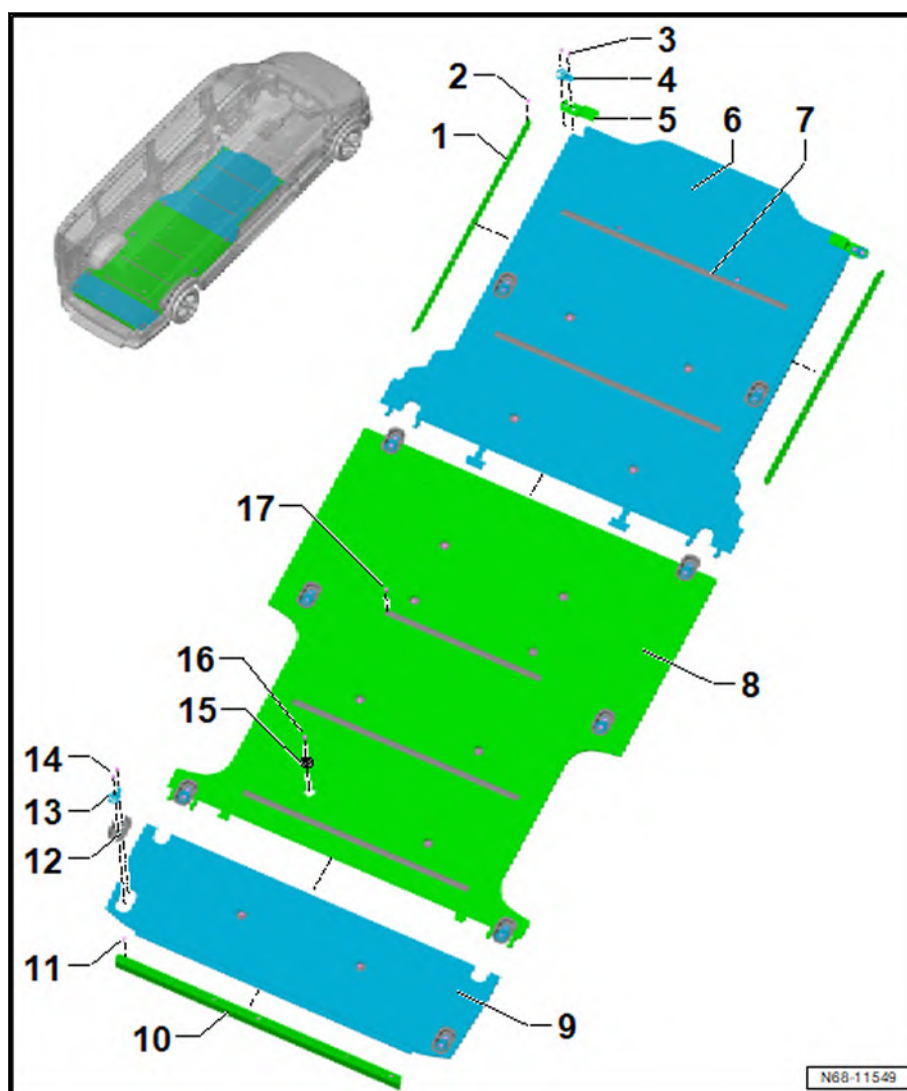
- ☐ Number depends on equipment level
- ☐ 20 Nm

15 - Caps

- ☐ Number depends on equipment level

16 - Bolts

- ☐ Number depends on equipment level
- ☐ 40 Nm



17 - Bolts

- ☐ Number depends on equipment level
- ☐ 40 Nm

4.1.3 Removing and installing universal floor

Special tools and workshop equipment required

- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



- ◆ Torque wrench -VAS 6253A-

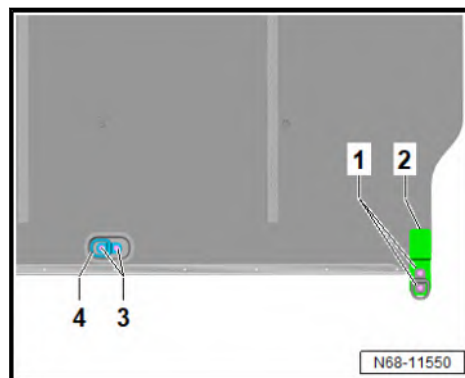


- ◆ Electric cutter -V.A.G 1561A-

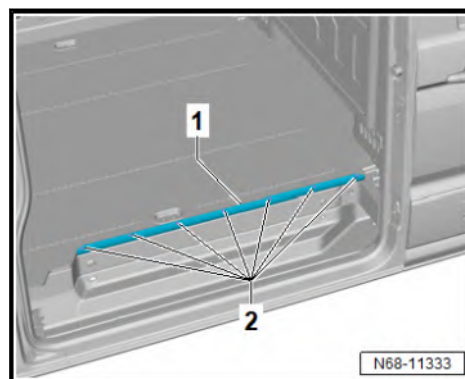




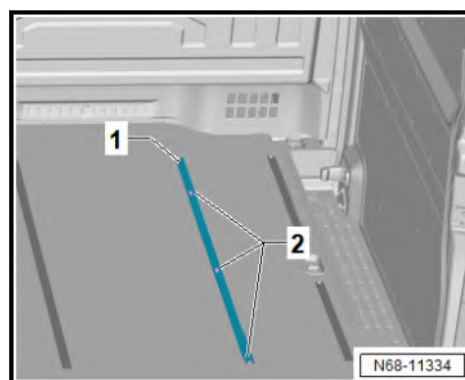
Removing



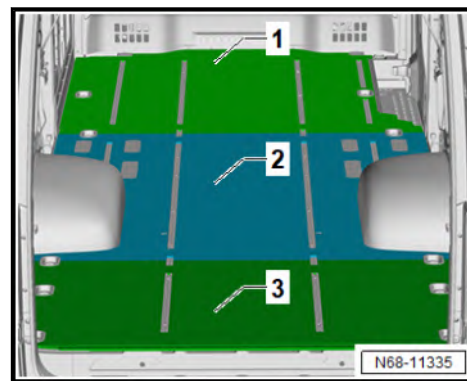
- Unscrew bolts -1-.
- Remove bracket -2-.
- Unscrew bolts -3-.
- Remove fastening rings -4-.
- Unscrew bolts -2-.



- Remove strip -1-.
- Unscrew bolts -2-.

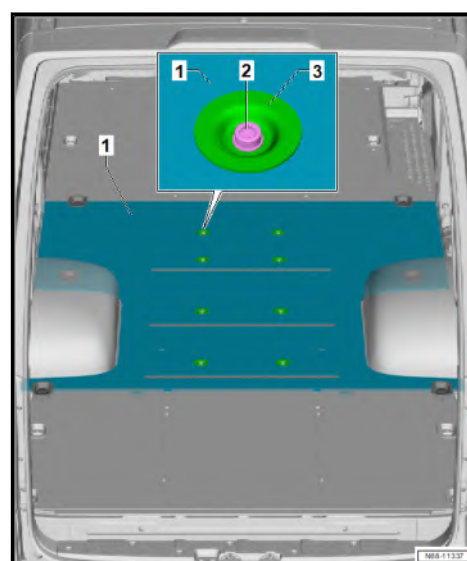


- ◆ Longitudinal rails -1- are permanently secured to universal floor.
- Detach universal floor -1-.



- Detach universal floor -2-.
- Detach universal floor -3-.

Installing



Install in reverse order of removal, observing the following:

- Make cut-outs for vehicle-dependent installations using electric cutter -V.A.G 1561A-.
- Insert cap -3-.
- Tighten bolt -2- for universal floor -1-.

Specified torques

- ◆ [⇒ o4.1.1 verview - universal floor, longitudinal rails", page 329](#)
- ◆ [⇒ o4.1.2 verview - universal floor, transverse rails", page 329](#)

4.2 Removable floor panel

[⇒ a4.2.1 nd installing removable floor panel", page 333](#)

4.2.1 Removing and installing removable floor panel

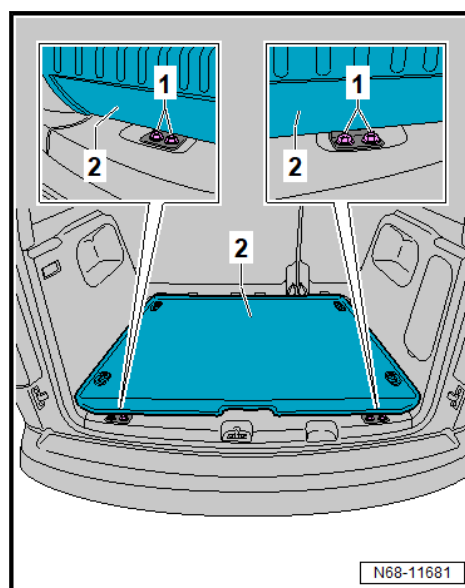
Special tools and workshop equipment required



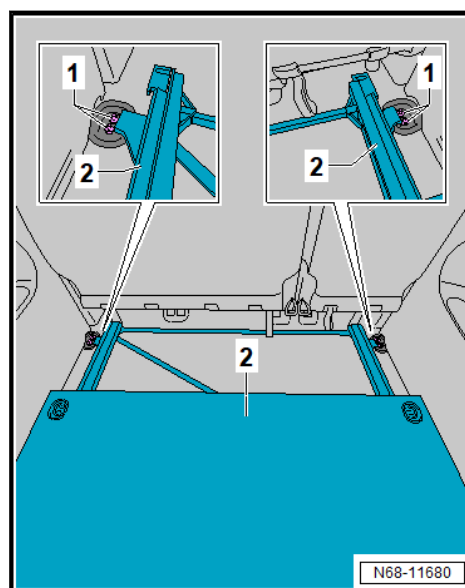
- ◆ Torque wrench, 6–50 Nm -V.A.G 1331-



Removing



- Unscrew nuts -1- on both sides.
- Pull out floor panel -2-.
- Unscrew nuts -1- on both sides.



- Remove load floor -2- with aid of second mechanic.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Nuts	10 Nm

4.3 Roof carrier in load compartment

⇒ [o4.3.1 verview - roof carrier in load compartment", page 335](#)

⇒ [a4.3.2 nd installing roof carrier in load compartment", page 335](#)

4.3.1 Assembly overview - roof carrier in load compartment

1 - Roof carrier

- ☐ Inner
- ☐ Qty. 2
- ☐ Front and rear
- ☐ Removing and installing ⇒ [page 335](#)

2 - Nuts

- ☐ Qty. 8
- ☐ Front and rear
- ☐ Left and right
- ☐ 14 Nm

3 - T-slot nuts with fastener

- ☐ Qty. 4
- ☐ Front and rear
- ☐ Left and right

4 - Fastening rails

- ☐ Roof
- ☐ Qty. 2
- ☐ Removing and installing ⇒ General body repairs, interior; Rep. gr. 68; Equipment; Removing and installing fastening rail

5 - Bolts

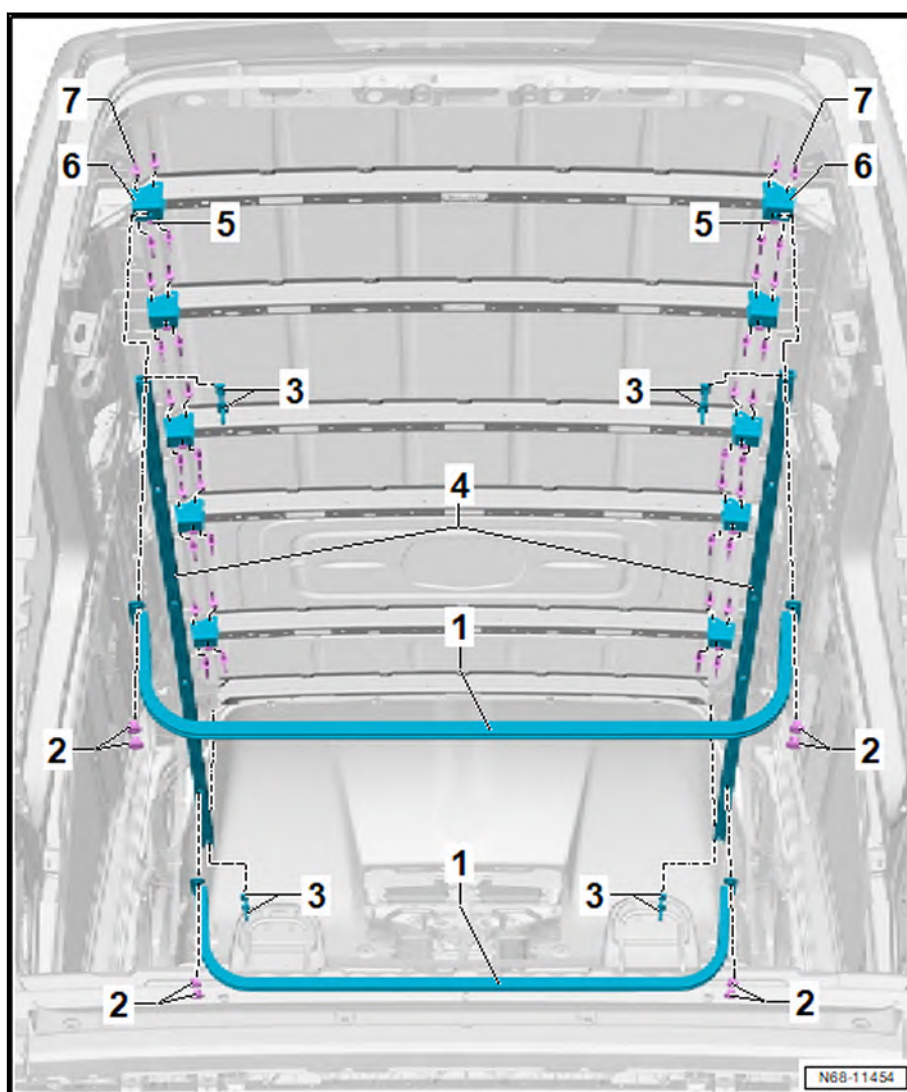
- ☐ Qty. 10
- ☐ 12 Nm

6 - Bracket

- ☐ Qty. 10

7 - Pop rivets

- ☐ Qty. 40

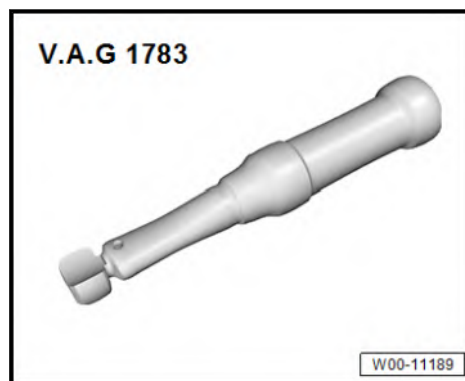


4.3.2 Removing and installing roof carrier in load compartment

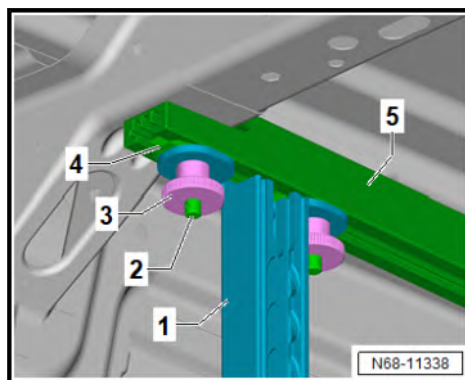
Special tools and workshop equipment required



- ◆ Torque wrench -V.A.G 1783-



Removing



- Unscrew knurled nut -3-.
- Remove inner roof carrier -1-.
- Remove T-slot nut with fastener -4- from C-rail -5-.

Installing

Installation is carried out in reverse order.

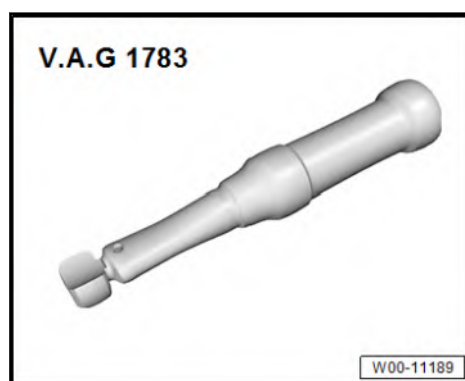
Specified torques

- ◆ [⇒ o4.3.1 verview - roof carrier in load compartment", page 335](#)

4.4 Removing and installing grab handle, A-pillar

Special tools and workshop equipment required

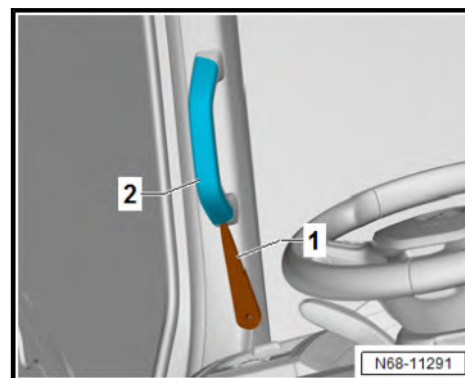
- ◆ Torque wrench -V.A.G 1783-



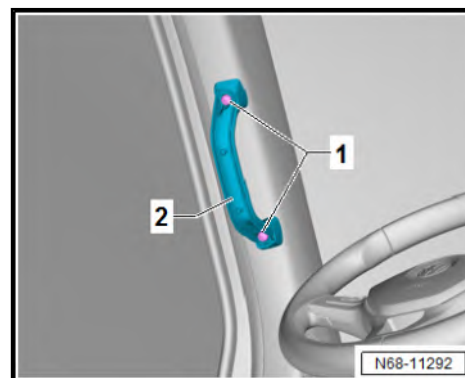
◆ Removal wedge -3409-



Removing



- Bring removal wedge -3409- -1- into position as shown.
- Lever off upper part of handle -2- using removal wedge -3409- -1-.
- Unscrew bolts -1-.



- Remove bottom part of handle -2-.

Installing

Install in reverse order of removal, observing the following:

Specified torque

Component	Specified torque
Bolts -1-	6 Nm



4.5 Partition panel (Holland double cab)

⇒ [o4.5.1 verview – partition panel”, page 338](#)

⇒ [a4.5.2 nd installing upper partition”, page 338](#)

⇒ [a4.5.3 nd installing lower partition”, page 341](#)

4.5.1 Assembly overview – partition panel

1 - Partition panel

- ☐ Top
- ☐ Removing and installing ⇒ [page 338](#)

2 - Bolts

- ☐ Qty. 18
- ☐ Tighten onto stop

3 - Cap

- ☐ For bolts and pop rivets
- ☐ Qty. 21

4 - Pop rivet

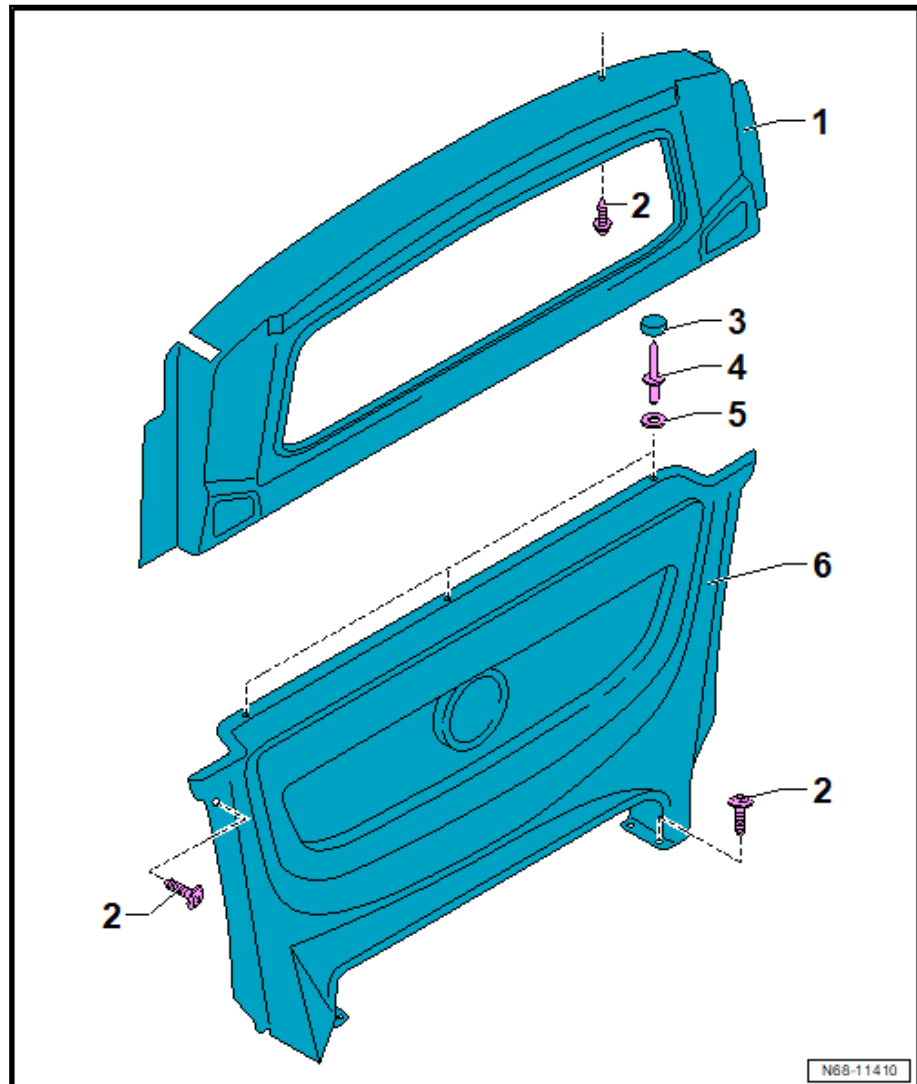
- ☐ Qty. 3 for connection between upper and lower sections of partition panel
- ☐ Qty. 4 for joining lower section of partition panel and bench seat trim

5 - Washer

- ☐ Qty. 7

6 - Partition panel

- ☐ Bottom
- ☐ Removing and installing ⇒ [page 341](#)

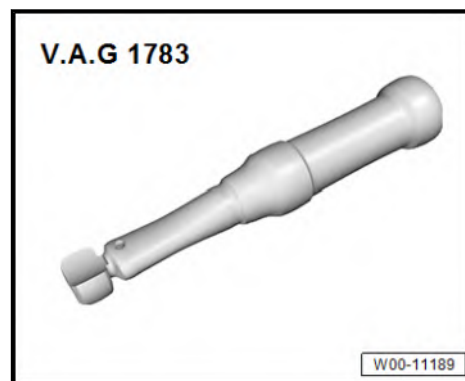


4.5.2 Removing and installing upper partition

Special tools and workshop equipment required



◆ Torque wrench -V.A.G 1783-



◆ Removal wedge -3409-



◆ Torque screwdriver -VAS 6494-



◆ Pop rivet pliers -VAS 5072A-

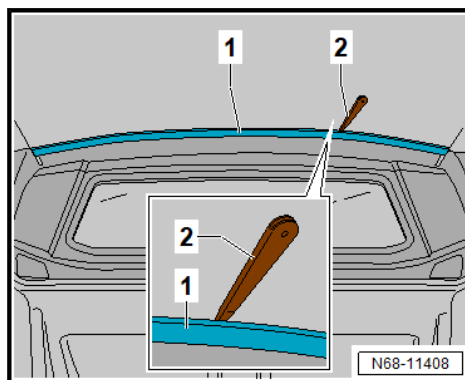




◆ Pin punch -VAS 5513-



Removing



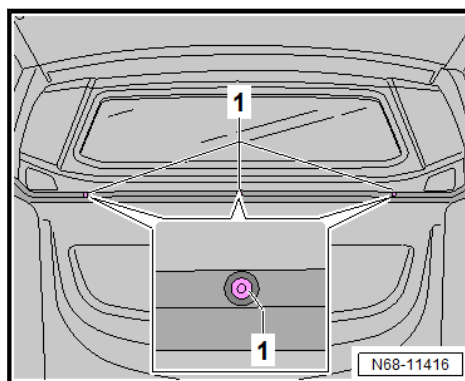
- Release trim strip -1- using removal wedge -3409- -2-.

CAUTION

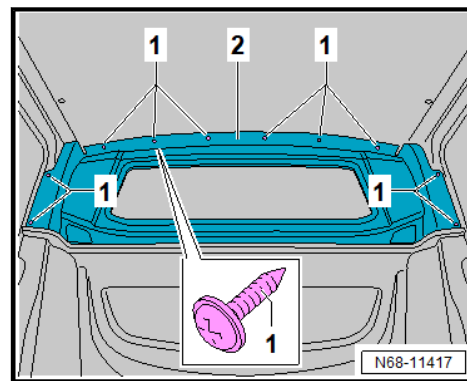
Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out rivets -1-.



- Drive out rivet shank with drift -VAS 5513-.
- Unscrew bolts -1-.



- Remove upper section of partition panel -2-.

Installing

Install in reverse order of removal, observing the following:

- If applicable, clean bonding surfaces.
- Draw in rivets with pop rivet pliers -VAS 5072A-.

Specified torques

- ◆ [⇒ o4.5.1 verview – partition panel", page 338](#)

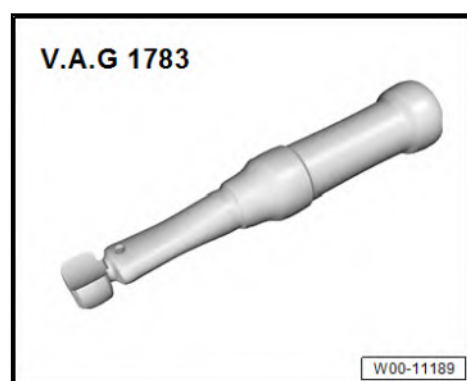
4.5.3 Removing and installing lower partition

Special tools and workshop equipment required

- ◆ Hot-air blower -V.A.G 1416-



- ◆ Torque wrench -V.A.G 1783-





◆ Torque screwdriver -VAS 6494-



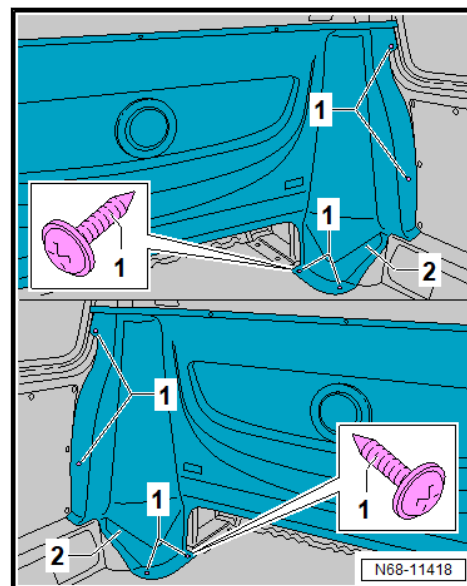
◆ Pop rivet pliers -VAS 5072A-



◆ Pin punch -VAS 5513-



Removing



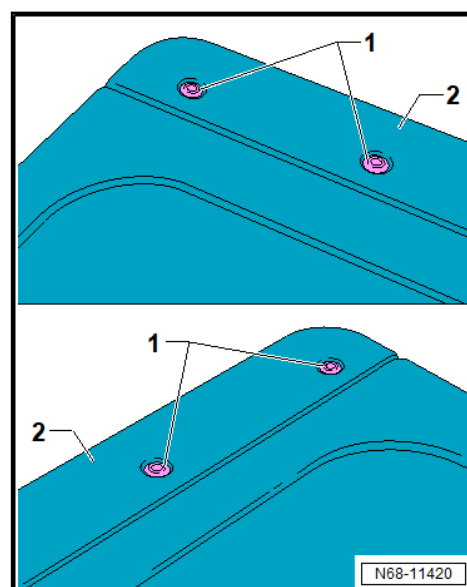
- Remove upper section of partition panel ➤ [page 338](#) .
- Unscrew bolts -1-. Detach partition panel -2- when doing this.

CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out rivets -1-.



- Drive out rivet shank with drift -VAS 5513-.
- Remove partition panel -2-.
- If necessary, heat up bonded joints of partition panel slightly (lukewarm) using hot air blower -V.A.G 1416-.



Installing

Install in reverse order of removal, observing the following:

- Install upper section of partition panel ⇒ [page 338](#) .
- Draw in rivets with pop rivet pliers -VAS 5072A-.
- If applicable, clean bonding surfaces.
- Apply new adhesive ⇒ Electronic parts catalogue (ETKA).

Specified torques

- ♦ ⇒ [o4.5.1 verview – partition panel", page 338](#)



5 Box body (Plus)

- ⇒ [o5.1 verview - box body \(Plus\)", page 345](#)
- ⇒ [a5.2 nd installing seat bench padding", page 350](#)
- ⇒ [a5.3 nd installing upper partition", page 351](#)
- ⇒ [a5.4 nd installing lower partition", page 352](#)
- ⇒ [a5.5 nd installing connection to side panel", page 353](#)
- ⇒ [a5.6 nd installing side panel", page 354](#)
- ⇒ [a5.7 nd installing sliding door step", page 357](#)
- ⇒ [a5.8 nd installing front and rear bench trim", page 358](#)
- ⇒ [a5.9 nd installing bench seat", page 360](#)
- ⇒ [a5.10 nd installing floor items", page 361](#)
- ⇒ [a5.11 nd installing headliner", page 362](#)
- ⇒ [a5.12 nd installing roof items", page 363](#)

5.1 Assembly overview - box body (Plus)

- ⇒ [o5.1.1 verview - upper partition and lower partition", page 345](#)
- ⇒ [o5.1.2 verview - bench seat trim, complete", page 346](#)
- ⇒ [o5.1.3 verview - bench seat, complete", page 347](#)

5.1.1 Assembly overview - upper partition and lower partition



1 - Partition panel

- ☐ Top
- ☐ Removing and installing ⇒ [page 351](#)

2 - Rivets

- ☐ Qty. 25

3 - Connecting element

- ☐ Right

4 - Spreader rivets

- ☐ Right
- ☐ Qty. 3

5 - Partition panel

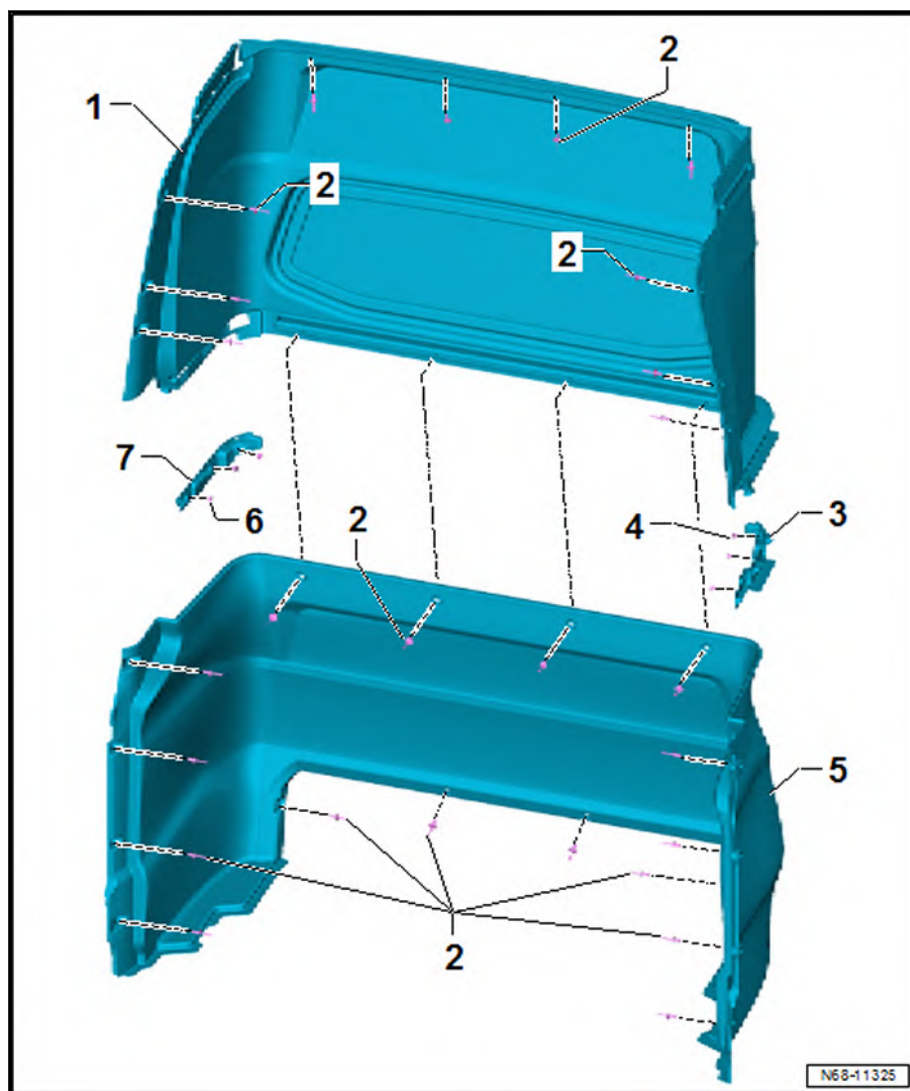
- ☐ Bottom
- ☐ Removing and installing ⇒ [page 352](#)

6 - Spreader rivets

- ☐ Left
- ☐ Qty. 3

7 - Connecting element

- ☐ Left



5.1.2 Assembly overview - bench seat trim, complete

1 - Side panel

- ☐ Left

2 - Spreader rivets

- ☐ Qty. 18

3 - Spreader rivets

- ☐ Qty. 6

4 - Rivets

- ☐ Qty. 7

5 - Bolts

- ☐ Qty. 4
- ☐ 10 Nm

6 - Washers

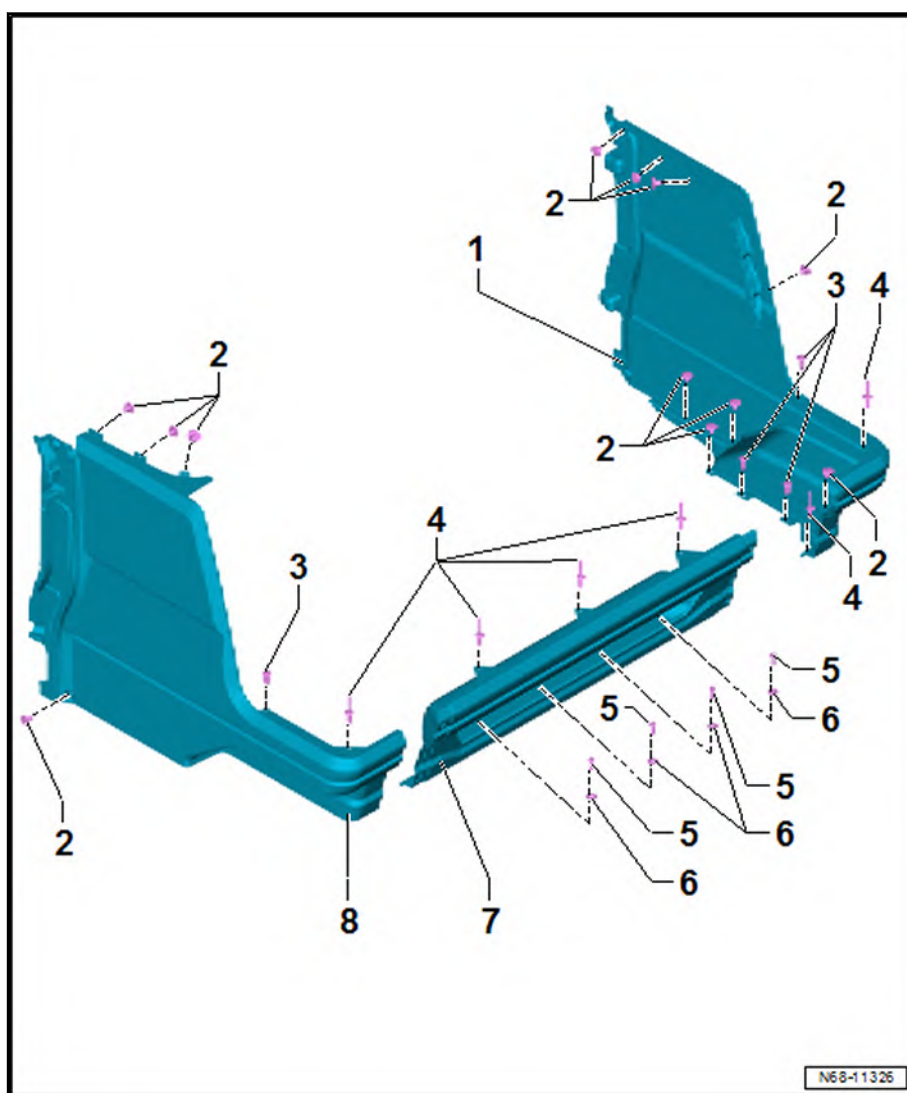
- ☐ Qty. 4

7 - Bench trim

- ☐ Front and rear
- ☐ Removing and installing ➔ [page 358](#)

8 - Side panel

- ☐ Right



5.1.3 Assembly overview - bench seat, complete



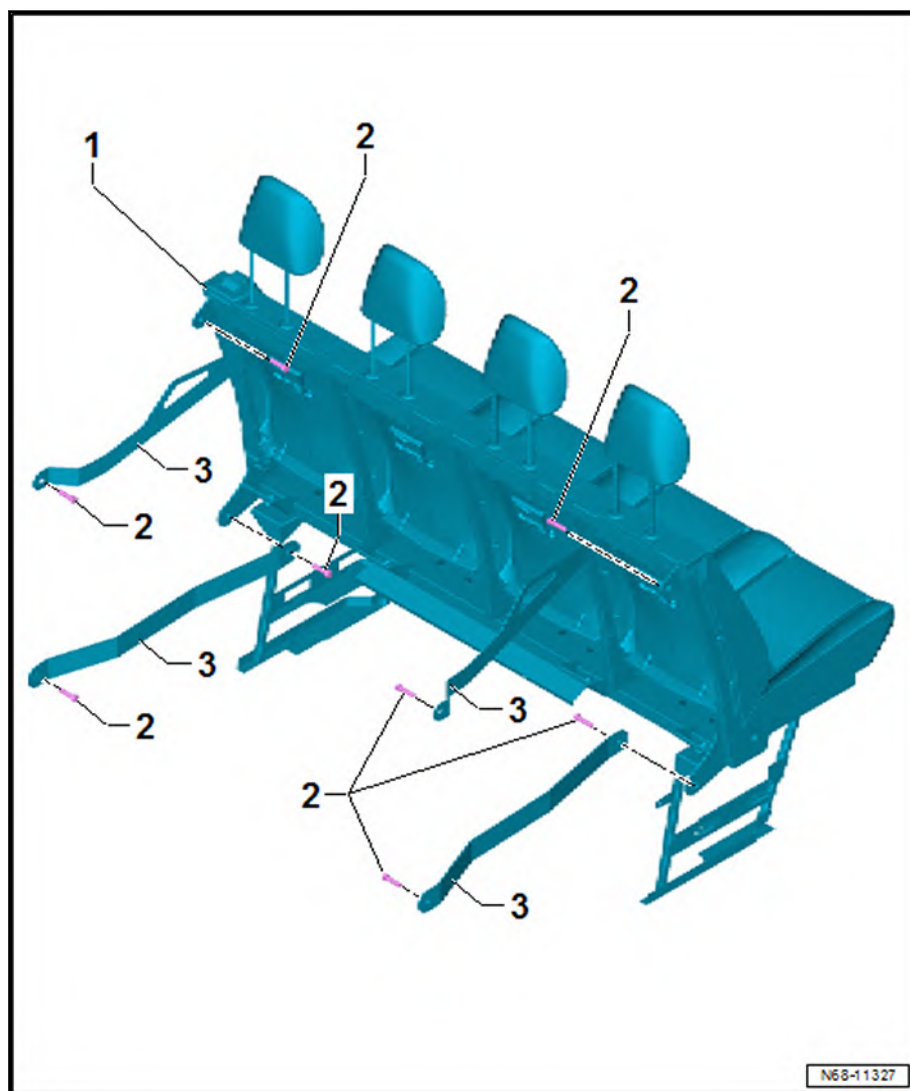
1 - Bench seat

2 - Bolts

- ☐ Qty. 8
- ☐ 100 Nm

3 - Connections

- ☐ Side panel
- ☐ Qty. 4
- ☐ Removing and installing ⇒ [page 353](#)



5.1.4 Assembly overview - floor, complete

1 - Step

- ☐ Right side panel

2 - Step

- ☐ Left sliding door

3 - Bolts

- ☐ Qty. 3 or 6
- ☐ 10 Nm

4 - Step

- ☐ Left side panel

5 - Spreader rivets

- ☐ Qty. 2 or 4

6 - Rivets

- ☐ Qty. 4

7 - Floor panel

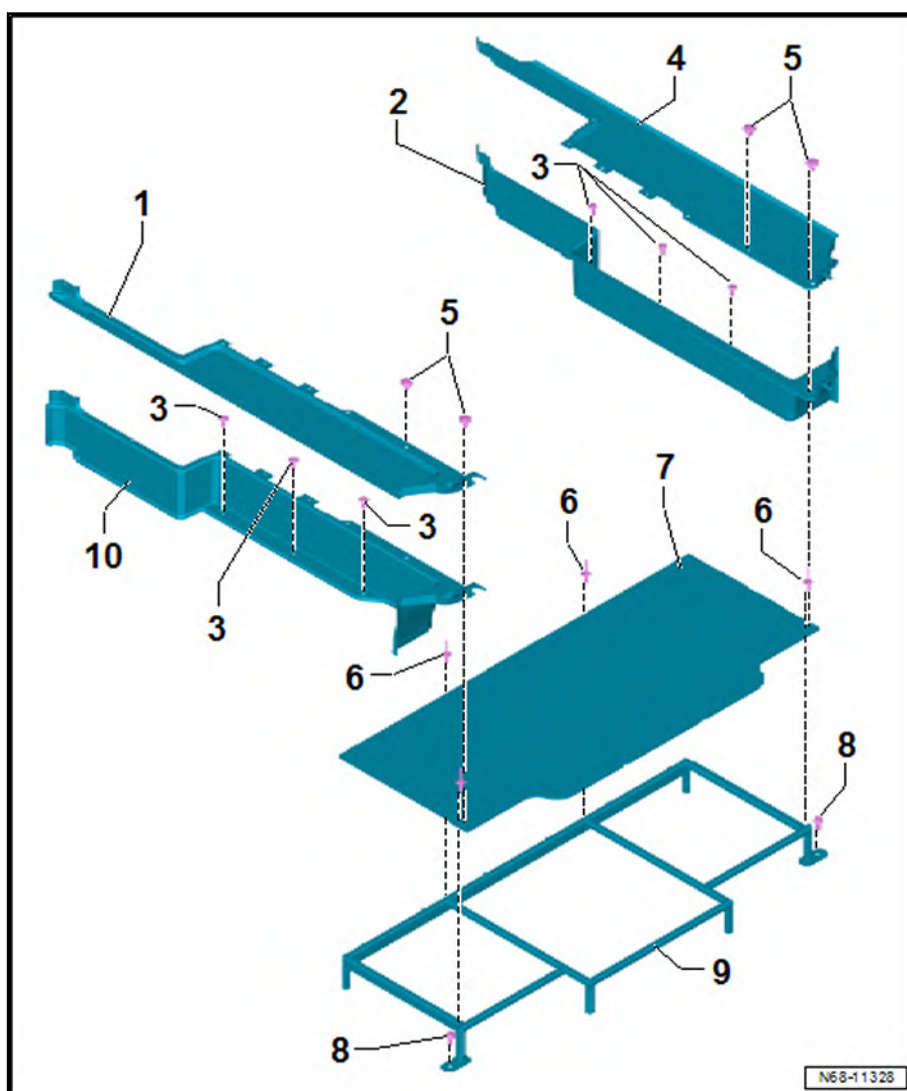
8 - Bolts

- ☐ Qty. 2
- ☐ 20 Nm

9 - Frame

10 - Step

- ☐ Right sliding door



5.1.5 Assembly overview - roof, complete



1 - Side panel

- ☐ Roof items, left

2 - Spreader rivets

- ☐ Left and right
- ☐ Qty. 4

3 - Headliner

4 - Side panel

- ☐ Roof items, right

5 - Rivets

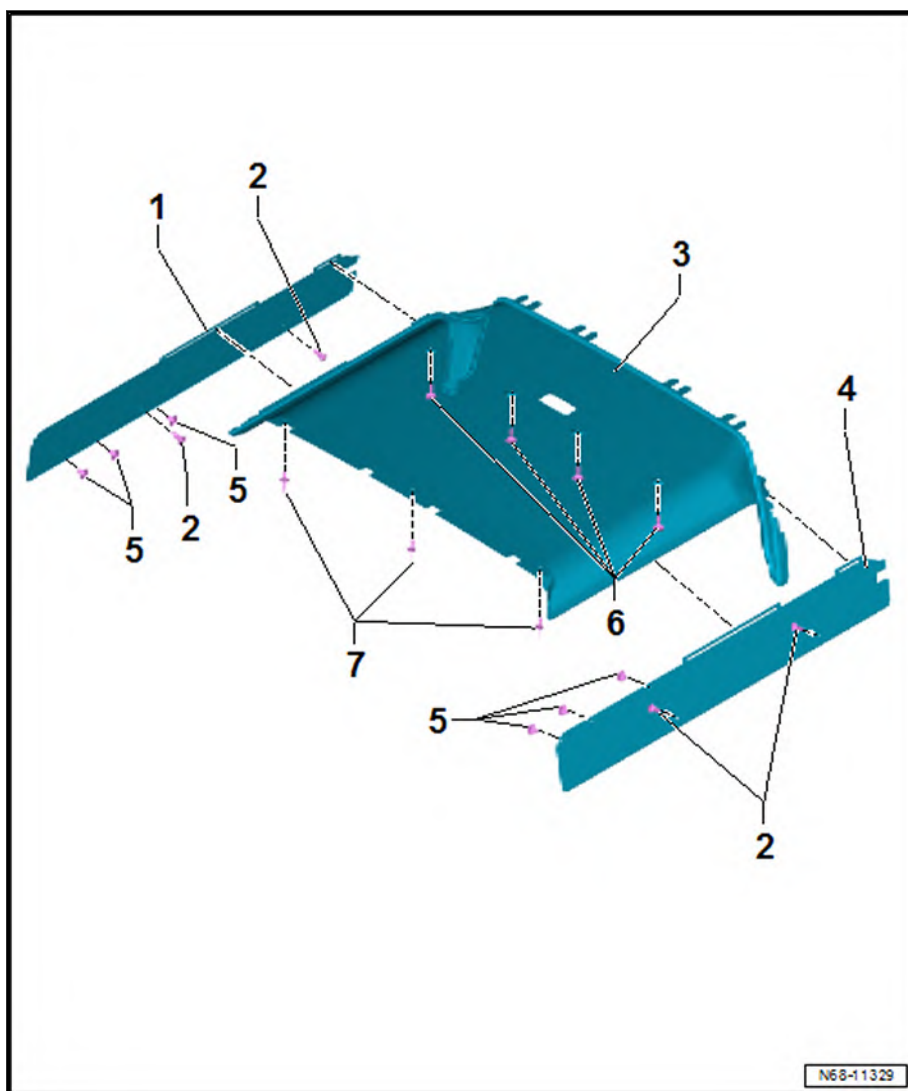
- ☐ Left and right
- ☐ Qty. 6

6 - Spreader rivets

- ☐ Qty. 4

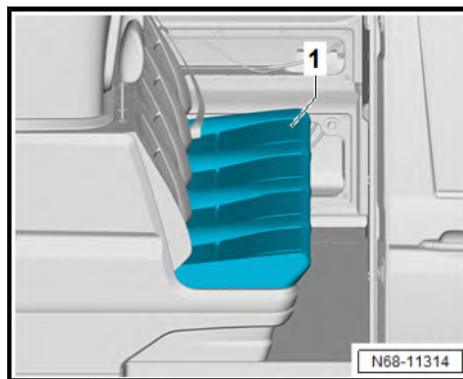
7 - Rivets

- ☐ Qty. 3



5.2 Removing and installing seat bench padding

Removing



- Lift up seat bench padding -1- at front.
- Remove seat bench padding.

Installing

Installation is carried out in reverse order.

5.3 Removing and installing upper partition

Special tools and workshop equipment required

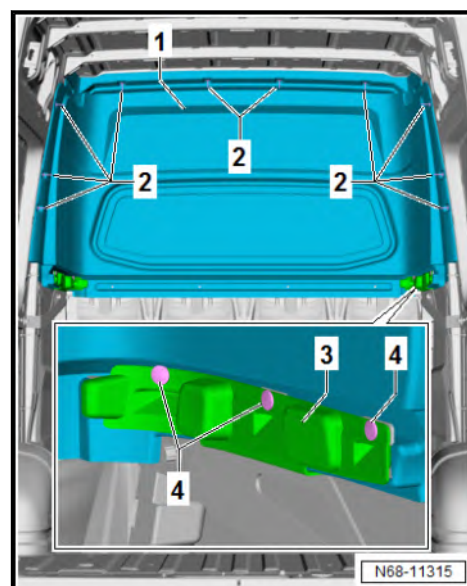
- ◆ Pop rivet pliers -VAS 5072A-



- ◆ Pin punch -VAS 5513-



Removing



- Remove rivet caps.



⚠ CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out rivet heads -2- using drill and drill bit.
- Drive out rivet shanks -2- with drift -VAS 5513-.
- Remove swarf.
- Drill out spreader rivets -4-.
- Remove connecting element -3-.
- Separate bonded joint between partition upper part -1- and partition lower part.
- Remove partition upper part -1-.

Installing

Install in reverse order of removal, observing the following:

- Use rivet pliers -VAS 5072A- to install new pop rivets.

5.4 Removing and installing lower partition

Special tools and workshop equipment required

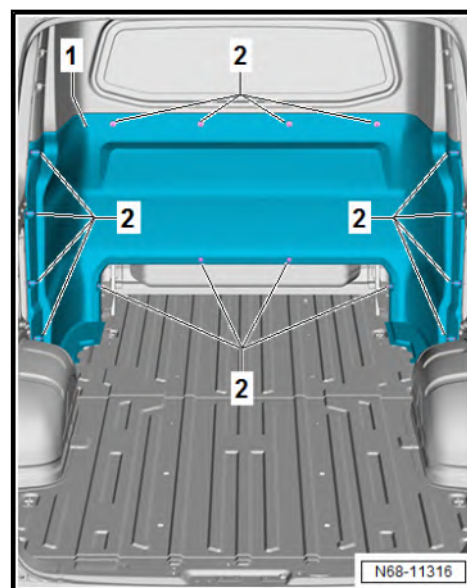
- ◆ Pop rivet pliers -VAS 5072A-



- ◆ Pin punch -VAS 5513-



Removing



- Remove rivet caps.

CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out rivet heads -2- using drill and drill bit.
- Drive out rivet shanks -2- with drift -VAS 5513-.
- Remove swarf.
- Separate bonded joint between partition lower part -1- and vehicle floor.
- Remove partition lower part -1-.

Installing

Install in reverse order of removal, observing the following:

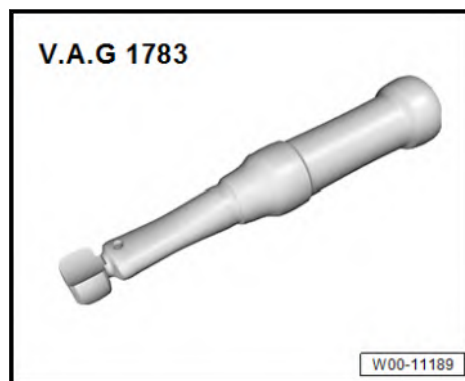
- Use rivet pliers -VAS 5072A- to install new pop rivets.

5.5 Removing and installing connection to side panel

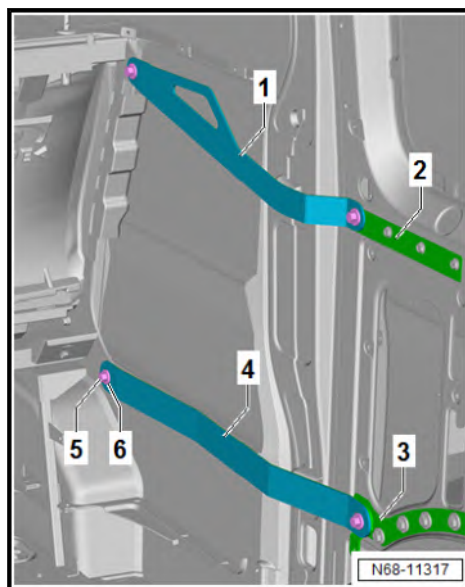
Special tools and workshop equipment required



- ◆ Torque wrench -V.A.G 1783-



Removing



- Unscrew bolts -5-.
- Remove washers -6-.
- Remove connection to side panel -1- from bracket -2-.
- Remove connection to side panel -4- from bracket -3-.

Installing

Install in reverse order of removal, observing the following:

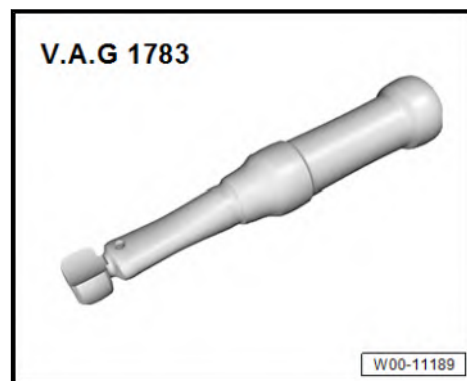
- ◆ ➔ [o5.1.3 verview - bench seat, complete", page 347](#)

5.6 Removing and installing side panel

Special tools and workshop equipment required



◆ Torque wrench -V.A.G 1783-



◆ Pop rivet pliers -VAS 5072A-

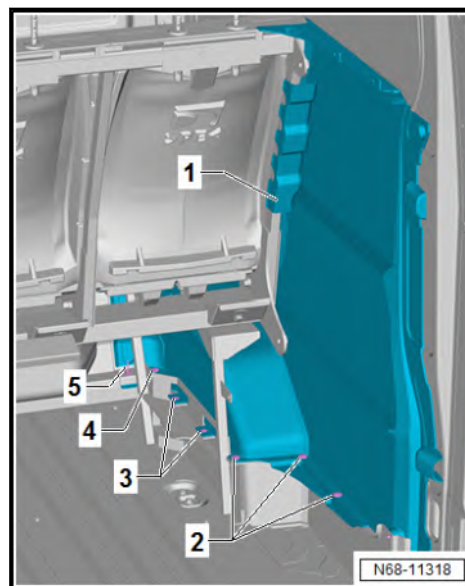


◆ Pin punch -VAS 5513-





Removing

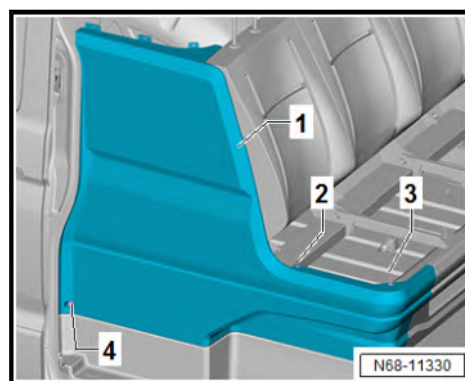


- Remove rivet caps.

CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
 - Wear protective gloves!
-
- Drill out rivet head -5- using drill and drill bit.
 - Drive out rivet shank -5- with drift -VAS 5513-.
 - Remove swarf.
 - Unscrew bolts -2-.
 - Unscrew bolts -3-.
 - Unscrew bolts -4-.
 - Remove rivet caps.



- Drill out rivet head -4- using drill and drill bit.
- Drive out rivet shank -4- with drift -VAS 5513-.
- Remove swarf.
- Unscrew bolts -2-.



- Unscrew bolts -3-.
- Remove side panel -1-.

Installing

Install in reverse order of removal, observing the following:

- Use rivet pliers -VAS 5072A- to install new pop rivets.

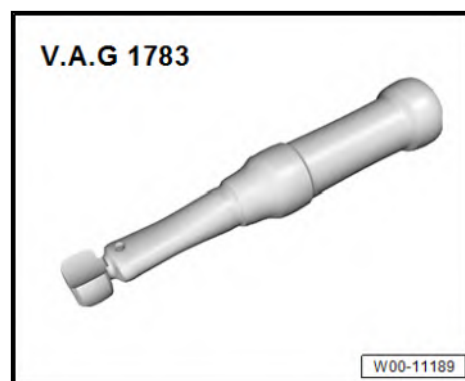
Specified torques

- ◆ ⇒ [o5.1.2 verview - bench seat trim, complete", page 346](#)

5.7 Removing and installing sliding door step

Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1783-



- ◆ Pop rivet pliers -VAS 5072A-

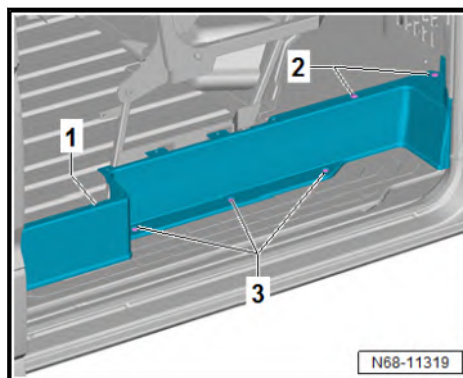


- ◆ Pin punch -VAS 5513-





Removing



- Remove rivet caps.



CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out rivet heads -2- using drill and drill bit.
- Drive out rivet shanks -2- with drift -VAS 5513-.
- Unscrew bolts -3-.
- Remove sliding door step -1-.

Installing

Install in reverse order of removal, observing the following:

- Use rivet pliers -VAS 5072A- to install new pop rivets.

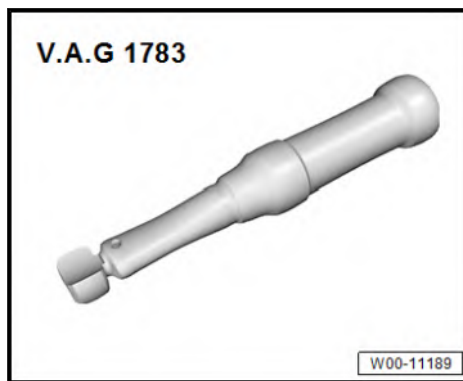
Specified torques

- ♦ ➔ [o5.1.4 verview - floor, complete", page 348](#)

5.8 Removing and installing front and rear bench trim

Special tools and workshop equipment required

- ♦ Torque wrench -V.A.G 1783-



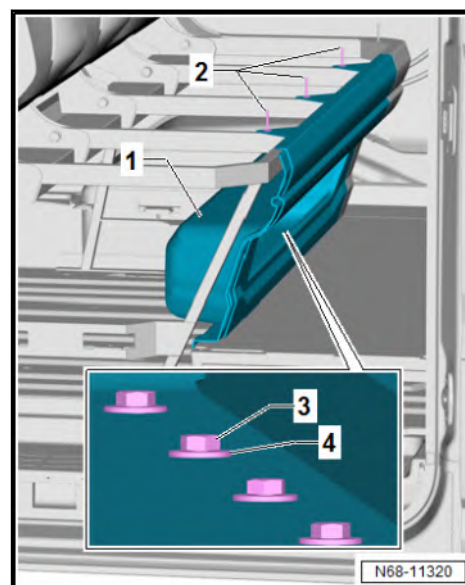
◆ Pop rivet pliers -VAS 5072A-



◆ Pin punch -VAS 5513-



Removing



- Remove rivet caps.

CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
 - Wear protective gloves!
-
- Drill out rivet heads -2- using drill and drill bit.
 - Drive out rivet shanks -2- with drift -VAS 5513-.



- Remove swarf.
- Unscrew bolts -3-.
- Remove washers -4-.
- Remove bench trim -1-.

Installing

Install in reverse order of removal, observing the following:

- Use rivet pliers -VAS 5072A- to install new pop rivets.

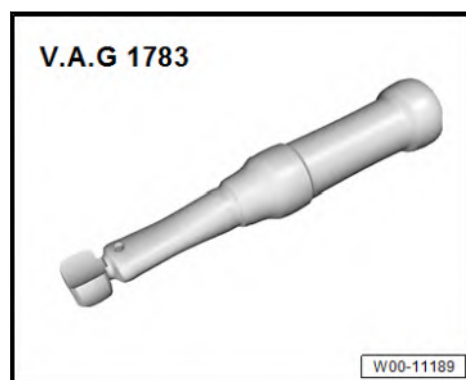
Specified torques

- ♦ ➔ [o5.1.2 verview - bench seat trim, complete", page 346](#)

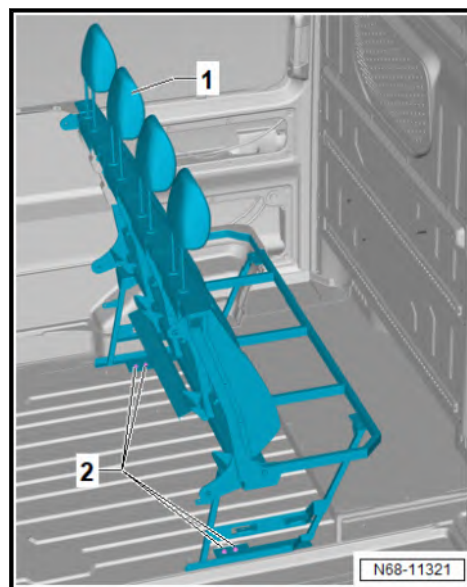
5.9 Removing and installing bench seat

Special tools and workshop equipment required

- ♦ Torque wrench -V.A.G 1783-



Removing



- Unscrew bolts -2-.
- Remove bench seat -1-.

Installing

Installation is carried out in reverse order.

Specified torques

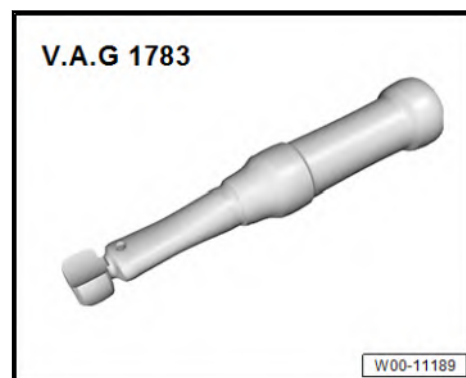
- ♦ ➔ [o5.1.3 verview - bench seat, complete", page 347](#)



5.10 Removing and installing floor items

Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1783-



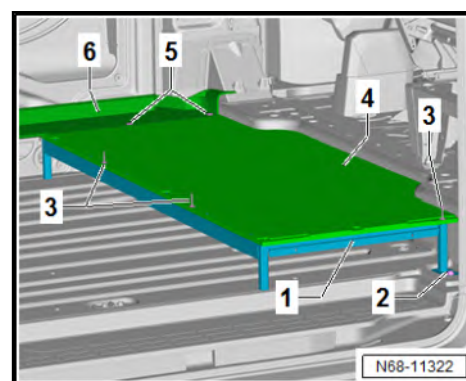
- ◆ Pop rivet pliers -VAS 5072A-



- ◆ Pin punch -VAS 5513-



Removing



- Unscrew bolts -2-.



- Remove rivet caps.

CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out rivet heads -3- and -5- using drill and drill bit.
- Drive out rivet shanks -3- and -5- with drift -VAS 5513-.
- Remove swarf.
- Remove step fixed side panel -6-.
- Remove base plate -4-.
- Remove frame -1-.

Installing

Install in reverse order of removal, observing the following:

- Use rivet pliers -VAS 5072A- to install new pop rivets.

Specified torques

- ♦ [⇒ o5.1.4 verview - floor, complete", page 348](#)

5.11 Removing and installing headliner

Special tools and workshop equipment required

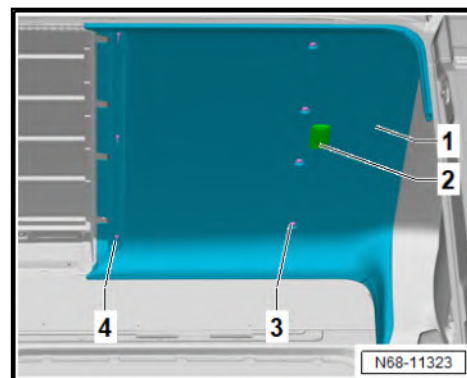
- ♦ Pop rivet pliers -VAS 5072A-



- ♦ Pin punch -VAS 5513-



Removing



- Unclip load compartment lighting -2-.
- Disconnect electrical connector of lighting.
- Remove load compartment light -2-.
- Remove rivet caps.

CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out rivet heads -3- and -4- using drill and drill bit.
- Drive out rivet shanks -3- and -4- with drift -VAS 5513-.
- Remove swarf.
- Remove headliner -1-.

Installing

Install in reverse order of removal, observing the following:

- Use rivet pliers -VAS 5072A- to install new pop rivets.

5.12 Removing and installing roof items

Special tools and workshop equipment required

- ◆ Pop rivet pliers -VAS 5072A-

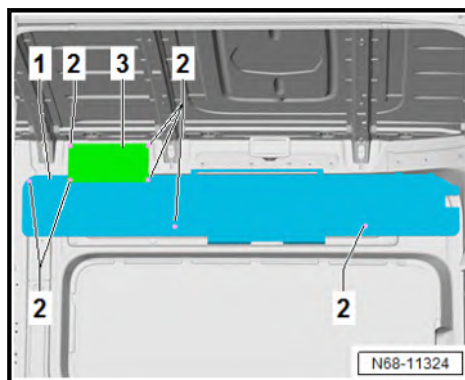




◆ Pin punch -VAS 5513-



Removing



- Remove rivet caps.



CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
 - Wear protective gloves!
-
- Drill out rivet heads -2- using drill and drill bit.
 - Drive out rivet shanks -2- with drift -VAS 5513-.
 - Remove swarf.
 - Remove roof side panel -1-.
 - Remove plate -3-.

Installing

Install in reverse order of removal, observing the following:

- Use rivet pliers -VAS 5072A- to install new pop rivets.



6 Tool cabinet

⇒ [o6.1 verview - tool cabinet", page 365](#)

⇒ [a6.2 nd installing tool cabinet", page 367](#)

6.1 Assembly overview - tool cabinet

⇒ [o6.1.1 verview - tool cabinet, Caddy, short wheelbase", page 365](#)

⇒ [o6.1.2 verview - tool cabinet, Caddy, long wheelbase", page 366](#)

6.1.1 Assembly overview - tool cabinet, Caddy, short wheelbase



1 - Retaining bracket

- Qty. 4

2 - Nuts

- Qty. 3
- 25 Nm

3 - T-slot plates

- Qty. 3

4 - Nuts

- Qty. 6
- 10 Nm

5 - Retaining bracket

6 - Tool cabinet

- Removing and installing ⇒ [page 367](#)

7 - Bracket

8 - Bolts

- Qty. 4
- 25 Nm

9 - Plates

- Qty. 4

10 - Bolt

- 10 Nm

11 - Bolts

- Qty. 2
- 10 Nm

12 - Nuts

- Qty. 2
- 25 Nm

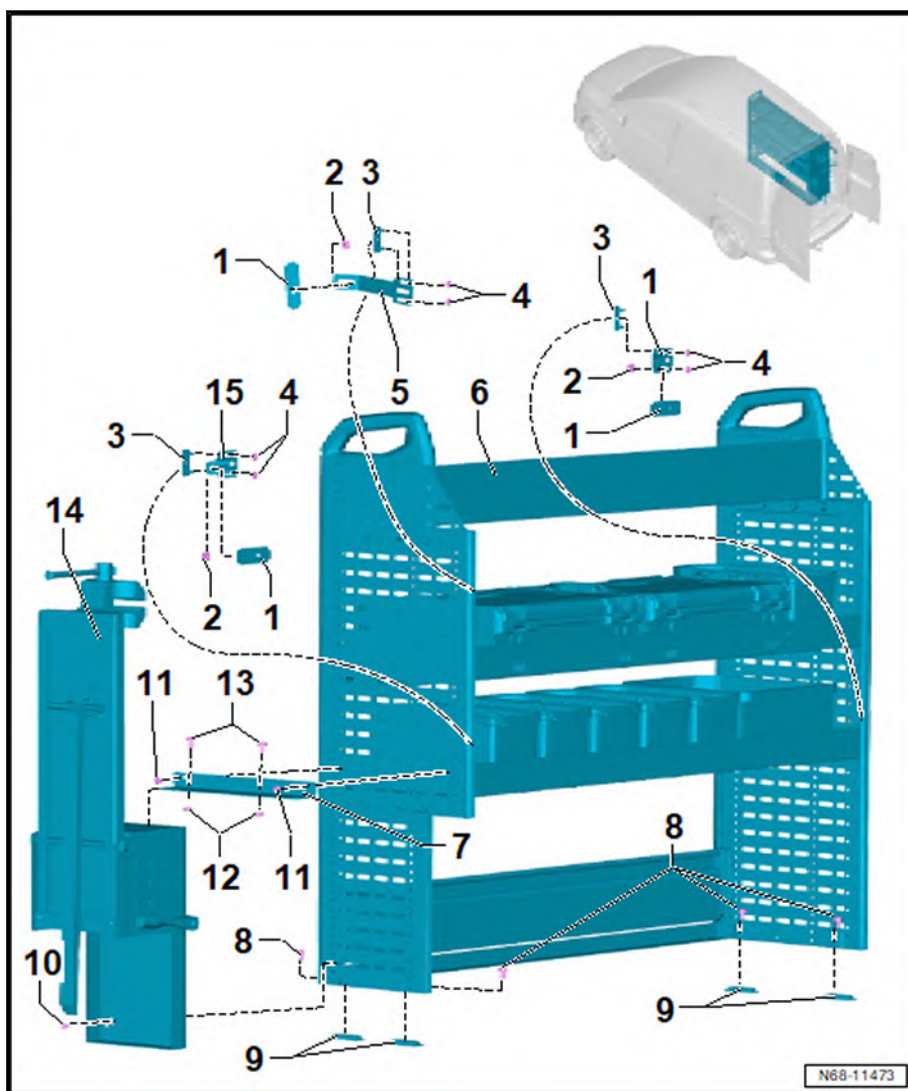
13 - Bolts

- Qty. 2
- 25 Nm

14 - Workbench

- With vice
- Removing and installing ⇒ [page 372](#)

15 - Retaining bracket



6.1.2 Assembly overview - tool cabinet, Caddy, long wheelbase

1 - Tool cabinet

- ☐ Removing and installing ➔ [page 367](#)

2 - Nuts

- ☐ Qty. 22
- ☐ 10 Nm

3 - Retaining bracket

4 - Nuts

- ☐ Qty. 2
- ☐ 25 Nm

5 - Retaining bracket

- ☐ Qty. 2

6 - T-slot plates

- ☐ Number depends on equipment level

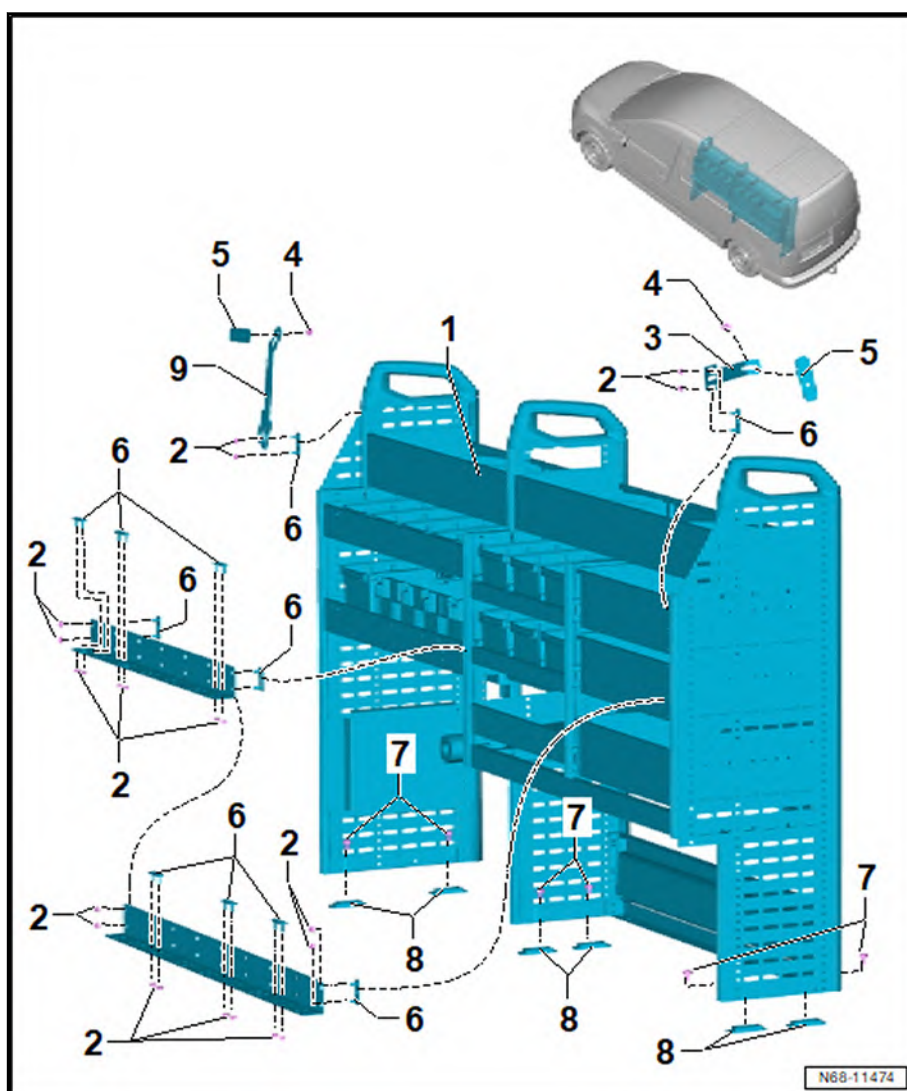
7 - Bolts

- ☐ Qty. 6
- ☐ 25 Nm

8 - Plates

- ☐ Qty. 6

9 - Retaining bracket



6.2 Removing and installing tool cabinet

➔ [a6.2.1 nd installing tool cabinet, Caddy, short wheelbase", page 367](#)

➔ [a6.2.2 nd installing tool cabinet, Caddy, long wheelbase", page 369](#)

➔ [a6.2.3 nd installing tool cabinet, Caddy, long wheelbase, right-hand side", page 370](#)

➔ [a6.2.4 nd installing workbench with vice", page 372](#)

6.2.1 Removing and installing tool cabinet, Caddy, short wheelbase

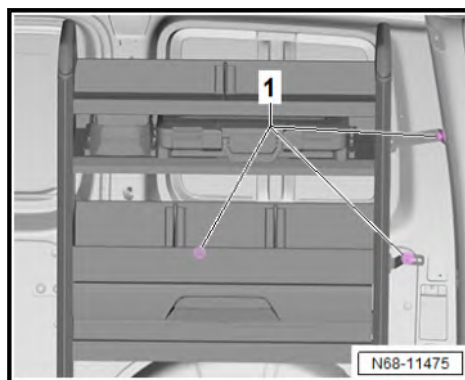
Special tools and workshop equipment required



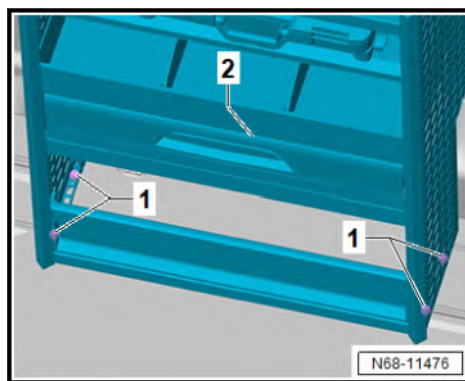
- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



Removing



- Remove drawer from tool cabinet.
- Unscrew nuts -1-.
- Unscrew bolts -1-.



- The following work steps require the assistance of a second mechanic.
- With the aid of a second mechanic, lift tool cabinet -2- out of vehicle.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [o6.1.1 verview - tool cabinet, Caddy, short wheelbase", page 365](#)



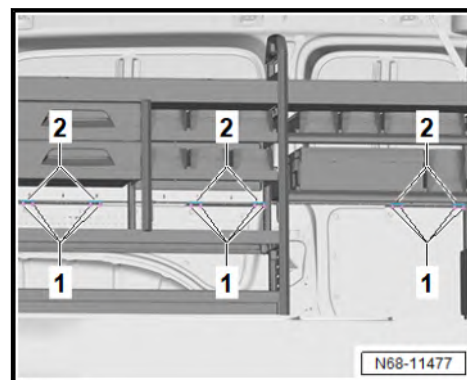
6.2.2 Removing and installing tool cabinet, Caddy, long wheelbase

Special tools and workshop equipment required

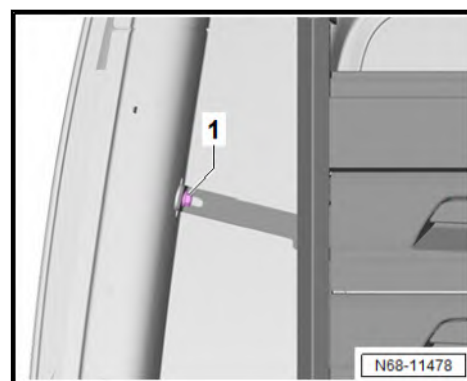
- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



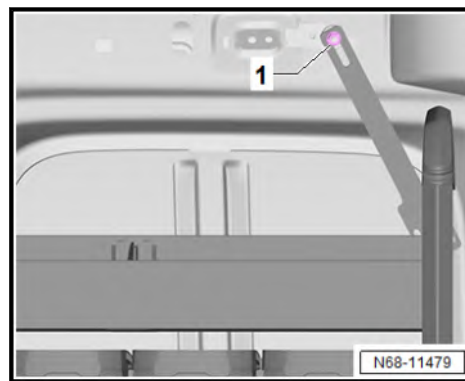
Removing



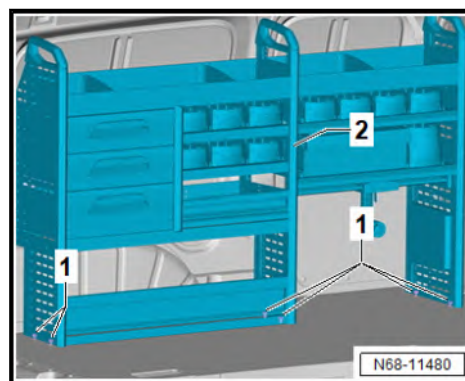
- Remove drawer from tool cabinet.
- Unscrew nuts -1-.
- Remove T-slot plates -2-.
- Unscrew nut -1-.



- Unscrew nut -1-.



- Unscrew bolts -1-.



- The following work steps require the assistance of a second mechanic.
- With the aid of a second mechanic, lift tool cabinet -2- out of vehicle.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ➔ [o6.1.2 verview - tool cabinet, Caddy, long wheelbase", page 366](#)

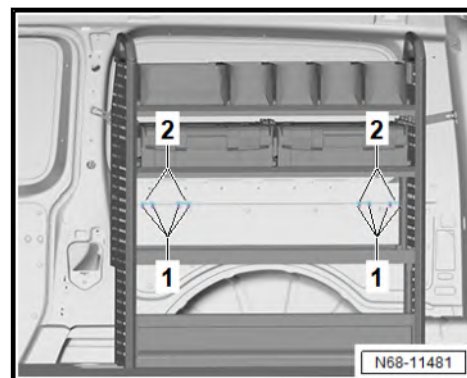
6.2.3 Removing and installing tool cabinet, Caddy, long wheelbase, right-hand side

Special tools and workshop equipment required

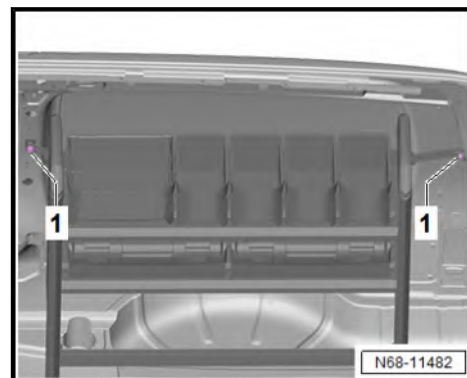
- ♦ Torque wrench 6-50 Nm -V.A.G 1331-



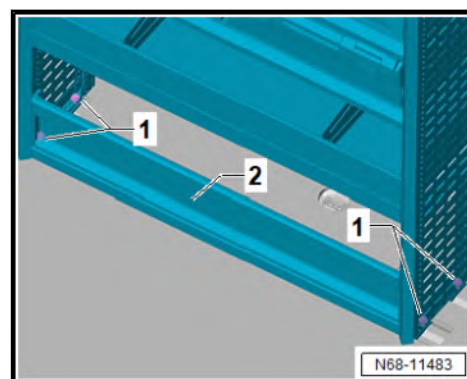
Removing



- Remove drawer from tool cabinet.
- Unscrew nuts -1-.
- Remove T-slot plates -2-.
- Unscrew nuts -1-.



- Unscrew bolts -1-.



- The following work steps require the assistance of a second mechanic.
- With the aid of a second mechanic, lift tool cabinet -2- out of vehicle.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➤ [o6.1.2 verview - tool cabinet, Caddy, long wheelbase", page 366](#)



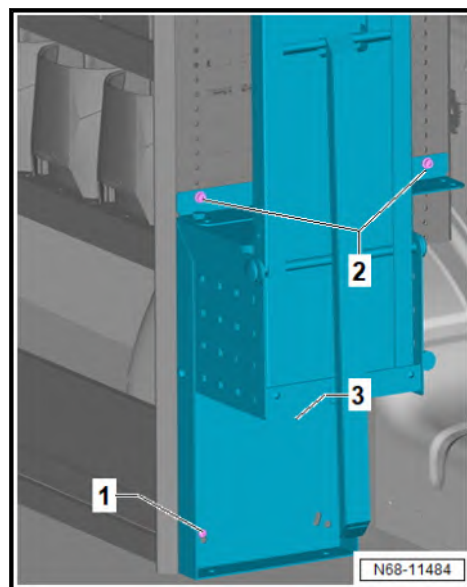
6.2.4 Removing and installing workbench with vice

Special tools and workshop equipment required

- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



Removing



- Unscrew bolt -1-.
- Unscrew bolts -2-.
- The following work steps require the assistance of a second mechanic.
- With the aid of a second mechanic, lift workbench -3- out of vehicle.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [o6.1 verview - tool cabinet", page 365](#)

7 Fastening rails

⇒ [o7.1 verview - fastening rails", page 373](#)

⇒ [a7.2 nd installing fastening rail", page 374](#)

7.1 Assembly overview - fastening rails

1 - Pop rivets

- ☐ Number depends on equipment level

2 - Fastening rail

- ☐ Left
- ☐ Caddy, short wheelbase
- ☐ Removing and installing ⇒ [page 374](#)

3 - Fastening rail

- ☐ Right
- ☐ Caddy, long wheelbase
- ☐ Removing and installing ⇒ [page 377](#)

4 - Fastening rail

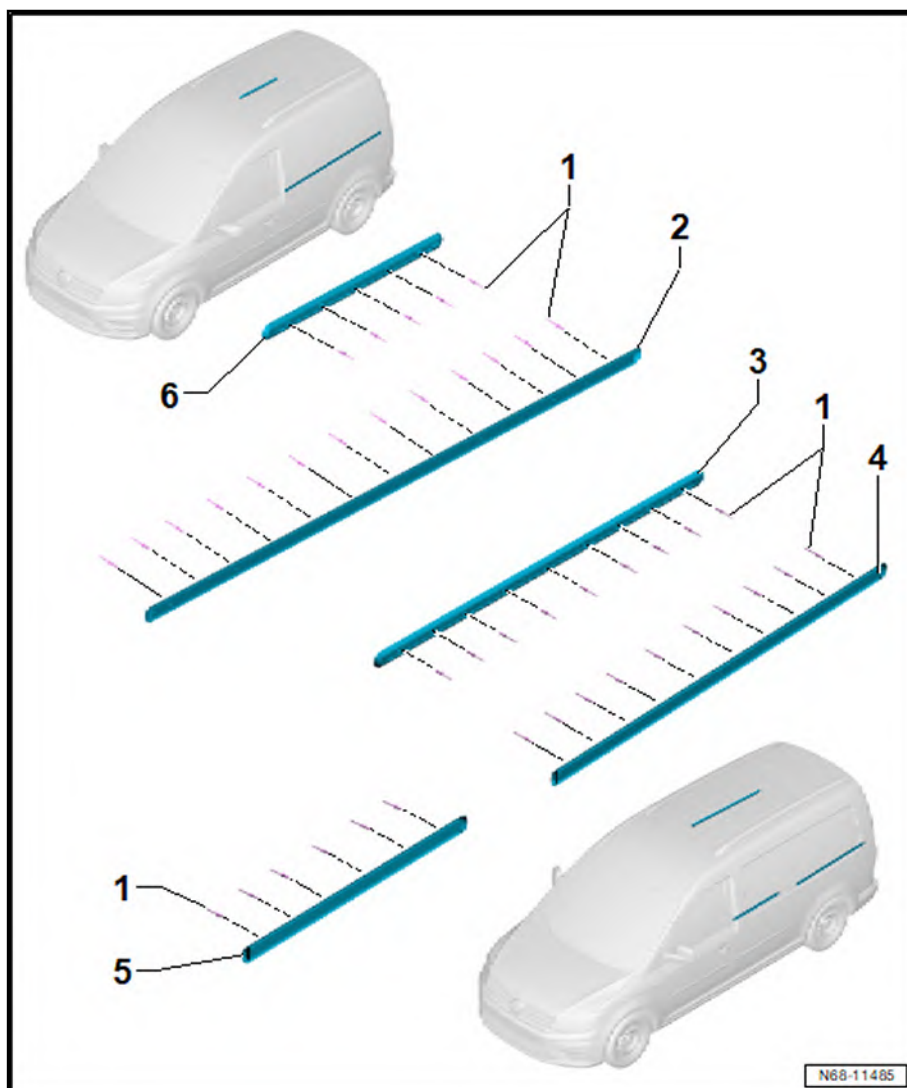
- ☐ Left
- ☐ Caddy, long wheelbase
- ☐ Removing and installing ⇒ [page 375](#)

5 - Fastening rail

- ☐ Left
- ☐ Caddy, long wheelbase
- ☐ Removing and installing ⇒ [page 375](#)

6 - Fastening rail

- ☐ Right
- ☐ Caddy, short wheelbase
- ☐ Removing and installing ⇒ [page 376](#)





7.2 Removing and installing fastening rail

⇒ [a7.2.1 nd installing fastening rail, Caddy, short wheelbase, left-hand side", page 374](#)

⇒ [a7.2.2 nd installing fastening rail, Caddy, long wheelbase, left-hand side", page 375](#)

⇒ [a7.2.3 nd installing fastening rail, Caddy, short wheelbase, right-hand side", page 376](#)

⇒ [a7.2.4 nd installing fastening rail, Caddy, long wheelbase, right-hand side", page 377](#)

7.2.1 Removing and installing fastening rail, Caddy, short wheelbase, left-hand side

Special tools and workshop equipment required

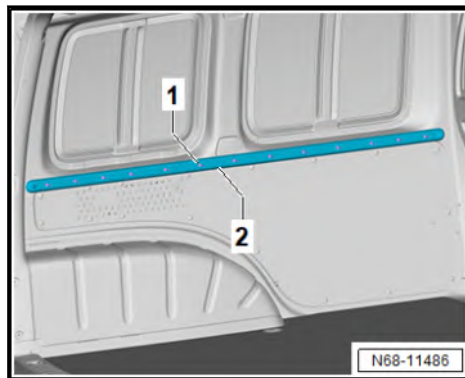
◆ Pop rivet pliers -VAS 5072A-



◆ Pin punch -VAS 5513-



Removing



⚠ CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out pop rivets -1-.
- Use roll pin drift -VAS 5513- to remove pop rivets -1-.
- Remove fastening rail -2-.

Installing

Install in reverse order of removal, observing the following:

- Drive in pop rivets using pop rivet pliers -VAS 5072A-.

7.2.2 Removing and installing fastening rail, Caddy, long wheelbase, left-hand side

Special tools and workshop equipment required

- ◆ Pop rivet pliers -VAS 5072A-

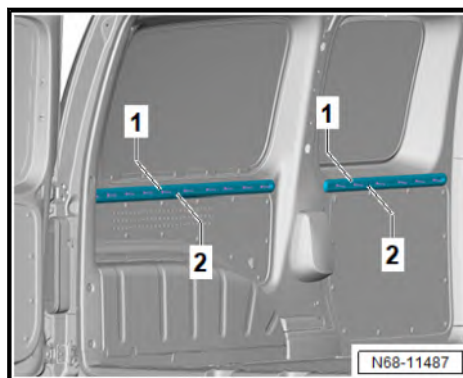


- ◆ Pin punch -VAS 5513-





Removing



CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out pop rivets -1-.
- Use roll pin drift -VAS 5513- to remove pop rivets -1-.
- Remove fastening rails -2-.

Installing

Install in reverse order of removal, observing the following:

- Drive in pop rivets using pop rivet pliers -VAS 5072A-.

7.2.3 Removing and installing fastening rail, Caddy, short wheelbase, right-hand side

Special tools and workshop equipment required

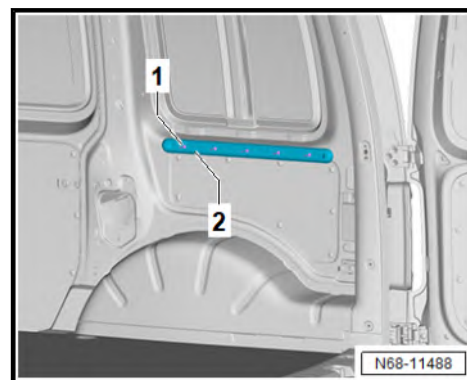
- ◆ Pop rivet pliers -VAS 5072A-



◆ Pin punch -VAS 5513-



Removing



CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out pop rivets -1-.
- Use roll pin drift -VAS 5513- to remove pop rivets -1-.
- Remove fastening rail -2-.

Installing

Install in reverse order of removal, observing the following:

- Drive in pop rivets using pop rivet pliers -VAS 5072A-.

7.2.4 Removing and installing fastening rail, Caddy, long wheelbase, right-hand side

Special tools and workshop equipment required



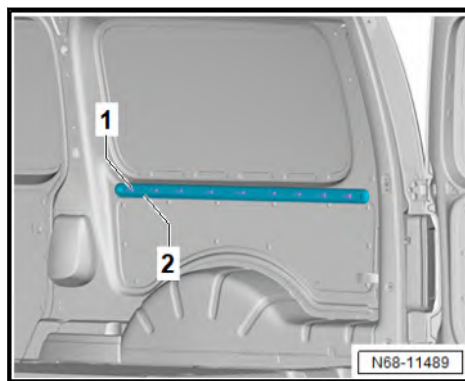
◆ Pop rivet pliers -VAS 5072A-



◆ Pin punch -VAS 5513-



Removing



CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out pop rivets -1-.
- Use roll pin drift -VAS 5513- to remove pop rivets -1-.
- Remove fastening rail -2-.

Installing

Install in reverse order of removal, observing the following:

- Drive in pop rivets using pop rivet pliers -VAS 5072A-.

8 Load bed covering, wood

⇒ [o8.1 verview - load bed covering, wood", page 379](#)

⇒ [a8.2 nd installing load bed covering, wood", page 379](#)

8.1 Assembly overview - load bed covering, wood

1 - Load bed covering

- ☐ Caddy, short wheelbase
- ☐ Removing and installing ⇒ [page 379](#)

2 - Load bed covering

- ☐ Caddy, long wheelbase
- ☐ Removing and installing ⇒ [page 381](#)

3 - Bolts

- ☐ Number depends on equipment level
- ☐ 10 Nm

4 - Spacer sleeves

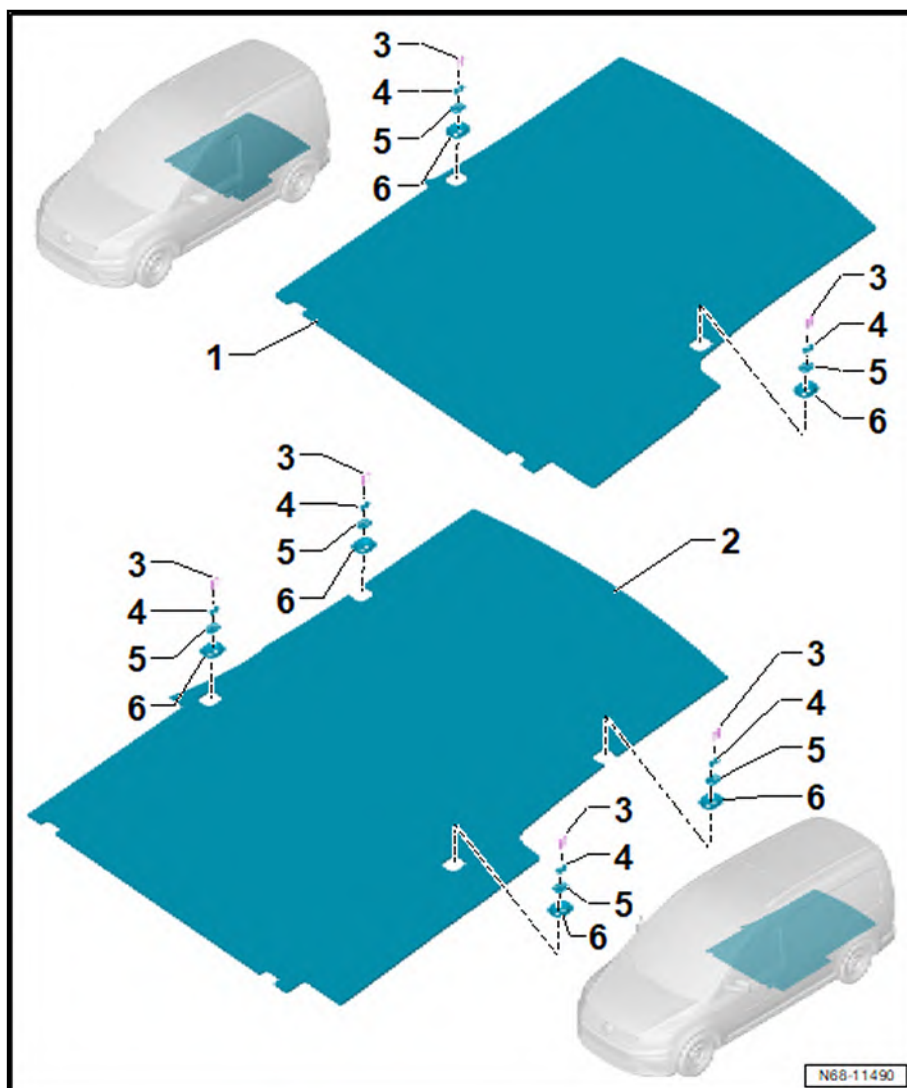
- ☐ Number depends on equipment level

5 - Fastening rings

- ☐ Number depends on equipment level

6 - Bracket

- ☐ For load bed
- ☐ Number depends on equipment level



8.2 Removing and installing load bed covering, wood

⇒ [a8.2.1 nd installing load bed covering, wood, Caddy, short wheelbase", page 379](#)

⇒ [a8.2.2 nd installing load bed covering, wood, Caddy, long wheelbase", page 381](#)

8.2.1 Removing and installing load bed covering, wood, Caddy, short wheelbase

Special tools and workshop equipment required

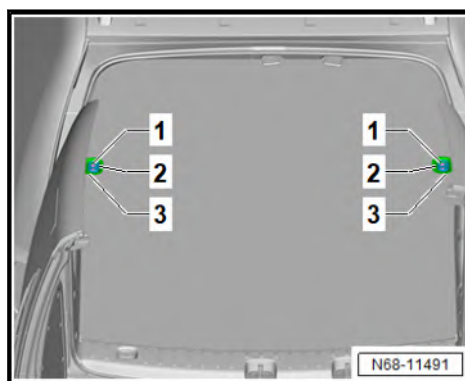


- ◆ Torque wrench 6-50 Nm -V.A.G 1331-

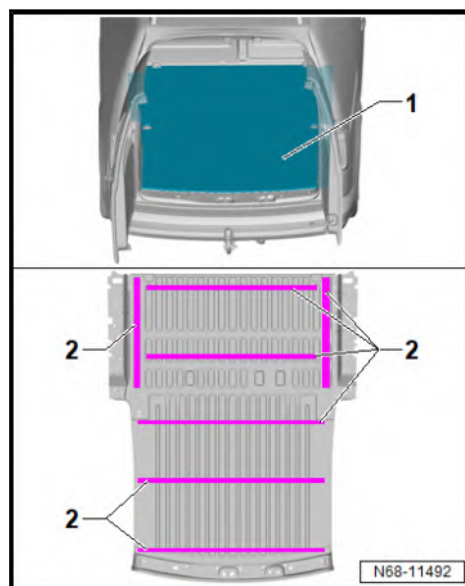


- ◆ Butyl sealing strip

Removing



- Remove lock carrier trim ⇒ General body repairs, interior; Rep. gr. 70; Luggage compartment trims; Removing and installing lock carrier trim.
- Unscrew bolts -1-.
- Remove fastening rings -2-.
- Remove bracket -3-.
- Lift load bed covering -1- together with a second mechanic. When doing this, carefully detach butyl sealing strip -2-.



- With the aid of a second mechanic, carefully remove load bed covering -1- from vehicle.

Installing

Install in reverse order of removal, observing the following:

- The bonding surfaces must be free of dust and other dirt.
- Clean bonding surfaces.
- Attach butyl sealing strip -2- in the areas shown.

Specified torques

- ◆ ⇒ [o8.1 verview - load bed covering, wood", page 379](#)

8.2.2 Removing and installing load bed covering, wood, Caddy, long wheelbase

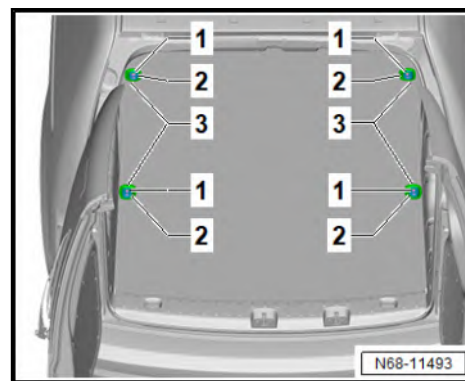
Special tools and workshop equipment required

- ◆ Torque wrench 6-50 Nm -V.A.G 1331-

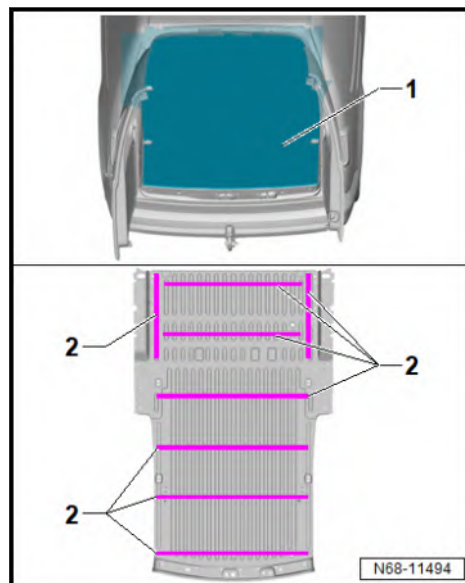


- ◆ Butyl sealing strip

Removing



- Remove lock carrier trim ⇒ General body repairs, interior; Rep. gr. 70; Luggage compartment trims; Removing and installing lock carrier trim.
- Unscrew bolts -1-.
- Remove fastening rings -2-.
- Remove bracket -3-.
- Lift load bed covering -1- together with a second mechanic. When doing this, carefully detach butyl sealing strip -2-.



- With the aid of a second mechanic, carefully remove load bed covering -1- from vehicle.

Installing

Install in reverse order of removal, observing the following:

- The bonding surfaces must be free of dust and other dirt.
- Clean bonding surfaces.
- Attach butyl sealing strip -2- in the areas shown.

Specified torques

- ♦ ➔ [o8.1 verview - load bed covering, wood", page 379](#)

9 C-pillar partition

⇒ [o9.1 verview - C-pillar partition", page 383](#)

⇒ [a9.2 nd installing C-pillar partition", page 384](#)

9.1 Assembly overview - C-pillar partition

1 - Partition panel

- ☐ Upper part
- ☐ Removing and installing ⇒ [page 384](#)

2 - Washers

- ☐ Qty. 11

3 - Bolts

- ☐ Qty. 2
- ☐ 5 Nm

4 - Caps

- ☐ Qty. 15

5 - Pop rivets

- ☐ Qty. 17

6 - Partition panel

- ☐ Lower part
- ☐ Removing and installing ⇒ [page 386](#)

7 - Assembly strut

- ☐ Removing and installing ⇒ [page 389](#)

8 - Lashing eyes

- ☐ Qty. 2

9 - Washers

- ☐ Qty. 4

10 - Bolts

- ☐ Qty. 4
- ☐ 10 Nm

11 - Bracket

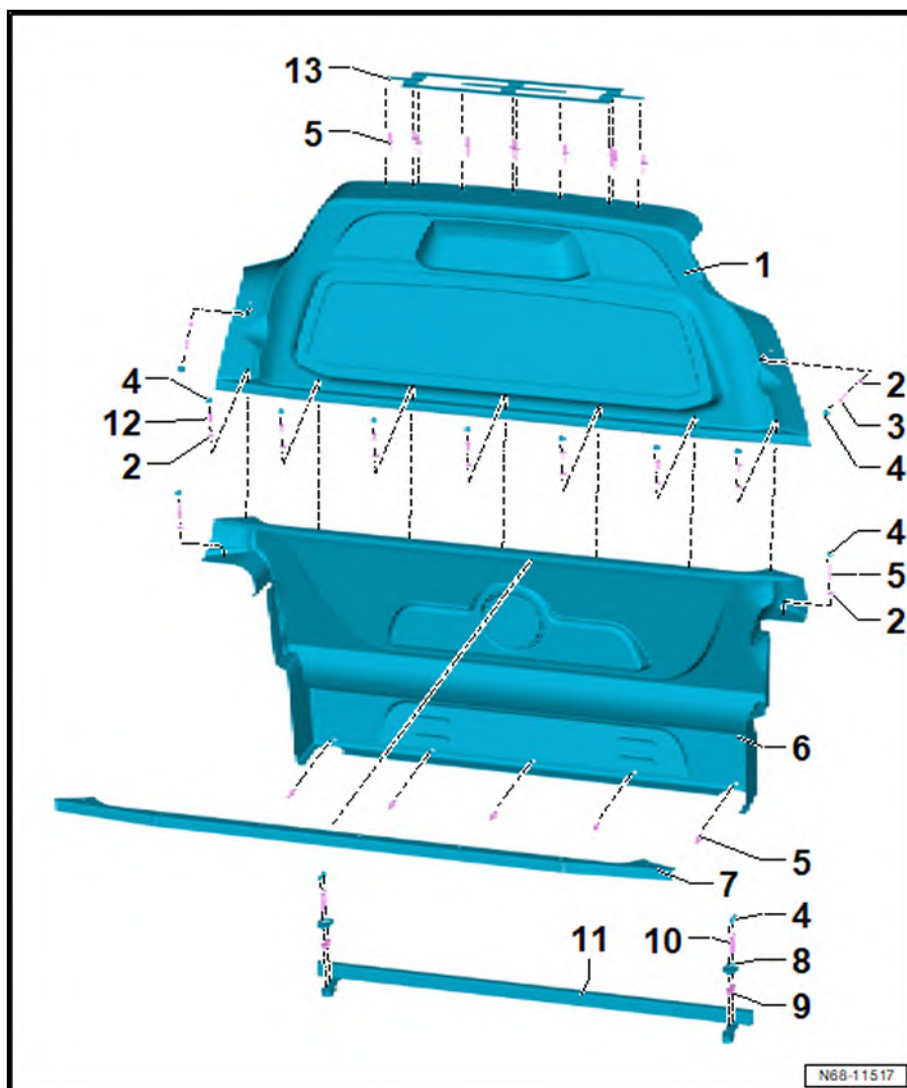
- ☐ Bottom
- ☐ Removing and installing ⇒ [page 388](#)

12 - Bolts

- ☐ Qty. 7
- ☐ 10 Nm

13 - Bracket

- ☐ Top
- ☐ Removing and installing ⇒ [page 387](#)





9.2 Removing and installing C-pillar partition

⇒ [a9.2.1 nd installing C-pillar partition, upper part", page 384](#)

⇒ [a9.2.2 nd installing C-pillar partition, lower part", page 386](#)

⇒ [a9.2.3 nd installing bracket, upper", page 387](#)

⇒ [a9.2.4 nd installing bracket, lower", page 388](#)

⇒ [a9.2.5 nd installing assembly strut", page 389](#)

9.2.1 Removing and installing C-pillar partition, upper part

Special tools and workshop equipment required

◆ Pop rivet pliers -VAS 5072A-



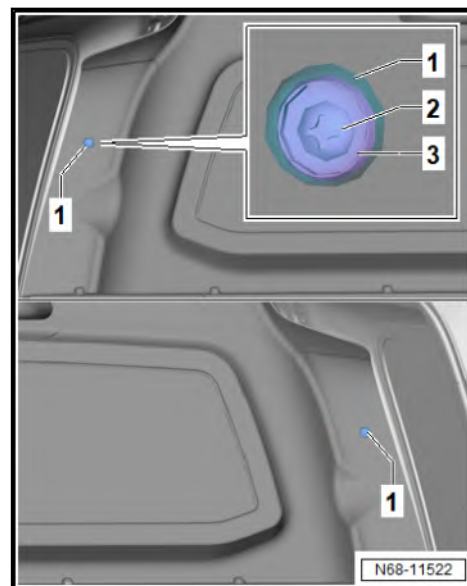
◆ Pin punch -VAS 5513-



◆ Torque wrench 6-50 Nm -V.A.G 1331-



Removing



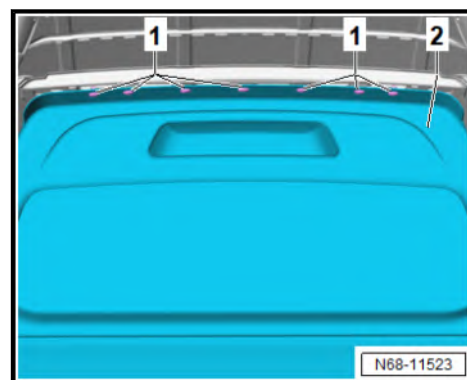
- Remove assembly strut ➤ [page 389](#) .
- Remove cover caps -1-.
- Unscrew bolts -2-. Remove washer -3- when doing this.

CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out pop rivets -1-.



- Use roll pin drift -VAS 5513- to remove pop rivets -1-.
- Remove partition panel -2-.

Installing

Install in reverse order of removal, observing the following:

- Drive in pop rivets using pop rivet pliers -VAS 5072A-.

Specified torques

- ♦ ➤ [o9.1 verview - C-pillar partition", page 383](#)



9.2.2 Removing and installing C-pillar partition, lower part

Special tools and workshop equipment required

- ♦ Pop rivet pliers -VAS 5072A-



- ♦ Pin punch -VAS 5513-



Removing

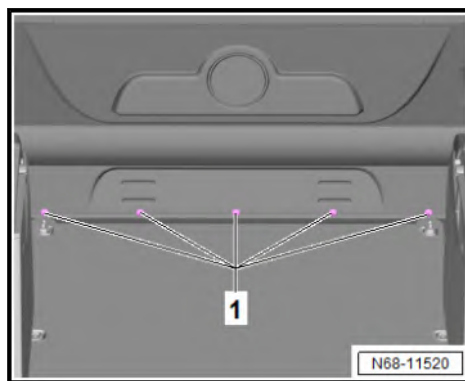
- Remove C-pillar partition ➔ [page 384](#) .

CAUTION

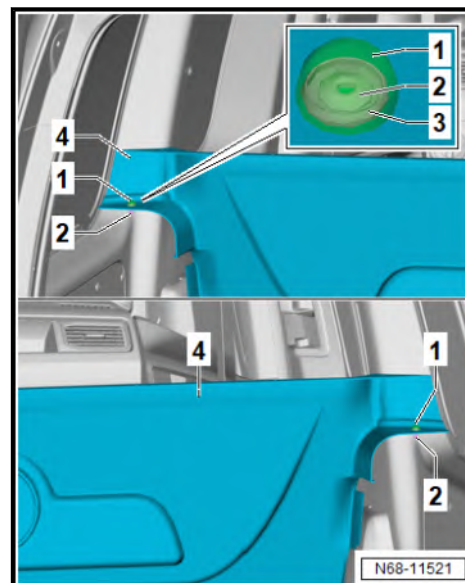
Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out pop rivets -1-.



- Use roll pin drift -VAS 5513- to remove pop rivets -1-.
- Remove cover caps -1-.



- Drill out pop rivets -2-.
- Use roll pin drift -VAS 5513- to remove pop rivets -2-. Remove washers -3-.
- Remove partition panel -4-.

Installing

Install in reverse order of removal, observing the following:

- Drive in pop rivets using pop rivet pliers -VAS 5072A-.

9.2.3 Removing and installing bracket, upper

Special tools and workshop equipment required

- ◆ Pop rivet pliers -VAS 5072A-

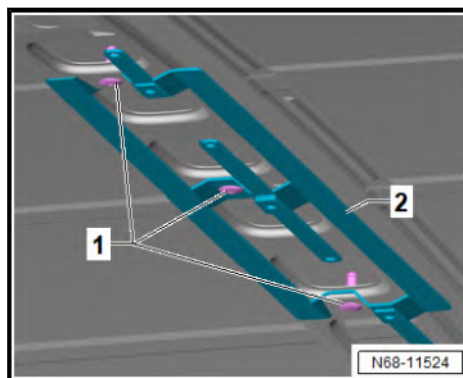


- ◆ Pin punch -VAS 5513-





Removing



- Remove C-pillar partition ⇒ [page 384](#) .

CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out pop rivets -1-.
- Use roll pin drift -VAS 5513- to remove pop rivets -1-.
- Remove retainer -2-.

Installing

Install in reverse order of removal, observing the following:

- Drive in pop rivets using pop rivet pliers -VAS 5072A-.

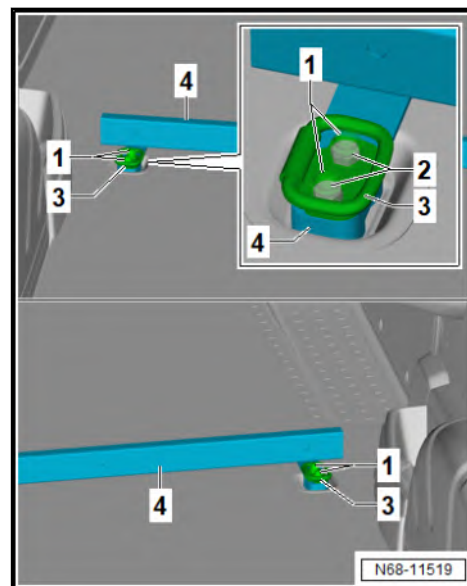
9.2.4 Removing and installing bracket, lower

Special tools and workshop equipment required

- ♦ Torque wrench 6-50 Nm -V.A.G 1331-



Removing



- Remove C-pillar partition, lower part ➤ [page 386](#) .
- Remove cover caps -1-.
- Unscrew bolts -2-.
- Remove lashing eyes -3-.
- Remove retainer -4-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➤ [o9.1 verview - C-pillar partition", page 383](#)

9.2.5 Removing and installing assembly strut

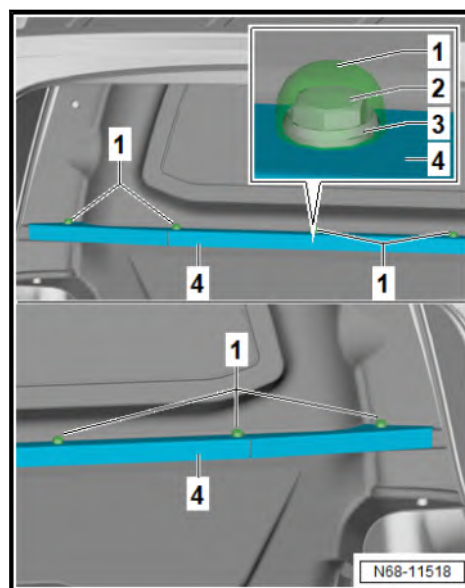
Special tools and workshop equipment required

- ◆ Torque wrench 6-50 Nm -V.A.G 1331-





Removing



- Remove cover caps -1-.
- Unscrew bolts -2-. Remove washer -3- when doing this.
- Remove assembly strut -4-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ⇒ [o9.1 verview - C-pillar partition", page 383](#)

10 Rear window protective grille

⇒ [o10.1 verview - rear window protective grille", page 391](#)

⇒ [a10.2 nd installing rear window protective grille", page 391](#)

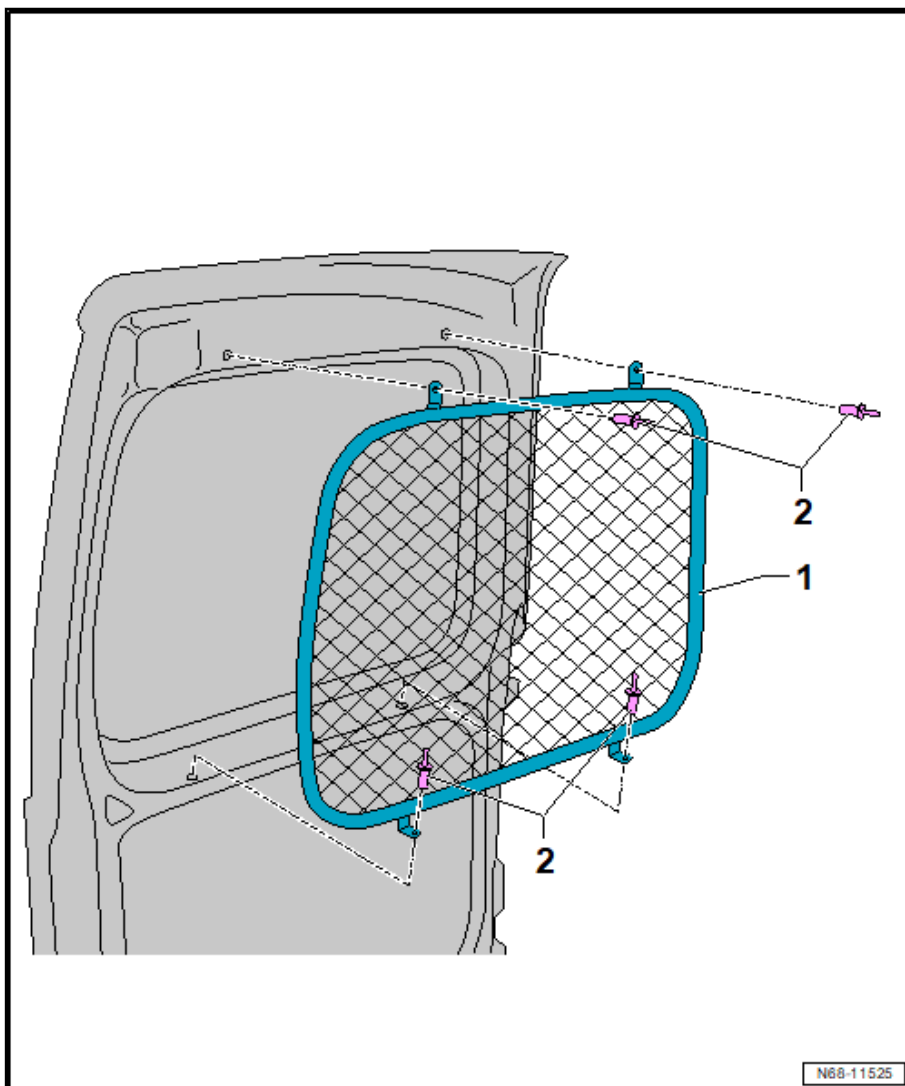
10.1 Assembly overview - rear window protective grille

1 - Protective grille

- ☐ Removing and installing ⇒ [page 391](#)

2 - Pop rivets

- ☐ Qty. 4



10.2 Removing and installing rear window protective grille

Special tools and workshop equipment required



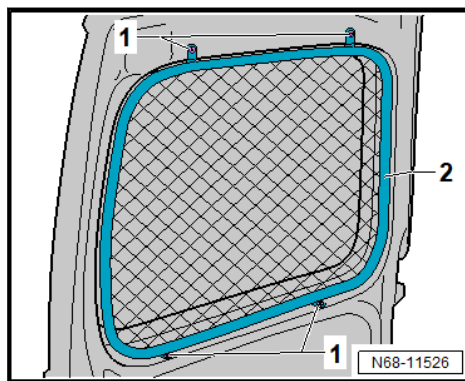
◆ Pop rivet pliers -VAS 5072A-



◆ Pin punch -VAS 5513-



Removing



CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out pop rivets -1-.
- Use roll pin drift -VAS 5513- to remove pop rivets -1-.
- Detach protective grille -2-.

Installing

Install in reverse order of removal, observing the following:

- Drive in pop rivets using pop rivet pliers -VAS 5072A-.

11 Tool box for load bed

⇒ [o11.1 verview - tool box", page 393](#)

⇒ [a11.2 nd installing tool box", page 393](#)

11.1 Assembly overview - tool box

1 - Nuts

- ☐ Qty. 2
- ☐ 20 Nm

2 - Bolts

- ☐ Qty. 8
- ☐ 20 Nm

3 - Bracket

- ☐ Qty. 2

4 - Bracket

- ☐ Qty. 2

5 - Tool box

- ☐ Removing and installing ⇒ [page 393](#)

6 - Plates

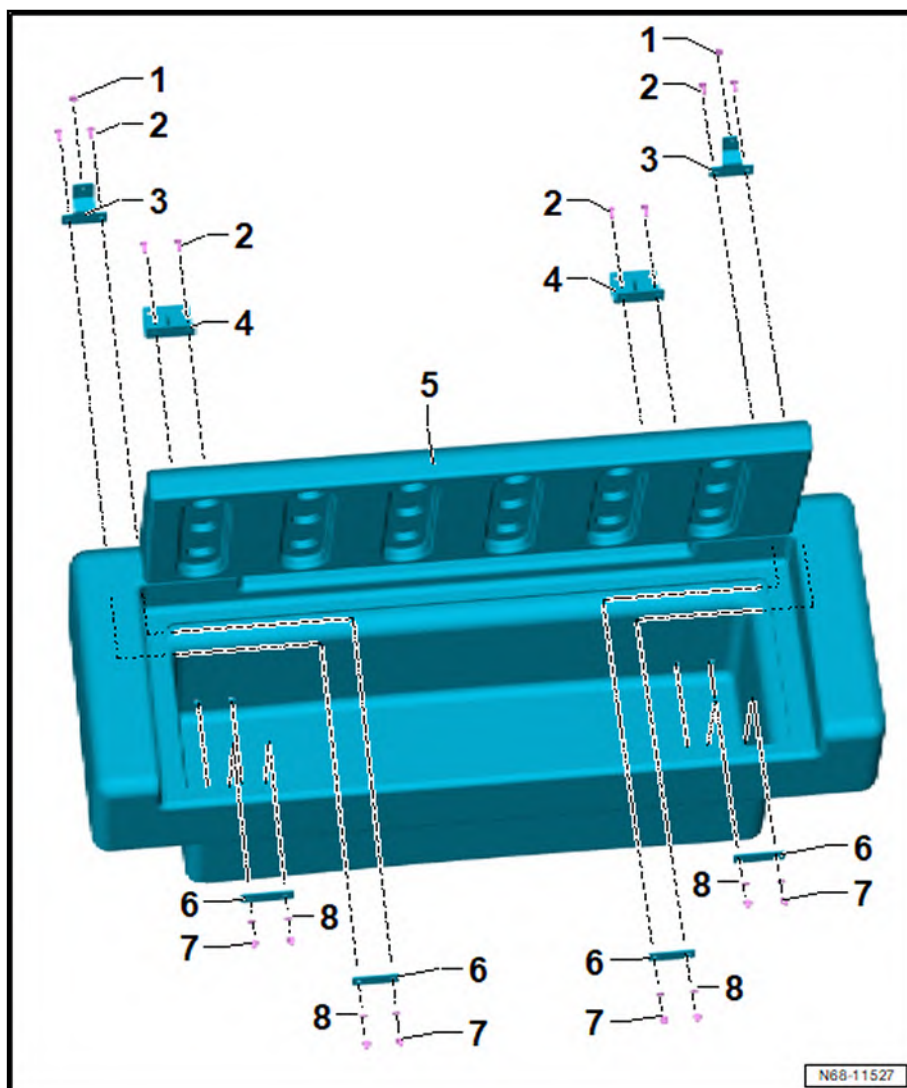
- ☐ Qty. 4

7 - Nuts

- ☐ Qty. 8
- ☐ 20 Nm

8 - Washers

- ☐ Qty. 8



11.2 Removing and installing tool box

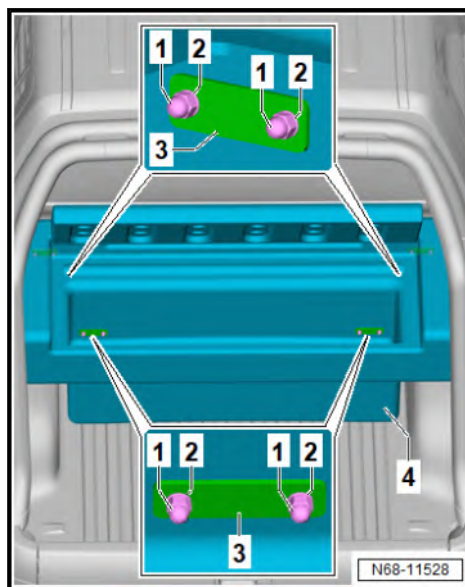
Special tools and workshop equipment required



- ◆ Torque wrench 6-50 Nm -V.A.G 1331-



Removing



- Unscrew nuts -1-.
- Remove washers -2-.
- Remove retainer -3-.
- Remove tool box -4-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➔ [o11.1 verview - tool box", page 393](#)

12 Wheel housing trim

⇒ [o12.1 verview – wheel housing trim”, page 395](#)

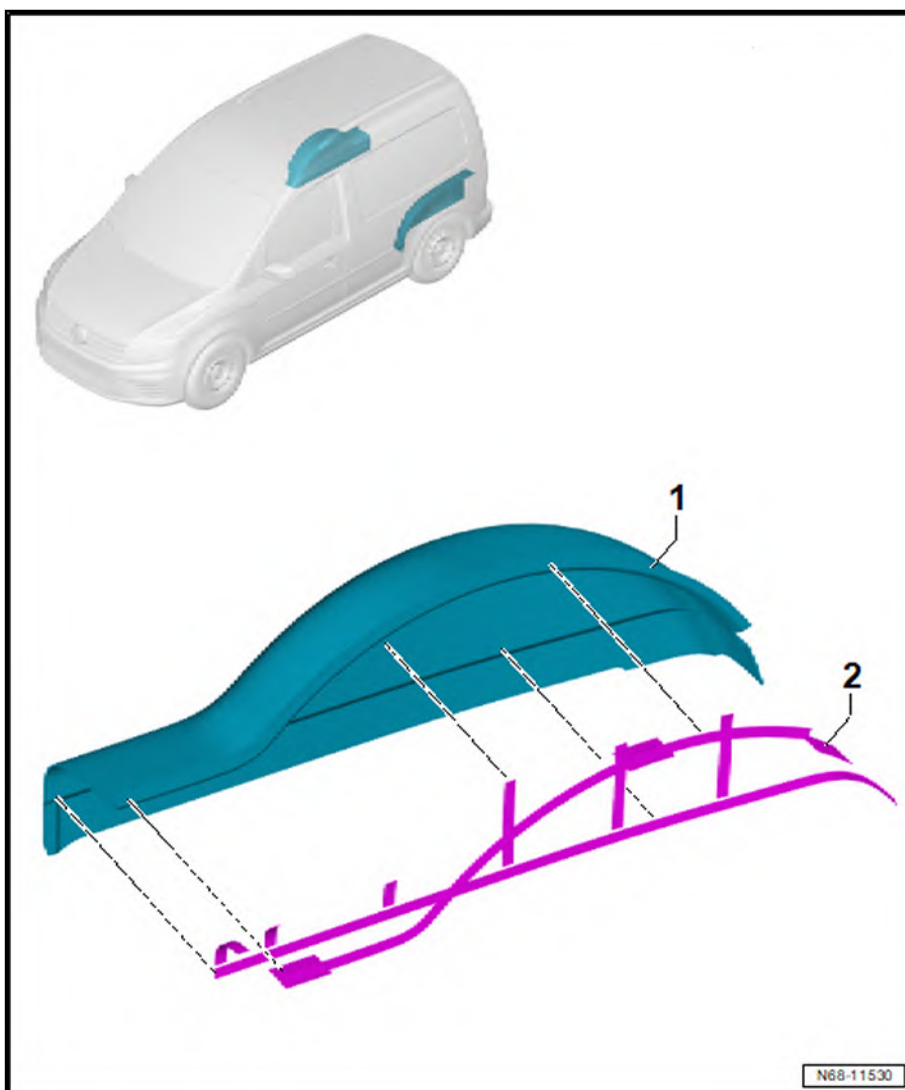
⇒ [a12.2 nd installing wheel housing trim”, page 395](#)

12.1 Assembly overview – wheel housing trim

1 - Trim

- ☐ Wheel housing
- ☐ Qty. 2
- ☐ Removing and installing ⇒ [page 395](#)

2 - Adhesive strips



12.2 Removing and installing wheel housing trim

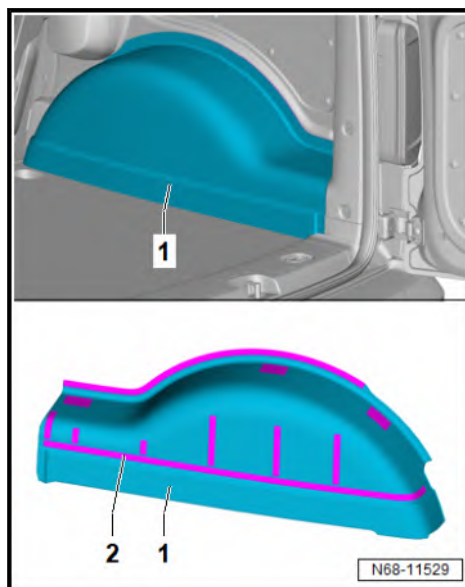
Special tools and workshop equipment required



◆ Hot-air blower -V.A.G 1416-



Removing



- Pull off wheel housing trim -1-.

Installing

Install in reverse order of removal, observing the following:

- Heat up bonding surfaces using hot air blower -V.A.G 1416-.
- Remove adhesive residue.
- Clean bonding surfaces using cleaning solution and a micro-fibre cloth.
- Observe flash-off time of cleaning solution.
- Do not touch adhesive surfaces once backing film has been removed.

Observe positions of adhesive strips -2-.

- Pull off protective film of adhesive strips.
- Fit wheel housing trim in position. Press down evenly on adhesive strips.

13 Stowage cases on wing door

⇒ [o13.1 verview - stowage cases", page 397](#)

⇒ [a13.2 nd installing stowage cases", page 398](#)

⇒ [a13.3 nd installing brackets for stowage cases", page 398](#)

13.1 Assembly overview - stowage cases

1 - Stowage cases

- ☐ Qty. 2
- ☐ Removing and installing ⇒ [page 398](#)

2 - Bolts

- ☐ Qty. 16
- ☐ 0.5 Nm

3 - Bracket

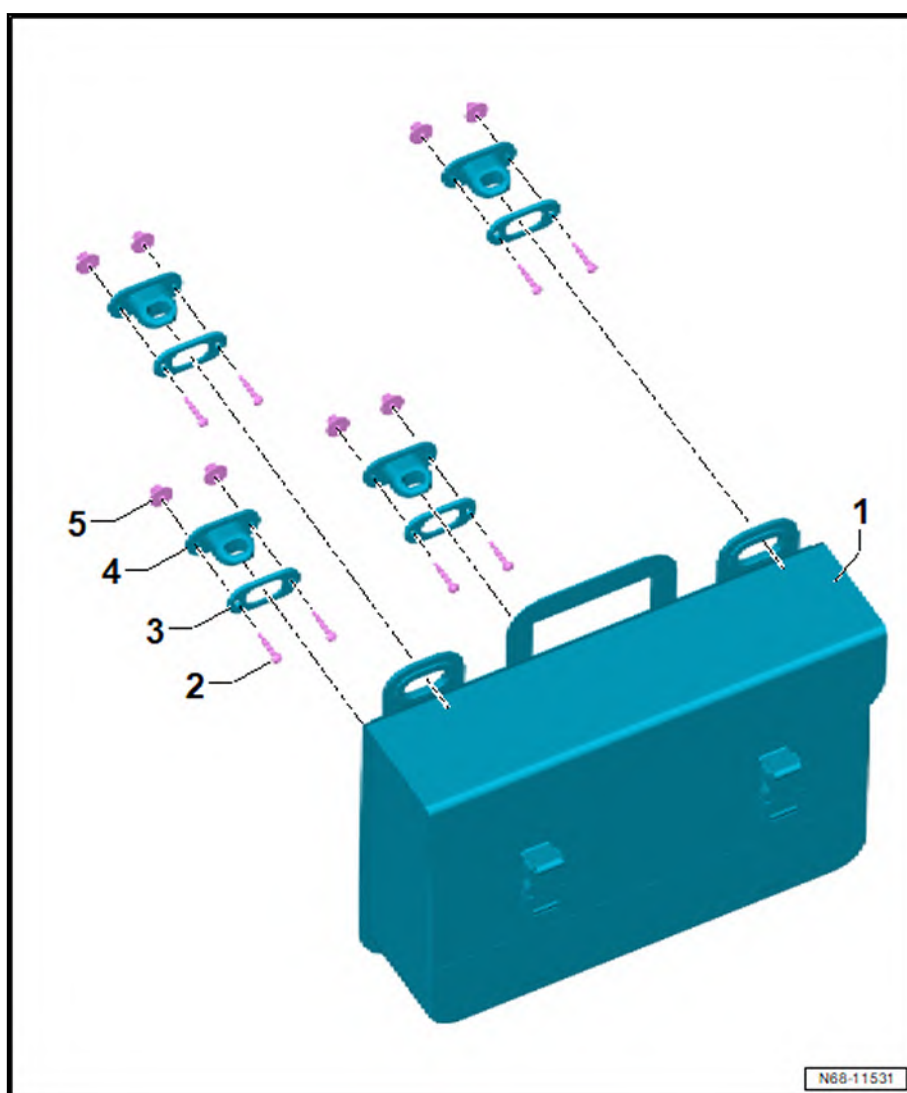
- ☐ Qty. 8

4 - Catches

- ☐ Qty. 8

5 - Nuts

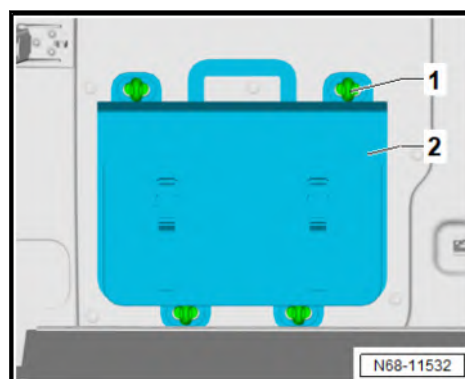
- ☐ Qty. 16
- ☐ 0.5 Nm





13.2 Removing and installing stowage cases

Removing



- Turn catches -1- 90°.
- Detach stowage case -2-.

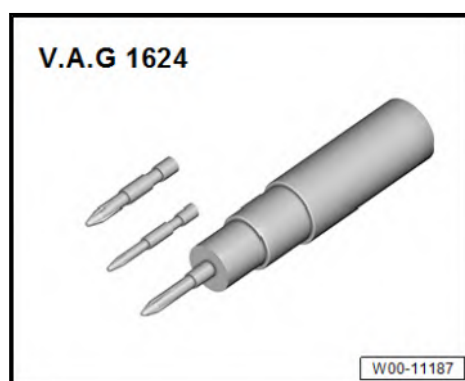
Installing

Installation is carried out in reverse order.

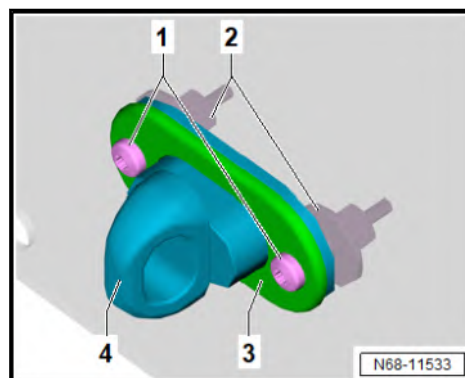
13.3 Removing and installing brackets for stowage cases

Special tools and workshop equipment required

- ◆ Torque screwdriver -V.A.G 1624-



Removing



- Remove stowage case ⇒ [page 398](#) .
- Remove trim for wing door ⇒ General body repairs, interior; Rep. gr. 70; Rear door trims; Removing and installing rear door trim.



- Unscrew bolts -1-. Unscrew nuts -2- when doing this.
- Detach cover -3-.
- Remove retainer -4-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ [⇒ o13.1 verview - stowage cases", page 397](#)



14 Floor ventilation, load bed

⇒ [o14.1 verview - floor ventilation", page 400](#)

⇒ [a14.2 nd installing floor ventilation", page 400](#)

14.1 Assembly overview - floor ventilation

1 - Bolts

- Qty. 4
- 2.5 Nm

2 - Grille

3 - Nuts

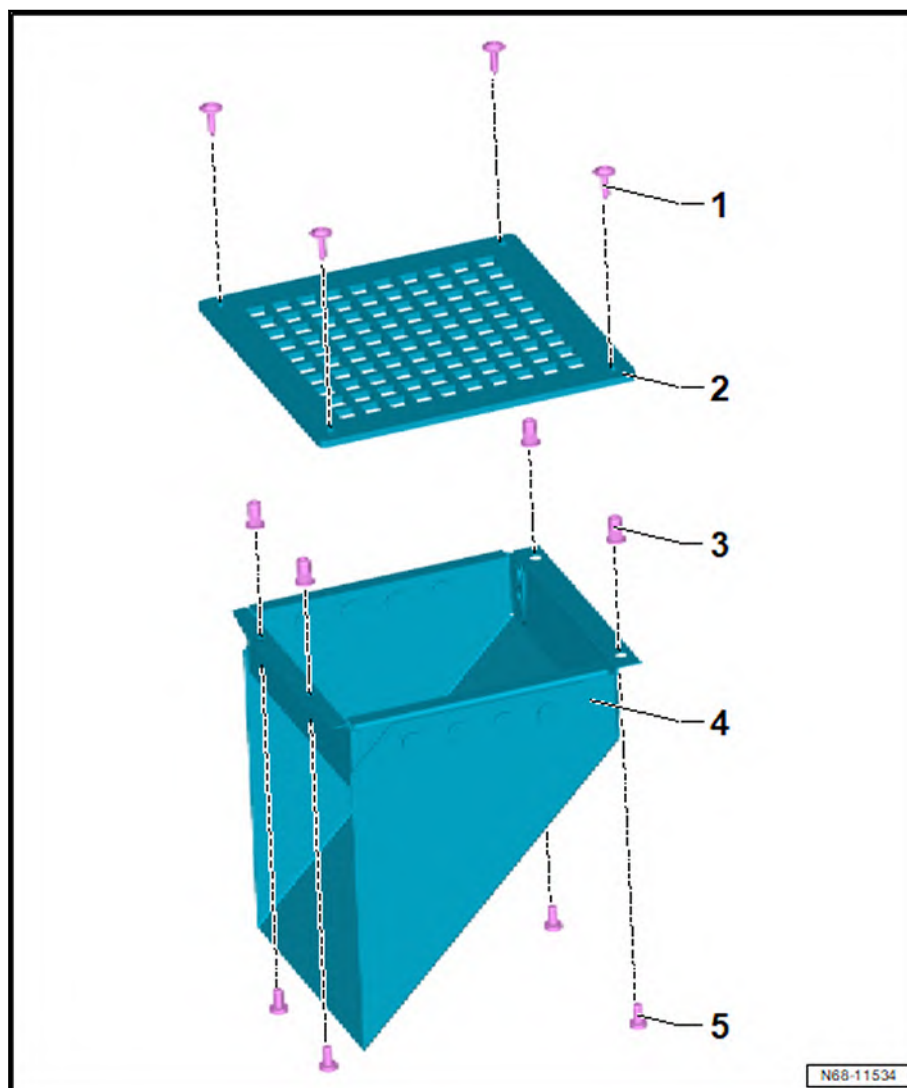
- Qty. 4

4 - Floor ventilation

- Removing and installing ⇒ [page 400](#)

5 - Bolts

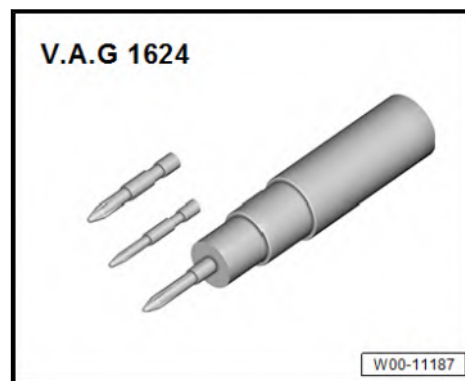
- Qty. 4
- 2.5 Nm



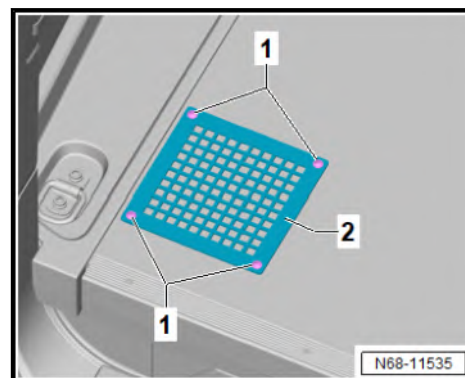
14.2 Removing and installing floor ventilation

Special tools and workshop equipment required

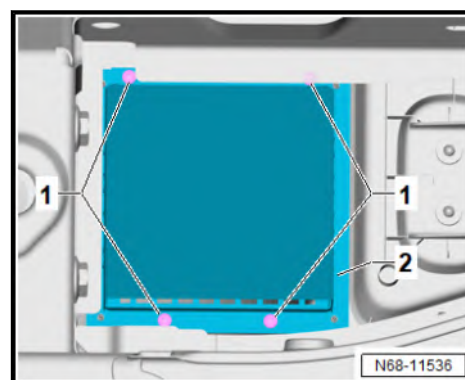
◆ Torque screwdriver -V.A.G 1624-



Removing



- Unscrew bolts -1-.
- Remove grille -2-.
- Unscrew bolts -1-.



- Remove floor ventilation -2-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➤ [o14.1 verview - floor ventilation", page 400](#)



15 High load floor with drawers

⇒ [o15.1 verview - high load floor with drawers", page 402](#)

⇒ [a15.2 nd installing load floor, upper part", page 404](#)

⇒ [a15.3 nd installing load floor, lower part", page 405](#)

⇒ [a15.4 nd installing frame for drawers", page 407](#)

15.1 Assembly overview - high load floor with drawers



1 - Load floor

- ☐ Left
- ☐ Upper part
- ☐ Removing and installing ➔ [page 404](#)

2 - Bolts

- ☐ Number depends on equipment level
- ☐ Turn onto stop

3 - Load floor

- ☐ Right
- ☐ Upper part
- ☐ Removing and installing ➔ [page 404](#)

4 - Bolts

- ☐ Number depends on equipment level
- ☐ Turn onto stop

5 - Bolts

- ☐ Number depends on equipment level
- ☐ Turn onto stop

6 - Edge protector

7 - Frame

- ☐ Removing and installing ➔ [page 407](#)

8 - Bolts

- ☐ Qty. 4
- ☐ 20 Nm

9 - Drawer

10 - Support

- ☐ Qty. 3

11 - Bolts

- ☐ Qty. 3
- ☐ 20 Nm

12 - Bolts

- ☐ Number depends on equipment level

13 - Bolts

- ☐ Number depends on equipment level
- ☐ 7.5 Nm

14 - Adapter plates

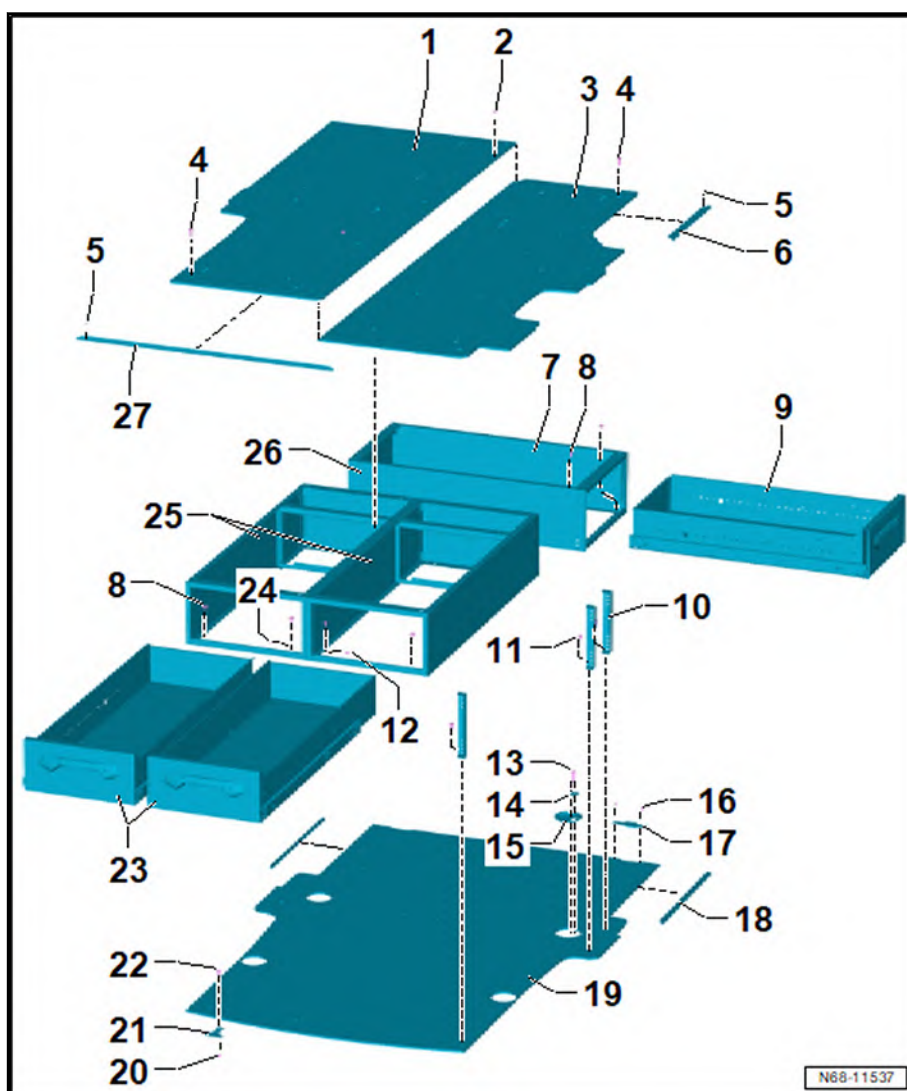
- ☐ Number depends on equipment level

15 - Fastening recesses

- ☐ Number depends on equipment level

16 - Bolts

- ☐ Qty. 4
- ☐ 1.5 Nm





17 - Bracket

- ☐ Qty. 2

18 - Edge protector

- ☐ Qty. 2

19 - Load floor

- ☐ Lower part
- ☐ Removing and installing ⇒ [page 405](#)

20 - Nuts

- ☐ Qty. 4

21 - Bracket

- ☐ Qty. 2

22 - Bolts

- ☐ Qty. 4
- ☐ Turn onto stop

23 - Drawers

- ☐ Qty. 2

24 - Nuts

- ☐ Number depends on equipment level
- ☐ 7.5 Nm

25 - Frame

- ☐ Qty. 2
- ☐ Removing and installing ⇒ [page 407](#)

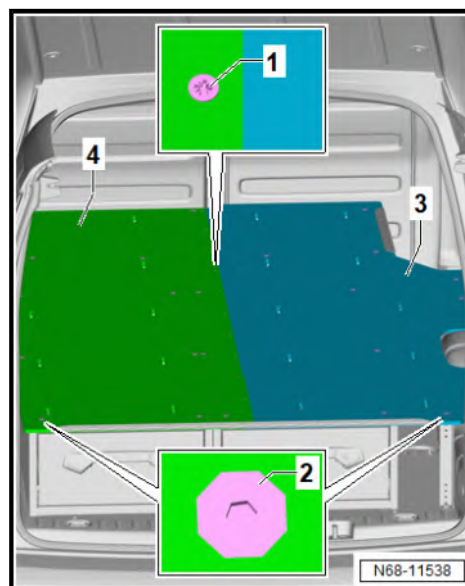
26 - Frame

- ☐ Removing and installing ⇒ [page 407](#)

27 - Edge protector

15.2 Removing and installing load floor, upper part

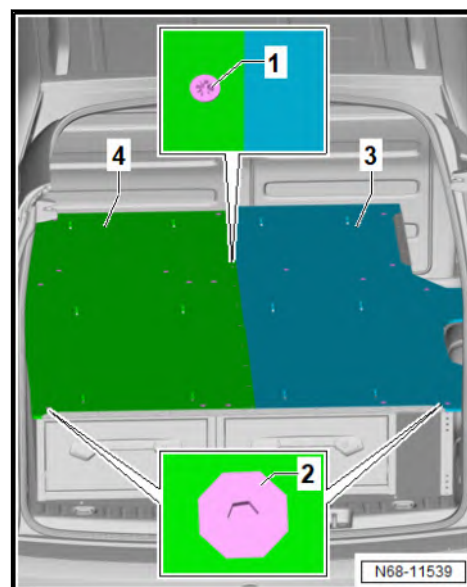
Removing



Vehicles with long wheelbase

- Unscrew bolts -1-.
- Unscrew bolts -2-.
- Remove load floor -4-.
- Remove load floor -3-.

Vehicles with short wheelbase



- Unscrew bolts -1-.
- Unscrew bolts -2-.
- Remove load floor -4-.
- Remove load floor -3-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ➤ [o15.1 verview - high load floor with drawers", page 402](#)

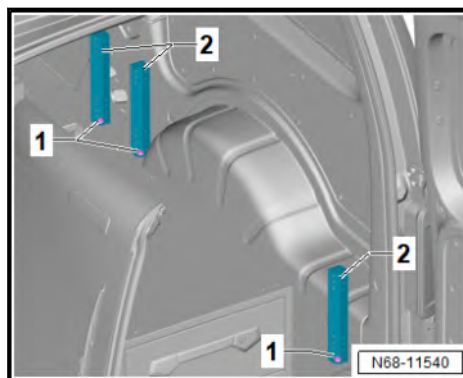
15.3 Removing and installing load floor, lower part

Torque wrench 6-50 Nm -V.A.G 1331-





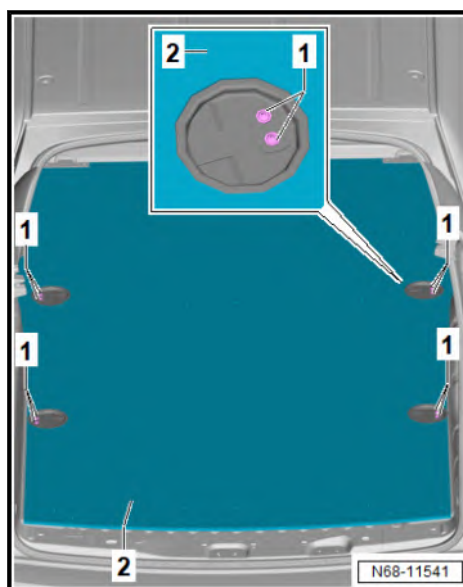
Removing



- Remove frame for drawers ⇒ [page 407](#) .
- Remove lock carrier trim ⇒ General body repairs, interior; Rep. gr. 70; Luggage compartment trims; Removing and installing lock carrier trim.

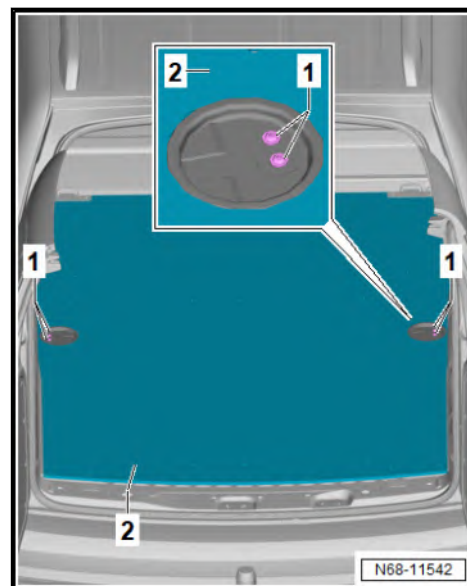
Vehicles with long wheelbase

- Unscrew bolts -1-.
- Remove supports -2-.
- Unscrew bolts -1-.



- The following work step must be performed with the aid of a second mechanic.
- Remove load floor -2- with aid of second mechanic. Detach butyl sealing strip when doing this.

Vehicles with short wheelbase



- Unscrew bolts -1-.
- The following work step must be performed with the aid of a second mechanic.
- Remove load floor -2- with aid of second mechanic. Detach butyl sealing strip when doing this.

Installing

Install in reverse order of removal, observing the following:

- The bonding surfaces must be free of dust and other dirt.
- Clean bonding surfaces.
- Reinstall butyl sealing strip.

Specified torques

- ♦ ➔ [o15.1 verview - high load floor with drawers", page 402](#)

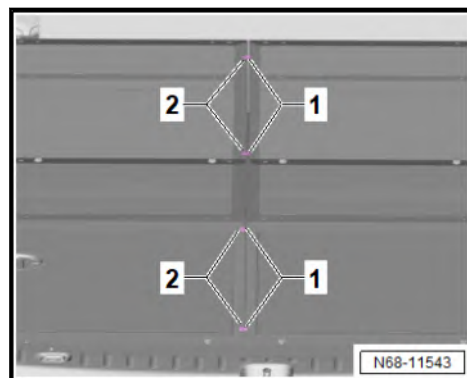
15.4 Removing and installing frame for drawers

Torque wrench 6-50 Nm -V.A.G 1331-

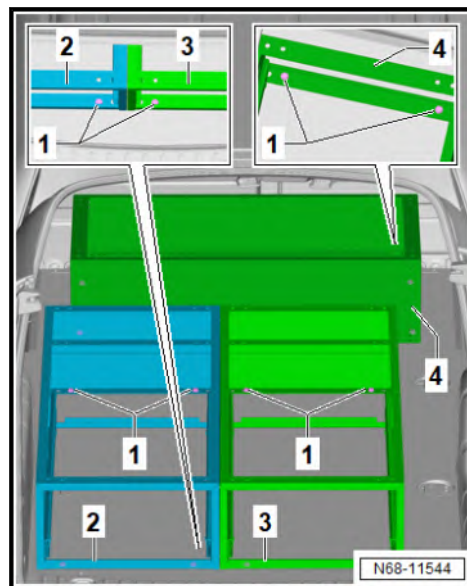




Removing



- Remove load floor upper part ➔ [page 404](#) .
- Remove drawers.
- Unscrew bolts -1-. Unscrew nuts -2- when doing this.
- Unscrew bolts -1-.



- Remove frame -2-, -3- and -4-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ ➔ [o15.1 verview - high load floor with drawers](#)”, [page 402](#)

16 Storage compartment on wheel housing

⇒ [o16.1 verview - storage compartment on wheel housing", page 409](#)

⇒ [a16.2 nd installing storage compartment", page 409](#)

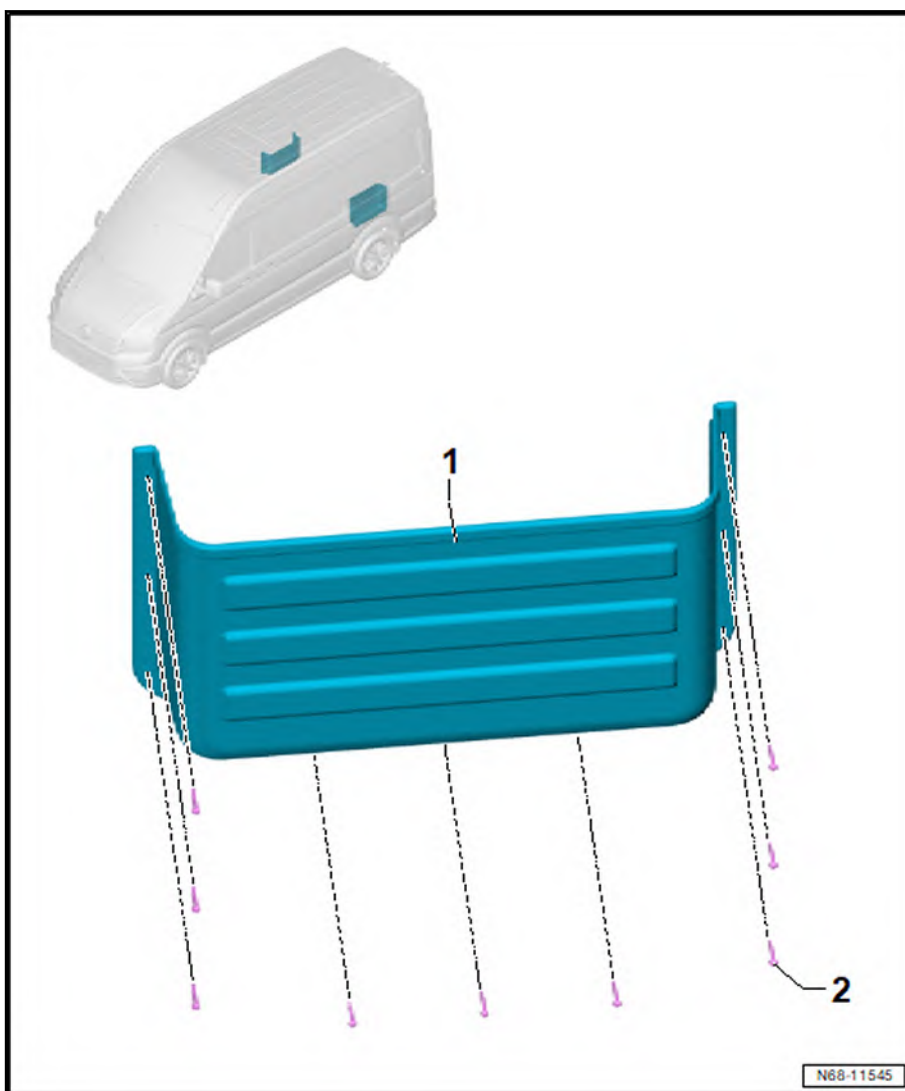
16.1 Assembly overview - storage compartment on wheel housing

1 - Storage compartment

- ☐ Qty. 2
- ☐ Removing and installing ⇒ [page 409](#)

2 - Bolts

- ☐ Qty. 18
- ☐ 0.5 Nm

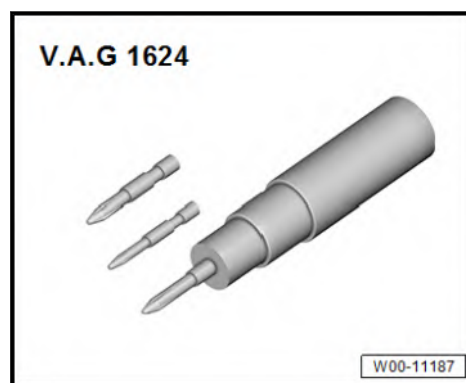


16.2 Removing and installing storage compartment

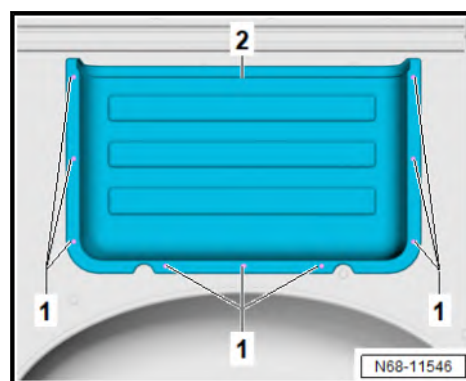
Special tools and workshop equipment required



◆ Torque screwdriver -V.A.G 1624-



Removing



- Unscrew bolts -1-.
- Remove stowage compartment -2-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [o16.1 verview - storage compartment on wheel housing", page 409](#)

17 Protective grille on wing door

⇒ [o17.1 verview - protective grille on wing door", page 411](#)

⇒ [a17.2 nd installing protective grille", page 411](#)

17.1 Assembly overview - protective grille on wing door

1 - Bolts

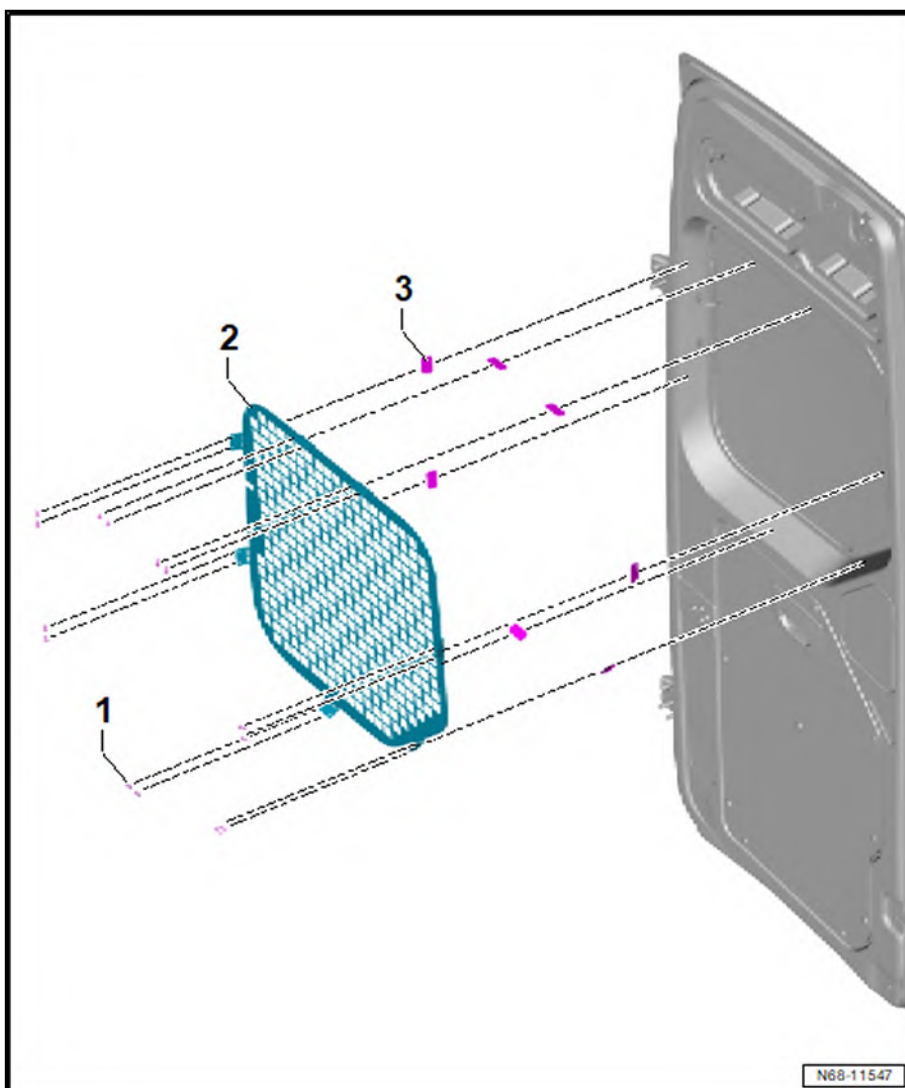
- ☐ Qty. 14 on each side
- ☐ 1 Nm

2 - Protective grille

- ☐ Qty. 2
- ☐ Removing and installing ⇒ [page 411](#)

3 - Protective pads

- ☐ Qty. 7 on each side
- ☐ Bonded

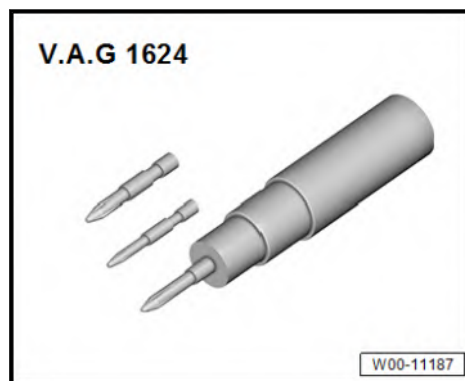


17.2 Removing and installing protective grille

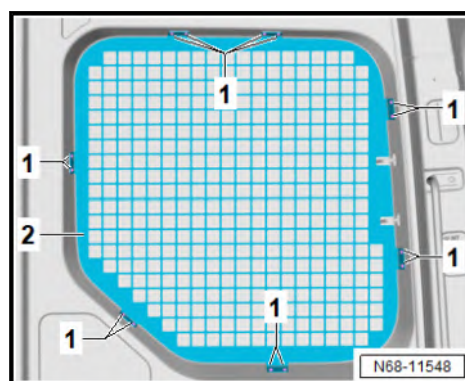
Special tools and workshop equipment required



◆ Torque screwdriver -V.A.G 1624-



Removing



- Unscrew bolts -1-.
- Detach protective grille -2-.

Installing

Install in reverse order of removal, observing the following:

- Check protective pads for damage.
- If necessary, renew protective pads.

Specified torques

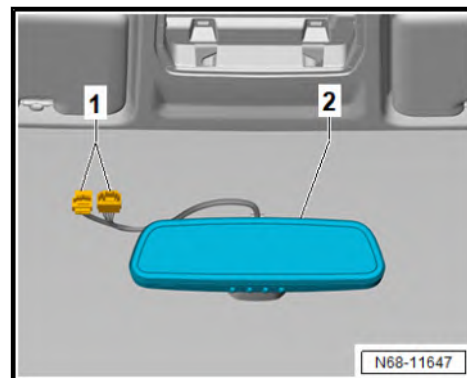
- ◆ ➔ [o17.1 verview - protective grille on wing door", page 411](#)

18 Interior mirror

⇒ a18.1 nd installing mirror taximeterG511", page 413

18.1 Removing and installing mirror taximeter -G511-

Removing



- Remove front interior light ⇒ Electrical system; Rep. gr. 96; Controls; Removing and installing front interior light/reading light.
- Disconnect electrical connectors -1-.
- Remove mirror taximeter -G511- -2- ⇒ General body repairs, interior; Rep. gr. 68; Interior mirror; Removing and installing interior mirror.

Removing

Install in reverse sequence of removal.



19 Interior mirror

19.1 Removing and installing interior mirror, mirror taximeter -G511-

- Remove mirror taximeter -G511- ⇒ [page 439](#) .

69 – Passenger protection

1 Seat belts

⇒ [a1.1 nd installing front three-point seat belt, bench seat \(Holland double cab\)](#), page 415

⇒ [a1.2 nd installing belt buckle, bench seat \(Holland double cab\)](#), page 416

1.1 Removing and installing front three-point seat belt, bench seat (Holland double cab)

Special tools and workshop equipment required

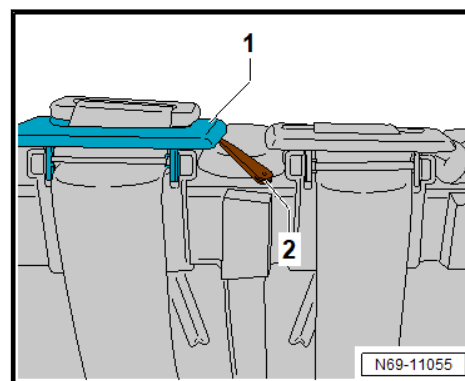
◆ Torque wrench, 40 to 200 Nm -V.A.G 1332A-



◆ Removal wedge -3409-



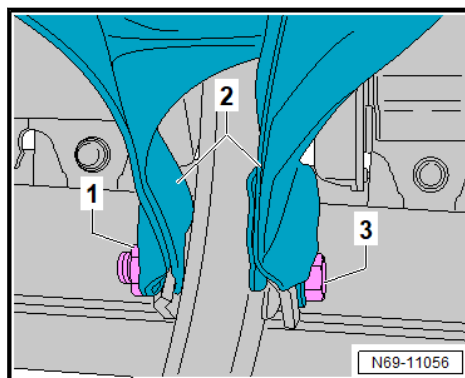
Removing



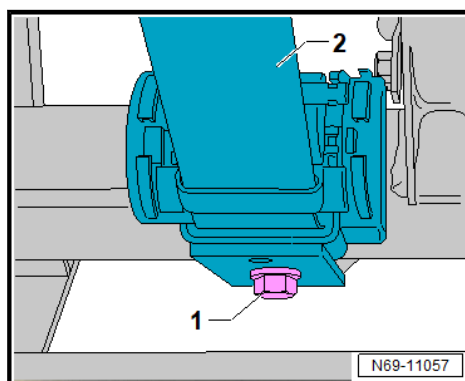
- Remove bench seat ⇒ [page 422](#) .
- Release belt guide -1- using removal wedge -3409- -2-.



- Unscrew nut -1-.



- Remove belt end fitting -2-.
- Remove bolt -3-.
- Unscrew bolt -1-.



- Remove three-point seat belt -2-.

Installing

Install in reverse order of removal, observing the following:

- Renew nuts and bolts after each removal.

Specified torques

Component	Specified torque
Always renew nut -1- after re-moving	40 Nm
Always renew bolt -1- after re-moving	40 Nm

1.2 Removing and installing belt buckle, bench seat (Holland double cab)

⇒ [a1.2.1 nd installing outer belt buckle”, page 416](#)

⇒ [a1.2.2 nd installing centre belt buckle”, page 417](#)

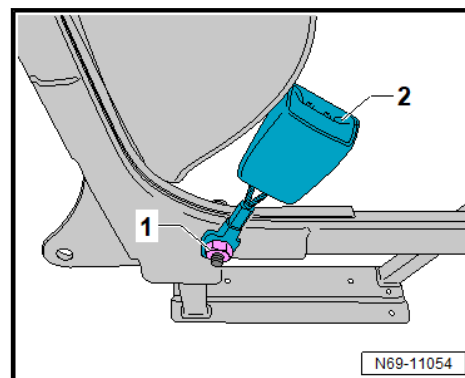
1.2.1 Removing and installing outer belt buckle

Special tools and workshop equipment required

- ◆ Torque wrench, 40 to 200 Nm -V.A.G 1332A-



Removing



- Remove bench seat ➔ [page 422](#) .
- Unscrew bolt -1-.
- If fitted, disconnect electrical connectors.
- Remove belt buckle -2-.

Installing

Install in reverse order of removal, observing the following:

- Renew nuts and bolts after each removal.

Specified torques

Component	Specified torque
Always renew bolt -1- after re-moving	40 Nm

1.2.2 Removing and installing centre belt buckle

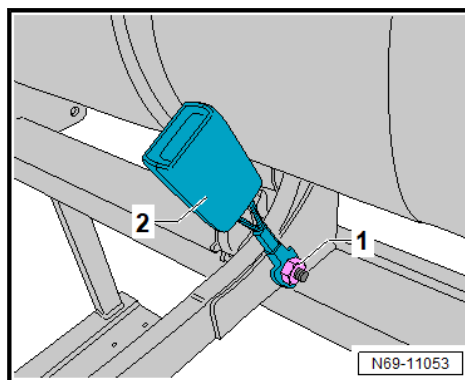
Special tools and workshop equipment required



- ◆ Torque wrench, 40 to 200 Nm -V.A.G 1332A-



Removing



- Remove bench seat from seat frame.
- Unscrew bolt -1-.
- If fitted, disconnect electrical connectors.
- Remove belt buckle -2-.

Installing

Install in reverse order of removal, observing the following:

- Renew nuts and bolts after each removal.

Specified torques

Component	Specified torque
Always renew bolt -1- after removing	40 Nm

70 – Trim, insulation

1 Interior trims

⇒ [a1.1 nd installing bench seat trim \(Holland double cab\)", page 419](#)

⇒ [a1.2 nd installing sliding door trim \(Holland double cab\)", page 419](#)

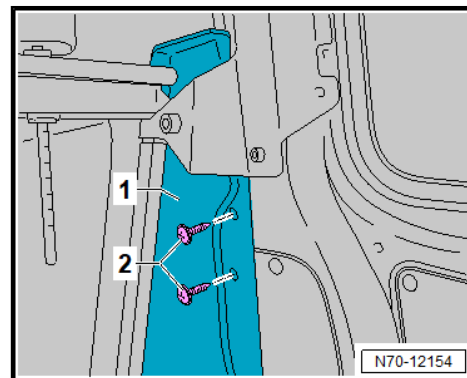
1.1 Removing and installing bench seat trim (Holland double cab)

Special tools and workshop equipment required

- ◆ Torque screwdriver -VAS 6494-



Removing



- Remove lower section of partition panel ⇒ [page 341](#) .
- Unscrew bolts -2- of bench seat trim -1-
- With the aid of a second mechanic, remove bench seat trim -1- from vehicle.

Installing

Install in reverse order of removal, observing the following:

Specified torques

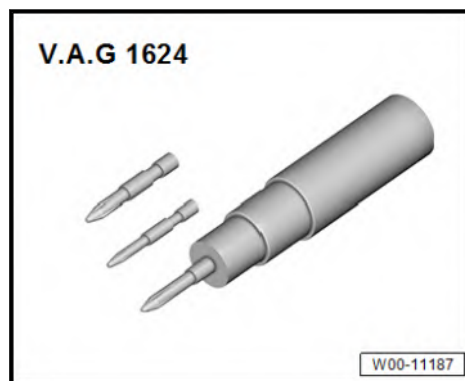
- ◆ ⇒ [o4.5.1 verview – partition panel", page 338](#)

1.2 Removing and installing sliding door trim (Holland double cab)

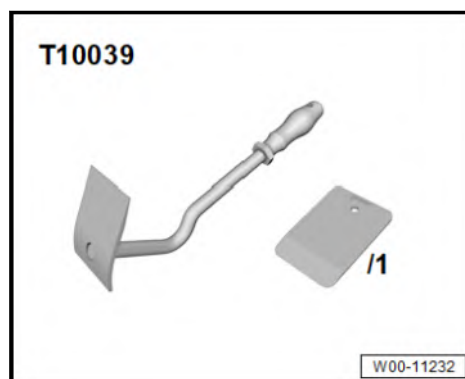
Special tools and workshop equipment required



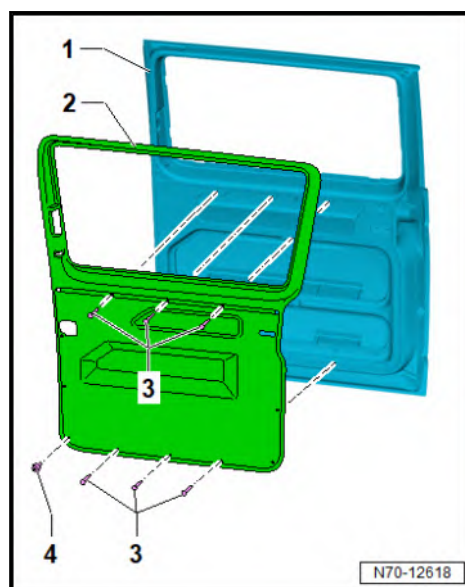
◆ Torque screwdriver -V.A.G 1624-



◆ Release lever -T10039-



Removing



- Remove clips -4-.
- If fitted, remove caps for bolts.
- Unscrew bolts -3-.
- Pull off trim -2- from sliding door -1-. Remove trim while doing so.

Installing

Install in reverse order of removal, observing the following:



Note

Before installing, check clips for damage, and renew them if necessary.

Torque settings

Component	Specified torque
Bolts	1.5 Nm

72 – Seat frames

1 Bench seats

⇒ a1.1 nd installing bench seat (Holland double cab)", page 422

1.1 Removing and installing bench seat (Holland double cab)

Special tools and workshop equipment required

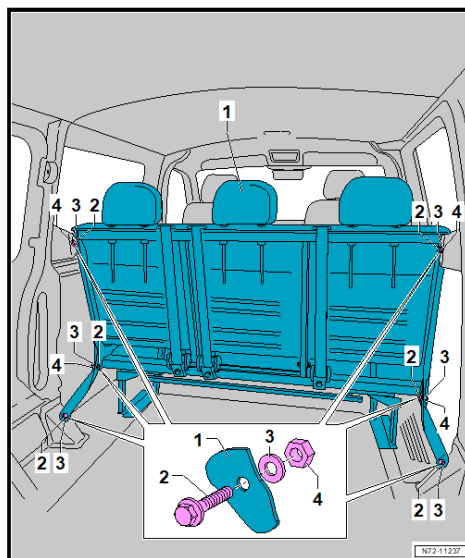
- ◆ Torque wrench, 40 to 200 Nm -V.A.G 1332A-



- ◆ Torque screwdriver -VAS 6494-



Removing



- Remove seat padding of bench seat.



- Remove lower section of partition panel ⇒ [page 341](#) .
- Remove bench seat trim from vehicle ⇒ [page 419](#) .
- Unscrew bolts -2-.
- Remove washers -3-.
- Remove nuts -4-.
- If fitted, disconnect electrical connectors.
- With the aid of a second mechanic, remove bench seat -1- from vehicle.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bench seat bolts -1-	60 Nm



74 – Seat - padding, covers

91 – Communication

1 Fitting locations

⇒ 01.1 fitting locations of components/control units in front section of the vehicle (taxi)", page 425

1.1 Overview of fitting locations of components/control units in front section of the vehicle (taxi)

1 - Taxi alarm off button - E486-

- ☐ Removing and installing ⇒ [page 531](#)

2 - Taximeter -G41-

- ☐ Depending on equipment
- ☐ Removing and installing ⇒ [page 439](#)

3 - Mirror taximeter -G511-

- ☐ Depending on equipment
- ☐ Removing and installing ⇒ [page 439](#)

4 - Two-way radio

- ☐ Preparation

5 - Roof aerial -R295-

- ☐ Removing and installing ⇒ [page 434](#)

6 - Taxi roof sign -J1066-

- ☐ Removing and installing ⇒ [page 513](#)

7 - Two-way radio loudspeaker -R25-

- ☐ Removing and installing ⇒ [page 447](#)

8 - Microphone 2 on driver side -R225-

- ☐ Removing and installing ⇒ [page 446](#)

9 - Button

- ☐ Depending on equipment

10 - Switch module for taxi functions -EX62-

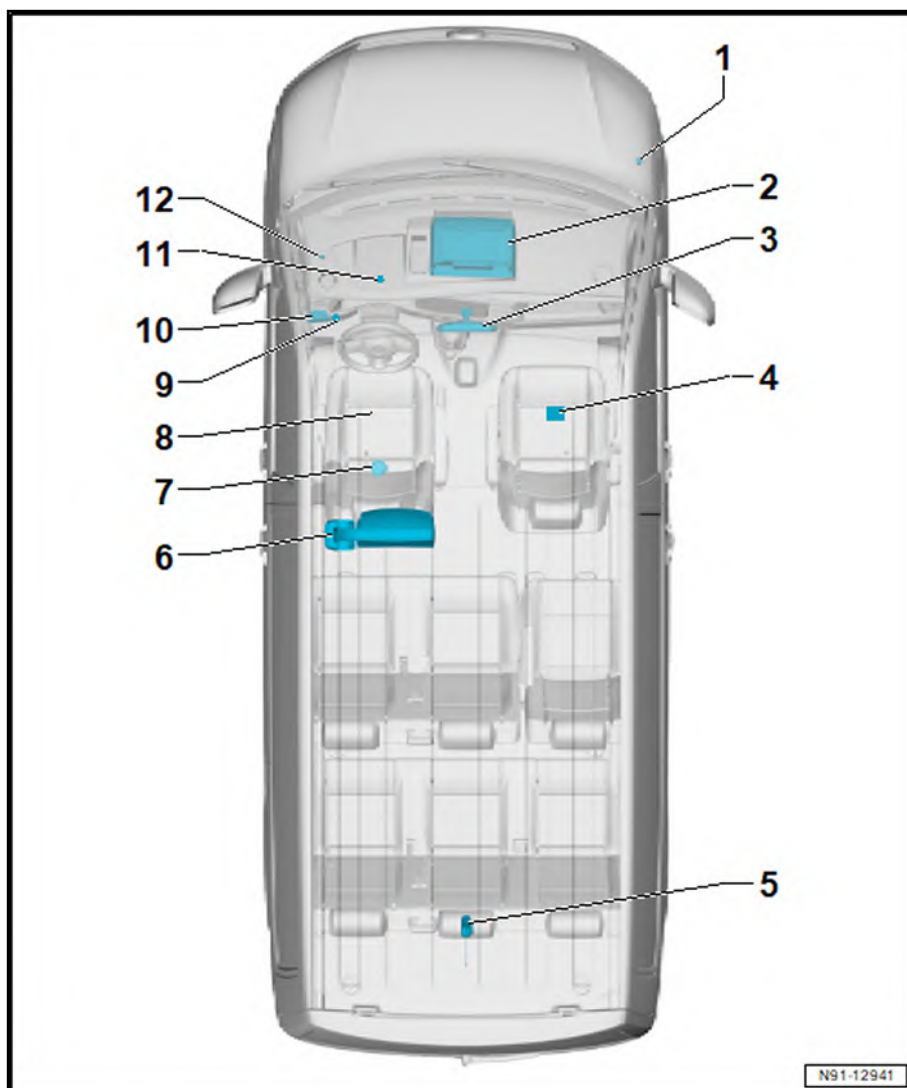
- ☐ Removing and installing ⇒ [page 531](#)

11 - Taxi alarm active button -E485-

- ☐ Removing and installing ⇒ [page 530](#)

12 - Passive taxi alarm button -E580-

- ☐ Removing and installing ⇒ [page 530](#)





2 Sound system

⇒ [o2.1 f fitting locations – sound system”, page 426](#)

⇒ [a2.2 nd installing two-way radio loudspeakerR25”, page 427](#)

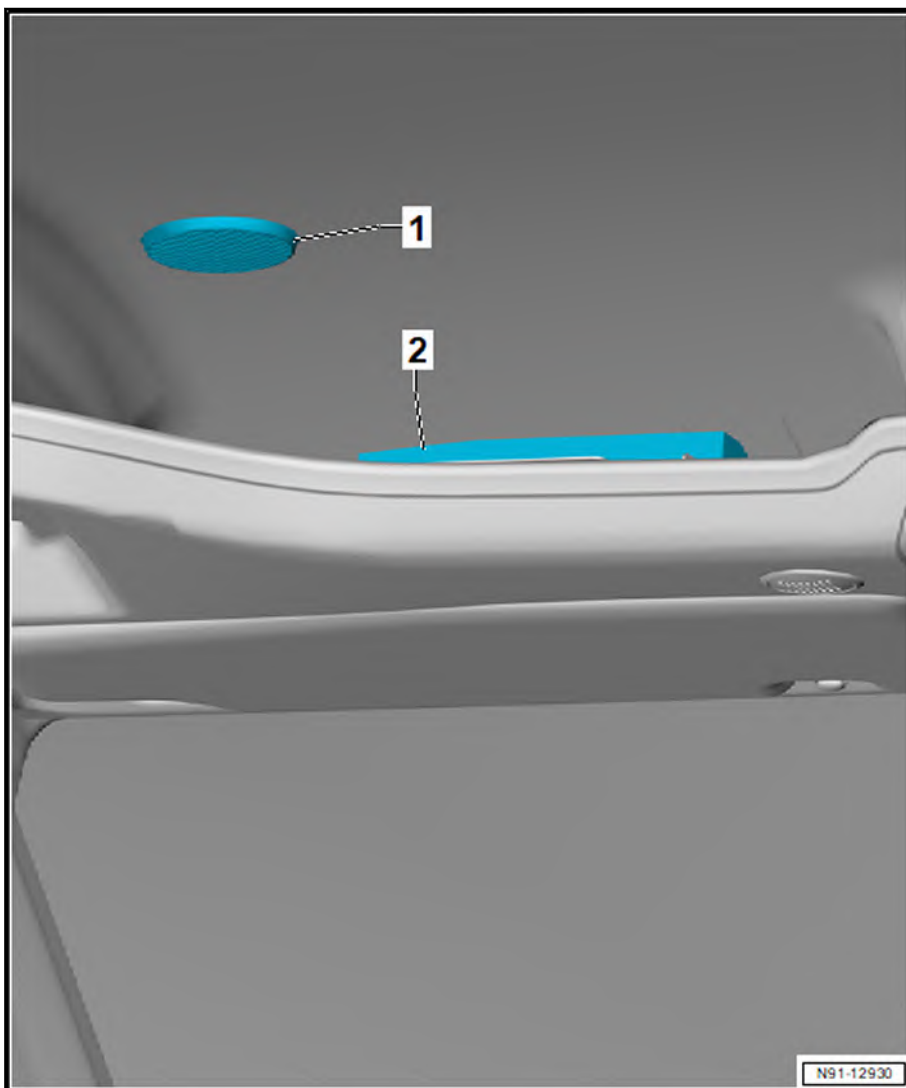
2.1 Overview of fitting locations – sound system

1 - Two-way radio loudspeaker -R25-

- ☐ Removing and installing ⇒ [page 427](#)

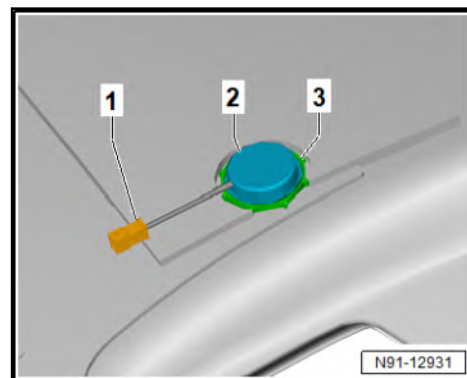
2 - Two-way radio -R8-

- ☐ Assembly overview ⇒ [page 443](#)
- ☐ Removing and installing ⇒ [page 444](#)



2.2 Removing and installing two-way radio loudspeaker -R25-

Removing



- Lower the moulded headliner ⇒ General body repairs, interior; Rep. gr. 70; Roof trims; Lowering the moulded headliner.
- Disconnect electrical connector -1-.
- Unscrew nut -3-.
- Detach two-way radio loudspeaker -R25- -2-.

Installing

Installation is carried out in reverse order.



3 Aerial systems

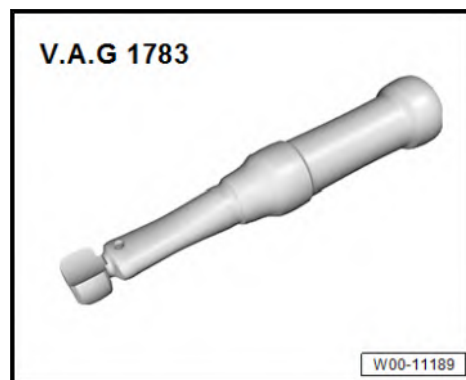
+

⇒ [a3.1 nd installing roof aerial R295", page 428](#)

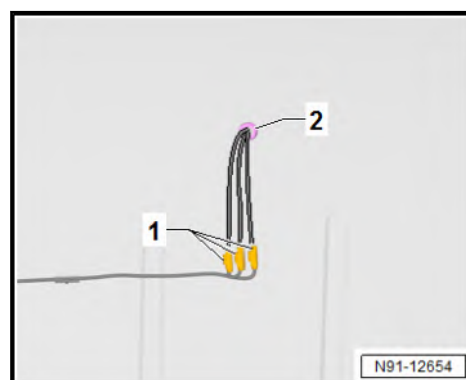
3.1 Removing and installing roof aerial - R295-

Special tools and workshop equipment required

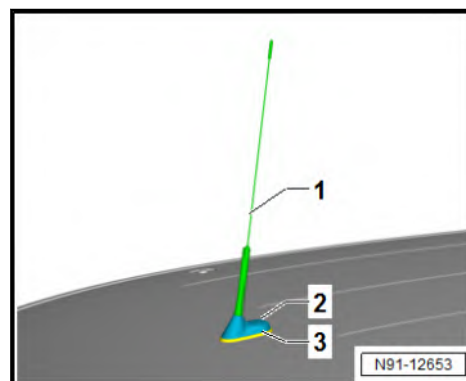
- ◆ Torque wrench -V.A.G 1783-



Removing



- Remove moulded headliner ⇒ General body repairs, interior; Rep. gr. 70; Roof trims; Removing and installing moulded headliner.
- Disconnect electrical connectors -1-.
- Unscrew nut -2-.
- Unscrew aerial rod -1-.



- Remove roof aerial -R295- -2- along with seal -3-.



Installing

Install in reverse order of removal, observing the following:

- Check seal -3- for damage, renew roof aerial -R295- if necessary.

Specified torques

Component	Specified torque
Nut -2-	7 Nm



4 Aerial systems

⇒ [a4.1 nd installing roof aerialRX5", page 430](#)

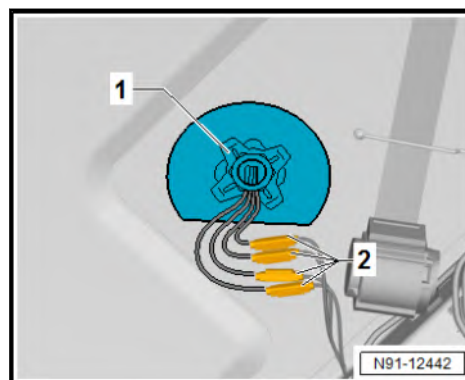
4.1 Removing and installing roof aerial -RX5-

Roof aerial -RX5-:

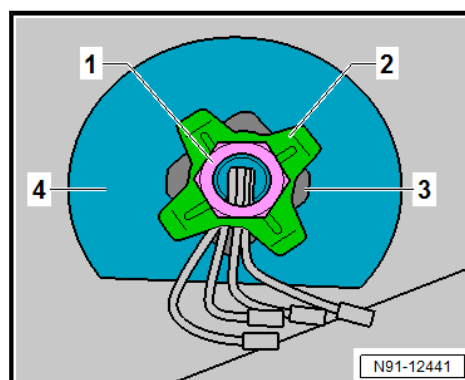
The following aerials are integral part of the roof aerial -RX5- and cannot be removed individually:

- ◆ GPS aerial -R50-
- ◆ Data transmission aerial -R180-
- ◆ GSM aerial -R205-
- ◆ Two-way radio aerial -R295-

Removing



- Lower the moulded headliner far enough for roof aerial -RX5- to become accessible ⇒ General body repairs, interior; Rep. gr. 70; Roof trims; Removing and installing moulded headliner.
- Disconnect electrical connectors -2- for roof aerial -RX5- -1-.
- Unscrew nut -1-.



- Remove clasp -2-.
- Remove washer -4-.
- Remove roof aerial -RX5- -3-.

Installing

Install in reverse order of removal, observing the following:



Specified torques

Component	Specified torque
Nut	4 Nm



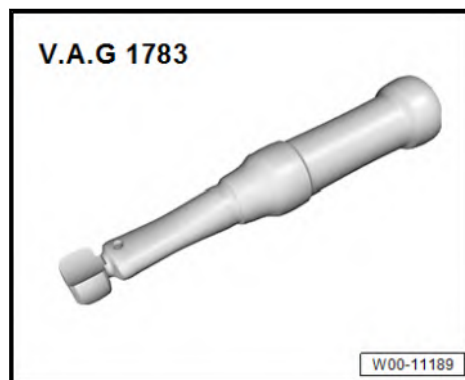
5 Aerial systems

⇒ a5.1 nd installing roof aerial R295", page 432

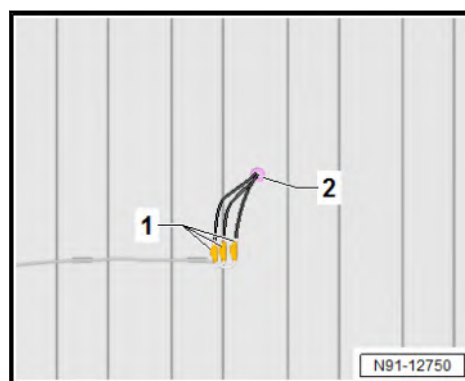
5.1 Removing and installing roof aerial - R295-

Special tools and workshop equipment required

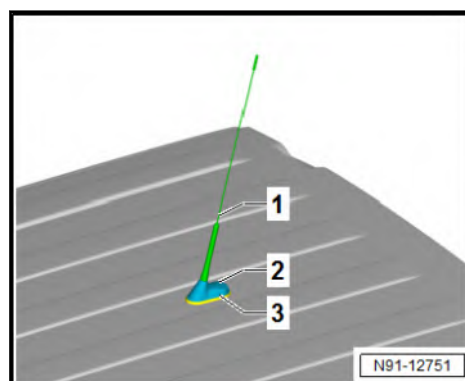
- ◆ Torque wrench -V.A.G 1783-



Removing



- Remove moulded headliner ⇒ General body repairs, interior; Rep. gr. 70; Roof trims; Removing and installing moulded headliner.
- Disconnect electrical connectors -1-.
- Unscrew nut -2-.
- Unscrew aerial rod -1-.



- Remove roof aerial -R295- -2- along with seal -3-.



Installing

Install in reverse order of removal, observing the following:

- Check seal -3- for damage, renew roof aerial -R295- if necessary.

Specified torques

Component	Specified torque
Nut -2-	7 Nm



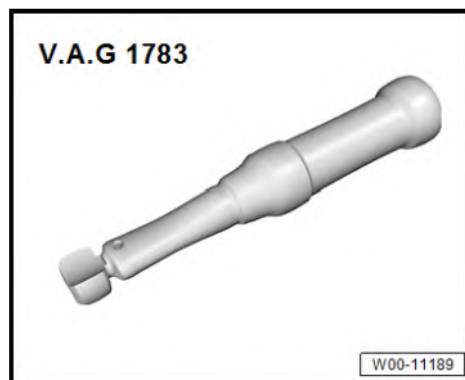
6 Aerial systems

⇒ [a6.1 nd installing roof aerial R295", page 434](#)

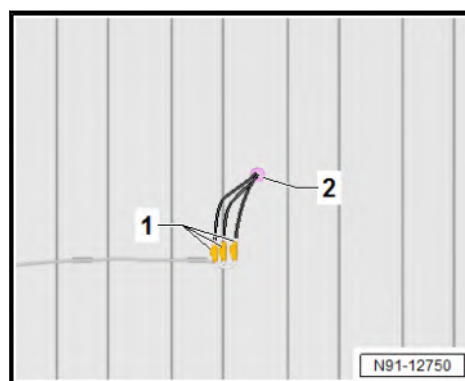
6.1 Removing and installing roof aerial - R295-

Special tools and workshop equipment required

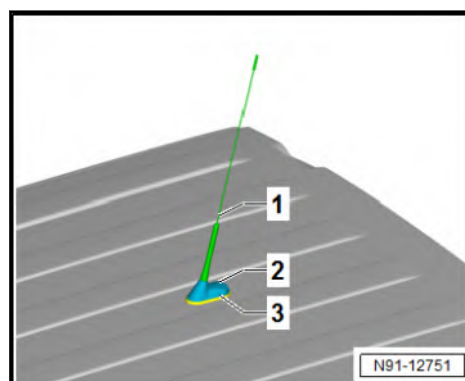
- ◆ Torque wrench -V.A.G 1783-



Removing



- Remove moulded headliner ⇒ General body repairs, interior; Rep. gr. 70; Roof trims; Removing and installing moulded headliner.
- Disconnect electrical connectors -1-.
- Unscrew nut -2-.
- Unscrew aerial rod -1-.



- Remove roof aerial -R295- -2- along with seal -3-. While doing this, route wiring harness from inside towards outside.



Installing

Install in reverse order of removal, observing the following:

- Check seal -3- for damage, renew roof aerial -R295- if necessary.

Specified torques

Component	Specified torque
Nut -2-	3.5 Nm



7 Taximeter

⇒ [a7.1 nd installing taximeterG41", page 436](#)

⇒ [a7.2 nd installing mounting for taximeterG41", page 437](#)

⇒ [a7.3 nd installing centre console cover with mounting for taximeterG41", page 438](#)

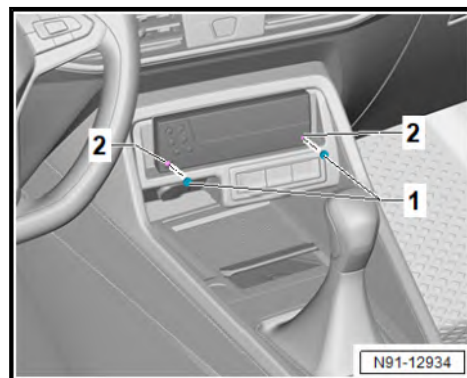
7.1 Removing and installing taximeter - G41-

Special tools and workshop equipment required

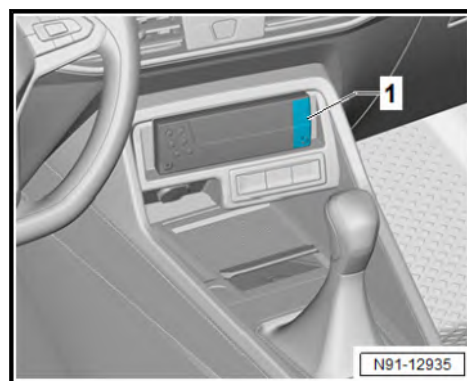
- ◆ Torque screwdriver -VAS 6494-



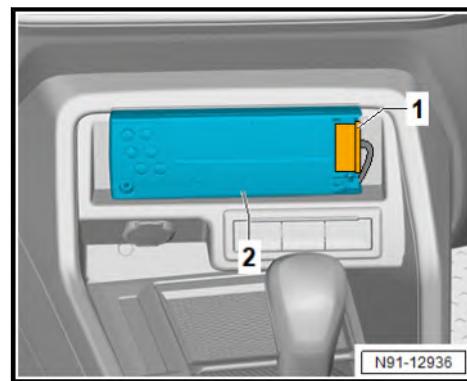
Removing



- Unclip caps -1-.
- Unscrew bolts -2-.
- Pull off cap -1-.



- Disconnect electrical connector -1-.



- Detach taximeter -G41- -2-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolts	1.5 Nm

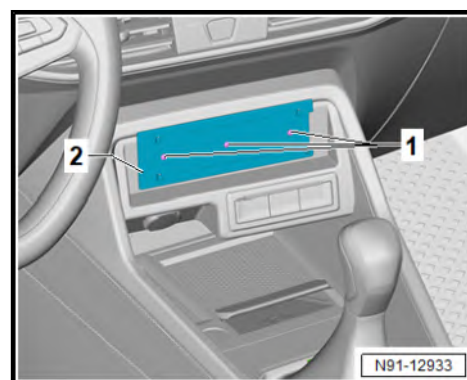
7.2 Removing and installing mounting for taximeter -G41-

Special tools and workshop equipment required

- ◆ Torque screwdriver -VAS 6494-



Removing



- Remove taximeter -G41- ➔ [page 436](#) .
- Unscrew bolts -1-.



- Remove mounting -2-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolts	1.5 Nm

7.3 Removing and installing centre console cover with mounting for taximeter -G41-

Removing

- Remove mounting for taximeter -G41- ➔ [page 437](#) .
- Remove centre console cover ➔ General body repairs, interior; Rep. gr. 68; Centre console; Removing and installing centre console cover.
- Disconnect electrical connectors to cover.

Installing

Install in reverse order of removal, observing the following:

- Route electrical connectors in such a way that noise is not generated through movements.

8 Taximeter

⇒ [a8.1 nd installing taximeterG41", page 439](#)

⇒ [a8.1.1 nd installing taximeter console", page 439](#)

⇒ [a8.2 nd installing mirror taximeterG511", page 439](#)

8.1 Removing and installing taximeter - G41-

Equipment version with console-mounted taximeter

- Remove additional console trim ⇒ Trims/insulation; Rep. gr. 70; Dash panel; Removing and installing additional console trim.
- If necessary, remove button plate ⇒ Trims/insulation; Rep. gr. 70; Dash panel; Removing and installing button plate support.
- Observe the taximeter manufacturer's specifications when removing or installing the taximeter.
- Remove taximeter -G41-. While doing this, disconnect electrical connectors.

Installing

Installation is carried out in reverse order.

8.1.1 Removing and installing taximeter console

Equipment version with console-mounted taximeter

- Remove taximeter ⇒ [page 439](#) .
- Remove tachograph console ⇒ Trims/insulation; Rep. gr. 70; Dash panel; Removing and installing tachograph console.

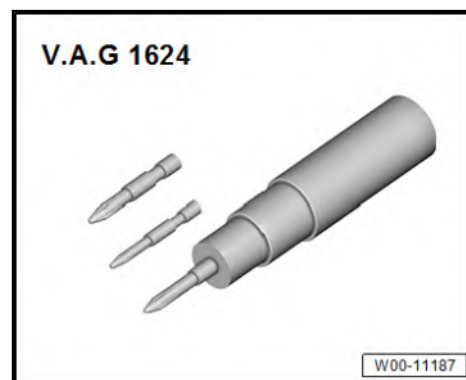
Installing

Installation is carried out in reverse order.

8.2 Removing and installing mirror taximeter -G511-

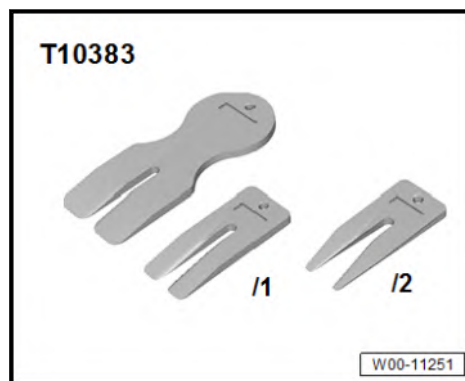
Special tools and workshop equipment required

- ◆ Torque screwdriver -V.A.G 1624-





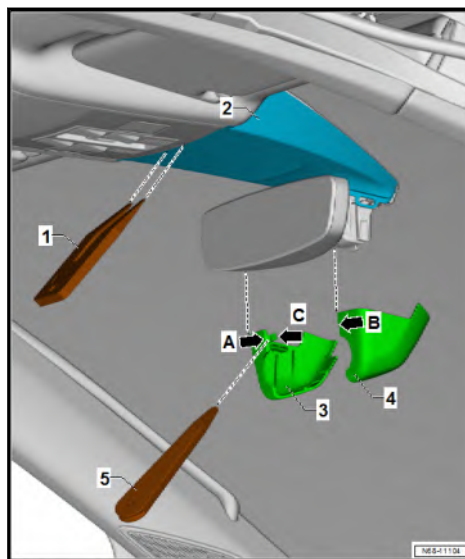
◆ Wedges -T10383-



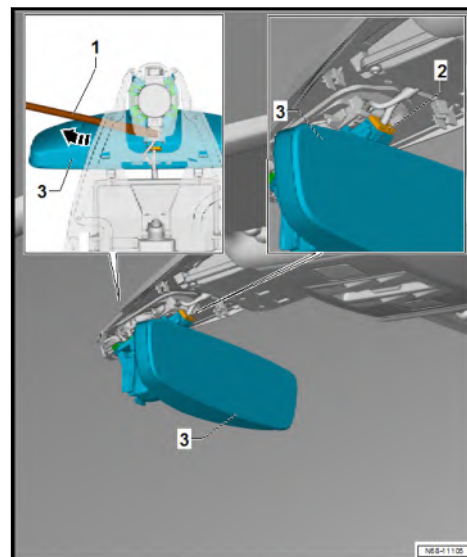
◆ Removal wedge -3409-



Removing



- Release retainer -C- in area marked by -arrow A- and -arrow B- using removal wedge -3409- -5-.
- Remove mirror base covers -3 and 4-.
- Unclip cover -2- using wedge -T10383/1- -1-.
- Detach cover -2-.
- Disconnect electrical connector -2-.



i Note

When performing the following step, ensure that the screwdriver/drift is inserted while turning but do not use as a lever.

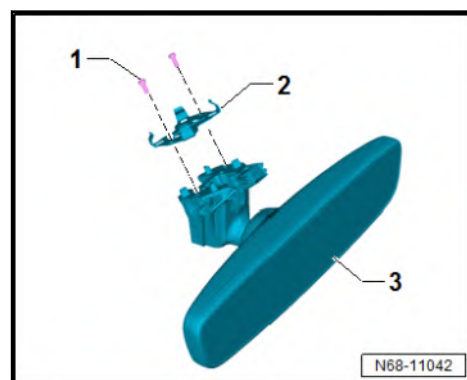
- Insert screwdriver or drift -1- with a diameter of exactly 4.5 mm into gap.
- Turn interior mirror -3- at mirror base approx. 20° in direction of -arrow-.
- Remove interior mirror -3-.

Installing

Install in reverse order of removal, observing the following:

i Note

- ◆ *Renew the spring clip each time the interior mirror has been removed ⇒ Electronic parts catalogue.*
- ◆ *Make sure that the components are from the same supplier.*
- Unscrew bolts -1-.



- Replace spring clip -2- on interior mirror -3- by a new part.



Note

Make sure that the mirror is aligned parallel to the retaining plate.

- Install interior mirror. In the process, make sure that the mirror engages audibly.

Specified torques

Component	Specified torque
Bolts -1-	1.5 Nm

9 Two-way radio

⇒ [o9.1 verview – two-way radio”, page 443](#)

⇒ [a9.2 nd installing two-way radioR8”, page 444](#)

⇒ [a9.3 nd installing two-way radio bracket”, page 444](#)

⇒ [a9.4 nd installing two-way radio unit microphoneR122”, page 445](#)

9.1 Assembly overview – two-way radio

1 - Two-way radio -R8-

- ❑ Removing and installing ⇒ [page 444](#)

2 - Washers

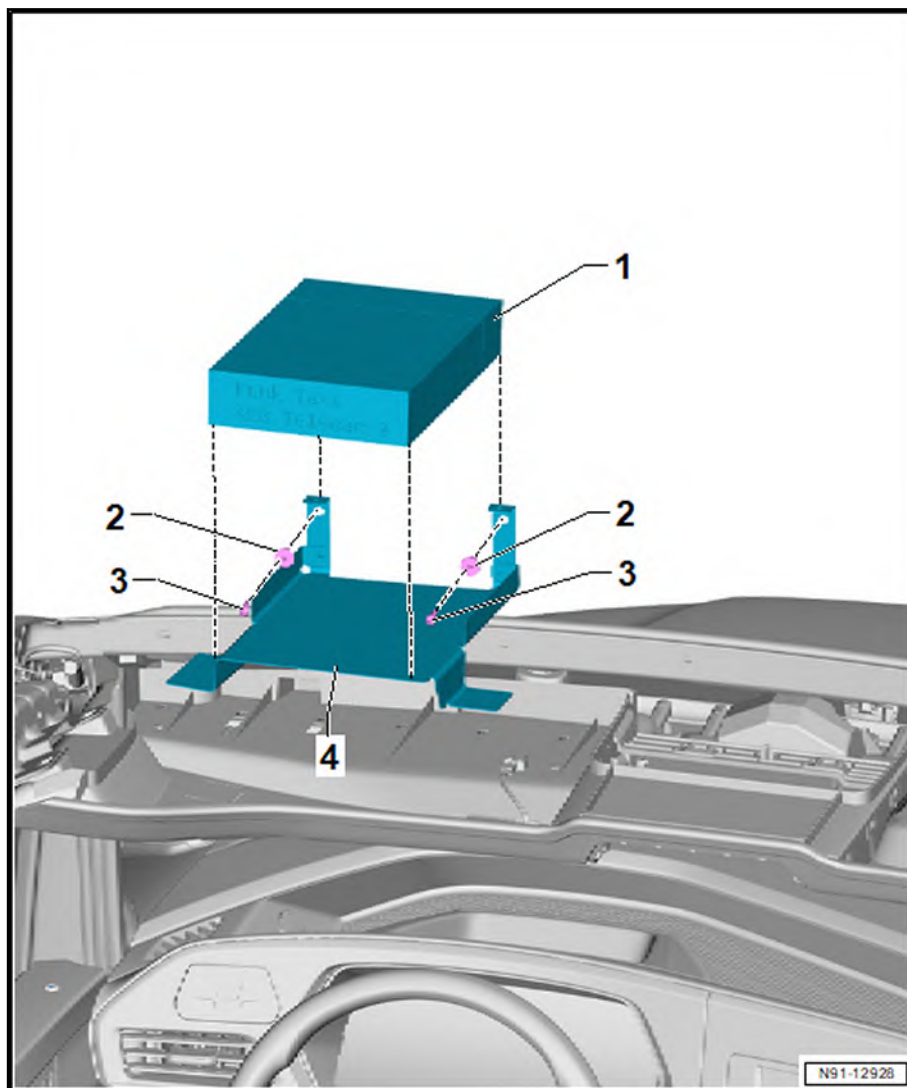
- ❑ Qty. 2

3 - Bolts

- ❑ Qty. 2
- ❑ 1.5 Nm

4 - Bracket

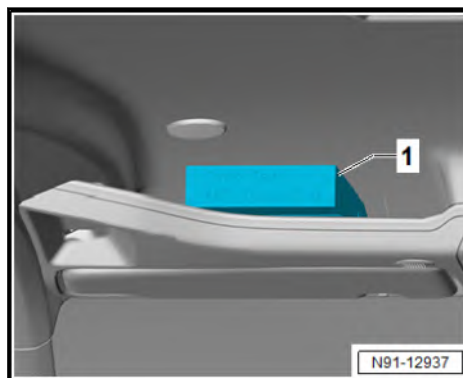
- ❑ Removing and installing ⇒ [page 444](#)





9.2 Removing and installing two-way radio -R8-

Removing



- Pull two-way radio -R8- -1- out of bracket.
- Disconnect electrical connectors.

Installing

Installation is carried out in reverse order.

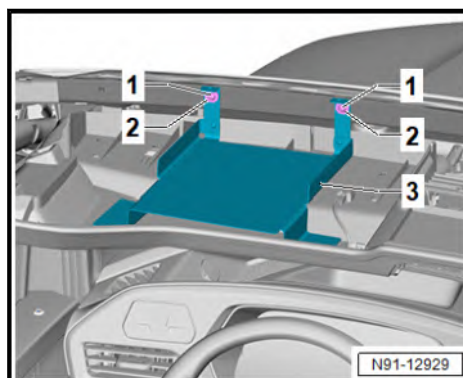
9.3 Removing and installing two-way radio bracket

Special tools and workshop equipment required

- ◆ Torque screwdriver -VAS 6494-



Removing



- Lower the moulded headliner ⇒ General body repairs, interior; Rep. gr. 70; Roof trims; Lowering the moulded headliner.
- Unscrew bolts -1-.



- Remove washers -2-.
- Remove retainer -3-.

Installing

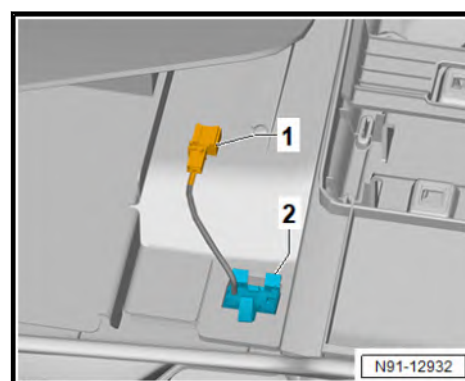
Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolts	1.5 Nm

9.4 Removing and installing two-way radio unit microphone -R122-

Removing



- Lower the moulded headliner ⇒ General body repairs, interior; Rep. gr. 70; Roof trims; Lowering the moulded headliner.
- Disconnect electrical connector -1-.
- Unclip two-way radio unit microphone -R122- -2-.

Installing

Installation is carried out in reverse order.



10 Two-way radio

⇒ [a10.1 nd installing two-way radioR8", page 446](#)

⇒ [a10.2 nd installing microphone 2 on driver sideR225", page 446](#)

⇒ [a10.3 nd installing two-way radio loudspeakerR25", page 447](#)

10.1 Removing and installing two-way radio -R8-

Removing



Note

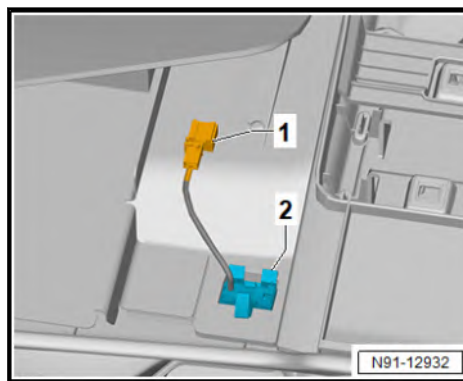
- ◆ *The two-way radio -R8- is installed by the user.*
- ◆ *The vehicle is only equipped with a preparation for the two-way radio. If applicable, observe the PR number.*
- Front passenger side; Removing and installing front seat
⇒ General body repairs, interior; Rep. gr. 72; Front seats; Removing and installing front seat
- Remove two-way radio -R8-. In the process, observe the specifications from the manufacturer of the two-way radio.

Installing

Installation is carried out in reverse order.

10.2 Removing and installing microphone 2 on driver side -R225-

Removing



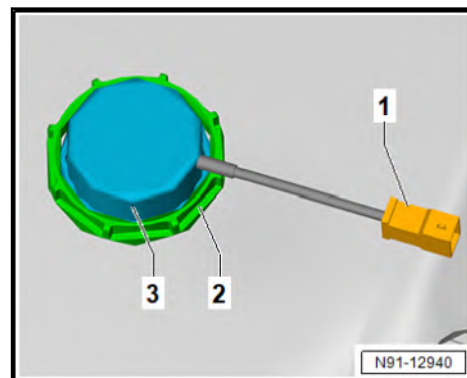
- Lower the moulded headliner at front ⇒ General body repairs, interior; Rep. gr. 70; Roof trims; Lowering the moulded headliner.
- Disconnect electrical connector -1-.
- Unclip microphone 2 on driver side -R225- -2-.

Installing

Installation is carried out in reverse order.

10.3 Removing and installing two-way radio loudspeaker -R25-

Removing



- Lower the moulded headliner at front ⇒ General body repairs, interior; Rep. gr. 70; Roof trims; Lowering the moulded headliner.
- Disconnect electrical connector -1-.
- Unscrew securing ring (threaded ring) -2-.
- Remove two-way radio loudspeaker -R25- -3- downwards.

Installing

Install in reverse order of removal, observing the following:

- Tighten securing ring (threaded ring) three full turns.



94 – Lights, bulbs, switches - exterior

1 Tipper body (Henschel)

⇒ [1.1, page 448](#)

⇒ [s1.2 system, general", page 450](#)

1.1 Lights

1.1.1 Removing and installing side marker lights, 3-way tipper

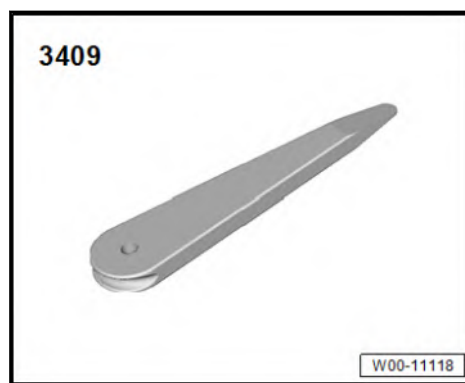
Removing and installing ⇒ [page 448](#) front side marker light bulb -M33-/-M34-

⇒ [page 449](#) Removing and installing side marker light bulb centre -M39-/-M40- and rear -M37-/-M38-

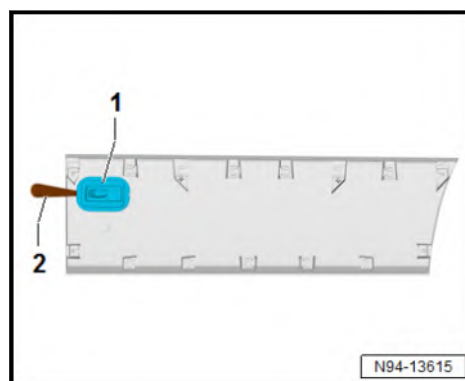
Removing and installing front side marker light bulb -M33-/-M34-

Special tools and workshop equipment required

- ◆ Removal wedge -3409-

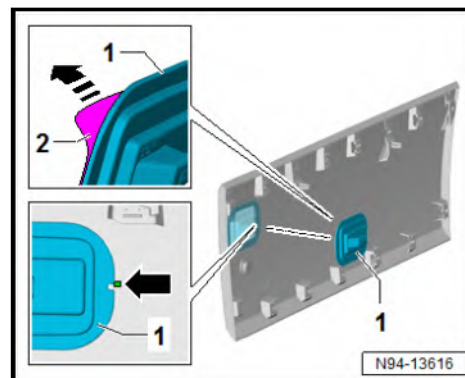


Removing



- Remove protective side strip of door ⇒ General body repairs, exterior; Rep. gr. 66; Mouldings/trims/extensions; Removing and installing protective side strip.
- Carefully lever side marker light -1- out of installation opening using removal wedge -3409- -2-.

Installing



Install in reverse order of removal, observing the following:

- Pull protective film -2- off side marker light -1-.
- Align groove of side marker light to marking on protective side strip -arrow-.
- Press on side marker light -1-.

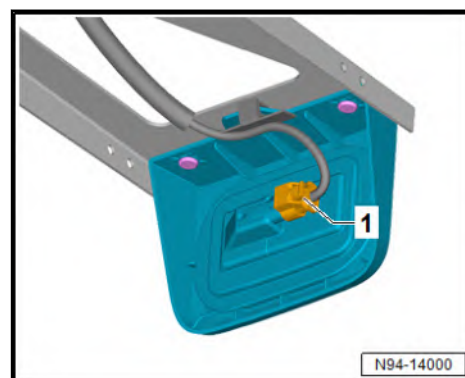
Removing and installing side marker light bulb at centre -M39-/-M40- and at rear -M37-/-M38-

Special tools and workshop equipment required

- ◆ Pop rivet pliers -VAS 5072A-

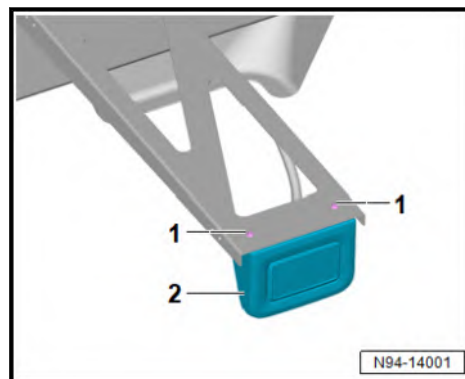


Removing



Rear side marker lights discontinued for vehicles as of June 2017 ➤ (on these vehicles, the end-outline marker lamps -M78-/-M79- are integrated in the tail light clusters)

- Disconnect connector -1-.
- Drill out pop rivets -1- using drill and drill bit of a diameter less than 4.5 mm.



- Remove side marker light -2-.
- Remove swarf.

Installing

Install in reverse order of removal, observing the following:

- Install pop rivets with pop rivet pliers -VAS 5072A-.

1.2 Electrical system, general

⇒ o1.2.1 ff the hydraulic unit - removing and installing limit switch", page 450

⇒ a1.2.2 nd installing start relay", page 452

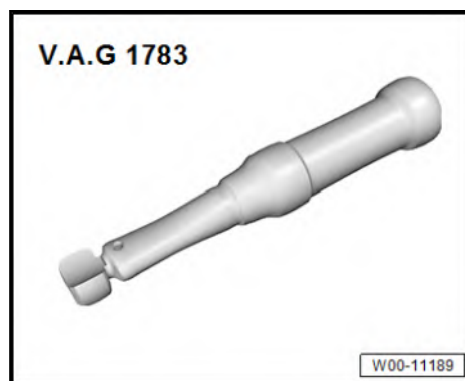
⇒ a1.2.3 nd installing main switch", page 454

⇒ f1.2.4 or 3-way tipper", page 455

1.2.1 Switching off the hydraulic unit - removing and installing limit switch

Special tools and workshop equipment required

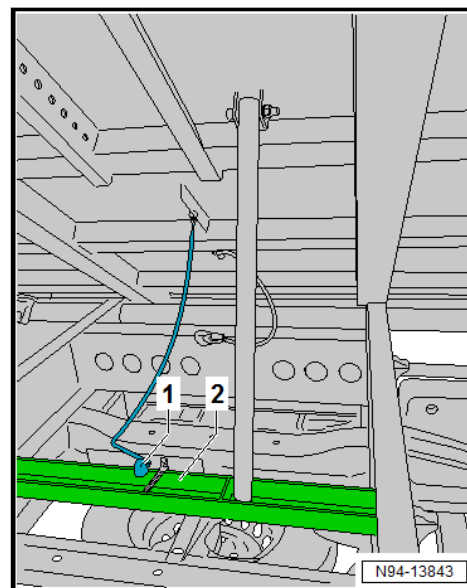
- ◆ Torque wrench -V.A.G 1783-



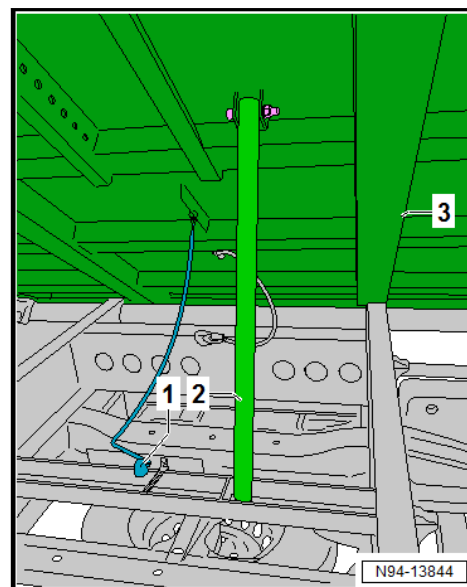
- ◆ Torque screwdriver -VAS 6494-



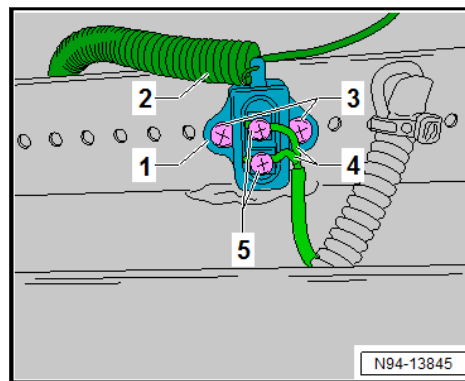
Removing



- ◆ The limit switch -1- for the hydraulic unit is located on the body frame -2-.
- Secure load bed -3- against falling using existing safety support -2-.



- Disconnect battery ⇒ Electrical system; Rep. gr. 27; Battery; Disconnecting and reconnecting battery.
- Detach cover -1-.
- Detach spring -2-.



- Unscrew bolts -5-.
- Detach wiring harness -4-.
- Unscrew bolts -3-.
- Remove limit switch -1-.

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolts -3-	3.2 Nm
Bolts -5-	1.5 Nm

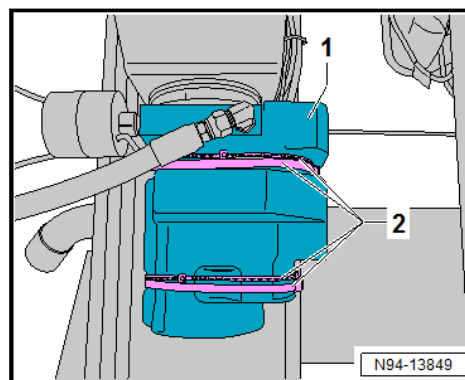
1.2.2 Removing and installing start relay

Special tools and workshop equipment required

- ♦ Torque wrench -V.A.G 1410A-

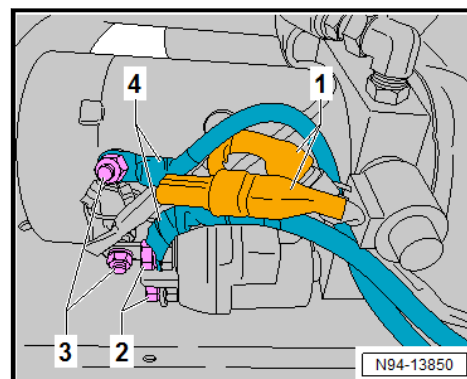


Removing

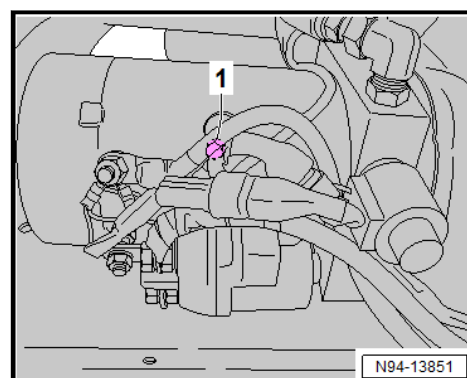


- Secure load bed against falling using provided safety support.

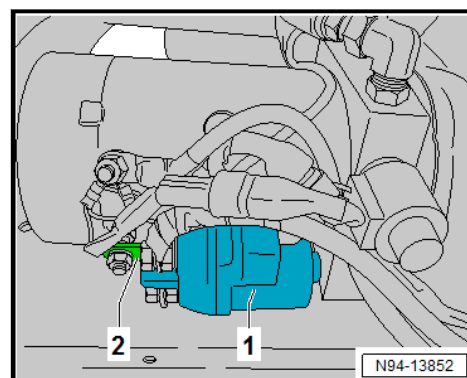
- Disconnect battery ⇒ Electrical system; Rep. gr. 27; Battery; Disconnecting and connecting battery.
- Cut through cable tie -2-.
- Detach cover -1-.
- Disconnect electrical connectors -1-.



- Unscrew nuts -2-.
- Unscrew nuts -3-.
- Remove wiring harnesses -4-.
- Unscrew bolt -1-.



- Remove earth strap -2-.



- Remove start relay -1-.

Installing

Install in reverse order of removal, observing the following:



Specified torques

Component	Specified torque
Bolt -1-	10 Nm
Nuts -2-	20 Nm
Nuts -3-	20 Nm

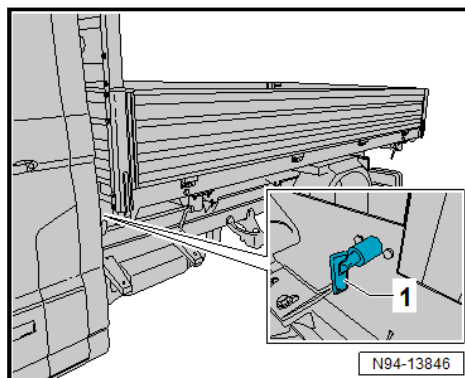
1.2.3 Removing and installing main switch

Special tools and workshop equipment required

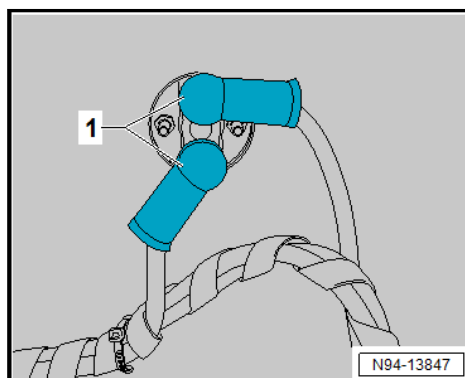
- ♦ Torque wrench 6-50 Nm -V.A.G 1331-



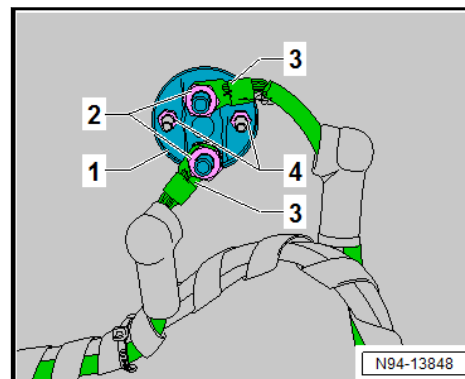
Removing



- Secure load bed against falling using provided safety support.
- Switch off main switch -1-.
- Disconnect battery ⇒ Electrical system; Rep. gr. 27; Battery; Disconnecting and reconnecting battery.
- Pull back caps -1-.



- Unscrew nuts -2-.



- Detach wiring harness -3-.
- Unscrew nuts -4-.
- Remove main switch -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Nuts -2-	41 Nm
Nuts -4-	8.5 Nm

1.2.4 Switch for 3-way tipper

- Remove tipper control switch ⇒ Electrical system; Rep. gr. 96; Lights and controls in dash panel; Removing and installing switch unit on left and right next to steering wheel.



2 Tipper body (Schoon)

⇒ [2.1 , page 456](#)

2.1 Electrics

⇒ [o2.1.1 ff the hydraulic unit - removing and installing limit switch", page 456](#)

⇒ [a2.1.2 nd installing start relay for 12-V Bosch Rexroth pumps", page 457](#)

⇒ [a2.1.3 nd installing main fuse", page 459](#)

⇒ [f2.1.4 or 3-way tipper", page 461](#)

2.1.1 Switching off the hydraulic unit - removing and installing limit switch

Special tools and workshop equipment required

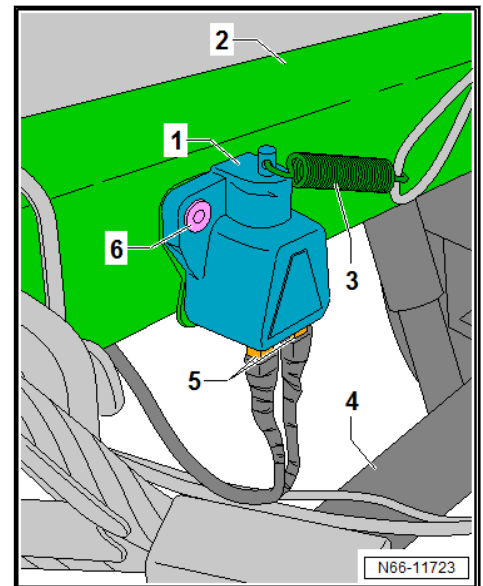
◆ Pop rivet pliers -VAS 5072A-



◆ Pin punch -VAS 5513-



Removing



- ◆ The limit switch -1- for the hydraulic unit is located on the body frame -2- above the rear axle -4-.
- Secure load bed against falling using provided safety support.
- Disconnect battery ⇒ Electrical system; Rep. gr. 27; Battery; Disconnecting and connecting battery.
- Disconnect electrical connectors -5-.
- Detach spring -3-.

CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out rivet head -6- using drill and drill bit.
- Drive out rivet shank -6- with drift -VAS 5513-.
- Remove swarf.
- Detach limit switch -1- from body frame -2-.

Installing

Install in reverse order of removal, observing the following:

- Use rivet pliers -VAS 5072A- to install new pop rivet.

2.1.2 Removing and installing start relay for 12-V Bosch Rexroth pumps

Special tools and workshop equipment required



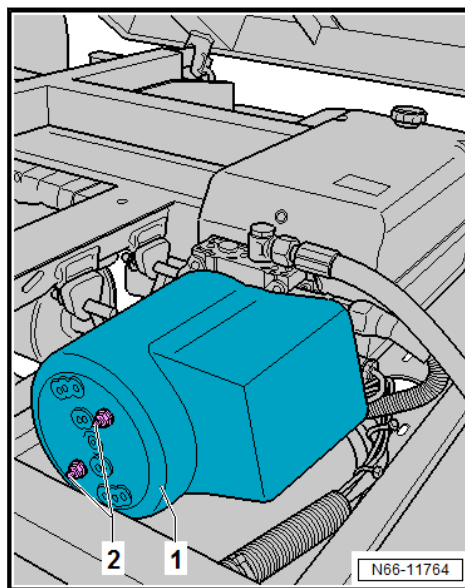
- ◆ Torque wrench -V.A.G 1410A-



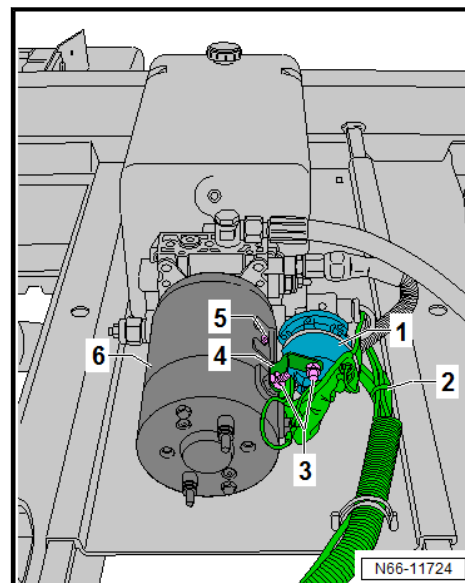
- ◆ Torque screwdriver -VAS 6494-



Removing



- Secure load bed against falling using provided safety support.
- Disconnect battery ⇒ Electrical system; Rep. gr. 27; Battery; Disconnecting and connecting battery.
- Unscrew nuts -2-.
- Detach cover -1-.
- Disconnect wiring harness -2-.



- Unscrew nuts -3-.
 - Remove earth strap -4-.
- Up to model year 2017, a hose clip has been used for securing the start relay instead of the bolt -5-.
- Unscrew bolt -5-.
 - Detach start relay -1- from hydraulic unit -6-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Nuts -2-	2.7 Nm
Earth strap nuts -3-	11 Nm
Bolt/hose clip for start relay -5-	2.7 Nm

2.1.3 Removing and installing main fuse

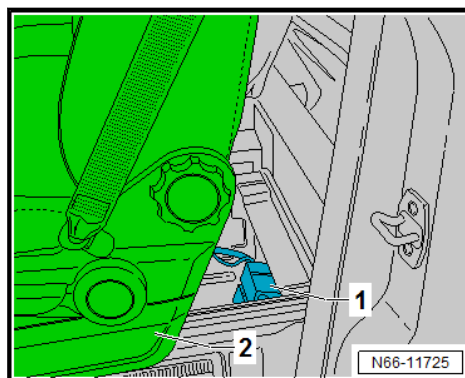
Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1410A-

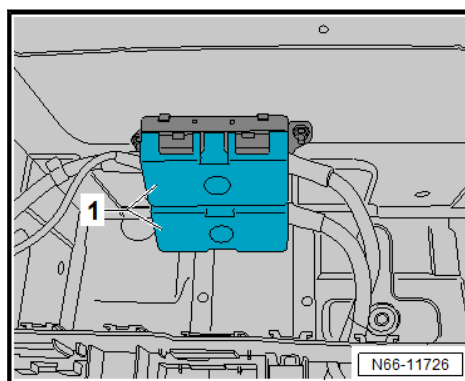




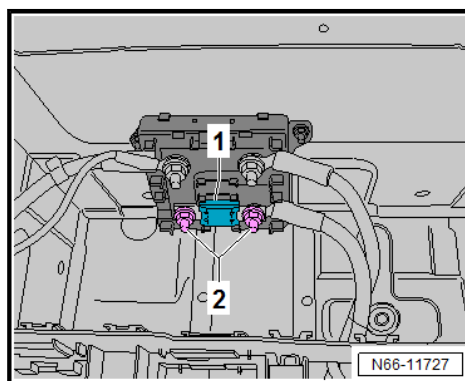
Removing



- Disconnect battery ⇒ Electrical system; Rep. gr. 27; Battery; Disconnecting and connecting battery.
- Move driver seat -2- to foremost position.
- Take off covers -1-.



- Unscrew nuts -2-.



- Remove main fuse -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Nuts -2-	8 Nm



2.1.4 Switch for 3-way tipper

- Remove tipper control switch ⇒ Electrical system; Rep. gr. 96; Lights and controls in dash panel; Removing and installing switch unit on left and right next to steering wheel.



3 Box body (Junge)

⇒ [a3.1 nd installing tail light MX3/MX4", page 462](#)

⇒ [a3.2 nd installing rear fog light L46/L47", page 463](#)

⇒ [a3.3 nd installing number plate lightX4/X5", page 464](#)

⇒ [a3.4 nd installing additional brake light M25", page 465](#)

⇒ [a3.5 nd installing upper marker lights", page 466](#)

⇒ [a3.6 nd installing side marker light
M33/M34/M37/M38/M39/M40 with bracket", page 466](#)

⇒ [a3.7 nd installing lower marker light", page 467](#)

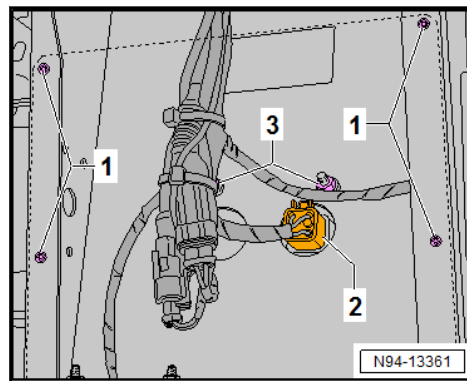
3.1 Removing and installing tail light - MX3-/-MX4-

Special tools and workshop equipment required

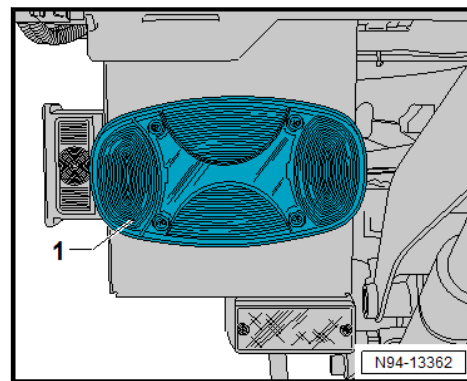
- ◆ Torque wrench -V.A.G 1410A-



Removing



- Turn light switch to "0" position.
- Unscrew bolts -1-.
- Remove trim.
- Disconnect connector -2-.
- Unscrew nuts -3-.
- Remove tail light cluster -MX3-/-MX4- -1- from box body.



Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Nuts -3-	5 Nm
Bolts -1-	Tighten onto stop

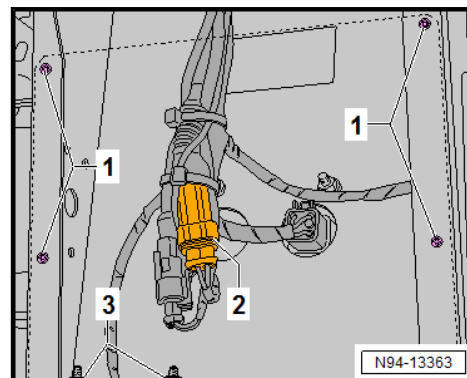
3.2 Removing and installing rear fog light -L46-/-L47-

Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1410A-



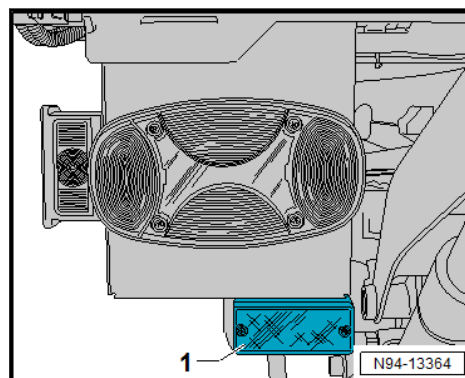
Removing



- Turn light switch to “0” position.
- Unscrew bolts -1-.
- Remove trim.
- Release cable tie.



- Disconnect connector -2-.
- Unscrew nuts -3-.
- Remove rear fog light -1- from box body.



Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolts -1-	Tighten onto stop
Nuts -3-	5 Nm

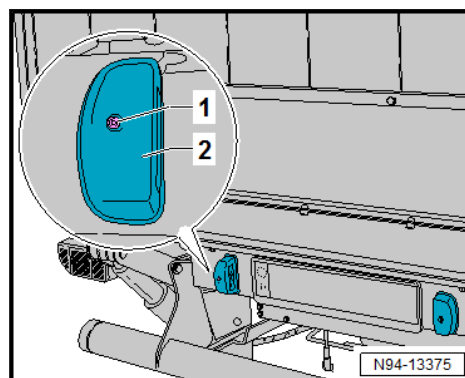
3.3 Removing and installing number plate light -X4-/-X5-

Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1410A-



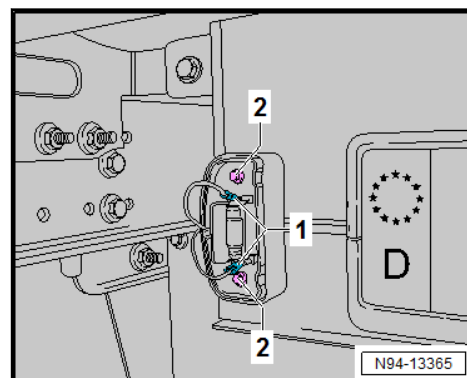
Removing



- Turn light switch to "0" position.



- Unscrew bolt -1-.
- Remove cap -2-.
- Separate electrical connectors -1- from connector contacts.



- Unscrew bolts -2-.
- Remove number plate light -X4-/-X5- from box body.

Installing

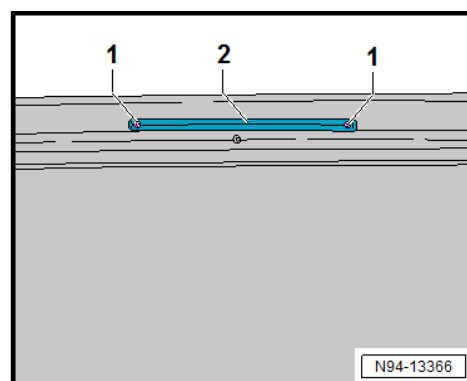
Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolt -1-	0.8 Nm
Bolts -2-	0.8 Nm

3.4 Removing and installing additional brake light -M25-

Removing



- Unscrew bolts -1-.
- Remove high-level brake light -2- from box body, separating electrical connector to do so.

Installing

Installation is carried out in reverse order.

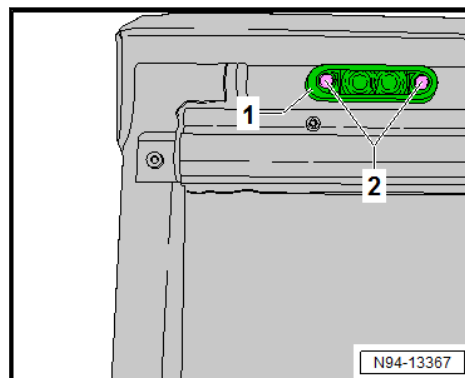
Specified torques

Component	Specified torque
Bolts -1-	Tighten onto stop



3.5 Removing and installing upper marker lights

Removing



- Turn light switch to “0” position.
- Unscrew bolts -2-.
- Remove upper marker light -1- from box body, separating electrical connector to do so.

Installing

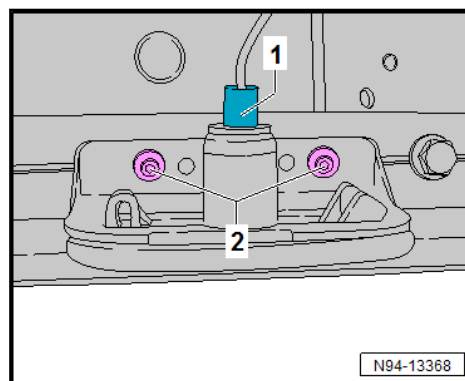
Installation is carried out in reverse order.

Specified torques

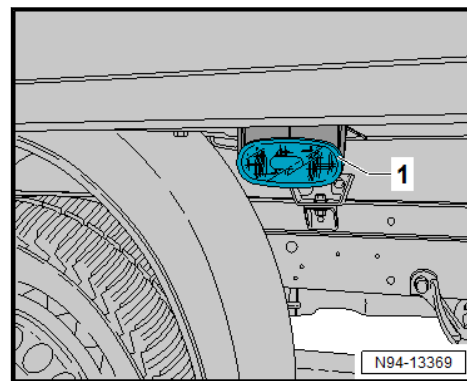
Component	Specified torque
Bolts -2-	Tighten onto stop

3.6 Removing and installing side marker light -M33-/-M34-/-M37-/-M38-/-M39-/-M40- with bracket

Removing



- Turn light switch to “0” position.
- Disconnect connector -1-.
- Unscrew bolts -2-.
- Remove side marker light -1- from box body.



Installing

Installation is carried out in reverse order.

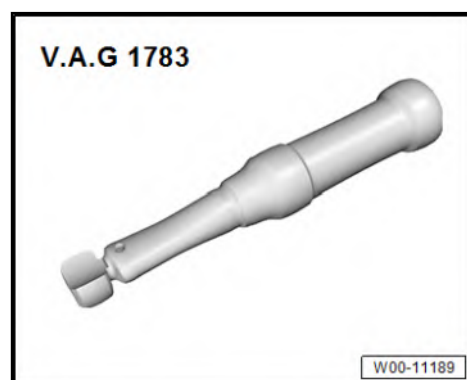
Specified torques

Component	Specified torque
Bolts -2-	Tighten onto stop

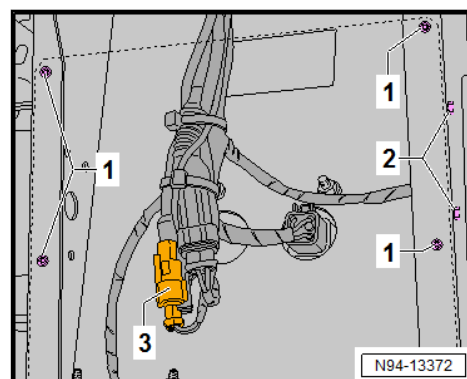
3.7 Removing and installing lower marker light

Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1783-



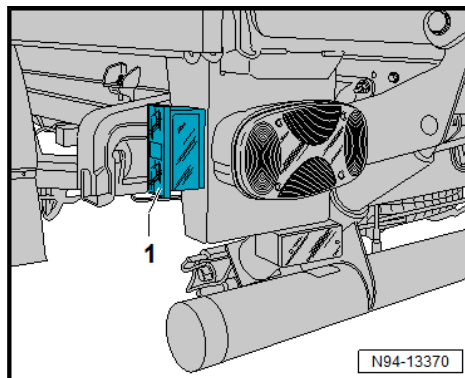
Removing



- Turn light switch to “0” position.
- Unscrew bolts -1-.
- Remove trim.
- Release cable tie.
- Disconnect connector -3-.



- Unscrew nuts -2-.
- Remove lower marker light -1- from box body.



Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Nuts -2-	3 Nm
Bolts -1-	Tighten onto stop

4 Box body (Spier)

⇒ [a4.1 nd installing tail light MX3/MX4", page 469](#)

⇒ [a4.2 nd installing number plate lightX4/X5", page 470](#)

⇒ [a4.3 nd installing high-level brake light", page 471](#)

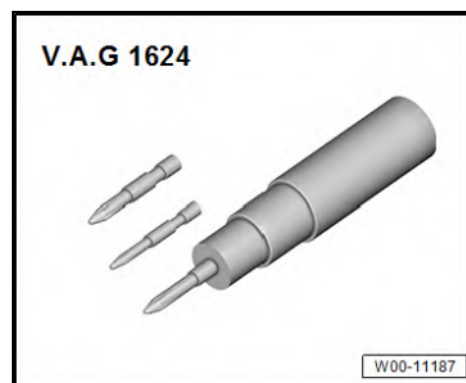
⇒ [a4.4 nd installing upper marker lights", page 471](#)

⇒ [a4.5 nd installing side marker light M33/M34/M37/M38/M39/M40", page 472](#)

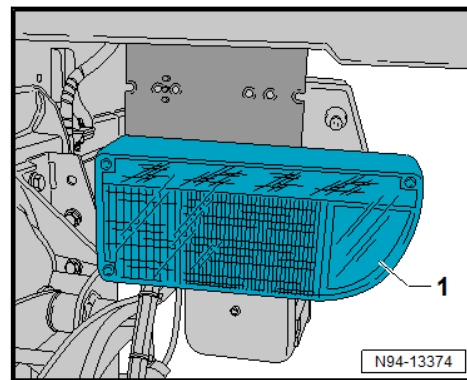
4.1 Removing and installing tail light - MX3-/-MX4-

Special tools and workshop equipment required

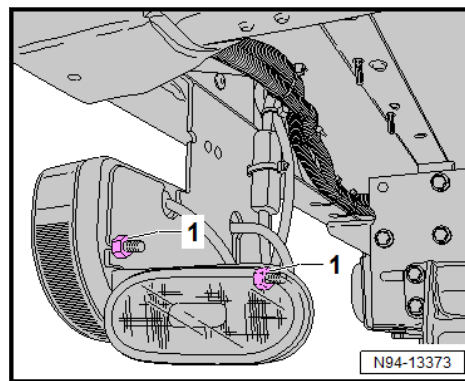
- ◆ Torque screwdriver -V.A.G 1624-



Removing



- Turn light switch to "0" position.
- Unscrew bolts from lens -1-.
- Remove lens -1- from tail light cluster.
- Separate electrical connector in tail light cluster and pull out.
- Unscrew nuts -1-.



- Remove tail light cluster -MX3-/-MX4- from box body.

Installing

Install in reverse order of removal, observing the following:

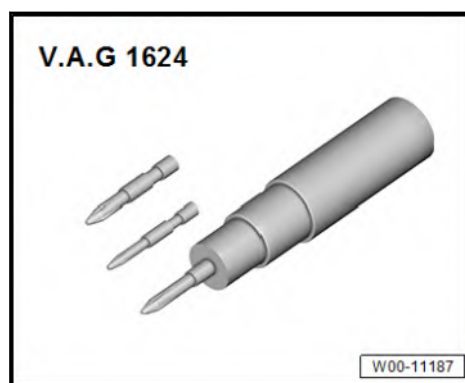
Specified torques

Component	Specified torque
Nuts -1-	1.5 Nm
Bolts -1-	0.5 Nm

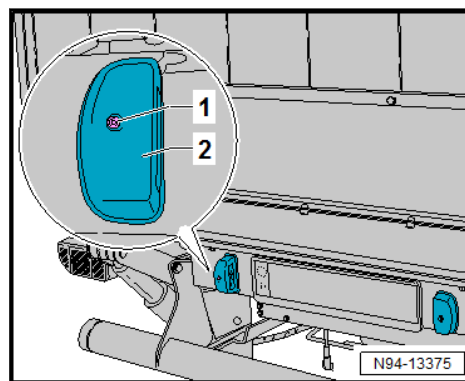
4.2 Removing and installing number plate light -X4-/-X5-

Special tools and workshop equipment required

- ◆ Torque screwdriver -V.A.G 1624-

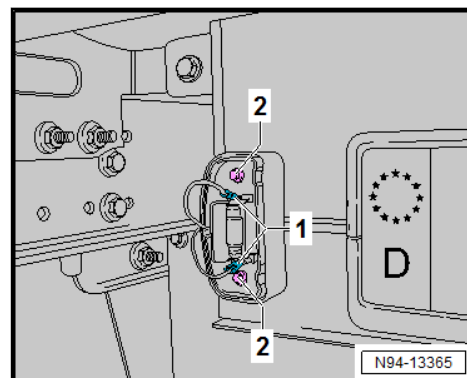


Removing



- Turn light switch to “0” position.
- Unscrew bolt -1-.
- Remove cap -2-.

- Separate electrical connectors -1- from connector contacts.



- Unscrew bolts -2-.
- Remove number plate light -X4/-X5- from box body.

Installing

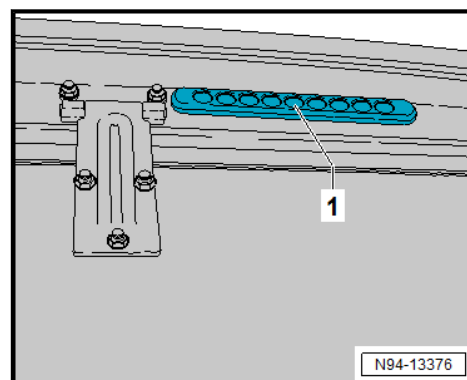
Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolt -1-	0.8 Nm
Bolts -2-	0.8 Nm

4.3 Removing and installing high-level brake light

Removing



- Detach brake light -1- from box body by suitable means.
- Remove brake light from box body, separating wiring harness when doing so.

Installing

Install in reverse order of removal, observing the following:

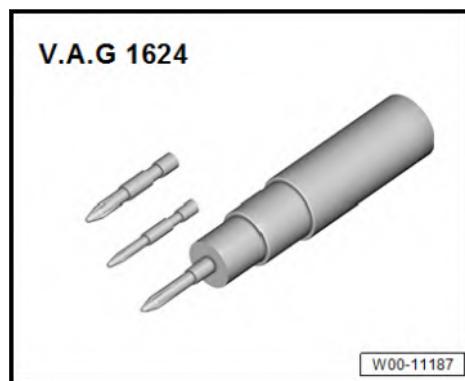
- Repair wiring harness ⇒ Electrical system, General information; Rep. gr. 97; Wiring; Repairing wiring harnesses and connectors.

4.4 Removing and installing upper marker lights

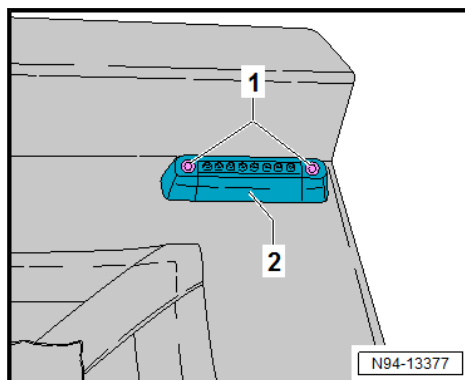
Special tools and workshop equipment required



◆ Torque screwdriver -V.A.G 1624-



Removing



- Turn light switch to “0” position.
- Unscrew bolts -1-.
- Remove upper marker light -2- from box body, cutting through wiring harness to do so.

Installing

Install in reverse order of removal, observing the following:

- Repair wiring harness ⇒ Electrical system, General information; Rep. gr. 97; Wiring; Repairing wiring harnesses and connectors.

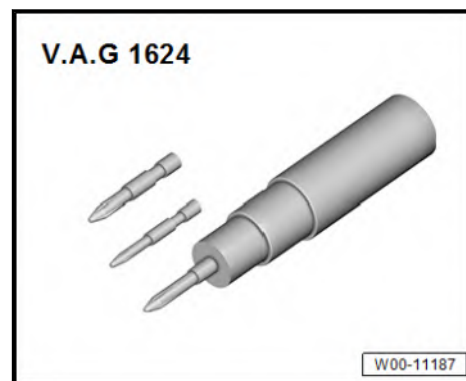
Specified torques

Component	Specified torque
Bolts -1-	0.3 Nm

4.5 Removing and installing side marker light -M33-/-M34-/-M37-/-M38-/-M39-/-M40-

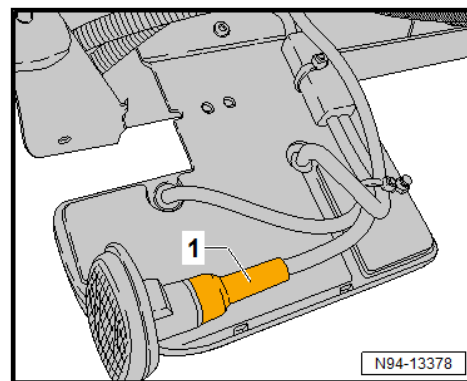
Special tools and workshop equipment required

◆ Torque screwdriver -V.A.G 1624-



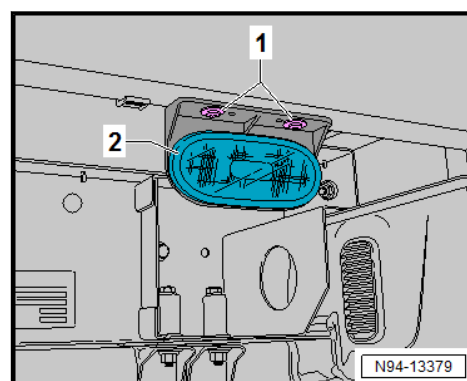
- ◆ Since there is no electrical connector on the marker light, the wiring harness must be cut through for complete renewal of the marker light with bulb holder.

Removing without bracket



- Turn light switch to "0" position.
- Cut through wiring harness -1-.
- Detach marker light from bracket.

Removing with bracket



- Turn light switch to "0" position.
- Cut through wiring harness.
- Unscrew bolts -1-.
- Remove side marker light -2- with bracket from box body.

Installing

Install in reverse order of removal, observing the following:



- Repair wiring harness ⇒ Electrical system, General information; Rep. gr. 97; Wiring; Repairing wiring harnesses and connectors.

Specified torques

Component	Specified torque
Bolts -1-	1 Nm

5 Box body (Schmitz Cargobull)

⇒ [a5.1 nd installing tail light cluster", page 475](#)

⇒ [a5.2 nd installing number plate light", page 476](#)

⇒ [a5.3 nd installing high-level brake light", page 478](#)

⇒ [a5.4 nd installing rear upper marker light", page 479](#)

⇒ [a5.5 nd installing front upper marker light", page 480](#)

⇒ [a5.6 nd installing side marker light", page 480](#)

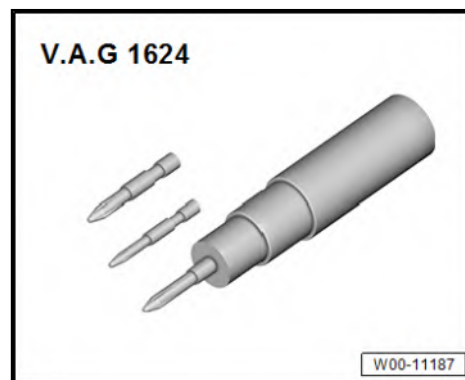
5.1 Removing and installing tail light cluster

Special tools and workshop equipment required

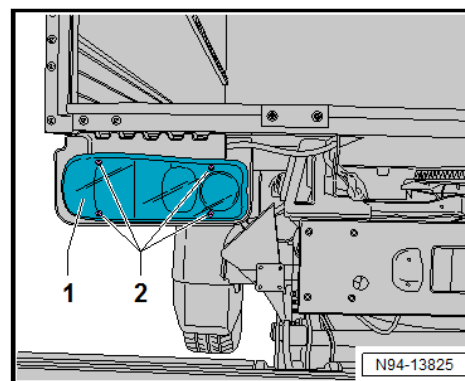
◆ Torque wrench -V.A.G 1410A-



◆ Torque screwdriver -V.A.G 1624-



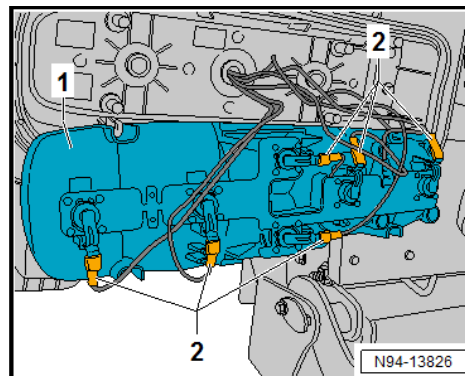
Removing



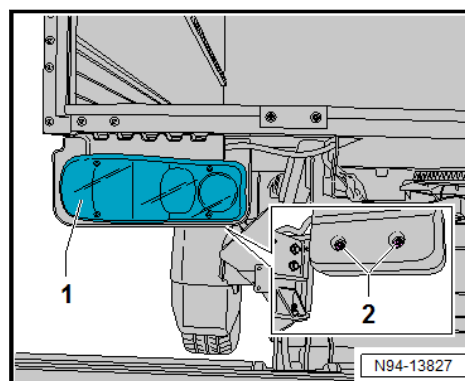
- Turn light switch to 0 position.
- Unscrew bolts -2-.



- Remove lens -1-.
- Remove reflector -1-.



- Mark electrical connectors -2-.
- Disconnect electrical connectors -2-.
- Unscrew nuts -2-.



- Remove tail light cluster -1-.

Installing

Install in reverse order of removal, observing the following:

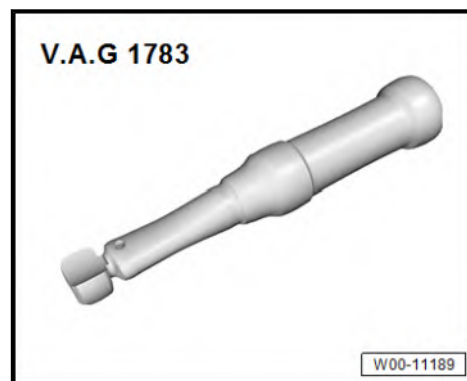
Specified torques

Component	Specified torque
Nuts -2-	5 Nm
Bolts securing lens -2-	1.5 Nm

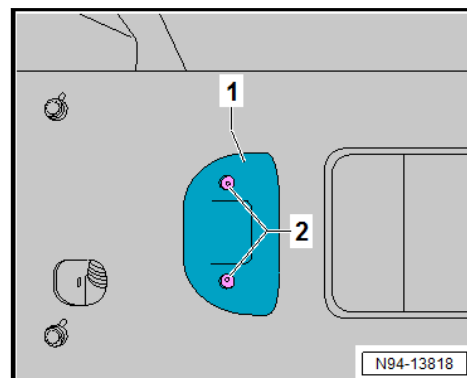
5.2 Removing and installing number plate light

Special tools and workshop equipment required

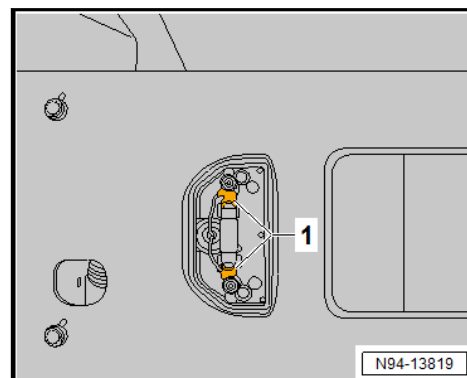
◆ Torque wrench -V.A.G 1783-



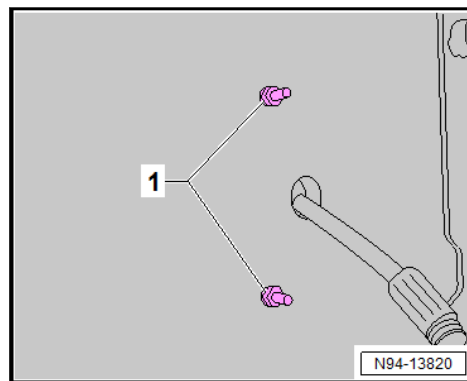
Removing



- Turn light switch to 0 position.
- Unscrew bolts -2-.
- Remove cap -1-.
- Disconnect electrical connectors -1-.



- Unscrew nuts -1-. Remove number plate light.



Installing

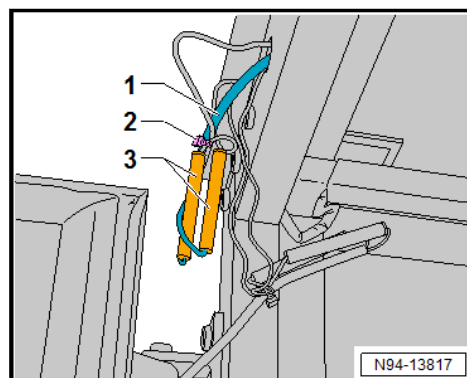
Install in reverse order of removal, observing the following:

Specified torques

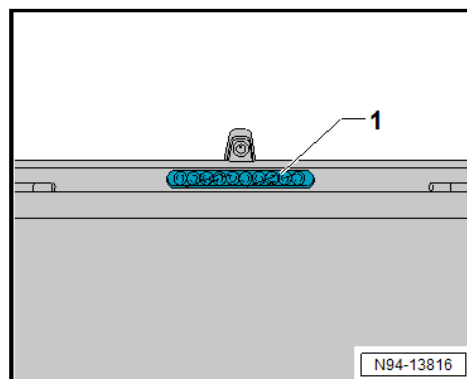
Component	Specified torque
Bolts -2-	2.5 Nm
Nuts -1-	2.5 Nm

5.3 Removing and installing high-level brake light

Removing



- Remove top left marker light ⇒ [page 479](#) .
- Pull out wiring harness -1-.
- Cut through pull relief -2-.
- Disconnect electrical connectors -3-.
- Detach high-level brake light -1- from box body by suitable means.



- Remove high-level brake light from box body. Pull out wiring harness when doing this.

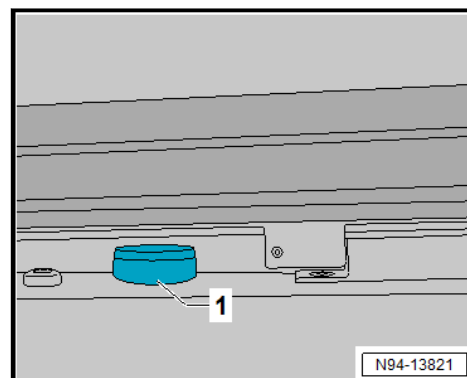
Installing

Install in reverse order of removal, observing the following:

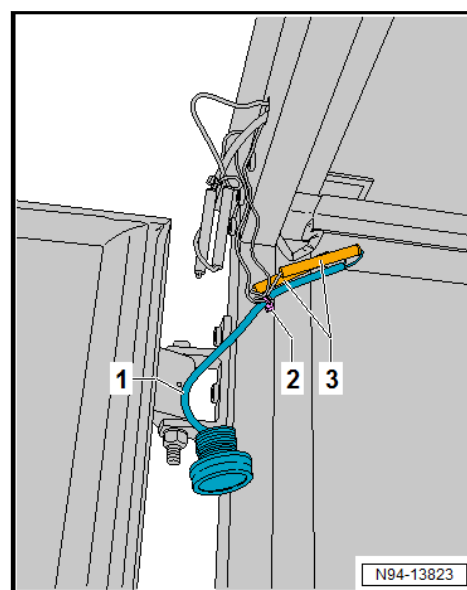
- Clean bonding surfaces.

5.4 Removing and installing rear upper marker light

Removing



- Turn light switch to 0 position.
- Remove upper roof marker light -1-.
- Pull out wiring harness -1-.



- Cut through pull relief -2-.
- Disconnect electrical connectors -3-.

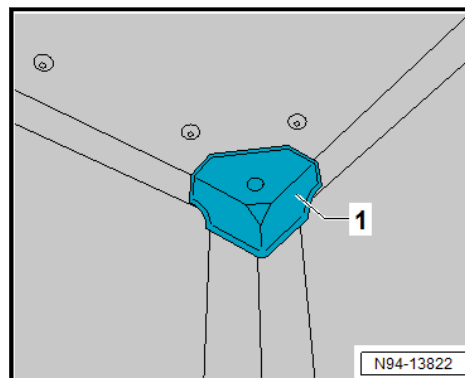
Installing

Installation is carried out in reverse order.

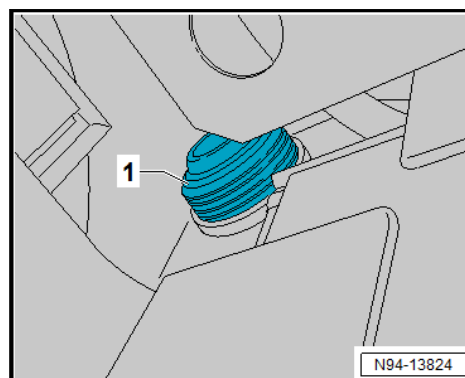


5.5 Removing and installing front upper marker light

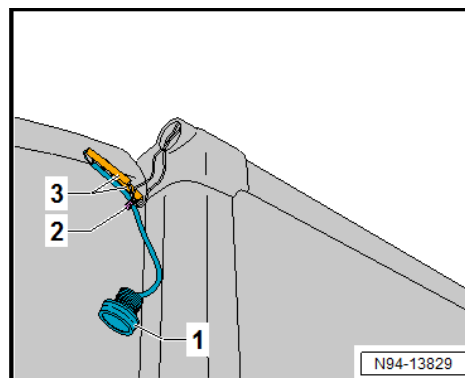
Removing



- Turn light switch to 0 position.
- Remove cap -1-.
- Push marker light -1- outwards.



- Pull out wiring harness -1-.



- Cut through pull relief -2-.
- Disconnect electrical connectors -3-.

Installing

Installation is carried out in reverse order.

5.6 Removing and installing side marker light

Special tools and workshop equipment required

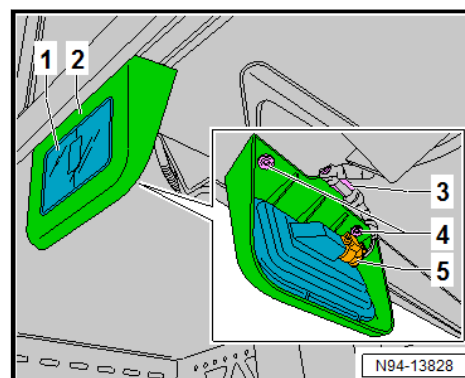
◆ Pop rivet pliers -VAS 5072A-



◆ Pin punch -VAS 5513-



Removing



- Turn light switch to 0 position.
- Unclip wiring harness -3-.
- Disconnect connector -5-.
- Detach side marker light -1- from bracket -2-.

CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out rivet head -4- using drill and drill bit.
- Drive out rivet shank -4- with drift -VAS 5513-.
- Remove swarf.



- Remove bracket -2-.

Installing

Install in reverse order of removal, observing the following:

- Use rivet pliers -VAS 5072A- to install new pop rivet.

6 Tail lift (Sörensen)

⇒ [a6.1 nd installing hazard warning lightM115/M116", page 483](#)

⇒ [a6.2 nd installing foot switchF558", page 484](#)

⇒ [a6.3 nd installing tail lift operating unitEX56", page 484](#)

⇒ [a6.4 nd installing tilt angle sensor", page 485](#)

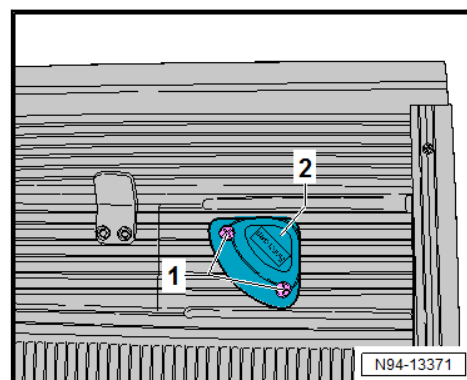
6.1 Removing and installing hazard warning light -M115/M116-

Special tools and workshop equipment required

- ◆ Torque screwdriver -VAS 6494-



Removing



- ◆ For the safety of approaching traffic, the tail lift platform must be clearly identifiable during operation by means of warning signs and flashing lights in accordance with paragraph 53b section 5 of the German road traffic act (StVZO).
- ◆ The hazard warning lights flash as soon as the tail lift is switched on and opened.
- ◆ The hazard warning light is connected to the wiring in the platform.
- Select off position on switch for tail lift operation ⇒ [page 522](#).
- Unscrew bolts -1-.
- Remove hazard warning light -M115/M116- -2- from tail lift, separating electrical connector to do so.

Installing

Install in reverse order of removal, observing the following:



Specified torques

Component	Specified torque
Bolts	2.5 Nm

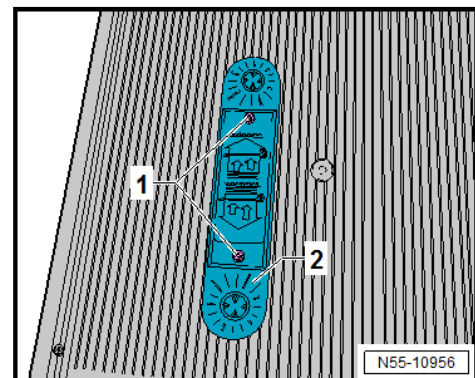
6.2 Removing and installing foot switch - F558-

Special tools and workshop equipment required

- ◆ Torque screwdriver -VAS 6494-



Removing



- Select off position on switch for tail lift operation ⇒ [page 522](#) .
- Unscrew bolts -1-.
- Remove foot switch -F558- -2- from tail lift, separating electrical connector to do so.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolts	2.5 Nm

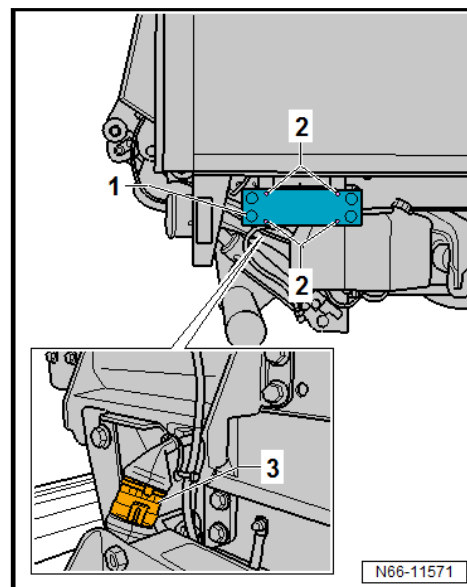
6.3 Removing and installing tail lift operating unit -EX56-

Special tools and workshop equipment required

- ◆ Torque screwdriver -VAS 6494-



Removing



- Select off position on switch for tail lift operation ⇒ [page 522](#).
- Cut through cable tie along wiring harness.
- Separate electrical connector -3- to tail lift operating unit -EX56- -1-.
- Unscrew bolts -2-.
- Remove tail lift operating unit -EX56- -1- from bracket.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolts	2 Nm

6.4 Removing and installing tilt angle sensor

Special tools and workshop equipment required



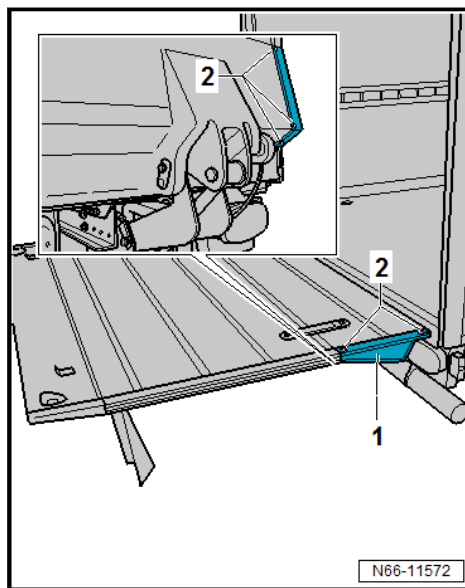
- ◆ Torque wrench -V.A.G 1410A-



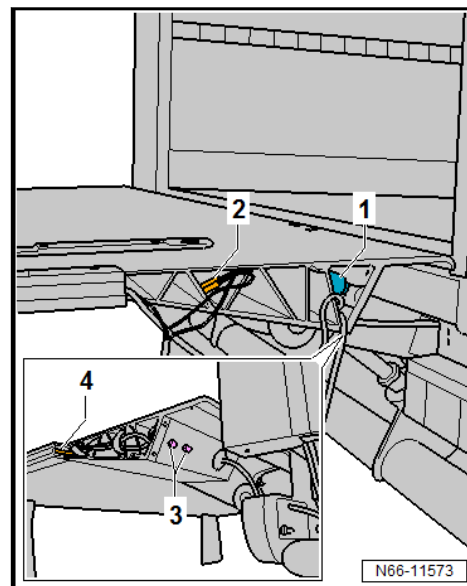
- ◆ Torque screwdriver -VAS 6494-



Removing



- Select off position on switch for tail lift operation ➤ [page 522](#).
- Unscrew bolts -2- from sidebar (aluminium profile) -1-.
- Remove sidebar (aluminium profile) -1- from platform.
- Separate electrical connector -2- from foot switch.



- Separate electrical connector -4- to right hazard warning light.
- Unscrew nuts -3- from tilt angle sensor -1-.
- Remove tilt angle sensor -1- from platform and disconnect the electrical connector to tail lift operating switch.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolts -2-	2.5 Nm
Nuts -3-	15 Nm



7 Tail lift (BÄR - Cargolift)

⇒ [a7.1 nd installing hazard warning lightM115/M116", page 488](#)

⇒ [a7.2 nd installing foot switchF556/F557", page 489](#)

⇒ [a7.3 nd installing operating unitEX56", page 490](#)

⇒ [a7.4 nd installing foot switch unitF558", page 491](#)

7.1 Removing and installing hazard warning light -M115/M116-

Special tools and workshop equipment required

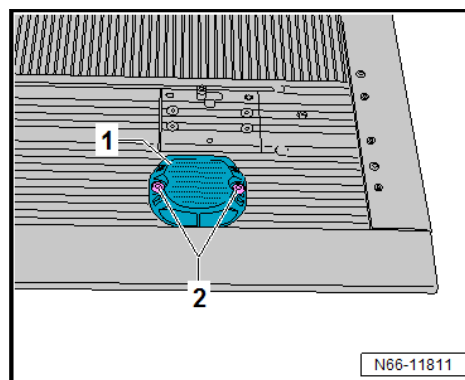
◆ Pop rivet pliers -VAS 5072A-



◆ Pin punch -VAS 5513-



Removing



– Switch off tail lift operation switch.

⚠ CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out rivet head -2- using drill and drill bit.
- Drive out rivet shank -2- with suitable drift -VAS 5513-.
- Remove swarf.
- Detach hazard warning light -M115/M116- -1-.
- While doing this, disconnect electrical connector.

Installing

Install in reverse order of removal, observing the following:

- Use rivet pliers -VAS 5072A- to install new pop rivets.

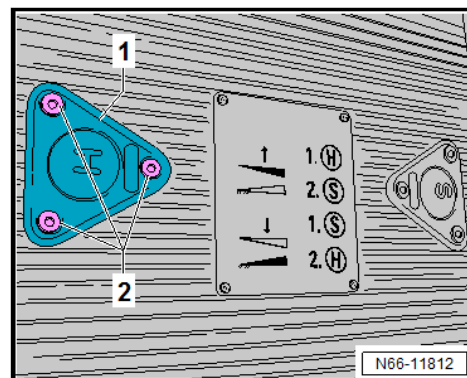
7.2 Removing and installing foot switch - F556/F557-

Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1410A-



Removing



- Open tail lift.
- Switch off tail lift operation switch.
- Unscrew bolts -2-.
- Remove foot switch -F556/F557- -1-.
- While doing this, disconnect electrical connector.



Installing

Install in reverse order of removal, observing the following:

- Route wiring harness in a loop. Make sure no kinks are formed.

Specified torques

Component	Specified torque
Bolts	10 Nm

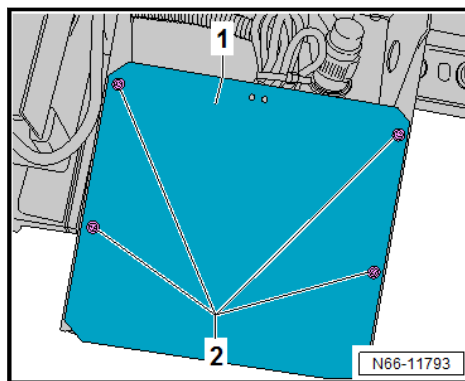
7.3 Removing and installing operating unit -EX56-

Special tools and workshop equipment required

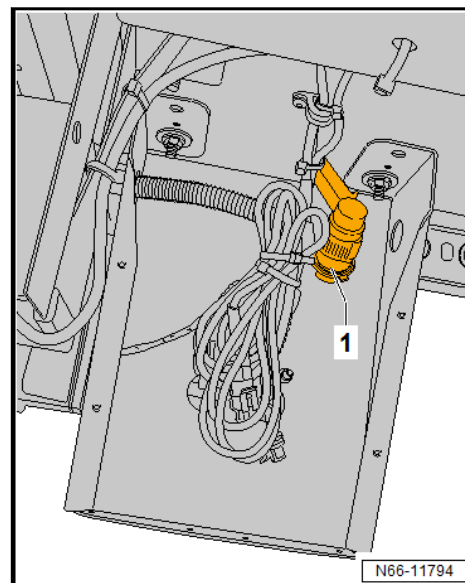
- ◆ Torque wrench -V.A.G 1410A-



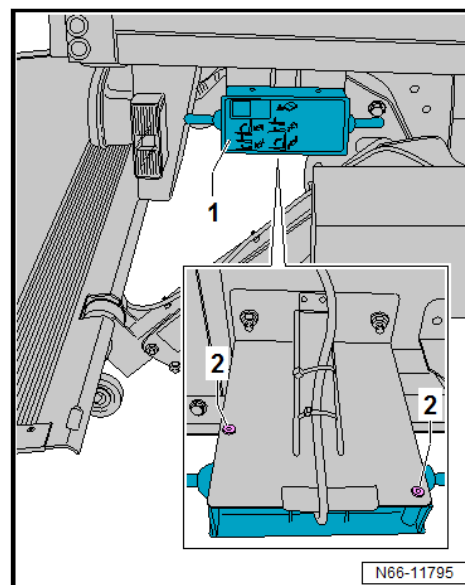
Removing



- Switch off tail lift operation switch.
- Cut through cable tie along wiring harness.
- Unscrew bolts -2-.
- Detach cover -1-.
- Disconnect connector -1-.



- Unscrew bolts -2-.



- Remove operating unit -EX56- -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

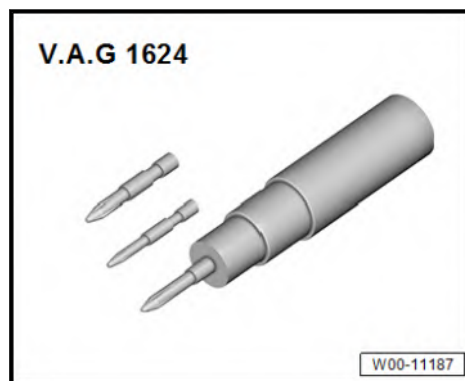
Component	Specified torque
Bolts	10 Nm
Bolts securing cover (qty. 4)	10 Nm

7.4 Removing and installing foot switch unit -F558-

Special tools and workshop equipment required



◆ Torque screwdriver -V.A.G 1624-



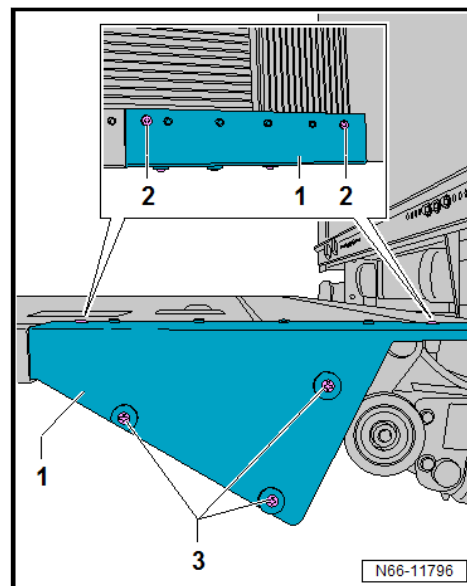
◆ Pop rivet pliers -VAS 5072A-



◆ Pin punch -VAS 5513-



Removing



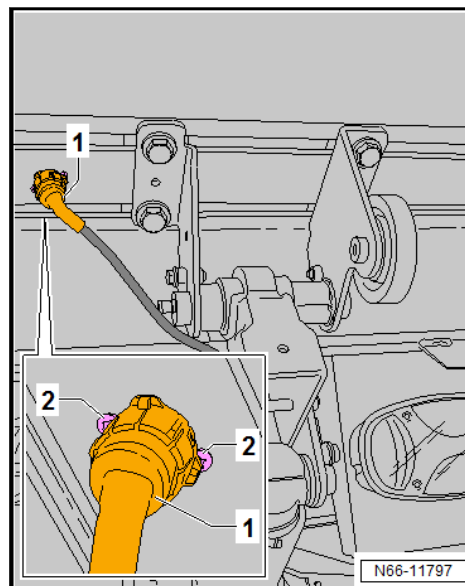
- Switch off tail lift operation switch.

CAUTION

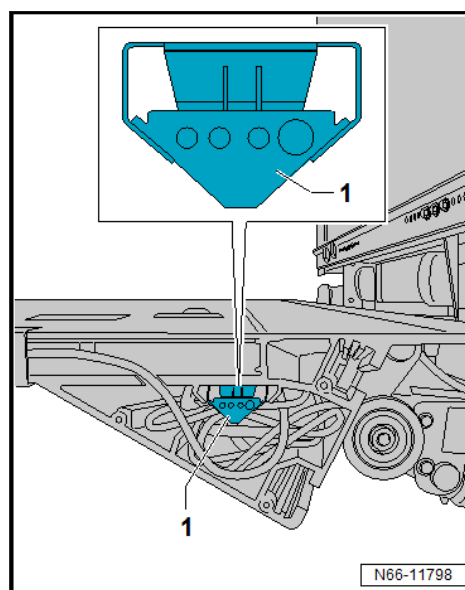
**Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.**

- **Wear protective goggles.**
- **Wear protective gloves!**

- Drill out rivet heads -2- and -3- using drill and drill bit.
- Drive out rivet shanks -2- and -3- with suitable drift -VAS 5513-.
- Remove swarf.
- Remove cover -1-.
- Disconnect electrical connectors of hazard warning switches.
- Separate electrical connector of foot switch unit -F558-.
- Disconnect connector -1-.



- Unscrew bolts -2-.
- Pull out foot switch unit -F558- -1- from tail lift.



Installing

Install in reverse order of removal, observing the following:

- Use rivet pliers -VAS 5072A- to install new pop rivets.

Specified torques

Component	Specified torque
Bolts	1 Nm

8 Tail lift (Palfinger)

⇒ [a8.1 nd installing hazard warning light", page 495](#)

⇒ [a8.2 nd installing foot controls", page 496](#)

⇒ [a8.3 nd installing tilt switch", page 496](#)

⇒ [a8.4 nd installing angle sensor", page 497](#)

⇒ [a8.5 nd installing control box", page 498](#)

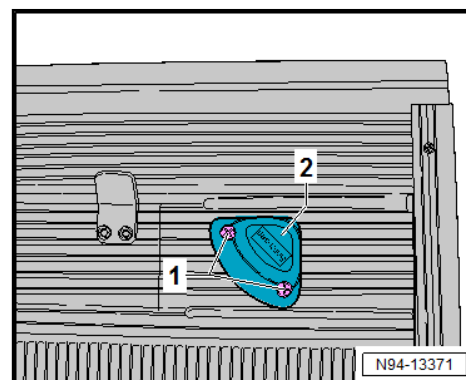
8.1 Removing and installing hazard warning light

Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1410A-



Removing



- Open tail lift.
- Switch off tail lift operation switch.
- Unscrew bolts -1-.
- Detach hazard warning light -2-.

Installing

Install in reverse order of removal, observing the following:

- Pay attention to markings on electrical connectors.

Specified torques

Component	Specified torque
Bolts securing hazard warning light	5 Nm



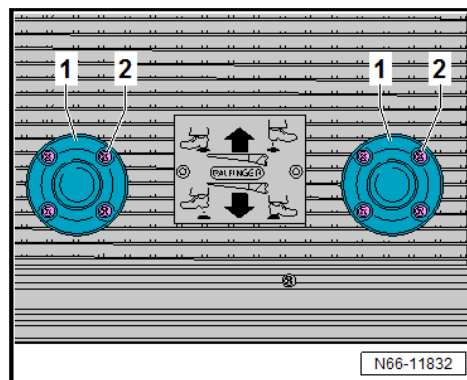
8.2 Removing and installing foot controls

Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1410A-



Removing



- Open tail lift.
- Switch off tail lift operation switch.
- Unscrew bolts -2-.
- Remove foot controls -1-.
- Separate electrical connector.

Installing

Install in reverse order of removal, observing the following:

- Pay attention to markings on electrical connectors.

Specified torques

Component	Specified torque
Bolts securing foot switch	5 Nm

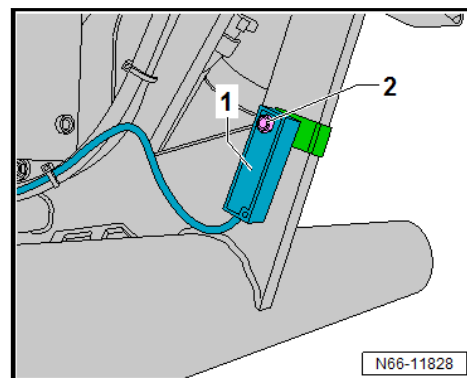
8.3 Removing and installing tilt switch

Special tools and workshop equipment required

◆ Torque wrench -V.A.G 1410A-



Removing



- Switch off tail lift operation switch.
- Detach through cable ties along wiring harness.
- Separate electrical connector.
- Unscrew nut -2-.
- Detach tilt switch -1-.

Installing

Install in reverse order of removal, observing the following:

- Adjust tilt switch ➔ [page 147](#) .

Specified torques

Component	Specified torque
Nut -2-	9 Nm

8.4 Removing and installing angle sensor

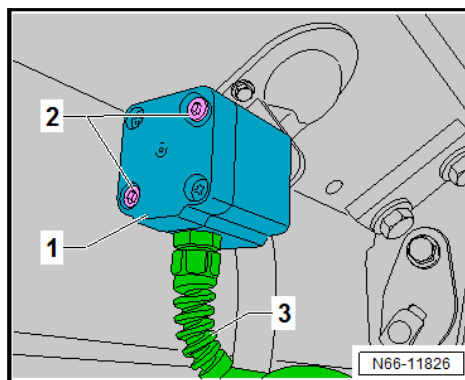
Special tools and workshop equipment required



- ◆ Torque wrench -V.A.G 1410A-



Removing



- Switch off tail lift operation switch.
- Detach through cable ties along wiring harness.
- Separate electrical connector.
- Unscrew bolts -2-.
- Detach angle sensor -1-.

Installing

Install in reverse order of removal, observing the following:

- Align wiring harness -3- downwards.

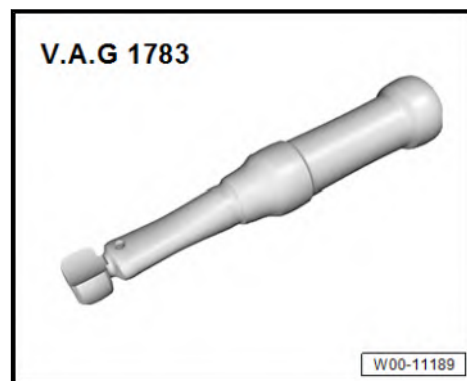
Specified torques

Component	Specified torque
Self-locking bolt	5 Nm

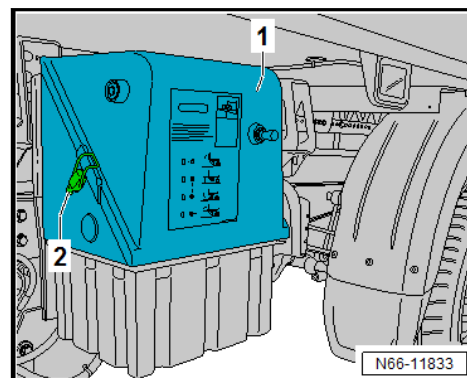
8.5 Removing and installing control box

Special tools and workshop equipment required

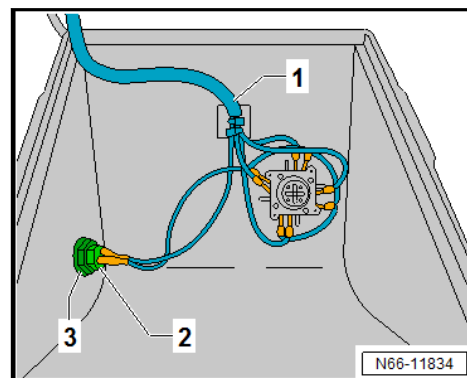
◆ Torque wrench -V.A.G 1783-



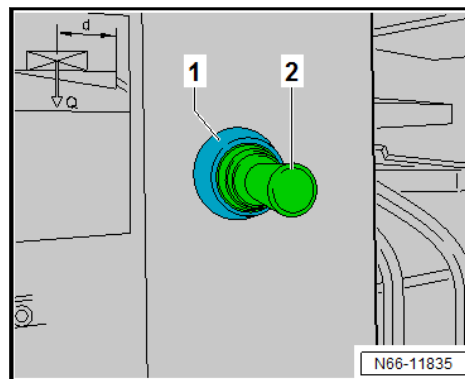
Removing



- Switch off tail lift operation switch.
- Open clip -2- on both sides.
- Detach cover -1-.
- Mark electrical connectors.



- Disconnect electrical connectors.
- Detach wiring harness -1-.
- Unscrew nut -3-.
- Detach button -2-.
- Unscrew nut -1-.



- Detach button -2-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Nut for button	3 Nm
Nut for tail lift operating button	5 Nm

9 Roof spoiler elevated side light

⇒ a9.1 nd installing roof spoiler elevated side light", page 501

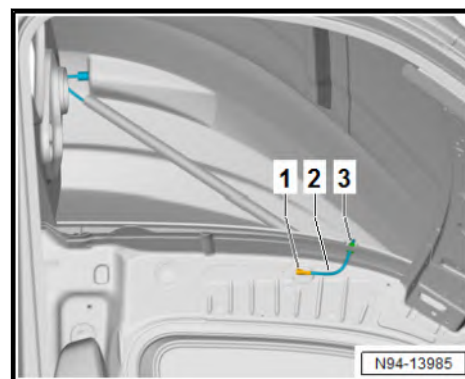
9.1 Removing and installing roof spoiler elevated side light

Special tools and workshop equipment required

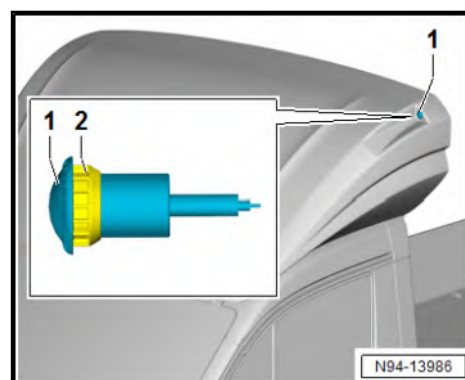
- ◆ Removal wedge -3409-



Removing



- Secure cord to electrical connector.
- Disconnect connector -2-.
- Remove elevated side light -1-.



- Pull out electrical connector -3-.
- Detach cord from electrical connector, and leave it in cable duct of roof spoiler.
- Remove rubber seal -4-.



Installing

Install in reverse order of removal, observing the following:

- Guide new electrical connector through cable duct of roof spoiler using cord.

10 Cross strut/tarpaulin lighting

⇒ a10.1 nd installing cross strut/tarpaulin lighting", page 503

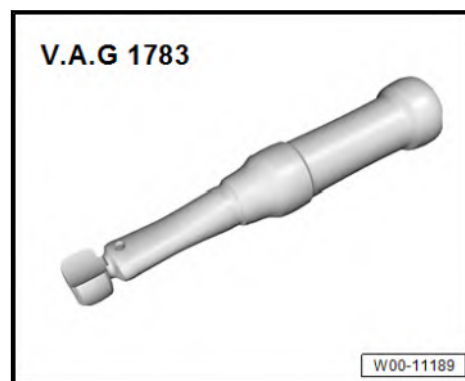
10.1 Removing and installing cross strut/tarpaulin lighting

Special tools and workshop equipment required

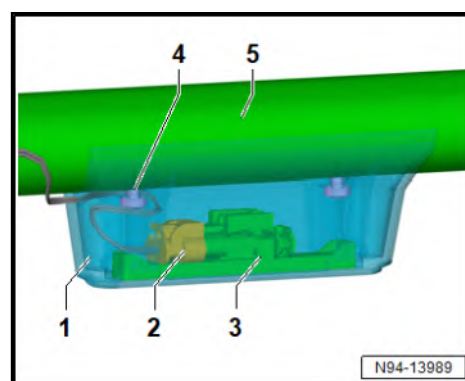
◆ Removal wedge -3409-



◆ Torque wrench -V.A.G 1783-



Removing



- Unclip cross strut light -3-.
- Disconnect connector -2-.
- Remove screws -4-.
- Detach bracket -1- from cross strut -5-.

Installing

Installation is carried out in reverse order.



Specified torques

Component	Specified torque
Bolts	3 Nm

11 Cross strut/tarpaulin wiring harness

⇒ [a11.1 nd installing cross strut/tarpaulin wiring harness", page 505](#)

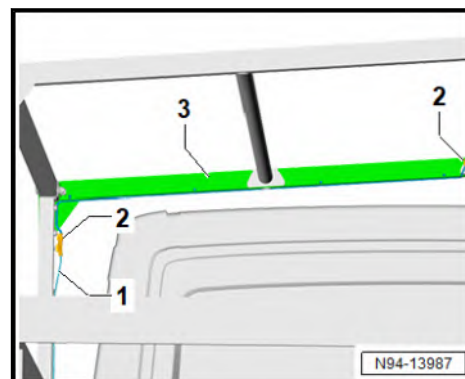
11.1 Removing and installing cross strut/tarpaulin wiring harness

Special tools and workshop equipment required

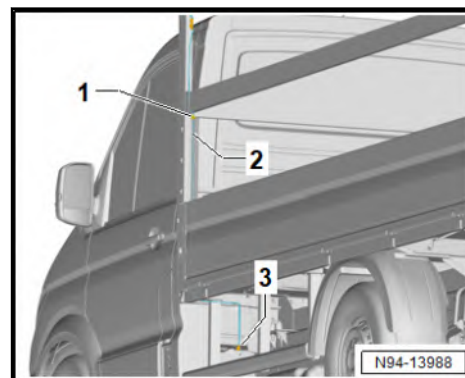
- ◆ Removal wedge -3409-



Removing



- Unclip line -1- from cross strut -3-.
- Separate electrical connectors -2-.
- Disconnect connector -1-.



- Remove rubber seals -2-.
- Pull wiring harness -3- out of foot.

Installing

Installation is carried out in reverse order.



12 LED rotating light

⇒ [a12.1 nd installing LED rotating light", page 506](#)

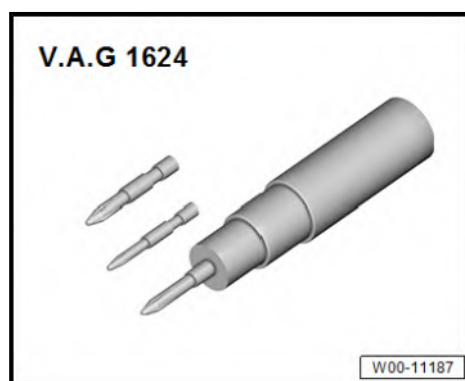
12.1 Removing and installing LED rotating light

Special tools and workshop equipment required

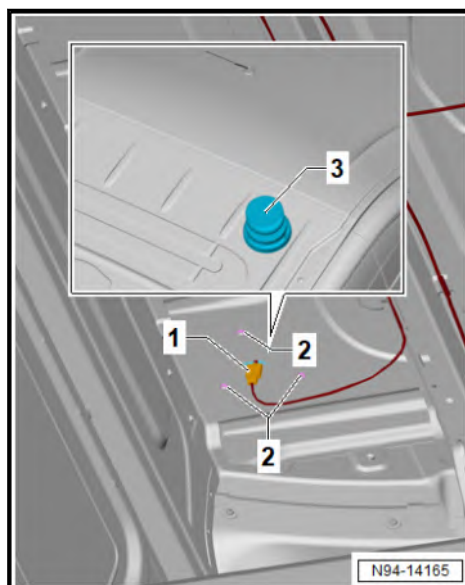
- ◆ Universal working platform



- ◆ Torque screwdriver -V.A.G 1624-



Removing



- ◆ The rotating light bulb -L271- has LEDs and cannot be renewed separately.



- If fitted, remove moulded headliner ⇒ General body repairs, interior; Rep. gr. 70; Roof trim; Removing and installing moulded headliner.
- Disconnect connector -1-.
- Unscrew nuts -2-.
- Using universal working platform, detach rotating light -3-.

Installing

Install in reverse order of removal, observing the following:

- Before installing, clean outer surface of roof in vicinity of rotating light.
- Check function of rotating light.

Specified torques

Component	Specified torque
Nuts -2-	3 Nm



13 Taxi roof sign

⇒ [o13.1 verview – taxi roof sign”, page 508](#)

⇒ [a13.2 nd installing taxi roof signJ1066”, page 509](#)

⇒ [a13.3 nd installing bracket for taxi roof signJ1066”, page 510](#)

13.1 Assembly overview – taxi roof sign

1 - Cover

Different designs depending on equipment version

2 - Taxi roof sign -J1066-

- ☐ Different designs depending on equipment version
- ☐ Removing and installing ⇒ [page 509](#)

3 - Suction cup

4 - Bracket

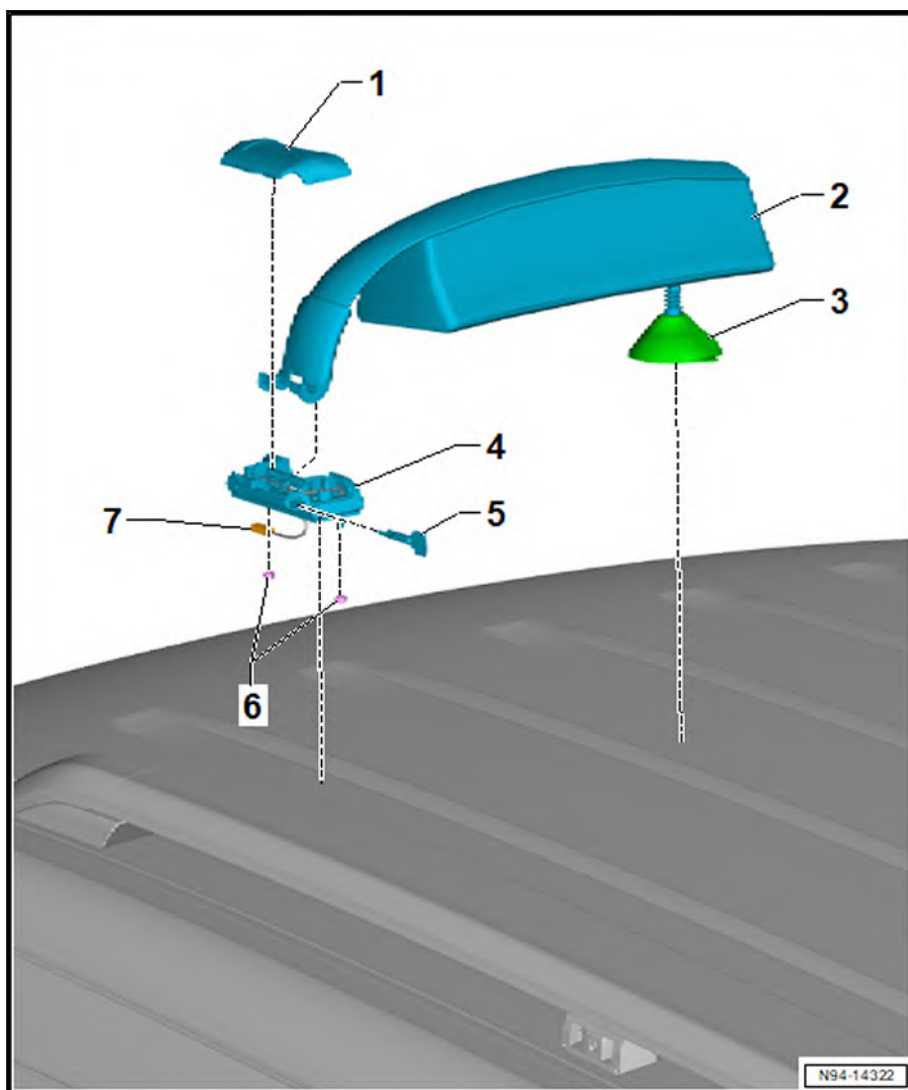
- ☐ Removing and installing ⇒ [page 510](#)

5 - Pin

6 - Nuts

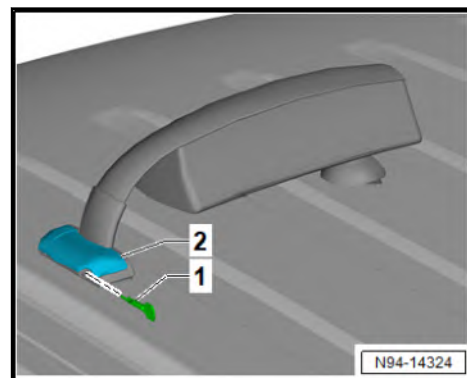
- ☐ Qty. 2
- ☐ 5 Nm

7 - Electrical connector



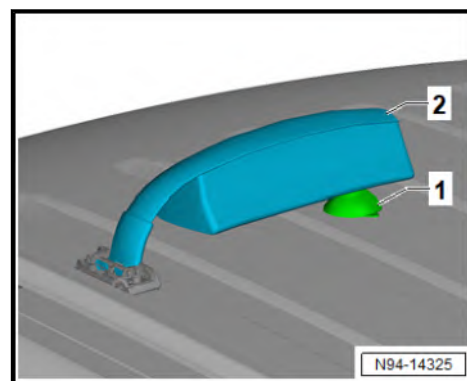
13.2 Removing and installing taxi roof sign -J1066-

Removing

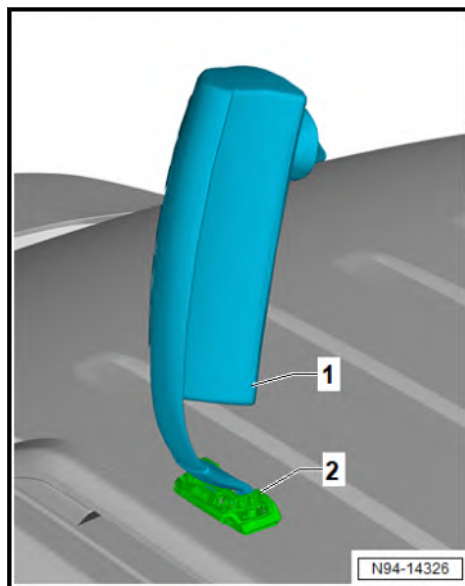


Note

- ◆ Due to different manufacturers of taxi roof signs, the installation procedure may vary slightly.
- ◆ If necessary, follow the special instructions for taxi equipment.
- Unlock anti-theft protection of taxi roof sign -J1066- using dedicated key -1-.
- Detach cover -2-.
- Vent suction cup -1-. While doing this, tilt taxi roof sign -J1066- -2- slightly.



- Swing taxi roof sign -J1066- -1- upwards.



- Guide taxi roof sign -J1066- -1- out of bracket -2-.
- Remove taxi roof sign -J1066- -1-.

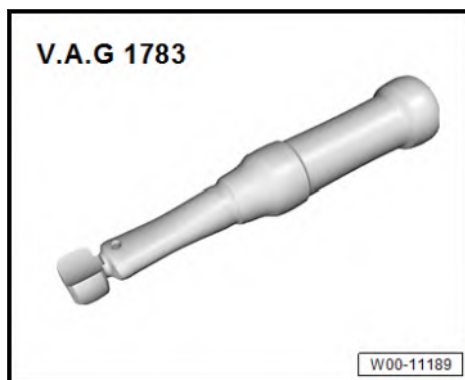
Installing

Installation is carried out in reverse order.

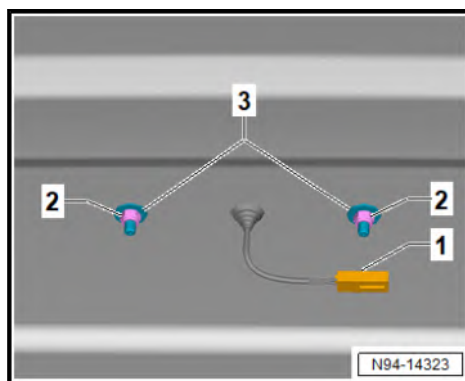
13.3 Removing and installing bracket for taxi roof sign -J1066-

Special tools and workshop equipment required

- ♦ Torque wrench -V.A.G 1783-



Removing



- Remove taxi roof sign -J1066- ⇒ [page 509](#) .



- Lower the moulded headliner ⇒ General body repairs, interior; Rep. gr. 70; Roof trims; Lowering the moulded headliner.
- Disconnect electrical connector -1-.
- Unscrew nuts -2-.
- Detach bracket -3- upwards.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [o13.1 verview – taxi roof sign”, page 508](#)



14 Taxi roof sign

⇒ [o14.1 verview – taxi roof sign”, page 512](#)

⇒ [a14.2 nd installing taxi roof signJ1066”, page 513](#)

⇒ [a14.3 nd installing bracket for taxi roof signJ1066”, page 514](#)

14.1 Assembly overview – taxi roof sign

1 - Cover

- ❑ Different designs depending on equipment version

2 - Taxi roof sign -J1066-

- ❑ Different designs depending on equipment version
- ❑ Removing and installing ⇒ [page 513](#)

3 - Suction cup

4 - Bracket

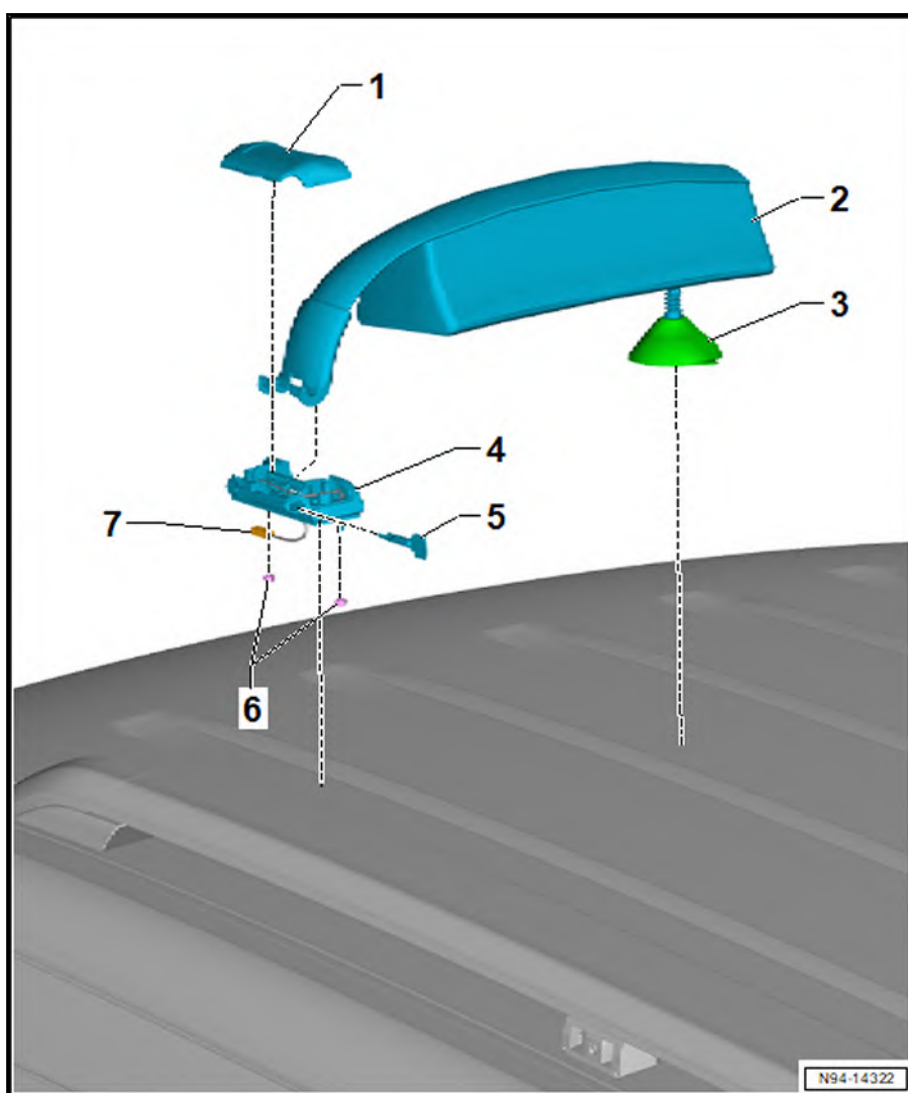
- ❑ Different designs depending on equipment version
- ❑ Removing and installing ⇒ [page 514](#)

5 - Pin

6 - Nuts

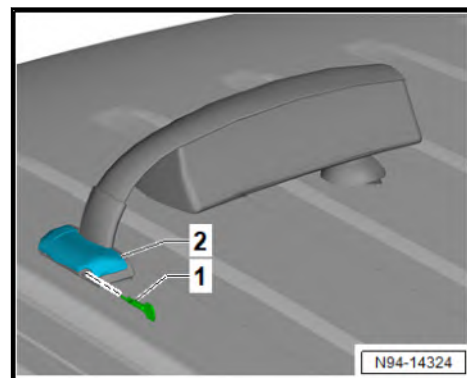
- ❑ Qty. 2
- ❑ 5 Nm

7 - Electrical connector



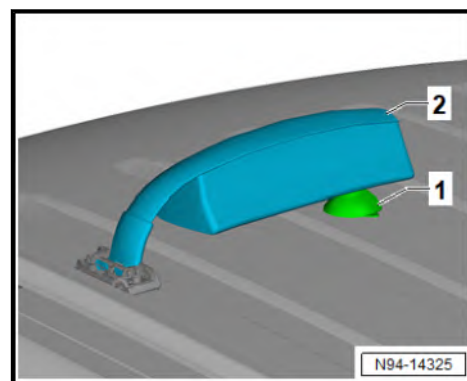
14.2 Removing and installing taxi roof sign -J1066-

Removing

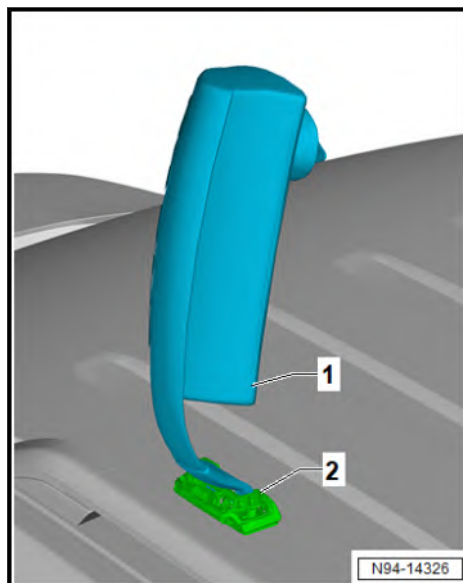


Note

- ◆ Due to different manufacturers of taxi roof signs, the installation procedure may vary slightly.
- ◆ If necessary, follow the special instructions for taxi equipment.
- Unlock anti-theft protection of taxi roof sign -J1066- using dedicated key -1-.
- Detach cover -2-.
- Vent suction cup -1-. While doing this, tilt taxi roof sign -J1066- -2- slightly.



- Swing taxi roof sign -J1066- -1- upwards.



- Guide taxi roof sign -J1066- -1- out of bracket -2-.
- Remove taxi roof sign -J1066- -1-.

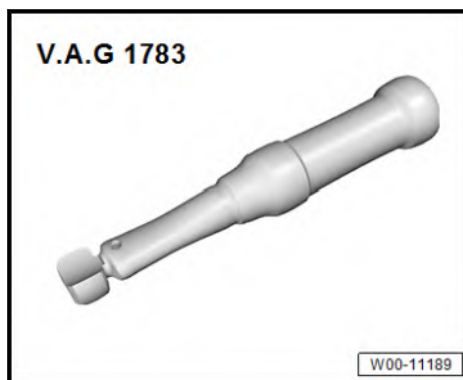
Installing

Installation is carried out in reverse order.

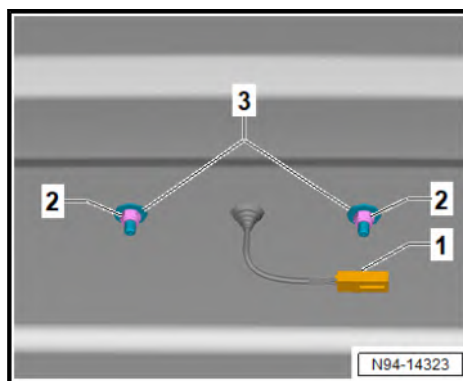
14.3 Removing and installing bracket for taxi roof sign -J1066-

Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1783-



Removing



- Remove taxi roof sign -J1066- ➔ [page 509](#) .



- Lower the moulded headliner ⇒ General body repairs, interior; Rep. gr. 70; Roof trims; Lowering the moulded headliner.
- Disconnect electrical connector -1-.
- Unscrew nuts -2-.
- Detach bracket -3- upwards.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [o13.1 verview – taxi roof sign”, page 508](#)



96 – Lights, bulbs, switches - interior

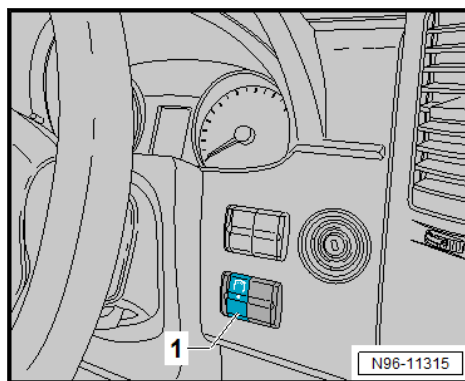
1 Box body (Junge)

⇒ [a1.1 nd installing load area illumination switch E481", page 516](#)

⇒ [a1.2 nd installing interior lighting", page 516](#)

1.1 Removing and installing load area illumination switch -E481-

Load area illumination switch -E481- -1- for box body is located on right next to steering wheel and is controlled from this point.

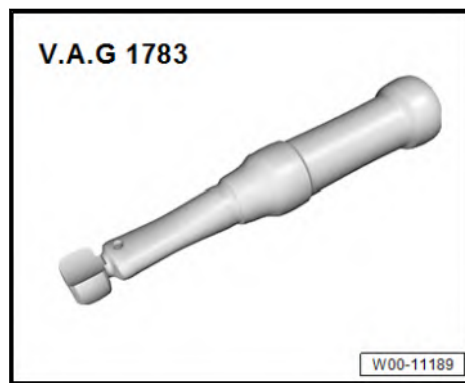


- Remove load area illumination switch -E481- for box body -1- ⇒ Electrical system; Rep. gr. 96; Lights and control elements in dash panel; Removing and installing left and right switch unit next to steering wheel.

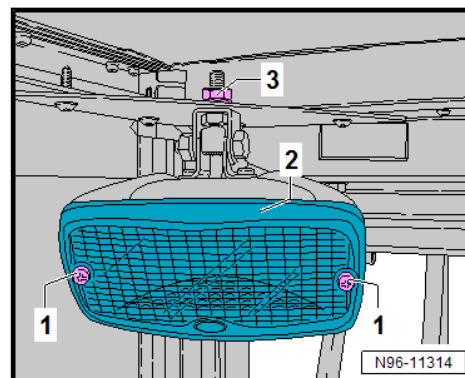
1.2 Removing and installing interior lighting

Special tools and workshop equipment required

- ♦ Torque wrench -V.A.G 1783-



Removing



- Switch off load area illumination switch -E481- for box body
⇒ [page 516](#) .
- Unscrew bolts -1-.
- Remove interior light lens -2-.
- Separate electrical connector in interior light.
- Unscrew nut -3-.
- Remove interior light from box body.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolts	2 Nm
Nut	8 Nm



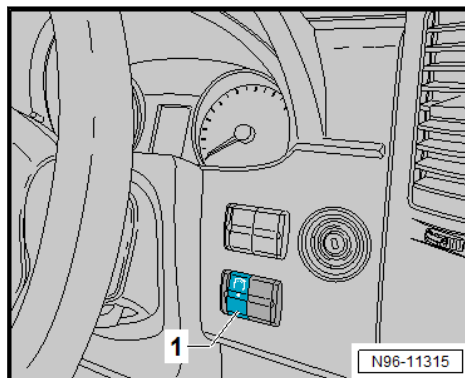
2 Box body (Spier)

⇒ [a2.1 nd installing load area illumination switch E481", page 518](#)

⇒ [a2.2 nd installing interior lighting", page 518](#)

2.1 Removing and installing load area illumination switch -E481-

Load area illumination switch -E481- -1- for box body is located on right next to steering wheel and is controlled from this point.



- Remove load area illumination switch -E481- for box body -1- ⇒ Electrical system; Rep. gr. 96; Lights and control elements in dash panel; Removing and installing left and right switch unit next to steering wheel.

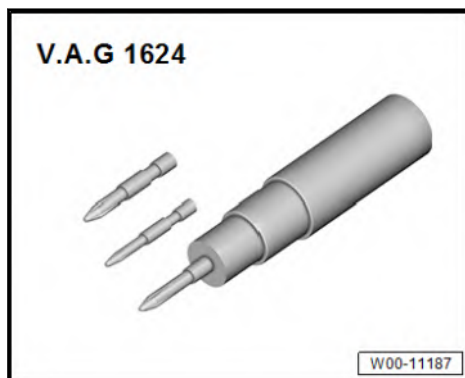
2.2 Removing and installing interior lighting

Special tools and workshop equipment required

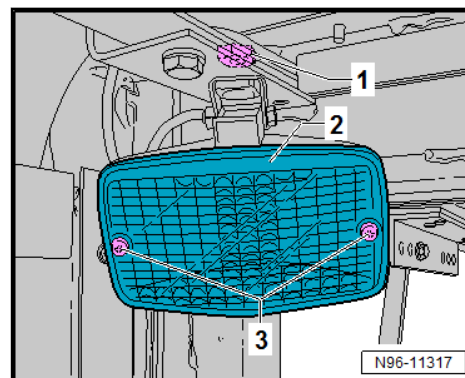
- ◆ Torque wrench -V.A.G 1410A-



- ◆ Torque screwdriver -V.A.G 1624-



Removing



- Switch off load area illumination switch -E481- for box body
⇒ [page 518](#) .
- Loosen bolts -3-.
- Remove interior light lens -2-.
- Separate electrical connector in interior light.
- Unscrew nut -1-.
- Remove interior light from box body.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Nut	15 Nm
Bolts securing lens	0.6 Nm



3 Box body (Schmitz Cargobull)

⇒ [a3.1 nd installing interior lighting", page 520](#)

3.1 Removing and installing interior lighting

Special tools and workshop equipment required

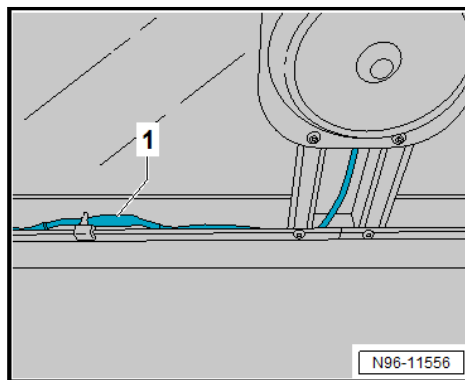
- ◆ Pop rivet pliers -VAS 5072A-



- ◆ Pin punch -VAS 5513-



Removing



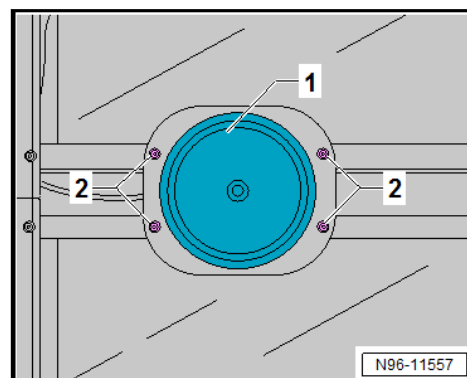
- Unclip wiring harness -1-.
- Pull out wiring harness -1- from front left corner.
- Unpin wiring harness -1-.
- Secure loop on wiring harness -1-.
- Pull out wiring harness -1-.

⚠ CAUTION

Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

- Drill out rivet heads -2- using drill and drill bit. Remove interior light -1- when doing this.



- Drive out rivet shanks -2- with suitable drift -VAS 5513-.
- Remove swarf.

Installing

Install in reverse order of removal, observing the following:

- Repair wiring harness ⇒ Electrical system, General information; Rep. gr. 97; Wiring; Repairing wiring harnesses and connectors.
- Use rivet pliers -VAS 5072A- to install new pop rivet.

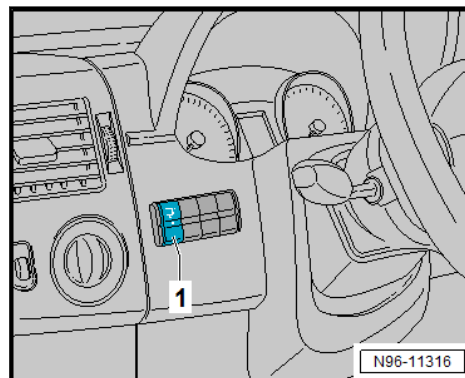


4 Tail lift (Sörensen)

⇒ [I4.1 lift switchE629", page 522](#)

4.1 Tail lift switch -E629-

Tail lift operating switch -1- is located on left next to steering wheel and is controlled from this point.



- Remove tail lift switch -E629- -1- ⇒ Electrical system; Rep. gr. 96; Lights and control elements in dash panel; Removing and installing left and right switch unit next to steering wheel.



5 Tail lift (BÄR - Cargolift)

⇒ [I5.1 lift switch E629", page 523](#)

5.1 Tail lift switch -E629-

- Remove tail lift switch -E629- ⇒ Electrical system; Rep. gr. 96; Lights and control elements in dash panel; Removing and installing left and right switch unit next to steering wheel.



6 Tail lift (Palfinger)

⇒ [I6.1 lift switch", page 524](#)

6.1 Tail lift switch

Tail lift operating switch is located on left next to steering wheel.

- Remove tail lift operating switch ⇒ Electrical system; Rep. gr. 96; Lights and controls in dash panel; Removing and installing switch unit on left and right next to steering wheel.

7 Controls

⇒ [o7.1 fitting locations - controls in dash panel, Taxi](#), page 525

⇒ [a7.2 nd installing interior light switch \(taxi\)E115](#), page 525

⇒ [a7.3 nd installing taxi sign switchE138](#), page 526

⇒ [a7.4 nd installing alarm system off switchE217](#), page 526

⇒ [a7.5 nd installing hands-free system buttonE487](#), page 526

7.1 Overview of fitting locations - controls in dash panel, Taxi

1 - Hands-free system button -E487-

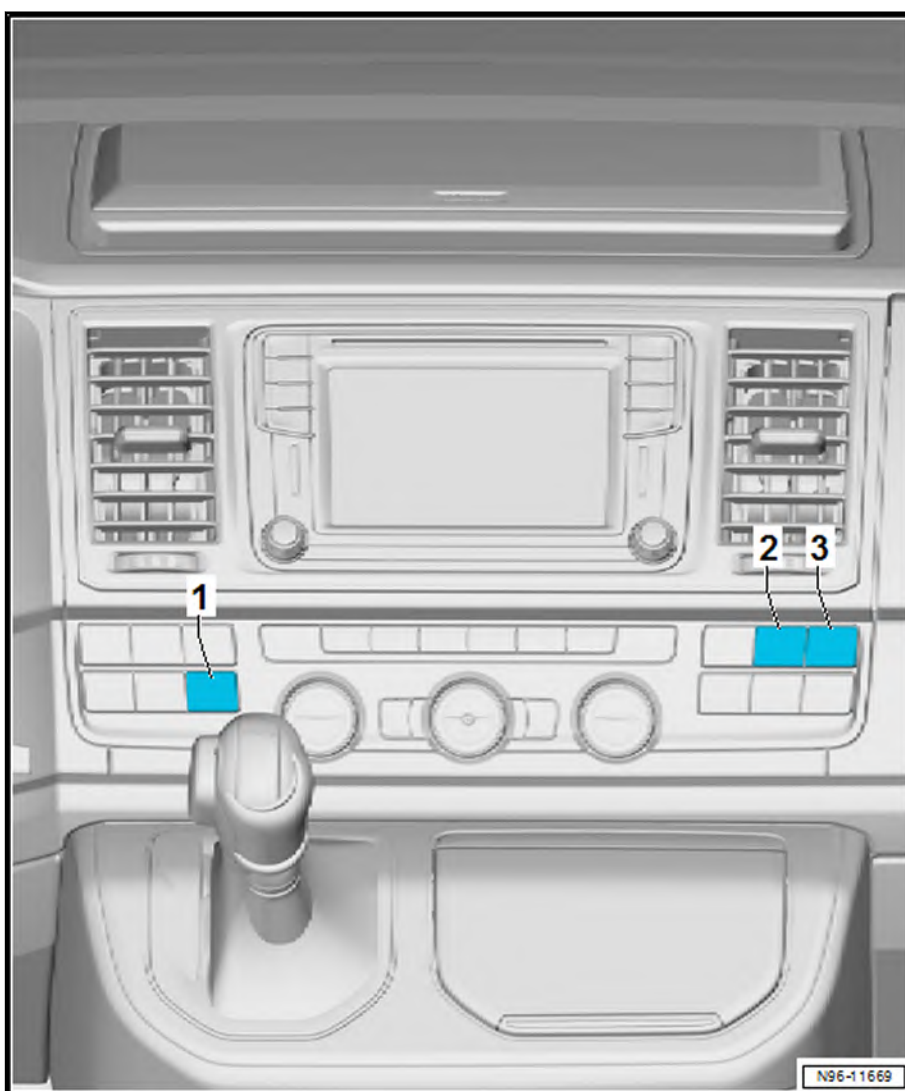
- ☐ Removing and installing ⇒ [page 526](#)

2 - Taxi sign switch -E138-

- ☐ Removing and installing ⇒ [page 526](#)

3 - Interior light switch (taxi) -E115-

- ☐ Removing and installing ⇒ [page 525](#)



7.2 Removing and installing interior light switch (taxi) -E115-

Removal and installation procedure is described in ⇒ Electrical system; Rep. gr. 96; Control elements; Removing and installing button in centre of dash panel.



7.3 Removing and installing taxi sign switch -E138-

Removal and installation procedure is described in ⇒ Electrical system; Rep. gr. 96; Control elements; Removing and installing button in centre of dash panel.

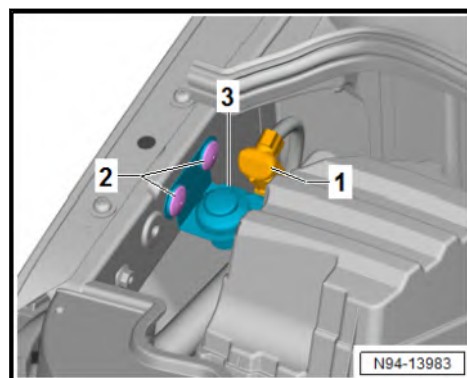
7.4 Removing and installing alarm system off switch -E217-

Special tools and workshop equipment required

- ◆ Removal wedge -3409-



Removing



- Disconnect connector -1-.
- Release clips -2- using removal wedge -3409-.
- Remove alarm system off switch -E217- -3-.

Installing

Installation is carried out in reverse order.

7.5 Removing and installing hands-free system button -E487-

Removal and installation procedure is described in ⇒ Electrical system; Rep. gr. 96; Control elements; Removing and installing button in centre of dash panel.

8 Control elements

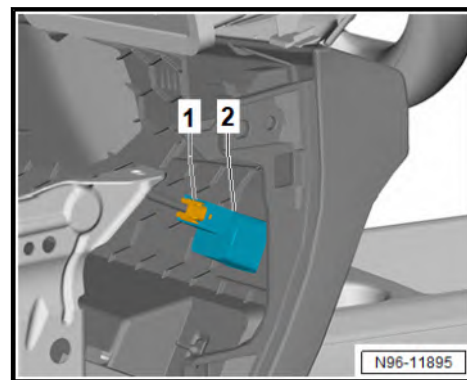
⇒ [a8.1 nd installing taxi alarm active buttonE485", page 527](#)

⇒ [a8.2 nd installing passive taxi alarm buttonE580", page 527](#)

⇒ [a8.3 nd installing switch module 2 in dash panel, next to steering wheelEX44", page 529](#)

8.1 Removing and installing taxi alarm active button -E485-

Removing



- Remove dash panel end cover ⇒ General body repairs, interior; Rep. gr. 70; Dash panel; Removing and installing dash panel end cover.
- Disconnect electrical connector -1-.
- Release taxi alarm active button -E485- -2-. While doing this, pull it out of trim.

Installing

Installation is carried out in reverse order.

8.2 Removing and installing passive taxi alarm button -E580-

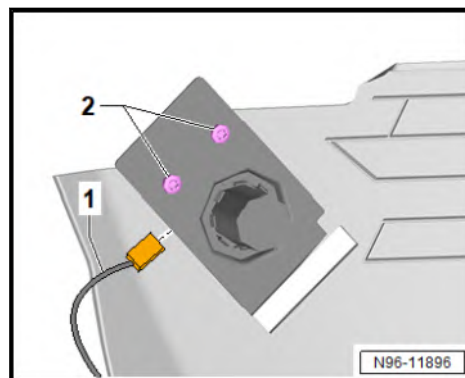
Special tools and workshop equipment required

- ◆ Torque screwdriver -VAS 6494-

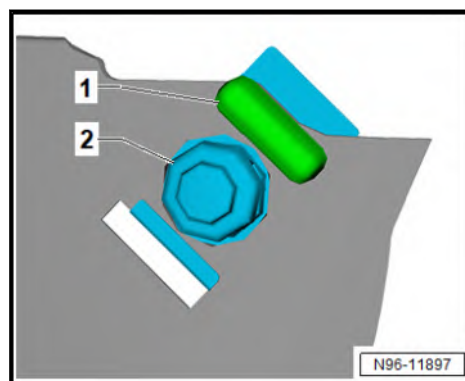




Removing



- Remove lower A-pillar trim ⇒ General body repairs, interior; Rep. gr. 70; Trims, interior; Removing and installing A-pillar trim.
- Disconnect electrical connector -1-.
- Unscrew bolts -2-.
- Remove fastener -1-.



- Detach passive taxi alarm button -E580- -2- together with mounting.

Installing

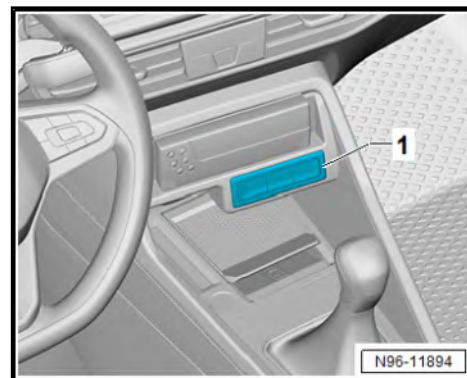
Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolts	1.5 Nm

8.3 Removing and installing switch module 2 in dash panel, next to steering wheel -EX44-

Removing



- Pull out switch module 2 in dash panel, next to steering wheel -EX44- -1-.
- Disconnect electrical connector.
- Remove switch module 2 in dash panel, next to steering wheel -EX44- -1-.

Installing

Installation is carried out in reverse order.



9 Control elements

⇒ a9.1 nd installing taxi alarm active buttonE485", page 530

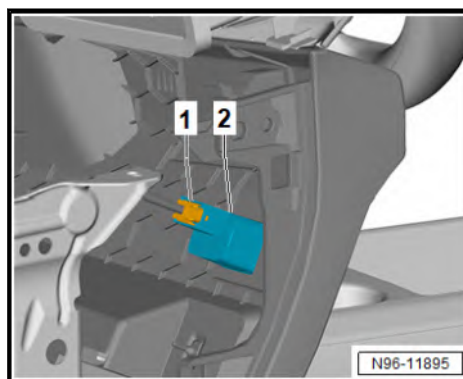
⇒ a9.2 nd installing passive taxi alarm buttonE580", page 530

⇒ a9.3 nd installing taxi alarm off buttonE486", page 531

⇒ a9.4 nd installing switch module for taxi functionsEX62", page 531

9.1 Removing and installing taxi alarm active button -E485-

Removing



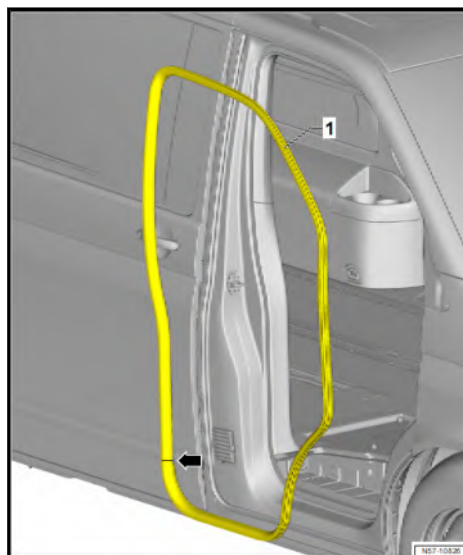
- Remove dash panel end cover ⇒ General body repairs, interior; Rep. gr. 70; Dash panel; Removing and installing dash panel end cover.
- Disconnect electrical connector -1-.
- Release taxi alarm active button -E485- -2-. While doing this, pull it out of trim.

Installing

Installation is carried out in reverse order.

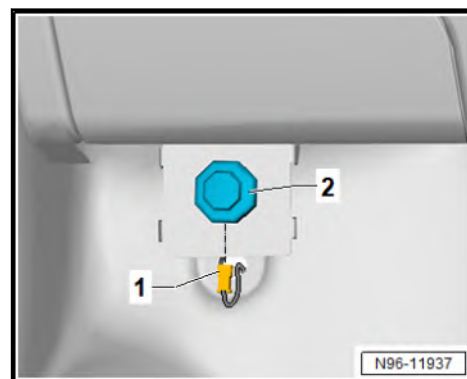
9.2 Removing and installing passive taxi alarm button -E580-

Removing





- Pull left inner door seal -1- in lower area off body flange.
- Do not separate coupling point -arrow-.
- Fold back floor covering in area of passive taxi alarm button -E580-.
- Disconnect electrical connector -1-.



- Detach passive taxi alarm button -E580- -2- together with mounting.

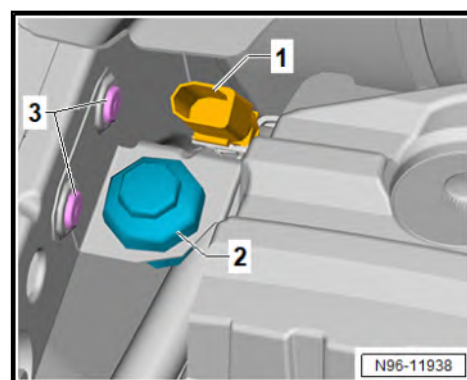
Installing

Install in reverse order of removal, observing the following:

- Bend locking lugs of bracket 90°.

9.3 Removing and installing taxi alarm off button -E486-

Removing



- Disconnect connector -1-.
- Remove clips -3-. While doing this, remove taxi alarm off button -E486- -2-.

Installing

Installation is carried out in reverse order.

9.4 Removing and installing switch module for taxi functions -EX62-

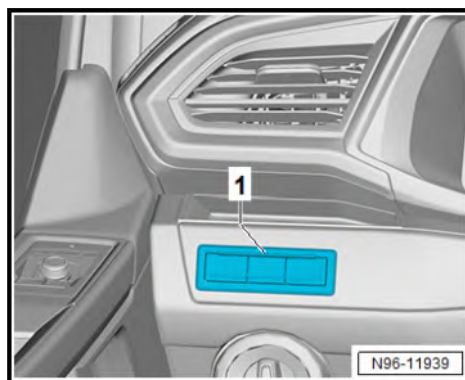
Special tools and workshop equipment required



◆ Removal wedge -VAS 3409-



Removing



- Pull out switch module for taxi functions -EX62- -1-. While doing this, disconnect electrical connectors.

Installing

Installation is carried out in reverse order.

10 Anti-theft alarm

⇒ [o10.1 verview - treble hornH2", page 533](#)

⇒ [a10.2 nd installing treble hornH2", page 533](#)

10.1 Assembly overview - treble horn -H2-

1 - Treble horn -H2-

- ❑ Removing and installing ⇒ [page 533](#)

2 - Bracket

- ❑ For treble horn -H2-
- ❑ Removing and installing ⇒ [page 534](#)

3 - Nut

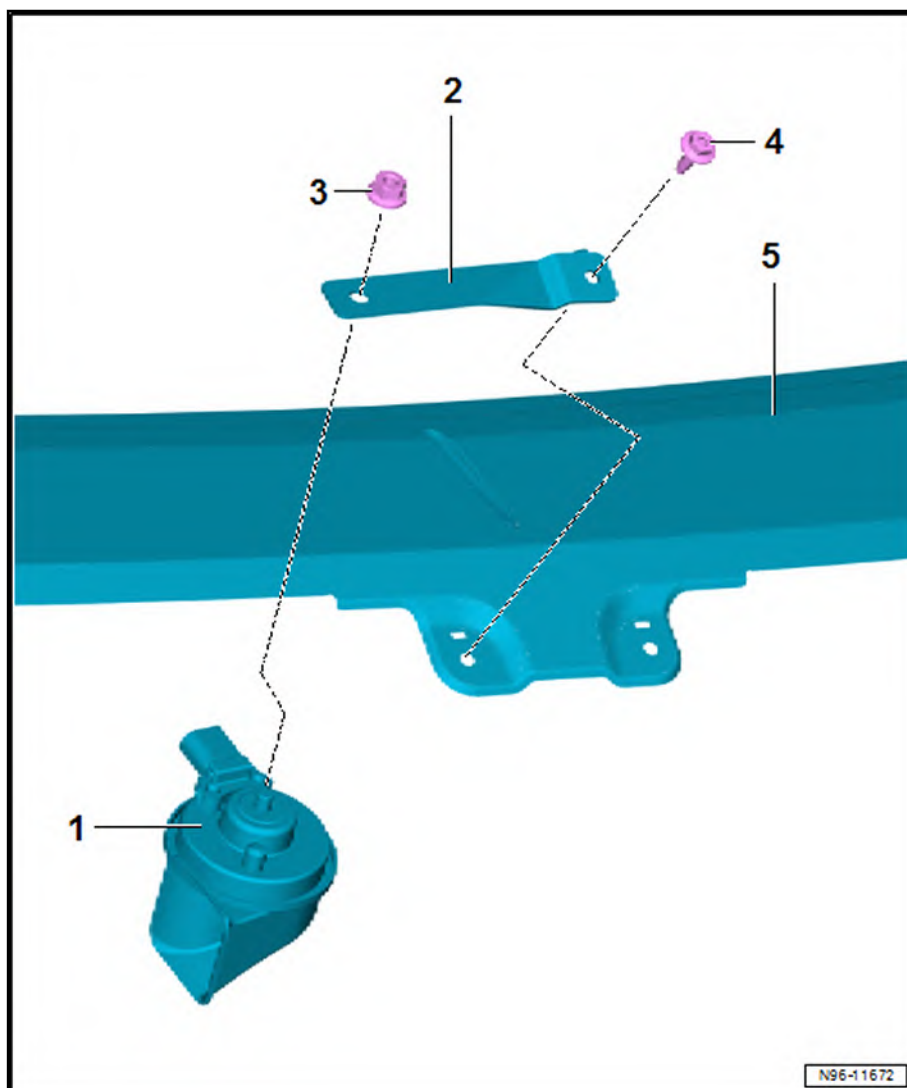
- ❑ Renew after removal
- ❑ 9 Nm

4 - Bolt

- ❑ 6 Nm

5 - Cowl panel

- ❑ Removing and installing ⇒ Body Repairs; Rep. gr. 51; Renewing cowl panel



10.2 Removing and installing treble horn - H2-

⇒ [a10.2.1 nd installing treble hornH2", page 533](#)

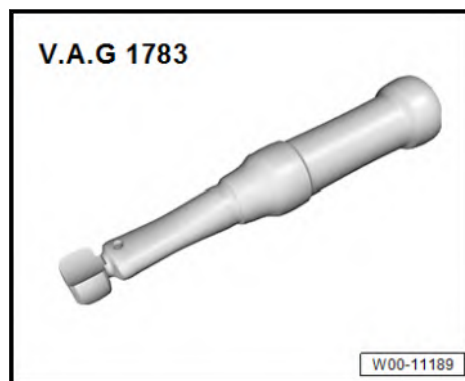
⇒ [a10.2.2 nd installing bracket for treble hornH2", page 534](#)

10.2.1 Removing and installing treble horn - H2-

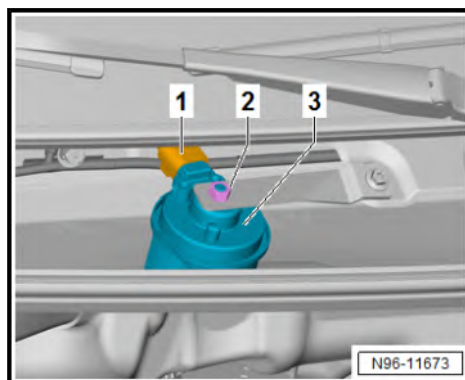
Special tools and workshop equipment required



- ◆ Torque wrench -V.A.G 1783-



Removing



- Remove plenum chamber cover ⇒ General body repairs, exterior; Rep. gr. 50; Plenum chamber cover; Removing and installing plenum chamber cover.
- Disconnect connector -1-.
- Unscrew nut -2-. Remove treble horn -H2- -3- when doing so.

Installing

Carry out installation in the reverse sequence of removal, observing the following:

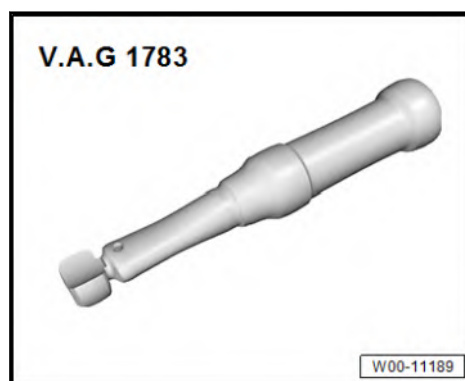
Specified torques

- ◆ ⇒ [o10.1 verview - treble hornH2", page 533](#)

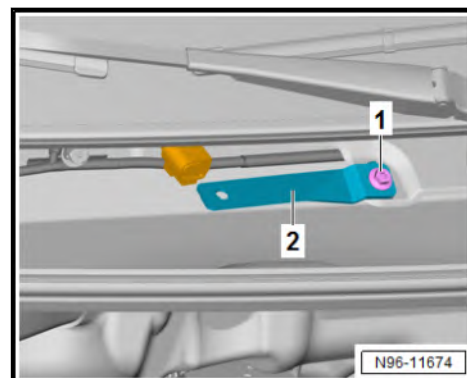
10.2.2 Removing and installing bracket for treble horn -H2-

Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1783-



Removing



- Remove treble horn -H2- ➤ [page 533](#) .
- Unscrew bolt -1-. When doing this, remove bracket for treble horn -H2- -2-.

Installing

Carry out installation in the reverse sequence of removal, observing the following:

Specified torques

- ♦ ➤ [o10.1 verview - treble hornH2", page 533](#)



11 LED spot lights

⇒ [o11.1 verview - LED spot lights", page 536](#)

⇒ [a11.2 nd installing LED spot lights", page 537](#)

11.1 Assembly overview - LED spot lights

⇒ [o11.1.1 verview - LED spot lights, cab", page 536](#)

⇒ [o11.1.2 verview - LED spot lights, passenger compartment", page 536](#)

11.1.1 Assembly overview - LED spot lights, cab

1 - Moulded headliner

2 - Electrical connectors

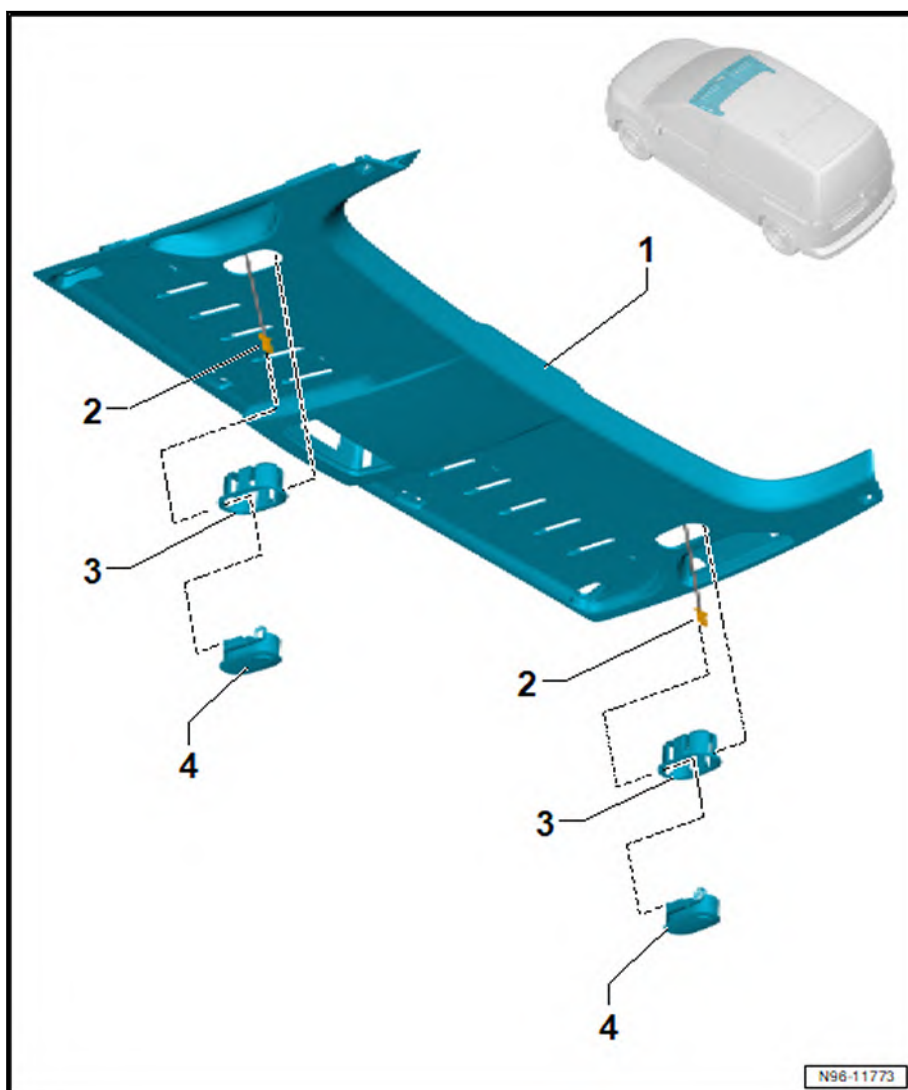
- For LED spot lights

3 - Mountings

- Qty. 2

4 - LED spot lights

- Qty. 2
- Cab
- Removing and installing ⇒ [page 537](#)



11.1.2 Assembly overview - LED spot lights, passenger compartment

1 - Moulded headliner

2 - Electrical connectors

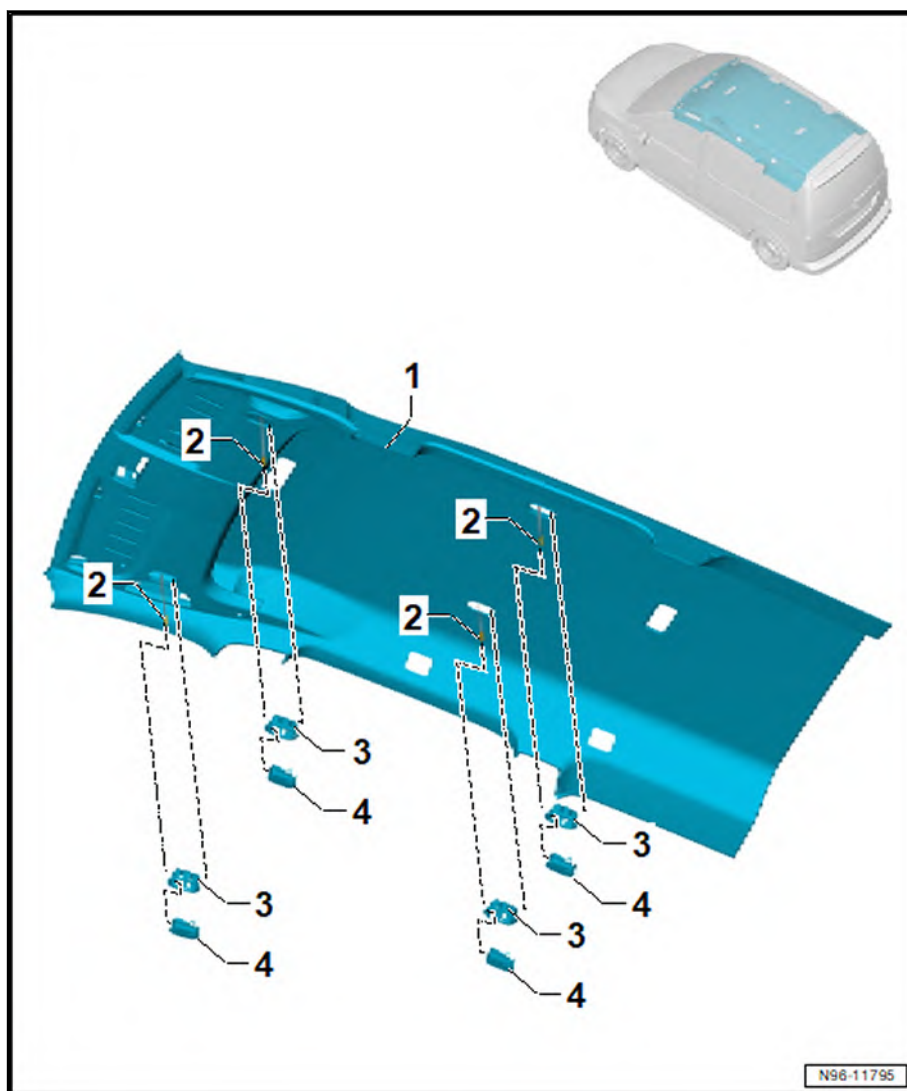
- ☐ For LED spot lights

3 - Mountings

- ☐ Qty. 4

4 - LED spot lights

- ☐ Qty. 4
- ☐ Cab and passenger compartment
- ☐ Removing and installing ➔ [page 537](#)



11.2 Removing and installing LED spot lights

➔ [a11.2.1 nd installing LED spot lights, cab", page 537](#)

➔ [a11.2.2 nd installing LED spot lights, passenger compartment", page 538](#)

11.2.1 Removing and installing LED spot lights, cab

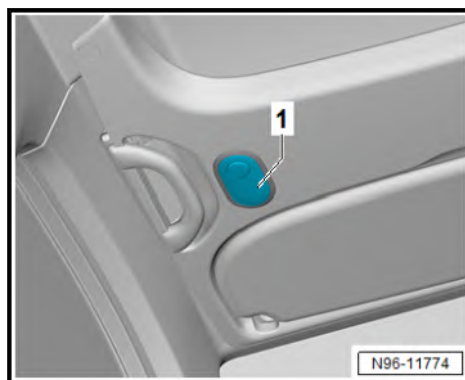
Special tools and workshop equipment required



◆ Scraper set -VAS 6845-



Removing



- Unclip LED spot light -1- using an appropriate wedge from scraper set -VAS 6845-.
- Separate electrical connector. Remove LED spot light.

Installing

Installation is carried out in reverse order.

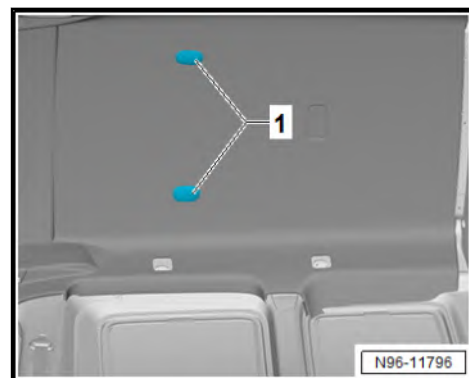
11.2.2 Removing and installing LED spot lights, passenger compartment

Special tools and workshop equipment required

◆ Scraper set -VAS 6845-



Removing



- Unclip LED spot light -1- using an appropriate wedge from scraper set -VAS 6845-.
- Separate electrical connector. Remove LED spot light -1-.

Installing

Installation is carried out in reverse order.



12 Cigarette lighter, socket

⇒ [o12.1 verview - DC/AC converter/socket", page 540](#)

⇒ [a12.2 nd installing DC/AC converter with socket, 12 V - 230 VU13", page 541](#)

⇒ [o12.3 f fitting locations - 230 V components", page 544](#)

⇒ [o12.4 verview - exterior socket, 230 V, 110 VU8", page 545](#)

⇒ [o12.5 verview - interior socket, 230 V, 110 VU11", page 546](#)

⇒ [o12.6 verview - 220 V automatic personnel protection circuit breakerS59", page 547](#)

⇒ [a12.7 nd installing interior socket, 230 V, 110 VU11 ", page 548](#)

⇒ [a12.8 nd installing 220 V automatic personnel protection circuit breakerS59", page 553](#)

⇒ [a12.9 nd installing exterior socket, 230 V, 110 VU8", page 554](#)

⇒ [a12.10 nd installing battery charger, 230 V or 110 V/12 VA10", page 556](#)

⇒ [a12.11 nd installing DC/AC converter with socket, 12 V-230 VU13", page 561](#)

12.1 Assembly overview - DC/AC converter/socket

1 - Glove compartment

- ☐ Assembly overview
⇒ General body repairs, interior; Rep. gr. 68; Compartments/covers; Assembly overview - glove compartment
- ☐ Removing and installing ⇒ General body repairs, interior; Rep. gr. 68; Compartments/covers; Removing and installing glove compartment

2 - Trim

- ☐ For DC/AC converter with socket, 12 V - 230 V -U13-

3 - Cover

- ☐ For DC/AC converter with socket, 12 V - 230 V -U13-

4 - Spreader rivets

- ☐ Qty. 4

5 - Spreader rivets

- ☐ Qty. 4

6 - Pop rivets

- ☐ Qty. 2
- ☐ Depending on the equipment, there may be two bolts fitted.
- ☐ Specified torque for bolts: 2 Nm

7 - Bolts

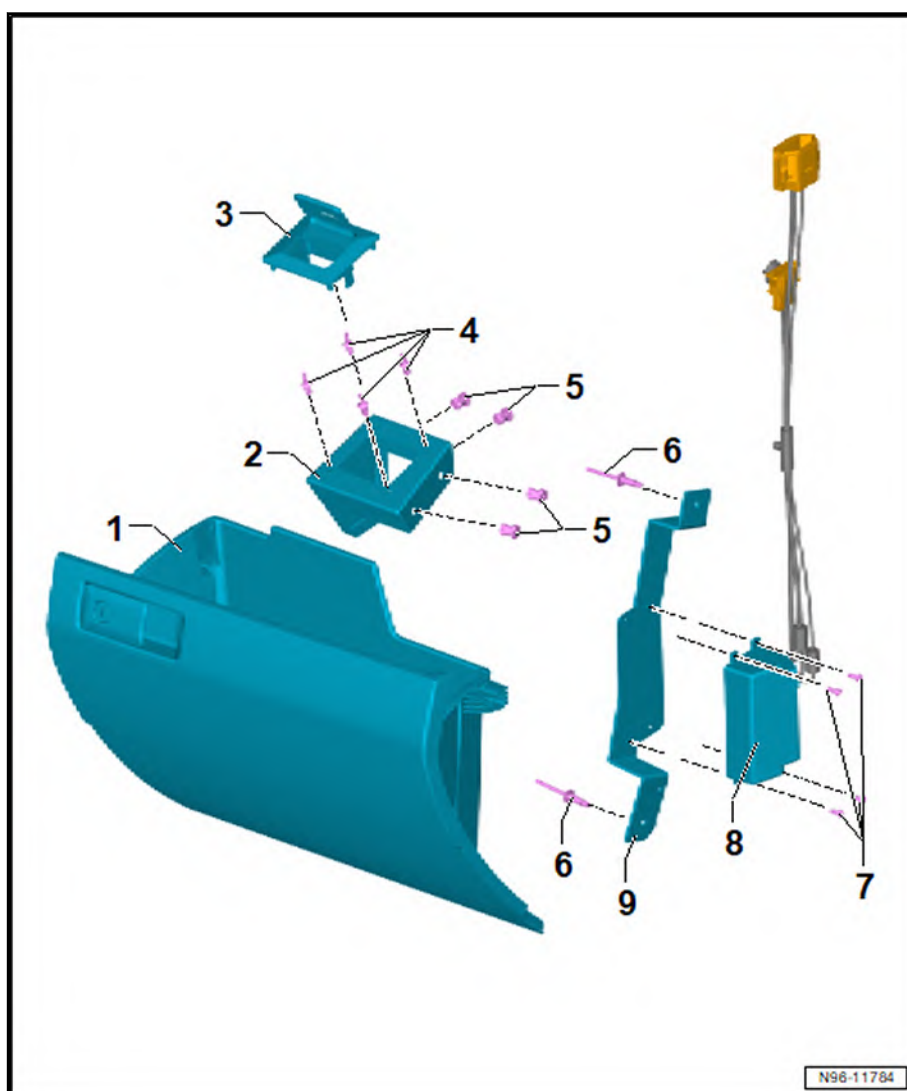
- ☐ Qty. 4
- ☐ 2 Nm

8 - DC/AC converter with socket, 12 V - 230 V -U13-

- ☐ Removing and installing ⇒ [page 541](#)

9 - Bracket

- ☐ For DC/AC converter with socket, 12 V - 230 V -U13-



12.2 Removing and installing DC/AC converter with socket, 12 V - 230 V -U13-

Special tools and workshop equipment required



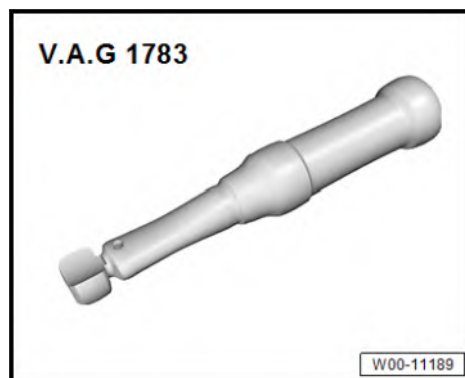
◆ Pop rivet pliers -VAS 5072A-



◆ Pin punch -VAS 5513-



◆ Torque wrench -V.A.G 1783-



Removing

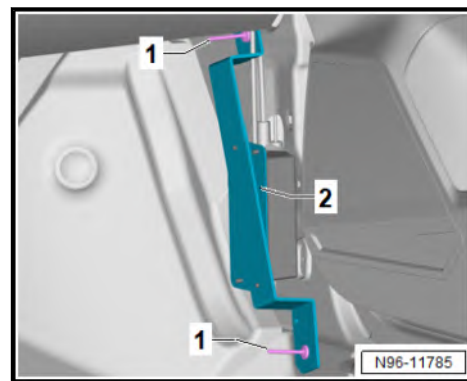
- Remove right dash panel end cover ⇒ General body repairs, interior; Rep. gr. 70; Dash panel; (right-side) Removing and installing dash panel end cover.
- Remove centre right A-pillar trim ⇒ General body repairs, interior; Rep. gr. 70; Interior trim; Removing and installing A-pillar trim.
- Remove front right sill panel moulding ⇒ General body repairs, interior; Rep. gr. 70; Trims, interior; Right Removing and installing sill panel moulding.

⚠ CAUTION

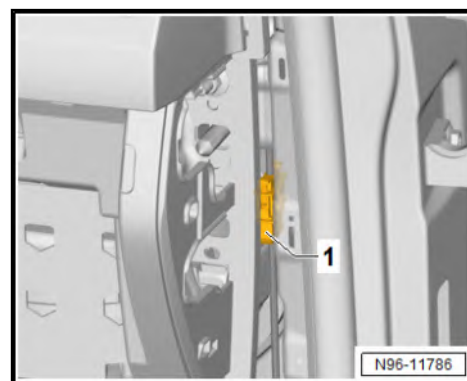
Risk of injury from swarf being flung into air.
Irritation and injury to skin and eyes possible.

- Wear protective goggles.
- Wear protective gloves!

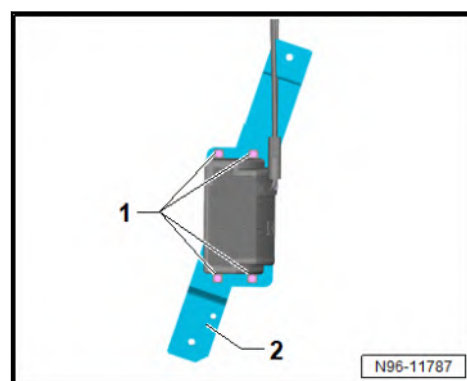
- Drill out pop rivets -1-.



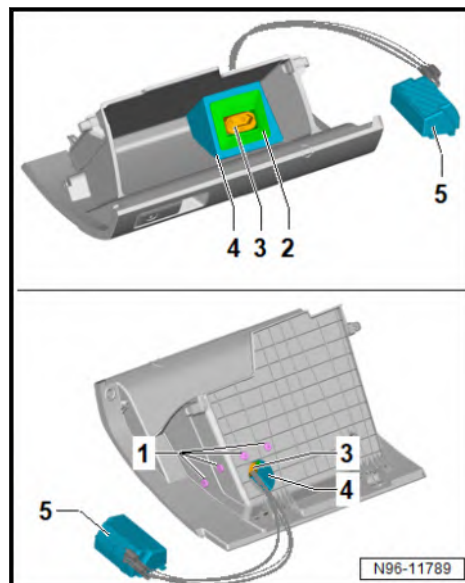
- Use roll pin drift -VAS 5513- to remove pop rivets -1-.
- If fitted, unscrew bolts.
- Remove bracket -2- with DC/AC converter with socket, 12 V - 230 V -U13- .
- Remove cover at top in front passenger footwell ⇒ General body repairs, interior; Rep. gr. 68; Compartments/covers; Removing and installing footwell cover.



- Remove glove compartment ⇒ General body repairs, interior; Rep. gr. 68; Compartments/covers; Removing and installing glove compartment.
- Disconnect connector -1-.
- Unscrew bolts -1-.



- Remove bracket -2-.
- Release spreader rivets -1-.



- Unclip cover -2-.
- Release socket -3- from trim -4-.
- Remove DC/AC converter with socket, 12 V - 230 V -U13- -5-.

Installing

Install in reverse order of removal, observing the following:

- Drive in pop rivets using pop rivet pliers -VAS 5072A-.

Specified torques

- ♦ ➔ [o12.1 verview - DC/AC converter/socket", page 540](#)

12.3 Overview of fitting locations - 230 V components

1 - Interior socket, 230 V, 110 V -U11-

- ☐ Rear left in vehicle
- ☐ Removing and installing ➔ [page 548](#)

2 - 220 V automatic personnel protection circuit breaker -S59-

- ☐ Rear left in vehicle
- ☐ Removing and installing ➔ [page 553](#)

3 - Exterior socket, 230 V, 110 V -U8-

- ☐ Rear left on vehicle
- ☐ Removing and installing ➔ [page 554](#)

4 - Charger, 230 V or 110 V/12 V -A10-

- ☐ Under driver seat
- ☐ Removing and installing ➔ [page 556](#)

5 - DC/AC converter with socket, 12 V - 230 V -U13-

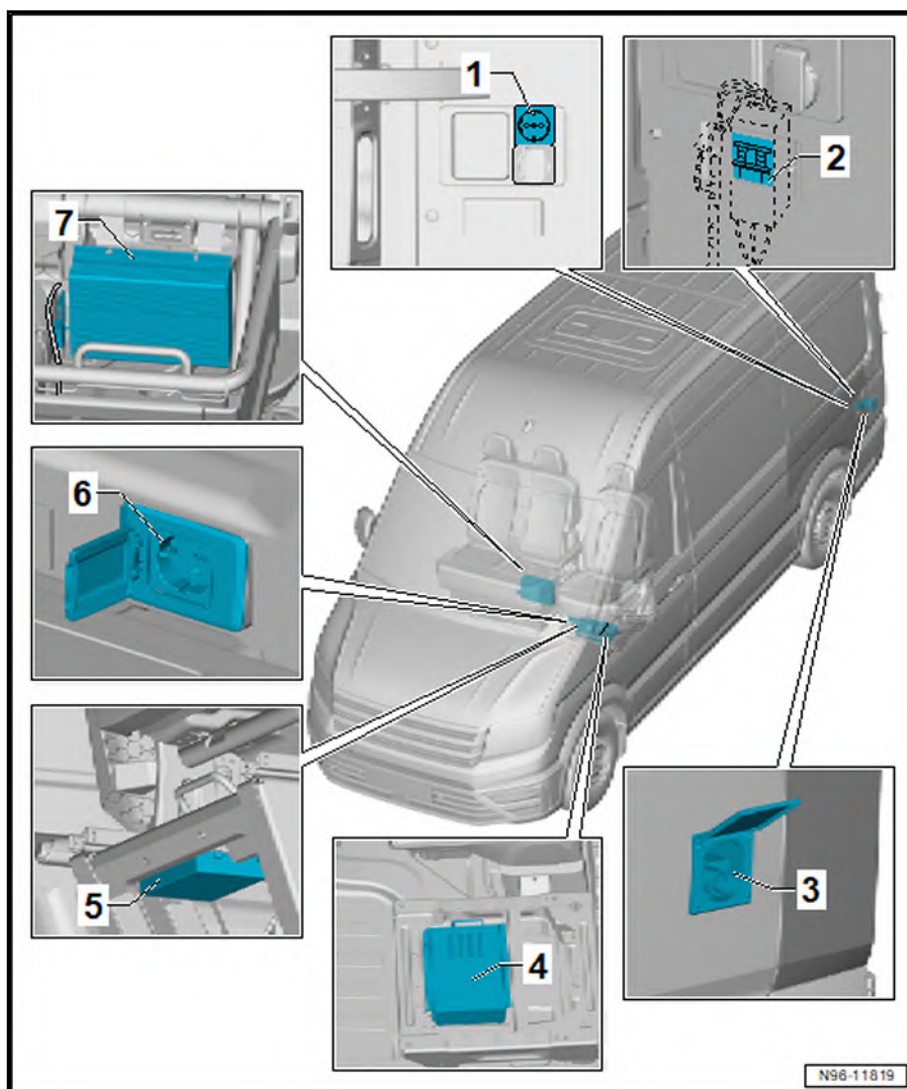
- ☐ Under driver seat
- ☐ Removing and installing ➔ [page 561](#)

6 - Interior socket, 230 V, 110 V -U11-

- ☐ On seat frame of driver seat
- ☐ Removing and installing ➔ [page 548](#)

7 - Charger, 230 V or 110 V/12 V -A10-

- ☐ Under bench seat on front passenger side
- ☐ Removing and installing ➔ [page 556](#)



12.4 Assembly overview - exterior socket, 230 V, 110 V -U8-



1 - Exterior socket, 230 V, 110 V -U8-

- ❑ Removing and installing ➔ [page 554](#)

2 - Socket adapter

- ❑ Removing and installing ➔ [page 554](#)

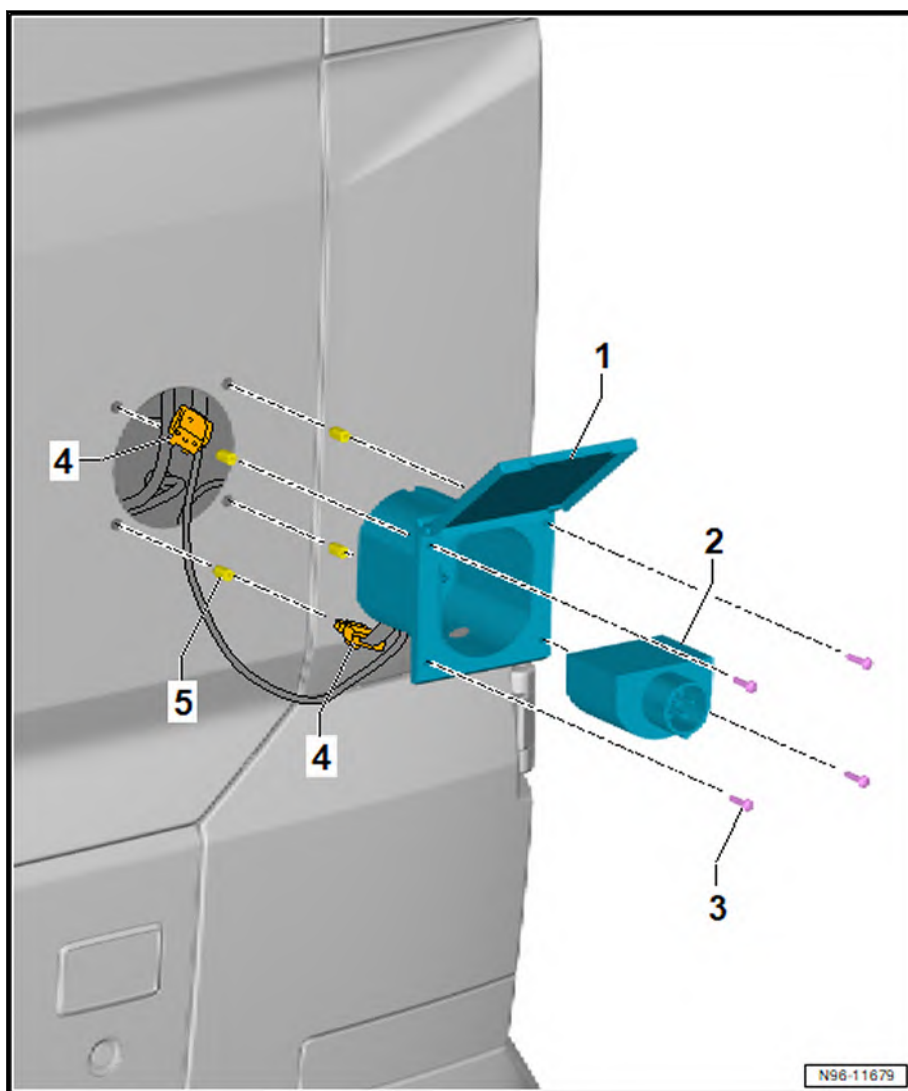
3 - Bolts

- ❑ Qty. 4
- ❑ 2 Nm

4 - Electrical connector

5 - Sleeves

- ❑ Qty. 4



12.5 Assembly overview - interior socket, 230 V, 110 V -U11-

1 - Interior socket, 230 V, 110 V -U11-

- ❑ Removing and installing ➔ [page 548](#)

2 - Bolts

- ❑ Qty. 4
- ❑ 2 Nm

3 - Frame

- ❑ Removing and installing ➔ [page 548](#)

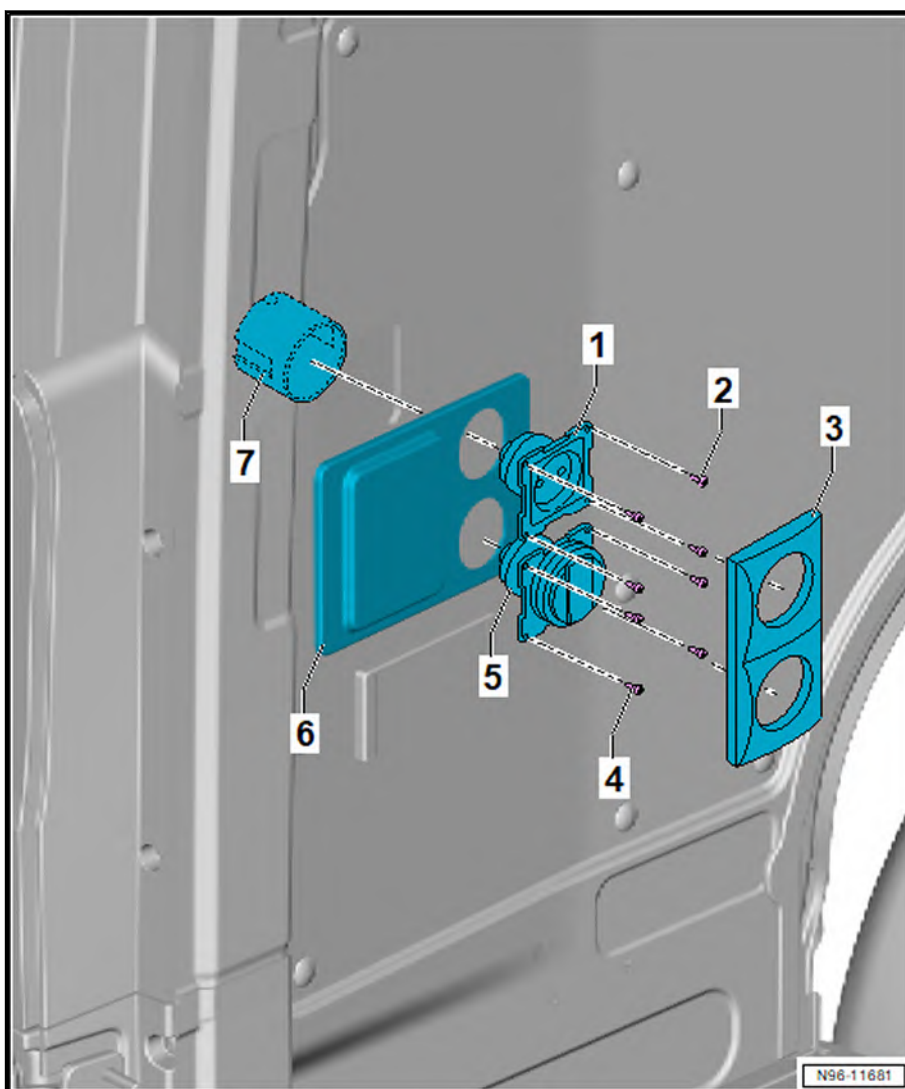
4 - Bolts

- ❑ Qty. 4
- ❑ 2 Nm

5 - 12 V socket -U5-

6 - Cap

7 - Protective cap



12.6 Assembly overview - 220 V automatic personnel protection circuit breaker -S59-



1 - Nuts

- ☐ Qty. 2
- ☐ 2 Nm

2 - Bracket

- ☐ Removing and installing ⇒ [page 553](#)

3 - 220 V automatic personnel protection circuit breaker -S59-

- ☐ Removing and installing ⇒ [page 553](#)

4 - Carrier plate

- ☐ Removing and installing ⇒ [page 553](#)

5 - Washers

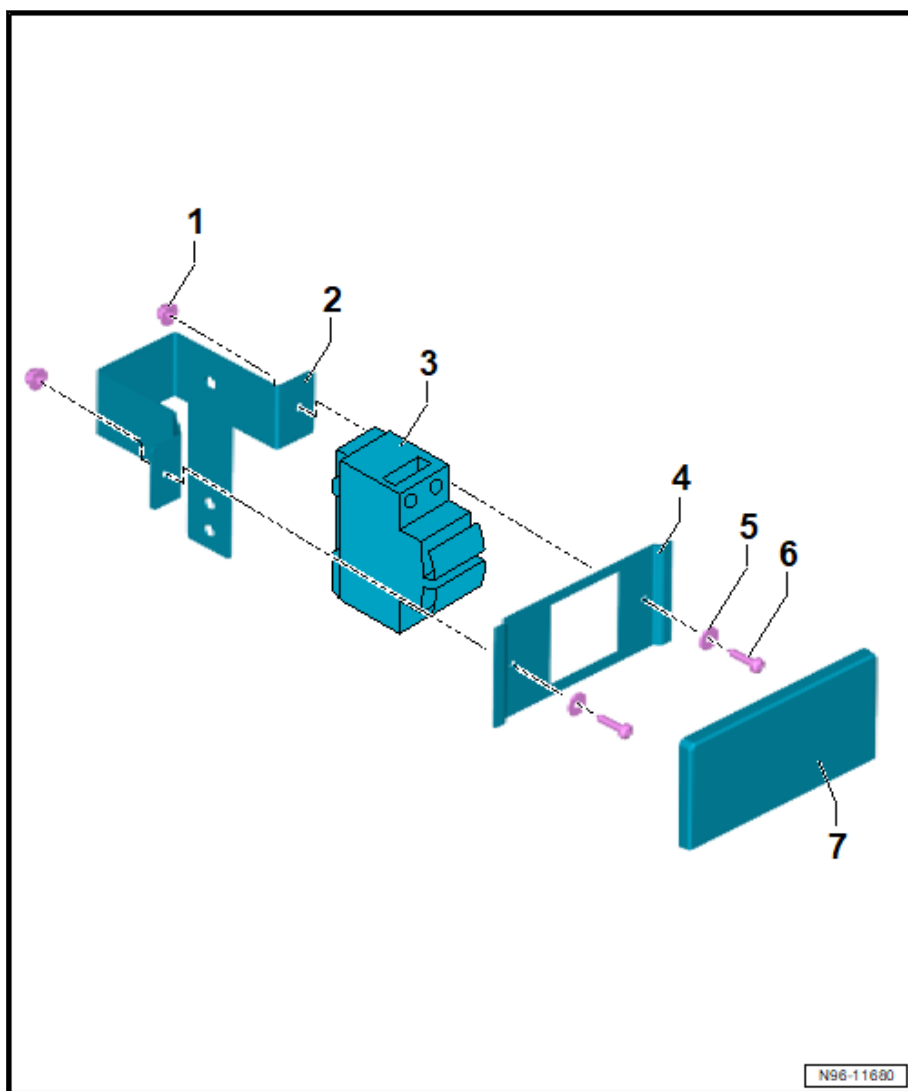
- ☐ Qty. 2

6 - Bolts

- ☐ Qty. 2
- ☐ 2 Nm

7 - Cap

- ☐ Removing and installing ⇒ [page 553](#)



12.7 Removing and installing interior socket, 230 V, 110 V -U11-

⇒ [a12.7.1 nd installing interior socket, 230 V, 110 VU11, rear left](#), [page 548](#)

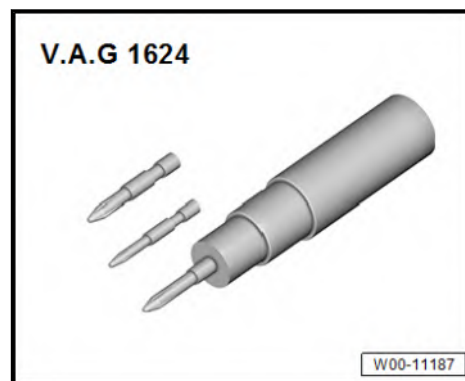
⇒ [a12.7.2 nd installing interior socket, 230 V, 110 VU11, on seat frame of driver seat](#), [page 551](#)

12.7.1 Removing and installing interior socket, 230 V, 110 V -U11-, rear left

Special tools and workshop equipment required



- ◆ Torque screwdriver -V.A.G 1624-



- ◆ Removal wedge -VAS 3409-

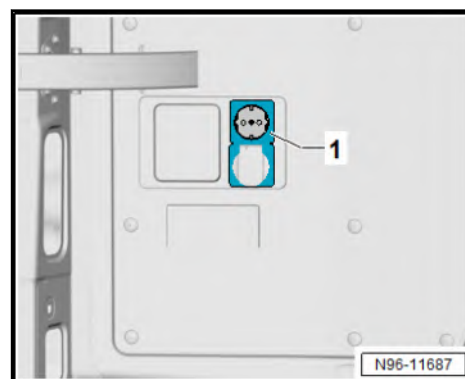


Removing

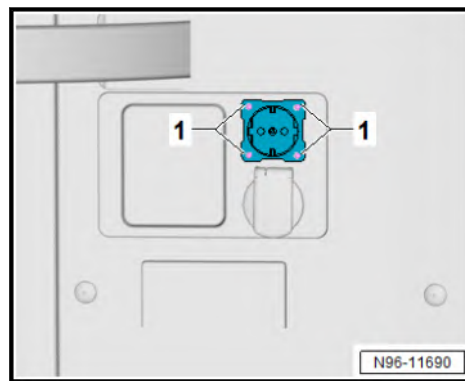
DANGER

Risk of fatal injury from electric shock

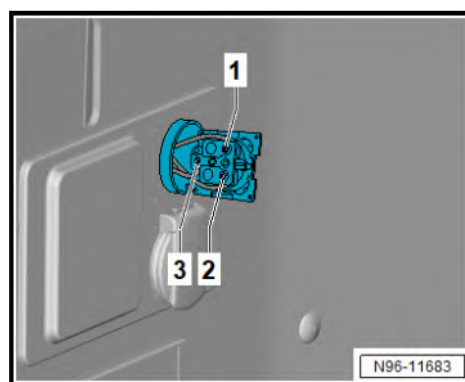
- On completion of repairs to the 110/230 V system the qualified electrician has a duty to perform tests after installation to ensure the safety of the system.
- If connected, pull out power supply cable 230/110 volts from exterior socket, 230 V, 110 V -U8-.
- Lever off frame -1- using removal wedge -VAS 3409-.



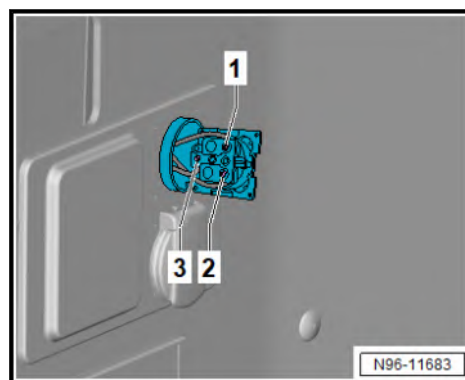
- Unscrew bolts -1-.



- Remove interior socket, 230 V, 110 V -U11-.
- Loosen bolts -1-, -2- and -3- and pull out cable connections from socket.



Installing



Install in reverse order of removal, observing the following:

- Connect cable connections to interior socket, 230 V, 110 V -U11- as follows:
 - 1 - Brown cable connection, phase
 - 2 - Blue cable connection, neutral conductor
 - 3 - Green/yellow cable connection, protective conductor

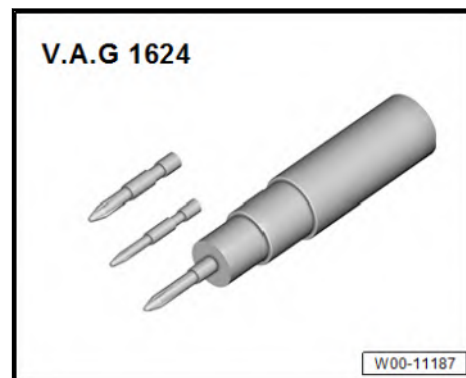
Specified torques

- ♦ [⇒ o12.5 verview - interior socket, 230 V, 110 VU11", page 546](#)

12.7.2 Removing and installing interior socket, 230 V, 110 V -U11-, on seat frame of driver seat

Special tools and workshop equipment required

- ◆ Torque screwdriver -V.A.G 1624-



- ◆ Removal wedge -VAS 3409-



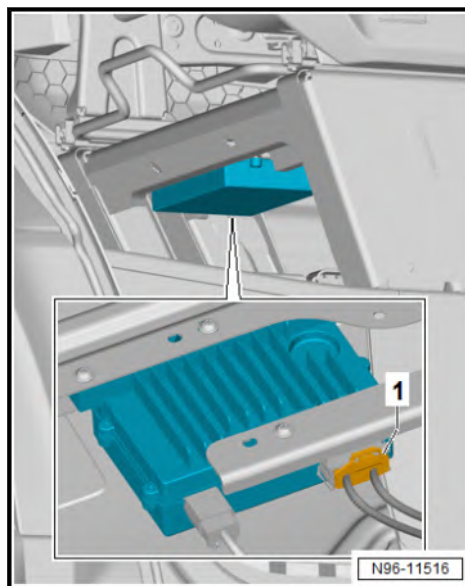
Removing

DANGER

Risk of fatal injury from electric shock

- On completion of repairs to the 110/230 V system the qualified electrician has a duty to perform tests after installation to ensure the safety of the system.

- ◆ Never open housing of DC/AC converter with socket, 12 V - 230 V -U13-.
- ◆ The complete unit must always be renewed if the connector, wiring, 230 V socket or the DC/AC converter are defective.
- Switch off ignition and all electrical consumers.

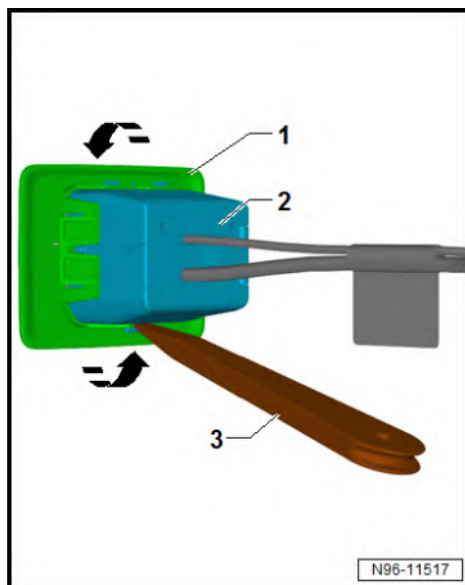


- Withdraw ignition key.
- If necessary, push back left front seat.

CAUTION

Risk of injury from residual voltage of the capacitors in the DC/AC converter.

- Observe the discharge time of 10 seconds for the capacitors.
- Remove cover of electric socket, remove DC/AC converter with socket, 12 V - 230 V -U13- ⇒ General body repairs, interior; Rep. gr. 72; Front seats; Removing and installing cover.
- Disconnect connector -1-.
- Remove 230 V socket -2- together with socket cover -1-.



- Use removal wedge -3409- -2- to lever out 230 V socket -3- in direction of -arrow-.

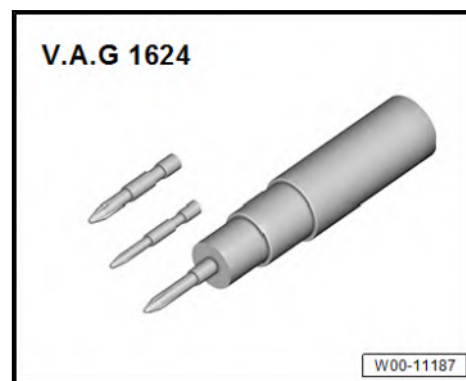
Installing

Installation is carried out in reverse order.

12.8 Removing and installing 220 V automatic personnel protection circuit breaker -S59-

Special tools and workshop equipment required

- ◆ Torque screwdriver -V.A.G 1624-

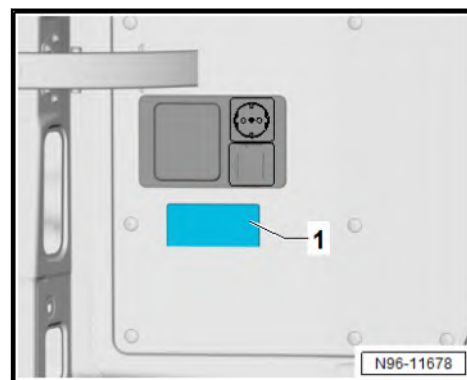


Removing

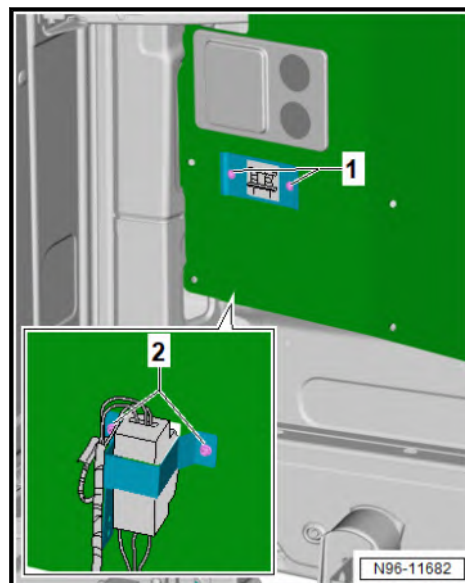
DANGER

Risk of fatal injury from electric shock

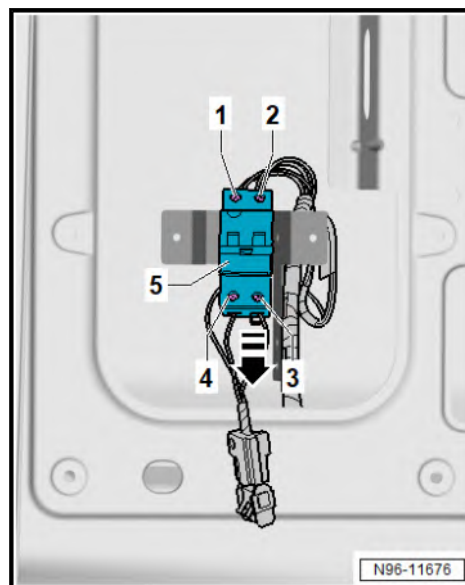
- On completion of repairs to the 110/230 V system the qualified electrician has a duty to perform tests after installation to ensure the safety of the system.
- If connected, pull out power supply cable 230/110 volts from exterior socket, 230 V, 110 V -U8-.
- Remove interior socket, 230 V, 110 V -U11- ➔ [page 548](#) .
- Remove 12 V socket -U5- ➔ Electrical system - general information; Rep. gr. 96; Lights, bulbs, switches - interior; 12 V socket.
- Remove rear left side trim ➔ General body repairs, interior; Rep. gr. 70; Trims, interior; Removing and installing side trim.
- Remove rear left side trim as far as cable connections from personal protection circuit breaker (220 V) -S59- allow.
- Push up cap -1-.



- Unscrew bolts -1-, counterholding nuts -2- on back of side trim to do so.



- Loosen bolt -1- and pull off brown cable connection.



- Loosen bolt -2- and pull off blue cable connection.
- Loosen bolt -3- and pull off blue cable connection.
- Loosen bolt -4- and pull off brown cable connection.
- Pull securing clip downwards in direction of -arrow- and remove 220 V automatic personnel protection circuit breaker -S59- -5-.

Installing

Install in reverse order of removal, observing the following:

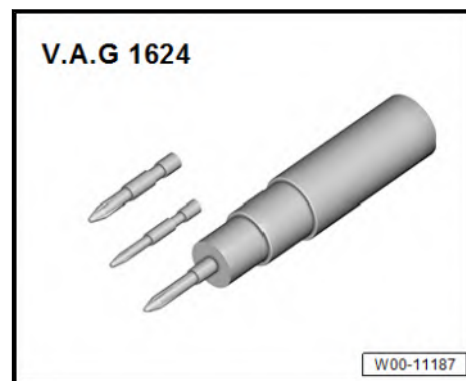
Specified torques

- ♦ ⇒ [o12.6 verview - 220 V automatic personnel protection circuit breakerS59", page 547](#)

12.9 Removing and installing exterior socket, 230 V, 110 V -U8-

Special tools and workshop equipment required

◆ Torque screwdriver -V.A.G 1624-

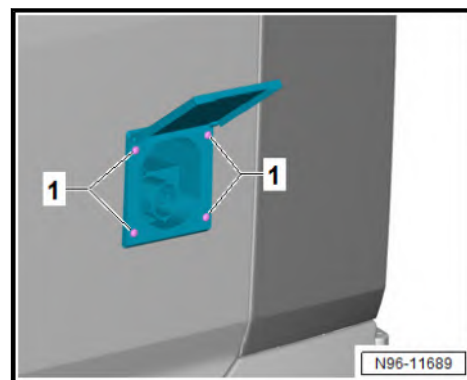


Removing

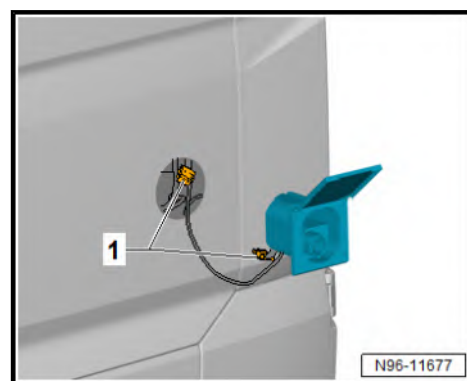
DANGER

Risk of fatal injury from electric shock

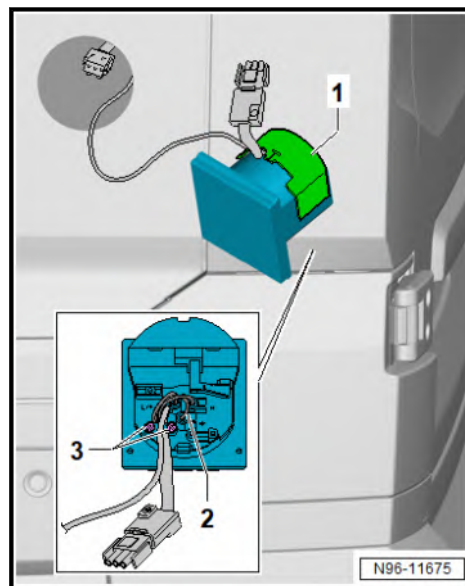
- On completion of repairs to the 110/230 V system the qualified electrician has a duty to perform tests after installation to ensure the safety of the system.
- If connected, pull out power supply cable 230/110 volts from exterior socket, 230 V, 110 V -U8-.
- Open flap of socket.
- Unscrew bolts -1-.



- Remove exterior socket, 230 V, 110 V -U8-.
- Disconnect connector -1-.



- Unclip cover -1-.



- Loosen bolt -2- on green/yellow cable connection.
- Loosen bolts -3- and pull out green/yellow cable connection from socket.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ♦ [⇒ a12.4 overview - exterior socket, 230 V, 110 VU8", page 545](#)

12.10 Removing and installing battery charger, 230 V or 110 V/12 V -A10-

[⇒ a12.10.1 nd installing battery charger, 230 V or 110 V/12 VA10, under driver seat", page 556](#)

[⇒ a12.10.2 nd installing battery charger, 230 V or 110 V/12 VA10, under bench seat on front passenger side", page 558](#)

[⇒ a12.10.3 nd installing bracket from battery charger, 230 V or 110 V/12 VA10, under bench seat on front passenger side", page 559](#)

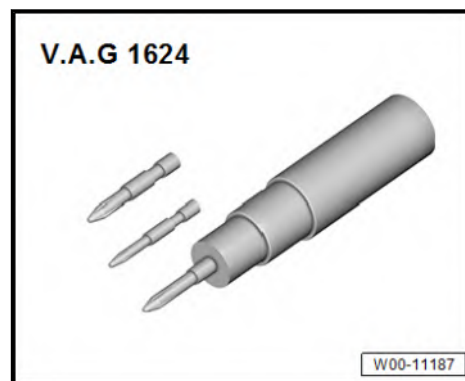
[⇒ a12.10.4 nd installing fuse on battery charger, 230 V or 110 V/12 VA10", page 560](#)

12.10.1 Removing and installing battery charger, 230 V or 110 V/12 V -A10-, under driver seat

Special tools and workshop equipment required



◆ Torque screwdriver -V.A.G 1624-



◆ Torque wrench -V.A.G 1410A-

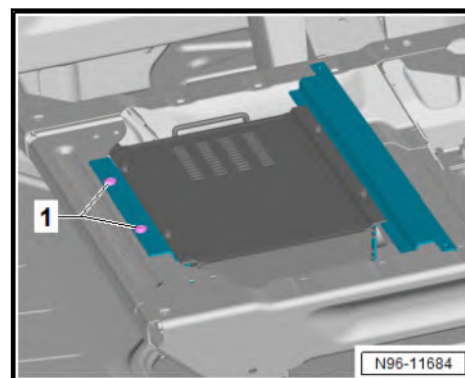


Removing

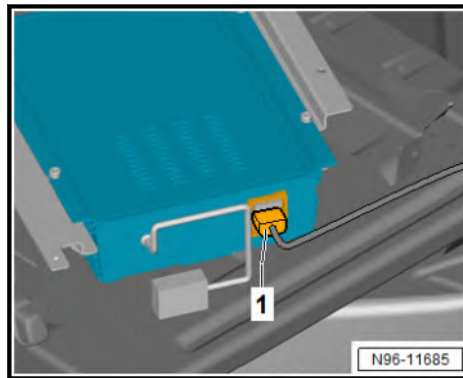
DANGER

Risk of fatal injury from electric shock

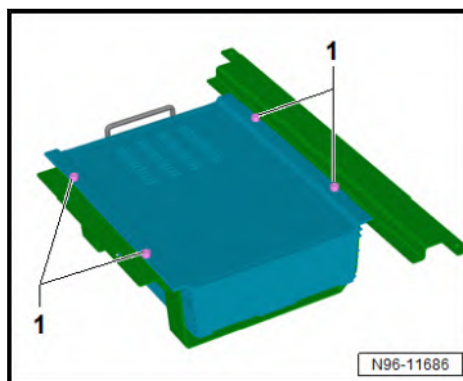
- On completion of repairs to the 110/230 V system the qualified electrician has a duty to perform tests after installation to ensure the safety of the system.
- If connected, pull out power supply cable 230/110 volts from exterior socket, 230 V, 110 V -U8-.
- Remove left front seat ⇒ General body repairs, interior; Rep. gr. 72; Front seats; Removing and installing front seat.
- Unscrew bolts -1-.



- Remove battery charger, 230 V or 110 V/12 V -A10- with bracket.
- Disconnect connector -1-.



- Unscrew bolts -1- while counterholding nuts from below.



Installing

Install in reverse order of removal, observing the following:

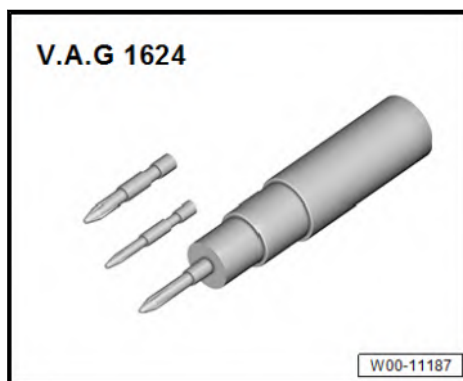
Specified torques

Component	Specified torque
Bolts on charging unit	6 Nm
Bolts on bracket	2 Nm

12.10.2 Removing and installing battery charger, 230 V or 110 V/12 V -A10-, under bench seat on front passenger side

Special tools and workshop equipment required

- ◆ Torque screwdriver -V.A.G 1624-

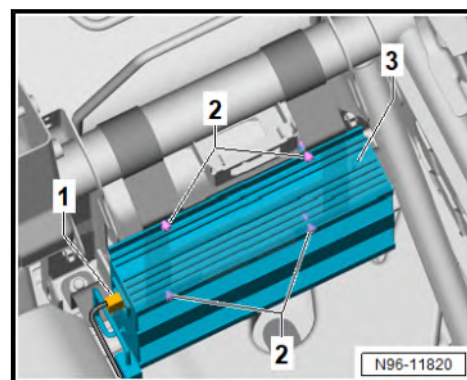


Removing

DANGER

Risk of fatal injury from electric shock

- On completion of repairs to the 110/230 V system the qualified electrician has a duty to perform tests after installation to ensure the safety of the system.
- If connected, pull out power supply cable 230/110 volts from exterior socket, 230 V, 110 V -U8-.
- Remove seat padding of bench seat ➔ General body repairs, interior; Rep. gr. 72; Front seats; Removing and installing front seat.
- Disconnect connector -1-.



- Unscrew bolts -2-.
- Remove battery charger, 230 V or 110 V/12 V -A10- -3-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolts -2-	4 Nm

12.10.3 Removing and installing bracket from battery charger, 230 V or 110 V/12 V -A10-, under bench seat on front passenger side

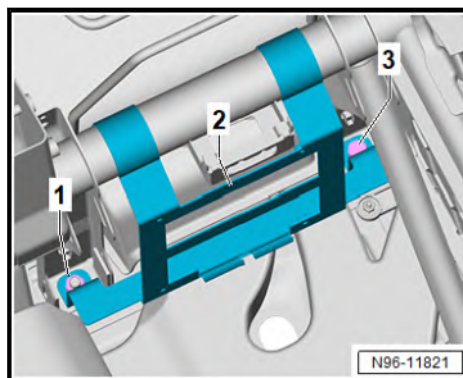
Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1410A-





Removing



- Remove battery charger, 230 or 110 Volt/12 Volt -A10- ➔ [page 558](#).
- Unscrew nut -1-.
- Unscrew bolt -3-.
- Remove retainer -2-.

Installing

Install in reverse order of removal, observing the following:

- Renew nut -1- and bolt -3- after removing.

Specified torques

Component	Specified torque
Bolt -1-	8 Nm
Nut -3-	8 Nm

12.10.4 Removing and installing fuse on battery charger, 230 V or 110 V/12 V - A10-

Removing

DANGER

Risk of fatal injury from electric shock

- On completion of repairs to the 110/230 V system the qualified electrician has a duty to perform tests after installation to ensure the safety of the system.

- If connected, pull out power supply cable 230/110 volts from exterior socket, 230 V, 110 V -U8-.

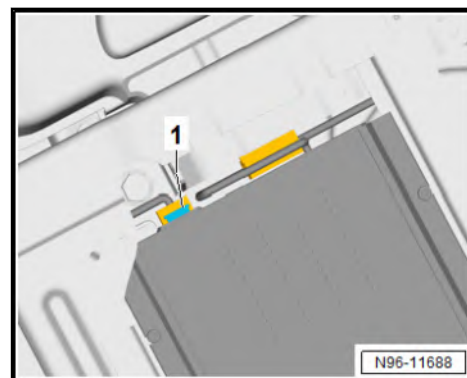
Battery charger under driver seat

- Push driver seat as far back as possible.

Battery charger under front bench seat

- Remove seat padding of bench seat ➔ General body repairs, interior; Rep. gr. 72; Front seats; Removing and installing front seat.

Continuation for all vehicles



- Pull out fuse -1-.

Installing

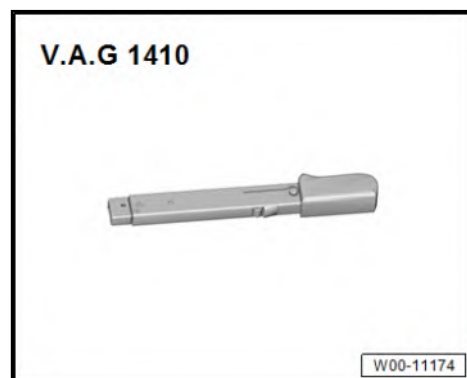
Install in reverse order of removal, observing the following:

- Corresponding fuse ⇒ Electronic parts catalogue (ETKA) or
⇒ webMANTIS for MAN TGE.

12.11 Removing and installing DC/AC converter with socket, 12 V-230 V -U13-

Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1410A-

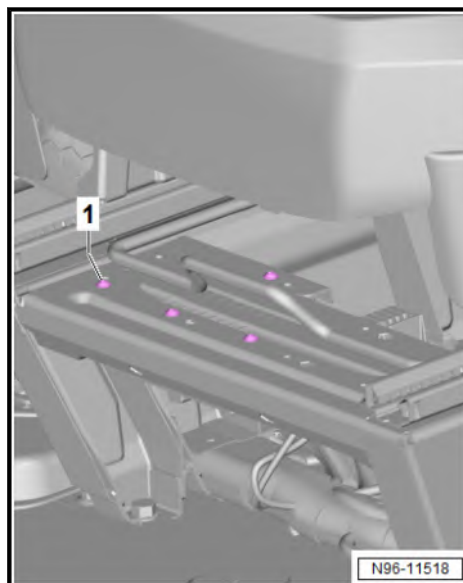


Removing

DANGER

Risk of fatal injury from electric shock

- On completion of repairs to the 110/230 V system the qualified electrician has a duty to perform tests after installation to ensure the safety of the system.
-
- ◆ Never open housing of DC/AC converter with socket, 12 V - 230 V -U13-.
 - ◆ The complete unit must always be renewed if the connector, wiring, 230 V socket or the DC/AC converter are defective.
 - Switch off ignition and all electrical consumers.

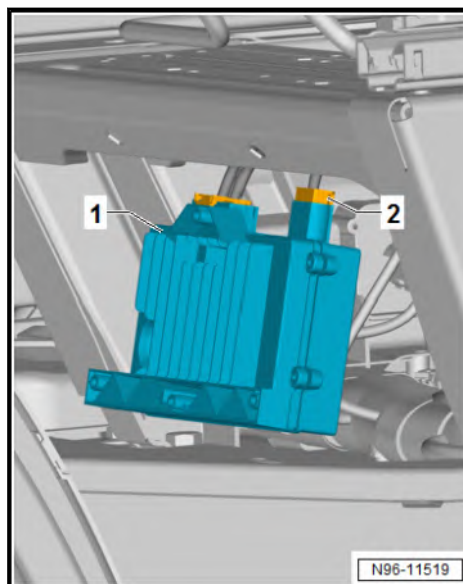


- Withdraw ignition key.
- If necessary, push back left front seat.
- Unscrew bolts -1-.

⚠ CAUTION

Risk of injury from residual voltage of the capacitors in the DC/AC converter.

- Observe the discharge time of 10 seconds for the capacitors.
- Disconnect connector -2-.



- Remove inverter -1-.

Installing

Install in reverse order of removal, observing the following:



Specified torques

Component	Specified torque
Bolts for DC/AC converter	8 Nm



97 – Wiring

1 Control units

⇒ [a1.1 nd installing special vehicle control unitJ608", page 564](#)

1.1 Removing and installing special vehicle control unit -J608-

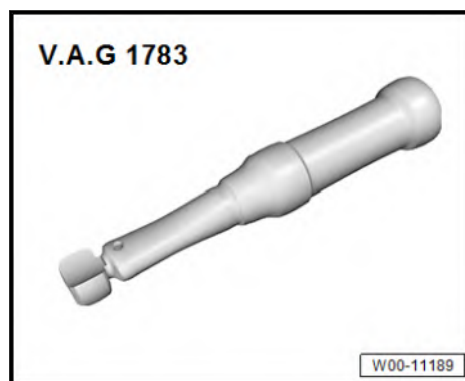
⇒ [a1.1.1 nd installing special vehicle control unitJ608", page 564](#)

⇒ [a1.1.2 nd installing bracket for special vehicle control unitJ608", page 566](#)

1.1.1 Removing and installing special vehicle control unit -J608-

Special tools and workshop equipment required

◆ Torque wrench -V.A.G 1783-



◆ Torque screwdriver -VAS 6494-



◆ Vehicle diagnostic tester

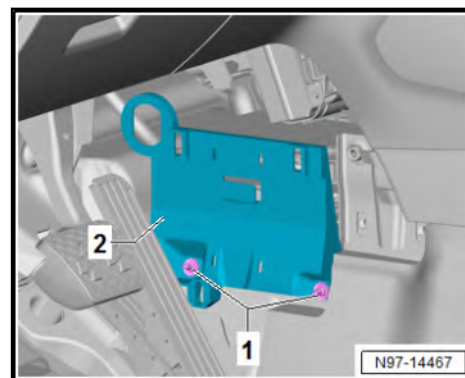
Removing

- Remove footwell cover on driver side ⇒ General body repairs, interior; Rep. gr. 68; Compartments/covers; Removing and installing footwell cover.
- Lay aside floor covering.

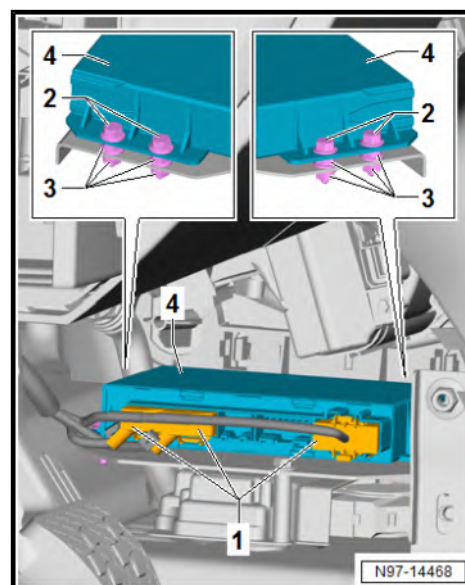
Procedure for replacement of special vehicle control unit -J608-

- Read coding with ⇒ Vehicle diagnostic tester.

Continued for all vehicles



- Unscrew nuts -1-.
- Remove retaining bracket -2-.
- Disconnect electrical connectors -1-.



- Unscrew bolts -2-. Counterhold nuts -3- while doing this.
- Remove special vehicle control unit -J608- -4-.

Installing

Install in the reverse order of removal, observing the following:

Procedure after replacement of special vehicle control unit - J608-

- Using ⇒ Vehicle diagnostic tester, enter coding and perform basic setting.

Specified torques

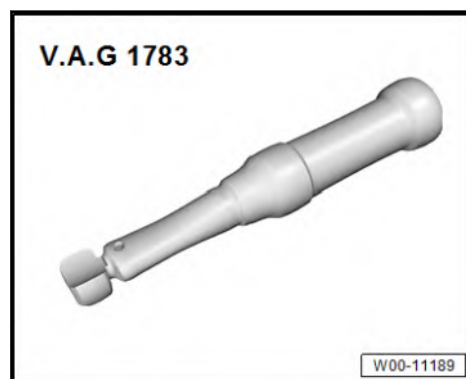
Component	Specified torque
Nuts -1- (plastic nuts)	1.5 Nm
Nuts -1- (metal nuts)	3.5 Nm
Bolts -2-	8 Nm



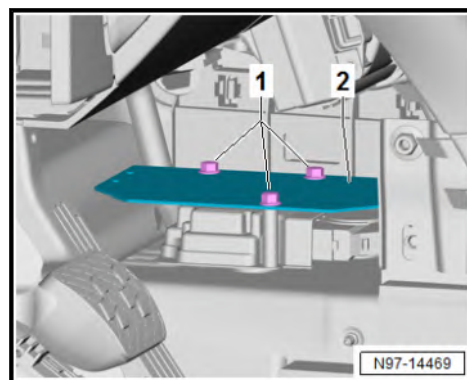
1.1.2 Removing and installing bracket for special vehicle control unit -J608-

Special tools and workshop equipment required

- ◆ Torque wrench -V.A.G 1783-



Removing



- Remove special vehicle control unit -J608- ➔ [page 564](#) .
- Unscrew bolts -1-.
- Remove bracket -2-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolts -1-	8 Nm

2 Control units

⇒ [a2.1 nd installing special vehicle control unitJ608", page 567](#)

2.1 Removing and installing special vehicle control unit -J608-

⇒ [a2.1.1 nd installing special vehicle control unitJ608, vehicles with single seat on front passenger side", page 567](#)

⇒ [a2.1.2 nd installing special vehicle control unitJ608, vehicles with bench seat on front passenger side", page 568](#)

⇒ [a2.1.3 nd installing bracket of special vehicle control unitJ608, vehicles with single seat on front passenger side", page 568](#)

⇒ [a2.1.4 nd installing bracket of special vehicle control unitJ608, vehicles with bench seat on front passenger side", page 569](#)

2.1.1 Removing and installing special vehicle control unit -J608-, vehicles with single seat on front passenger side

Special tools and workshop equipment required

- ◆ Vehicle diagnostic tester

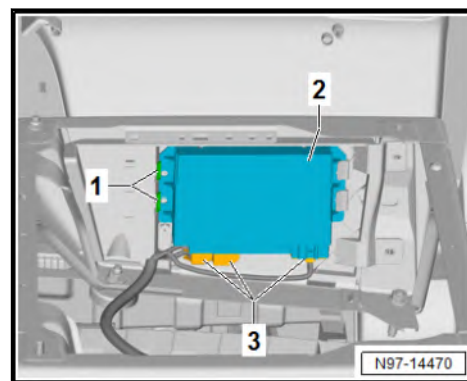
Removing

- Remove front passenger seat ⇒ General body repairs, interior; Rep. gr. 72; Front seats; Removing and installing front seat.

Procedure for replacement of special vehicle control unit -J608-

- Read coding with ⇒ Vehicle diagnostic tester.

Continued for all vehicles



- Release attachment points -1-. While doing so, detach special vehicle control unit -J608- -2-.
- Disconnect electrical connectors -3-.
- Remove special vehicle control unit -J608- -2-.

Installing

Install in reverse order of removal, observing the following:

Procedure after replacement of special vehicle control unit -J608-

- Using ⇒ Vehicle diagnostic tester, enter coding and perform basic setting.



2.1.2 Removing and installing special vehicle control unit -J608-, vehicles with bench seat on front passenger side

Special tools and workshop equipment required

- ◆ Vehicle diagnostic tester

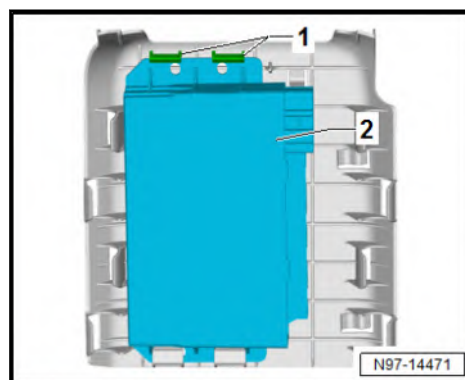
Removing

- Folded up seat padding of double bench seat.

Procedure for replacement of special vehicle control unit -J608-

- Read coding with ⇒ Vehicle diagnostic tester.

Continued for all vehicles



Remove bracket of special vehicle control unit -J608- ⇒ [page 569](#).

- Release attachment points -1-. While doing so, detach special vehicle control unit -J608- -2-.

Installing

Install in reverse order of removal, observing the following:

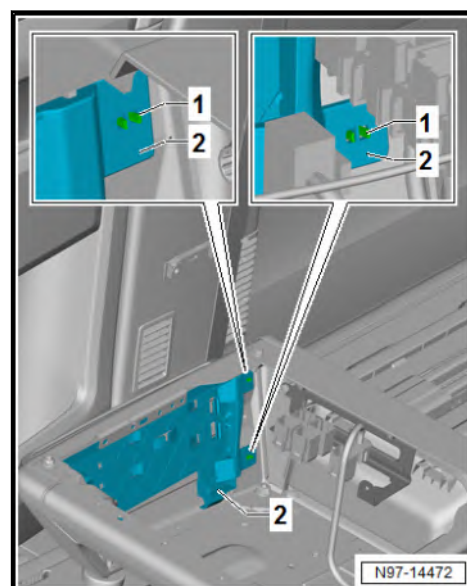
Procedure after replacement of special vehicle control unit -J608-

- Using ⇒ Vehicle diagnostic tester, enter coding and perform basic setting.

2.1.3 Removing and installing bracket of special vehicle control unit -J608-, ve-

ehicles with single seat on front passenger side

Removing



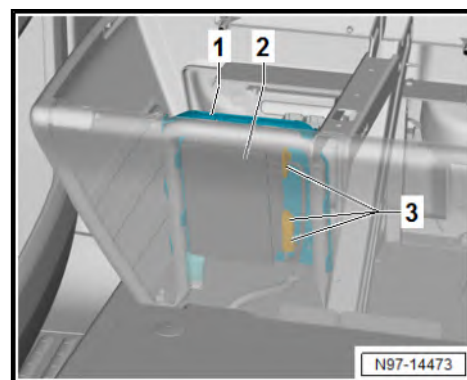
- Remove special vehicle control unit -J608- ➔ [page 567](#) .
- Release locking lugs -1-. While doing so, detach bracket of special vehicle control unit -J608- -2-.

Installing

Installation is carried out in reverse order.

2.1.4 Removing and installing bracket of special vehicle control unit -J608-, vehicles with bench seat on front passenger side

Removing



- Pull off bracket -1- from frame together with special vehicle control unit -J608- -2-.
- Disconnect electrical connectors -3-.
- Remove bracket -1- together with special vehicle control unit -J608- -2-.

Installing

Installation is carried out in reverse order.